



GE Healthcare

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Rev. 3



LOGIQ 9 Basic User Manual

R7.x.x

Operating Documentation

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Regulatory Requirement

This product complies with regulatory requirements of the following European Directive 93/42/EEC concerning medical devices.



This manual is a reference for the LOGIQ 9. It applies to all versions of the R7.x.x software for the LOGIQ 9 ultrasound system.



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Rev. 2	July 26, 2006	R7.0.0 Update
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Chapter 8	Rev. 3		

Please verify that you are using the latest revision of this document. Information pertaining to this document is maintained on GPC (GE Medical Systems Global Product Configuration). If you need to know the latest revision, contact your distributor, local GE Sales Representative or in the USA call the GE Ultrasound Clinical Answer Center at 1 800 682 5327 or 1 262 524 5698.

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Regulatory Requirements

Conformance Standards

The following classifications are in accordance with the IEC/EN 60601-1:6.8.1:

- According to 93/42/EEC Medical Device Directive, this is Class IIa Medical Device.
- According to IEC/EN 60601-1, Equipment is Class I, with BF or CF Applied Parts.
- According to CISPR 11, LOGIQ 9 with CRT is Group 1, Class B ISM Equipment; LOGIQ 9 with LCD is Group 1, Class A (commercial use only).
- According to IEC 60529, the footswitch rate IPx8 is suitable for use in surgical rooms.

This product complies with the regulatory requirement of the following:

- Council Directive 93/42/EEC concerning medical devices: the CE label affixed to the product testifies compliance to the Directive.

The location of the CE marking is shown in Chapter 2 of this manual.

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Conformance Standards (continued)

- International Electrotechnical Commission (IEC).
 - IEC/EN 60601-1 Medical Electrical Equipment, Part 1 General Requirements for Safety.
 - IEC/EN 60601-1-1 Safety requirements for medical electrical systems.
 - IEC/EN 60601-1-2 Electromagnetic compatibility - Requirements and tests.
 - IEC/EN 60601-1-4 Programmable electrical medical systems.
 - IEC 60601-2-37 Medical electrical equipment. Particular requirements for the safety of ultrasonic medical diagnostic and monitoring equipment.
- International Organization of Standards (ISO)
 - ISO 10993-1 Biological evaluation of medical devices.
- Underwriters' Laboratories, Inc. (UL), an independent testing laboratory.
 - UL 60601-1 Medical Electrical Equipment, Part 1 General Requirements for Safety.
- Canadian Standards Association (CSA).
 - CAN/CSA 22.2, 601.1 Medical Electrical Equipment, Part 1 General Requirements for Safety.
- NEMA/NIUM Acoustic Output Display Standard (NEMA US-3, 1998).
- Medical Device Good Manufacturing Practice Manual issued by the FDA (Food and Drug Administration, Department of Health, USA).

Certifications

- General Electric Healthcare Ultrasound is ISO 9001 and ISO 13485 certified.

Original Documentation

- The original document was written in English.

Software License Acknowledgements

- WindowBlinds™ OCX © Stardock®
- Roxio® Powered

Country Specific Approval

- JAPAN

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Chapter 1

Introduction

This chapter consists of information concerning indications for use/contraindications, contact information and how this documentation is organized.

System Overview

Attention

This manual contains necessary and sufficient information to operate the system safely. Advanced equipment training may be provided by a factory trained Applications Specialist for the agreed-upon time period.

Read and understand all instructions in this manual before attempting to use the LOGIQ 9 system.

Keep this manual with the equipment at all times. Periodically review the procedures for operation and safety precautions.

Documentation

LOGIQ 9 documentation consists of three manuals:

- The Basic User Manual (TRANSLATED) and Online Help (TRANSLATED) provides information needed by the user to operate the system safely. It describes the basic functions of the system, safety features, operating modes, measurements/calculations, probes, and user care and maintenance.

NOTE: *Probe information displayed on screen examples does not necessarily reflect the probes available on your ultrasound system. Please refer to the Probes chapter for a listing of available probes and features.*

- The Advanced Reference Manual (ENGLISH ONLY) contains data tables, such as OB and Acoustic Output tables.
- The Quick Guide (TRANSLATED) provides descriptions of basic system features and operation. It is intended to be used in conjunction with the Basic User Manual in order to provide the information necessary to operate the system safely. Quick Cards may also be provided with additional feature information.
- The User Guide is a condensed user instruction guide (translated into Swedish, Danish, Russian, Greek, Dutch, Finnish, Norwegian, and Polish).
- AIUM Booklet

NOTE: *The documentation kit provides the Quick Guide and Release Notes on paper and electronically and the Basic User Manual and Advanced Reference Manual are only provided in electronic format. The media includes English and all translations. Paper documentation may be ordered by using a form in the Quick Guide.*

The LOGIQ 9 manuals are written for users who are familiar with basic ultrasound principles and techniques. They do not include sonographic training or detailed clinical procedures.

Principles of Operation

Medical ultrasound images are created by computer and digital memory from the transmission and reception of mechanical high-frequency waves applied through a transducer. The mechanical ultrasound waves spread through the body, producing an echo where density changes occur. For example, in the case of human tissue, an echo is created where a signal passes from an adipose tissue (fat) region to a muscular tissue region. The echoes return to the transducer where they are converted back into electrical signals.

These echo signals are highly amplified and processed by several analog and digital circuits having filters with many frequency and time response options, transforming the high-frequency electrical signals into a series of digital image signals which are stored in memory. Once in memory, the image can be displayed in real-time on the image monitor. All signal transmission, reception and processing characteristics are controlled by the main computer. By selection from the system control panel, the user can alter the characteristics and features of the system, allowing a wide range of uses, from obstetrics to peripheral vascular examinations.

Transducers are accurate, solid-state devices, providing multiple image formats. The digital design and use of solid-state components provides highly stable and consistent imaging performance with minimal required maintenance. Sophisticated design with computer control offers a system with extensive features and functions which is user-friendly and easy to use.

Indications for Use

The LOGIQ 9 is intended for use by a qualified physician for ultrasound evaluation. Specific clinical applications and exam types include:

- Fetal/Obstetrics
- Gynecology
- Obstetric
- Abdominal (including GYN)
- Pediatric
- Small Organ (including breast, testes, thyroid)
- Neonatal Cephalic
- Adult Cephalic
- Peripheral Vascular
- Musculo-skeletal Conventional and Superficial
- Urology (including prostate)
- Transrectal
- Transvaginal
- Intraoperative (abdominal, thoracic, vascular and neurological)

Contraindication

The LOGIQ 9 ultrasound system is not intended for ophthalmic use or any use causing the acoustic beam to pass through the eye.

Prescription Device

CAUTION: United States law restricts this device to sale or use by, or on the order of a physician.

Contact Information

Contacting GE Medical Systems Ultrasound

For additional information or assistance, please contact your local distributor or the appropriate support resource listed on the following pages:

INTERNET

<http://www.gehealthcare.com>

http://www.gehealthcare.com/usen/ultrasound/products/probe_care.html

USA

GE Healthcare TEL: (1) 800-437-1171
Ultrasound Service Engineering FAX: (1) 414-721-3865
9900 Innovation Drive
Wauwatosa, WI 53226

Clinical Questions

For information in the United States, Canada, Mexico and parts of the Caribbean, call the Customer Answer Center
TEL: (1) 800-682-5327 or (1) 262-524-5698

In other locations, contact your local Applications, Sales or Service Representative.

Service Questions

For service in the United States, call GE CARES

TEL: (1) 800-437-1171

In other locations, contact your local Service Representative.

Accessories

Catalog Requests

To request the latest GE Accessories catalog or equipment brochures in the United States, call the Response Center

TEL: (1) 800-643-6439

In other locations, contact your local Applications, Sales or Service Representative.

Contacting GE Medical Systems Ultrasound (continued)

Placing an Order To place an order, order supplies or ask an accessory-related question in the United States, call the GE Access Center

TEL: (1) 800-472-3666

In other locations, contact your local Applications, Sales or Service Representative.

CANADA GE Healthcare TEL: (1) 800-664-0732
Ultrasound Service Engineering
9900 Innovation Drive
Wauwatosa, WI 53226
Customer Answer Center TEL: (1) 262-524-5698

LATIN & SOUTH AMERICA GE Healthcare TEL: (1) 262-524-5300
Ultrasound Service Engineering
9900 Innovation Drive
Wauwatosa, WI 53226
Customer Answer Center TEL: (1) 262-524-5698

EUROPE GE Ultraschall TEL: 0130 81 6370 toll free
Deutschland GmbH & Co. KG TEL: (33) 130.831.300
Beethovenstrasse 239 FAX: (49) 212.28.02.431
Postfach 11 05 60
D-42655 Solingen

ASIA GE Ultrasound Asia (Singapore) TEL: 65-291 8528
Service Department - Ultrasound FAX: 65-272-3997
298 Tiong Bahru Road #15-01/06
Central Plaza
Singapore 169730

JAPAN
GE Yokogawa Medical Systems TEL: (81) 426-48-2950
Customer Service Center FAX: (81) 426-48-2902

Contacting GE Medical Systems Ultrasound (continued)

ARGENTINA	GEME S.A. TEL: (1) 639-1619 Miranda 5237 FAX: (1) 567-2678 Buenos Aires - 1407
AUSTRIA	GE GesmbH Medical Systems Austria TEL: 0660 8459 toll free Prinz Eugen Strasse 8/8 FAX: +43 1 505 38 74 A-1040 WIEN TLX: 136314
BELGIUM	GE Medical Systems Benelux TEL: 0 800 11733 toll free Gulkenrodestraat 3 FAX: +32 0 3 320 12 59 B-2160 WOMMELGEM TLX: 72722
BRAZIL	GE Sistemas Medicos TEL: 0800-122345 Av Nove de Julho 5229 FAX: (011) 3067-8298 01407-907 Sao Paulo SP
DENMARK	GE Medical Systems TEL: +45 4348 5400 Fabriksparken 20 FAX: +45 4348 5399 DK-2600 GLOSTRUP
FRANCE	GE Medical Systems TEL: 05 49 33 71 toll free 738 rue Yves Carmen FAX: +33 1 46 10 01 20 F-92658 BOULOGNE CEDEX
GERMANY	GE Ultraschall TEL: 0130 81 6370 toll free Deutschland GmbH & Co. KG TEL: (49) 212.28.02.207 Beethovenstrasse 239 FAX: (49) 212.28.02.431 Postfach 11 05 60 D-42655 Solingen
GREECE	GE Medical Systems Hellas TEL: +30 1 93 24 582 41, Nikolaou Plastira Street FAX: +30 1 93 58 414 G-171 21 NEA SMYRNI
ITALY	GE Medical Systems Italia TEL: 1678 744 73 toll free Via Monte Albenza 9 FAX: +39 39 73 37 86 I-20052 MONZA TLX: 3333 28
LUXEMBOURG	TEL: 0800 2603 toll free

Contacting GE Medical Systems Ultrasound (continued)

MEXICO	GE Sistemas Medicos de Mexico S.A. de C.V. Rio Lerma #302, 1° y 2° Pisos TEL: (5) 228-9600 Colonia Cuauhtemoc FAX: (5) 211-4631 06500-Mexico, D.F.
NETHERLANDS	GE Medical Systems Nederland B.V. TEL: 06 022 3797 toll free Atoomweg 512 FAX: +31 304 11702 NL-3542 AB UTRECHT
POLAND	GE Medical Systems Polska TEL: +48 2 625 59 62 Krzywickiego 34 FAX: +48 2 615 59 66 P-02-078 WARSZAWA
PORTUGAL	GE Medical Systems Portuguesa S.A. TEL: 05 05 33 7313 toll free Rua Sa da Bandeira, 585 FAX: +351 2 2084494 Apartado 4094 TLX: 22804 P-4002 PORTO CODEX
RUSSIA	GE VNIIEM TEL: +7 095 956 7037 Mantulinskaya Ul. 5A FAX: +7 502 220 32 59 123100 MOSCOW TLX: 613020 GEMED SU
SPAIN	GE Medical Systems Espana TEL: 900 95 3349 toll free Hierro 1 Arturo Gimeno FAX: +34 1 675 3364 Poligono Industrial I TLX: 22384 A/B GEMDE E-28850 TORREJON DE ARDOZ
SWEDEN	GE Medical Systems TEL: 020 795 433 toll free PO-BOX 1243 FAX: +46 87 51 30 90 S-16428 KISTA TLX: 12228 CGRSWES
SWITZERLAND	GE Medical Systems (Schweiz) AG TEL: 155 5306 toll free Sternmattweg 1 FAX: +41 41 421859 CH-6010 KRIENS
TURKEY	GE Medical Systems Turkiye A.S. TEL: +90 212 75 5552 Mevluk Pehliran Sodak FAX: +90 212 211 2571 Yilmaz Han, No 24 Kat 1 Gayretteppe ISTANBUL

Contacting GE Medical Systems Ultrasound (continued)

UNITED KINGDOM	GE Medical Systems TEL: 0800 89 7905 toll free Coolidge House FAX: +44 753 696067 352 Buckingham Avenue SLOUGH Berkshire SL1 4ER
OTHER COUNTRIES	NO TOLL FREE TEL: international code + 33 1 39 20 0007

Manufacturer

GE Medical Systems
Ultrasound & Primary Care
Diagnostics, LLC
9900 Innovation Drive
Wauwatosa, WI 53226
U.S.A

Chapter 2

Safety

*Describes the safety and regulatory information
pertinent for operating this ultrasound system.*

Safety Precautions

Precaution Levels

Icon description

Various levels of safety precautions may be found on the equipment and different levels of concern are identified by one of the following flag words and icons which precede the precautionary statement.



Indicates that a specific hazard is known to exist which through inappropriate conditions or actions will cause:

- Severe or fatal personal injury
- Substantial property damage.



Indicates that a specific hazard is known to exist which through inappropriate conditions or actions may cause:

- Severe personal injury
- Substantial property damage.



Indicates that a potential hazard may exist which through inappropriate conditions or actions will or can cause:

- Minor injury
- Property damage.

NOTE: *Indicates precautions or recommendations that should be used in the operation of the ultrasound system, specifically:*

- *Maintaining an optimum system environment*
- *Using this Manual*
- *Notes to emphasize or clarify a point.*

Hazard Symbols

Icon Description

Potential hazards are indicated by the following icons:

Table 2-1: Potential Hazards

Icon	Potential Hazard	Usage	Source
	• Patient/user infection due to contaminated equipment.	• Cleaning and care instructions • Sheath and glove guidelines	ISO 7000 No. 0659
	• Electrical micro-shock to patient, e.g., ventricular	• Probes • ECG, if applicable • Connections to back panel	
	• Console, accessories or optional storage devices that can fall on patient, user, or others. • Collision with persons or objects may result in injury while maneuvering or during system transport. • Injury to user from moving the console.	• Moving • Using brakes • Transporting	
	• Patient injury or tissue damage from ultrasound radiation.	• ALARA, the use of Power Output following the 'as low as reasonably achievable' principle	
	• Risk of explosion if used in the presence of flammable anesthetics.	• Flammable anesthetic	
	• Patient/user injury or adverse reaction from fire or smoke. • Patient/user injury from explosion and fire.	• Replacing fuses • Outlet guidelines	

Important Safety Considerations

The following topic headings (Patient Safety, and Equipment and Personnel Safety) are intended to make the equipment user aware of particular hazards associated with the use of this equipment and the extent to which injury can occur if precautions are not observed. Additional precautions may be provided throughout the manual.



Improper use can result in serious injury. The user must be thoroughly familiar with the instructions and potential hazards involving ultrasound examination before attempting to use the device. Training assistance is available from GE Medical Systems if needed.

The equipment user is obligated to be familiar with these concerns and avoid conditions that could result in injury.

Patient Safety

Related Hazards



WARNING

The concerns listed can seriously affect the safety of patients undergoing a diagnostic ultrasound examination.

Patient identification

Always include proper identification with all patient data and verify the accuracy of the patient's name and ID numbers when entering such data. Make sure correct patient ID is provided on all recorded data and hard copy prints. Identification errors could result in an incorrect diagnosis.

Diagnostic information

Equipment malfunction or incorrect settings can result in measurement errors or failure to detect details within the image. The equipment user must become thoroughly familiar with the equipment operation in order to optimize its performance and recognize possible malfunctions. Applications training is available through the local GE representative. Added confidence in the equipment operation can be gained by establishing a quality assurance program.



CAUTION

Allowing the machine to transmit acoustic output with the probe not in use (or in its holder) can cause the transducer to build up heat. Preset Auto Freeze to freeze the image when the machine is not in use.

Related Hazards (continued)

Mechanical hazards

The use of damaged probes or improper use and manipulation of intracavity probes can result in injury or increased risk of infection. Inspect probes often for sharp, pointed, or rough surface damage that could cause injury or tear protective barriers. Never use excessive force when manipulating intracavity probes. Become familiar with all instructions and precautions provided with special purpose probes.

The use of damaged probes can result in injury or increased risk of infection. Inspect probes often for sharp, pointed, or rough surface damage that could cause injury or tear protective barriers. Become familiar with all instructions and precautions provided with special purpose probes.



Electrical Hazard

A damaged probe can also increase the risk of electric shock if conductive solutions come in contact with internal live parts. Inspect probes often for cracks or openings in the housing and holes in and around the acoustic lens or other damage that could allow liquid entry. Become familiar with the probe's use and care precautions outlined in *Probes and Biopsy*.



CAUTION

Ultrasound transducers are sensitive instruments which can easily be damaged by rough handling. Take extra care not to drop transducers and avoid contact with sharp or abrasive surfaces. A damaged housing, lens or cable can result in patient injury or serious impairment or operation.

Related Hazards (continued)



Ultrasound can produce harmful effects in tissue and potentially result in patient injury. Always minimize exposure time and keep ultrasound levels low when there is no medical benefit. Use the principle of ALARA (As Low As Reasonably Achievable), increasing output only when needed to obtain diagnostic image quality. Observe the acoustic output display and be familiar with all controls affecting the output level. See the *Bioeffects* section of the *Acoustic Output* chapter in the *Advanced Reference Manual* for more information.



Do not use with Defibrillator.

This equipment does not have a defibrillator approved applied part.

Training

It is recommended that all users receive proper training in applications before performing them in a clinical setting. Please contact the local GE representative for training assistance.

ALARA training is provided by GE Application Specialists. The ALARA education program for the clinical end-user covers basic ultrasound principles, possible biological effects, the derivation and meaning of the indices, ALARA principles, and examples of specific applications of the ALARA principle.

Equipment and Personnel Safety

Related Hazards



This equipment contains dangerous voltages that are capable of serious injury or death.

If any defects are observed or malfunctions occur, stop operating the equipment and perform the proper action for the patient. Inform a qualified service person and contact a Service Representative for information.

There are no user serviceable components inside the console. Refer all servicing to qualified service personnel only.



Only approved and recommended peripherals and accessories should be used.

All peripherals and accessories must be securely mounted to the LOGIQ 9.



The LOGIQ 9 is not intended to be used as a storage device; backup of the Patient and Image Database is your institution's responsibility. GE is NOT responsible for any lost patient information or for lost images.



The concerns listed below can seriously affect the safety of equipment and personnel during a diagnostic ultrasound examination.



Risk of explosion if used in the presence of flammable anesthetics.



This equipment provides no special means of protection from high frequency (HF) burns that may result from using an electrosurgical unit (ESU). To reduce the risk of HF burns, avoid contact between the patient and ultrasound transducer while operating the ESU. Where contact cannot be avoided, as in the case of TEE monitoring during surgery, make sure the transducer is not located between the ESU active and dispersive electrodes and keep the ESU cables away from the transducer cable.

Related Hazards (continued)



To avoid injury:

- Do not remove protective covers. No user serviceable parts are inside. Refer servicing to qualified service personnel.
- To assure adequate grounding, connect the attachment plug to a reliable (hospital grade) grounding outlet (having equalization conductor ⚡).
- Never use any adaptor or converter of a three-prong-to-two-prong type to connect with a mains power plug. The protective earth connection will loosen.
- Do not place liquids on or above the console. Spilled liquid may contact live parts and increase the risk of shock.
- Plug any peripherals into the LOGIQ 9 AC power outlet.



Do not use this equipment if a safety problem is known to exist. Have the unit repaired and performance verified by qualified service personnel before returning to use.



The system must be supplied from an adequately rated electrical circuit. The capacity of the supply circuit must be as specified.



For patient and personnel safety, be aware of biological hazards while performing invasive procedures. To avoid the risk of disease transmission:

- Use protective barriers (gloves and probe sheaths) whenever possible. Follow sterile procedures when appropriate.
- Thoroughly clean probes and reusable accessories after each patient examination and disinfect or sterilize as needed. Refer to *Probes and Biopsy* for probe use and care instructions.
- Follow all infection control policies established by your office, department or institution as they apply to personnel and equipment.

Related Hazards (continued)



Contact with natural rubber latex may cause a severe anaphylactic reaction in persons sensitive to the natural latex protein. Sensitive users and patients must avoid contact with these items. Refer to package labeling to determine latex content and FDA's March 29, 1991 Medical Alert on latex products.



Allowing the machine to transmit acoustic output with the probe not in use (or in its holder) can cause the transducer to build up heat. Always turn off acoustic output or freeze the image when not in use. The system's acoustic output remains transmitting when the user controls are being used.

The system is equipped with an Auto Freeze feature which disables acoustic output and freezes the image when the machine is not in use.



When you move the Control Panel up/down with the monitor, place BOTH hands on the Control Panel. Touching other moving parts other than the Control Panel may cause personal injury.



To avoid injury or system damage, NEVER place any object or liquid on the monitor.



Archived data is managed at the individual sites. Performing data backup (to any device) is recommended.

Related Hazards (continued)



Do not unpack the LOGIQ 9. This must be performed by qualified service personnel only.



DO NOT load non-system software on the system computer.



To minimize accidental loss of data, perform EZBackup and Backup on a regular basis.

1. First, perform EZBackup to save the images.
2. Next, perform Backup at Utility -> Backup/Restore. Enable the following checkboxes under Backup:
 - Patient Archive
 - Report Archive
 - User defined configuration
 - Voice Training

Related Hazards (LCD Monitor)



- **DO NOT** place a finger, hand or any object on the joint of the monitor or monitor arm to avoid injury when moving the monitor and monitor arm.
- To avoid result of injury or system damage, **NEVER** place any object or liquid on the monitor, whether in the home or flip down/transport position.
- **DO NOT** place any object on the ventilation slots on the upper rear side of the monitor cabinet. Blocking the ventilation slots prevents proper airflow and may result in fire, electric shock, or equipment damage.
- Keep small objects or liquids away from the LCD monitor. Small objects or liquid may accidentally fall through the ventilation slots into the cabinet or spill into the cabinet which may result in fire, electric shock, or equipment damage.

If an object or liquid falls/spills into the cabinet, unplug the system immediately. Call a Service Representative for information.

- **DO NOT** scratch or press on the panel with any sharp objects, such as a pencil or pen, as this may result in damage to the panel.
- To avoid injury or damage, make sure nothing is within the range of motion before moving the monitor and monitor arm. This includes both objects and people.



- Pay attention to the monitor arm position to avoid hitting it against anyone or anything.
- Before moving the system to another location, be sure to lock LCD/LCD arm in transport position.

Device Labels (LOGIQ 9)

Label Icon Description

The following table describes the purpose and location of safety labels and other important information provided on the equipment.

Table 2-2: Label Icons

Label/Icon	Purpose/Meaning	Location
Identification and Rating Plate	• Manufacturer's name and country of origin • Date of manufacture • Model and serial numbers • Electrical ratings (Volts, Amps, phase, and frequency)	See 'Warning Label Locations' on page 2-30 for more information.
Type/Class Label	Used to indicate the degree of safety or protection.	
IP Code (IPX8)	Indicates the degree of protection provided by the enclosure per IEC60 529. Can be used in operating room environment.	Footswitch
	Type BF Applied Part (man in the box) symbol is in accordance with IEC 60878-02-03.	Probe and PCG marked Type BF
	"ATTENTION" - Consult accompanying documents" is intended to alert the user to refer to the operator manual or other instructions when complete information cannot be provided on the label.	Various
	"CAUTION" - Dangerous voltage" (the lightning flash with arrowhead) is used to indicate electric shock hazards.	Inside of console
	"Mains OFF" indicates the power off position of the mains power breaker.	See the Console Overview section for location information.
	"Mains ON" indicates the power on position of the mains power breaker.	See the Console Overview section for location information.

Table 2-2: Label Icons

Label/Icon	Purpose/Meaning	Location
	"ON" indicates the power on position of the power switch. CAUTION: This Power Switch DOES NOT ISOLATE Mains Supply. "Standby" indicates the power standby position of the power switch. CAUTION: This Power Switch DOES NOT ISOLATE Mains Supply.	See the Console Overview section for location information.
	"Protective Earth" indicates the protective earth (grounding) terminal.	Internal
	"Equipotentiality" indicates the terminal to be used for connecting equipotential conductors when interconnecting (grounding) with other equipment. Connection of additional protective earth conductors or potential equalization conductors is not necessary in most cases and is only recommended for situations involving multiple equipment in a high-risk patient environment to provide assurance that all equipment is at the same potential and operates within acceptable leakage current limits. An example of a high-risk patient would be a special procedure where the patient has an accessible conductive path to the heart such as exposed cardiac pacing leads.	Rear of console
	Alternating Current symbol is in accordance with IEC 60878-01-14.	Rear Panel, Circuit breaker label of console and front panel (if applicable).
	This symbol indicates that waste electrical and electronic equipment must not be disposed of as unsorted municipal waste and must be collected separately. Please contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.	Rear Panel

Label Icon Description (continued)

Classifications

Type of protection against electric shock

Class I Equipment (*1)

Degree of protection against electric shock

Type BF Applied part (*2) (for PCG, Probes marked with BF symbol)

Continuous Operation

System is Ordinary Equipment (IPX0)

Footswitch is IPX8

***1. Class I Equipment**

EQUIPMENT in which protection against electric shock does not rely on BASIC INSULATION only, but includes an earth ground. This additional safety precaution prevents exposed metal parts from becoming LIVE in the event of an insulation failure.

***2. Type BF Applied Part**

TYPE BF APPLIED PART providing a specified degree of protection against electric shock, with particular regard to allowable LEAKAGE CURRENT.

Table 2-3: Type BF Equipment

	Normal Mode	Single fault condition
Patient leakage current	Less than 100 microA	Less than 500 microA

EMC (Electromagnetic Compatibility)

NOTE: *This equipment generates, uses and can radiate radio frequency energy. The equipment may cause radio frequency interference to other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, this product complies with emissions limits for a Group 1, Class B Medical Devices Directive as stated in EN 60601-1-2 (except for LCD option, which complies with Group 1, Class A Medical Devices Directive as stated in EN 60601-1-2). However, there is no guarantee that interference will not occur in a particular installation.*

NOTE: *If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):*

- *reorient or relocate the affected device(s)*
- *increase the separation between the equipment and the affected device*
- *power the equipment from a source different from that of the affected device*
- *consult the point of purchase or service representative for further suggestions.*

NOTE: *The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.*

NOTE: *To comply with the regulations on electromagnetic interference for a Class B FCC Device, all interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference in violation of the FCC regulations.*

EMC (Electromagnetic Compatibility) (continued)

EMC Performance

All types of electronic equipment may characteristically cause electromagnetic interference with other equipment, either transmitted through air or connecting cables. The term EMC (Electromagnetic Compatibility) indicates the capability of equipment to curb electromagnetic influence from other equipment and at the same time not affect other equipment with similar electromagnetic radiation from itself.

Proper installation following the service manual is required in order to achieve the full EMC performance of the product.

The product must be installed as stipulated in 4.2, Notice upon Installation of Product.

In case of issues related to EMC, please call your service personnel.

The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.

EMC Performance (continued)



Do not use devices which intentionally transmit RF signals (cellular phones, transceivers, or radio controlled products), other than those supplied by GE (wireless microphone, broadband over power lines, for example) unless intended for use with this system, in the vicinity of this equipment as it may cause performance outside the published specifications.

Keep power to these devices turned off when near this equipment.

Medical staff in charge of this equipment is required to instruct technicians, patients and other people who may be around this equipment to fully comply with the above regulation.

Portable and mobile radio communications equipment (e.g. two-way radio, cellular/cordless telephones, wireless computer networks), other than those supplied by GE (wireless microphone for VoiceScan), should be used no closer to any part of this system, including cables, than determined according to the following method:

Table 2-4: Portable and mobile radio communications equipment distance requirements

Frequency Range:	150 kHz - 80 MHz	80 MHz - 800 MHz	800 MHz - 2.5 GHz
Calculation Method:	$d = [3.5/V_1]$ square root of P	$d = [3.5/E_1]$ square root of P	$d = [7/E_1]$ square root of P
Where: d= separation distance in meters, P = rated power of the transmitter, V_1 =compliance value for conducted RF, E_1 = compliance value for radiated RF			
If the maximum transmitter power in watts is rated	The separation distance in meters should be		
5	2.6	2.6	5.2
20	5.2	5.2	10.5
100	12.0	12.0	24.0

Notice upon Installation of Product

Separation distance and effect from fixed radio communications equipment: field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast transmitter cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the ultrasound system is used exceeds the applicable RF compliance level as stated in the immunity declaration, the ultrasound system should be observed to verify normal operation. If abnormal operation is observed, additional measures may be necessary, such as re-orienting or relocating the ultrasound system or using an RF shielded examination room may be necessary.

1. Use either power supply cords provided by GE Medical Systems or ones designated by GE Medical Systems. Products equipped with a power source plug should be plugged into the fixed power socket which has the protective grounding conductor. Never use any adaptor or converter to connect with a power source plug (e.g. three-prong-to-two-prong converter).
2. Locate the equipment as far away as possible from other electronic equipment.
3. Be sure to use only the cables provided by or designated by GE Medical Systems. Connect these cables following the installation procedures (e.g. wire power cables separately from signal cables).
4. Lay out the main equipment and other peripherals following the installation procedures described in the Option Installation manuals.

General Notice

1. Designation of Peripheral Equipment Connectable to This Product.

The equipment indicated in Chapter 18 can be hooked up to the product without compromising its EMC performance.

Avoid using equipment not designated in the list. Failure to comply with this instruction may result in poor EMC performance of the product.

2. Notice against User Modification

The user should never modify this product. User modifications may cause degradation in EMC performance.

Modification of the product includes changes in:

- a. Cables (length, material, wiring, etc.)
 - b. System installation/layout
 - c. System configuration/components
 - d. Securing system parts (cover open/close, cover screwing)
3. Operate the system with all covers closed. If a cover is opened for some reason, be sure to shut it before starting/resuming operation.
 4. Operating the system with any cover open may affect EMC performance.

Peripheral Update for EC countries

The following is intended to provide the users in EC countries with updated information concerning the connection of the LOGIQ 9 to image recording and other devices or communication networks.

The LOGIQ 9 has been verified for overall safety, compatibility and compliance with the following on-board image recording devices:

- Sony UP-D895 Digital Printer
- Sony UP-D897 Digital Printer
- Sony UP-D21MD Color Printer
- Sony UP-D23MD Color Printer
- Sony UP-D50 Color Printer
- Sony UP-D55 Color Printer
- Wired/Wireless Microphone
- Panasonic DVD Video Recorder (LQ-MD800)
- Sony SVO9500MD2 NTSC/MDP2 PAL Video Cassette Recorder
- Mitsubishi HS-MD3000U NTSC VCR and HS-MD3000E PAL VCR
- Hewlett Packard 970C, 990, 6122 Inkjet Printers
- Hewlett Packard 5650, 6540 Deskjet Printers
- Flock of Birds (for Tru3D)
- USB Flash Drive (Memory Stick) NDL 256MByte 066E0690
- Tally 8006, Xerox 8200/8400 Network Printers
- External MO Drive, Fujitsu DynaMO 1300 U2 Pocket Magnetic Optical (MO) Drive
- Hard Disk Drive, LaCie USB Hard Disk Drive, p/n300765

The LOGIQ 9 has also been verified for compatibility, and compliance for connection to a local area network (LAN) via the rear panel Ethernet connection, provided the LAN components are IEC/EN 60950 compliant.

Connection may also be made to a CE Marked and IEC/EN 60950 compliant modem using one of the serial or USB ports on the system.

The LOGIQ 9 may also be used safely while connected to devices other than those recommended above if the devices and their specifications, installation, and interconnection with the system conform to the requirements of IEC/EN 60601-1-1.

Peripheral Update for EC countries (continued)

General precautions for installing an alternate on-board device would include:

1. The added device must have appropriate safety standard conformance and CE Marking.
2. The total power consumption of the added devices, which connect to the LOGIQ 9 and are used simultaneously, must be less than or equal to the rated supply of the LOGIQ 9.
3. There must be adequate heat dissipation and ventilation to prevent overheating of the device.
4. There must be adequate mechanical mounting of the device and stability of the combination.
5. Risk and leakage current of the combination must comply with IEC/EN 60601-1-1.
6. Electromagnetic emissions and immunity of the combination must conform to IEC/EN 60601-1-2.

General precautions for installing an alternate off-board, remote device or a network would include:

1. The added device(s) must have appropriate safety standard conformance and CE Marking.
2. The added device(s) must be used for their intended purpose having a compatible interface.
3. Signal or mains isolation devices and additional protective earth may be needed to assure compliance with IEC/EN 60601-1-1.



The connection of equipment or transmission networks other than as specified in the user instructions can result in an electric shock hazard or equipment malfunction. Substitute or alternate equipment and connections requires verification of compatibility and conformity to IEC/EN 60601-1-1 by the installer. Equipment modifications and possible resulting malfunctions and electromagnetic interference are the responsibility of the owner.

Declaration of Emissions

This system is suitable for use in the following environment. The user must assure that it is used only in the electromagnetic environment as specified.

LOGIQ 9 with CRT

Table 2-5: Declaration of Emissions (with CRT)

Emission Type	Compliance	Electromagnetic Environment
CISPR 11 Radio Frequency (RF) Emissions	Group 1 Class B	This system uses RF energy only for its internal function. Therefore, RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. It is suitable for use in all establishments.
CISPR-11 Conducted Emissions	Class B	230Vac 50/60Hz, 100Vac 50/60Hz

LOGIQ 9 with LCD Option

Table 2-6: Declaration of Emissions (with LCD Option)

Emission Type	Compliance	Electromagnetic Environment
CISPR 11 Radio Frequency (RF) Emissions	Group 1 Class A	This system uses RF energy only for its internal function. Therefore, RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. It is suitable for use in all establishments, other than domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
CISPR-11 Conducted Emissions	Class B	230Vac 50/60Hz, 100Vac 50/60Hz

Declaration of Immunity

This system is suitable for use in the following environment. The user must assure that the system is used according to the specified guidance and only in the electromagnetic environment listed.

Table 2-7: Declaration of Immunity

Immunity Type	Equipment Capability	Regulatory Acceptable Level	EMC Environment and Guidance
IEC 61000-4-2 Static discharge (ESD)	± 9 kV contact ± 12 kV air	± 9 kV contact ± 12 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%. Mains power quality should be that of a typical commercial and/or hospital environment. If the user requires continued operation during power mains interruptions, it is recommended that the system be powered from a UPS or a battery. NOTE: UT is the a.c. mains voltage prior to application of the test level. Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial and/or hospital environment. Separation distance to radio communication equipment must be maintained according to the method below. Interference may occur in the vicinity of equipment marked with the symbol: 
IEC 61000-4-4 Electrical fast transient/burst	± 2 kV for mains ± 1 kV for SIP/SOP	± 2 kV for mains ± 1 kV for SIP/SOP	
IEC 61000-4-5 Surge Immunity	± 1 kV differential ± 2 kV common	± 1 kV differential ± 2 kV common	
IEC 61000-4-11 Voltage dips, short interruptions and voltage variations on mains supply	< 50 _T (> 95% dip) for 0.5 cycle; 400 _T (60 0ip) for 5 cycles; 700 _T (30 0ip) for 25 cycles; < 50 _T (>95% dip) for 5 sec	< 50 _T (> 95% dip) for 0.5 cycle; 400 _T (60 0ip) for 5 cycles; 700 _T (30 0ip) for 25 cycles; < 50 _T (>95% dip) for 5 sec	
IEC 61000-4-8 Power frequency (50/60 Hz) magnetic field	6 A/m	6 A/m	
IEC 61000-4-6 Conducted RF	3 V _{RMS} 150 kHz - 80 MHz	3 V _{RMS} 150 kHz - 80 MHz	
IEC 61000-4-3 Radiated RF	3 V/m 80 MHz - 2.5 GHz	3 V/m 80 MHz - 2.5 GHz	
IEC 61000-3-2 Harmonic Disturbance Emissions	IEC Limits	230Vac	
IEC 61000-3-3 Voltage Fluctuations Emissions	IEC Limits	230Vac 50Hz, 100Vac 50Hz	
NOTE: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people. If noise generated from other electronic equipment is near the probe's center frequency, noise may appear on the image. Good power line isolation is required.			

Patient Environmental Devices



Figure 2-1. Patient Environmental Devices

- | | |
|---|---------------------------------------|
| 1. Peripheral Device (Signals I/O Port, Power In) | 15. Power In |
| 2. Front Panel (Signal I/O Port, Power Out) | 16. Signals I/O Port |
| 3. VoiceScan USB Port | 17. Power In |
| 4. USB Flash Drives (2) | 18. InSite (Signal I/O Port) |
| 5. CD/DVD Drive | 19. Power Telephone Line |
| 6. Non-Imaging Probes (if applicable) | 20. Power Line (AC~) |
| 7. Imaging Probes | 21. Ground Line |
| 8. Probe Port | 22. Power Cable with Protective Earth |
| 9. Peripheral Devices | 23. Footswitch |
| 10. Rear Panel | |
| 11. Signals I/O Port | |
| 12. Power Out | |
| 13. Signals I/O Port | |
| 14. Footswitch Connector | |

Acceptable Devices

The Patient Environmental devices shown on the previous page are specified to be suitable for use within the PATIENT ENVIRONMENT.



DO NOT connect any probes or accessories without approval by GE within the PATIENT ENVIRONMENT.

See 'Peripheral Update for EC countries' on *page 2-21 for more information.*

Unapproved Devices



DO NOT use unapproved devices.

If devices are connected without the approval of GE, the warranty will be INVALID.

Any device connected to the LOGIQ 9 must conform to one or more of the requirements listed below:

1. IEC standard or equivalent standards appropriate to devices.
2. The devices shall be connected to PROTECTIVE EARTH (GROUND).

Accessories, Options, Supplies



Unsafe operation or malfunction may result. Use only the accessories, options and supplies approved or recommended in these instructions for use.

Acoustic Output

Located on the upper right section of the system display monitor, the acoustic output display provides the operator with real-time indication of acoustic levels being generated by the system. See the *Acoustic Output chapter* in the *Advanced Reference Manual* for more information. This display is based on NEMA/AIUM Standards for Real-time Display of Thermal and Mechanic Acoustic Output Indices on Diagnostic Ultrasound Equipment.

Acoustic Output Display Specifications

The display consists of three parts: Thermal Index (TI), Mechanical Index (MI), and a relative Acoustic Output (AO) value. Although not part of the NEMA/AIUM standard, the AO value informs the user of where the system is operating within the range of available output.

The TI and MI are displayed at all times. The TI display starts at a value of 0.0 and increments in steps of 0.1. The MI display values between 0 and 0.4 increment in steps of 0.01 and for values greater than 0.4, increments in steps of 0.1.

Thermal Index

Depending on the examination and type of tissue involved, the TI parameter will be one of three types:

- **Soft Tissue Thermal Index (TIS).** Used when imaging soft tissue only, it provides an estimate of potential temperature increase in soft tissue.
- **Bone Thermal Index (TIB).** Used when bone is near the focus of the image as in the third trimester OB examination, it provides an estimate of potential temperature increase in the bone or adjacent soft tissue.
- **Cranial Bone Thermal Index (TIC).** Used when bone is near the skin surface as in transcranial examination, it provides an estimate of potential temperature increase in the bone or adjacent soft tissue.

Acoustic Output Display Specifications (continued)

Mechanical Index	MI recognizes the importance of non-thermal processes, cavitation in particular, and the Index is an attempt to indicate the probability that they might occur within the tissue.
Changing the Thermal Index Type	You can select the displayed TI type on Utility -> Imaging -> B-Mode. This preset is application dependent so each application could specify a different TI type. Display precision is ± 0.1 and accuracy is $\pm 50\%$. Accuracy of the power output displayed value on the Touch Panel is $\pm 10\%$.

Controls Affecting Acoustic Output

The potential for producing mechanical bioeffects (MI) or thermal bioeffects (TI) can be influenced by certain controls.

Direct. The Acoustic Output control has the most significant effect on Acoustic Output.

Indirect. Indirect effects may occur when adjusting controls. Controls that can influence MI and TI are detailed under the Bioeffects portion of each control in the Optimizing the Image chapter.

Always observe the Acoustic Output display for possible effects.

Best practices while scanning



HINTS

Raise the Acoustic Output only after attempting image optimization with controls that have no effect on Acoustic Output, such as Gain and TGC.

NOTE: *Refer to the sections in Chapter 5 for a complete discussion of each control.*



WARNING

Be sure to have read and understood control explanations for each mode used before attempting to adjust the Acoustic Output control or any control that can effect Acoustic Output.



Acoustic Output Hazard

Use the minimum necessary acoustic output to get the best diagnostic image or measurement during an examination. Begin the exam with the probe that provides an optimum focal depth and penetration.

Acoustic Output Default Levels

In order to assure that an exam does not start at a high output level, the LOGIQ 9 initiates scanning at a reduced default output level. This reduced level is preset programmable and depends upon the exam category and probe selected. It takes effect when the system is powered on or **New Patient** is selected.

To modify acoustic output, adjust the Power Output level on the Touch Panel.

Warning Label Locations



Figure 2-2. Label Location (a)

Table 2-8: Label Location Explanations

- | | |
|---|---|
| <ol style="list-style-type: none"> 1. Possible shock hazard. Do not remove covers or panels. No user serviceable parts are inside. Refer servicing to qualified service personnel. 2. Prescription Device (For U.S.A. Only) 3. The equipment weighs approximately 200 kg (434 lbs). To avoid possible injury and equipment damage when transporting from one area of use to another: <ul style="list-style-type: none"> • Be sure the pathway is clear. • Limit movement to a slow careful walk. • Use two or more persons to move the equipment on inclines or long distance. 4. The side panel should only be removed by a field service engineer, using the appropriate tools. 5. The CE Mark of Conformity indicates this equipment conforms with the Council Directive 93/42/EEC. 6. WEEE Label 7. Type BF Applied Part (Man in the Box). | <ol style="list-style-type: none"> 8. Identification and Rating Plate Country Specific Label, China Example 9. ETL Label: NRTL Listing and Certification Mark is used to designate conformance to nationally recognized product safety standards. The Mark bears the name and/or logo of the testing laboratory, product category, safety standard to which conformity is assessed, and a control number. 10. CISPR CAUTION: The LOGIQ 9 with CRT monitor conforms to the CISPR11, Group 1, Class B of the international standard for Electromagnetic disturbance characteristics. The LOGIQ 9 with LCD monitor conforms to the CISPR11, Group 1, Class A of the international standard for Electromagnetic disturbance characteristics. <ul style="list-style-type: none"> • (Not Shown) Control Panel Warning Label. DO NOT move the system until you have secured the Control Panel. |
|---|---|

Chapter 3

Preparing the System for Use

Describes the site requirements, console overview, system positioning/transporting, powering on the system, adjusting the display monitor, probes and operator controls.

Site Requirements

Introduction

NOTE: *Only qualified physicians or sonographers should perform ultrasound scanning on human subjects for medical diagnostic reasons. Request training, if needed.*

Do not attempt to install the system alone. General Electric, Affiliate, or Distributor Field Engineers and Application Specialists will install and setup the system. See 'Contact Information' on *page 1-7 for more information.*

The LOGIQ 9 does not contain any operator serviceable internal components. Ensure that unauthorized personnel do not tamper with the unit.

Perform regular preventive maintenance. See 'System Care and Maintenance' on *page 18-10 for more information.*

Maintain a clean environment. Turn off, and if possible, disconnect the system before cleaning the unit. See 'Cleaning the system' on *page 18-12 for more information.*

Never set liquids on the unit to ensure that liquid does not drip into the control panel or unit.

Before the system arrives

NOTICE

This medical equipment is approved, in terms of the prevention of radio wave interference, to be used in hospitals, clinics and other institutions which are environmentally qualified. The use of this equipment in an inappropriate environment may cause some electronic interference to radios and televisions around the equipment.

Ensure that the following is provided for the new system:

- A separate power outlet with a 20 amp circuit breaker for 120 VAC for 120V area, 7.5 amp circuit breaker for 220-240VAC for 220/240V area or 15 amp circuit breaker for 100 VAC for Japan.
- Take precautions to ensure that the console is protected from electromagnetic interference.

Precautions include:

- Operate the console at least 15 feet away from motors, typewriters, elevators, and other sources of strong electromagnetic radiation.
- Operation in an enclosed area (wood, plaster or concrete walls, floors and ceilings) helps prevent electromagnetic interference.
- Special shielding may be required if the console is to be operated in the vicinity of radio broadcast equipment.

Environmental Requirements

The system should be operated, stored, or transported within the parameters outlined below. Either its operational environment must be constantly maintained or the unit must be turned off.

Table 3-1: System Environmental Requirements

	Operational	Storage	Transport (<16hrs.)
Temperature	10 - 35 degrees C 50 - 95 degrees F	-10 - 60 degrees C 14 - 140 degrees F	-40 - 60 degrees C -40 - 140 degrees F
Humidity	30 - 85% non-condensing	30 - 90% non-condensing	30 - 90% non-condensing
Pressure	700 - 1060hPa	700 - 1060hPa	700 - 1060hPa



WARNING

To ensure the UPS battery remains fully charged, DO NOT turn off the breaker switch or disconnect the system power cord from the AC outlet, except for maintenance or during a portable exam.

Console Overview

Console graphics

The following are illustrations of the console:

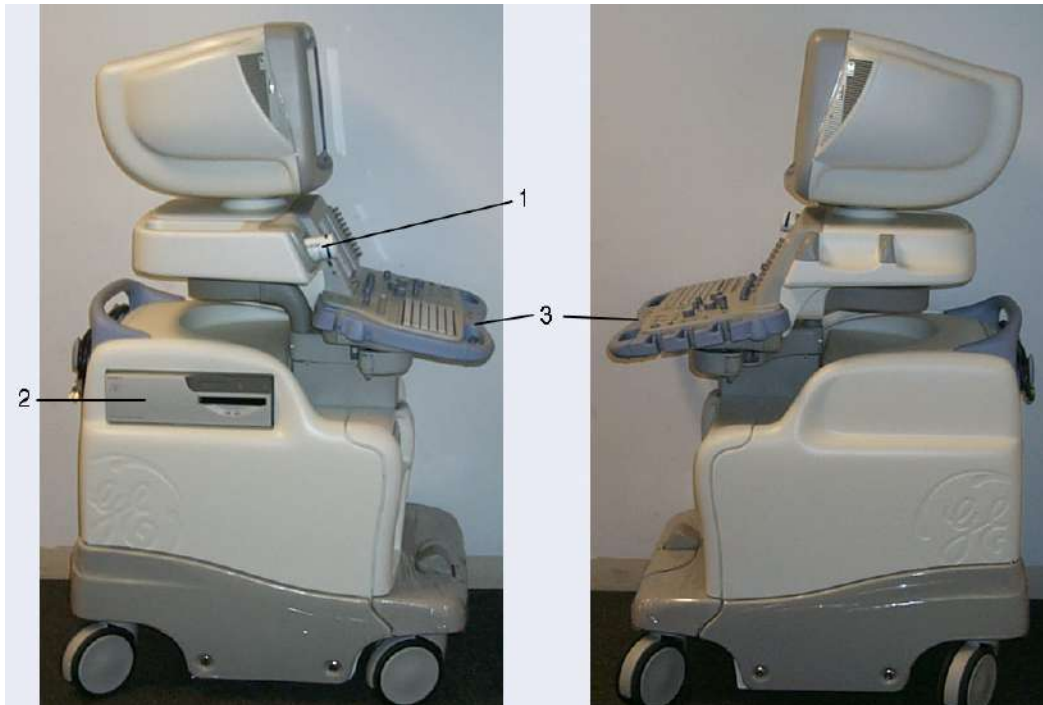


Figure 3-1. LOGIQ 9 System (right and left side views)

1. Probe and Gel Bottle Holder
2. Storage area of Peripheral Device
3. Storage area of Peripheral Device (located beneath the control panel)



DO NOT place probes or the footswitch into the peripheral device's storage area.

Console graphics (continued)



Figure 3-2. LOGIQ 9 System (front and back views)

1. Optional B/W Video Page Printer
2. DVD-R/CD-R Drive
3. USB Ports
4. CW and 4D Probe Connectors
5. Probe Connector
6. DVR/VCR Recorder
7. Connector Panel (Additional USB Port, Network Connection, Footswitch Connection, etc.)



For compatibility reasons, use only GE approved probes, peripherals or accessories.

DO NOT connect any probes or accessories without approval by GE.

External Drives (DVD-R/CD-R, USB Drives, Optional MO)

The external DVD-R/CD-R drive (Standard) is located below the keyboard. An optional MO Drive is also available. Approved USB Hard Disk and USB Flash Drives may be used for Save As, MPEGVue, and Backup/Restore in the powered system USB ports.

Use the DVD-R/CD-R to perform software upgrades, image archiving, or service diagnostics.

USB Drives are an ESD sensitive device. Only use USB Drives recommended by GE.



Make sure that the DVD/CD disk tray and/or MO disk are securely placed in each device during system operation. Mechanical damage may occur if other objects hit the tray or MO disk.

External MO Drive

The LOGIQ 9 supports the portable, external Fujitsu DynaMO 1300 U2 Pocket Magnetic Optical (MO) drive to read old media and to use up unused MO disks.

NOTE: *The MO drive is intended to be a temporary storage solution only and NOT for everyday use. For permanent storage, ALWAYS use the DVD drive that is shipped with your system.*

Connecting the Portable MO Drive

Primary Site for MO Drive: Side Peripheral Storage Area next to image printer. To connect the portable MO drive to the LOGIQ 9 system:

1. Connect the USB cable to the back of the MO drive. The MO drive is powered by the USB port; therefore, no power cord is needed.
2. Set the MO drive inside the side Peripheral Storage Area, next to the image printer. Make sure the MO drive sits far enough inside the storage area, but close enough to reach while plugged into the USB port.



Figure 3-3. MO Drive Storage Area

NOTE: *If you have a printer that takes up the entire storage area, see Alternative Site for MO drive for instructions.*

**Connecting the
Portable MO Drive
(continued)**

3. Connect the USB cable to the USB port on the back of the system.



Figure 3-4. MO Drive USB Connection

Connecting the Portable MO Drive (continued)

Alternative Site for MO Drive. Above Front Peripheral Storage Area. To connect the portable MO drive to the LOGIQ 9 system:

1. Connect the USB cable to the back of the MO drive.
2. Set the MO drive above the Front Peripheral Storage Area, far enough away from the edge.
3. Connect the USB cable to the USB port on the front of the system.



Figure 3-5. Alternate MO Drive Location



DO NOT connect the MO drive to the USB port adjacent to the top of the Control Panel.

If you intend to use the MO drive semi-permanently, attach the MO drive to the system using velcro, or other secure method (not supplied by GE).

DO NOT lock the Control Panel into place while the MO Drive is located above the Front Peripheral Storage Area.

DO NOT load any CDs that accompany the MO device.

Storage areas

Several convenient storage areas are provided within the console as shown in the figure below. Use them to store gel, options, probe cables, accessories, etc.



Figure 3-6. Storage Areas

1. Gel Storage Area
2. Media Storage
3. Probe Cable Maintenance
4. Footswitch



DO NOT place probes or the footswitch into the peripheral device's storage area.

Peripheral/Accessory Connection

Peripheral/Accessory Connector Panel

LOGIQ 9 peripherals and accessories can be properly connected using the rear connector panel located behind the rear door. Only the B/W Printer can be connected to the front accessory panel.

Located on the rear panel are video input and output, audio input and output, camera rec/expose, Insite, Ethernet, power and control connections for VCR, printer and service tools. See the Connector Panel figure for more specifics.



Each outer (case) ground line of peripheral/accessory connectors are **Earth Grounded**.

Signal ground lines are Not Isolated.



For compatibility reasons, use only GE approved probes, peripherals or accessories.

Be sure to plug the power supply cable of the peripherals to the connector panel of the system to avoid unexpected electrical shock.

DO NOT connect any probes or accessories without approval by GE.



The connection of equipment or transmission networks other than as specified in these instructions can result in electric shock hazard. Alternate connections will require verification of compatibility and conformity to IEC/EN 60601-1-1 by the installer.

Peripheral/Accessory Connector Panel (continued)



Figure 3-7. Connector Panel, Circuit Breaker, and Power Cord

Table 3-2: Peripheral/Accessory Connector Panel Descriptions

No.	Item	Type of connector
1.	Audio In/Out	RCA
2.	Footswitch	
3.	Camera Color/Camera B/W	
4.	Insite2	RJ-11 Modular, 6-pin
5.	Composite Color/B/W	
6.	S-Video Out	S-Terminal, 4-pin
7.	VGA	Shrunked D-Sub, 15-pin
8.	Service	Universal Serial Port
9.	Ethernet	RJ-45 Modular, 8-pin
10.	Power	IEC Couplers
11.	Equipotentiality	Terminal for connecting equipotential conductors when interconnecting with other equipment

Front Panel

CD/DVD drive, 4D and CW probe connectors, 2 USB ports, and the hard drive access LED indicator are located beneath the Black/White Printer.



Figure 3-8. Front Panel

Footswitch (Option)

Use only the recommended optional multi-functional footswitch.

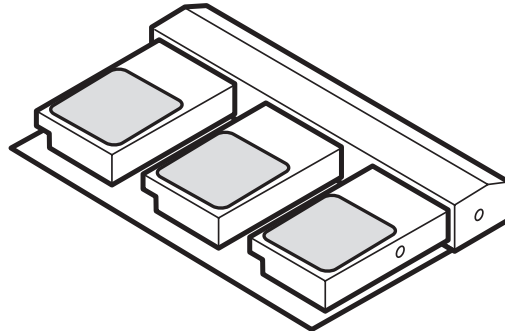


Figure 3-9. Footswitch

The footswitch may be used as select keys, video recording, etc.

The three switches have different function assignments depending on the current application. See Table 16-40 for more information.

The footswitch connection is located below the operator control panel.



DO NOT place or store the footswitch in the side pocket or the Peripheral Device storage area on the system. This is to prevent damaging the footswitch from falling.

Powering the System

Connecting and Using the System

To connect the system to the electrical supply:

1. Ensure that the wall outlet is of the appropriate type.
2. Ensure that the power switch is turned off.
3. Unwrap the power cable. Make sure to allow sufficient slack in the cable so that the plug is not pulled out of the wall if the system is moved slightly.
4. Attach the power plug to the system and secure it in place by using the retaining clamp.



Figure 3-10. Power Plug

- a. Retaining clamp for power plug



Ensure that the retaining clamp for the power plug is fixed firmly.

Connecting and Using the System (continued)

5. Push the power plug securely into the wall outlet.



Use caution to ensure that the power cable does not disconnect during system use.

If the system is accidentally unplugged, data may be lost.



To avoid risk of fire, the system power must be supplied from a separate, properly rated outlet. See 'Before the system arrives' on page 3-3 for more information.

Under no circumstances should the AC power plug be altered, changed, or adapted to a configuration rated less than specified. Never use an extension cord or adapter plug.

To help assure grounding reliability, connect to a "hospital grade" or "hospital only" grounded power outlet.

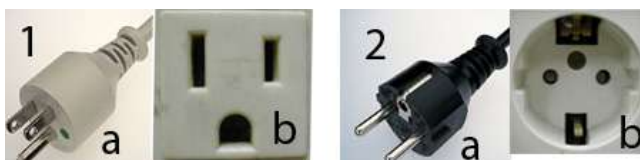


Figure 3-11. Example Plug and Outlet Configurations

1. 100-120 VAC, 1200 VA
Plug and Outlet Configuration
2. 220-240 VAC, 1200 VA
Plug and Outlet Configuration

Acclimation Time

After being transported, the unit requires one hour for each 2.5 degree increment its temperature is below 10 degree C or above 40 degree C.

Table 3-3: System Acclimation Time Chart

Degree C	60	55	50	45	40	35	30	25	20	15	10
Degree F	140	131	122	113	104	95	86	77	68	59	50
hours	8	6	4	2	0	0	0	0	0	0	0
Degree C	5	0	-5	-10	-15	-20	-25	-30	-35	-40	
Degree F	41	32	23	14	5	-4	-13	-22	-31	-40	
hours	2	4	6	8	10	12	14	16	18	20	

Power On



Press the **Power On/Off** switch to turn the power on. The circuit breaker must also be in the on position. For circuit breaker location, See 'Circuit breaker' on *page 3-26 for more information*.



Figure 3-12. Power On/Off Switch Location

1. Power On/Off Switch

Power Up Sequence

The system is initialized. During this time:

- System diagnostics run and start-up status is reflected on the monitor.



Figure 3-13. Power Up Graphic Sequence



HINTS

If problems occur, freeze the image and take a picture for reference. This will help if there is a need to call for service.

- Probes are initialized for immediate operation.

NOTE: *If no probe is connected, the system goes into freeze mode.*

- Peripheral devices are activated on power up.

After initialization is complete, all lighted buttons on the Control Panel light and the default B-Mode screen is displayed in the monitor display.

Password Protection

Login Personal IDs and associated passwords can be preset in the LOGIQ 9.

If the Use Auto Logon preset is blank, you are prompted to login.



Figure 3-14. Operator Login Window

1. **Operator:** Select the Operator.
2. **Password:** Enter Operator's password (optional).
3. Select type of Logon or Cancel.
 - **Emergency:** Data stored only for the duration of the current examination.
 - **OK:** Standard logon
 - **Cancel:** Cancel logon

Logoff To logoff, press the **Power On/Off** switch momentarily and a SYSTEM-EXIT window appears.

Standby Mode

Use Standby Mode to decrease the power off/on time while transporting your system for a portable exam.

To place the system in standby mode:

1. Press the **Power On/Off** switch at the front of the system once. The System-Exit window displays.
2. Use the **Trackball** to select Standby.

NOTE: *The Monitor Display and Touch Panel turn black and the keyboard turns off. The Power On/Off switch indicator lights up.*



You need to wait at least one minute after the monitor blacks out before unplugging the power cable. The system is still in the process of going into Standby Mode after the monitor blacks out.

3. Unplug the system from the wall. The system beeps.

NOTE: *DO NOT press the Power On/Off switch while the system is unplugged. This will turn the system off.*

4. Transport the system as needed and plug it back in within 20 minutes. If the system is unplugged for longer than 20 minutes, the battery is discharged and the system shuts down.

See 'Moving the System' on page 3-27 for more information.

Standby Mode (continued)

5. Press and immediately release the **Power On/Off** switch to come out of Standby Mode. This takes about 45 seconds.

NOTE: If the Power On/Off switch is pressed too long, the system shuts down instead of coming out of Standby Mode.

NOTE: It takes about four hours to fully recharge the battery.

Power Off

To power off the system:

1. When you shutdown the system, enter the scan screen and lightly press the **Power On/Off** switch at the front of the system once. The System-Exit window is displayed.

NOTE: *DO NOT press and hold down the Power On/Off switch to shutdown the system. Instead, lightly press the Power On/Off switch and select Shutdown.*

2. Using the **Trackball**, select Shutdown.

The shutdown process takes a few seconds and is completed when the Power On/Off switch illumination is turned amber.

NOTE: *If the system has not fully shut down in 60 seconds, press and hold down the On/Off switch until the system shuts down.*

3. Disconnect the probes.

Clean or disinfect all probes as necessary. Store them in their shipping cases to avoid damage.



DO NOT turn off the circuit breaker before the Power On/Off switch LED is amber.

Data may be lost or system software damaged if the circuit breaker is turned off before the Power On LED is amber.

NOTE: *When the Uninterruptible Power Supply (UPS) is installed, battery replacement may be necessary if the circuit breaker is turned off for long periods of time (3 to 6 months).*

Additional System Information

For optimum system operation, we recommend that you restart the system at least once every 24-hour period. If you shut down the system at the end of the day, no other action is needed.

**Crash Recovery
Instructions**

In cases where the system detects an internal error, the system may reboot on its own. If this happens, the system automatically returns to the start-up screens after a period of time equal to a normal power-on sequence, taking approximately four minutes. All images and measurements, except for generic worksheets, are preserved in the system. When the system returns, the system alerts you that unsaved images are still in the system from the previous patient. Respond to the prompt to continue the current patient. Check that all images and measurements have been preserved in the system. Then resume the exam.

If the system fails to respond to your commands within a typical period of time, you need to manually reset the system. Simply hold down the power switch to initiate a normal power down sequence. After the system has completely shut down (power switch amber), restart the system using the standard power-up sequence. All images and measurements, except for generic worksheets, are preserved in the system. When the system has fully powered up, the system alerts you that unsaved images are still in the system from the previous patient. Respond to the prompt to continue the current patient. Check that all images and measurements have been preserved in the system. If you do not have any images on the clipboard, the patient must be retrieved from the database. Then resume the exam.

Circuit breaker

The Circuit Breaker is located at the rear panel of the system. **On** supplies main power to all internal systems. **Off** removes main power from all internal systems. The circuit breaker automatically shuts off power to the system in case of a power overload.

If a power overload occurs:

1. Turn off all peripheral devices.
2. Reactivate the Circuit Breaker switch.

The Circuit Breaker switch should stay in the **On** position; **DO NOT** hold the switch in the **On** position. If the Circuit Breaker switch remains **On**, follow the Power On procedure previously described.

NOTE: *If the Circuit Breaker switch does **not** remain in the **On** position or trips again:*

1. Disconnect the Power Cable.
2. Call Service immediately.

DO NOT attempt to use the system.

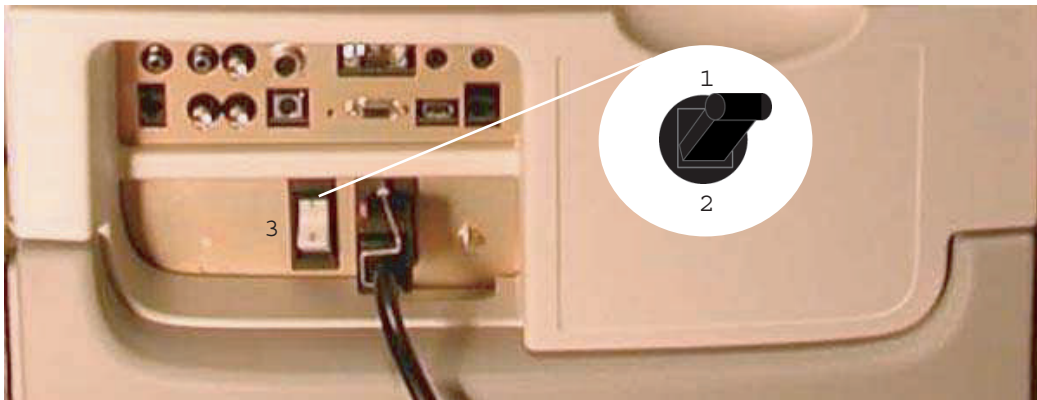


Figure 3-15. Location of Circuit Breaker

1. Circuit Breaker On position
2. Circuit Breaker Off position
3. Circuit Breaker location

System Positioning/Transporting

Moving the System

When moving or transporting the system, follow the precautions below to ensure the maximum safety for personnel, the system, and other equipment.

Before moving the system

1. Press the **Power On/Off** switch to power off. See 'Power Off' on *page 3-24 for more information*.
2. Unplug the power cord.
3. All cables from off-board peripheral devices (IIE camera, external printer, VCR, etc.) must be disconnected from the console.
4. Disconnect the footswitch from the console.
5. Ensure that no loose items are left on the console.
6. Wind the power cable around the cable hook below the rear handle or wrap the cord in a bundle and store it behind the rear storage panel.



Please ensure that the peripheral devices, which are installed in the storage areas, are secured with the supplied straps.

Before moving the system (continued)

To prevent damage to the Power Cord, **DO NOT** pull excessively on the cord or make sharp bends while wrapping.



Figure 3-16. Location of rear handle

a. Handle

7. Connect all probes to be used while off site. Ensure that probe cables are out of the way from the wheels and not protruding beyond the console.

NOTE: *If more than four (4) probes are intended to be used, store the additional probes securely.*

8. Store all other probes in their original cases or in soft cloth or foam to prevent damage.
9. Store sufficient gel and other essential accessories in the provided space.



DO NOT place probes or the footswitch into either the side pocket or the peripheral device's storage area when moving the system. These are not storage spaces for probes or the footswitch.

10. Adjust the monitor and control panel to its lowest position possible by using the release lever located underneath the control panel. Make sure the keyboard is locked in place.
11. Unlock the wheels.

When moving the system

1. Always use the rear handle grips to move the system.
2. Take extra care when moving the system long distances and on inclines. Ask for help if necessary.

Avoid ramps that are steeper than ten degrees to avoid tipping over the system.

NOTE: *Wheel chair ramps are usually less than five degrees.*

Utilize additional care and personnel when moving on steep incline (>5 degrees) or loading into a vehicle for transport.



DO NOT attempt to move the console using any cables or fixtures, such as the probe connectors.

3. Use the foot brake (pedal), located on the bottom of the system in the front, when necessary.
4. Do not let the system strike walls or door frames.
5. Use extra care when crossing door or elevator thresholds.
6. Once the destination is reached, lock the wheels.



The system with CRT weighs approximately 185 kg (408 lbs); with LCD 171 kg (376 lbs). To avoid possible injury and equipment damage:

- Be sure the pathway is clear.
- Limit movement to a slow careful walk.
- Use two or more persons to move the system on inclines or long distances.

Transporting the System

Use extra care when transporting the system using vehicles. In addition to the instructions used when moving the system (See *'Moving the System' on page 3-27 for more information.*), also perform the following:

1. Only use vehicles that are designed for transport of the LOGIQ 9 system.
2. Load and unload the system to a vehicle parked on a level surface.
3. Ensure that the transporting vehicle can handle the weight of the system plus the passengers.
4. Ensure that the load capacity of the lift (a minimum of 200 kg [434 lbs] is recommended) is capable of handling the weight of the system.
5. Ensure that the lift is in good working order.
6. Secure the system while it is on the lift so that it cannot roll. Use either wood chocks, restraining straps, or other similar types of constraints. Do not attempt to hold it in place by hand.

NOTE: *Strap the system below its handle so that the system does not break loose.*



Never ride on the lift with the system. A person's weight coupled with the weight of the system may exceed the load capacity of the lift.

7. Employ two to three persons to load and unload safely from a vehicle.

Transporting the System (continued)

8. Load the unit aboard the vehicle carefully and over its center of gravity. Keep the unit still and upright.

NOTE: Do not lay the unit down on its side.

9. Ensure that the system is firmly secured while inside the vehicle. Any movement, coupled with the weight of the system, could cause it to break loose.
10. Secure system with straps or as directed otherwise to prevent motion during transport.
11. Prevent vibration damage by driving cautiously. Avoid unpaved roads, excessive speeds, and erratic stops or starts.

Wheels

Examine the wheels frequently for any obvious defects that could cause them to break or bind.

Front wheels

Swivel and brake lock.

Back wheels

Swivel, swivel lock and brake lock.

Setting the front wheels lock

The pedal is located between the front wheels of the unit and enables the user to control the movement of the wheels.

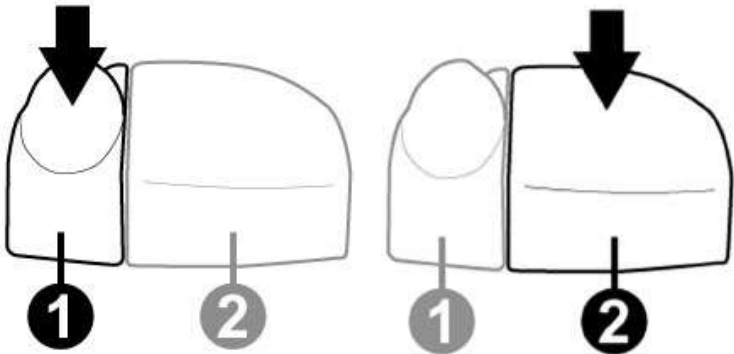


Figure 3-17. Front Pedal

Table 3-4: Front Pedal Functions

	Pedal position	Function
1.	Left—Swivel Lock	Swivel Lock. To move the system, press the Swivel Lock to set the wheels in a forward-to-back position. First move the system in the direction you will be going, then press the Swivel Lock. To release the Swivel Lock, press the Park key.
2.	Right—Parking Brake	Park. To set the brake, press the right pedal. To release the brake, press the right pedal.

Setting the front wheels lock (continued)

The pedal which is located between the rear wheels of the unit enables the user to control the movement of the wheels.



Figure 3-18. 1 Rear Pedal

Table 3-5: Rear Pedal Functions

	Pedal Position	Function
1.	Left and Right	Swivel lock. Move the swivel lock to the right or left to set for each wheel.



When two or more people are releasing the swivel/pivot wheel controls with the front and rear pedals, take extra precaution to prevent unexpected movement which could result in possible toe injuries.

Moving the LOGIQ 9 (with an LCD Monitor)

Before moving the system (with an LCD monitor)



When the system is not in use AND/OR before moving/transporting the system, make sure that the monitor arm locks firmly and flip down the monitor to prevent system damage.

To lock the monitor arm:

1. Lock the arm at the bottom at the base of the arm. Move arm until it locks into place.
2. Lock the arm at joint. Move the arm until it locks into place.
3. Make sure the monitor arm does not move, then flip down the monitor so that the display is face down.

To unlock the monitor arm:

1. Unlock the arm at the bottom at the base of the arm. Move arm to desired position.
2. Unlock the arm at joint. Move the arm until to desired position.

Before transporting the system (with an LCD monitor)



- **DO NOT** attempt to move the system with the LCD monitor by pulling cables or belts placed around the monitor and/or monitor arm.
- **DO NOT** restrain the LOGIQ 9 at the LCD monitor or rotating arm using a belt as shown. Always secure the LCD at the body part of the console.

Adjusting the LCD Monitor Position

LOGIQ 9 Option

Grab the bottom of the LCD monitor (refer to Figure 3-19) when you adjust the position of the monitor and monitor arm.



Figure 3-19. Positioning the LCD monitor



CAUTION

DO NOT grab the speakers when moving or adjusting the monitor and monitor arm. When flipping up the monitor from the flip down status, you can grab the upper corner of the monitor.

Adjusting the LCD Monitor Position (continued)

Range of Movement

- Tilt Adjustable Monitor: Tilt Angle: Up 10 degrees, Down 90 degrees
- Range of Movement: Up 89mm, Down 211mm
Left and Right from center: max855mm
- Monitor Arm Neck: plus or minus 90 degrees
- Monitor Arm Rotary: plus or minus 150 degrees
- Monitor Neck Swivel: plus or minus 90 degrees



- To avoid injury or damage, make sure nothing is within the range of motion before moving the monitor and monitor arm. This includes both objects and people.
- Pay attention to the monitor arm position to avoid hitting it against anyone or anything.

Adjusting the Display Monitor

Rotate, tilt, raise and lower the monitor

The monitor position can be adjusted for easy viewing.

- The monitor can be rotated around its central pivot point.
- The monitor can be tilted for the optimum viewing angle.
- The monitor and console panel can be raised or lowered for the best viewing height.

NOTE: *When moving the LOGIQ 9, lower and lock the monitor and console panel to its lowest possible position to improve stability.*



DO NOT place objects on the monitor.

Keyboard Console Safety



DO NOT move the system if the keyboard console is in the free (unlocked) position.

Ensure that the hands of the patient are away from the console arm when moving the keyboard console.

Avoid ramps that are steeper than 10 degrees.

Use two or more persons to move the system over long distances or on inclines.



Figure 3-20. Warning Label

To raise/lower the floating keyboard, push in the left-handle-bar keyboard lever while raising/lowering the keyboard. To move the keyboard, pull the lever beneath the handle bars to position the keyboard.



Figure 3-21. Floating Keyboard

Brightness and Contrast

Adjusting the monitor's contrast and brightness is one of the most important factors for proper image quality. If these controls are set incorrectly, the Gain, TGC, Dynamic Range and even Power Output may have to be changed more often than necessary to compensate.

The proper setup displays a complete gray scale. The lowest level of black should just disappear into the background and the highest white should be bright, but not saturated.

Brightness and Contrast (continued)

To adjust the contrast:

1. Press the Toggle button on the display monitor once.
2. Press the Adjustment (+) button to increase contrast. Press the Adjustment (-) button to decrease contrast.

The amount of contrast is shown on a slide bar on the screen.

To adjust the brightness:

1. Press the Toggle button on the display monitor twice.
2. Press the Adjustment (+) button to increase brightness. Press the Adjustment (-) button to decrease brightness.

The amount of brightness is shown on a slide bar on the screen.

NOTE: On the LCD monitor, the brightness adjustment comes before contrast.

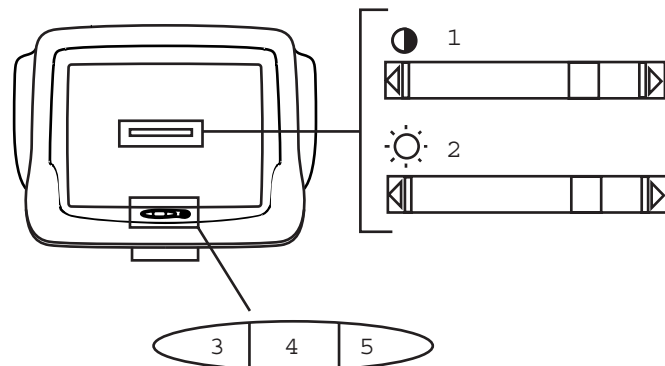


Figure 3-22. Brightness and Contrast

1. Contrast Indicator
2. Brightness Indicator
3. Adjustment (-) Button
4. Toggle Button
5. Adjustment (+) Button

Brightness and Contrast (continued)

Record any changes to the final brightness and contrast settings and leave this information with the system. Generally speaking, do not change the controls once they have been set. Once set, the display then becomes the reference for the hard copy device(s).

NOTE: *After readjusting the monitor's Contrast and Brightness, readjust all preset and peripheral settings.*

For instructions on calibrating the brightness/contrast on the monitor, please refer to *Customizing Your System*.

Table 3-6: Typical settings to adjust the LCD Monitor

Room Condition	Contrast	Brightness
Dark Room for Radiology/Cardiology	85	35
Dim Room for Radiology/Cardiology	90	55
Bright Room	100	100

LOGIQ 9 LCD Gamma Setting

You can adjust the Gamma Setting on the LCD for the following:

- Higher Gamma Setting: Suppresses mid- and low-level grays, making the image appear crisper and cleaner, at the expense of low-level grays.
- Lower Gamma Setting: Increases mid- and low-level grays, making the image appear softer. May result in a flat, noisy image.

NOTE: *These LCD adjustments DO NOT affect the images sent to PACS systems.*

To adjust the Gamma Setting on the LCD,

1. You use the controls at the bottom of the LCD to adjust the gamma settings.

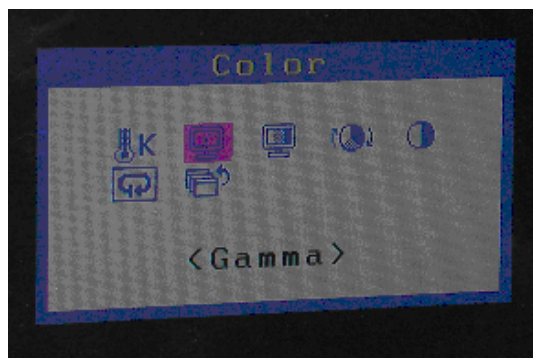


Table 3-7: LCD Controls

1. Left Arrow	2. Brightness/Contrast/Select	3. Right Arrow
---------------	-------------------------------	----------------

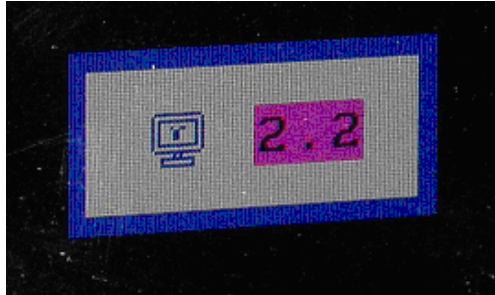
LOGIQ 9 LCD Gamma Setting (continued)

2. First you may want to move the on-screen menu moved to a location off the image.
 - a. Press and hold down the Brightness/Contrast/Select control for about 30 seconds. A menu appears that initially says "Brightness 55%," but then changes to the Advanced Menu in about another 30 seconds.
 - b. Press the Right Arrow control until you get to the "Etc." icon to get to the "Others" menu.
 - c. Press the Brightness/Contrast/Select control to select Menu Position.
 - d. Press the Left or Right Arrow control to move the menu to the corner of the display.
 - e. Press the Brightness/Contrast/Select control to save the position.
 - f. Press the "Return" icon to return to the Advanced Menu.
3. Press the Right Arrow control until you reach the Gamma Menu.



LOGIQ 9 LCD Gamma Setting (continued)

4. Press the Brightness/Contrast/Select control to enter the Gamma Menu.



5. Press the Right Arrow control to increase the Gamma setting; press the Left Arrow control to decrease the Gamma setting.
6. When you have completed making the adjustment, press the Brightness/Contrast/Select control to save the setting.
7. Press the "Return" icon to return to the Advanced Menu.
8. Press the Brightness/Contrast/Select control to "Exit" from the Advanced Menu.

Manual Degauss

Degaussing refers to the process of removing magnetic-field effects from the monitor. Operation of the monitor within a magnetic field may adversely effect color purity. Degaussing can be used to correct this problem. The monitor automatically degausses at power on.

To activate the manual degauss:

1. Press the <<< or >>> monitor adjustment buttons while the brightness/contrast adjustment indicator is not displayed on the monitor. To cancel the manual degauss, press the <<< or >>> adjustment button again.
2. Press the toggle button for brightness and contrast.
3. Press the <<< or >>> Adjustment button.

NOTE: *Monitor degaussing (demagnetizing) is done automatically when the system is turned on.*

Speakers

Audio is provided by speakers located on the sides of the display monitor.

- Audio Doppler operation
- Audio playback of videotaped scan sessions
- Audio error notification.

Probes

Introduction

Only use approved probes.

All imaging probes can be plugged into any of the probe ports.

Refer to the *Probes* chapter.

Selecting probes

- Always start out with a probe that provides optimum focal depths and penetration for the patient size and exam.
- Begin the scanning session by choosing the correct application and preset for the examination by selecting **Preset**.
- Begin the scan session using the default Power Output setting for the probe and exam.

NOTE: *Selecting a new probe unfreezes the image.*

Connecting the Probe

Probes can be connected at any time, regardless of whether the console is powered on or off. To ensure that the ports are not active, place the system in the image freeze condition.

To connect a probe:

1. Place the probe's carrying case on a stable surface and open the case.
2. Carefully remove the probe and unwrap the probe cord.
3. DO NOT allow the probe head to hang free. Impact to the probe head could result in irreparable damage. Use the integrated cable management hook to wrap the cord.

Inspect the probe before and after each use for damage or degradation to the housing, strain relief, lens, seal and connector. DO NOT use a transducer which appears damaged until functional and safe performance is verified. A thorough inspection should be conducted during the cleaning process.

4. Turn the connector locking handle counterclockwise.
5. Align the connector with the probe port and carefully push into place.

Before inserting the connector into the probe port, inspect the probe connector pin. If the pin is bent, do not use the probe until it has been inspected and repaired/replaced by a GE Service Representative.

6. Turn the connector locking handle clockwise to secure the probe connector.
7. Carefully position the probe cord so it is free to move and is not resting on the floor.



Fault conditions can result in electric shock hazard. Do not touch the surface of probe connectors which are exposed when the probe is removed. Do not touch the patient when connecting or disconnecting a probe.

Connecting the 4D Probe

To connect the 4D probes, follow the steps described earlier for Connecting the Probe plus these additional 4D probe steps:

1. Attach the 4D cable to the probe and to the probe connector on the Front Panel. Connect the smaller end of the 4D cable to the probe connector, just above the connector locking handle.
2. Connect the larger end of the 4D cable to the probe connector.



Figure 3-23. 4D Probe Connection

- a. Probe Connector
- b. 4D Cable
- c. Connector Cable Clip (holds cable in place)
- d. Probe Connector and Probe Locking Handle

Using the 4D Probe Holder

The 4D probe holder provides additional stability for 4D probes when they are resting in the permanent probe holders. The 4D probe holder fits inside the permanent probe holders.



DO NOT insert a 4D probe into a permanent probe holder without the special 4D probe holder. 4D probes are especially top-heavy and do not fit securely into the permanent probe holders.

To connect the 4D probe holder:

1. Line up the opening of the 4D probe holder with the opening of one of the permanent probe holders. These openings must be lined up for the 4D probe holder to fit properly into the permanent holders.



Figure 3-24. 4D Probe Holder Aligned with Permanent Probe Holder

2. Insert the 4D probe holder into the permanent holder. You may have to use a little force to ensure that the 4D holder is pushed all the way into the permanent holder.

NOTE: *If the 4D probe holder doesn't fit properly into the permanent holder, gently remove the 4D probe holder, line up the openings, and try again.*

**Using the 4D
Probe Holder
(continued)**

3. When the 4D probe holder is secure, insert the 4D probe into the holder.



Figure 3-25. 4D Probe Holder With 4D Probe

Connecting the CW Probe

Probes can be connected at any time, regardless of whether the console is powered on or off. To ensure that the ports are not active, freeze the image.

To connect a probe:

1. Carefully remove and unwrap the probe cord.
2. DO NOT allow the probe head to hang free. Impact to the probe head could result in irreparable damage. Use the integrated cable management hook to wrap the cord.
3. Attach the probe adapter to the probe connector on the Front Panel.
4. Carefully position the probe cord so it is free to move and is not resting on the floor.

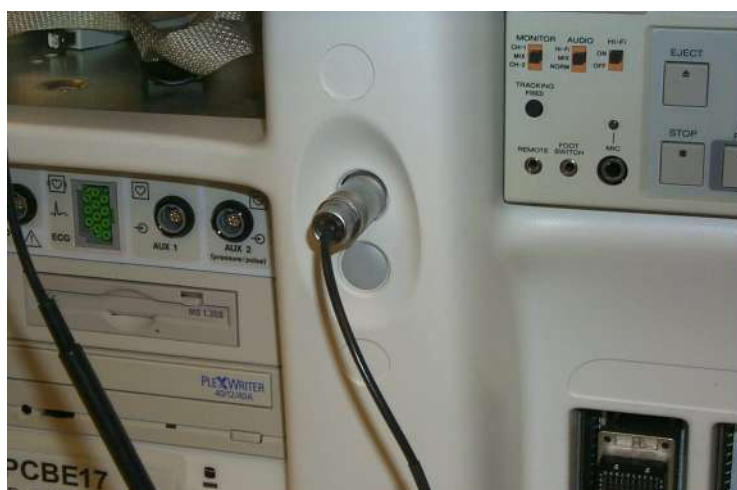


Figure 3-26. CW Probe Connector

Cable Handling

Take the following precautions with probe cables:

- Keep free from wheels
- Do not bend the cable acutely
- Avoid crossing cables between probes.

Activating the Probe

To activate the probe, select the appropriate probe from the probe indicators on the Touch Panel.

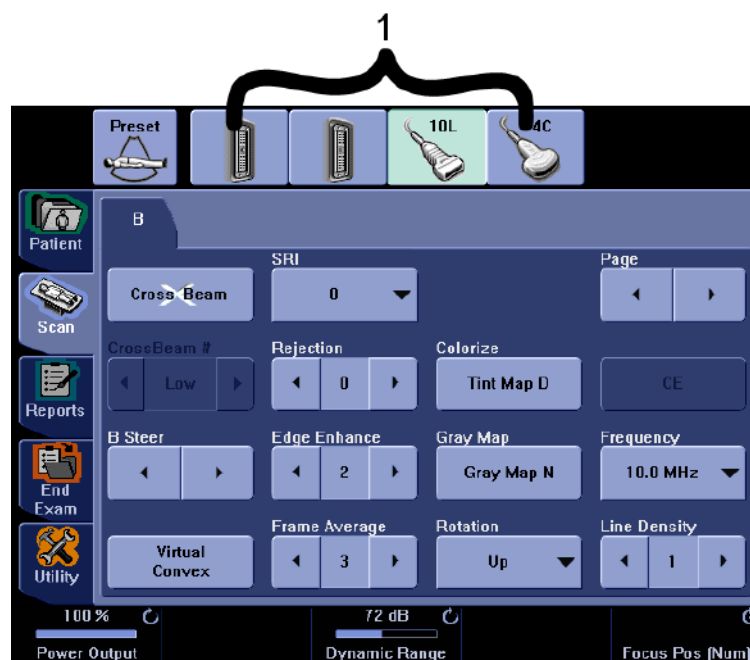


Figure 3-27. Probe Indicators

1. Probe Indicators

The probe's default settings for the mode and selected exam are used automatically.

Deactivating the Probe

When deactivating the probe, the probe is automatically placed in standby mode.

To deactivate a probe:

1. Ensure the LOGIQ 9 is in freeze mode. If necessary, press the **Freeze** key.
2. Gently wipe the excess gel from the face of the probe.
3. Carefully slide the probe around the right side of the keyboard, toward the probe holder. Ensure that the probe is placed gently in the probe holder.

Disconnecting the Probe

Probes can be disconnected at any time. However, the probe should not be active when disconnecting the probe.

1. Ensure the probe is deactivated. Deactivate by selecting another probe or pressing Freeze.
2. Move the probe locking handle counterclockwise.
3. Pull the probe and connector straight out of the probe port.
4. Carefully slide the probe and connector away from the probe port and around the right side of the keyboard.
5. Ensure the cable is free.
6. Be sure that the probe head is clean before placing the probe in its storage box or a wall hanging unit.

Disconnecting the 4D Probe

After step 2 above, disconnect the 4D cable, then pull the probe and connector straight out of the probe port.

Disconnecting the CW Pencil Probe

1. Pull out the probe connector carefully.
2. Ensure the probe cable is free.
3. Be sure that the probe head is clean before placing the probe in its case.

Transporting Probes

Secure the probe in its holder for moving short distances.

When transporting a probe a long distance, store it in its carrying case.

Storing the Probe

It is recommended that all probes be stored in the provided carrying case or in the wall rack designed for probe storage.

Carrying case:

- First place the probe connector into the carrying case.
- Carefully wind the cable into the carrying case.
- Carefully place the probe head into the carrying case. **DO NOT** use excessive force or impact the probe head.



DO NOT store probes in either the side pocket or the peripheral device's storage area on the system. To avoid damage, store the probe in its carrying case.

Operator Controls

Control Panel Map

Controls are grouped together by function for ease of use. See the callouts for this figure on the following page.



Figure 3-28. Control Panel

Control Panel Map (continued)

1. Gel Warmer
2. Touch Panel Brightness Control
3. Touch Panel and Controls
4. TGC
5. Video
6. Audio On/Off Volume
7. Additional Feature Keys
8. Keyboard
9. Mode/Gain Keys
10. Imaging/Measurement Keys
11. Depth
12. Left/Right Keys
13. Freeze and Print Keys
14. Reverse
15. Probe and Cord Holder
16. 4D Probe Holder Attachment



Place the gel bottle in the gel holder nozzle first to avoid damaging the control panel.

Key Illumination

All the keys on the front panel, except for the keyboard, and some rotaries have a two-level backlight capability. The following backlighting indicates availability.

Keyboard

The standard alpha-numeric keyboard has some special functions.

Esc	Exit current display screen.
Help (F1 Key)	Enter Online help / user manual.
Arrow (F2 Key)	Annotation arrow.
Eject (F3 Key)	Eject media.
Spooler (F4 Key)	Activates DICOM Job Spooler screen.
Home/Set Home (F7 Key)	Move annotation cursor to home position; shift+key to set current annotation cursor position as the new home position.
Text1/Text2 (F8 Key)	Switch between user text annotation overlays.
Grab Last (F9 Key)	Activate the last selected data for edit.
Word Delete (F10 Key)	Erase word associated with comment cursor.

Touch Panel

The Touch Panel contains exam function and mode/function specific controls.

Exam Function Controls

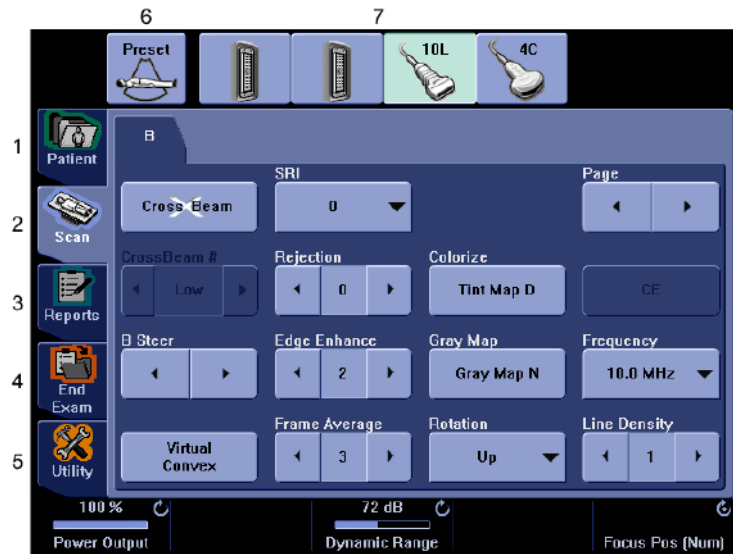


Figure 3-29. Exam Function Controls

1. Patient: Enters Patient screen
2. Scan: Enters scanning mode screen
3. Reports: Activates default report and Touch Panel of report choices.
4. End Exam: Activates Image Management and Touch Panel with end of exam options.
5. Utility: Activates system configuration menus.
6. Preset: Selects the application to use.
7. Probe Indicator: Indicates and selects the probes.

NOTE: Different menus are displayed depending on which Touch Panel is selected.

At the bottom of the Touch Panel, there are five combination rotary dials/push buttons. The functionality of these rotaries changes, depending upon the currently-displayed menu. Press the button to switch between controls (as with Focus Position/Number), or rotate the dial to adjust the value.

Mode/Function Specific Controls

In general, the key name is indicated at the top of the key. There are different types of Touch Panel keys as illustrated below:

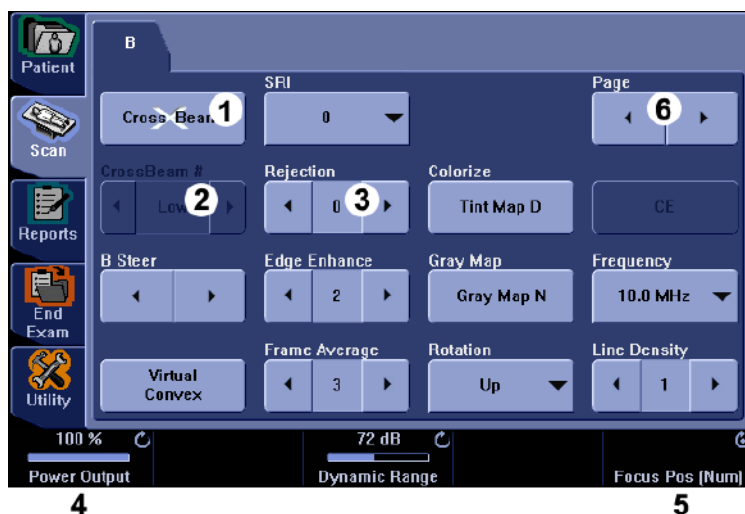


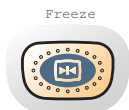
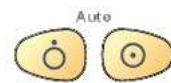
Figure 3-30. Mode/Function Specific Controls

1. Press to toggle control on/off.
2. Progress/Select keys are used for controls that have three or more choices.
3. Progression keys are used to assess the impact of the control on the image progressively.
4. Rotate the knob below the Touch Panel to set values.
5. Press the knob below the Touch Panel to select additional control, then rotate the knob to set values.
6. Press to move to the next Touch Panel page.

Mode, Display and Record

This group of controls provides various functions relating to the display mode, display orientation, image recording/saving, freeze, gain and Cine scroll.

The Mode Controls select the desired display mode or combinations of display modes.



- During dual display modes the **L** and **R** keys activate the Left or Right displayed image.
 - B Pause freezes the B-Mode image while keeping the Doppler spectrum display active.
 - Auto is used to:
 - initiate auto optimize
 - turn off the auto mode.
 - B Flow activates/deactivates the B Flow scan.
 - Depth controls the image display depth.
 - The Reverse key toggles the left/right orientation of the scan image.
 - Record keys are used to activate/print the designated recording device (e.g. video page printer, multi-image camera, image archive option).
 - The Freeze key is used to stop the acquisition of ultrasound data and freeze the image in system memory. Pressing **Freeze** a second time continues live image data acquisition.
 - To activate a specific mode, press the appropriate mode key.
- Each mode has its own gain control via the larger gray knob surrounding the mode key.

Measurement and Annotation

This group of controls performs various functions related to making measurements, annotating and adjusting the image information.

Comment



- The Comment key enables the image text editor and displays the annotation library Touch Panel.

Clear



- The Clear key is generally used to erase functions, such as annotations/comments and measurements. Pressing the Clear key again exits the selected function.

Body Pattern

Ellipse



- The Body Pattern/Ellipse control has a dual purpose:
 - When the Body Pattern/Ellipse control is pressed, it enables the Body Pattern Touch Panel and displays the default pattern on the screen. When body patterns are active, the knob rotates the probe position indicator.
 - When the Body Pattern/Ellipse control is adjusted, it activates the ellipse measurement function after the first distance measurement has been set and the second caliper is activated.

Press Set to fix the measurement after the ellipse adjustment is complete. The measurement is then displayed in the measurement result window.

Measure



- The Measure key is used in all types of basic measurements. It also becomes the “mouse” arrow for making selections along with the Set key (to fix or finish a selection). When the Measure key is pressed, the measurement Touch Panel is displayed.

Measurement and Annotation (continued)

 Zoom



- The **Zoom** control has a dual purpose:
 - When the Zoom control is pressed, it activates the write zoom function.
 - When the Zoom control is adjusted, it is used to activate the read zoom function.

M/D Cursor



- The M/D Cursor key enables Trackball control of the M-Mode or Doppler cursor line (not angle correction) or the CFM window in real-time Color Flow Mode.

Scan Area

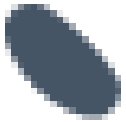


- The Scan Area key enables Trackball control of the B-Mode image area size and position in B/W, the CFM window size and position in Color mode and the Zoomed area size and position in Zoom.

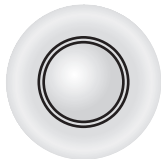
Set



- The Set key is used for various functions, but is generally used to fix or finish an operation (e.g. to fix a measurement caliper).



- The programmable Set key allows you to program the key's functionality (Pointer, Set or Freeze). The Program Key Mapping preset is located on the Utility -> System -> General preset menu and requires system reboot after reprogramming.



- The Trackball is used with almost every key function in this group. Trackball control depends on the last key function pressed.

Monitor Display

Monitor Display

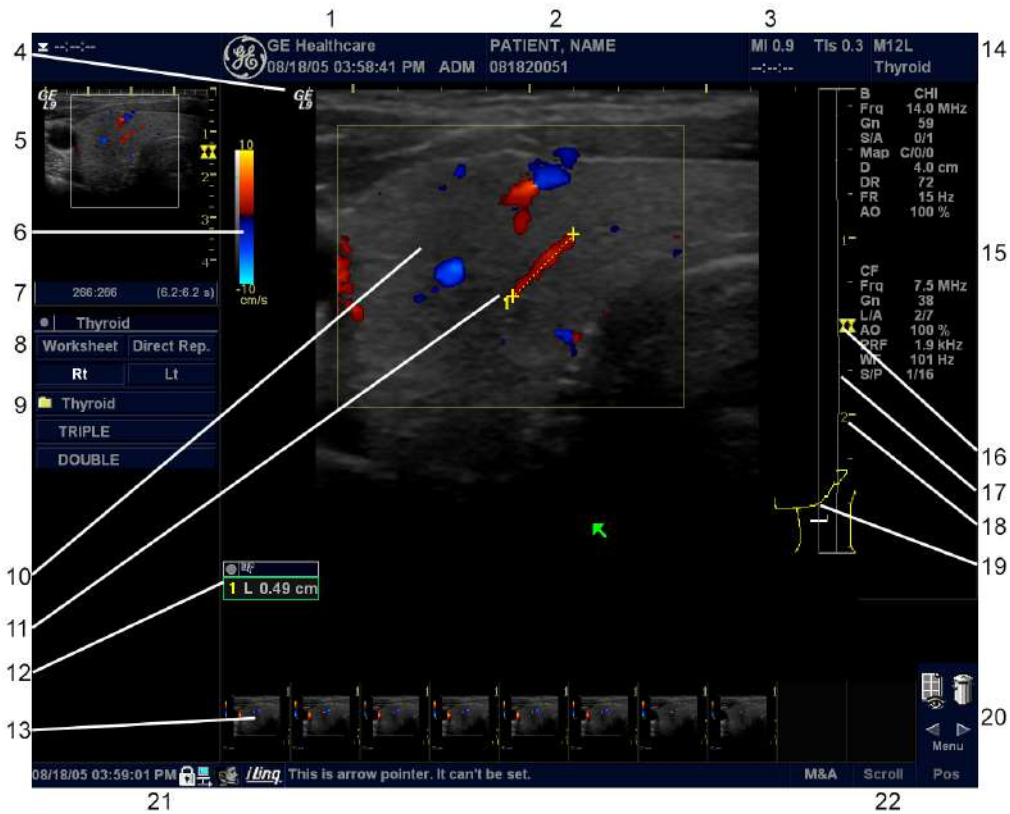


Figure 3-31. Monitor Display Tour

Monitor Display (continued)

1. Institution/Hospital Name, Date, Time, Operator Identification.
2. Patient Name, Patient Identification.
3. Power Output Readout
4. GE Symbol: Probe Orientation Marker.
5. Image Preview.
6. Gray/Color Bar.
7. Cine Gauge.
8. Worksheet/Direct Report
9. Measurement Summary Window.
10. Image.
11. Measurement Calipers.
12. Measurement Results Window.
13. Image Clipboard.
14. Probe Identifier. Exam Preset.
15. Imaging Parameters by Mode.
16. Focal Zone Indicator.
17. TGC.
18. Depth Scale.
19. Body Pattern.
20. Image Management Icons: Active Images, Delete, Previous/Next Image and Menu.
21. Current date and time, Caps Lock: (lit when on), network connection indicator (PC=connected, PC with X=not connected), Human face (VoiceScan), iLinq, system messages display.
22. Trackball Functionality Status: Scroll, M&A (Measurement and Analysis), Position, Size, Scan Area Width and Tilt.

Chapter 4

Preparing for an Exam

Describes how to begin an exam.

Beginning an Exam

Introduction

Begin an exam by entering new patient information.

The operator should enter as much information as possible, such as:

1. Dataflow
2. Exam category
3. Patient ID
4. Patient name
5. Exam Information

The patient's name and ID number is retained with each patient's image and transferred with each image during archiving or hard copy printing.



To avoid patient identification errors, always verify the identification with the patient. Make sure the correct patient identification appears on all screens and hard copy prints.

Beginning a New Patient

Pressing the **Patient** Touch Panel key displays the Patient Screen on the monitor. This screen is also accessible from the **Patient** key on the control panel.

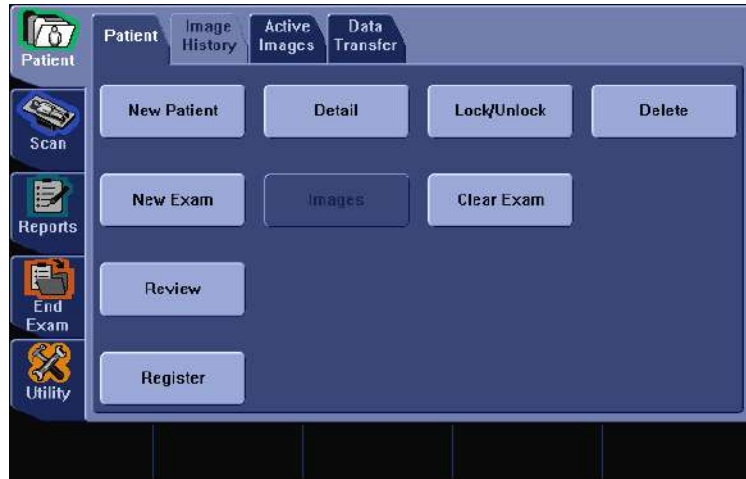


Figure 4-1. Patient Touch Panel

New Patient should be selected at the beginning of each new patient exam. End Exam should be pressed at the end of each exam. Pressing **New Patient** automatically stores all patient data, annotations, measurements, calculations and worksheets of the last exam. Images are not automatically stored. A warning message appears regarding images being erased or saved. It is always best to select “Select All” if you are unsure of saving. Exams and patients can always be erased later.

NOTE: *If the current patient does not exist, selecting **Patient View** activates the Patient Search screen.*

NOTE: *If the current patient exists, selecting **Exam View** activates the Exam History screen.*

Patient Screen

The screenshot shows the GE Healthcare Patient Screen interface. On the left is a vertical toolbar with icons for Patient, Data Transfer, Image History, and Active Images. Below these are buttons for 'New Patient', 'Register', 'Details', 'EZBackup', 'EZMove', 'Dataflow', and 'Exit'. The main area is divided into sections for patient information, exam selection, and a patient list. Numbered callouts point to specific elements: 1 points to the toolbar icons; 2 points to the 'New Patient' button; 3 points to the 'EZBackup' button; 4 points to the 'Dataflow' section; 5 points to the 'Exit' button; 6 points to the patient information fields (Patient ID, Last Name, First Name, Middle Name, DOB, Age, Sex); 7 points to the 'OB' category tab; 8 points to the exam information fields (LMP, Gravida, Para, AB, Ectopic, Accession #, Exam Description, Ref. Physician, Operator); and 9 points to the 'Patient View' tab and the patient list table.

Patient ID	Last Name	First Name	Birthdate	Sex	Last Exam	Img. size
12345	Smith	Mary	10/14/1945	F	08/09/2005 13:38	None

Figure 4-2. Patient Screen (Example: Category OB)

1. Image Management
2. Function Selection
3. EZBackup/EZMove
4. Dataflow Selection
5. Exit
6. Patient Information
7. Category Selection
8. Exam Information
9. Patient View

Patient Screen (continued)

Enter Patient Data with the alphanumeric keyboard.

To navigate through the Patient Entry menu, use the **Tab** key or **Trackball** and **Set** to move and fix the cursor.

The Patient Screen details are:

1. Image Management

- Patient—Provides a search and creation of patient. (currently selected)
- Image History—Provides a list of images per exam for the currently selected patient.
- Active Images—Provides preview of the currently selected exam.
- Data Transfer—Provides an interface to handle patient data from a remote device.

2. Function Selection

- New Patient—Used to clear patient entry screen in order to input a new patient's data into the database.
- Register—Used to enter new patient information into the database prior to the exam.

NOTE: *If you are using the auto-generate Patient ID feature, do not select Register.*

Patient Screen (continued)

- Details—Select the Detail box to activate/deactivate the exam details. Exam details include Indications, Comments, Admission Number, Performing Physician's Telephone Number, Referring Physician's Telephone Number, Operator Telephone Number and Exam Description.

NOTE: *Select preset at Exam Description to use as the identifier in DICOM.*

The screenshot displays a software interface for patient information. At the top, there are input fields for Patient ID, Last Name, DOB, Patient Phone #, First Name, Age, Address, Middle Name, Sex (radio buttons for Female and Male), and Comments. Below this is a tabbed interface with tabs labeled ABD, OB, GYN, CARD, VAS, UR, SMP, and PED. The CARD tab is currently selected. Under the CARD tab, there are fields for Height (cm), Weight (kg), BSA (m²), Accession #, Exam Description (a dropdown menu), Perf Physician (a dropdown menu), Perf Phone #, Admission #, Ref Physician (a dropdown menu), Ref Phone #, Indications (a dropdown menu), Operator (a dropdown menu with 'ADM' selected), Comments, and Operator Phone #. At the bottom right, there are two buttons: 'Images' and 'Clear'.

Figure 4-3. Detail Window

3. EZBackup/EZMove—One-step method to backup (move and delete patient images) to an external media. See Chapter 16 for more details.

Patient Screen (continued)

4. Dataflow Selection—Select the appropriate dataflow.

NOTE: *If you use a DVD-R or DVD-RAM, select DICOM CD Read in Dataflow.*

If you place the cursor on the icon, the pop-up menu displays disk capacity.



Figure 4-4. Dataflow Pop-up



WARNING

Ensure you have selected a dataflow. If No Archive is selected, no patient data is saved. A Ø appears next to Dataflow if No Archive is selected.

5. Exit—Used to exit Patient Menu.
6. Patient Information
 - Patient ID Number
 - Patient Name—Last, First and Middle
 - DOB (Birthdate)
 - Age (automatically calculated when birthdate is input)
 - Sex
7. Category Selection

Select from 8 exam application categories: Abdomen, Obstetrics, Gynecology, Cardiology, Vascular, Urology, Small Parts or Pediatrics.

When a category is selected, the measurement and category presets are displayed.

Patient Screen (continued)

8. Exam Information

Shows the Current/Active Exam information. Information pertinent to the selected exam category appears in the window. All possible information needs to be entered.

- Images—Displays the selected exam's images.



Figure 4-5. Images

- Clear—Clears existing data.
- Past Exam (only for OB)—Input past exam data (register the patient before using).

The screenshot shows a software window titled 'Fetus A'. In the top right corner, there are 'Prev' and 'Next' buttons. Below them, it says 'Page 1 of 2'. The main title of the window is 'Input Past Exam'. Below this title, there are six input fields arranged in two rows. The first row has labels: 'Exam Date', 'EFW', 'BPD', 'HC', 'AC', and 'FL'. The second row has labels: '(mm/dd/yyyy)', 'g', 'cm', 'cm', 'cm', and 'cm'. Below these labels, there are six 'Hadlock' buttons. Below the buttons is a large empty grid area. At the bottom of the grid, there is a text label: 'Past Exam Data is used for plotting on Fetal Trend Graph'. In the bottom right corner, there are 'Cancel' and 'Exit to Save' buttons.

Figure 4-6. Input Past Exam

Patient Screen (continued)

9. Patient View—Lists the patients in the database.

NOTE: Use **Ctrl + A** to select all from the Data Transfer list/Patient view.

NOTE: When you double-click the patient on the patient list with the **Set** key, the Review screen or New Exam entry screen displays. Select Review or New Exam in Utility -> Connectivity -> Miscellaneous -> Double click on patient list to start preset.

- Search key—select search item from Patient ID, Last Name, First Name, Birthdate, Sex, Exam Date, Exam Today, Accession Number, Exam Description, Exam Date Before, Exam Date Between, Exam Date After, Locked (Y, N) or Archived (Y, N).

NOTE: If “Exam Date Between” is selected, the Input Dialog displays and you can select the date from the displayed calendar.

NOTE: Archived means that the exam was backed up to external media by EZBackup or Export.

- String—enter appropriate information.
If you select Locked (Y, N) or Archived (Y, N) for the Search key, enter Y (Yes) or N (No).

NOTE: If “Exam Date Between” is used for the Search key, the From and To dates are separated by a “-” (dash) in the Search String.

- Clear—Clears the entered string.
- Show All—Displays all patients in the database.

Patient Screen (continued)

- Show XX/XX—Displays the searched (selected) patients (number of patients currently displayed on the list/ number of patients in the database). Displays a maximum of 102 patients.
- Review—Select Patient/Exam for review.
- Resume Exam—Continues the exam for that patient if you select the last exam of the day.



Figure 4-7. Patient List with Resume Exam

- NOTE:**
- New Exam—Creates a new exam on a current patient.
 - Delete—Deletes Patient/Exam.
“Delete” is only displayed when you login as Administrator.
 - Hide—Hides the Patient List Section.

Patient Screen (continued)

- Lock/Unlock—Locks the exam/patient. Prevents move and delete functions.

To lock, select the exam or patient to be locked and select **Lock**.



Figure 4-8. Lock the exam

If you select the patient, all exams are locked. If you select one exam, the selected exam is locked and the lock icon displays in the patient ID cell.

To unlock, select the locked exam or patient and select **Unlock**.



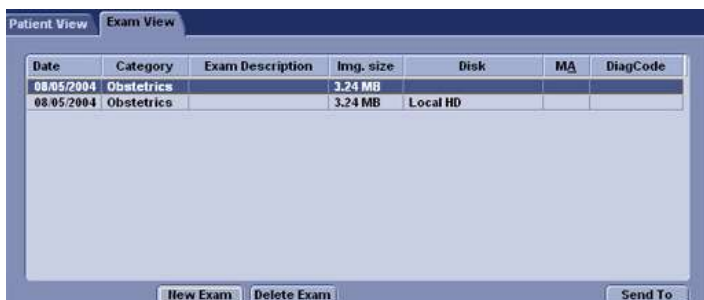
Figure 4-9. Unlock the exam

Patient Screen (continued)

Exam View—Displays the Exam History of the selected patient.

- Disk - Displays the disk name on which you saved the exam's image data. If "+" displays behind the disk name, the data is saved on two or more disks.

See 'Reviewing the Image' on page 4-22 for more information.



The screenshot shows a software window titled 'Patient View' with a sub-tab 'Exam View'. Inside, there is a table with the following data:

Date	Category	Exam Description	Img. size	Disk	MA	DiagCode
08/05/2004	Obstetrics		3.24 MB			
08/05/2004	Obstetrics		3.24 MB	Local HD		

Below the table are three buttons: 'New Exam', 'Delete Exam', and 'Send To'.

Figure 4-10. Exam View

NOTE:

The system can display the Detail Mode instead of Exam View when you select the patient on the patient list and press Review or Register. If the Detail Mode preset on Utility -> Connectivity -> Miscellaneous menu is selected, the Detail Mode displays.



To avoid patient identification errors, always verify the identification with the patient. Make sure the correct patient identification appears on all screens and hard copy prints.

Patient Screen (continued)



To maintain optimum performance and to safeguard patient data, keep the total number of patients in the database below 1,000.

To reduce the total number of patients in the database, perform the following procedure.

1. Before starting EZBackup, select "Unlock All" on the Utility -> Admin -> Logon screen.

Prepare the unformatted CD-R, DVD-R or MOD before EZBackup.

NOTE: *Formatted CD-R, DVD-R or MOD cannot be used for EZBackup.*

2. Backup all patient data by EZBackup. Before beginning EZBackup, set the "Backup Files Older Than in Days" to "0" on the Utility -> System -> Backup/Restore -> Backup/Move screen to backup all data.

Deselect (uncheck) "Move Files after Backup".

3. Go to the patient screen, select the patients/exams to delete so the remaining number is less than 1,000. Select "Delete" to delete the selected data.

NOTE: *Removing image data with the "Move" function does not reduce the patient number in the database.*

Scanning a New Patient

When starting a new patient's exam, ensure you do the following:

1. Press **Patient**.
2. Press **New Patient** on the Patient menu.
3. If there are images on the clipboard, a pop-up menu appears. Specify whether you want to store images permanently, delete images, or go to active images.
4. Choose the exam category.
5. Verify the dataflow.

NOTE: *DO NOT use the removable media Data Flows on the New Patient menu.*

NOTE: *The system can display a warning dialog when the patient is registered to "No Archive". If the "Warn register to No Archive" preset is selected in the Utility -> Connectivity -> Miscellaneous menu, a warning displays. A different dataflow for permanent storage of patient data should be selected.*

6. Fill in patient information.

NOTE: *You can also select a patient from the patient database at the bottom of the Patient menu if the patient has a patient ID.*

Columns drive the ordering of the patients displayed. The column that you select drives the order of the displayed patient database.

7. Select **Register**. Enter Past OB Exam information, if desired.
8. Select the probe from the Touch Panel to start scanning (or select Exit, Esc, Scan, or Freeze).
9. Perform the exam.

Scanning a New Patient (continued)

10. Store the raw data to the clipboard.

To store the still image, press **Freeze** and run the cineloop using the **Trackball**. Select the frame and press **P1** (or the assigned Printer key).

To store the cineloop, press **Freeze** and run the cineloop using the **Trackball**. Select the start/end frame and run the selected loop. Press **P1** (or the assigned Print key).

11. When you have completed the study, press **End Exam**. The image management screen displays. Select the images (still frame or cineloop) you want to store or select **Select All** to store all images. Select **Permanent Store** to store the images permanently.

NOTE: *Return to the patient screen automatically from the scan screen when you select OK from the "ID is not unique" warning message.*

Entering a Patient List

All patient information can be entered before starting an exam.

1. Press **Patient** to display Patient Screen.
2. Press **New Patient** to erase the current patient data.
3. Enter the Patient ID.
4. Enter the patient and exam information.
5. Press **Register**.
6. Repeat above steps as required.

Select the patient from the Patient List and select **Resume Exam** to start an exam.

Patient List Printout

You can print the patient list on a connected printer.

1. Move the cursor to the patient list and select any patient.
2. Press the left **Set** key. The pop-up menu displays.

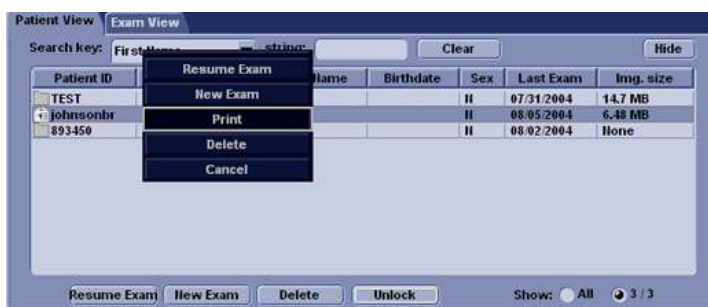


Figure 4-11. Print the patient list

3. Select **Print** and press **Set**.

Starting a new exam on an existing patient

1. Press **Patient**.
2. Select the patient from the Patient List.
3. Select **New Exam**.
4. A new exam is created. Enter the data and begin the scan.

Scanning without entering any patient data

To scan a patient without entering any patient data until the end of the exam:

1. Press **Scan**.
2. Scan the patient and save images to clipboard without patient. You receive a warning that states "A patient must be selected for permanent storage of image." Press OK.
3. Press **Patient** to display the Patient Search screen.
4. Enter the Patient ID.
5. Enter patient data and exam information as necessary.
6. Press **Active Images**.
7. Press **Permanent Store**.

Retrieving and editing archived information

Searching for an existing patient

1. Press **Patient** to display the Patient Screen. The Patient Search Screen is displayed.

GE Healthcare

Patient ID: 234561 Last Name: Smith DOB: 01/23/1944
First Name: Eileen Age: 57Y
Middle Name: Sex: ☐ female ☐ male

ABD OB GYN CARD VAS UR SM P PED

Height: 0.0 cm Accession #:
Weight: 0.00 kg Exam Description:
BSA: m^2 Ref.Physician:
Operator: ADM

Images Clear

Patient View Exam View

Date	Category	Exam Description	Img. size	MA	Disk
08/10/2005	Gynecology		None	☒	
08/15/2005	Gynecology		None		
08/16/2005	Abdomen		None		

EZBackup EZMove

Dataflow: Local Archive - Int. HD

Exit New Exam Delete Exam Send To

Figure 4-12. Patient Search Screen

2. Select the search key. Enter the search string.

NOTE: When the number of patients on a hard disk is in the hundreds, it takes time to search for a patient or switch to another screen. In this case, do one of the following:

- Uncheck the Auto Search function at the Utility -> Miscellaneous screen.
- Do not use "Show All".
- Delete unnecessary patient data.

Searching for an existing patient (continued)

- An appropriate patient is displayed in the patient list.

Select **Review** to review the exam history of this patient.

Select **New Exam** to create a new exam for this patient.

NOTE: The preset “Double click on patient list to start”, located on the Utility -> Connectivity -> Miscellaneous screen, allows you to display either the Review or New Exam screen by double clicking the **Set** key on the patient name.

Select **Delete** to delete this patient.

NOTE: “Delete” is only displayed when you login as Administrator.

NOTE: If you select the patient and press the left **Set** key, the pop-up menu displays.



Figure 4-13. Archived Patient

If you select the patient of the day, **Resume Exam** displays to continue the exam.

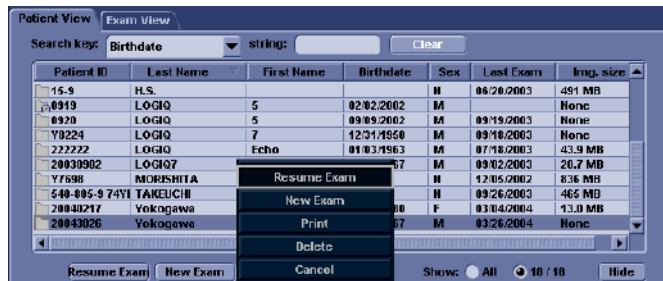


Figure 4-14. Patient of the Day

Reviewing the patient exam

To review the patient exam,

1. Search the patient by using Patient Search Field.
2. Move the cursor to the patient and press **Set**.
3. Select **Review** or the Exam View tab.
4. The Patient Exam screen displays.

GE Healthcare

Patient ID: 12345 Last Name: Smith DOB: 10/14/1945
First Name: Mary Age: 59Y
Middle Name: Anne Sex: ☐ female ☐ male

AED OB GYN CARD VAS UR SMP PED

LMP: 10/14/1995 Gravida: Accession #:
Para: Exam Description:
AB: Ref Physician:
Ectopic: Operator: ADM
Images Clear

Patient View Exam View

Date	Category	Exam Description	Img. size	MA	Disk
08/09/2005	Abdomen		None	☒	
08/09/2005	Gynecology		None		

EZBackup EZMove

Dataflow: Local Archive - Int. HD

Exit New Exam Delete Exam Send To

08/09/05 01:50:51 PM iLing

Figure 4-15. Patient Exam Screen

5. The exam list displays in the Exam History field.

OR

Reviewing the patient exam (continued)

1. Search the patient on the patient screen.
2. Place the cursor on the desired patient in the Patient View and double click the **Set** key. The Exam View displays.
3. Place the cursor on the desired exam and press **Set**.
4. Select **Scan**. The scan screen displays. Place the cursor on the desired image in the clipboard and press **Set**. The image displays on the scan screen.

Reviewing the Image

1. Select the exam which includes the image to review.
2. Select **Active Images**.
3. Select the image and press **Review** or double click on the image. The image is displayed.



Figure 4-16. Active Images Screen

Reviewing the Image (continued)

Another method to review the image:

1. Select the patient.
2. Select **Image History**.
3. Select the appropriate button which shows the old exam by date and storage location. 'Active Exam' is displayed on the button.



Figure 4-17. Image History Screen

4. Move the cursor to the required image for review.
5. Select **Review**. The selected image (maximum of 4 images) is displayed.

Reviewing the Image (continued)

If the image data is saved on a disk and you do not insert the disk when displaying the Image History page, a triangle icon displays instead of a thumbnail.

Place the cursor on the icon. The disk name displays under the preview window. Insert the appropriate disk.

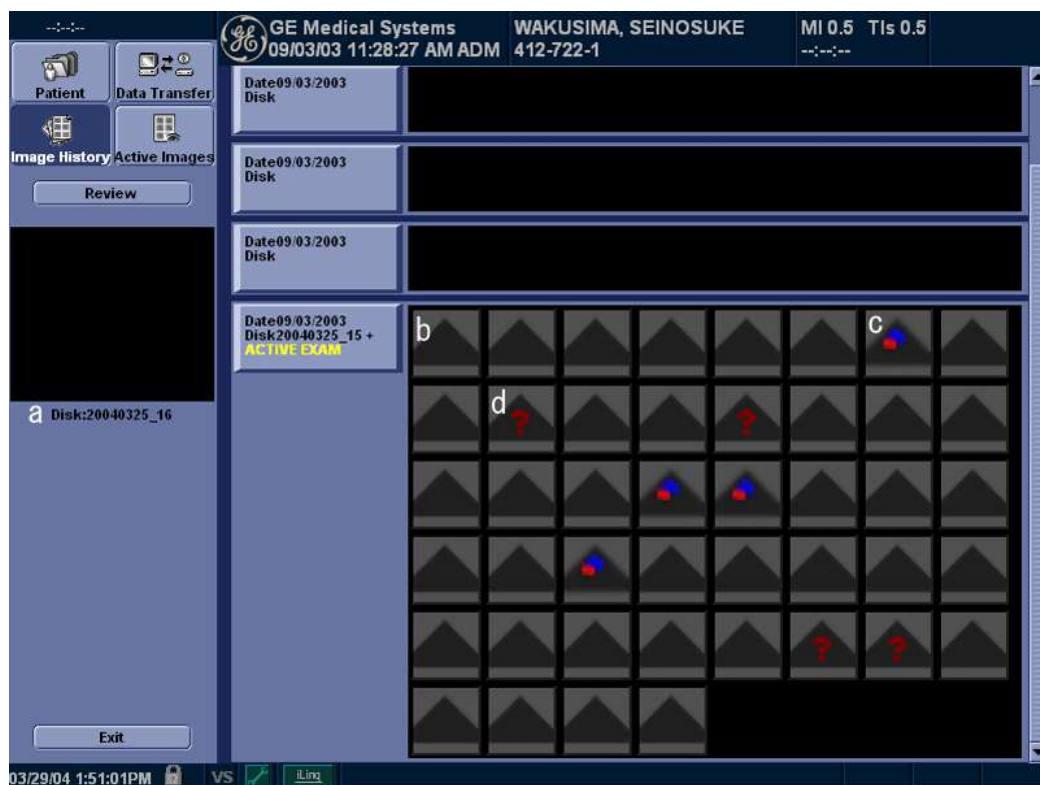


Figure 4-18. Image History (without disk in drive)

- a. Name of the disk displayed under the preview window
- b. Raw data B-Mode image
- c. Raw data Color image
- d. Image which does not have raw data (screen capture image)

Deleting the existing patient/exam/image



Deleting the existing patient

Before deleting a patient or image from the Patient Screen, make sure you have already saved the data with EZBackup/ EZMove or Export. Verify the media before deletion.

1. Search and select the patient in the patient list with the **Ctrl** or **Shift** keys.
2. Select **Delete**. The confirmation dialog box displays.
OR

Press the left **Set** key. A pop-up menu displays. Select **Delete**. The confirmation dialog box displays.



Figure 4-19. Select the patient in the patient list

3. Select OK to delete or Cancel.

Deleting the existing patient/exam/image (continued)

Delete multiple patients from the patient list

1. Select the multiple patients to be deleted from the patient list.
2. Select **Delete**. The confirmation dialog box displays.
OR
Press the left **Set** key. A pop-up menu displays. Select **Delete**. The confirmation dialog box displays.



Figure 4-20. Select deleted patients in the patient list

Deleting the existing exam

3. Select **OK** to delete or **Cancel**.
1. Search and select the patient in the patient list.
2. Select **Review**.
3. The patient exam screen displays. Select the exam to be deleted.
4. Select **Delete**. The confirmation dialog box displays.
5. Select OK to delete or Cancel.

Deleting the existing image

1. Search and select the patient in the patient list.
2. Select **Review**. The patient exam screen displays.
3. Select the exam which contains the image to be deleted.
4. Select **Active Images** to display the image list.
5. Select the image to delete and select **Delete**. The confirmation dialog box displays.
6. Select **Yes** to delete or **No** to cancel.

Changing Patient Information or an Exam

If patient information needs to be edited, pressing **Patient** enables the Patient Screen for modifying information.

If the patient is still active, you can go to the New Patient page and choose the tab that is desired. The exam changes in the Exam View area.

If the exam category needs to be changed, pressing **New Exam** allows modification of the Patient Screen without erasing accumulated patient images, measurements, annotations, calculations and worksheets.

1. Display the Patient screen by pressing **Patient**.
2. Select patient from the Patient list. The system automatically searches to see if the patient is already in the database.
 - Select Search key (Patient Data: ID, First Name, Last Name, Birthdate, Sex or Exam Date).
 - Enter search string (for example, initial letter of Patient Name)
3. The appropriate patient is displayed.

If patient information needs to be edited or the exam category changed, use the New Exam feature. Pressing **New Exam** allows modification of the Patient Screen without erasing accumulated patient images, measurements, annotations, calculations and worksheets.
4. To have the database shown in its entirety, **Backspace** on the Search string and all patient names appear.
5. Select Register to register the new exam.

A new exam is automatically created on that patient unless an exam already exists on that day for that patient.
6. To display the patient information on the title bar, press the **Esc** key, the **B-Mode** key or **Register**.

Select the Preset and appropriate probe Touch Panel keys, if necessary.

Selecting an Application Preset and a probe

Selecting an Application Preset

The exam category preset that best describes the desired exam to be performed is chosen after the exam category is selected. The factory default preset selections are displayed on the Touch Panel.

A typical example for the Abdomen exam category is shown below.



Figure 4-21. Abdomen Preset Selection Touch Panel

Use these parameters as a starting point for the exam.

User-Defined Application Presets

There are four user-defined application presets that can be set on the system.

To set up User-Defined Application Presets,

1. From the Touch Panel, select the **Preset** icon at the top of the Touch Panel. The Preset Touch Panel appears:



Figure 4-22. Preset Touch Panel

User-Defined Application Presets (continued)

2. Press **Save**. A pop-up menu appears: The Create New Application menu appears.



Figure 4-23. Create New User Pop-up Menu

3. Select **Create New**. The Create New Application menu appears.



Figure 4-24. Create New User Application Menu

NOTE: *The new user application is based upon the current exam category and application, plus any modifications you have made, including the comment library and M&A calcs.*

User-Defined Application Presets (continued)

4. Type the name of the new application. Specify which User Touch Panel key you want to use and press **Save**.

NOTE: *The name of the new application cannot include spaces or symbols. However, the name can include numbers and letters.*



Figure 4-25. Create New User Application Menu

The new application now appears on the Touch Panel.



Figure 4-26. New User Application

NOTE: *You can set up to four (4) user-defined application presets for each exam category.*

User-Defined Application Presets (continued)

5. To view/edit the parameters for the user-defined preset, press Utility --> Imaging.

If you change the settings for this application, make sure to save the changes via Save --> Overwrite (user application).

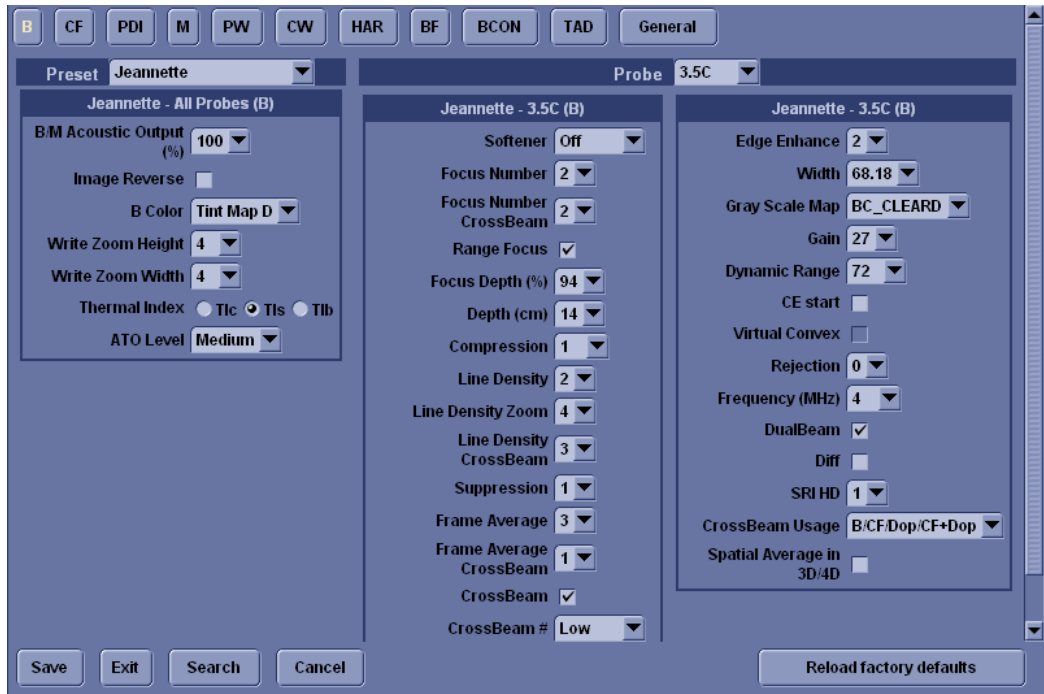


Figure 4-27. Imaging Presets for User-Defined Application

NOTE: If you select Reload Factory Defaults for the User-Defined application that you created, the settings for this user-defined application revert back to the factory settings for the exam category and application it was based upon.

NOTE: Currently, there is no way to delete a user-defined application.

Selecting a probe

Select the appropriate probe from the probe indicator. Probes can be used in more than one exam category/application.

- Always start out with a probe that provides optimum focal depths and penetration for the patient size and exam.
- Begin the scan session using the default Power Output setting for the probe and exam.

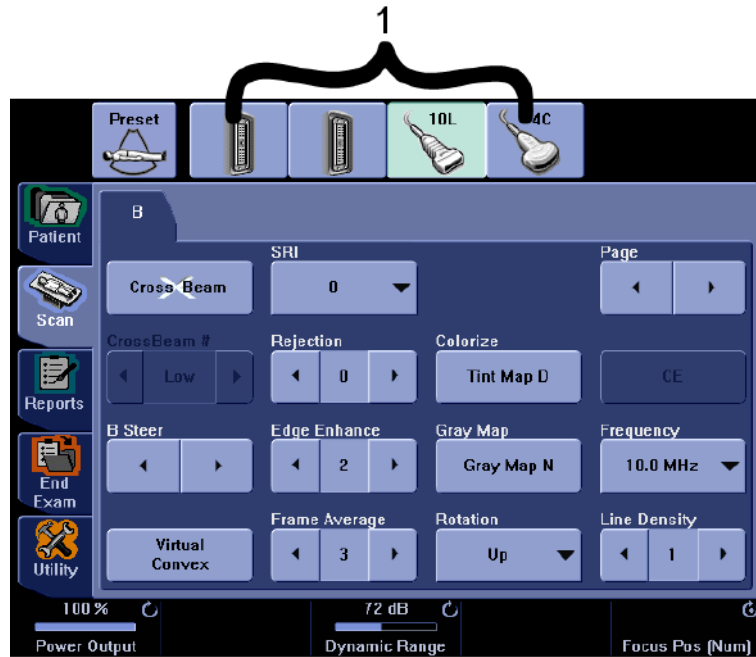


Figure 4-28. Probe Indicators (Example)

1. Probe Indicators

Ending a Patient Exam

When you have completed the study, press **End Exam**, End Current Patient or New Patient, then Store All Images Permanently.

Chapter 5

Optimizing the Image

Describes how to adjust the image. This chapter is broken into the following sections: B-Mode, M-Mode, Color Flow Mode, Doppler Mode, and 3D Mode.

Optimizing B-Mode

Intended Uses

B-Mode is intended to provide two-dimensional images and measurement capabilities concerning the anatomical structure of soft tissue.

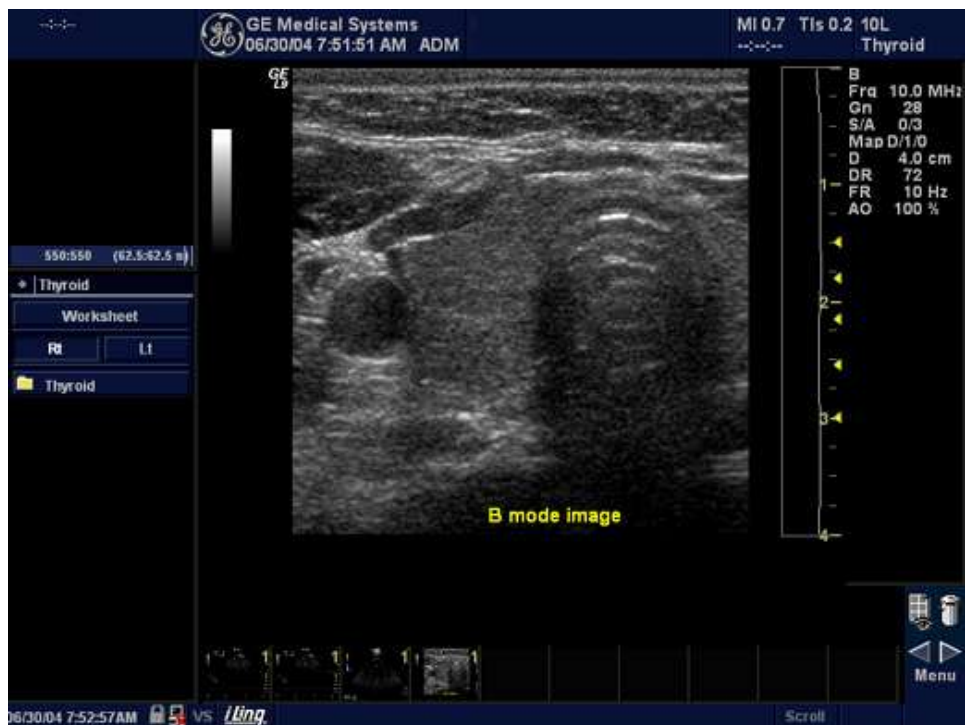


Figure 5-1. B-Mode Display

Typical B-Mode Exam Protocol

A typical examination using B-Mode might proceed

1. Record exam-related patient information. Verify system setup (probes and presets).
2. Position the patient and the console for optimum operator and patient comfort. Perform the scan.
3. Complete the study by collecting all the data.

B-Mode Touch Panel



Figure 5-2. B-Mode Touch Panel, Page 1

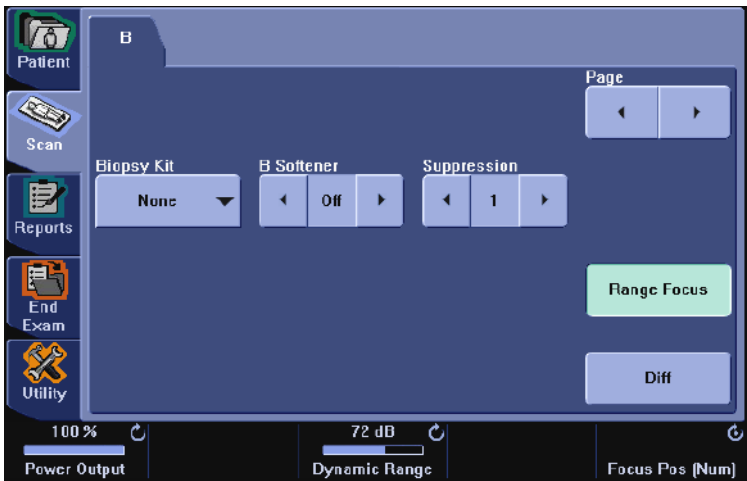


Figure 5-3. B-Mode Touch Panel Page 2

B-Mode Scanning Hints



HINTS

These B-Mode controls produce the following results:

Auto Optimize. Improves imaging performance while reducing optimization time. Available in B-Mode, Color Flow Mode, and Doppler Mode.

Coded Harmonics. Enhances near field resolution for improved small parts imaging as well as far field penetration.

B Flow. Provides a more intuitive representation of non-quantitative hemodynamics in vascular structures.

Frequency. Changes system parameters to best optimize for a particular patient type.

Gray Map. Affects the presentation of B-Mode information. Choose the gray map prior to making other adjustments. There is an interdependency between gray maps, gain, and dynamic range. If you change a map, revisit gain and dynamic range settings.

Dynamic Range. Affects the amount of gray scale information displayed. If you increase the gain, you may want to decrease the Dynamic Range.

Edge Enhance. Affects the amount of border crispness.

Frame Average. Smooths the image by averaging frames. Affects the amount of speckle reduction.

B Softener. Affects the amount of lateral smoothing.

TGC. Adjust TGC to adjust Gain in specific areas.

Focus Number/Position. The best focusing is at the focal zone location. Put focal zone(s) at the area of interest. Be conscious of where the focal zones are. Focal zones must be moved to track at the center of the anatomy of interest.

Scan Area. Sizes region of interest. Adjust the Scan Area to the smallest reasonable size to maximize frame rate.

B-Mode Scanning Hints (continued)

Table 5-1: Additional B-Mode Scanning Hints

If...	Then...
Image Too Grainy	1. Increase B-Mode Dynamic Range. 2. Increase Frame Averaging. 3. Decrease B-Mode Edge Enhance. 4. Activate the smoothing filter.
Image Too Soft	1. Decrease B-Mode Dynamic Range. 2. Increase B-Mode Edge Enhance. 3. Decrease Frame Averaging. 4. Deactivate the smoothing filter.
Image Too Noisy	1. Decrease Gain 2. Decrease B-Mode Dynamic Range. 3. Increase Frame Averaging. 4. Increase B-Mode Edge Enhance. 5. Select a Gray Map with more contrast.
Improve Uniformity	1. Increase the number of focal zones. 2. Decrease the region of interest size. 3. Adjust the TGC to compensate for attenuation.
Cystic Imaging	1. Decrease Gain. 2. Increase B-Mode Dynamic Range. 3. Reduce the image width. 4. Increase the number of focal zones. 5. Position the focal zones correctly.
Technically Difficult Patients	You may also want to adjust the Frequency in addition to these steps: 1. Select the proper probe (larger patient, lower frequency). 2. Increase the Acoustic Power, if necessary. 3. Maintain a lower B-Mode Dynamic Range. 4. Decrease the region of interest size for faster frame rates.

Depth

Description	Depth controls the distance over which the B-Mode images anatomy. To visualize deeper structures, increase the depth. If there is a large part of the display which is unused at the bottom, decrease the depth.
Adjusting	Each adjustment cycles you to the next Depth setting. Imaging and display parameters adjust automatically. To increase/decrease, adjust Depth .
Preset	You can reverse how the Depth control functions on the Utility -> System -> General page. You can set the depth by probe and application on the Utility --> Imaging page.
Values	Depth increments vary by probe and application. Depth displays on the monitor in centimeters. Depth values are returned to the factory or user preset value when you change the following: Probe, Exam Category/Exam Calc, or New Patient.
Benefits	Depth adjusts your field of view. It increases your field of view to look at larger or deeper structures; it decreases your field of view to look at structures near the skin line.
Affect on other controls	After adjusting the depth, you may need to adjust the TGC and focus. Changing Depth, • Clears Cine memory. • Erases real-time calculations graphics on the display (but not the completed results on the worksheet page).
Bioeffects	Changing the depth may change the TI and/or MI. Observe the output display for possible effects.



HINTS

Make sure enough space is left below the anatomy of interest to demonstrate shadowing or enhancement.

Gain

Description B-Mode Gain increases or decreases the amount of echo information displayed in an image. It may have the effect of brightening or darkening the image if sufficient echo information is generated.

Adjusting Gain values vary depending on the probe; they are not associated with a particular position of the knob.

NOTE: TruAccess allows you to change the gain on a frozen image. B-Mode gain is independent of M-Mode and Doppler and Color Flow Gain. Changing the M-Mode Gain while in M-Mode does not affect the B-Mode image gain.

To decrease/increase, rotate **Gain**.

Preset You can set gain by probe and application on the Utility --> Imaging page.

Values Gain displays on the monitor in Gn (dB). Maximum gain varies by probe. Gain values vary by probe, application, and frequency setting.

NOTE: Maximum gain is factory preset to an optimum setting to eliminate noise in the display.

Gain values are returned to the factory or user preset value when you change the following: Probe, Exam Category/Exam Calc, or New Patient.

Benefits Gain allows you to balance echo contrast so that cystic structures appear echo-free and reflecting tissue fills in.

Affect on other controls After you adjust the Power Output, you may need to adjust the gain. Generally speaking, if you increase the Power Output, you need to decrease the gain; if you decrease the Power Output, you need to increase the gain. Gain and TGC interact by adding together.

Bioeffects Gain has no affect on Power Output. However, with increased gain, the power output level can usually be reduced to produce an equivalent image quality.

NOTE: Always optimize gain before increasing the Power Output.

Focus

Description	Increases the number of focal zones or moves the focal zone(s) so that you can tighten up the beam for a specific area. A graphic caret corresponding to the focal zone position(s) appears on the right edge of the image.
Adjusting	To increase/decrease the number of focal zones, adjust Focus Number. <i>NOTE: Focus Position and Focus Number share the same control. Press the control to toggle between Focus Position and Focus Number.</i> To move the focal zone to the near/far field, adjust Focus Position. Focal zones adjust automatically when you adjust the depth.
Preset	You can reverse how the Focus control functions on the Utility -> System -> General and VoiceScan pages. <i>NOTE: Each adjustment cycles you to the next setting.</i> You can set the focus (Depth % and Number) by probe and application on the Utility --> Imaging page.
Values	Focus zone number and position vary depending on the depth, zoom, probe, application, and frequency setting selected. Focal zone number values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calc, or New Patient.
Benefits	Focus optimizes the image by increasing the resolution for a specific area.
Affect on other controls	Changing the focal number affects the frame rate. The greater number of focal zones, the slower the frame rate.
Bioeffects	Changing the focal zone may change the TI and/or MI. Observe the output display for possible effects.

Auto Optimize / Auto TGC

Description	Auto Optimize (Auto) lets you optimize the image based upon a the actual B Mode image data (Auto Tissue Optimize, ATO). The preset levels (Low, Medium, and High) allow you to pick a preference for the contrast enhancement in the resulting image. Low does the least amount of contrast enhancement, high does the most. Auto is available in single or multi image, on live, frozen or CINE images (in B-Mode only), and while in zoom, in Color Flow Mode, and in Spectral Doppler. Auto in Color Flow Mode automatically adjusts the overall color gain. If you find that the gain adjustment that is automatically performed consistently results in more or less gain than you expect, then the Auto Optimize Adjustment allows you to adjust the result of the Auto feature (-5 to 5 range) so that the result more consistently matches your expectation. Auto in PW Doppler Mode optimizes the spectral data. Auto adjusts the Velocity Scale/PRF (live imaging only), baseline shift, dynamic range, and invert (if preset). Upon deactivation, the spectrum is still optimized. Auto TGC allows the system to automatically adjust internal TGC curves to achieve uniform brightness throughout the image. Auto TGC is supported on all probes in B-Mode and Harmonics. TruAccess allows Auto TGC to work on archived images.
Benefit	Auto can be found in reduced optimization time and a more consistent and accurate optimization process.
Adjusting	To activate, press Auto.

Auto Optimize / Auto TGC (continued)

Preset	To set up Auto, preset via Utility --> Imaging --> B-Mode --> Auto Optimize Mode. To set up ATO (Auto Tissue Optimize), preset via Utility --> Imaging --> B-Mode --> Auto Optimize Mode. Select ATO, Auto TGC, or Both (ATO + Auto TGC). Specify the ATO Level: Low, Medium, or High. Specify the CF Auto Optimize Adjustment from -5 to 5.
Values	Auto and Auto TGC are active until you deactivate it or when you change the following: Probe, Exam Category, Exam Calc, or New Patient.
Affect on other controls	You may need to adjust the Gain.

M/D Cursor

Description	Displays the M/D-Mode cursor on the B-Mode image.
Adjusting	To activate/deactivate the M/D-Mode cursor, press M/D Cursor . Trackball to position M/D-Mode cursor.
Values	On/Off.
Benefits	Lets you position the cursor before you go into M-Mode or Doppler Mode so that you can make optimum use of the larger B-Mode image.

B Flow

Description	B Flow is intended to provide a more intuitive representation of non-quantitative hemodynamics in vascular structures. All B-Mode measurements are available with B Flow active: depth, distance along a straight line, % stenosis, volume, trace, circumference, and enclosed area.
Activating	To activate/deactivate B Flow, press B Flow. Doppler Mode is available while in B Flow; however, M-Mode and Color Flow/PDI Modes are not available. See Chapter 17 for probe availability.
Using B Flow	To optimize the image: Adjust the gain, optimize the image for frame rate/resolution via Line Density, adjust the focal zone, zoom the image, colorize the image, adjust the field of view via Scan Area, select Auto.
Scanning Hints	B Flow is especially intuitive when viewing blood flow, for acute thrombosis, parenchymal flow, and jets. B Flow helps you visualize complex hemodynamics and highlights moving blood and tissue. B Flow was implemented using Digitally Encoded Ultrasound (DEU). There are no artifacts such as bleeding, blooming, or aliasing. The greater the speed, the better the image scatter density and size. If the scan direction is the same as the flow direction, then the image scatter is elongated; if the scan direction is the opposite as the flow direction, then the image scatter is tighter. Therefore, have the scan direction opposite to that of flow direction. Switch the way you hold the probe, with the probe orientation marker inferior to maintain correct orientation on the monitor. Flow starts from where the focal zone is located. Turn off the background when imaging the kidney, liver, and spleen. Keep the focal zone as close to flow as possible. It is beneficial to narrow the sector width and increase the frame rate.

B Flow (continued)

Benefits	B Flow is less angle-independent, no velocity-aliasing artifacts, and provides better resolution when compared with Color Doppler Mode, and displays a full field of view. B Flow is therefore a more realistic (intuitive) representation of flow information, allowing you to view both high and low velocity flow at the same time.
Affect on other controls	When you activate B Flow, the image remembers the imaging parameters set while in B-Mode. When you optimize the frame rate via Line Density, you compromise the resolution and when you optimize the resolution, you compromise the frame rate. B Flow is not available in 3DView.
Bioeffects	When you select B Flow, the PRF increases. Adjusting the following controls may affect the Power Output: Focal Zone Position, Depth, Zoom, Line Density, Scan Area, Frequency, or angle steer. Observe the output display for possible effects.

CrossXBeam

Description

CrossXBeam is the process of combining three or more frames from different steering angles into a single frame. CrossXBeam is available on Convex and Linear probes.

Adjusting

CrossXBeam combines frames steered from a minimum of three different angles and up to a maximum of nine steer angles.

To activate CrossXBeam, press the **CrossXBeam** key on the Touch Panel.

To adjust the number of frames being compounded, press the **CrossXBeam#** Touch Panel key. Select Low, Medium, High or Max.

You can preset B-Mode CrossXBeam:

- To activate upon entering B-Mode/Color Flow Mode/PW Doppler Mode
- Preset the CrossXBeam #
- Preset Focus Number CrossXBeam
- Preset Line Density CrossXBeam
- Frame Average CrossXBeam
- CrossXBeam Usage (B, B/CF, B/Dop, B/CF/Dop, or B/CF/Dop/CF+Dop)

These presets can be made via the Utility --> Imaging page.

CrossXBeam (continued)

Values

All linear and certain curved convex probes are supported. Multiple focal zones are supported. B-Mode CrossXBeam is available while in B-Mode, Color Flow, or PW Doppler Mode. Steering is optimized by probe. The displayed compound image depth is equal to the image depth of the non-steered frame.

The following features are supported in CrossXBeam:

- Harmonic Imaging
- CINE Loops acquired using CrossXBeam.
- Read Zoom
- All Measurement and Analysis packages
- Virtual Convex (Linear probe)
- SRI HD, Line Density, Depth, Map, TGC, Gain, Edge Enhance, Frame Averaging, Compression, Colorize, Acoustic Output, Rejection, Rotation, B-Softener, Auto, Reverse, Dynamic Range, Biopsy, LOGIQView, 3D Imaging

The following controls are not available:

- Anatomical M-Mode
- B Flow

Benefits

The combined single image has the benefits of reduced speckle noise, reduced clutter, and continuity of specular reflectors. Therefore, this technique can improve contrast resolution with increased conspicuity of low contrast lesions, better detection of calcifications, biopsy needle visualization, and cystic boundary definition.

SRI HD (High Definition Speckle Reduction Imaging)

Description

SRI HD (High Definition Speckle Reduction Imaging) is an adaptive algorithm to reduce the unwanted effects of speckle in the ultrasound image. Image speckle usually appears as a grainy texture in otherwise uniform areas of tissue. Its appearance is related to image system characteristics, rather than tissue characteristics, so that changes in system settings, such as probe type, frequency, depth, and others, can change the appearance of the speckle. Too much speckle can impair image quality and make it difficult to see the desired detail in the image. Likewise, too much filtering of speckle can mask or obscure desired image detail. Extra care must be taken to select the optimal SRI HD level.

SRI HD is available in B-Mode and Color Flow imaging and may be used with any transducer or clinical application when image speckle appears to interfere with the desired image detail.



Figure 5-4. B-Mode Image before SRI (left) and After SRI (right)

SRI HD (High Definition Speckle Reduction Imaging) (continued)

Adjusting	To activate SRI HD, adjust the SRI HD Touch Panel.
Preset	You can also set presets via Utility--> Imaging--> B-Mode, B Flow, or Contrast.
Values	0-4 (0=Off). We recommend that you select the SRI HD level by observing the enhanced image in side-by-side dual image comparison with the original, unprocessed image. Dual display mode is activated by pressing the L and R keys simultaneously. In selecting the level of SRI HD, you must observe the effects of SRI HD in the desired region of interest and should make a real-time comparison with the original image. The optimal level depends on the clinical situation and improves with experience. Observing the original and SRI HD-processed images together helps to determine whether too much or too little SRI HD has been applied. Dual image mode for SRI HD can also be activated on a stored CINE Loop. This allows you to always see the original, unprocessed or enhanced image by going into the Dual display mode and to change the SRI HD settings when reviewing the CINE Loop.
Benefits	Smooths the image when image speckle interferes with the desired image detail.

LOGIQ 9 Coded Harmonic Imaging (CHI)

Description	Harmonic imaging utilizes Digitally Encoded Ultrasound (DEU). Coded Harmonics enhances near field resolution for improved small parts imaging as well as far field penetration.
Adjusting	To activate Coded Harmonic imaging, Press Harmonics .
	<i>NOTE: Multi frequency is not active when the image is frozen.</i>
Values	On/Off. See Chapter 17 for probe availability. 'CHI' appears next to the active probe indicator.
	<i>NOTE: Changing multi frequency resets those parameters which are presettable by frequency to their preset values for the current harmonic frequency.</i>
	Multi frequency values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Application, or New Patient.
Benefits	Coded Harmonics diminishes low frequency high amplitude noise and improves imaging technically difficult patients. Coded Harmonics may be especially beneficial when imaging isoechoic lesions in shallow-depth anatomy in the breast, liver, and hard-to-visualize fetal anatomy. Coded Harmonics may improve B-Mode image quality without introducing a contrast agent.
Affect on other controls	You may need to change the field of view (Depth). DO NOT use contrast agents while in Coded Tissue Harmonics.
Bioeffects	Activating multi frequency mode may change the TI and/or MI. Observe the output display for possible effects.

Frequency

Description	Multi Frequency mode lets you downshift to the probe's next lower frequency or shift up to a higher frequency.
Adjusting	To select a new frequency, 1. Select Frequency. 2. Select the desired frequency. <i>NOTE: Frequency change is not active when the image is frozen.</i> The selected frequency appears in the upper, right-hand portion of the monitor display. <i>NOTE: Changing frequency resets those parameters which are presettable by frequency to their preset values for the current frequency.</i>
Values	Vary, depending on the probe and application. Frequency values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	This optimizes the probe's wide band imaging capabilities at multiple frequencies to image at greater depths.
Bioeffects	Activating multi frequency mode may change the TI and/or MI. Observe the output display for possible effects.

Virtual Convex

Description	On Linear and Sector probes, Virtual Convex provides a larger field of view in the far field. Virtual Convex is always active with Sector probes.
Adjusting	To activate/deactivate Virtual Convex, select Virtual Convex .
Values	On/Off.
Benefits	Virtual Convex allows for a wider field of view. Available in B-Mode, Color Flow Mode, and Doppler Mode. CrossXBeam is available on Virtual Convex with linear probes.
Bioeffects	Activating Virtual Convex may change the TI and/or MI. Observe the output display for possible effects.
NOTE:	<i>Color Flow can be steered when in straight fire, but not in Virtual Convex.</i>

TGC

Description	TGC amplifies returning signals to correct for the attenuation caused by tissues at increasing depths. TGC slide pots are spaced proportionately to the depth. The area each pot amplifies varies as well. A TGC curve may appear on the display (if preset), matching the controls that you have set (except during zoom). You can choose to deactivate the TGC curve on the image.
Adjusting	To decrease/increase TGC, move slide pot to the left/right.
NOTE:	<i>TGC adjusts automatically when using zoom.</i>
Values	When you change the depth, TGC is rescaled across the new depth range. Each pot is proportionately scaled across the depth.
Preset	On/Off -- preset via Utility --> System --> System Imaging.
Benefits	TGC balances the image so that the density of echoes is the same throughout the image.

Scan Area

Description	You can widen or narrow the size of the sector angle to maximize the image's region of interest (ROI).
Adjusting	To narrow/widen the angle, press Scan Area until <i>Width</i> is highlighted in the message line and move the Trackball left/right to decrease/increase the angle size. Then press Scan Area to set the ROI.
Values	Varies, depending upon the probe (not applicable to linear probes) and application.
Benefits	Increase the sector angle to see a wide field of view; decrease the sector angle when you need to have a faster frame rate, as in fetal heart.
Affect on other controls	Changing the sector angle affects the frame rate. The narrower the sector angle, the faster the frame rate.
Bioeffects	Changing the sector angle may change the TI and/or MI. Observe the output display for possible effects.

Tilt

Description	You can steer the sector angle to get more information without moving the probe while in B-Mode, M-Mode, Doppler Mode, and Color Flow Mode. <i>Tilt</i> is not available on Linear probes.
Adjusting	To steer the angle to the left/right, press Scan Area until <i>Tilt</i> is highlighted, then move the Trackball to the left/right.
Values	Varies, depending on the probe.
Benefits	Allows you to move a reduced sector angle laterally, without moving the probe. Beneficial in GYN.
Bioeffects	Steering the sector angle may change the TI and/or MI. Observe the output display for possible effects.

B Steer

Description	You can slant the B-Mode or Color Flow linear image left or right to get more information without moving the probe. The angle steer function only applies to linear probes.
Adjusting	To slant the linear image to the left/right, select B Steer .
Values	Linear probes can be steered left, center, or right up to a maximum of 15 degrees, depending on the probe. B Steer values are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Preset	Reverse Steer Controls via Utility --> System --> General.
Benefits	Provides a Doppler cursor angle suitable for linear probe orientation. Beneficial in Peripheral Vascular to image carotids.
Bioeffects	Activating angle steer may change the TI and/or MI. Observe the output display for possible effects.

Reverse

Description	Flips the image 180 degrees left/right.
Adjusting	To flip the image 180 degrees, select Reverse.
Values	The image rotates in 180 degrees left/right. Reverse settings vary by probe and application. Reverse settings are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Used for anatomical correctness.



When reading a reverse image, be careful to observe the probe orientation to avoid possible confusion over scan direction or left/right image reversal.

Dynamic Range (Compression)

Description	Dynamic Range controls how echo intensities are converted to shades of gray, thereby increasing the adjustable range of contrast. The Dynamic Range control name changes to Compression on frozen images.
Adjusting	To increase/decrease, adjust Dynamic Range.
Values	The settings cycle in 3dB steps from 30 dB to 120 dB. The current value displays. Dynamic Range values vary by probe, application, and frequency setting. Dynamic Range levels are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, New Patient, or Multi Frequency.
Benefits	Dynamic Range is useful for optimizing tissue texture for different anatomy. Dynamic Range should be adjusted so that the highest amplitude edges appear as white while lowest levels (such as blood) are just visible.
Affect on other controls	Dynamic range operates in realtime, Freeze, CINE, and CINE Timeline, but is not available while in VCR playback. It also affects Gain.

Range Focus

Description	Improves the near/mid field image quality, borders/interfaces, increases contrast and detail resolution across the image, and allows for less filling in the vessels.
Adjusting	On/Off via Utility--> Imaging--> B-Mode preset page. Then activate via the 2nd B-Mode Touch Panel. Available in B-Mode and Harmonics.
Bioeffects	Activating Range Focus may change the TI and/or MI. Observe the output display for possible effects.

Line Density

Description	Optimizes B-Mode frame rate or spatial resolution for the best possible image.
Adjusting	Press Line Density right to increase resolution or left to increase frame rate.
Values	Varies by probe.
	<i>NOTE: Not available in timeline.</i>
	Values vary by probe and application. Line Density values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	A lower line density is useful in fetal heart beat, adult cardiac applications and in clinical Radiology applications requiring significantly higher frame rates. A higher line density is useful in obtaining very high resolution, e.g., thyroid, testicles.
Affect on other controls	Line density changes the vector density and frame rate.
Bioeffects	Activating color flow line density may change the TI and/or MI. Observe the output display for possible effects.

Map

Description The system supplies B, M, and Doppler Mode system maps.

Adjusting To select a map, press **Gray Map** or **Clear Map**. A map window displays. The image reflects the map as you go through the selections.

Values Gray maps gradually change from least contrasty or softest to most contrasty. Clear Maps provide a more transparent map.

Map values vary by probe, application, and multi frequency setting. Map values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.

Map	Description
A	Assigns a greater amount of gray scale pixel values to bright reflectors in the image. Useful when imaging abdomen, liver, kidney, OB, pelvic, etc.
B	Assigns a greater amount of gray scale pixel values to bright reflectors in the image. Useful when the abdomen, liver, kidney, OB, pelvic, etc.
C	Assigns a greater amount of gray scale pixel values to bright reflectors in the image. Useful when imaging the abdomen, liver, kidney, OB, pelvic, etc.
D	Assigns a lesser amount of gray scale pixel values to bright reflectors in the image, compared to Maps B and C. Useful when imaging bright carotid plaque reflectors.
E	Assigns a equal amounts of gray scale pixel values to all reflectors in the image. Useful when imaging tissue with bright reflectors, e.g., cyst with septations or calcifications.
F	Assigns a lesser amount of gray scale pixel values to bright reflectors in the image. Useful when imaging arteries and grafts.
G	Assigns a lesser amount of gray scale pixel values to weak reflectors in the image than Map F. Useful when imaging small parts.
H	Assigns an s-shape to gray scale pixel values. Useful when imaging anatomical signals with less tissue differentiation, e.g., tendon, vein, carotid, thyroid, breast, etc.
I	S-shaped map. This map highlights tissue differentiation for a certain band of signals. Useful when imaging structures where you want more contrast, e.g., renal, tendon, vein, carotid, thyroid, breast, etc.
K	S-shaped map. This map highlights tissue differentiation for a certain band of signals. Useful when imaging structures where you want more contrast, e.g., renal, tendon, vein, carotid, thyroid, breast, etc.
L	S-shaped map. This map highlights tissue differentiation for a certain band of signals. Useful when imaging structures where you want more contrast, e.g., cardiology.
Q	Musculoskeletal Map.

Frame Average

Description	Temporal filter that averages frames together, thereby using more pixels to make up one image. This has the effect of presenting a smoother, softer image.
Adjusting	To adjust frame averaging, adjust Frame Average on the Touch Panel.
Values	The current value displays on the Touch Panel. Frame Average values vary by probe, application, and multi frequency setting. Frame Average values are returned to the preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Smooths the image.

Coded Excitation (CE)

Description	Coded Excitation improves image resolution and penetration in the far field.
Adjusting	To activate CE, press CE.
Values	On and Off.
Benefits	Allows you to use a higher frequency.
Affect on other controls	Activating CE affects the frame rate.
Bioeffects	Changing the focal zone may change the MI. Observe the output display for possible effects.

NOTE: *Coded Excitation can be turned on by default at Utility -> Imaging -> B-Mode menu. Select "CE Start".*

Colorize

Description

Colorize is the colorization of a conventional B-Mode image or Doppler Spectrum to enhance the user's ability to discern B, M, and Doppler Mode intensity valuations. Colorize is NOT a Doppler Mode.

NOTE: *You can colorize realtime or CINE images or Timeline CINE, but not VCR images.*

Colorizes the gray scale image to enhance the eye's discrimination capability.

Spectrum Colorize colorizes the spectrum as a function of power using the inverse of the Colorize map for the signal intensity in each Doppler line.

Colorize enhances the visibility of the spectrum's characteristics and enhances your ability to identify spectral broadening and the edge contours of the spectrum used to define the peak frequency/velocity.

The gray bar displays while Colorize is activated.

Adjusting

To activate Colorize,

1. Select **Colorize** on the Touch Panel.
2. **Trackball** to cycle through available maps.
3. Press **Set** to select.

To deselect, select a gray map.

Edge Enhance

Description	Edge Enhance brings out subtle tissue differences and boundaries by enhancing the gray scale differences corresponding to the edges of structures. Adjustments to M-Mode's edge enhancement affects the M-Mode only.
Adjusting	To cycle through settings, adjust Edge Enhance on the Touch Panel.
Values	0-5, with 0 = Off and 5 = High. The current value displays on the Touch Panel. Values vary by probe, application, and multi frequency setting. Values are returned to the preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Edge Enhance cleans out the B-Mode image/M-Mode timeline by subduing some of the gray scale in order to highlight the vessel wall or organ. This is helpful when you cannot differentiate between the chambers of the heart.
Affect on other controls	Edge Enhance operates in realtime only; not in Freeze or CINE.

Rotation

Description	Flips the image 180 degrees up/down.
Adjusting	To flip the image vertically, select Rotation .
Values	Up/down. Values vary by probe and application. Values are returned to the preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Beneficial in transvaginal and transrectal scanning.



When reading an rotated image, be careful to observe the probe orientation to avoid possible confusion over scan direction or left/right image reversal.

Rejection

Description	Selects a level below which echoes will not be amplified (an echo must have a certain minimum amplitude before it will be processed).
Adjusting	Adjust to the right to increase and left to decrease.
Values	0-5
Benefits	Allows for the elimination from the display of low level echoes caused by noise.

B Softener

Description	Affects the amount of lateral smoothing. It's similar to dynamic range by softening the appearance/edges of the image.
Values	Off, Low, Medium, High

Suppression

Description	Suppresses the noise in the image.
Values	0-5

LOGIQ 9 Contrast Imaging



Please be advised that Contrast imaging MAY NOT BE available on your system. Contrast agents for radiology use are undergoing clinical trial and therefore, not yet available in the United States.

Contrast Imaging Overview

Contrast Imaging Option NOT Available in the United States

See Chapter 17 for probe availability.

By adjusting the acoustic output, you can enhance either contrast harmonics or stimulated acoustic emission (SAE).

Table 5-2: Available Contrast Settings

Label	Description	Clinical Use
Coded Harmonic Angio - Res and Pen	Coded Harmonic Angio Imaging	Vascular, early phase High MI imaging. Use Res (resolution) for most patients; use Pen (penetration) for deep lesions or for technically difficult patients.
TruAgent Detection	Utilizes Power Doppler techniques that are designed for optimal detection of contrast agent signals.	Late phase imaging for bubble destruction
Coded PI 1 and 2	Coded Phase Inversion	Low MI Harmonic imaging for perfusion, tumor flow characteristics

Benefits

Injected contrast agents re-emit incident acoustic energy at a harmonic frequency much more efficiently than the surrounding tissue. Blood containing the contrast agent stands out brightly against a dark background of normal tissue.

Clinical Use

Possible clinical uses are to detect and characterize tumors of the liver, kidney, and pancreas and to enhance flow signals in the determination of stenosis or thrombus.

Contrast Imaging Overview (continued)

Activation Press the **Contrast** key on the Control Panel. B-Mode Contrast activates in the B fundamental frequency.

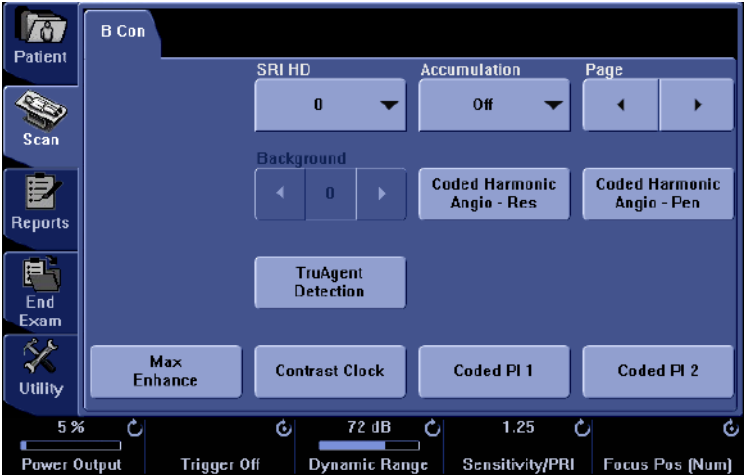


Figure 5-5. B-Mode Contrast Imaging Touch Panel, Page 1

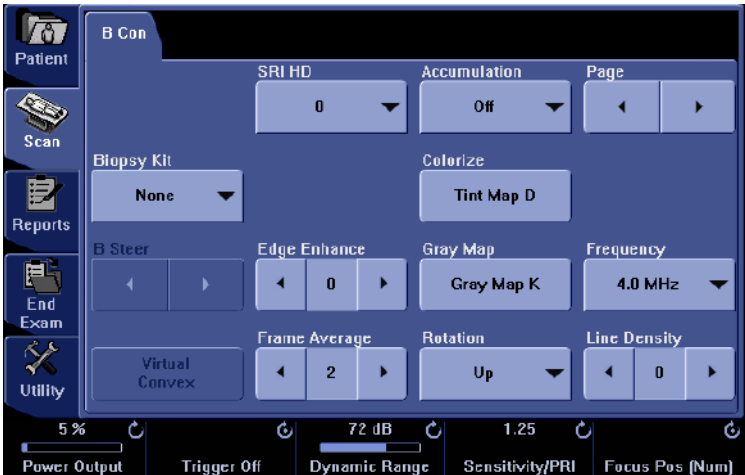


Figure 5-6. B-Mode Contrast Imaging Touch Panel, Page 2

Contrast Imaging Overview (continued)

To activate TruAgent Detection, press the Touch Panel key.

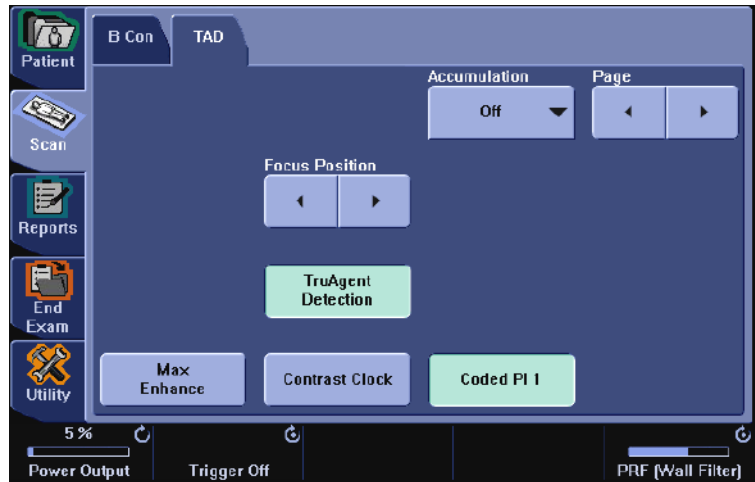


Figure 5-7. TruAgent Detection Contrast Imaging, Page 1

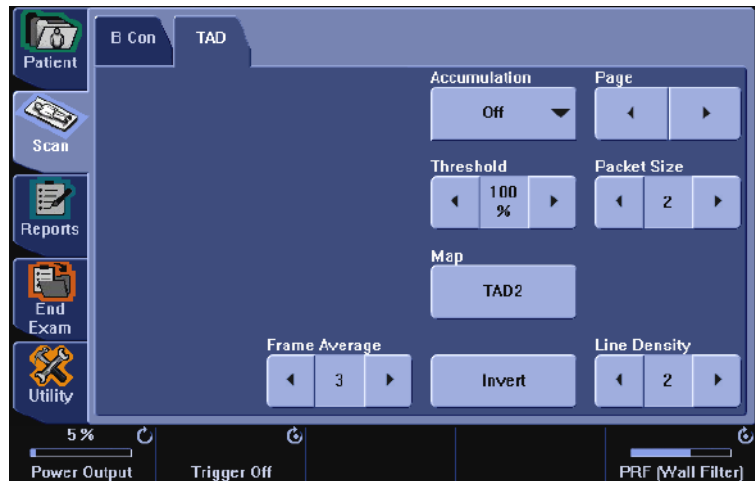


Figure 5-8. TruAgent Detection Contrast Imaging, Page 2

Contrast Imaging Overview (continued)

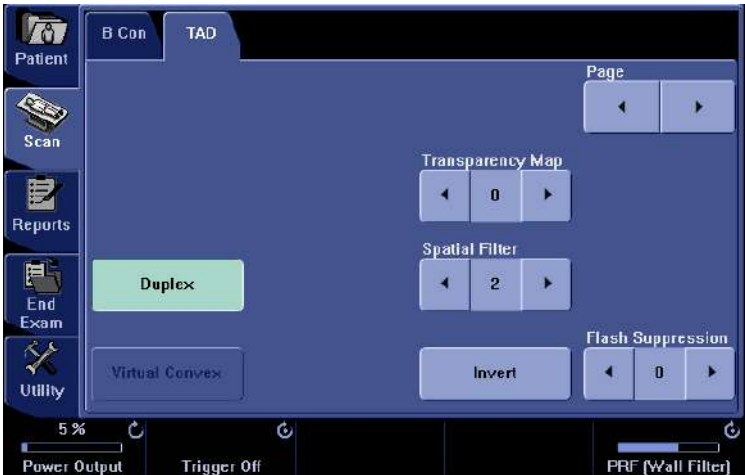


Figure 5-9. TruAgent Detection Contrast Imaging, Page 3

Contrast Imaging Overview (continued)

Affect on other controls	The default acoustic output adjusts for contrast imaging and the Power Output key provides more subtle gradations for use while in contrast imaging. When you exit Contrast Imaging, the system returns the acoustic output to its original setting. Most system controls are available (Depth, ROI, Zoom, Colorize, etc.). However, some controls are not available (Coded Excitation, Anatomical M Mode, Rejection, B Softener, and Suppression). Controls adjusted while in Contrast Imaging retain these values when you exit Contrast Imaging (except for post-processing controls). When you reactivate Contrast Imaging, your last selected contrast technique is remembered (except for TruAgent Detection, which returns to B Fundamental).
Bioeffect	Activating Contrast Imaging may change the TI and/or MI. Observe the output display for possible effects. The Power Output key has been enhanced to provide more subtle gradations for use while contrast imaging. The Mechanical Index displays values less than 0.1.
Scanning Hints	Place focal zone just below the target anatomy. You can visualize perfusion in interval imaging -- see the contrast agent that washes in-between the frames.
Feature Availability	Multi Image and 3D are available; LOGIQView is not available.
Technique Availability by Probe	Use the table in Chapter 17 to determine which contrast technique is available by probe.

Static 3D with Contrast (Option)

Static 3D is available while in Contrast imaging mode. Static 3D with Contrast is supported in Coded Phase Inversion (CPI). Static 3D with Contrast is supported on the 4D3C-L probe.

1. Activate Contrast imaging.
2. Adjust the image, as necessary.
3. Activate the Contrast Clock and Max Enhance (if you want the acoustic output set at 100%). Or, you can activate these when you inject the contrast agent.

NOTE: *To restart the Contrast Clock, you need to turn it off, then back on.*

4. Activate 3D--> Static 3D. Specify the quality, angle, and render mode (Sectional, Render, and VCI Static are supported). If you have not activated Max Enhance, now would be a good time to do so.

Wait to inject the contrast agent until after you have set up your pre-mode data. Adjust the ROI and other parameters before injecting the agent.

5. Activate the Contrast Clock and inject the contrast agent, then press 'L' to activate the Static 3D acquisition as soon as the contrast agent washes in.

The system automatically 'stores' the sweep (to internal memory, not to the clipboard).

NOTE: *You may want to set up a Fast Key to automate these keystrokes. See Chapter 6 for more information.*

Static 3D with Contrast (Option) (continued)

6. To store this VOI, press a print key. All CINE and volume data loops are saved in 4D when you press P1. Store loops from Static 3D or from the Volume Review tab in order to store the complete volume.

NOTE: You can store a single Static 3D Contrast VOI by pressing one of the print keys. Therefore, you would press 'L', then press 'Print' to store a volume of interest.

NOTE: You can also configure the system to perform a count down for the contrast injection via Utility-->System-->System Imaging.

7. At the completion of the sweep, the Static 3D tab appears.

NOTE: The Contrast Timer on the Volume Review images represents a timestamp of when the image was captured relative to when the timer was started.

8. When you press the 'L' key again, you are returned to the Pre-Mode state and the data is flushed for the next volume sweep.

Quick 3D (Option)

Quick 3D allows you to perform a 4D volume sweep and store in one step. This feature is available on the 4D3C_L probe in Coded Phase Inversion (CPI) in 4D pre-mode.

1. Activate Contrast imaging.
2. Adjust the image, as necessary.
3. Activate 3D--> 4D. You are now in 4D Pre-Mode.
4. Press 'L.' The sweep is performed and stored automatically. If you have your system set to 'beep' when you press a print key, then you will hear each time that the sweep is stored.
5. To exit, press 'L.' The best time to exit is after the sweep has been stored (or right after the beep).

B Contrast Presets

	You access the B Contrast Imaging and TruAgent Imaging Presets via Utility --> Imaging --> BCON or TAD tabs.
Acoustic Output Presets	To set the acoustic output value for B Contrast Imaging, press the down arrow next to <i>Acoustic Output (%)</i> to select the value. To set the acoustic output value for the background image while in TruAgent Detection, press the down arrow next to <i>Acoustic Output for Fundamental (%)</i> or <i>Acoustic Output for CFPI Fundamental (%)</i> to set the value.
Time Delay	To set the Trigger time delay, press the down arrow next to <i>Time Delay</i> .
Background	To adjust the level of the background anatomy, press the down arrow next to <i>Background</i> when scanning in Coded Harmonic Angio mode.
Additional Presets	Adjust other B Contrast Imaging presets (Line Density, Frame Average, etc.) via Utility --> Imaging --> BCon. See 'Imaging Presets' on page 16-45 for more information.

TruAgent Detection Presets

	DualView TruAgent Detection is the default when you activate Contrast imaging.
Default Map	To set the default TruAgent map, press the down arrow next to <i>Default Map</i> .
Acoustic Output	To set the acoustic output value for TruAgent Detection Imaging, press the down arrow next to <i>Acoustic Output (%)</i> to set the value.
Additional Presets	Adjust other TruAgent Detection Imaging presets (Line Density, Frame Average, etc.) via Utility --> Imaging --> TAD. See 'Imaging Presets' on page 16-45 for more information.

SRI HD

Description	SRI HD (High Definition Spectral Reduction Imaging) is an adaptive algorithm to reduce the unwanted effects of speckle in the ultrasound image.
Values	0-4, 0=Off.
Benefit	Smooths the image when image speckle interferes with the desired image detail.

Accumulation

Description	Accumulation enhances the displayed flow in an image.
Values	8 settings: 0=Off, 6 Various levels, or frames, of accumulation, and Infinite. If Accumulation is turned off, then Frame Averaging is used; if Accumulation is set to a value, then Accumulation is used.
Availability	Available in Contrast, Color Flow, PDI, and B Flow.
Benefit	Accumulation detects the maximum signal and holds it for the level specified.

Max (Maximum) Enhance

Description	Sets the acoustic output to its maximum setting (100%)
Values	On/Off. When you activate Max Enhance, the acoustic output is set to 100% and the acoustic output control knob is disabled. When you deactivate Max Enhance, the acoustic output is returned to its previous setting. Max Enhance is deactivated by the system when you turn it off, change probes, change the contrast technique, or exit Contrast.
Benefits	This control provides quick transition to High MI imaging. This allows the user a quick one-button push to destroy the agent. Useful when the user is interested in the bubble wash-in characteristics of the anatomy being scanned.

Contrast Clock (Timer)

Description	You can use the Contrast Clock by activating it at the time of injection and deactivating it at the end of the exam. OR -- You can use the Contrast Clock to enable a countdown for the contrast injection. Set the countdown time at Utility-->System-->System Imaging, Countdown Time for Contrast.
Values	On/Off. You deactivate the Contrast Clock via the Touch Panel knob or by starting a new patient.
Display	There are two areas on the screen where the Contrast Clock displays: on the image and on the lower, left-hand portion of the display. The timer on the image freezes when you freeze the image (the timer updates when you unfreeze the image). However, the timer located on the lower, left-hand portion of the display continues to display over a freeze, probe change, mode change, multi image, and zoom. The timer also appears on CINE Loops and archived images.
Benefits	The Contrast Clock measures the time since injection.

Contrast Trigger

Description	The Contrast Trigger scans images at set intervals, delaying imaging according to the time delay that you specify.
Adjusting	Press Trigger Off (knob beneath the Touch Panel) to activate the time delay. When you activate the trigger, the Time Delay can be controlled by turning the knob. The Contrast Trigger is deactivated if you go into a Timeline Mode, activate B Flow or Harmonics, or exit Harmonics.
Values	Values are 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 seconds.

Sensitivity PRI

Description	Sensitivity PRI is used to adjust the sensitivity of the Contrast Agent. By increasing the sensitivity, you lower the frame rate; by decreasing the sensitivity, you raise the frame rate.
Adjusting	Adjust the dial left/right.
Values	1, 1.25, 1.5, 2 and 3.
Benefits	By using Sensitivity PRI, sensitivity can be changed for each anatomy of interest.

Coded Harmonic Angio Example

How to perform a Coded Harmonic Angio study:

1. Select a probe that supports Contrast.
2. Select **Contrast**.
3. Select **Coded Harmonic Angio** to access CHA mode.
4. Adjust the focus position.
5. Start the recording, if needed.
6. Prepare the contrast agent, as described by the manufacturer, just before the injection.
7. Select **Contrast Clock** to start the clock simultaneously with the injection.

NOTE: *If the Acoustic Output value does not change from 100%, activate **Max Enhance**.*

NOTE: *DO NOT press the CF or PDI keys during the CHA scan. If you press these keys, exit from Contrast mode and reset CINE.*

8. Press the **Trigger** control to activate Intermittent mode. Adjust the control to set the intermittent time.
9. Observe the vascular phase for about one minute after injection, then press **Freeze**.
10. Select **Start/End** frames to define the CINE Loop, then save to the clipboard. Store the single frame image to the clipboard, if needed.
11. Scan and store the post vascular (portal) phase and the late phase, if needed.

NOTE: *If you want to check the ROI by Low MI before scanning the post vascular phase and the late phase, turn off **Coded Harmonic Angio**.*

Time Intensity Curve (TIC) Analysis

The basic TIC process works as follows:

1. Scan the patient after injecting the contrast agent.
2. Watch the agent flow through the anatomy of interest.
3. When the desired contrast effect has been visualized, freeze the image and select a range of images for analysis.
4. Position an ROI (region of interest) on one of those images where the contrast effect is visible.
5. The system then calculates the mean pixel intensity within that ROI for all frames in the user designated loop and plots the resulting data as a function of time.

You can also choose to fit this data to one of several mathematical functions. The fundamental idea is that the contrast effect flowing through the organ of interest can be modeled mathematically, and details of the wash in and washout of the agent can be gleaned by analyzing the numerical parameters of the mathematical model.

Activating TIC

1. Scan the patient in Contrast mode or select a desired cine loop from the stored images.

NOTE: *Images from the current scan session (already in CINE) or from a saved image loop can be used for TIC analysis.*

NOTE: *TIC analysis is only available if the user has selected an image loop. If the user has selected a saved still image (just one frame), TIC analysis is not available.*

2. Press **Freeze**.

NOTE: *The TIC package is only available when the system is in FREEZE mode.*

3. **TIC Analysis** displays on the Cine Touch Panel.

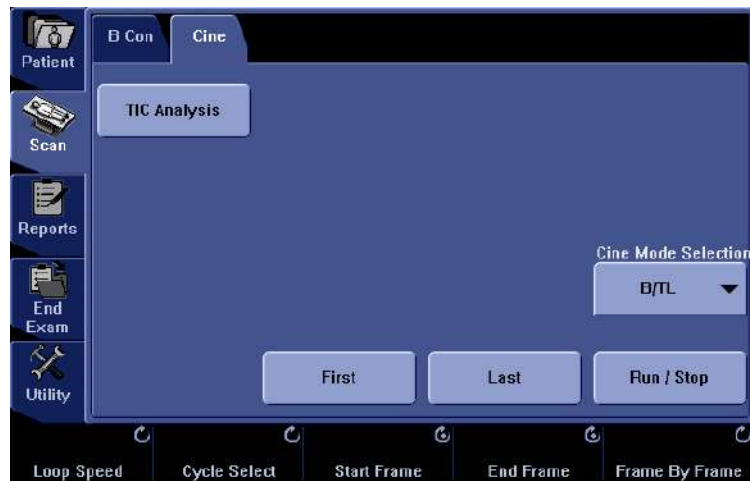


Figure 5-10. Cine Touch Panel

NOTE: *Subsequent TIC processing is done on the images stored in CINE memory.*

Activating TIC (continued)

4. Select **TIC Analysis**. The TIC Analysis screen and Touch Panel displays. To toggle the trackball function between TIC and Scroll, press **Scan Area**.

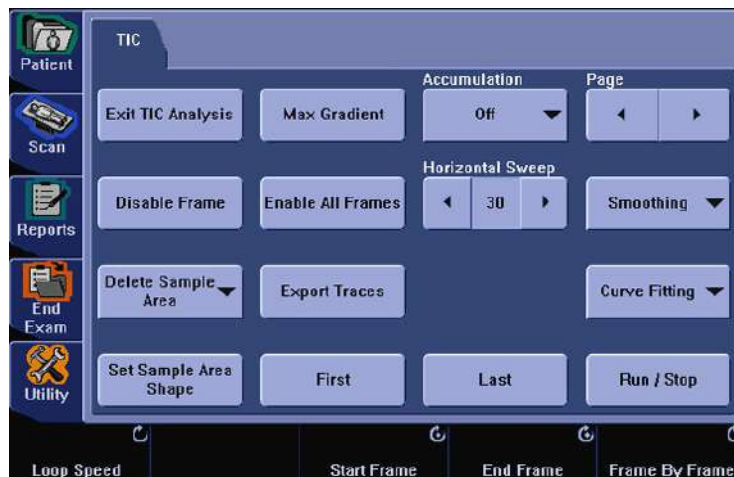


Figure 5-11. TIC Analysis Touch Panel, Page 1

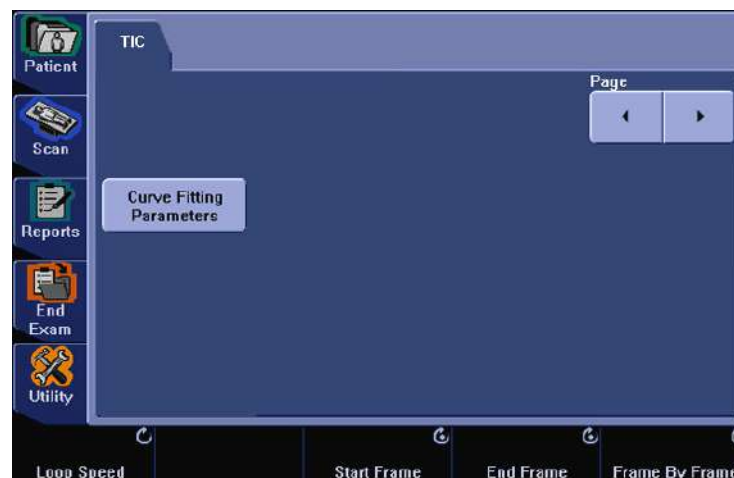


Figure 5-12. TIC Analysis Touch Panel, Page 2

Activating TIC (continued)

Table 5-3: TIC Touch Panel Description

Parameter	Description
Max Gradient	Between the CINE Start and End frame, displays the time and gradient that becomes the maximum gradient.
Accumulation	Enhances the flow in an image.
Disable Frame	The current frame is excluded from the CINE Loop display.
Enable All Frames	Re-enables disabled frames.
Horizontal Sweep	Allows you to increase/decrease the time interval over which to plot the TIC curve.
Smoothing	Smooths the trace displayed by applying a filter over a defined time window. Both the filter type and time window are user-selectable. The type of filter available depends on the analysis signal displayed.
Delete Sample Area	Removes selected sample area from the CINE Loop window and accompanying trace in the Analysis window. The Trackball marker must be pointed at an anchored sample area.
Export Trace	Saves trace data in ASCII format, readable in spreadsheet programs. If present, trace data for physiological traces are also exported.
Curve Fitting	Toggles between Wash-In, Wash-Out, and Off.
Set Sample Area Shape	Enables resizing of a selected sample area by setting height, width, and tilt angle. The Trackball marker must be pointed at an anchored sample area.

NOTE: *The CINE Touch Panel is available to use when in TIC Analysis mode. CINE Touch Panel controls retain their traditional functions. While the CINE Touch Panel is on the top level, the Trackball and other imaging controls perform as they usually do in standard CINE mode.*

Exiting TIC Analysis

There are several methods to exit TIC Analysis.

- Select **Exit TIC Analysis** on the TIC Touch Panel.
- Press **Freeze** to unfreeze and resume scanning.
- Press any other button that returns the system to real-time scanning.

TIC Analysis Screen Description

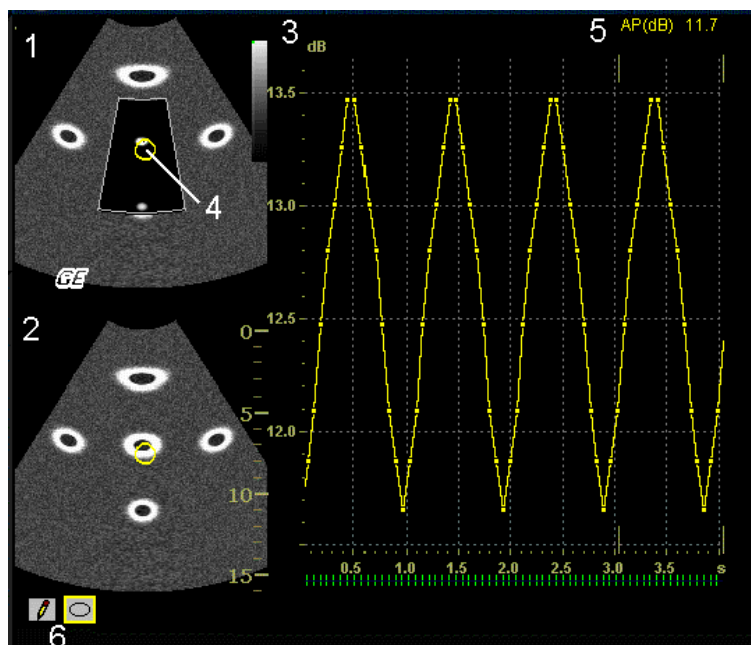


Figure 5-13. TIC Analysis Screen (TAD mode)

1. Contrast Cineloop Window
Sample Area: Indicates sampling position of the intensity (contrast) trace. The sample area is color-coded: the first sample area is yellow, the second green, etc.
2. B Cineloop Window
Sample Area: Indicates sampling position of the intensity (B) trace. The sample area is color-coded: the first sample area is yellow, the second green, etc.

TIC Analysis Screen Description (continued)

3. Displays time-intensity curve.
 - Y axis: Intensity scale (logarithmic) (db) or linear acoustic units (AU).
 - X axis: Time(s) or Dt(s), elapsed time from previous frame.
 - ECG (where available -- not shown), Frame Marker: displays ECG trace (where available), the current frame marker and the start and stop markers for the cineloop.
 - Time at cursor position.
 - Intensity (dB or AU) at cursor position.
 - Intensity (dB or AU) at frame marker position (color coded)
4. Sample Area
5. Time at cursor position and velocity at cursor position.
6. Sample Area Tools.
 - Pencil Icon: Creates a sample area based on freehand drawing.
 - Shape Icon: Creates a sample area with a pre-defined circular/ellipse shape.

System Menu

The System menu on the monitor display can be used instead of the Touch Panel.

Analysis Window
System Menu

Position the cursor over the analysis window and press the left **Set** key. The Analysis Window system menu displays at the cursor position.

NOTE: The system menu is dependent on mode.



Figure 5-14. Analysis Window System Menu (Contrast Mode)

Reference Image
System Menu

Position the cursor over the reference image and press the left **Set** key. The reference image system menu displays at the cursor position.

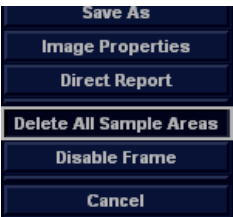


Figure 5-15. Reference Image System Menu

ROI System Menu Position the cursor on the ROI and press the left **Set** key. The ROI system menu displays at the cursor position.



Figure 5-16. ROI System Menu

Frame Marker System Menu Position the cursor on the Frame Marker in the Analysis Window and press the left **Set** key. The Frame Marker system menu displays at the cursor position.

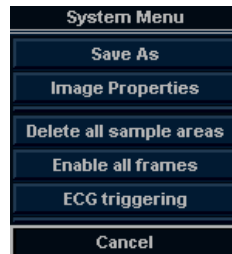


Figure 5-17. Frame Marker System Menu

Selecting TIC Analysis Image Range

A range of frames is selected for the TIC analysis in Cine mode (before accessing TIC Analysis). Only the frames in this range are used for the TIC analysis.

If a range is not selected prior to accessing the TIC Analysis, the system uses the default Cine start and end frames as the default TIC start and stop frames.

1. The first frame in the analysis series is selected by adjusting the **Start Frame** control to the desired frame
OR
using the **Trackball** or the **Frame by Frame** control to select the desired first frame and then selecting the **Start Frame** control.
2. The last frame in the analysis series is selected by adjusting the CINE **Last Frame** control to the desired frame
OR
using the **Trackball** or the **Frame by Frame** control to select the desired last frame and then selecting the **Last Frame** control.

Generating a Trace

Trace from a pre-defined sample area

1. If the trackball assignment is not on QA, press **Scan Area** until QA highlights.
2. If necessary, select the sample area Ellipse ROI button (shape icon on the monitor display).
3. Move the cursor to one of the Cineloop windows using the **Trackball**.
4. Press **Set** to anchor the sample area.

In this frame, the sample area is marked with an anchor. If the cineloop has more than one heart cycle, a sample area will also be anchored in the corresponding frame in the next heart cycle.

The trace is updated accordingly in the Analysis window.

Trace from freehand sample area

1. Select the Freehand ROI button (pencil icon on the monitor display).
2. Move the cursor to one of the Cineloop windows using the **Trackball**.
3. Press and hold down the **Set** key while drawing a sample area using the **Trackball**.
4. Release the **Set** key.

The sample area is automatically closed and the trace is updated accordingly in the Analysis window.

Generating a Trace (continued)

Manual tracking of the sample area (dynamic anchored sample area)

1. Place a sample area over a region of interest. Note the anatomical location of the sample area.
2. Scroll to a new frame using the **Trackball**.
3. Press **Scan Area** until the QA trackball assignment is selected.
4. Move the cursor to the sample area using the **Trackball**.
5. Press **Set**. The sample area is unanchored.
6. Drag the sample area to the corresponding anatomical location in the new frame.

When the sample area is anchored in more than one frame, linear interpolation is performed so that the sample area is smoothly moved between the anchored positions in the selected frames when running the cineloop.

7. Press **Scan Area** until the scroll trackball assignment is selected.
8. Using the **Trackball**, scroll through the cineloop and control that the sample area follows the moving anatomical structure.
9. Add anchored sample areas in several frames to obtain a more accurate displacement of the sample area.

Generating a Trace (continued)

Moving a dynamic anchored sample area

1. Press **Scan Area** until the scroll trackball assignment is selected.
2. Using the **Trackball**, browse through the cineloop to display one of the frames where the sample area was anchored.

NOTE: In these frames, the sample area is marked with an anchor.

3. Press **Scan Area** until the QA trackball assignment is selected.
4. Move the cursor to the sample area using the **Trackball**.
5. Press **Set**. The sample area is unanchored.
6. Drag the sample area to a new location.
7. Press **Set** to anchor the sample area to the new location.

If you want to move the sample area to the same depth, select **Move (same depth)** from the System Menu.

Copy, move, and paste a Sample Area

To copy and paste the ROI,

1. Move the cursor over the ROI and press the left Set key. The system menu displays.
2. Select **Copy sample area**.
3. Move the cursor to the desired location for the copied ROI and press the left Set Key. The system menu displays.
4. Select **Paste sample area**.

To copy and move the ROI,

1. Move the cursor over the ROI and press the left Set key. The system menu displays.
2. Select **Copy & move**. Or if you want to move to the same depth as the original ROI, select **Copy & move (same depth)**.
3. Move the copied ROI using the **Trackball**. Press **Set** to fix the position.

Generating a Trace (continued)

Zooming in the Analysis window

To zoom:

1. In the Analysis window, press and hold down the **Set** key while dragging the cursor to define the zooming area.
2. Release the **Set** key.

To unzoom:

1. Press the left **Set** key in the Analysis window. The system menu displays.
2. Select **Unzoom**.

Manipulating the Sample Area

Up to eight ROIs can be saved on the reference image, with the corresponding eight traces plotted simultaneously on the graph. Each ROI display has a different color, and its corresponding trace data is plotted using that same color.

Once eight ROIs have been saved, the system does not automatically generate an active ROI when the cursor is positioned over the displayed reference image.

The saved ROIs can be a mixture of elliptical and freehand ROIs.

When the user repositions an ROI, the old trace data is erased from the plot and the trace data for the new position replotted.

- Elliptical ROIs that are being repositioned shall have their traces automatically recalculated and replotted in the trace plot window as they are being moved.
- Freehand ROIs that are being repositioned shall not have traces calculated and plotted until the new position is finalized and **Set** is pressed a second time.

If the ROI position on the last frame of the selected image range is moved, the corresponding ROIs on all frames are repositioned to match the last frame.

The user shall also have the capability of setting separate ROI positions on different frames of the contrast images, and the system shall linearly interpolate the ROI positions for the frames in between the selected frames.

Setting the default sample area shape

1. Select **Set sample area shape**. The Information Box displays.

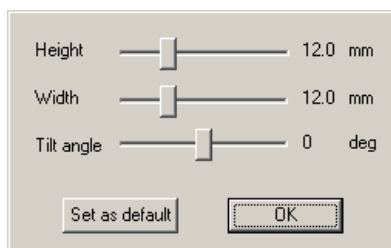


Figure 5-18. Sample Area Information Box

2. Select Height, Width and Tilt angle.
3. Select **Set as default**. The current ROI size is set as the default for subsequent Ellipse ROIs.

Sample Area Shapes

There are two different methods for determining the shapes of the sample area.

Ellipse ROI

1. Select the ellipse icon (shape icon on the monitor display).
2. When the trackball positions the image display cursor over the reference image(s), an elliptical ROI is automatically generated and displays on the reference image(s).
3. The average intensity value inside the ellipse is calculated for every image in the image analysis range and plotted in the image display area.
4. The last generated or selected ellipse is considered the active ROI, and its trace plot automatically updates as the user repositions it on the reference image. Old traces are erased.
5. When scanning with an elliptical ROI, press **Set** to fix the ROI position and freeze its corresponding trace on the plot. A new active ROI is generated whose position is manipulated by the trackball and whose time-intensity curve traces will be plotted as before, while the previous ROI and trace remain fixed at the points they were saved at.

NOTE: *Elliptical ROIs can be positioned in any manner that keeps their center within the image boundaries. In the case that part of the ROI is outside the image boundary, only data from within the image boundary is used for calculating the mean intensity value.*

NOTE: *You can change the size of the Ellipse ROI by adjusting the Ellipse control.*

Sample Area Shapes (continued)

Freehand ROI

1. Select Freehand icon (pencil icon on the monitor display). Use the **Trackball** to position the caliper on the reference image at the start point. Press **Set** to fix the start point.
2. Trace the outline of the desired ROI by moving the caliper with the **Trackball**.
3. When a suitable ROI has been drawn, press **Set** a second time.

The system automatically links the start point to the end point by drawing a straight line between them. The caliper is then free for repositioning for another freehand ROI.

NOTE: *You cannot go outside the image boundary when drawing a freehand ROI.*

Reshaping a Sample Area

To reshape the sample area:

1. Position the cursor on the ROI to reshape and press the left **Set** key.
2. The ROI system menu displays. Select **Set sample area shape**.

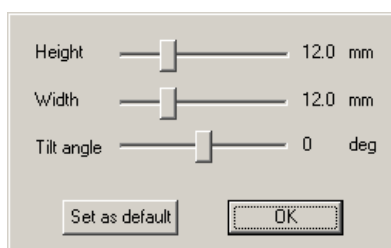


Figure 5-19. Sample Area Information Box

3. Adjust Height, Width and Tilt angle.
4. Press **OK**. The selected ROI size changes.

Labeling a Sample Area

The sample area label is used to identify data associated with the sample area when exporting.

1. Position the cursor on the ROI to label and press the left **Set** key.
2. The ROI system menu displays. Select **Label sample area**. The Label Dialog box displays.

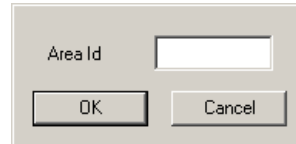


Figure 5-20. Label Dialog Box

3. Enter a name for the sample area.
4. Select **OK**.

Deleting a Sample Area

Sample ROIs and their corresponding traces can be deleted using **Delete Sample Area**.

1. Select **Delete Sample Area**; a pull-down menu displays.



Figure 5-21. Delete Sample Area pull-down menu

2. Select **Current sample** to delete the currently active ROI. Select **Delete all** to delete all currently set ROIs and all of their traces.

NOTE: *The corresponding traces for the deleted ROIs are erased from the plot.*

NOTE: *Deleting an ROI causes the ROIs to be deleted from all frames in the analysis loop.*

TIC Plot Control

The following controls are user configurable presets which are configurable via the Utility Menu or through the pull-down menu in TIC Analysis mode. When using the pull-down menu:

1. Position the cursor over the analysis window and press the left **Set** key. The system menu displays at the cursor position.
2. Select the appropriate parameter.

Vertical Unit

When analyzing the contrast data, the Y-axis can be set to display either longarithmic scale (dB) or linear, acoustic units (AU) for both tissue intensity (2D) or Angio intensity data.

To toggle between dB and acoustical display units for the Y-axis.

- **dB**—The traditional log compressed B-mode data is used to calculate the time-intensity curve values.
- **Acoustic**—The system reverse the log compression function to provide un-log compressed data for the TIC analysis.

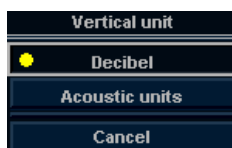


Figure 5-22. Vertical Unit Pop-up menu

TIC Plot Control (continued)

Vertical auto-scaling

The system can be configured to display the full unit range or a range according to the maximum and minimum values of the displayed trace(s) (auto-scaling function). In addition, the auto-scaling function can be set to be live update (updates while the sample area is moved) or delayed (updated when the sample area is anchored).

- **Delayed**—The system automatically rescale the vertical axis of the trace graph only when a new ROI is saved, to account for changing input dynamic range.
- **On**—The system automatically rescale the vertical axis of the trace graph every time the currently selected (active) ROI is moved.
- **Off**—Disable any automatic scaling of the vertical axis. There is user-defined system defaults on the system preset page for the fixed vertical scale to be used for the plot.

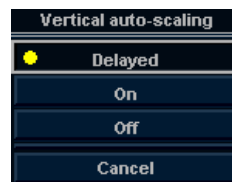


Figure 5-23. Vertical Autoscale Pop-up menu

Line Style

- **Solid**—Setting the results in a plotted trace that does not display small boxes at the data points
- **Squares**—Setting the results in a plot where small squares are displayed at each data point, and the squares are linked together by lines.

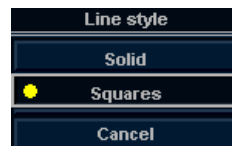


Figure 5-24. Line style Pop-up menu

TIC Plot Control (continued)

Horizontal Scale Set the horizontal unit as time scaling (s) or time interval (dt) between frames.

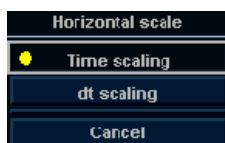


Figure 5-25. Horizontal Scale Pop-up menu

NOTE: *The Horizontal Scale menu displays for CHA and Phase Inversion.*

Horizontal Sweep Horizontal Sweep allows you to increase or decrease the time interval over which to plot the TIC curve.

The Horizontal Sweep control can range from TBD on the short side to the time interval between the user selected first and last frame. The default is the user selected image range. If the user has not yet selected a first and last frame, the first and last default frames from the displayed CINE loop are used.

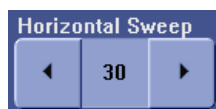


Figure 5-26. Horizontal Sweep

Smoothing

The system can smooth the traces displayed by applying a filter over a defined time window. The type of filter available is depending on the analysis signal displayed.

1. Select **Smoothing**.

OR

Position the cursor over the analysis window and press the left **Set key**. The System menu is displayed at the cursor position. Select **Smoothing**.

NOTE: *When smoothing is turned on, it applies to all traces in the plot window.*

2. The smoothing filter list displays. Select the appropriate parameter.



Figure 5-27. Smoothing Menu

NOTE: *If 'dt' is selected for Horizontal Scale, you cannot use Smoothing.*

Analysis Signal

Analysis Signal toggles the trace display between velocity, displacement or gray scale intensity curves.

Curve Fit

1. Select **Curve Fit**.
OR
Position the cursor over the analysis window and press the left **Set** key. The system menu displays at the cursor position. Select **Curve Fit**.
2. The Curve Fit selection list displays.

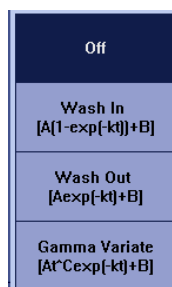


Figure 5-28. Curve Fit Selection List

- **Off**—Remove the fitted curves from the plot and the fit parameters from the display.
- **Wash-in**—Used to find and estimate the local perfusion rate using the contrast agent. Exponential wash-in is described by the function:

$$Y(t) = A(1-\exp(-kt))+B, \text{ where :}$$

- A (dB or AU) is the intensity from the contrast agent.
- B (dB or AU) is the intensity at time $t=0$ (defined as the time of the left marker). This corresponds to the tissue (baseline) signal if no contrast is present at the selected starting point.

NOTE: $A + B = \text{contrast} + \text{tissue} = \text{plateau level}.$

- k (1/s) is a time constant.

Curve Fit (continued)

- **Wash-out**—Used to find and estimate a local wash-out rate. Exponential wash-out is described by the function:

$$Y(t) = A \exp(-kt) + B, \text{ where :}$$

- A (dB or AU) is the intensity from the contrast agent.
- B (dB or AU) is the intensity from the tissue = baseline signal.

NOTE: $A + B$ is the initial intensity level.

- k (1/s) is a time constant.

- **Gamma variate**

$$Y(t) = At^C \exp(-kt) + B$$

**Parameters of
Gamma curve
fitting**

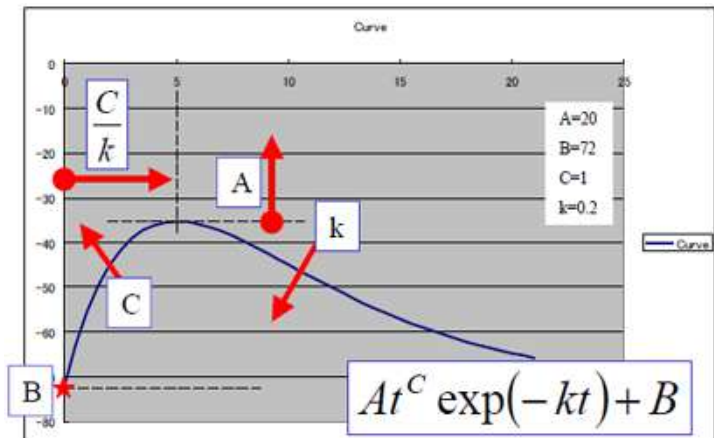


Figure 5-29. Gamma Curve

- t^C : Increasing function ($C > 0$) for “Wash-in”.
For larger C, the intensity increases quickly before the peak.
- $\exp(-kt)$: Decreasing function ($k > 0$) for “Wash-out”.
For larger k, the intensity decreases quickly after the peak.
- B: Intercept intensity at $t=0$.
- The peak intensity of the curve is affected by all parameters. Larger A, larger B, larger C, and smaller k make larger peak. The peak time is calculated by C/k .
- MSE: Mean Square Error

If the MSE is small, the difference of actual data and the fitted curve is small.

Max Gradient

Displays the time and gradient that becomes the maximum gradient between the CINE start and end frame.

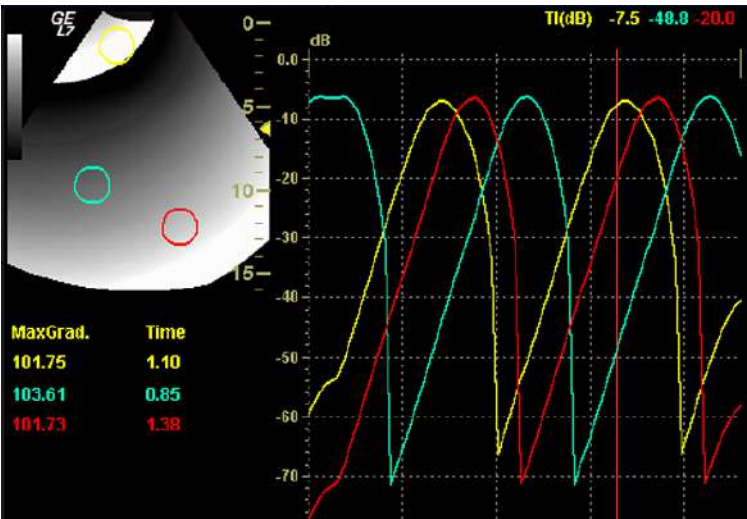


Figure 5-30. Max Gradient

Gradient

Gradient is displayed on the screen instead of Intensity (db or AU). The gradient calculates from 7 points (includes previous and next frames).

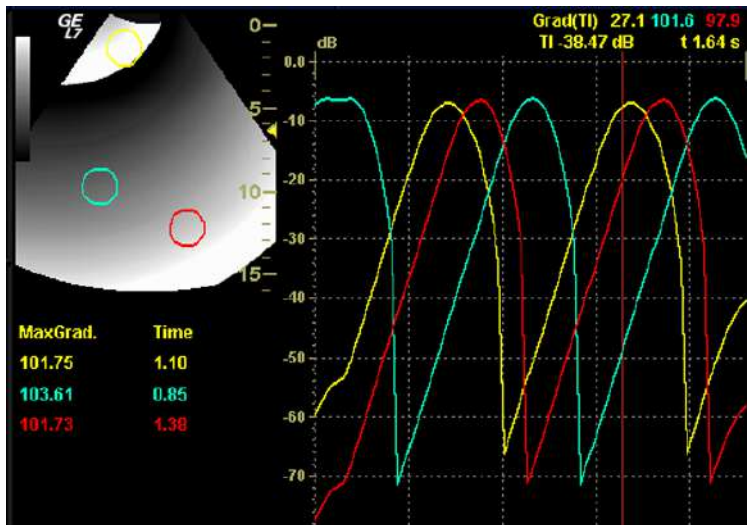


Figure 5-31. Gradient

Set Start/End Frame for Curve Fit per ROI position

1. Generate TIC and perform a Curve Fit.

In this state, the Curve Fit graph is drawn from Cine Start Frame to Cine End Frame for all the ROIs.

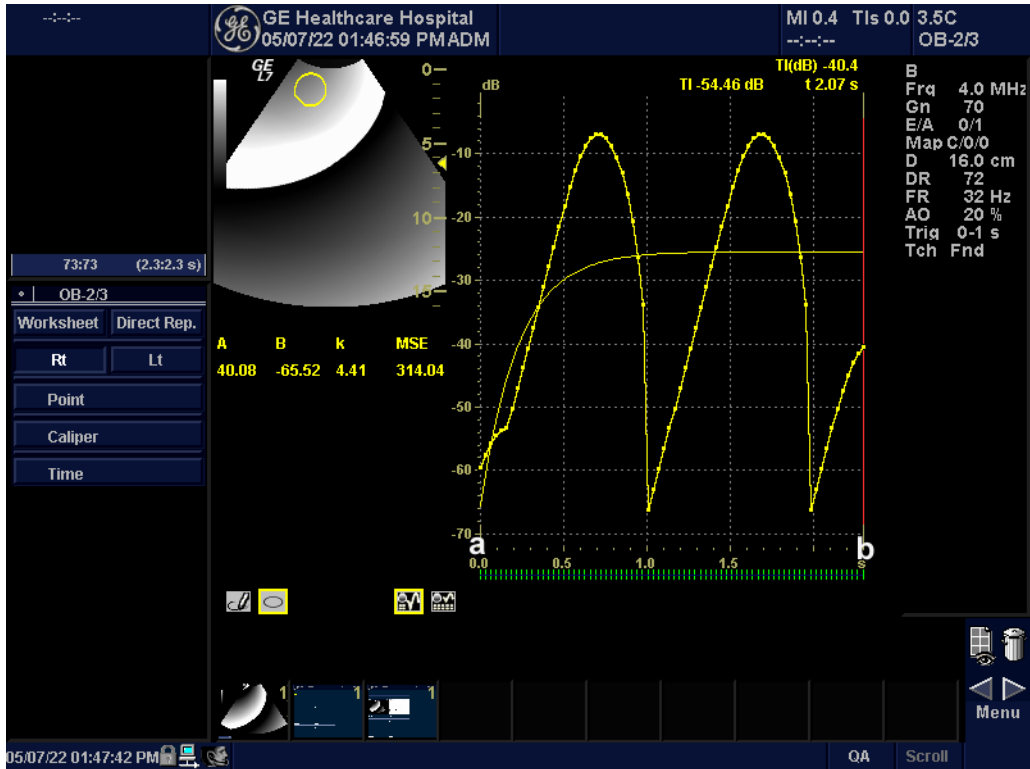


Figure 5-32. Curve Fit screen

- a. Cine Start Frame
- b. Cine End Frame

Set Start/End Frame for Curve Fit per ROI position (continued)

2. Select the Start Frame as with Cine or move the cursor to the desired position on the graph and select the right **Set** key.
3. Move the arrow pointer on the ROI (hand cursor appears) and select the unmarked key (opposite the **Set** key). The system menu appears.
4. Select **Set curve fit start frame** from the menu.
5. Select the End Frame as with Cine or move the cursor to the desired position on the graph and select the right **Set** key.
6. Move the arrow pointer on the ROI (hand cursor appears) and select the unmarked key (opposite the **Set** key). The system menu appears.
7. Select **Set curve fit end frame** from the menu. The ROI colored line displays.

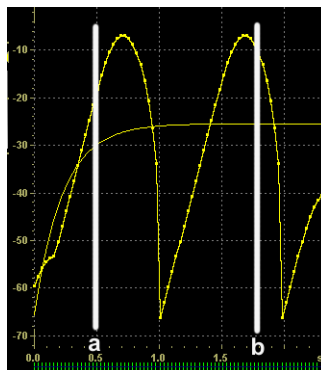


Figure 5-33. New Start Frame and End Frame (Example)

- a. Start Frame
 - b. End Frame
8. Repeat the above procedures as necessary.

The system retains the start/end frame per ROI while TIC is active. Once the TIC menu is closed, the settings are lost.

Display/Hide Calculation Values

You can specify Time to Peak, Area under the curve, Curve gradient, and Arrival time values for Wash-in, Wash-out, and Gamma Variate curve fitting values.

1. Select **Curve Fitting Parameters** on the Touch Panel. The Curve Fitting Parameters dialog appears.

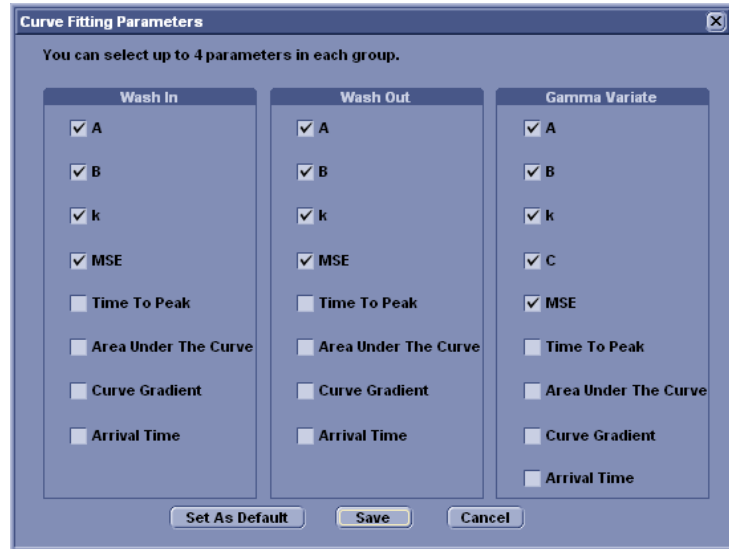


Figure 5-34. Curve Fitting Parameters Dialog

2. Select a maximum of 4 parameters to display for each curve fitting.
 - Save as default: saves as a system preset.
 - Save: saves as temporary.
 - Cancel

NOTE: *If you select more than 4 parameters and select “Save” or “Save as default,” you will be prompted to select only 4 parameters.*

Display/Hide Calculation Values (continued)

- 3. The selected parameter displays below the image with the curve fitting active. Press the display button to display all parameters; or press the hide button to hide all parameters.



Figure 5-35. TIC image with the selected parameters

Disabling/Enabling the frame

Disabling the frame from the frame marker

Frame disabling excludes the actual frame from the cineloop display. Frame disabling is available only with contrast data.

To disable One Frame :

1. Use the Trackball to move the cursor to the frame maker to disable.
2. Press **Set** to disable the frame.
3. The frame marker is changed from green to red to indicate the frame has been disabled.

NOTE:

The disabled frame is no longer displayed in the reference window when scrolling through CINE memory.

Disabling multi-frames from the frame marker

1. Use the trackball to move the cursor to the first frame maker to disable.
2. Press and hold down **Set**
3. Move the cursor with the Trackball to the last frame to be disabled and release Set.

The marker is turn red and the data from that frame is removed from the trace and any subsequent trace processing.

Disabling a frame from the cineloop window

1. Use the trackball to move the cursor to the cineloop window.
2. Press the left **Set** key. The system menu displays.
3. Select **Disable frame**.

The current frame is disabled and the corresponding frame marker displays red.

To enable the frames

To re-enable all deleted frames :

1. Position the cursor on the Frame Marker line and press the left **Set** key. The system menu is displayed at the cursor position.
2. Select **Enable all frames**.
3. All disabled frames are re-enabled.

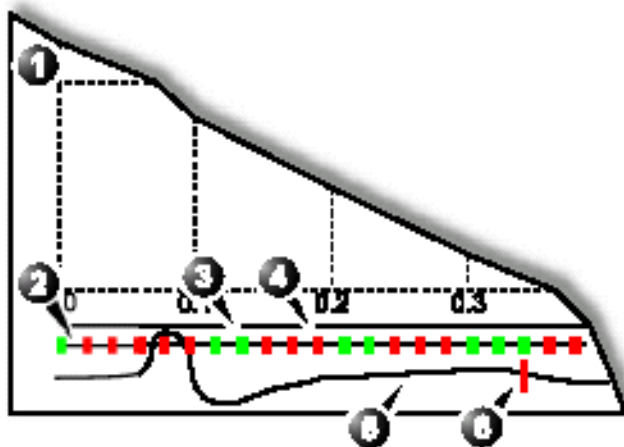


Figure 5-36. Frame makers

1. Analysis Window
2. Frame markers axis
3. Enabled frame (Green)
4. Disabled frame (Red)
5. ECG (where available)
6. Current frame

Exporting Traces (Saving the Trace Data)

You can save the trace data to an external file.

1. Select **Export Traces** to save the trace data.

OR

Position the cursor to the Analysis window and press the left **Set** key. The system menu displays. Select **Export traces**.

2. The following window displays.

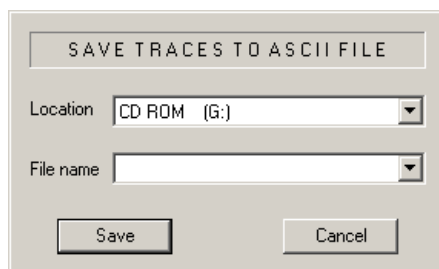


Figure 5-37. Export Trace Window

- Location : Select Location which to save.
 - Filename : Enter the filename. (Only Text)
 - Comment : Enter the comment if necessary.
3. Select **OK** to save the data and return to the TIC Analysis screen.
 - All displayed ROI traces are saved in the exported file.
 - The fit parameters are included in the trace file if the user has done a curve fit.

NOTE: *The Smoothed trace is the one saved is the user has applied a smoothing filter.*

NOTE: *Only data from the user selected image range is included in the exported trace file.*

Exporting Traces (Saving the Trace Data) (continued)

NOTE: Data for disabled frames are not be included in the exported trace file.

NOTE: No trace results are saved in the standard image database.

NOTE: Trace results are now shown on the Worksheet.

Annotating the TIC Data

The user can annotate both the reference image and the trace plot displays. Use **Comment** key to type the annotation. See Chapter 6 for reference.

Printing TIC Data

Press the appropriate print key in the TIC mode.

The system capture a single still frame which consists of the plot, the reference image and user annotation.

LOGIQView

Description

LOGIQView provides the ability to construct and view a static 2D image which is wider than the field of view of a given transducer. This feature allows viewing and measurements of anatomy that is larger than what would fit in a single image. Examples include scanning of vascular structures and connective tissues in the arms and legs.

LOGIQView constructs the extended image from individual image frames as the operator slides the transducer along the surface of the skin in the direction of the scan plane. The quality of the resulting image is somewhat user-dependent and requires some additional skill and practice to develop proper technique and become fully proficient.

LOGIQView is not available for the following:

- Multi Image
- Timeline Modes
- B Flow Mode
- Color Flow Mode
- PDI Mode

Benefits

The user can look at a larger region of interest within one field of view that is wider than any given probe would normally provide.

Clinical Use

LOGIQView is intended for scanning areas too large to fit on one image.

Using LOGIQView

To perform an exam using LOGIQView,

1. Perform a detailed examination of the anatomy/pathology. Optimize parameters for tissue texture and visible window PRIOR TO activating LOGIQView.
2. Press the LOGIQView key on the Control Panel.
3. To start acquiring the image, press 'L' (the left split screen key).

LOGIQView (continued)



HINTS

When you scan, scan slowly and in a uniform motion lengthwise, end-to-end (with or against the probe orientation marker). LOGIQView acquires images via leading edge vectors (and does not acquire slices, as in CINE). The image is being stored as you perform the scan and you can watch the LOGIQView as it is being acquired.

4. To restart the scan, press 'L' again. You can back up the probe, realign it, then go forward to redo a portion of the scan.
5. To complete the scan, press 'R' or Freeze (or allow the scan to auto complete. The LOGIQView is then displayed, scaled to fit entirely on the screen.
6. Perform measurements and record images.

NOTE: *Measurement error is within 5% of the distance you measured for all linear probes.*

Uniform Motion

The quality and usefulness of LOGIQView images is affected by transducer motion. Incorrect technique can contribute to image distortion.

Guidance and precautions for uniform motion:

- Continuous contact is required throughout the length of the extended image. DO NOT lift the transducer from the skin surface.
- Always keep the transducer perpendicular to the skin surface. DO NOT rock the transducer.
- Keep the motion within the same scan plane, if possible. DO NOT slide the transducer laterally.
- Lateral turning (change in direction to follow anatomical structure) can be accommodated with slower motion. DO NOT make abrupt changes in direction.
- The system accommodates a reasonable range of motion velocity. DO NOT make abrupt changes in speed of motion. Deeper scans generally require reduced speed.

Bioeffects

Activating LOGIQView has no effect upon Acoustic Output values.

Optimizing M-Mode

Intended Use

M-Mode is intended to provide a display format and measurement capability that represents tissue displacement (motion) occurring over time along a single vector.

Introduction

M-Mode is used to determine patterns of motion for objects within the ultrasound beam. The most common use is for viewing motion patterns of the heart.

Typical exam protocol

A typical examination using M-Mode might proceed as follows:

1. Get a good B-Mode image. Survey the anatomy and place the area of interest near the center of the B-Mode image.
2. Press **M/D Cursor**.
3. Trackball to position the mode cursor over the area that you want to display in M-Mode.
4. Press M-Mode.
5. Adjust the Sweep Speed, TGC, Gain, Power Output, and Focus Position, as needed.
6. Press **Freeze** to stop the M trace.
7. Record the trace to disk or to the hard copy device.
8. Press **Freeze** to continue imaging.
9. To exit, press M-Mode.

M-Mode Display

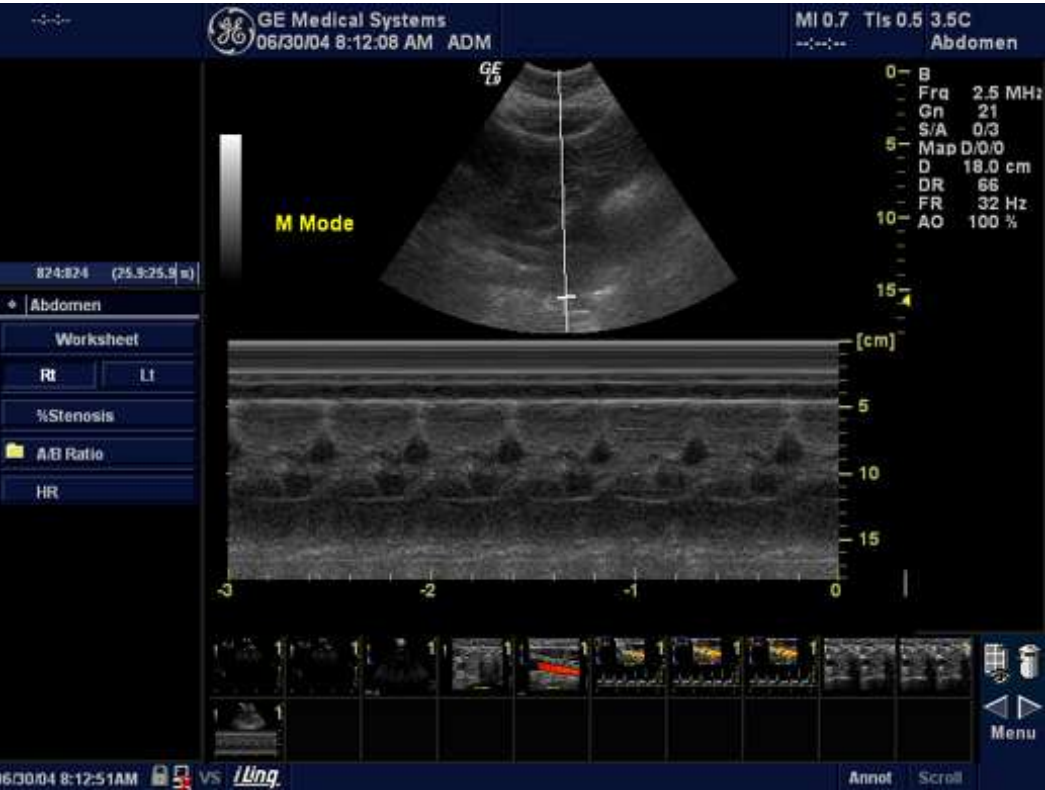


Figure 5-38. M-Mode Display

M-Mode Display	Description, Format, Values
M-Mode Gain	Displays Overall M-Mode Gain as ##dB.
Depth Scale Indicator	Image depth, in cm.
Time Scale	Eight speeds: 0-7. Major marker = 1 second.
M-Mode Cursor	Displays where the M-Mode Cursor is positioned.

M-Mode Touch Panel



Figure 5-39. M-Mode Touch Panel, Page 1



Figure 5-40. M-Mode Touch Panel, Page 2

Scanning Hints



HINTS

These M-Mode controls produce the following results:

Edge Enhance. Affects border sharpness.

Sweep Speed. Controls speed of M-Mode update.

Dynamic Range. Affects the amount of gray scale information displayed.

Sweep Speed

Description

Changes the speed at which the timeline is swept.

Available in M-Mode and Doppler Mode.

Available in M Color Flow Mode.

Adjusting

To increase/decrease, select **Sweep Speed**.

Values

Each selection represents a different sweep time.

Sweep Speed values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.

Benefits

You can speed up or slow down the timeline to see more or fewer occurrences over time.

Anatomical M-Mode

Description

Anatomical M-Mode gives you the ability to manipulate the cursor at different angles and positions. The M-Mode display changes according to the position of the cursor.



Figure 5-41. Anatomical M Mode

Activating

To activate Anatomical M-Mode:

Press the **Anatomical M** Touch Panel control.

Anatomical M-Mode (continued)

Adjusting	Use the Touch Panel control and Trackball to position the M cursor over the required area of the B-Mode image. • Adjust the Touch Panel control to move the M cursor to the desired area. • Press Scan Area to angle the M cursor. • Move the Trackball left/right to move the cursor line left/right. • Move the Trackball up/down to move the M cursor up/down.
Values	On/Off.
Benefits	Color Flow Mode and Color M-Mode are Doppler Modes intended to add color-coded qualitative information concerning the relative velocity and direction of fluid motion within the B-Mode or M-Mode image.
Bioeffects	Changing the Packet Size, PRF, and ROI size may change the TI and/or MI. Observe the output display for possible effects.

Optimizing Color Flow

Intended Use

Color Flow Mode is a Doppler Mode intended to add color-coded qualitative information concerning the relative velocity and direction of fluid motion within the B-Mode image.

Introduction

A typical examination using Color Flow Mode,

1. Follow the same procedure as described under B-Mode to locate the anatomical area of interest.
2. After optimizing the B-Mode image, add Color Flow.
3. Move the color flow area of interest as close to the center of the image as possible.
4. Optimize the color flow parameters so that a high frame rate can be achieved and appropriate flow velocities are visualized.
5. Press **Freeze** to hold the image in memory.
6. Record color flow images as necessary.
7. If more definitive information is needed about flow, utilize the procedures described under Doppler Mode.

Activating Color Flow

To activate Color Flow Mode,

1. Press **CF** (Color Flow). The CF window appears over the B-Mode image. Move the **Trackball** to move the CF window.



Figure 5-42. Color Flow Mode Display

Uses

Color Flow is useful to see flow in a broad area. Color Flow allows visualization of flow in the CF ROI, whereas Doppler Mode provides spectral information in a smaller area.

Color Flow is also sometimes used as a stepping stone to Doppler. You use Color Flow to locate flow and vessels prior to activating Doppler.

Exiting Color Flow

To exit Color Flow, select **CF-Mode** or **B-Mode**.

Color Flow and Power Doppler Scanning Hints



HINTS

Color Flow Mode controls produce the following results:

Line Density. Trades frame rate for sensitivity and spatial resolution. If the frame rate is too slow, reduce the size of the region of interest, select a different frame rate setting, or reduce the packet size.

Wall Filter. Affects low flow sensitivity versus motion artifact.

Color **Threshold.** Percentage of gray scale level where color Doppler is overwritten.

Frame Average. Affects temporal smoothing and color Doppler 'robustness.'

Flash Suppression. Flash Suppression is a proprietary algorithm for motion artifact control.

Packet Size. Affects the amount of color Doppler sensitivity versus frame rate.

Scan Area. Increases/decreases size of color window.

Focus Position. The best focusing is at the focal zone location. Put focal zone(s) at the area of interest.

Color Flow Mode Touch Panel

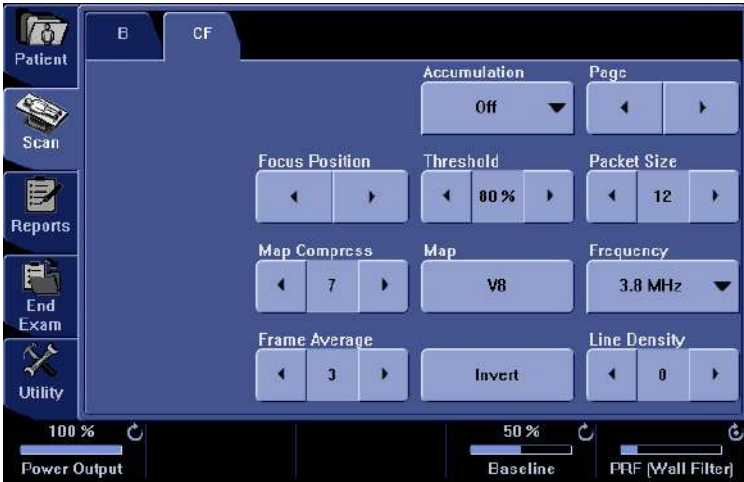


Figure 5-43. Color Flow Mode Touch Panel, Page 1

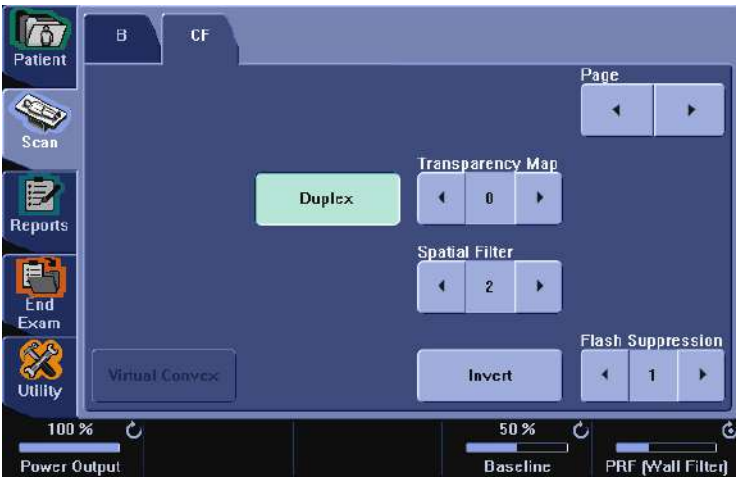


Figure 5-44. Color Flow Mode Touch Panel, Page 2

Gain

Description	Gain amplifies the overall strength of echoes processed in the Color Flow window or spectral Doppler timeline.
Adjusting	Gain values change depending on the probe and application; they are not associated with a particular position of the button. To decrease/increase Gain, adjust the dial to the left/right.
Values	Values vary by probe, application, and multi frequency setting. Gain displays as dB. Gain values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, New Patient, or Multi Frequency.
Benefits	Allows you to control the amount of color within a vessel or to fill in or clean out spectral information.
Bioeffects	Gain has no affect on Power Output. However, with increased Gain, the power output level can usually be reduced to produce an equivalent image quality.

PRF (Pulse Repetition Frequency)

Description	Increases/decreases the PRF on the color bar.
Adjusting	To raise/lower the velocity scale, press PRF/Wall Filter until you reach PRF, then adjust PRF up/down.
Values	PRF is in kHz. PRF values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Imaging of higher velocity flow requires increased scale values to avoid aliasing.
Affect on other controls	Changing the PRF may affect Power Output, frame rate, and wall filter. When you adjust the velocity scale, CINE memory is cleared.
Bioeffects	Changing the PRF range may change the TI and/or MI. Observe the output display for possible effects.

Wall Filter

Description	Filters out low flow velocity signals. It helps get rid of motion artifacts caused from breathing and other patient motion.
Adjusting	To raise/lower the wall filter, press PRF/Wall Filter until you reach Wall Filter, then adjust Wall Filter up/down.
Values	Values vary, depending upon probe, application, and packet size. The wall filter is displayed numerically on the monitor (Hz). Wall Filter values vary by probe and application and are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Gets rid of excess, unnecessary low frequency signals caused by motion.

Color Scan Area

Description	Color Scan Area sizes and positions the color window.
Adjusting	The window grows from the center of the color flow window. To adjust the size, press Scan Area (Size appears in the Trackball Status area on the monitor display) then move the Trackball left/right, up/down. To adjust the position, press Scan Area (Pos appears in the Trackball Status area on the monitor display) then move the Trackball to position the color window.
Values	Sector and Convex Probes. Ranges from 5 degrees to full B-Mode image. Linear Probe. Ranges from 5mm to full B-Mode image.
Benefits	Increase the color window to see a larger area; decrease the color window to improve frame rate and spatial resolution.
Affect on other controls	The smaller the color window, the faster the frame rate and vice versa.
Bioeffects	Sizing the color window may change the TI and/or MI. Observe the output display for possible effects.

Invert (Color Invert)

Description Lets you view blood flow from a different perspective, e.g., red away (negative velocities) and blue toward (positive velocities). You can invert a real-time or frozen image.

NOTE: Invert reverses the color map, NOT the color PRF.

Adjusting To reverse the color flow, press **Invert (Color Invert)**.

In Triplex, both Color Flow and Doppler Mode velocity scales are inverted.

Values Invert and non-Invert. Values vary by probe and application. Invert values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.

Benefits Allows you to view blood flow according to your personal preference, without flipping the probe.

Baseline

Description Changes the Color Flow or Doppler spectrum baseline to accommodate higher velocity blood flow. Minimizes aliasing by displaying a greater range of forward flow with respect to reverse flow, or vice versa.

Baseline adjusts the alias point. The default baseline is at the midpoint of the color display and at the midpoint of the color bar reference display.

Adjusting To adjust the baseline, adjust **Baseline** up/down, as necessary.

Values Zero velocity follows the baseline. The total PRF range remains the same. Values vary by probe and application.

Baseline values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.

Benefits Unwraps the alias during color flow imaging. Higher velocities can be displayed without reversal of colors.

Angle Steer

Description	You can slant the ROI of the Color Flow linear image left or right to get more information without moving the probe. The Angle Steer function only applies to linear probes.
Adjusting	To slant the linear image to the left/right, adjust Angle Steer .
Values	Linear probes can be steered left (20 degrees), center, or right (20 degrees). Angle Steer values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New patient
Benefits	Provides a Doppler cursor angle suitable for linear probe orientation. Beneficial in Peripheral Vascular to image carotids.
Bioeffects	Activating angle steer may change the TI and/or MI. Observe the output display for possible effects.

Accumulation

Description	Accumulation enhances the flow in an image.
Values	8 settings: 0=Off, 6 Various levels, or frames, of accumulation, and Infinite. If Accumulation is turned off, then Frame Averaging is used; if Accumulation is set, then Accumulation is used.
Availability	Available in Contrast, Color Flow, PDI, and B Flow.
Benefit	Accumulation detects the maximum signal and holds it for the level specified.

Map Compress

Description	When you increase the value, high velocity elements in the map are compressed so that the map darkens. When you decrease the value, low velocity elements in the map are compressed so that the map lightens. The effect is visible in the color bar.
Adjusting	Select Map Compress to adjust.
Values	0-14. Center value is 7.
Benefits	Changes the gradation in the map.

Color Flow Line Density

Description	Optimizes the Color Flow frame rate or spatial resolution for the best possible color image.
Adjusting	To adjust the line density, select the Line Density Touch Panel key.
Values	The current frame rate is shown on the Touch Panel. Frame Rate/Resolution values vary by: Probe, Exam Category, Exam Calcs, New Patient, and Frequency. Settings are returned to factory or user preset value when you change any of the above.
Benefits	Low line density is useful in fetal heartbeat, adult cardiac applications, and clinical Radiology applications which require significantly higher frame rates. High resolution is useful in situations where very small vessels are being imaged, e.g., thyroid, testicles.
Affect on other controls	Line density changes the vector density and frame rate.
Bioeffects	Modifying line density may change the TI and/or MI. Observe the output display for possible effects.

Map

Description	Allows you to select a specific color map. After you have made your selection, the color bar displays the resultant map.
Adjusting	After you activate Color Flow, the Color Flow Touch Panel displays. To cycle through available maps, select Map , move the Trackball to view available maps, and press Set to select.
Values	Velocity Maps (V). Flow shown as blue away/red toward the probe. Velocity Variance Maps (VV). Provides a measure of turbulence (stenosis). Adds green to velocity maps. Power Doppler Imaging Maps (P). PDI Map P7 is used for Directional PDI.
Benefits	Shows the direction of the flow and highlights the higher velocity flows.

Threshold

Description	Threshold assigns the gray scale level at which color information stops.
Adjusting	To increase/decrease the gray scale threshold, select Threshold left/right.
Values	The settings cycle through various values: 0%-100% of the gray scale. High values display more color; low values displays more B-Mode gray scale data. The Color Threshold level is displayed on the Touch Panel. Values vary by probe and application and are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Limits color flow overlay to low level echoes inside vessel walls. Helps minimize color 'bleeding' outside vessel walls.

Frame Average

Description	Averages color frames.
Adjusting	To smooth temporal averaging, select Frame Average . The selected value displays on the Touch Panel.
Values	Frame Average values vary by probe and application. The values are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Higher frame averaging keeps the color displayed longer for increased flow visualization while lower frame averaging provides greater flow dynamics.
Affect on other controls	Trades off between frame rate and color quality. As the color quality increases, the frame rate may decrease and as the frame rate increases, the color image quality decreases.

Transparency Map

Description	Brings out the tissue behind the color data.
Adjusting	Select Transparency Map to adjust
Values	0-4
Benefits	Helps demonstrate the tissues behind the color.

Spatial Filter

Description	Smooths out the color, makes it look less pixely.
Adjusting	Select Spatial Filter to adjust.
Values	0-5.
Benefits	Smooths the image.

Duplex/Triplex

Description	Duplex allows two modes to be active at the same time; Triplex allows three modes to be active at the same time. • B + PW or B + CW or B + CF (Duplex) • B + PW + CF or B + CW + CF (Triplex) B Pause pauses the image while keeping the CW / PW timeline active. When Duplex/Triplex is OFF, either the image or timeline is active. B Pause then switches the active side between the image and the timeline. With Duplex/Triplex active, M/D Cursor functions the same as the Duplex/Triplex control by cycling Duplex/Triplex on/off.
Adjusting	Select Duplex/Triplex .
Preset	You can preset Duplex in Utility --> Imaging --> General.
Benefits	Allows the user to have multiple modes active at the same time.

Flash Suppression

Description	Activates/deactivates Flash Suppression, a motion artifact elimination process.
Adjusting	To adjust, select the next page and then select Flash Suppression . Touch Panel key.
Values	0 through 4. Values are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Beneficial to suppress flash.

Packet Size

Description	Controls the number of samples gathered for a single color flow vector.
Adjusting	To increase/decrease the packet size, select Packet Size left/right.
Values	Values vary by probe and application and are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient. Values are displayed on the monitor display as P.
Benefits	Allows you to improve the color sensitivity and accuracy of color averaging (increase packet size) or frame rate (decrease packet size), as needed.
Affect on other controls	When you decrease the packet size, you increase the frame rate at the expense of image quality. When you increase the packet size, you improve image quality at the expense of frame rate.
Bioeffects	Changing packet size may change the TI and/or MI. Observe the output display for possible effects.

Power Doppler Imaging (PDI)

Description

Power Doppler Imaging (PDI) is a color flow mapping technique used to map the strength of the Doppler signal coming from the flow rather than the frequency shift of the signal. Using this technique, the ultrasound system plots color flow based on the number of reflectors that are moving, regardless of their velocity. PDI does not map velocity, therefore it is not subject to aliasing.



Figure 5-45. Power Doppler Imaging Display

Adjusting

Press **PDI**. The color flow window appears over the B-Mode image. Move the Trackball to move the CF window. To exit, press **PDI** or select a new mode.

Values

On/Off.

Twelve power (P0-P6 and P8-P12) and one Directional PDI map (P7) is available.

Benefits

Since PDI does not display velocity, it does not alias.

Power Doppler Imaging (PDI) (continued)

Affect on other controls

When PDI is activated, the following controls are adjusted: PRF is set to a lower setting. Color Map is set to a power map. Line Density is adjusted. Wall Filter is set to a lower setting. Threshold is set to 100%. Frame Averaging is adjusted. Packet Size is adjusted.

NOTE: *These controls are reset to their previous values upon exiting PDI.*



HINTS

When changing maps, higher gain settings may be needed.

P-1. The expanded map is designed to expand or increase the dynamic range of data displayed in a PDI image to show both low and high flow states. It is especially useful for highlighting very low flow states just above the noise floor as in renal vessels.

P-3. The compressed map is designed to compress or decrease the dynamic range of the PDI image data to highlight high flow states and suppress noise.

P-7. Directional PDI displays the direction of flow while in Power Doppler Imaging. Gives all the benefits of PDI while also providing directional information not available in traditional PDI. Used in applications where sensitivity and angle independence is desired, but directional information is also required. Flow towards the transducer is red; flow away is dark blue to light blue.

PDI (Power Doppler Imaging) Mode Touch Panel

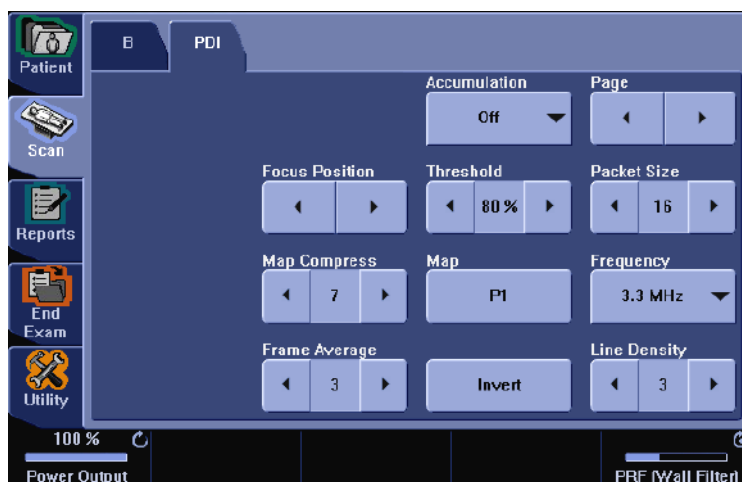


Figure 5-46. PDI Mode Touch Panel, Page 1

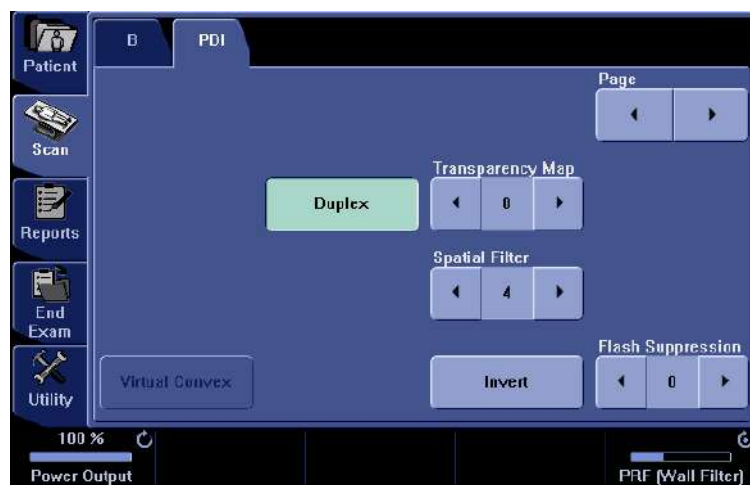


Figure 5-47. PDI Mode Touch Panel, Page 2

Directional Power Doppler

You can select the P7 Directional Power Doppler map while in PDI.

NOTE: *If you store a PDI image and recall it, you can still switch to the Directional Power Doppler map and vice versa. However, an image stored as non-directional then switched to directional just adds direction to a non-directional map and vice versa.*



HINTS

If the image is aliasing while in Directional Power Doppler, increase the PRF and reduce the Wall Filter.

Optimizing M Color Flow

M Color Flow Mode

Description

M Color Flow is used for fetal cardiac applications. Color Flow overlays color on the M-Mode image using velocity and variance color maps. The Color Flow wedge overlays the B-Mode image and M-Mode timeline.

The Color Flow maps available in M-Mode are the same as in Color Flow Mode. The size and position of the Color Flow window in B-Mode determines the size and position of the Color Flow window in M-Mode.

All M-Mode measurements are available with M Color Flow active: depth, distance along a straight line, % stenosis, volume, trace, circumference, enclosed area, distance, time, slope, and heart rate.

Activating

To activate M Color Flow Mode, press M (M-Mode). Then press CF (Color Flow) - or - press CF, then press M.

To toggle between M Color Flow controls and Color Flow controls, press the appropriate Touch Panel Mode tab.

Values

On/Off.

Benefits

Color Flow Mode and Color M-Mode are Doppler Modes intended to add color-coded qualitative information concerning the relative velocity and direction of fluid motion within the B-Mode or M-Mode image.

M Color Flow Mode (continued)

Bioeffects Changing the Sweep Speed, Packet Size, Frame Rate/Resolution, Zoom, PRF, and ROI size may change the TI and/or MI. Observe the output display for possible effects.

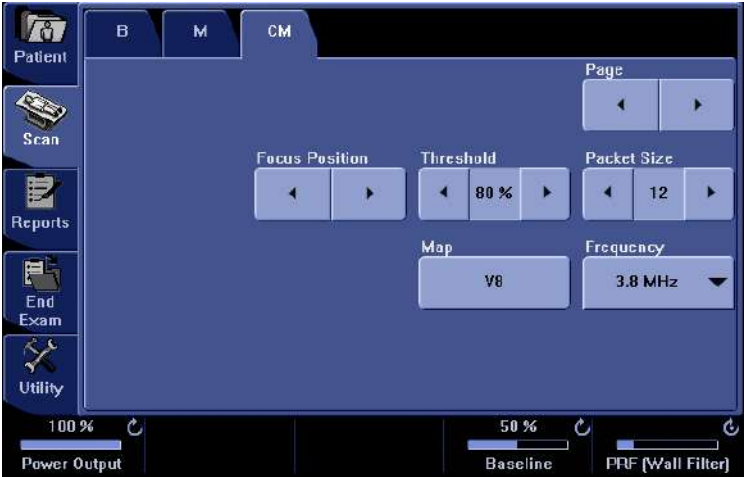


Figure 5-48. M Color Flow, Page 1



Figure 5-49. M Color Flow, Page 2

Optimizing Spectral Doppler

Intended Use

Doppler is intended to provide measurement data concerning the velocity of moving tissues and fluids. PW Doppler lets you examine blood flow data selectively from a small region called the sample volume.

Typical Use - PW Doppler

In Pulsed Wave Doppler (PW) Mode, energy is transmitted from the ultrasound probe into the patient, as in B-Mode. However, the received echoes are processed to extract the difference in frequency between the transmitted and received signals. Differences in frequencies can be caused by moving objects in the path of the ultrasound signal, such as moving blood cells. The resultant signals are presented audibly through the system speakers and graphically on the system display. The X axis of the graph represents time while the Y axis represents the shift in frequency. The Y axis can also be calibrated to represent velocity in either a forward or reverse direction.

PW Doppler is typically used for displaying the speed, direction, and spectral content of blood flow at selected anatomical sites. PW Doppler operates in two different modes: conventional PW and High Pulse Repetition Frequency (HPRF).

PW Doppler can be combined with B-Mode for rapidly selecting the anatomical site for PW Doppler examination. The site where PW Doppler data is derived appears graphically on the B-Mode image (Sample Volume Gate). The sample volume gate can be moved anywhere within the B-Mode image.

Typical exam protocol

A typical examination using PW Doppler Mode might proceed as follows:

1. Connect the appropriate probe, leaving the probes in their respective holders.
2. Position the patient for the examination.
3. Press **Patient**. Enter the appropriate patient data using the appropriate exam category.
4. Select the preset, application and probe to be used.
5. Locate the anatomy to be examined. Get a good B Mode image. Press CF to help locate the vessel you wish to examine.
6. Press **M/D Cursor** to display the sample volume cursor and gate.
or
Press **PW**. The PW Doppler spectrum appears and the system operates in combined B+Doppler Mode. Adjust **Volume** to adjust Doppler audio. The Doppler signal is heard through the speakers.
7. Position the sample volume cursor by moving the **Trackball** left and right. Position the sample volume gate by moving the **Trackball** up and down. Size the gate by clicking **SV Length**.
8. Optimize the PW Doppler spectrum, as necessary. Refer to the *Doppler Optimization* section of this chapter for more information.
9. Press **B Pause** to toggle between real time B-Mode with Doppler Mode (with audio).
10. Sample along the whole length of the vessel. Make sure that the probe is parallel to flow. Listen, then look, when positioning the sample volume cursor.
11. Press **Freeze** to hold the trace in memory and stop imaging. Activate CINE Timeline, as necessary. See 'Activating CINE' on *page 6-7 for more information*.
12. Perform measurements and calculations, as necessary. Refer to the Measurements and Calculations chapter for more information.

Typical exam protocol (continued)

13. Record results by pressing the appropriate print key, depending on the setup of your recording devices.
14. Press **Freeze** to resume imaging.
15. Repeat the above procedure until all relevant flow sites have been examined.
16. Replace the probe in its respective holder.

Activating Doppler Mode

To activate PW Doppler Mode, press PW.

The Doppler spectrum displays along with the B-Mode image. The cursor changes to a Doppler cursor.

You can now position and size the sample volume gate to get a velocity. Use Doppler Audio to listen for when the sample volume gate is positioned over an area of flow.

B Pause toggles between real time B-Mode with Doppler Mode and real time spectral display.

Use Doppler is used to examine blood flow information.

To exit PW Doppler Mode, press PW.

Activating Triplex Mode

To activate Triplex Mode, press CF. Press PW.

The Doppler spectrum displays along with the Color Flow and B-Mode image. The cursor changes to a Doppler cursor.

You can now position and size the sample volume gate to get a velocity. Use Doppler Audio to listen for when the sample volume gate is positioned over an area of flow.

Use Triplex Mode is used to examine blood flow information.

To Exit PW Doppler Mode, press CF, then PW.

Spectral Doppler Display

Time zero (the start of the trace) appears on the left side of the graph. As time progresses, the trace moves to the right. The baseline of the graph (representing zero velocity, zero frequency shift, or no detected flow), appears as a solid line running horizontally across the display. By convention, movement toward the probe is positive and movement away from the probe is negative. Positive frequencies or velocities appear above the baseline. Negative frequencies or velocities appear below the baseline.

Typically, blood flow is not uniform but is composed of a mix of blood cells moving at different velocities and in different directions. Thus, the display is composed of a spectrum as gray scale values. Strong signals are displayed as bright while weak signals are displayed as varying shades of gray.

HPRF (High Pulse Repetition Frequency) is invoked when you are operating in PW Doppler Mode and conditions activate HPRF (when the velocity scale factor or sample volume gate depth exceeds certain limits). When HPRF is active, multiple sample volume gates appear along the Doppler mode cursor. Doppler information can be received from any of the multiple sample volume gates. The Doppler signals from all the gates are added together and displayed in one spectrum.

Information about the PW Doppler display is automatically written on the screen and updated when scanning parameters are changed.

This chapter includes:

- A discussion of PW Doppler.
- Activating Pulsed Wave Doppler.
- Optimizing the Doppler spectrum.

Imaged Doppler

When you view the PW Doppler spectrum and a B-Mode image with a 3.5 MHz probe, this means that you are viewing the image at 3.5 MHz and 'Dopplering' at 2.5 MHz.

Doppler Mode Display



Figure 5-50. PW Doppler Mode Display

Table 5-4: Doppler Mode Display Explanations

Doppler Display	Description, Format, Values
PRF	Pulse repetition frequency, displayed as PRF in kHz.
Wall Filter	Wall filter size, displayed as WF in Hz.
Doppler Gain*	Displays as GN in decibels (dB).
Sample Volume Depth	Displays (in Cm) when Doppler cursor is present.
Doppler Angle (AC ##)	Indicates angle in degrees between the Doppler mode cursor and the angle correction indicator. Displays when Doppler cursor is present. The Doppler Angle displays in red when the angle exceeds 60°. Velocities obtained when the angle is greater than 80° are displayed as asterisks (***).
Spectral Invert	INVERT appears when the spectral trace is inverted and the plus/minus signs (+/-) are reversed.

Table 5-4: Doppler Mode Display Explanations

Doppler Display	Description, Format, Values
HPRF	HPRF mode is used when detected velocities exceed the processing capabilities of the currently selected PW Doppler scale or when the selected anatomical site is too deep for the selected PW Doppler scale.
Time Scale	Each selection represents a different sweep time.
Angle Correct	Indicates flow direction.
Sample Volume Gate	Indicates sample volume box. Each probe defaults to a specific range gate.
Doppler Velocity Scale	Flow direction has a positive and negative indicator, noted in centimeters per second (cm/sec). When the velocity scale is less than 10 cm/s, it is displayed to the first decimal point (4.6 rather than 5 cm/s). The Doppler velocity scale adjust as you adjust the PRF.

Doppler Mode Scanning Hints



HINTS

The best Doppler data is collected when parallel to flow, with orientation also parallel to the anatomic target; whereas, the best B-Mode image data is collected perpendicular to the anatomic target. Therefore, you don't usually get both an ideal B-Mode image and ideal Doppler data simultaneously.

Control overview

Dynamic Range. Affects the amount of Doppler amplitude data displayed.

Wall Filter. Removes the noise caused by vessel or heart wall motion at the expense of low flow sensitivity.

Sweep Speed. Controls speed of spectral update.

Doppler Mode Touch Panel

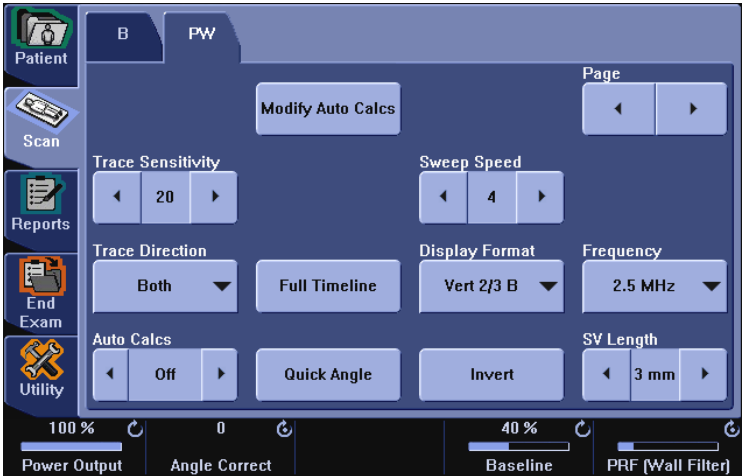


Figure 5-51. PW Mode Touch Panel, Page 1

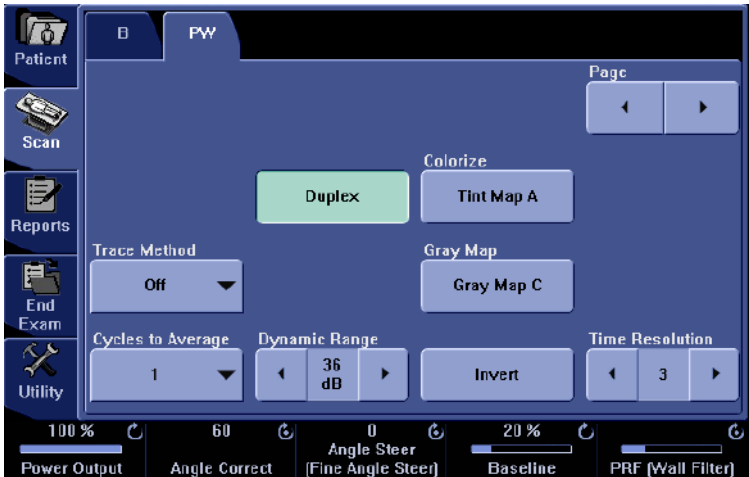


Figure 5-52. PW Mode Touch Panel, Page 2

B Pause

Description	Toggles between simultaneous and update presentation while viewing the timeline.
Adjusting	To activate, press B Pause to toggle between simultaneous and update. Doppler Mode does not restart each time the image is updated; however, a black bar may appear with a lightning bolt signalling a break in the timeline.
Values	On/Off.
Benefits	Update increases the Spectral Doppler display quality.
Bioeffects	Activating Update may change the TI and/or MI. Observe the output display for possible effects.

Doppler Sample Volume Gate Position (Trackball)

Description	Moves the sample volume gate on the B-Mode's Doppler Mode cursor. The gate is positioned over a specific position within the vessel.
Adjusting	To move Doppler Mode cursor position, move Trackball left or right until positioned over the vessel. To move sample volume gate position, move Trackball up or down until positioned inside the vessel.
Values	Defaults to 50% of the depth and can move continuously throughout the field of view.
Benefits	Positions the sample volume gate to sample blood flow.
Bioeffects	Changing the sample volume gate position may change the TI and/or MI. Observe the output display for possible effects.

Doppler Sample Volume Length

Description	Sizes the sample volume gate.
Adjusting	To increase/decrease the gate size, adjust SV Length on the Touch Panel. Hold down key to continuously size gate. You can adjust the sample volume gate length whenever the sample volume gate appears on the display. <i>NOTE: Adjustments to the sample volume gate size are made from the center point of the sample volume position.</i>
Values	Values vary by probe and application. Sample volume gate size values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	A smaller gate produces accurate sampling results because it is more sensitive. You can also enlarge the gate for sampling large vessels or areas.
Bioeffects	Changing the sample volume gate size may change the TI and/or MI. Observe the output display for possible effects.

PRF

Description	Adjusts the velocity scale to accommodate faster/slower blood flow velocities. Velocity scale determines pulse repetition frequency. If the sample volume gate range exceeds single gate PRF capability, the system automatically switches to high PRF mode. Multiple gates appear, and HPRF is indicated on the display.
High PRF	High Pulse Repetition Frequency (HPRF) is a special operating mode of PW Doppler. In HPRF mode, multiple energy pulses are used. This allows higher velocities to be detected without causing aliasing artifacts. HPRF mode is used when detected velocities exceed the processing capabilities of the currently selected PW Doppler scale or when the selected anatomical site is too deep for the selected PW Doppler scale. The pulse repetition frequency (PRF) is displayed to the left of the spectrum in frames per second. <i>NOTE: Ensure that only one gate overlays a blood vessel at a time. Otherwise, signals from more than one flow area are superimposed.</i>
Adjusting	To raise/lower, adjust the PRF on the Touch Panel (PRF and Wall Filter share the same control. Press the control to toggle between PRF and Wall Filter). The display updates velocity scale parameters after you adjust the velocity scale.
Values	Velocity Scale values vary by probe and application. In Triplex, when you change the velocity scale in Color Flow, the Doppler Mode velocity scale is also updated if Triplex is on. Velocity Scale values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.

PRF (continued)

Benefits	Blood flow information is not cut off due to the effect of aliasing.
Affect on other controls	When you raise the velocity scale, the spectral waveform may decrease in size; when you lower the velocity scale, the spectral waveform may increase in size. Changes in the spectrum are relative to changes in the velocity scale, that is, it sizes accordingly. When you adjust the velocity scale, CINE memory is cleared. Adjustments may affect sample volume size and Doppler wall filter.
Bioeffects	Changing the velocity range may change the TI and/or MI. Observe the output display for possible effects.

Angle Correct

Description	Estimates the flow velocity in a direction at an angle to the Doppler vector by computing the angle between the Doppler vector and the flow to be measured.
	<i>NOTE: When the Doppler Mode Cursor and angle correct indicator are aligned (the angle is 0), you cannot see the angle correct indicator.</i>
Adjusting	Flow toward the probe is mapped above the baseline and vice versa. To adjust the angle relative to the probe face, adjust Angle Correct to the left/right. The velocity scale changes when you adjust angle correct.
Values	1 increments from 0 to 90. The possible range of operation is from 0 degrees to 90 degrees in either direction. For optimum velocity measurements, the angle of incidence should be between 45-60 for vascular applications. Angle Correct values vary by probe and application. Angle Correct values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Optimizes the accuracy of the flow velocity. This is especially useful in vascular applications where you need to measure velocity.

Quick Angle

Description	Quickly adjusts the angle by 60 degrees.
Adjusting	To quickly adjust the angle, press Quick Angle .
Values	0 degrees and 60 degrees.

Wall Filter

Description	Insulates the Doppler signal from excessive noise caused from vessel movement.
Adjusting	To increase/decrease, select Wall Filter , then adjust Wall Filter on the Touch Panel (PRF and Wall Filter share the same control. Press the control to toggle between PRF and Wall Filter). Each adjustment cycles you to the next setting.
Values	Values vary, depending upon the probe and application. The current value displays on the Touch Panel and the monitor. Wall Filter values are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, New Patient.
Benefits	Gets rid of excess, unnecessary information. Cleans out low level noise above and below the baseline so you don't see or hear it on the spectrum.
Affect on other controls	Wall filter may be changed by changes to the velocity scale.

Baseline

Description	Adjusts the baseline to accommodate faster or slower blood flows to eliminate aliasing.
Adjusting	Baseline adjusts the point in the spectrum where the velocity trace is at zero. The default baseline is at the midpoint of the spectrum. To shift the baseline, adjust Baseline on the Touch Panel. The baseline displays as a solid line running across the spectrum. The baseline is raised and lowered in equal increments, depending on the current Doppler scale factor. The control does not wrap when the maximum baseline shift (in either direction) has been reached.
Values	12% increments, with 0 being the center of the display and +100% being the top edge of the display and -100% being the bottom edge of the display. Baseline values vary by probe and application and are returned to the factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Unwraps the alias. Rearranges the velocity scale without changing the velocity scale. Readjusts the positive and negative velocities limit without changing the total velocity range.

M/D Cursor

Description	Displays the Doppler Mode cursor on the B-Mode image.
Adjusting	To activate/deactivate the Doppler Mode cursor, press M/D Cursor. Trackball to position sample volume graphic. The M/D Cursor key backlights.
Values	On/Off.
Benefits	Lets you position the cursor before you go into Doppler Mode.

Angle Steer and Fine Angle Steer

Description	You can slant the ROI of the Color Flow linear image left or right to get more information without moving the probe. The angle steer function only applies to linear probes.
Adjusting	To slant the linear image to the left/right, adjust Angle Steer . Press Angle Steer to access Fine Angle Steer .
Values	Linear probes can be steered left (20 degrees), center, or right (20 degrees). Angle Steer values are returned to factory or user preset value when you change: Probe, Exam Category, Exam Calcs, or New Patient. Fine Angle Steer steers the Doppler cursor left/right between 0 and 30 degrees in 2 degree increments.
Benefits	Provides a Doppler cursor angle suitable for linear probe orientation. Beneficial in Vascular applications.
Bioeffects	Activating angle steer may change the TI and/or MI. Observe the output display for possible effects.

Audio Volume

Description

Controls audio output.

Adjusting

To adjust the audio, adjust the **Volume** control. The volume defaults to the last Doppler audio volume setting.



Audio sounds change rapidly, often abruptly. Increase the volume in small steps to avoid startling the patient.

Values

Usually, a one-third adjustment of the dial gives you the best volume. The volume decreases/increases logarithmically.

Benefits

An audio representation of the flow within a vessel can be used to evaluate proper probe angle and position.

Invert

Description

Vertically inverts the spectral trace without affecting the baseline position.

Adjusting

To invert the spectral trace, press **Invert**. The plus (+) and minus (-) signs on the velocity scale reverse when the spectrum is inverted.

Positive velocities display below the baseline.

Values

Forward/reverse. The trace corresponds to flow direction (positive flow is forward flow toward the probe or negative flow is reverse flow away from the probe). The invert setting is returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient. In Triplex, both Color Flow and Doppler Mode velocity scales are inverted.

Benefits

If you change the probe angle to accommodate anatomy, blood flow still moves in the same direction, but the Doppler information will be reversed. It is easier in cases like this to invert the spectrum instead of reversing the probe orientation.

Dynamic Range (Compression)

Description	Dynamic range controls how echo intensities are converted to shades of gray, thereby increasing the range of contrast you can adjust.
Adjusting	You access Doppler dynamic range from the Doppler Mode Touch Panel. Each click cycles you to the next setting. To increase/decrease, press Dynamic Range .
Values	The settings cycle in 4 dB increments. The current value displays on the Touch Panel. Dynamic Range values vary by probe and application and are returned to the factory or user preset value when you change the following: Probe, Exam Category, Exam Calcs, or New Patient.
Benefits	Optimizes the image's texture and smoothness by increasing or decreasing the amount of gray scale.
Affect on other controls	Dynamic range operates only in realtime; Compression is available while frozen, in CINE, or CINE Timeline. Neither Dynamic range nor Compression is available during VCR playback.

Spectral Trace (Trace Method)

Description	Traces the average mean and peak velocities in realtime or frozen images.
Adjusting	Select the spectral trace method from Trace Method . To activate the Spectral Trace Touch Panel, press To get a peak trace, click MAX TRACE. A green trace displays on the spectrum. To get a mean trace, click MEAN TRACE. A blue trace displays on the spectrum.
Selecting Trace Position	Trace direction key lets you trace the cardiac cycle in the timeline above, below, or above and below (composite) the baseline.
Benefits	Lets you trace the cardiac cycle.

Cycles to Average

Description	The average value over a number of cycles (from 1-5). For example, if you set the number of cycles at 3, values would be averaged for 3 cycles. If you have 5 cycles on the display, the PS would be the average of 3 of the 5 cycles, which are identified by a line drawn over the 3 averaged cycles.
Values	1-5, available for live and frozen images.

Trace Sensitivity

Description	Adjust the trace to follow the waveform for signal strength.
Adjusting	Select the appropriate Touch Panel control to adjust the value.
Values	From 0 to 40.
Benefits	If the signal is very faint, increasing the Trace Sensitivity will allow the system to trace that signal strength.

Auto Update in Duplex/Triplex

Description	Auto Update allows you to activate the B-Mode or B+Color Flow Mode image by using the Trackball while B Pause is active in Duplex/Triplex. The B-Mode or B+Color Flow Mode image is active for a short time while you move the M/D Cursor.
Preset	You can set the auto update response time in Utility --> Imaging --> PW --> Auto Update/Response Time.
Value	Off, Long, Mid, or Short.
	<i>NOTE: The value sets the response time for the cursor position to settle. After you move the M/D Cursor, the system waits for the time specified (Long, Mid, or Short), then restarts the timeline.</i>
Benefits	Switching the active image between B-Mode or B+Color Flow Mode and the Spectrum without pressing B Pause enables you to track the best sample gate position.

Trace Direction

Description	Specifies trace direction.
Adjusting	Select Trace Direction .
Values	Above, Below, or Both.
Benefits	You can select where on the waveform to perform the trace, above, below, or both.

Full Timeline

Description	Expands display to full timeline display.
Adjusting	Select Full Timeline .
Values	On/Off.
Benefits	Allows you to view just the timeline, to see more detail.

Display Format

Description	Changes the horizontal/vertical layout between B-Mode and M-Mode, or timeline only.
Adjusting	Select Display Format .
Values	Vertical 1/3, 1/2, or 2/3 B-Mode, Horizontal 1/4 or 1/2 B-Mode, or Timeline Only.
Benefits	You can select how to have your Doppler timeline and anatomy displayed.

Time Resolution

Description	Adjusts image appearance so that if you select a lower setting, the image appears smoother; if you select a higher setting, the image appears sharper.
Adjusting	Select Time Resolution .
Values	1-3
Benefits	Time Resolution = 3 (high) smooths the Timeline; Time Resolution = 1 (low) increases the time resolution (sharpens the image).

Spectral Average

Description	Averages the spectrum in a vertical direction and timeline direction
Adjusting	Select Spectral Aver. to select the values.
Values	Off, Low, Med (Medium), High and Very High.
Benefits	This has the effect of presenting a smoother and softer image.

Modify Auto Calcs

Description	Activates the menu to select which calculations are automatically calculated.
Adjusting	Select Modify Auto Calcs to activate Modify Auto Calcs Touch Panel.
Benefits	Provides flexibility.

Auto Calcs

Description	Activates the calculation automatically which you select in the Modify Auto Calculation when the system is in a state of freeze or live.
Adjusting	Select the appropriate Touch Panel control to adjust the value.
Values	• Live: Auto calculation activates when the system in a state of live. • Freeze: Auto calculation activates when you press Freeze. • Off

LOGIQ 9 Continuous Wave Doppler (CWD)

CW allows examination of blood flow data all along the Doppler Mode cursor rather than from any specific location. CW gathers samples along the entire Doppler beam. Range-gated CW allows information to be gathered at specific locations.

Requires a CWD probe and probe adapter and supports the P5D and P8D probes.

NOTE: *There is no orientation marking for the CW probes.*

Activating CW Doppler

To activate CW Doppler Mode:

- Ensure that the appropriate CW probe is connected.
- Select the probe from the Touch Panel. The **CW** key is automatically selected.

The Doppler Spectrum appears, along with the CW Touch Panel.

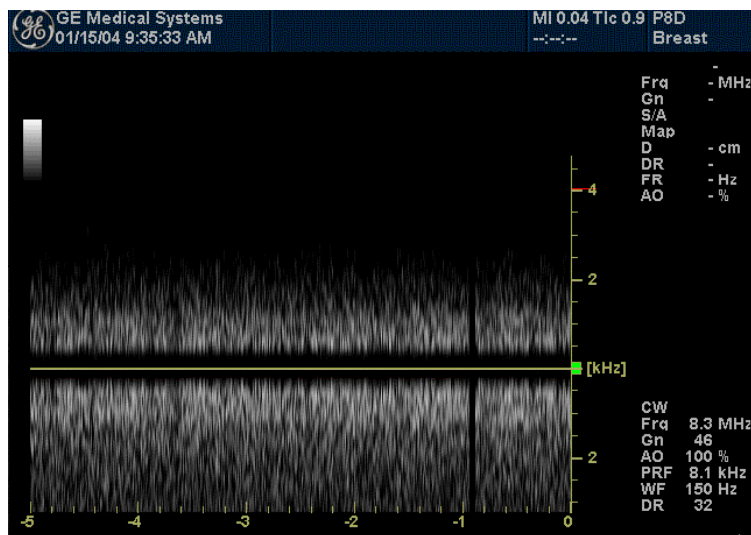


Figure 5-53. CW Screen

The following CW parameters are displayed: Frequency, Gain, Acoustic Output, PRF, Wall Filter, and Dynamic Range.

Activating CW Doppler (continued)

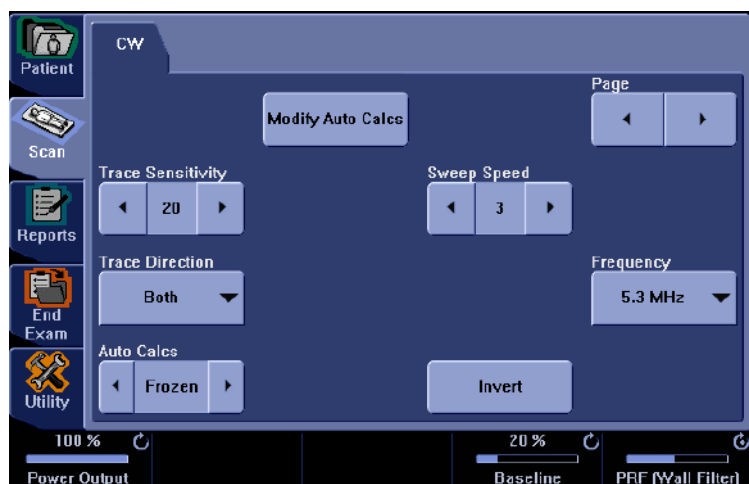


Figure 5-54. CWD Mode Touch Panel, Page 1

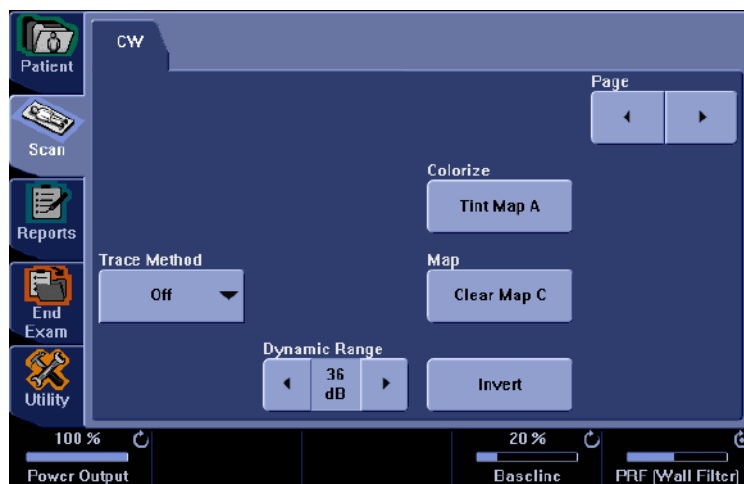


Figure 5-55. CWD Mode Touch Panel, Page 2

Exiting CW Doppler

CW Doppler Mode, press **CW**.

Using 3D

Overview



DO NOT scan any patient with implanted pacemaker or other medical device that is susceptible to magnetic interference when using the 3D sensor device for performing 3D measurements. The magnetic fields emitted from the device may interfere with the pacemaker operation.

There are three 3D Packages:

Table 5-5: 3D Package Options

3D Type	Description	Sensor/No Sensor	Available Tabs
Easy 3D	Designed for rendering B Mode images, e.g., Baby Face scans.	No sensor	3D Acquisition, Easy 3D, Movie
Advanced 3D	Designed for rendering B Mode and Color Flow Mode images, e.g., vessel trees.	No sensor	3D Acquisition, Easy 3D, Advanced 3D, Movie
Tru 3D	Designed for rendering any image sequence	Sensor	3D Acquisition, Easy 3D, Advanced 3D, Movie, Tru 3D, Volume Measurement

3D Acquisition

Acquiring a 3D Scan

To acquire a 3D scan,

1. Optimize the B-Mode image. Ensure even gel coverage.
2. Press the **3D/4D** key. Two screens appear.
3. To start acquiring the image, press 'L' (the left split screen key).
4. To perform a parallel scan, scan evenly. To perform a sweep (fan) scan, rock the probe once. Note the distance of the scan.
5. The 3D volume of interest (VOI) is dynamically assembled on the right side of the screen.

NOTE: *Set appropriate values for Acq Mode and Scan Plane. Also, set the scan distance before scanning.*

NOTE: *If the image stops before you're done scanning, start acquiring the 3D volume of interest again by pressing **L** again.*

6. To complete the 3D scan, press 'R' (the right split screen key).

NOTE: *You can also press Freeze, but then you need to also press the 3D/4D key to obtain the final render.*

3D Notes

Adjust the 3D dataset brightness with B-Mode or Color Flow Mode Gain.

Use Colorize to change the color of the active dataset.

Use Zoom to increase the zoom factor of the active dataset.

Manipulating the Volume of Interest

Imagine you are able to manipulate the 3D volume of interest (VOI) in your hand. The 3D VOI is a tangible anatomical object that you can see and manipulate easily using the Trackball and Set control panel keys.

Practice positioning the pointer at different places within the 3D VOI. Highlight different colors (white, red, yellow, or green). Press Set to select a VOI for manipulation. Use the hand to manipulate the 3D VOI.

Rotating the 3D VOI Left/Right or Forward/Backward

You can rotate it left to right or right to left. You can rotate it forward/backward. Press right Set key when the white pointer finger is positioned on the white box. Move the closed white hand to manipulate the 3D VOI.

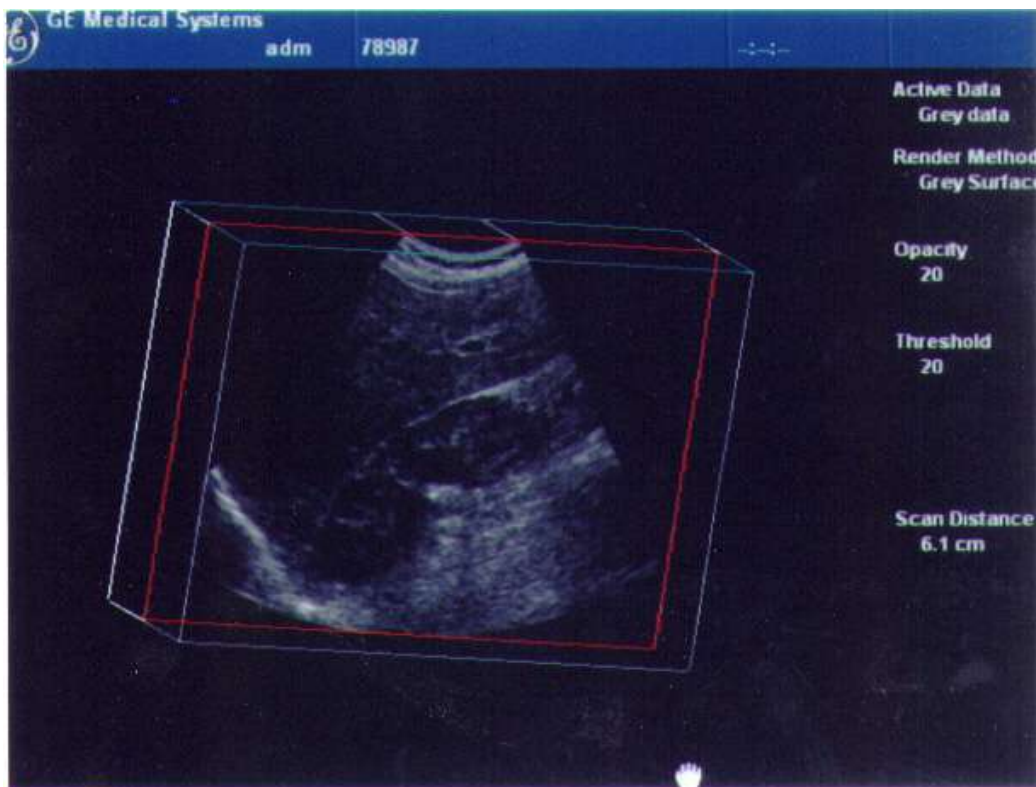


Figure 5-56. Manipulating the 3D Volume of Interest (White Hand)

**Moving Through
the 3D VOI**

You can move through the 3D VOI using the red hand. Press Set when the red pointer finger is positioned on the red box. Move the closed red hand to move through the 3D VOI.

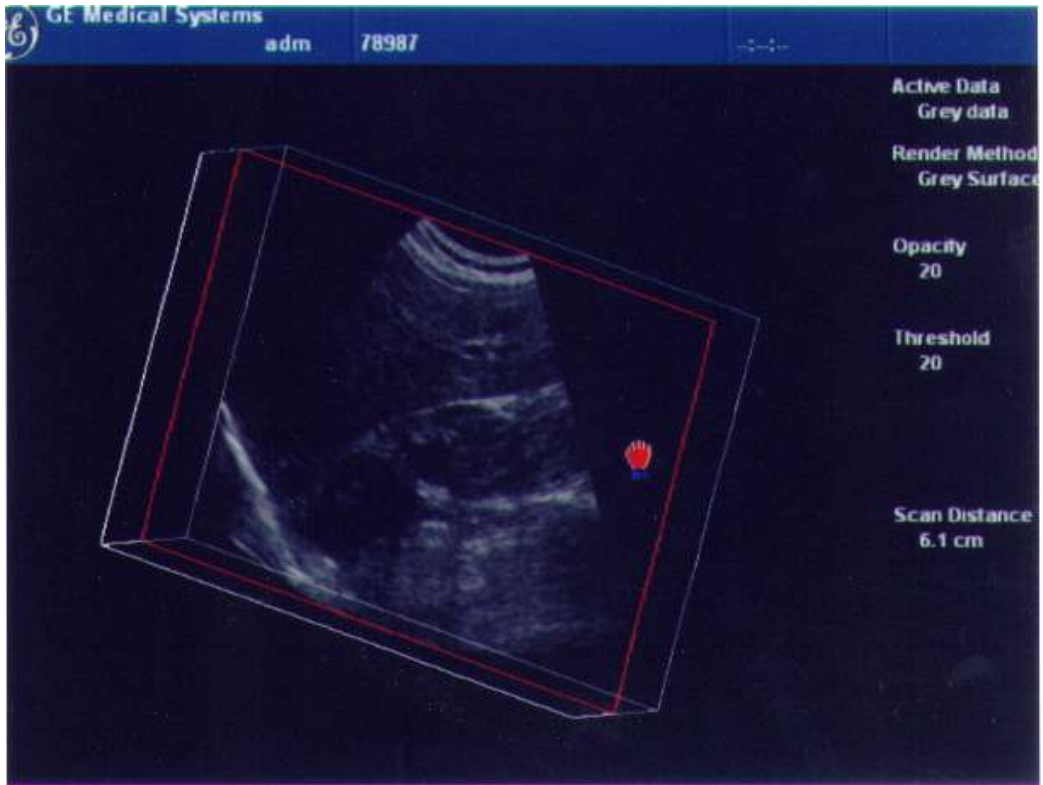


Figure 5-57. Moving through a 3D Volume of Interest (Red Hand)

NOTE: Any plane in the volume can be made active (highlighted with red box) by clicking on it.

Viewing Specific Portions of the Anatomy

You can pull back tissue to view specific portions of anatomy using the yellow hand. Press Set when the yellow pointer finger is positioned on the yellow box. Move the closed yellow hand to manipulate the 3D VOI.

NOTE: *This actually moves an edge. A yellow hand appears only when the pointer is on an edge of the VOI.*

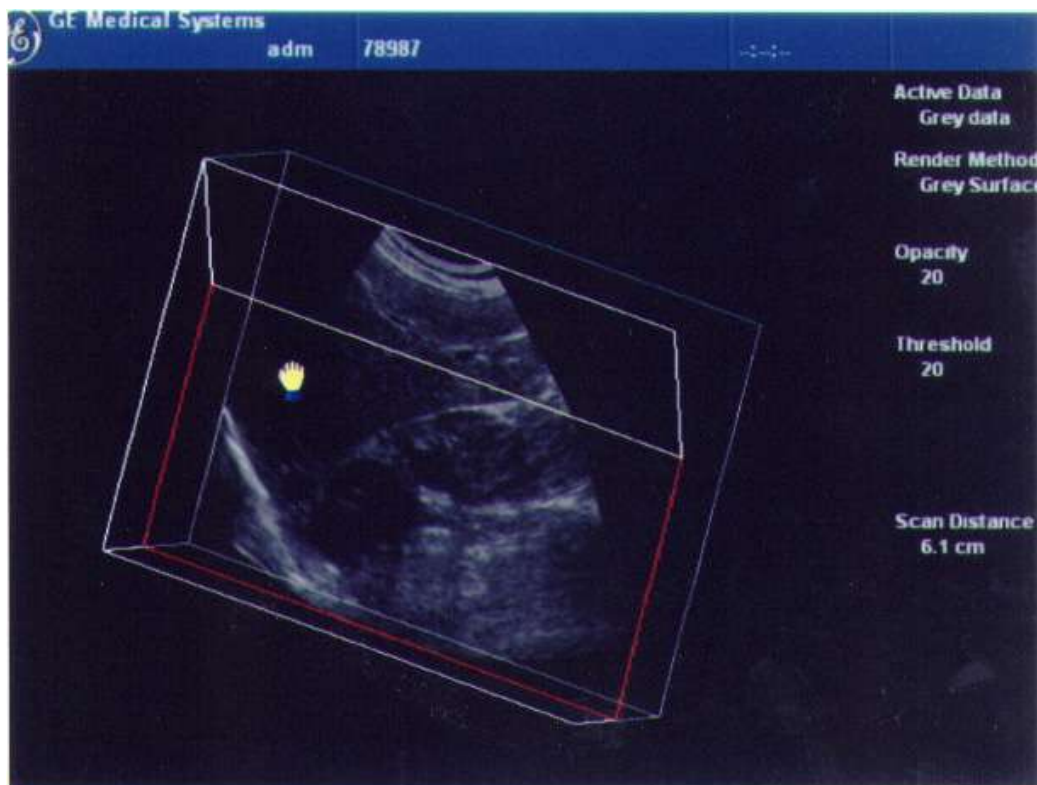


Figure 5-58. Manipulating the Edge of a 3D Volume of Interest (Yellow Hand)

**Pulling Back a
Corner of the VOI
to View Specific
Anatomy**

You can pull back a corner to view specific portions of anatomy using the green hand. Press Set when the green pointer finger is positioned on the green box. Move the closed green hand to manipulate the 3D VOI.

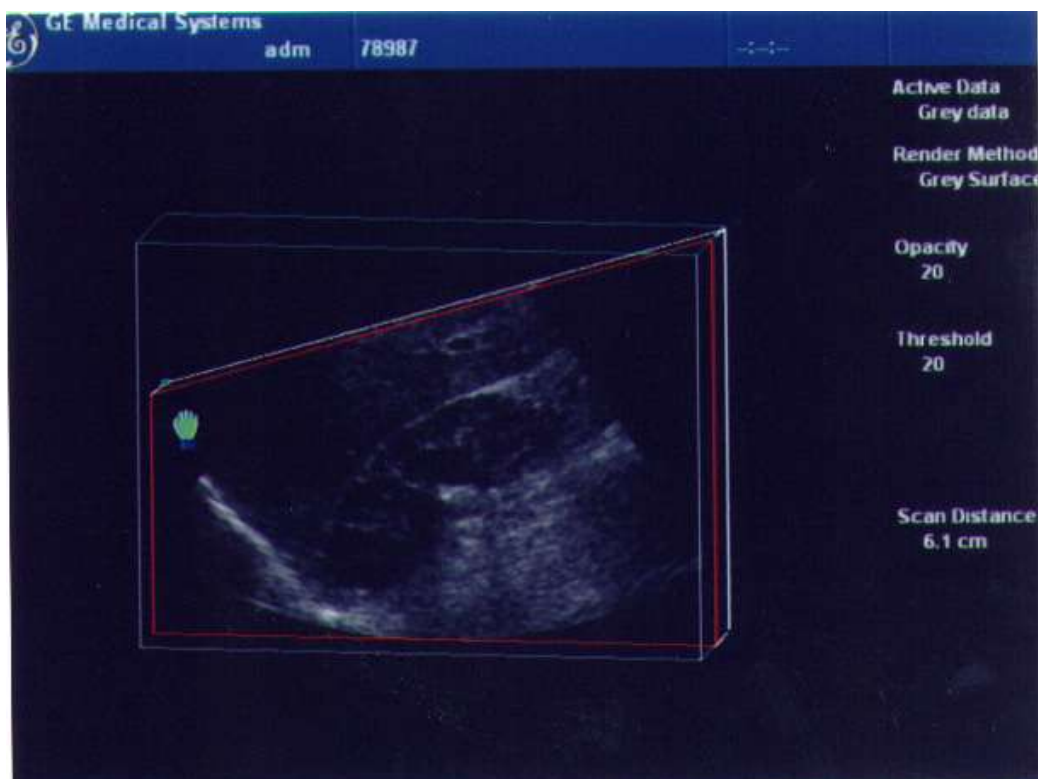


Figure 5-59. Manipulating a Corner of the 3D Volume of Interest (Green Hand)

3D Acquisition Parameter Description



Figure 5-60. 3D Acquisition

Table 5-6: 3D Acquisition Description and Instructions for Use

3D Parameter	Description
Application Presets	Selections: None, OB - Baby Face, Vascular None. No application preset applied. OB - Baby Face. After having scanned in this mode, certain rendering parameters are set automatically. The gray surface mode is activated and the texture mode is switched off. The gray surface mode values for opacity and threshold are set automatically according to the datasets histogram. Vascular. Available only with Advanced 3D or Tru 3D package. After having scanned in this mode, certain rendering parameters are set automatically. The color image is rendered in the texture mode. The values for opacity and threshold of the texture mode are set automatically according to the datasets histogram. The B-Mode image is rendered in the gray surface mode. Opacity and threshold values are defined according to the histogram.

Table 5-6: 3D Acquisition Description and Instructions for Use

3D Parameter	Description
Acquisition Mode	Selections: Sensorless Parallel, Sensorless Sweep, With Sensor Sensorless Parallel. In this mode the probe must be moved during 3D data acquisition without angling it. You should scan the object you want to render in 2-4 seconds. The speed at which you scan should be constant. No sensor is mounted on the probe. • Since the time for post-processing depends on the acquired number of frames, it is recommended that you check the frame rate. Low frame rates result in fewer acquired frames for the 3D dataset which results in intensive post-processing (interpolation). Therefore, low frame rate = long post-processing. Sensorless Sweep. In this mode the probe must be moved to a position where you can clearly see a middle cut of the object you want to scan and render. Tilt the probe to about 30 degrees until the object you want to scan disappears. Start the acquisition and tilt the probe over a distance of around 60 degrees until the object disappears again. The entire scan time should be around 2-4 seconds. During the sweep, the probe may not be moved parallel, just tilted. No sensor is mounted on the probe. With Sensor. Only available if you are using the Tru 3D package. In this mode the probe can be moved in different ways over the object you want to render. The probe can be moved parallel and angled during the same 3D acquisition. The sensor system registers any movement. It is recommended not to move in one direction and back again during one scan. It is also not recommended to turn the probe around its axis. The scanning speed does not have to be constant, but you should not change the velocity too much. In this mode the sensor has to be mounted on the probe. Before starting an acquisition, take care that the transmitter is positioned correctly during data acquisition and that the transmitter cannot move. The transmitter has to be placed in such an orientation that the probe with the receiver attached is always in the forward hemisphere of the transmitter during scanning. If the distance between the receiver and the transmitter exceeds 70 cm, a beep tone signals that the data acquisition has been interrupted. The patient MAY NOT move during the acquisition. The sensor device consists of an electromagnetic field transmitter and a field receiver. The field transmitter generates an alternating spheric electromagnetic field in a strength of up to five times the earth's magnetic field, depending on the distance between the transmitter and the receiver. For 3D data acquisition, the probe with the attached position sensor can be moved freely in an area of 0.7 cm around the transmitter. During data acquisition, the electromagnetic sensor device generates with a frequency of about 100 Hz a set of three translation and three angle values. These values describe the position of the ultrasound probe in space.
Scan Plane	Selections: Front to Back, Side to Side Front to Back. After having scanned in this mode, the rendered dataset is shown in a frontal view. For acquiring a fetal face in sagittal cuts, use this mode. Side to Side. After having scanned in this mode, the rendered dataset is shown from a side view. For acquiring a fetal face in coronal cuts, use this mode.
3D	Starts the rendering process.
Scan Distance	Adjusts the distance covered during the scan. Depending on the real width of a scan acquired during a sensorless 3D acquisition, the volume of interest's width can be enlarged or reduced. You can adapt the form of a fetal face if the baby's head looks oval instead of round. The assumed default width of a parallel scan is 6 cm; of a fan scan 60 degrees.

Easy 3D

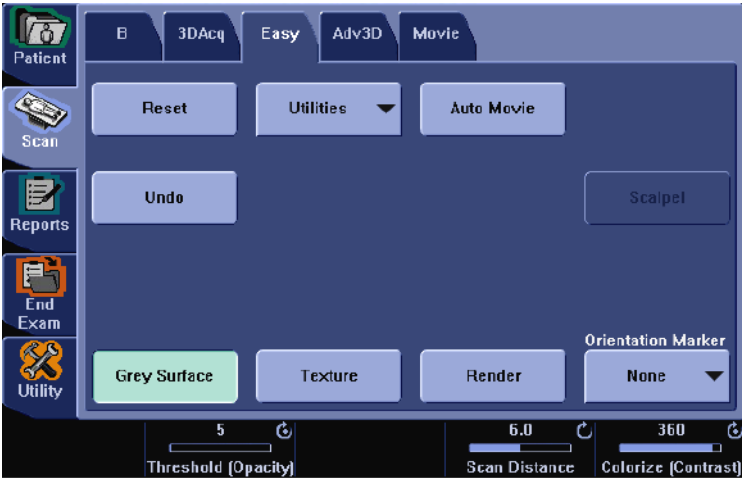


Figure 5-61. Easy 3D Touch Panel

Descriptions and instructions for using Easy 3D follow:

Table 5-7: Easy 3D Description and Instructions for Use

3D Parameter	Description
Reset	Resets the 3D volume of interest back to its original orientation.
Utilities	Select Average Off, Average Light, Average Medium, or Average Strong.
Undo	Undoes any manipulation you have done to your 3D dataset.
Auto Movie	Initializes the calculation and display of a 3D movie. A rotation of 30 degrees left and right around the actual image position (either the default position after acquisition or the position that was manually defined by manipulating the 3D volume of interest) is shown. For this 60 degree rotation, eleven images in steps of 6 degrees are calculated.
Scalpel	Structures, for example a part of the placenta hiding the view to a fetal face, can be cut out in a rendered image. All visible structures can be cut out. The option of 'erase inside' deletes all structures inside the marked region. The option of 'erase outside' deletes all structures outside the marked region. The region in the rendered image is marked with the right Set key. To define the contour of the region, press the right Set key for each vertex. To close the contour, double click the right Set key. As long as a contour is not closed, it can be traced back with the left Set key. The cut out process can be undone by the Undo Last function. As soon as the Apply button is pressed, a new dataset is generated.
Gray Surface	Activates the gray surface rendering mode. It leads to a transparent appearance of the object, generated by displaying only a surrounding shell of structures.
Texture	Activates the texture or photorealistic rendering mode. It creates a photorealistic appearance of the object. The shading depends on the orientation of the surface of the object. If both Texture and Gray Surface mode are switched on, the mixture percentage of both modes can be defined.

Table 5-7: Easy 3D Description and Instructions for Use

3D Parameter	Description
Render	Changes between the rendered image view and the view of a volume of interest. The volume of interest shows the acquired ultrasound images transformed into an isotropic rectangular coordinate system. The volume of interest can be manipulated as described above.
Threshold/Opacity	Threshold defines which gray values are used for rendering and which are considered noise. Opacity defines how strict Threshold is used for discrimination. A low opacity value creates a firmer appearance of the surface. A high opacity value leads to a transparent appearance of the rendered image.
Colorize/Contrast	Colorizes the 3D render or adds contrast to the 3D rendered image.

Advanced 3D

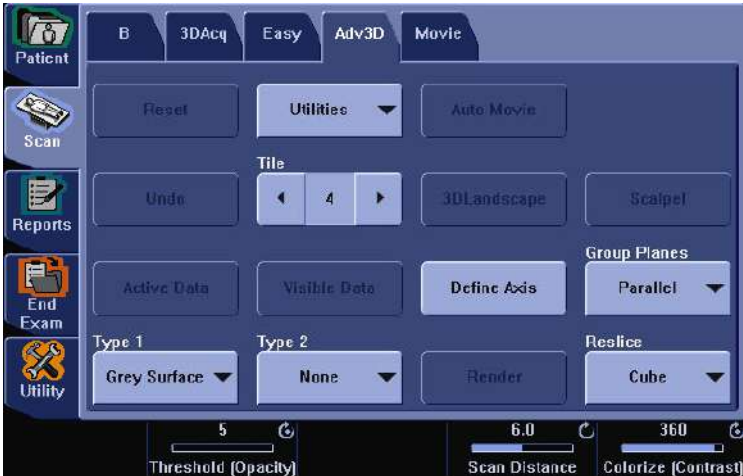


Figure 5-62. Advanced 3D

Descriptions and instructions for using Advanced 3D follow:

Table 5-8: Advanced 3D Description and Instructions for Use

3D Parameter	Description
Tile	The display can be divided into 1, 2, 4, or 6 windows. Switching to a lower number of windows keeps the images from left to right.
3D Landscape	Shows a combination of 2D slices and a 3D rendered image. After a color acquisition you can combine the 2D B-Mode image slices with a 3D rendered color image. This mode allows stepping through the B-Mode images along a vessel structure. The 2D slice can be moved with the right Set key. The Trackball symbol has to be positioned inside the 2D plane.
Active Data	Manipulations of rendering parameters only have an effect on the data defined as Active Data. After having selected Active Data, a list of data is displayed, Gray Data, Red, Blue. Choose the data to be manipulated.
Visible Data	After selecting Visible Data, a list of data is displayed, Gray Data, Red, Blue. Choose the data you want to display. For example, if only color data is chosen, the B-Mode image is switched off in the rendered image and only vessel structures are displayed.
Main	Three orthogonal cuts (with colored frames) of the acquired volume of interest are shown after pressing Main. The volume of interest shows the acquired ultrasound images transformed into an isotropic rectangular coordinate system. On the left top of the image window a complete volume of interest is displayed. It shows the position of the three orthogonal planes in the VOI. A green point displayed in each plane defines the point of intersection of the three planes. This point can be set to different positions in the planes by a double pressing the right Set key. A plane can be moved parallel in the VOI by pressing the right Set key on the position of the green point and moving the Trackball up and down inside the plane.

Tru 3D

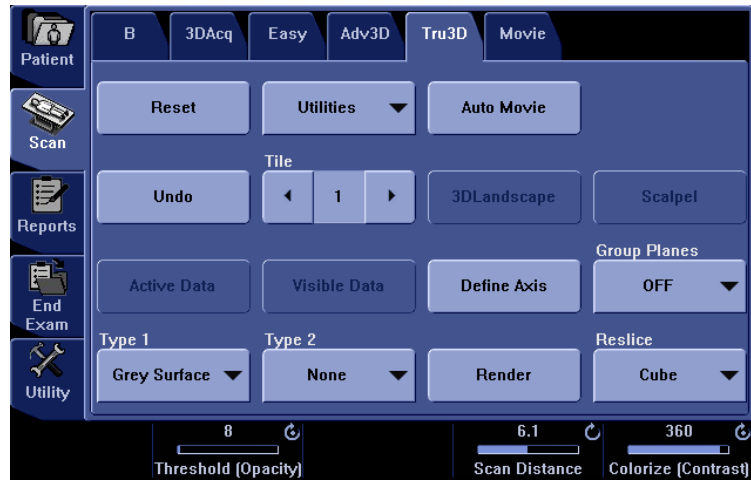


Figure 5-63. Tru3D

Descriptions and instructions for using Tru 3D follow:

Table 5-9: Tru 3D Descriptions and Instructions for Use

3D Parameter	Description
Rotate	The displayed rendered image or non-rendered VOI is rotated around the x or y axis, or 90 degrees right or down.
Define Axis	For certain display and measurement modes (Angular Plane Mode, Angular Volume Measurement Mode), an axis in the volume of interest is required. To define the axis, set the start point by using the Trackball to position one end of the axis and pressing the right Set key, then positioning the other end of the axis and pressing the right Set key.

Table 5-9: Tru 3D Descriptions and Instructions for Use

3D Parameter	Description
Group Plane	Selections: Off, Main, Parallel, Angular Off. A VOI or a rendered image is displayed. The Render button changes between the rendered image view and the view of the VOI. The VOI shows the acquired Ultrasound images transformed into an isotropic rectangular coordinate system. Main. Three orthogonal cuts (with colored frames) of the acquired VOI are displayed after pressing Main. The VOI shows the acquired ultrasound images transformed into an isotropic rectangular coordinate system. On the left top of the image a complete VOI is displayed. It shows the position of the three orthogonal planes in the VOI. A green point displayed in each plane defines the point of intersection of the three planes. This point can be set to different positions in the planes by double clicking the right Set key. A plane can be moved parallel in the VOI by pressing the right Set key on the position of the green point and moving the Trackball up and down inside the plane. Parallel. In this mode all displayed VOIs get the orientation of the last modified cube. Normally four VOIs are displayed. It is possible to display six VOIs by increasing the number of displayed cubes in the Tile area. Between the first and the last VOI, the selected planes are parallel and equidistant. A modification on the plane in one VOI results in a parallel modification of the planes in all other VOIs. Angular. Before starting the Long Axis Rotation Mode, make sure that a long axis has correctly been defined in the VOI (see Define Axis above). The function starts in the long axis display mode. In the upper, left-hand corner a short axis cut is shown which gives an overview of the orientation of the long axis planes. To move these planes, press and hold down the right Set key while moving the Trackball.
Type 1/2	Defines the rendering modes. Selections: Gray Surface, Texture, Maximum Intensity, Minimum Intensity, Average Intensity, and None. If both Type 1 and Type 2 rendering modes are switched on, the mixture of both modes can be defined. Gray Surface. Activates the gray surface rendering mode. It leads to a transparent appearance of the object, generated by displaying only a surrounding shell of anatomical structures. Adjust Threshold and Opacity as well. Texture. Activates the texture of photorealistic rendering mode. It creates a photorealistic appearance of the object. The shading depends on the orientation of the surface of the object. Adjust Threshold and Opacity as well. Maximum Intensity. Transparent appearance of the object. Generated by displaying the maximum gray values in the VOI. Minimum Intensity. The rendered image is generated by displaying the lowest gray values in the VOI that exceed the defined threshold. Dark anatomical structures, like cysts, can be shown in this mode. Average Intensity. Transparent appearance of the object. Generated by a summation of the gray values. None for Type 2. No second rendering mode is used in addition to the Type 1 rendering mode.
Reslice	Selections: Cube, 2D Plane, and 2D Cubic Plane Cube. The VOI shows the acquired ultrasound images transformed into an isotropic rectangular coordinate system. This mode allows you to work simultaneously with six cut planes. 2D Plane. The marked cut planes under Reslice Cube (red border) is displayed without any perspective distortions, e.g., parallel to the screen. 2D Cubic Plane. Only one cut plane view is shown in a perspective displayed VOI. The cut plane can be moved freely without any limitations.

Performing a Sensor Scan

To perform a sensor scan,

1. Attach the sensor cables to the front of the Ultrasound system.
2. Attach the probe bracket to the probe.
3. Insert the receiver into the bracket on the probe.
4. Position the magnet near the patient, within 18 inches of the probe, next to the patient on the bed.
5. Acquire the 3D scan.

NOTE: *3D sensor brackets are available on the 3.5C, M12L, M7c, E8c, 10L, and 4S probes.*

Movie 3D



Figure 5-64. Movie 3D

Descriptions and instructions for using Movie 3D follow

Table 5-10: Movie 3D Descriptions and Instructions for Use

3D Parameter	Description
Manual	An animated rotation of the rendered image can be calculated and displayed by this function. Using this function, you first need to define the start and end position of the rotation. To define this, move the VOI to the start position, the press Define Start. Move the VOI to the end position and press Define End.
Movie 360 Degrees	The calculation and display of a complete rotation around the axis, defined by the Axis button, starts in steps of 15 degrees.
Auto Movie	Initializes the calculation and display of a 3D movie. A rotation of 30 degrees left and right around the actual image position (either the default position after acquisition or the position that was manually defined by manipulating the 3D volume of interest) is shown. For this 60 degree rotation, eleven images in steps of 6 degrees are calculated.
Axis	All rotations (Auto Move and Movie 360) are calculated as rotations around the specified axis (X, Y, or Z).
Movie Speed	You can adjust the speed of any 3D rotation.
Pause	Stops and restarts the rotation. As soon as Pause is pressed, the different rotation steps can be displayed by moving the Trackball.

Volume Measurement

The average measurement error rate for distance measurements is 2% and for volume measurements is 2.3%.

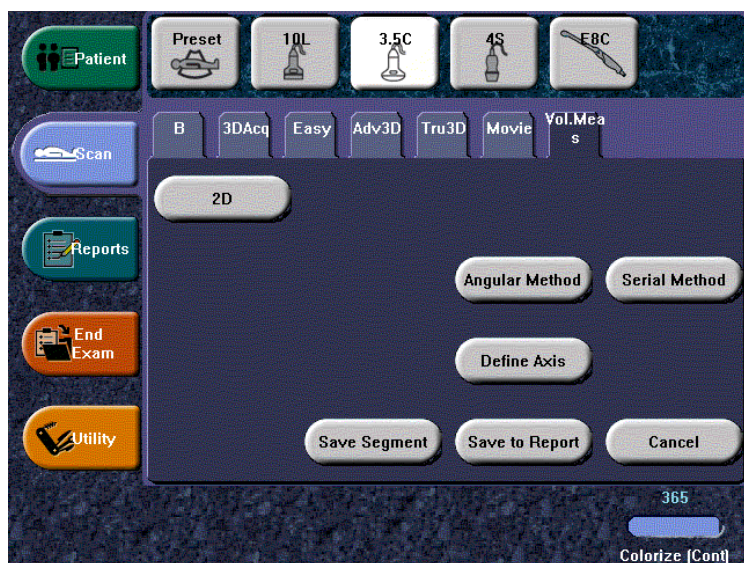


Figure 5-65. 3D Volume Measurement

Table 5-11: Volume Measurement Descriptions and Instructions for Use

3D Parameter	Description
2D	Type of 2D measurement: Distance, Angle, Circumference, Area Distance. Set the start and end distance using the right Set key. Angle. An angle is measured by marking two lines. Press the right Set key once marks the beginning of the first line. Press the right Set key again to mark the end of the first line and simultaneously the beginning of the second line (the intersection of two lines). Press the right Set key a third time to close the angle measurement. Circumference. The circumference of an area is measured by marking the contour of the anatomy by a polygon contour. Each point of the polygon is marked by pressing the right Set key. A double click of the right Set key closes the circumference. As long as the contour is not finished, each point can be deleted by a single press of the left Set key. Area. An area is measured by marking the contour of the anatomy by a polygon contour. The area of an area is measured by marking the contour of the anatomy by a polygon contour. Each point of the polygon is marked by pressing the right Set key. A freehand contour can be drawn by pressing and holding down the right Set key and moving the Trackball. Double click the right Set key to complete the area. As long as the contour is not finished, each point can be deleted by a single press of the left Set key.

Table 5-11: Volume Measurement Descriptions and Instructions for Use

3D Parameter	Description
Angular Method	This function allows you to mark any volume of interest in the dataset to measure its volume or to perform a segmentation of the object. The volume of a 3D object is determined by drawing the contour in several planes, which are rotated around an axis defined by the user. The contours are used to calculate the volume of the object. To determine the volume of an object based on the multiplanar Simpson rule, you need to define the rotation axis via Define Axis. When you press Angular Method, six cut planes are displayed. The Long Axis is marked in blue. You measure the volume by marking the contour of the anatomy. The contour can be marked in different ways: Polygon, Spline, Ellipse, Rectangle, or Rotate. Polygon. Each point of the polygon is marked with a single press of the right Set key. A freehand contour can be drawn by pressing and holding down the right Set key and moving the Trackball. Double click the right Set key to complete the area. As long as the contour is not finished, each point can be deleted by a single press of the left Set key. Spline. An area can be marked by single points positioned by pressing the right Set key. A double click on the right Set key closes the spline. The position of the points defining the contour can be changed by clicking on the point and moving it by pressing the right Set key and using the Trackball. As long as the contour is not finished, each point can be deleted by a single press of the left Set key. Ellipse. When you select this mode, a circle is displayed. You can move the circle by pressing and holding down the right Set key while moving the circle with the Trackball. Press the right Set key to set the chosen position. To manipulate the shape of the circle, move the edges of the circle while pressing the right Set key. Rectangle. When you select this mode, a rectangle is displayed. You can move the rectangle by pressing and holding down the right Set key while moving the rectangle with the Trackball. Press the right Set key to set the chosen position. To manipulate the shape of the rectangle, move the edges of the circle while pressing the right Set key. Rotate. Using the rotate function, you can rotate an area around the Z axis. When you select this function, the Trackball symbol changes as soon as it is positioned to an edge of a region. The region can then be rotated by pressing and holding down the right Set key. A region can be selected and deleted by pressing the Clear key. To save a measurement, press Save Segment, Save to Report, or Cancel.
Serial Method	The Serial Method allows you to mark any volume of interest in the dataset to measure its volume or perform a segmentation of the object. A volume definition is done by defining areas at different depths. Before starting the volume measurement, you need to select a plan showing a cut in which the object can be clearly defined. When you press Serial Method, the display window shows two different views. The left side displays the active plane as a single plane. The VOI on the right side is displayed in cubic mode. In the right VOI select the cut plane position where the measurement process should start. In the left plane, mark the object of interest by designating one of the area definition modes (Curve, Ellipse, Rectangle, Polygon, Rotate). After you have completed defining the first area, the depth of the VOI should be changed on the right side. To change the depth, position the Trackball symbol inside the plane to be moved. press and hold down the right Set key while moving the Trackball backwards. By defining the contour of an object at different depths, its volume can be calculated by summing up the defined slices. To save a measurement, press Save Segment, Save to Report, or Cancel.

Table 5-11: Volume Measurement Descriptions and Instructions for Use

3D Parameter	Description
Save Segment	After measuring a volume, you can use the defined volume for segmentation by pressing Save Segment. Segmentation means that a new dataset is created with voxel information based on the defined volume. A dataset containing only voxels inside the measured volume is created. The original dataset is saved additionally to the segmented data. The segmented data can be chosen in the Active Data or Visible Data list.
Save to Report	After measuring a volume, press Save to Report to register the measurement result in a database that is used for report generation.

Measurement examples are shown below:

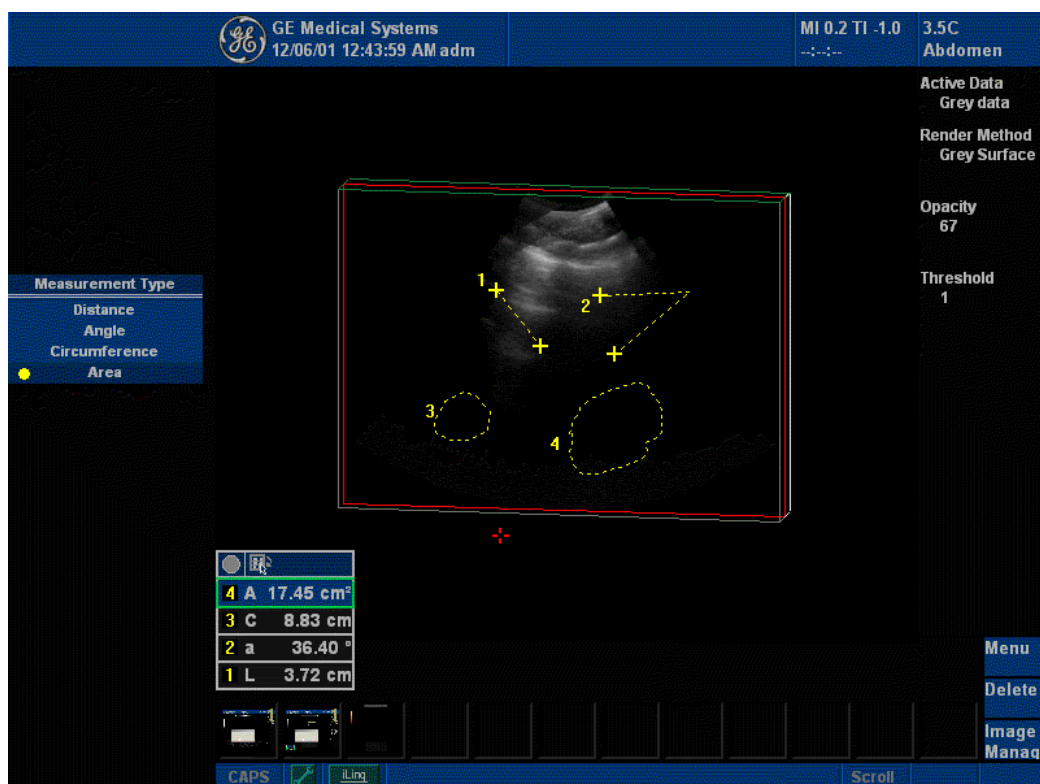


Figure 5-66. Example of 3D Measurements

Volume Measurement (continued)

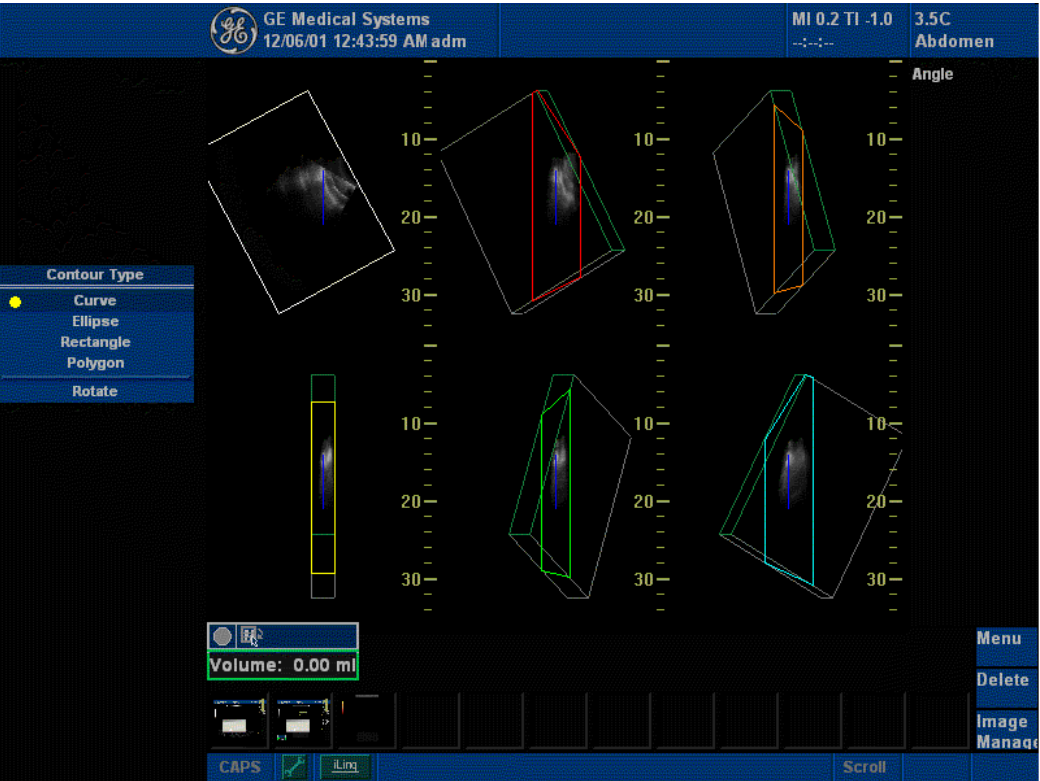


Figure 5-67. Example of Angular Volume Measurement

Volume Measurement (continued)

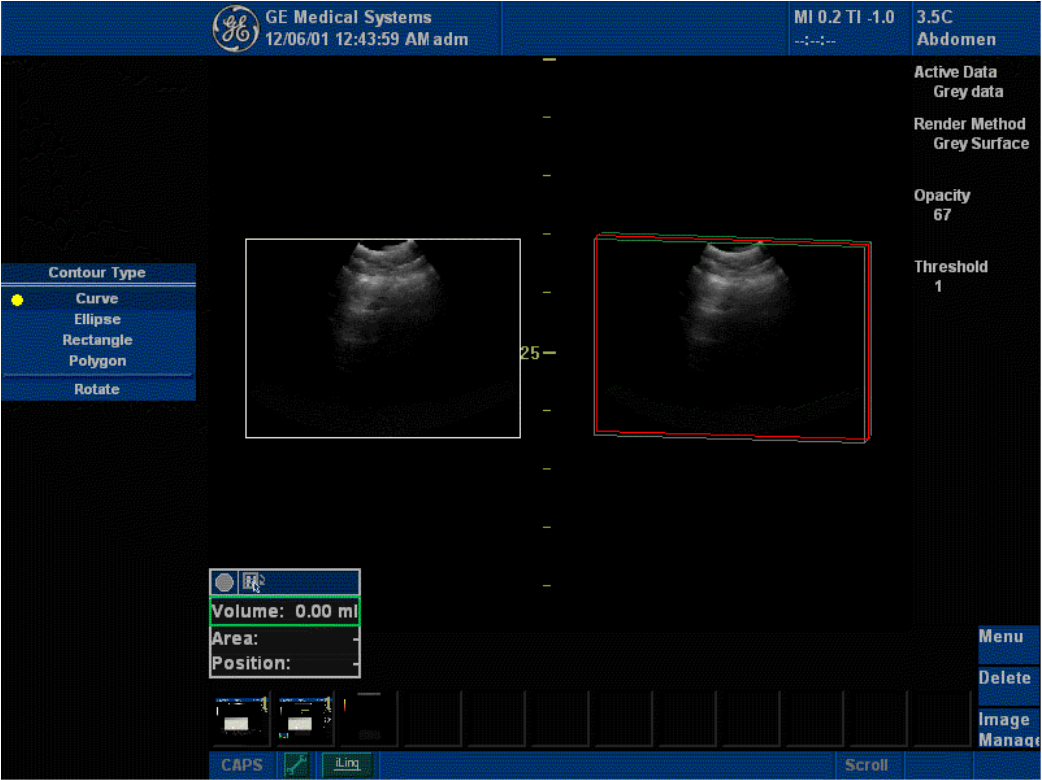


Figure 5-68. Example of Segment Volume Measurement

Using 4D

4D Introduction

4D provides continuous, high volume acquisition of 3D images. 4D adds the dimension of “movement” to a 3D image by providing continuous, real-time displays. With 4D, you can apply rendering techniques to smooth out the appearance of an anatomical structure, for example, a baby’s spine.

You can perform the following types of volume acquisitions within the 4D feature:

Table 5-12: 4D Package Options

4D Type	Description	Acquisition Mode
4D	Designed for continuous volume acquisition of a 3D image.	B, 4D
Static 3D	Designed for single volume acquisition of a 3D image.	B, 3D

Features supported with 4D

The following features are supported with 4D:

- Most B-Mode controls
- Speckle Reduction Imaging (SRI)
- Annotations
- Measurements and Calculations

The following post-processing controls are available with 4D:

- CINE
- Zoom

4D Principles of Operation

The acquisition of volume starts with a 2D image using special probes designed for performing 3D sweeps and 4D scans. The volume box defines the region of interest to be used for the volume sweep.

Volume sweep refers to the range of the sweep of the 2D image to be transformed into a rendered, 3D or 4D image. Static 3D acquisition involves a single volume sweep. 4D involves multiple, continuous volume sweeps.

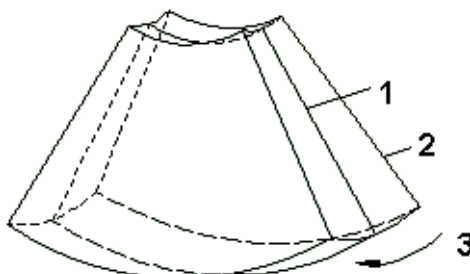


Figure 5-69. Volume Sweep

1. Central 2D scan
2. Start 2D scan
3. Range of VOI sweep

When you initiate a volume sweep, you can adjust the angle of the volume.

What is Interactive 3D Rendering?

Interactive 3D rendering allows you to visualize certain structures and to view and analyze different sections of the volume.

Region of Interest (ROI) / Render Box

The Region of Interest (ROI) - also referred to as the Render Box in rendering - contains the section of the volume you want to render. Therefore, objects that are not inside of the box are not included in the render process and are cut out (this is important in surface mode to allow a free line of sight). This may or might not be the entire VOI.

You can adjust the view direction of the ROI.

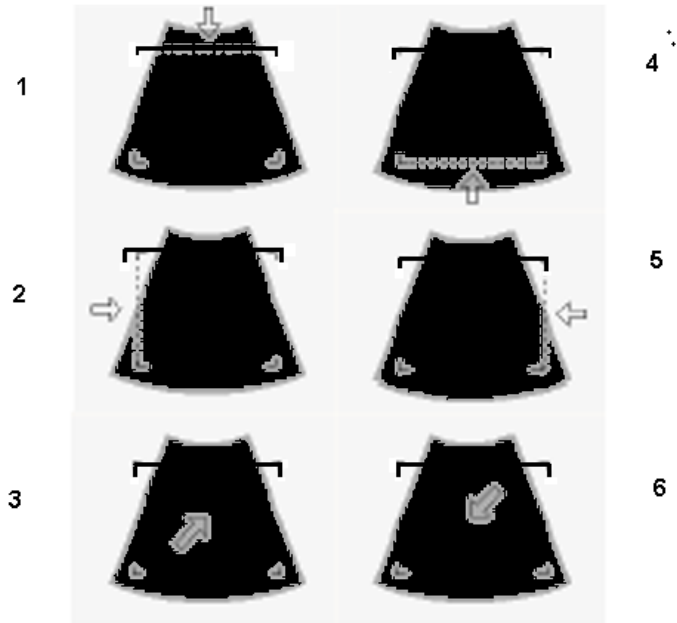


Figure 5-70. Render View Directions

- | | |
|---------------|---------------|
| 1. Up/Down | 4. Down/Up |
| 2. Left/Right | 5. Right/Left |
| 3. Front/Back | 6. Back/Front |

Region of Interest (ROI) / Render Box (continued)

Render View In Render view, only the rendered image displays - no reference images.



Figure 5-71. Monitor Display in Render View - Dual Window Display

Image Orientation

Orientation of Image in Sectional view

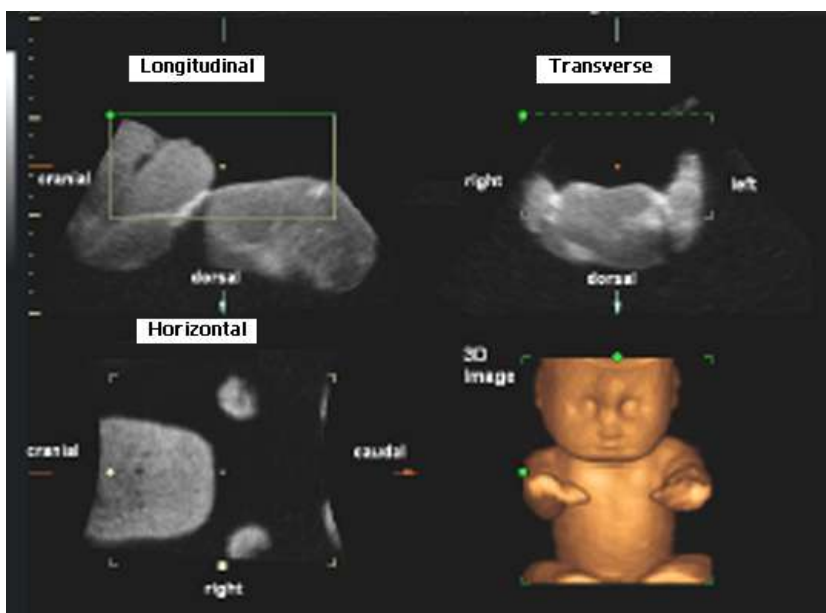


Figure 5-72. Orientation of Image in Sectional View (the resulting image in read mode)

Principle of Sectional Planes

Sectional planes represent three different planes of the same 3D volume. There are three separate planes, A (Longitudinal), B, (Transverse) and C (Coronal).

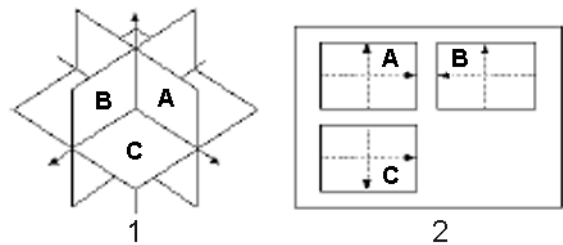


Figure 5-73. Illustration of Sectional Planes

The intersected lines of each plane is color-coded: AB=blue, AC=red, and BC=yellow.

Table 5-13: Orientation of intersected lines

Section/Field	A	B	C
Intersection of line AB	V	V	H
Intersection of line AC	H	N	H
Intersection of line BC	N	V	V
V=Vertical, H=Horizontal, N=Normal			

The presentation of three orthogonal sectional planes is different from the conventional patient orientation in 2D sonography.

NOTE: *Whenever you select the usual, longitudinal section of the patient to display in field A, the conventional orientation for longitudinal and transverse sections is valid.*

Reference Images

Reference images are the individual image displays within the corresponding sectional plane. Reference image A represents sectional plane A; reference image B represents sectional plane B, and reference image C represents sectional plane C.

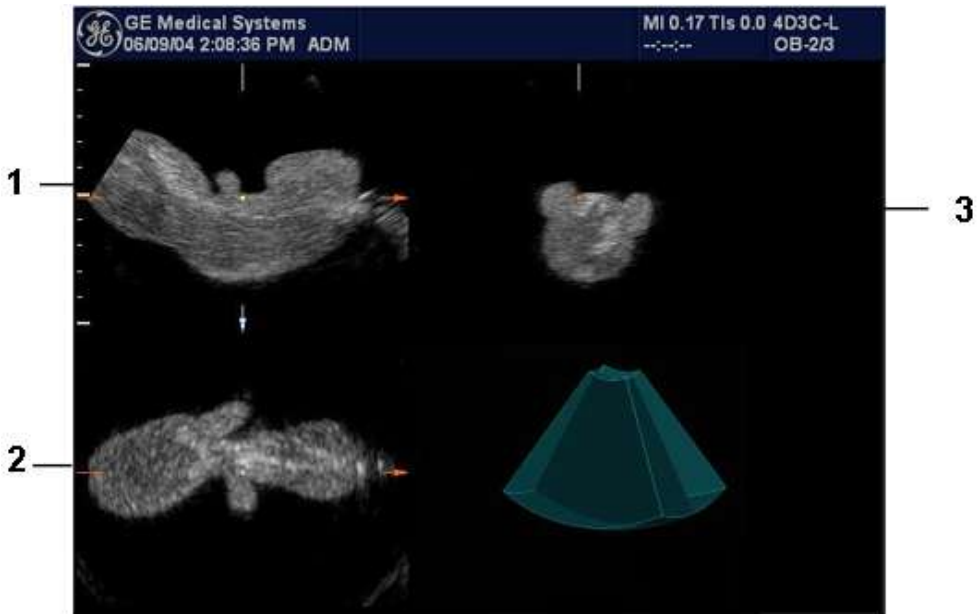


Figure 5-74. Monitor Display of Reference Images in Sectional View

1. Reference Image A (Longitudinal)
2. Reference Image B (Transverse)
3. Reference Image C (Coronal)

Orientation Help. When you view a 4D image on the display, it's sometimes difficult to recognize the orientation. To help, the system displays a three-dimensional drawing to illustrate the orientation. This drawing displays **ONLY** in sectional view.

Reference Images (continued)

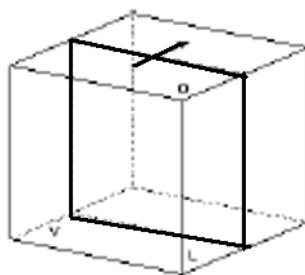


Figure 5-75. Reference Image A

For Reference image A, the transducer beam migrates from the FRONT to the REAR through the volume body.

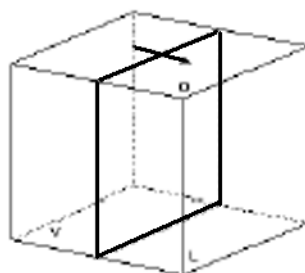


Figure 5-76. Reference Image B

For Reference image B, the transducer beam migrates from the LEFT to the RIGHT through the volume body.

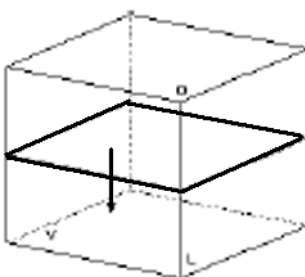


Figure 5-77. Reference Image C

For Reference image C, the transducer beam migrates from the TOP to the BOTTOM through the volume body.

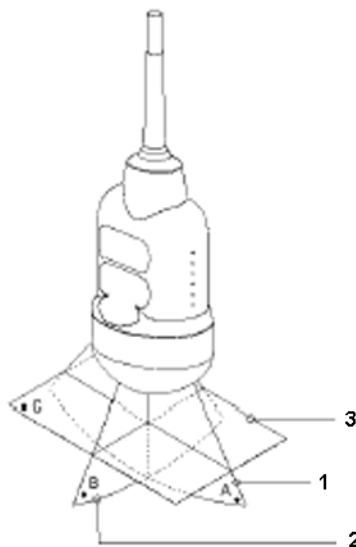
Reference Images (continued)**Examples of Probe
Orientation with
Reference Planes**

Figure 5-78. Abdominal Probe Orientation

1. Image Plane A
2. image Plane B
3. Image Plane C

Reference Images (continued)

Examples of Probe Orientation with Reference Planes



Figure 5-79. Small Part Probe Orientation

1. Image Plane A
2. Image Plane B
3. Image Plane C

Reference Images (continued)

Examples of Probe Orientation with Reference Planes

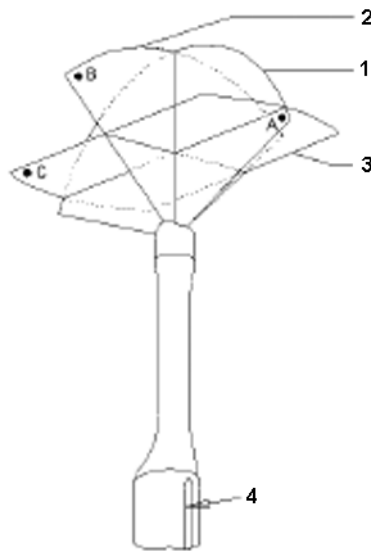


Figure 5-80. Endocavity Probe Orientation

1. Image Plane A
2. Image Plane B
3. Image Plane C
4. Groove

4D Operational Controls

Control Panel Overview

When you enter 3D/4D mode, the behavior of some of the Control Panel buttons changes. For example, in 3D/4D mode, you use the PW-Mode, CF-Mode, M-Mode buttons (along with Depth) to manipulate the Volume of Interest (VOI).

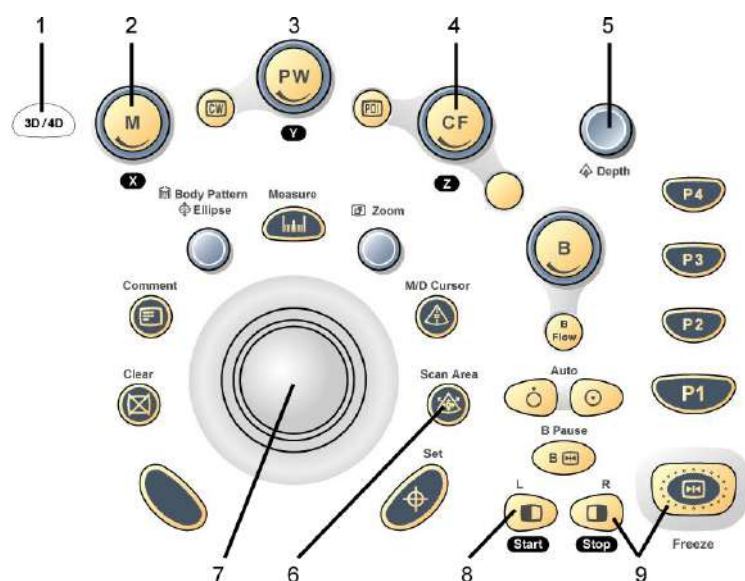


Figure 5-81. Control Panel Buttons

1. 3D/4D key, used to activate/deactivate 3D/4D.
2. M key, M-Mode key used to rotate about the X axis.
3. PW key, PW-Mode key used to rotate about the Y axis.
4. CF Key, CF-Mode key used to rotate about the Z axis.
5. Depth key, used to cut through the image to view deeper slices.
6. Scan Area key, used to adjust the size and position of the VOI.
7. Trackball, used to move the VOI.
8. L (left) key, used to begin a 4D acquisition.
9. Freeze or R (right) key, used to freeze a 4D image.

4D Monitor Display

Below is an example of a 4D display. Imaging parameters are displayed in the upper right-hand portion of the display. The 4D specific parameters are Quality (Q), Volume Angle (A) and Volume Rate (VR). The Status Bar contains instructions on the tasks you can perform at each stage of the 4D imaging process. Remember to take a look at the Status Bar as needed.



Figure 5-82. 4D Monitor Display

4D Touch Panel Overview

The following is the first Touch Panel that appears when you press 3D/4D.

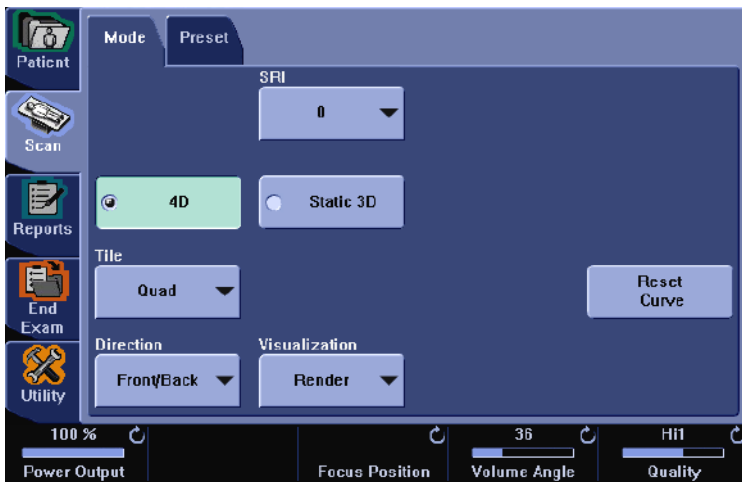


Figure 5-83. First Touch Panel Displayed When Entering 3D/4D Mode

Common 4D Touch Panel Controls

Most 4D Touch Panel screens contain some similar controls. Refer to the table below for descriptions of these controls. Controls that are unique to or that contain slightly different functionality are described in their respective sections.

Table 5-14: Common 4D Touch Panel Controls

Preset Parameter	Description
SRI	Allows you to apply SRI (Speckle Reduction Imaging) to the selected reference image.
Tile	You can divide the display into 1, 2, or 4 windows for Render view (Render = On) and 1 or 4 windows for Sectional view (Render = Off).
Reset Curve	Resets the three-point curve to a straight line.
Direction	Adjusts the view direction of the ROI.
Visualization	Sectional, Render, VCI-Static, or Tomographic Ultrasound Imaging (TUI). Render view displays one rendered image, or reference image(s) and rendered image.
Volume Angle	Sets the range of the volume sweep.
Quality	Balances speed with line density. Hi2 combines the highest density with the slowest speed; Low combines the lowest density with the highest speed.

4D Presets

Real-Time 4D/Static 3D Presets

1. When you enter 3D/4D mode, press the **Preset** tab.



Figure 5-84. Real-Time 4D/Static 3D Presets

2. Select one of the preset settings for data acquisition and display. Presets are defined in the preset file and differ by application.

Table 5-15: Common 4D Touch Panel Controls

Preset Parameter	Description
Save	Selections: Overwrite, Create New, Cancel. Overwrite . Overwrite the application preset file with the changes you just made. Create New . Create a new user application preset file based upon the current exam category and application. Cancel . Cancel without saving preset parameters.
Abdomen, Obstetric, Gyn, Surface, Skeleton, Default, Peri. Vasc., Small Part, Contrast	Reloads the presets for the selected application.
User1, User2, User3, User4	Used to define new user presets for a given application.

Static 3D Presets

1. When you enter 3D/4D mode, press **Static 3D**, then the **Preset** tab.



Figure 5-85. Static 3D Presets

2. Select one of the preset settings for data acquisition and display. Presets are defined in the preset file and differ by application.

Performing a 4D Scan

Visualizations

4D provides two types of views for displaying and working with images: Sectional and Render.

Sectional View Sectional view contains one display for each sectional plane.

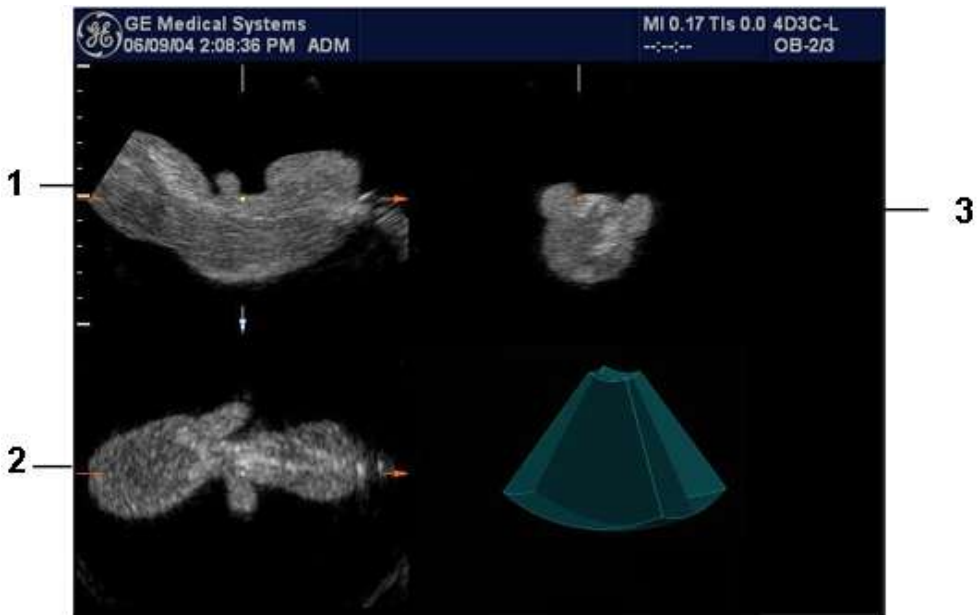


Figure 5-86. Monitor Display in Sectional View

1. Sectional Image A
2. Sectional Image B
3. Sectional Image C

Visualizations (continued)

Render View The LOGIQ 9 continuously displays the 4D rendered image.



Figure 5-87. Monitor Display in Single Tile Render View

NOTE: When the tile selection is single, only the rendered 4D image appears. When the tile selection is quad, the sectional images are located in 3 quadrants with the rendered 4D image in the fourth.

Render View (continued)



Figure 5-88. Quad Tile Render View

Orientation Help

When you view a 4D image on the display monitor, it's sometimes difficult to recognize the orientation. To help, the system displays a three-dimensional drawing to illustrate the orientation. This drawing displays **ONLY** in sectional view.

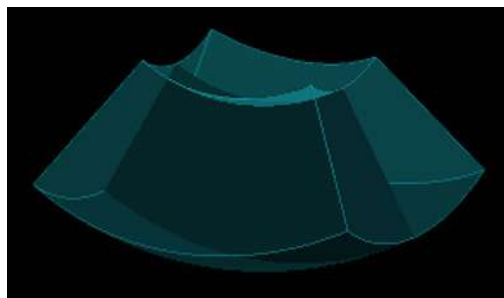


Figure 5-89. Orientation Help Graphic

Acquiring and Rendering a 4D VOI

Starting with a 2D Image

To create a 4D image, you start with an optimized 2D image. The 2D image serves as the mid-line for the resulting 4D image.

1. Connect the appropriate 4D-compatible probe, leaving the probes in their respective holders. Follow the guidelines in Chapter 3 for connecting probes.

NOTE: *If the appropriate 4D probe is not connected, the original 3D Touch Panel appears.*

2. Obtain a 2D image. Optimize the image as usual.

Entering 3D/4D Mode

In 3D/4D mode, you choose the type of scan you want to perform: 4D or Static 3D.

1. Press **3D/4D** to enter 3D/4D mode. The first time you press 3D/4D, the system is in B Pre mode.

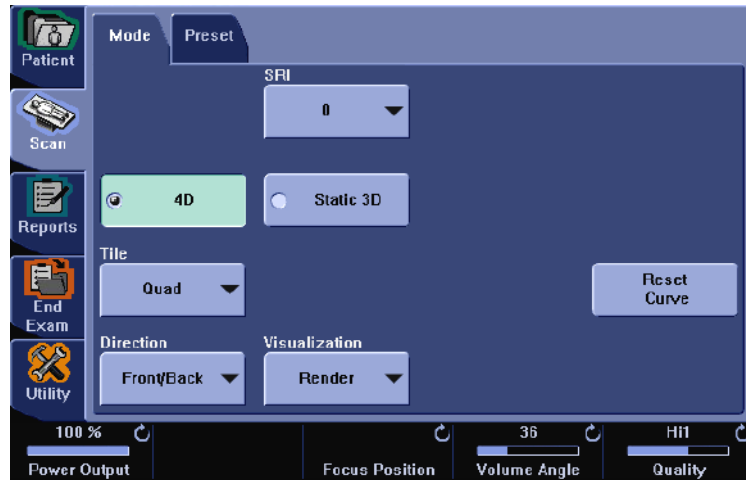


Figure 5-90. First Touch Panel Displayed When Entering 3D/4D Mode

NOTE: *The location of the number of focal zones might change when you enter 3D/4D mode, since the number of zones is pre-determined by the default ROI.*

Entering 3D/4D Mode (continued)

The default acquisition mode varies by application. If you are in OB, the default acquisition mode is Real-Time 4D; for all other applications, Static 3D is the default acquisition mode. When you enter pre-mode, an ROI graphic appears on the monitor display that defines the initial ROI (Region of Interest) of the volume.

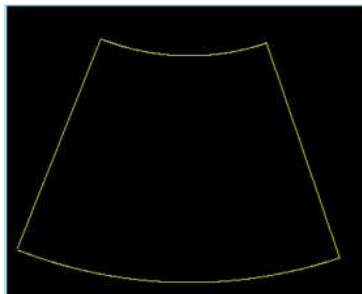


Figure 5-91. ROI Graphic

2. Press the **Preset** tab. Select one of the preset settings for data acquisition and display. Presets are defined in the preset file and differ by application.

Quick Acquisition Steps

1. Connect the appropriate 4D-compatible probe, leaving the probes in their respective holders. Follow the guidelines in Chapter 3 for connecting probes.
2. Select a 4D probe from the probe indicator.
3. Obtain a 2D image. Optimize the image as usual.
4. Press **3D/4D**. An ROI graphic appears. 4D is selected.
5. Define the Volume of Interest (VOI) to be scanned. Use the **Trackball** to move the VOI and the **Scan Area** button to re-size and re-position the VOI. Only the area defined within the VOI is rendered.
6. Adjust the volume angle. This defines the range of the volume sweep. A small sweep angle results in a lower number of slices with a high volume rate.
7. To begin 4D acquisition, press the **L** key.

You DO NOT have to hold the probe steady during data acquisition.

During data acquisition, you can manipulate the VOI to see different views of the image. To rotate the VOI left or right, use the **PW** control. To rotate the VOI forward or backward, use the **CF** control. To rotate the VOI in a circular motion, use the **M** control.

To return to 3D/4D pre-mode, press **L**.

NOTE:

If the volume size is too large, the message "Volume Size Too Big - Quality Degraded" displays in the status bar. The system changes the quality automatically to below the upper limit and displays the proper Quality value in the information window.

8. Set Render to On.
9. To complete the acquisition, press **Freeze** or **R**.
10. Store the image.

4D

4D provides continuous, high volume acquisition of 3D images. You can apply rendering techniques to smooth out the appearance of an anatomical structure, for example, a baby face.

4D imaging contains two main viewing modes: Sectional and Render. Sectional mode displays three separate representations of the image: Longitudinal (original 2D image), Transverse (elevational), and Coronal (horizontal). Render mode displays one rendered 4D image.

Acquiring a 4D Volume of Interest (VOI)

Once you have acquired an optimized 2D image, you can perform a 4D scan to acquire the 4D image.

During 4D image acquisition:

- Frame Averaging is disabled.
- You cannot change the transmit frequency.
- To change the position of the focal zone, adjust the depth of the VOI.
- You cannot change the number of focal zones.

Acquiring a 4D Volume of Interest (VOI) (continued)

To acquire a 4D VOI:

1. Press **4D**.
2. Make sure the VOI is defined appropriately. If necessary, adjust the volume angle. This defines the range of the volume sweep. A small sweep angle results in a lower number of slices with a higher volume rate.
See Manipulating the Volume of Interest (VOI) for more information.
3. To begin 4D acquisition, press the **L** key. The system will perform continuous sweeps across the VOI. You do not have to hold the probe steady during a 4D scan.

To return to 4D pre-mode, press **L**.

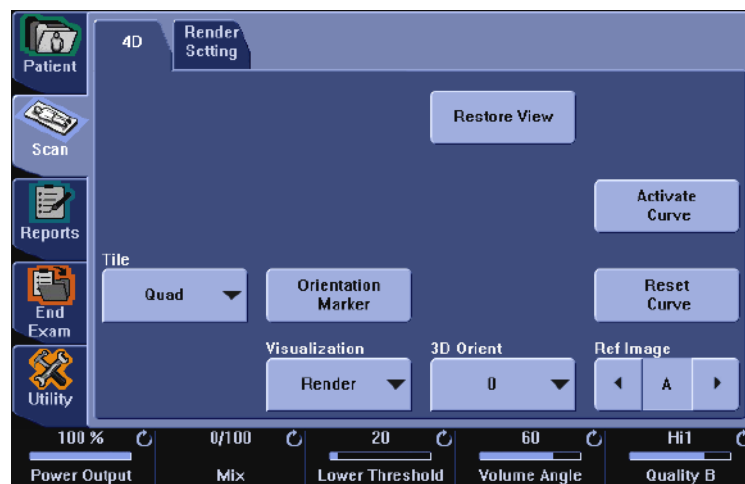


Figure 5-92. 4D Tab - Data Acquisition

4. Set the Render to On.

Sectional VOI Acquisition

Sectional view provides three separate views of the same image: Longitudinal (original image), Transverse (elevational), and Coronal (horizontal).

- 1. In the 4D tab, Render defaults to On (Render mode). Change Render to Off for Sectional view.
- 2. To select a reference image, use the Ref Image control on the Touch Panel. The reference image selected contains the focus for control panel keys, allowing you to manipulate or optimize that image.

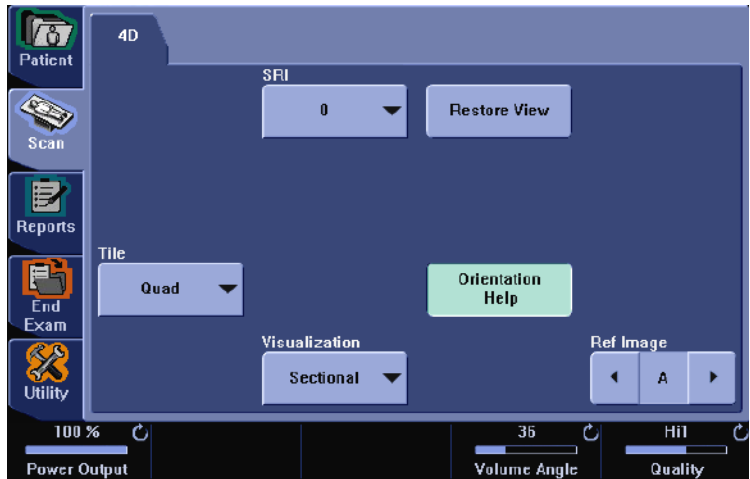


Figure 5-93. 4D Data Acquisition Sectional View

Table 5-16: 4D Data Acquisition Parameters

4D Parameter	Description
SRI HD	Allows you to apply SRI HD (Speckle Reduction Imaging) to the selected reference image. Values: 0-6 (0=Off). See the SRI (Speckle Reduction Imaging) section for more information.
Tile	Selections: Single, Quad. You can divide the display into 1 or 4 windows.
Reset Curve	Resets all parameters back to the original values or chosen presets.
Direction	Up/Down
Visualization	Sectional, Render, VCI-Static, or Tomographic Ultrasound Imaging (TUI). Render view displays one rendered image, or reference image(s) and rendered image.
Volume Angle	Sets the range of the volume sweep.
Quality B	Selections: Hi2, Hi1, Mid2, Mid1, Low. Used to balance speed with line density. Hi2 combines the highest density with the slowest speed. Low combines the lowest density with the highest speed.

Render VOI Acquisition

Rendering allows you to distinguish subtle anatomical detail. You can render all areas of an VOI, or just certain regions of the VOI. The region you define for rendering is referred to as the Render Box.

1. Define the area you want to render. For example, if you have an image of an entire fetus, you might only want the fetal face to be rendered. Therefore, you would define the fetal face as the VOI.
2. Set Render to On.



Figure 5-94. Example of Rendered 4D Image

Render VOI Acquisition (continued)

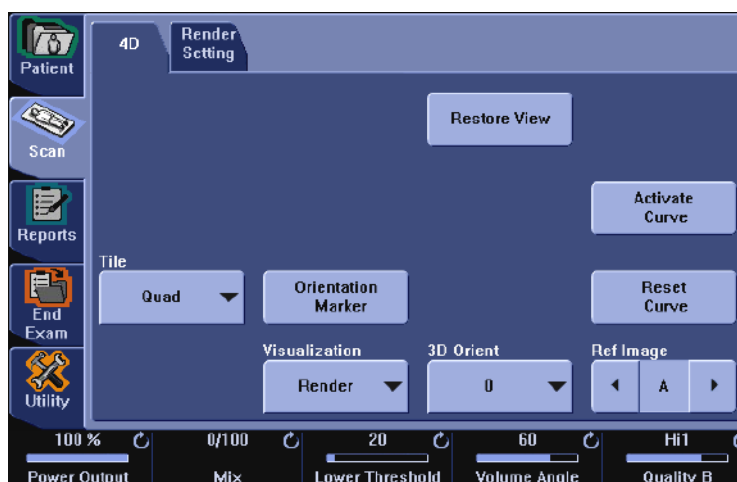


Figure 5-95. 4D Tab (Data Acquisition) Render View

Table 5-17: 4D Data Acquisition Parameters - Render Mode

4D Parameter	Description
SRI	Allows you to apply SRI (Speckle Reduction Imaging) to the selected reference image. Values: 0-6 (0=Off). See the SRI (Speckle Reduction Imaging) section for more information.
Restore View	Select to reset all parameters back to the original values or chosen presets.
Tile	Selections: Single, Dual, Quad You can divide the display into 1, 2, or 4 windows.
Visualization	Sectional, Render, VCI-Static, or Tomographic Ultrasound Imaging (TUI). Render view displays one rendered image, or reference image(s) and rendered image.
3D Orient	When selected, changes the orientation of the image on the monitor display. Selections include: 0 degrees, 90 degrees, 180 degrees, and 270 degrees.
Ref Image	Use to select the reference image that has focus for use with the control panel keys and Trackball . This control is enabled only if Tile is set to Quad.
Volume Angle	Sets the range of the volume sweep.
Quality	Selections: Hi2, Hi1, Mid2, Mid1, Low. Used to balance speed with line density. Hi2 combines the highest density with the slowest speed. Low combines the lowest density with the highest speed.
Activate Curve	Define a three-point curved surface for the render window using the Trackball.
Reset Curve	Reset the three-point curve to a straight line.
Mix	Selections: 0-100% in increments of 2. Allows you to mix a Rend Mode 1 mode with a Rend Mode 2 mode. Always select two modes.
Lower Threshold	Selections: 0-255. Sets a lower threshold below which weaker echoes are removed.

Render VOI Acquisition (continued)

1. Select the **Render Setting** tab.

The Render Setting tab allows you select and combine gray-scale and color rendering modes.



HINTS

If you are using Surface modes, we recommend that you adjust the Lower Threshold to recognize border structures more clearly.

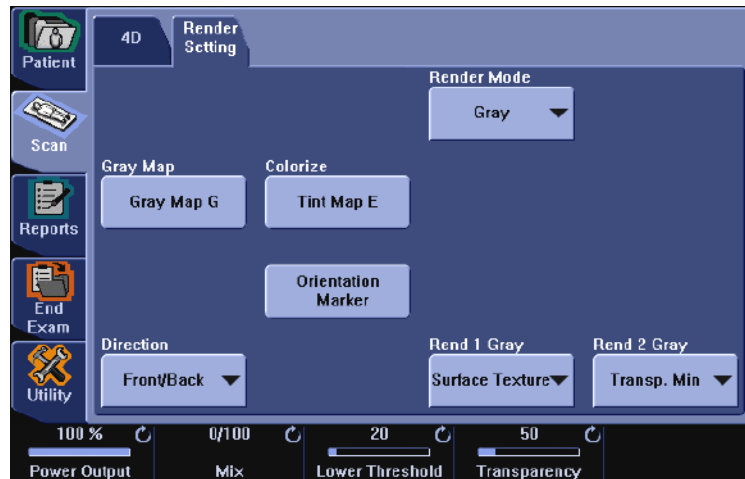


Figure 5-96. Render Setting Tab (4D Data Acquisition)

Table 5-18: 4D (Data Acquisition) Render Parameters

4D Parameter	Description
Direction	The ROI determines the region that is rendered during 4D acquisition. You can change the direction in which this ROI is viewed. Selections: Up/Down, Down Up, Left/Right, Right/Left, Front/Back, Back/Front.
Gray Map	Displays the gray map selections on the display monitor. Select maps using the Trackball.
Colorize	Displays the tint map selections on the display monitor. Select maps using the Trackball.
Render Mode	Select Gray or Inversion. If you select Inversion, inverts the gray values of the rendered image (e.g., image information that was black becomes white and vice versa).

Table 5-18: 4D (Data Acquisition) Render Parameters (Continued)

4D Parameter	Description
Render 1 / Render 2	Allows you to combine render mode values from render mode 1 and render mode 2. Select the render map combination from the upper-left portion of the monitor display. Select map combinations using the Trackball. Render Mode 1 Selections: Surface Smooth, Surface Texture, Transp Max, Transp X-ray, Transp Min. Surface Smooth - Surface displays in a smoothed texture mode, which means that the gray values of the surface are identical with the gray values of the original 2D scan. Surface Texture - Surface displays in texture mode, which means that the gray values of the surface are identical with the gray values of the original 2D scan. Transp Max. - Displays the maximum intensity of gray values in the ROI. This is helpful for viewing bony structures. Transp X-Ray - Displays all gray values in the ROI. Trans Min. - Displays the minimum number of gray values in the ROI. This is helpful for viewing vessels and hollow structures. Render Mode 2 Selections: Surface Smooth, Transp Max, Light, Transp X-ray, Transp Min, Gradient Light. Surface Smooth - Surface displays in a smoothed texture mode, which means that the gray values of the surface are identical with the gray values of the original 2D scan. Light - Surface displays in light mode. Structures in the near field are brighter; structures in the far field are darker. Gradient Light - Surface displays as if it is illuminated from a spot light source. This is helpful if the displayed surface is surrounded by hypoechoic structures (for example, liquids). Transp Max. - Displays the maximum intensity of gray values in the ROI. This is helpful for viewing bony structures. Transp X-ray - Displays all gray values in the ROI. Transp Min. - Displays the minimum number of gray values in the ROI. This is helpful for viewing vessels and hollow structures.
Transparency	Selections: 20 to 250. Sets the transparency of the image. The higher the number, the more transparent the gray scale information.

**Tomographic
Ultrasound
Imaging (TUI)**

Tomographic Ultrasound Imaging (TUI) is a visualization mode which presents data as parallel slices (planes) through the dataset. This method of visualization is consistent with CT and MRI. The distance between the different planes can be adjusted.

1. Select TUI as the Visualization mode.
2. Press 'L' to start acquisition.
3. If in 4D, press 'R' to end the acquisition. This step is not required in Static 3D.

The reference image + the number of specified slices appears. The reference image always displays and indicates which slices you are currently viewing as solid lines.

4. Adjust the number of slices and slice distance.
You can adjust the number of slices by using the Slices rotary. You can adjust the distance between the slices using the Slice Distance rotary.
5. Move forward/backward through the slices via Prev/Next Slice.

Press Adjust Slice on the Touch Panel. Select a slice by using the Trackball, then press Set. This slice is marked with an "X". To print this slice, press a print key.

Anatomical Markers in 4D

Anatomical markers are available in 4D.

To add an anatomical marker in 4D,

- Rotate the Body Pattern/Ellipse key. A person pattern overlays the 4th quadrant on the display.

To change the person pattern,

- Press the Touch Panel key. From left to right, select either Transrectal, Neonatal Head, Face (down), or Face (up).

To rotate the person pattern,

- Rotate the Body Rotation Touch Panel key (0, 45, 90, 135, 180, 225, 270, or 315 degrees).

To tilt the probe,

- Rotate the Probe Tilt Touch Panel key (0, 45, 90).

To align the marker with the probe's orientation marker.

- Rotate the Body Pattern/Ellipse key.

To position the marker on the anatomy,

- Move the Trackball to position the marker at the appropriate anatomy.

The display is also annotated:

- Cr = Cranial
- Ca = Caudal
- A = Anterior
- P = Posterior
- L = Left
- R = Right

Manipulating the Volume of Interest (VOI)

Imagine you are able to manipulate the 4D volume of interest (VOI) in your hand. The 3D/4D ROI is a tangible anatomical object that you can see and manipulate easily using the **Trackball** and other control panel keys.

If the monitor display is in Sectional view, select the desired reference image before you manipulate the image.

Rotating the 4D VOI Left/Right or Forward/Backward

You can rotate it left to right or right to left. You can rotate it forward/backward. To rotate left, turn the **PW** control clockwise; to rotate right, turn the **PW** control counter-clockwise. To rotate the image forward, turn the **CF** control counter-clockwise; to rotate the image backward, turn the **CF** control clockwise.

To speed up the rotation, press the **PW** or **CF** controls before you turn them. To slow down the rotation, press **PW** or **CF** again.

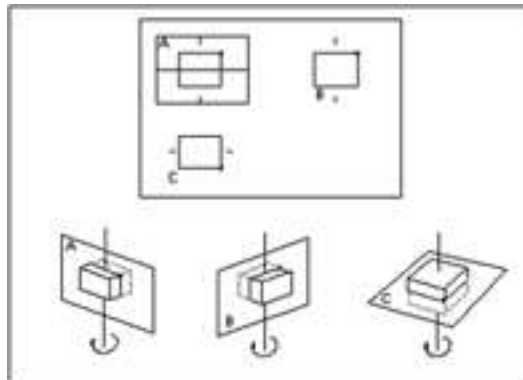


Figure 5-97. Clockwise turn of the PW Control

Manipulating the Volume of Interest (VOI) (continued)

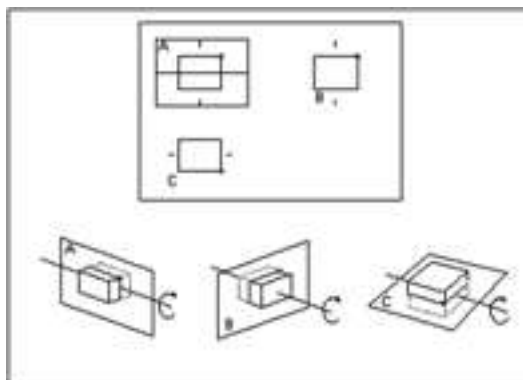


Figure 5-98. Clockwise turn of the CF Control

Rotating the 4D Image in a Circular Motion

You can rotate the VOI in a circular motion, moving left or right. To rotate left, turn the **M** control counter-clockwise; to rotate right, turn the M control clockwise.

To speed up the rotation, press the **M** key before you turn it. To slow down the rotation, press **M** again.

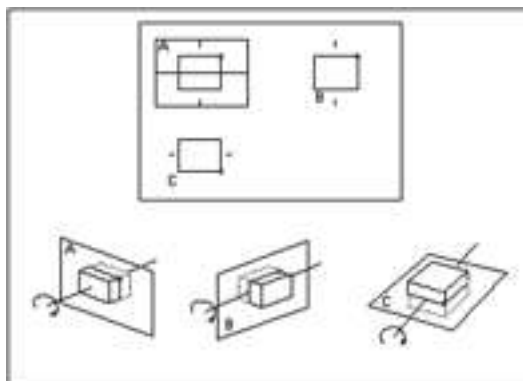


Figure 5-99. Clockwise turn of the M Control

Manipulating the Volume of Interest (VOI) (continued)

Moving through the VOI

To move through the image to view a particular slice, press **Depth**.

This allows a displacement of the center of rotation along the intersection lines of the sectional planes A, B, and C. The displacement of the center of rotation leads to the display of parallel sectional images. See 'Reference Images' on *page 5-153 for more information*.

Zooming the Image

Press **Zoom** to zoom on the image.

Moving the VOI Position

To move the position of the VOI, move the **Trackball** left, right, up and down, as needed.

Resizing the VOI

To re-size the VOI, use the **Scan Area** control panel button. See the Scan Area section in this chapter.

Stopping 4D Image Acquisition

To stop acquiring a 4D image, press **Freeze** or R if you are in Render view or just **Freeze** if you are in Sectional view.

4D VOI Post-Processing

When you press **Freeze** or **R**, one of the following Touch Panel displays, depending on whether you are in Render view or Sectional view.

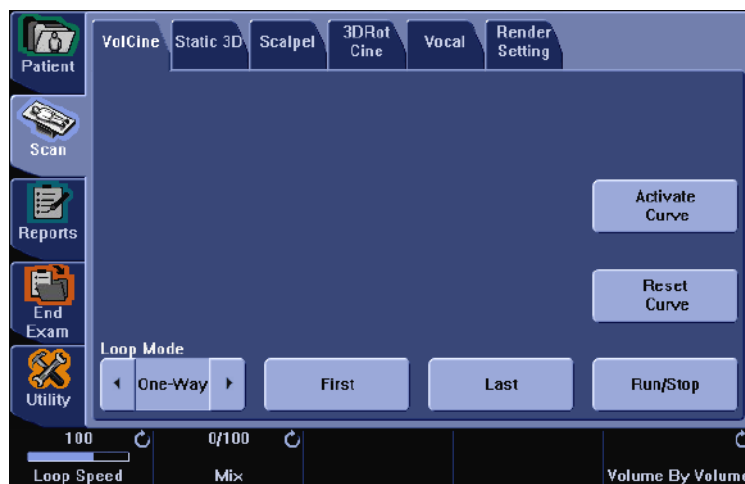


Figure 5-100. 4D (After Acquisition) Touch Panel - Render View



Figure 5-101. 4D (After Acquisition) Touch Panel - Sectional View

Volume CINE The system constantly stores CINE images so you can play back and review those images. CINE is useful for focusing on images during the specific part of the heart cycle or to view short segments of a scan session.

To activate CINE in 4D:

1. Press **Freeze**.
2. Select the **VolCine** tab.

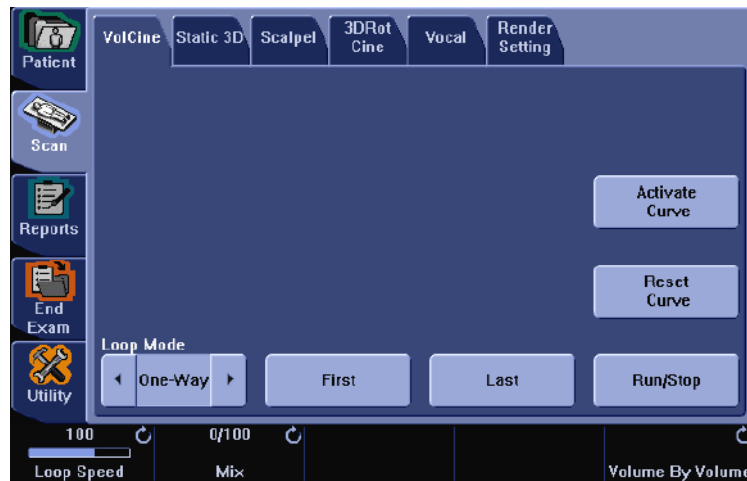


Figure 5-102. 4D Volume Cine

Table 5-19: 4D Cine Parameters

Preset Parameter	Description
Loop Mode	Selections include: One Way, BiDirectional (two-way). One Way - plays one loop sequence forward. BiDirectional - plays the sequence forward and backward.
First	Displays the first volume in the CINE loop.
Last	Displays the last volume in the CINE loop.
Run/Stop	Starts and stops the CINE loop.
Loop Speed	Values include: 10% - 200%, in increments of 10. Adjust the loop speed of the CINE.
Volume by Volume	Used to select an individual volume in the CINE loop.

Volume CINE (continued)

3. If you were in Render visualization mode when you entered 4D CINE mode, you can also press **L** to start and stop the CINE loop.

If you were in Sectional visualization mode when you entered 4D CINE mode, you can press **R** to start and stop the CINE loop.

4. To re-start real-time 4D acquisition, press **Freeze**.

- Static 3D** You can create a single sweep, single volume static 3D image.
- Performing a Static 3D Scan**
1. Connect the appropriate 4D-compatible probe, leaving the probes in their respective holders. Follow the guidelines in Chapter 3 for connecting probes.
 2. Select a 4D probe from the probe indicator.
 3. Obtain a 2D image. Optimize the image as usual.
 4. Press **3D/4D**.
 5. Press **Static 3D**. Set Render to On.
 6. Set the Volume of Interest (VOI) to be rendered. Use the **Trackball** to move the VOI and **Scan Area** to re-size the ROI.
 7. Adjust the volume angle. This defines the range of the volume sweep. A small sweep angle results in a lower number of slices with a high volume rate.
 8. Set the probe down on the patient, making sure the probe is held steady. Press the **L** key to start acquisition.
- NOTE:** *During 3D acquisition, the CW, PW, M, and Depth control panel keys are not available.*
- NOTE:** *When the 3D acquisition begins, the Touch Panel appears blank for a brief moment.*
9. Hold the probe steady until the system stops automatically. You will know the acquisition has stopped when the **Touch Panel** changes to display the Render Setting, 3D Rotational Cine, and Scalpel tabs.
To stop the acquisition manually, press the **R** key.
 10. Save the image.
 11. To further manipulate the 3D image, press **Static 3D**.
The Touch Panel that displays depends on the visualization mode selected prior to freeze, Sectional, Render, or VCI Static.

Static 3D
Sectional View

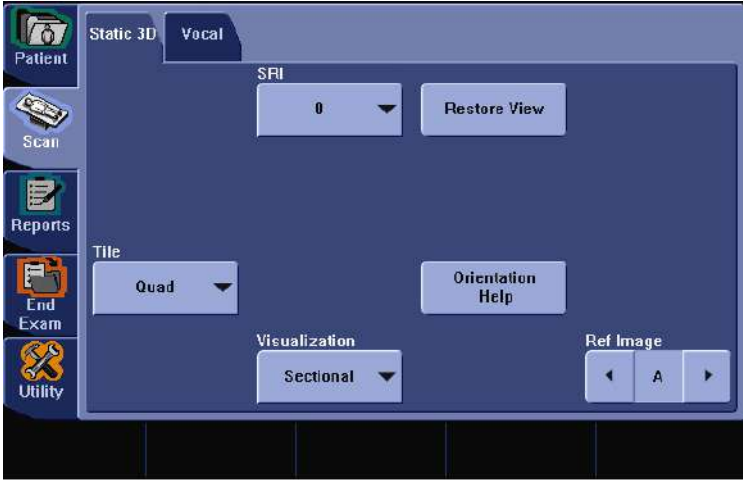


Figure 5-103. 3D (After Acquisition) Touch Panel - Sectional View

Table 5-20: 3D After Acquisition Parameters - Sectional View

Preset Parameter	Description
Orientation Help	The system displays a 3-dimensional drawing to illustrate the orientation. This drawing displays only in sectional view.

Static 3D Render View

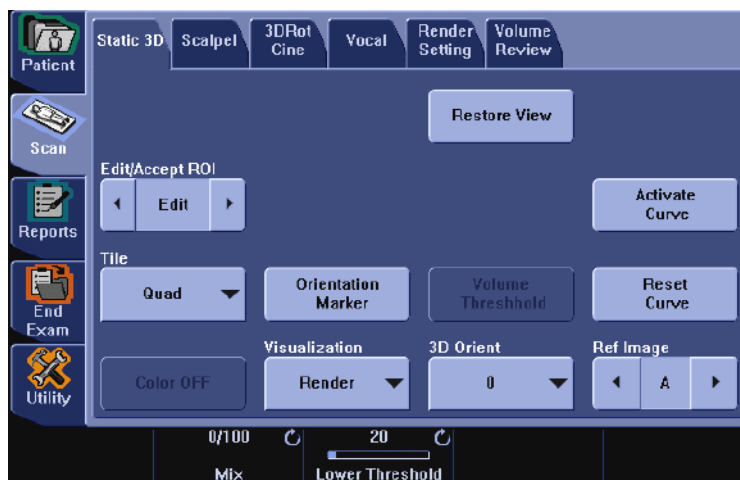


Figure 5-104. 3D (After Acquisition) Touch Panel - Render View

Table 5-21: 3D After Acquisition Parameters - Render View - Page 1

Preset Parameter	Description
Edit/Accept ROI	Selections include Edit, Accept. Edit - Select to adjust the size of the Region of Interest (ROI). Accept - accepts the active 3D image. Returns the rotation, mix, 3D orientation, reference image, and lower threshold to their original values
3D Orient	When selected, changes the orientation of the image on the monitor display. Selections include: 0 degrees, 90 degrees, 180 degrees, and 270 degrees.

VCI Static

- Introduction

VCI (Volume Contrast Imaging) allows you to sweep smaller slices of data with a higher volume rate. The resulting image shows an average, integrated gray value of the tissue contained within the ROI. VCI improves the contrast resolution and signal/noise ratio. It also reduces image speckle. This may facilitate finding diffuse lesions in organs.
- Touch Panel

The data is represented as in Static 3D - Sectional Planes. However, the three planes are VCI renderings (tissue information of a thick slice) computed from the 3D dataset.

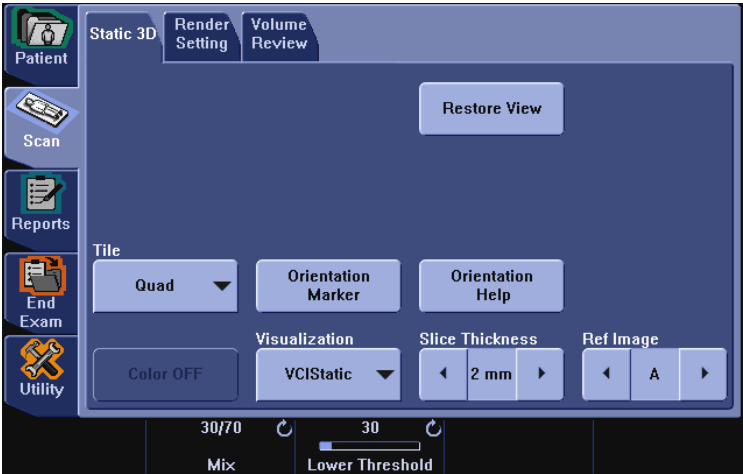


Figure 5-105. 3D (After Acquisition) Touch Panel VCI Static

Table 5-22: VCI Static View

Preset Parameter	Description
Slice Thickness	Select the slice thickness

Static 3D Color

To view Static 3D Color,

1. Acquire the anatomy you want to view in B-Mode.
2. Activate Color Flow.
3. Activate 3D--> Static 3D. Select 'Render' as the Visualization.
4. Press 'L' to render the image.

The color is rendered as Glass Body. You can adjust the Render Mode:

- **Gray.** Gray, no color.
- **Inversion.** Inverts the grayscale.
- **Color.** Just color.
- **Glass Body.** Mix of color and grayscale.

To remove the color, select Color Off from the Touch Panel.

Volume Review

You can post process VOI (the entire volume of interest) and scroll through acquisition planes via Volume Review. This allows you to review all of the frames within this volume set.

In Volume Review, start in Run/Stop so that every frame in the volume is displayed.

NOTE: If you adjust the screen while in Volume Review, these changes ARE NOT reflected on the Static 3D volume.

Scalpel Scalpel allows you to edit/cut sections of a 3D image. Scalpel is available only on a rendered image.

- 1. Press **Scalpel**.



Figure 5-106. Scalpel

Table 5-23: Scalpel Parameters

Preset Parameter	Description
Cut Mode	Selections: Inside Contour, Outside Contour, Inside Box, Outside Box, Eraser Big, Eraser Small. Inside Contour, Outside Contour - Allows you to trace the portion of the image you want to cut. Trace Outside removes all portions of the image that fall outside your traced region. Trace Inside removes all portions of the image that fall inside your traced region. Inside Box, Outside Box - Displays a box you can use to define the portion of the image you want to cut. Outside Box removes all portions of the image that fall outside the box. Inside Box removes all portions of the image that fall inside the box. Eraser Big, Eraser Small - Provides a big and small eraser tool you can use to define the portion of the image to cut by hand. Available only if Depth is Full.
Cut Depth	Selections: Full, Define. Full - The entire depth of the selected region will be cut. Define - Allows you to define the depth to cut using the Depth control panel knob.
Undo Last	Undoes the last cut only.
Redo	Select to redo scalpel.
Undo All	Undoes all cuts since you entered Scalpel mode.
Done	Select when scalpel is complete.

- Scalpel**
(continued)
2. Select the cut mode.
 3. Use the **Trackball** and **Set** key to define the portion of the image to cut. Press **Set** to start, move the **Trackball** to define the region, then press **Set** again to cut the image. The portion is removed.

To undo the last cut, select **Undo Last**.

To undo all cuts in the current session, select **Undo All**.

NOTE: *With the cut image displayed, if you attempt to switch to the Static 3D tab to edit the ROI, the following warning message appears: Scalpel changes will be lost. Do you want to continue? [Yes/No].*

3D Rotation CINE 3D Rotation CINE allows you to view the 3D image from various angles.

To activate rotation CINE in 3D:

1. Press **Freeze**.
2. Select the 3DRot Cine tab.

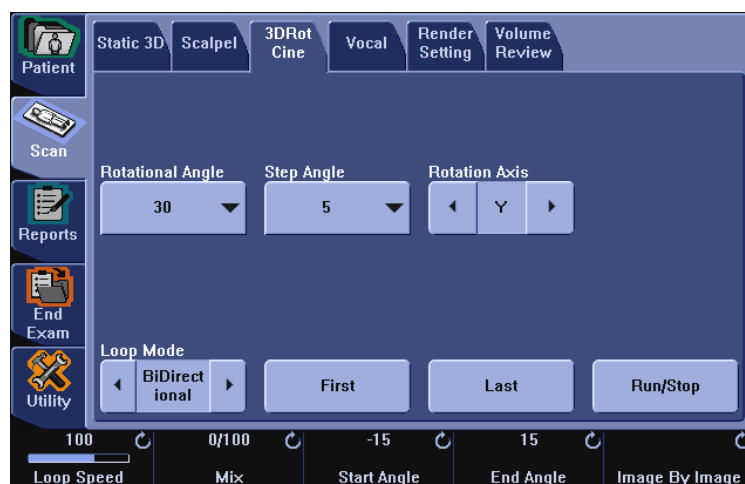


Figure 5-107. 3D Rotation

NOTE: If you press Run/Stop to start the CINE sequence without changing the Calc Cine Sequence or other parameters, the system calculates a sequence and displays a message that a time-sensitive sequence is being performed.

Table 5-24: 3D Rotation Cine Parameters

Preset Parameter	Description
Rotational Angle	Sets the rotational angle of the 3D image over which the CINE loop is played. Typical values are 30, 45, 60, 90, 180 and 360 degrees.
Step Angle	Sets the step angle between individual frames in the CINE loop.
Rotation Axis	Sets the axis about which the CINE loop is calculated. Selections X and Y.
Loop Mode	Selections include: One Way, BiDirectional (two-way). One Way - plays one loop sequence forward. BiDirectional - plays the sequence forward and backward.
First	Displays the first volume in the CINE.
Last	Displays the last volume in the CINE.
Run/Stop	Starts and stops the CINE sequence.
Start Angle	Used to select the starting angle in the CINE loop range. The default Start Image is calculated from the rotational angle as: $-1 \times \text{Rotational angle} / 2$ If you adjust the Start Image, the Rotational Angle is re-set to be the value of the adjusted Start Image.

Table 5-24: 3D Rotation Cine Parameters (Continued)

Preset Parameter	Description
End Angle	Used to select the ending angle in the CINE loop range. The default End Image is calculated from the rotational angle as: $\text{Rotational angle} / 2$. If you adjust the End Image, the Rotational Angle is re-set to be the value of the adjusted End Image.
Image by Image	Used to select an individual image in the CINE loop.

- VOCAL** You use VOCAL (Virtual Organ Computer-aided Analysis) to visualize and calculate the volume of anatomical structures, such as a tumor lesion, cysts, and the prostate. VOCAL is available after a Static 3D or Real-Time 4D acquisition.
1. Press **Vocal**. Specify the volume calculation method (**Manual**, **Contour Detect**, **SemiAuto Detect**, or **Sphere**). Select the reference image you want to use to perform the trace by selecting **Ref Image** A, B, or C. Press **Start**.
 2. Trace the anatomy using the **Trackball**. Press **Set** to start and end the trace. You must go across the dotted line for the trace to take effect (it turns yellow). The trace is performed on each image slice, separated by the rotational step angle. Rotate the **Rot. Ref** dial until you have completed the total of the required rotations (for example, if you've selected 30 degrees, you need to complete six traces if you've selected Manual). After you've completed the trace target, the **Calc Volume** button is active for you to press. The calculated VOCAL image appears in the lower, right-hand corner of the display. You are now in the edit state.
 3. Edit as necessary. You can apply a shell, adjust its thickness, navigate through the reference angles, or restart the VOCAL.

VOCAL (continued)

VOCAL Touch Panel states are described in the following tables.

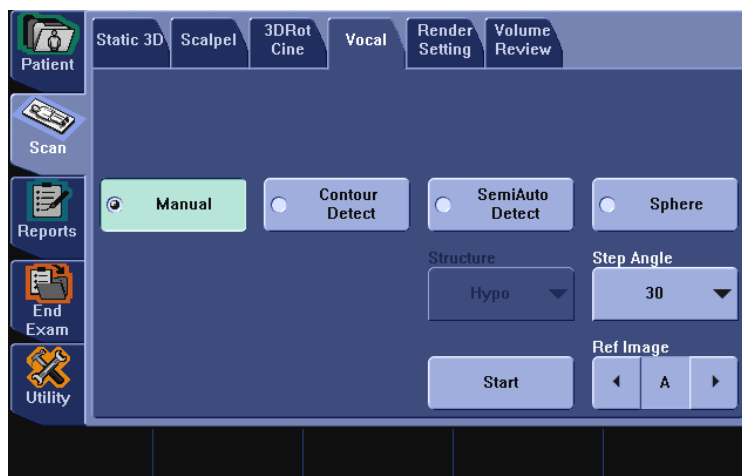


Figure 5-108. VOCAL Setup Touch Panel

Table 5-25: VOCAL Parameters on Setup Touch Panel

Parameter	Description
Manual	When you select the Manual method, you need to perform a manual trace on each of the rotation angles.
Contour Detect	When you select the Contour Detect method, you need to perform a manual trace on each of the rotation angles.
SemiAuto Detect	When you select the Semi Automatic Detect method, you need to perform a trace on only two rotation angles. The system applies an algorithm to define the traces.
Structure	Structure is only available with the SemiAuto Detect method. Select Hypo, Cystic, or Hyper/Iso.
Sphere	Calculates the volume based on the pole settings.
Pole 1	Adjust the upper contour point (green arrow) of the structure.
Pole 2	Adjust the lower contour point (green arrow) of the structure.
Rotational Step Angle	Specify the angular spacing between contour traces. Typical values are 6, 9, 15, and 30 degrees. The number of planes varies by this formula: 180 degrees / selected rotational step angle.
Ref Image	Use this to select the image you want to use to perform the trace.
Start	Press Start when you're ready to perform the trace.
Rot.Ref ## Back/Next	Select Next/Back to move to the next image for contour definition in the rotation step.

VOCAL
(continued)



Figure 5-109. VOCAL Calculate Volume Touch Panel

Table 5-26: VOCAL Calculate Volume Touch Panel

Parameter	Description
Calc Volume	Press Calc Volume to initiate the VOCAL image calculation.
Clear	Press Clear to remove the trace from the image.
Restart Vocal	Press Restart Vocal to return to the initial VOCAL state.

VOCAL (continued)



Figure 5-110. VOCAL Edit Touch Panel

The Shell Modes allows you to construct a shell or contour “around” the structure of interest, which enables you to distinguish between the contour of the targeted structure and the contours the inside and outside of the structure.

Table 5-27: VOCAL Parameters on Edit Touch Panel

Parameter	Description
Shell Off	Select Shell Off if you do not want a shell around the VOCAL image.
Inside	Select Inside if you want a shall inside the volume.
Outside	Select Outside if you want a shell outside the volume.
Symmetric	Select Symmetric if you want half of the shell thickness inside and half outside the volume's perimeter.
Shell Thickness	Adjust to vary the thickness of the shell.

Storing 4D Images

You store 4D images exactly as you would your 2D images. Because 4D images contain more data, they also require more space. Pay close attention to the size of the volumes.

- Still image: Store as Raw Data
- CINE: Store as DICOM

Chapter 6

Scanning/Display Functions

*Describes additional ways in which to adjust the image.
In addition, describes additional ways to get useful
information electronically.*

Zooming an Image

Introduction

Zoom is used to magnify a zoom region of interest (ROI). The system adjusts all imaging parameters accordingly. You can also zoom frozen images.

Zoom

Bioeffect

Zooming an image changes the frame rate which tends to change thermal indices. The position of the focal zones may also change which may cause the peak intensity to occur at a different location in the acoustic field. As a result, the MI may change.



Acoustic
Output
Hazard

Observe the output display for possible effects.

To zoom an image, adjust Zoom. A reference image appears in the upper, left-hand section of the display.

To exit zoom, adjust Zoom until the reference zoom image is removed or press **B**-Mode.

Read vs. Write Zoom

Read Zoom To activate Read Zoom, turn the **Zoom** knob.

Available in pre and post-processing.

Write Zoom To activate Write Zoom, press the **Zoom** knob.

With Write Zoom, the Ultrasound line density and/or sampling frequency increases, giving a better resolution.

Available only in pre-processing.

You can preset the write zoom window size (height and width) on Utility -> Imaging -> B-Mode.

NOTE: *The difference between Read Zoom and Write Zoom can be described in relation to photography. With a photograph, Read Zoom manipulates the negative and enlarges the picture; whereas Write Zoom uses a telephoto lens to bring the image closer before taking the picture.*



HINTS

First use the Read Zoom (turn knob) to get to the area of interest, then use Write Zoom (press knob).

Split Screen

Overview

To activate a dual split screen, press **L**. To activate a quad display, press and hold down **L**.

To switch between active images, press **L/R**.

To deactivate, press **R** until the screen changes.



HINTS

If you press and hold down the L or R key, the quad screen activates. This is useful, for example, when measuring AFI.



HINTS

While using CFM or PDI, press the L and R keys simultaneously to display B and B+CFM, or B and B+PDI in real-time on the left and right side.

It is useful to observe the ROI in B-Mode.

Freezing an Image

Introduction

Freezing a real-time image stops all movement and allows you to measure and print the image.

You can also freeze the image during VCR playback.

NOTE: *While the image is frozen, all Power Output is suspended.*

NOTE: *Selecting a new probe unfreezes the image*

Freezing an image

To freeze an image,

1. Press Freeze. The key backlights.

If you are in a mixed mode, both screen formats stop immediately. Deactivating Freeze restarts both modes and places a black bar on the trace to indicate the time discontinuity.

To reactivate the image,

1. Press Freeze again.

NOTE: *Deactivating Freeze erases all measurements and calculations from the display (but not from the worksheet).*

Use the Trackball to start CINE after pressing Freeze.

Freezing an Image (Footswitch Option)

You can also freeze the image via the footswitch.

Post processing

You can use the following controls to process a frozen B-Mode image.

- Map
- Zoom
- Rotation
- Reverse
- Rejection
- Compression
- B Softener
- Focus Width
- Gain

You can use the following controls to process a frozen Color Flow or Doppler Mode image.

- Angle Correct
- Invert
- CF
- Threshold
- Packet Size
- Flash Suppression
- Dynamic Range

NOTE: *Color cannot be added to a black and white image that has been frozen.*

Auto Optimize. Optimizes the B-Mode image or Doppler Spectrum.

Using CINE

Introduction

CINE images are constantly being stored by the system and are available for playback or manual review via CINE.

Timeline data is continually stored at four times the display width of timeline data (and updates the corresponding B-Mode images).

You can view CINE as a continuous loop via CINE Loop or manually review CINE images frame by frame via the Trackball.

Data in CINE is available until new data is acquired. CINE is stored on the system's memory and can be archived as well.

CINE is useful for focusing on images during a specific part of the heart cycle or to view short segments of a scan session.

Activating CINE

To activate CINE,

1. Press Freeze.
2. Move the Trackball.

CINE and Monitor Display

The CINE display (located on the left-hand side of the monitor) indicates which frame you are viewing of the whole loop (62:123), as well as the time at which this frame occurs within the loop (1.6:3.2 s).

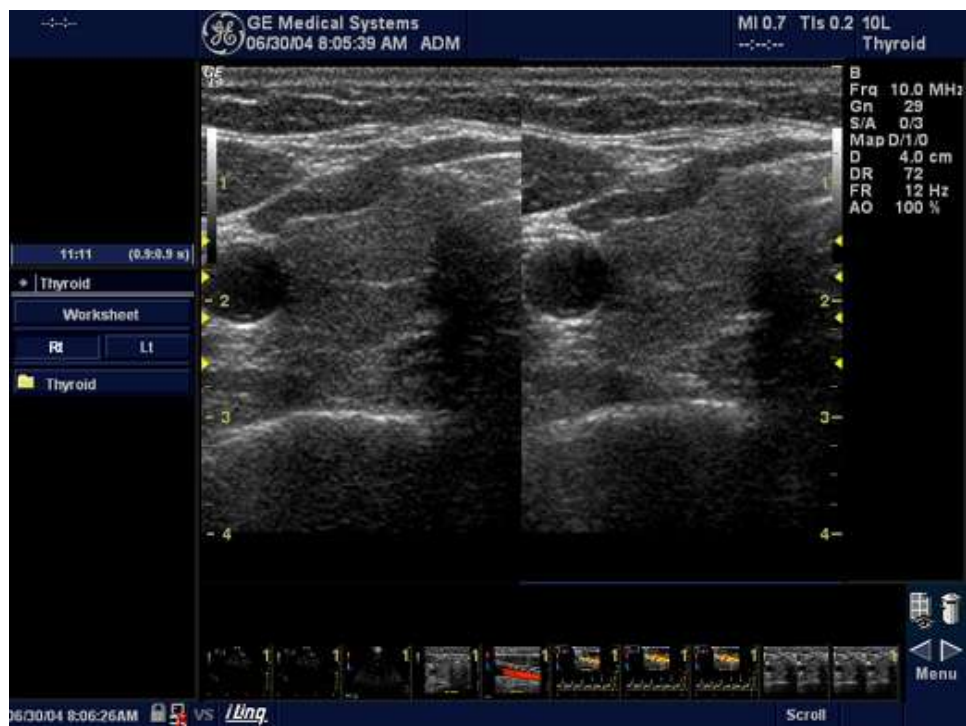


Figure 6-1. CINE Display

Using CINE

The following Touch Panel appears:

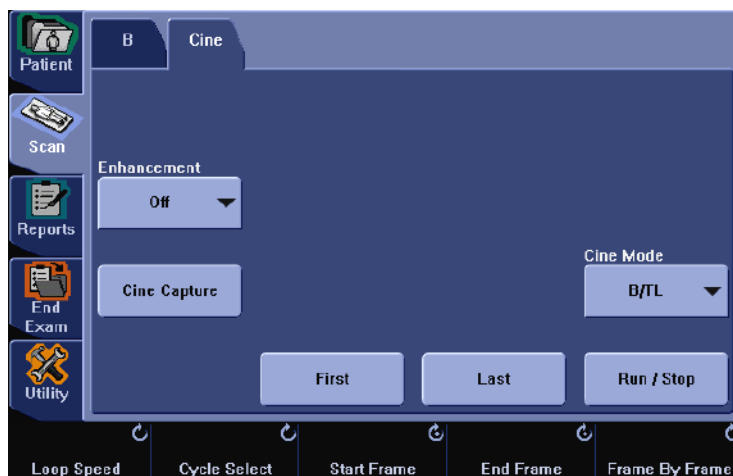


Figure 6-2. CINE Touch Panel

Adjusting CINE Loop Speed

Adjust the Loop Speed control at the bottom of the Touch Panel.

Cine Loop Start/Stop

1. Press **Freeze**.
2. Rotate the **Trackball** horizontally to display the CINE Loop.
3. Select **Run/Stop** to playback the loop.
4. Select **Run/Stop** again to stop the CINE Loop.

Select the start/end frame

1. Rotate Start Frame clockwise/counterclockwise to select the start frame.
2. Press Start Frame when the frame shown is the desired start frame.
3. Rotate End Frame clockwise/counterclockwise to select the end frame.
4. Press End Frame when the frame shown is the desired end frame.

Cine Capture

Selecting **Cine Capture** searches through all images between the start frame and end frame and displays each peak or the highest velocity/tissue power. Adjust the start frame and end frame points to limit the image frames used in the process.

1. Display the CINE loop which is in memory or recalled from archive.

NOTE: *Cine Capture applies only to 2D images (B, CF, PDI, Contrast, TAD, etc).*

NOTE: *On 2D duplex modes (B/CF, B/PDI, B/TAD, etc), Cine Capture is not applied to the background B-Mode image, even if the CF/PDI/TAD display is turned off.*

2. Run the cineloop.
3. Select **Cine Capture** on the Touch Panel to display the captured image.
A character 'C' displays on the screen instead of the frame average level.
4. If necessary, save the captured image.
5. Press **Cine Capture** again to turn Cine Capture off.

NOTE: *Cine Capture can be used on exported files by using the Save As function. You can save the stillimage (jpeg) and cine loop (avi) by using Save As.*

NOTE: *Cine Capture does not effect the TIC Analysis.*

Enhancement

Enhancement executes the enhancement process to the cine capture images.

- NOTE:**
1. Displays the cine capture image using **Cine Capture**.
Enhancement applies only to Cine Capture image.
 2. Select **Enhancement**. A character 'CE' displays on the screen instead of the frame average level.
 - Shade Fw/Shade Rv
Select Fw (Forward) that brings start frame or Rv (Reverse) that brings last frame of cine to the front. The system has two thresholds for Shade process. Comparison will be made frame by frame after the thresholds have been met. If the two thresholds are satisfied, no comparison will be made with the rest of the frame.
Using ShadeFw/RV, lower intensity echoes in the near frame will be masked by higher intensity echoes in the far frame. ShadeFw/Rv makes it possible to show lower intensity echoes in the near frame despite the fact that there may be a higher echo projection in the far frame. Therefore, the anteroposterior position of the blood vessel is clearly displayed.

Enhancement (continued)

- Enhance1/Enhance2/Enhance3

The part corresponding to the data of the selected frame is enhanced and superimposed on the entire cine capture images. So You can easily understand the spatial relationships with B-mode image and the flow appearance.

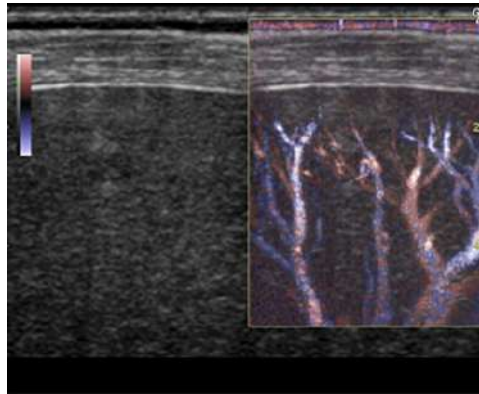


Figure 6-3. Enhancement image example

- If necessary, save the enhanced image.

NOTE: You can save the still image (JPEG) and cine loop (AVI) by using *Save As*.

Disconnecting B-Mode CINE from Timeline CINE

To review the B-Mode CINE Loop only, select **Cine Mode Selection** and select **B Only**.

To review the Timeline CINE Loop only, select **Cine Mode Selection** and select **TL Only**.

To return to linked B-Mode and Timeline CINE Loop review, select **Cine Mode Selection** and select **B/TL**.

Annotating an Image

Introduction

The comment function provides the capability to type the comments of free text and/or insert the pre-defined comments from the comment library. It also provides the user with arrow markers to point to parts of the image.

Pressing the **Comment** key or any keys on the alphanumeric keyboard initiates the comment mode. This assigns the trackball function to controlling the cursor and displays the comment library on the Touch Panel menu.

Introduction (continued)

In comment mode, text can be added by using the comment library or by typing from the alphanumeric keyboard.

Comments can be erased by powering down, when you press **Clear** or **New Patient**, or when preset via Utility--> Comments.

In addition, the display's home position can be changed (preferred comment area) for each display so that all subsequent comments begin in the same spot.

Pressing the **F7 (Cursor Home)** key returns to the user specified position or factory default position.

A new cursor home position is established by placing the cursor in the desired position and pressing **Shift+Home**.

Comment Mode is activated by pressing the **Comment** key. Comment Mode can also be automatically activated by typing from the alphanumeric keyboard.

NOTE: *In this case, the cursor begins at the same location where the comment mode was exited.*

After activating the comment mode, a vertical bar type cursor appears on the screen. Use the **Trackball** to move the cursor.

The factory default color for comments is yellow. The color selection can be changed to any of the colors available on the system. The choices are white, yellow, bright red, orange, etc.

To indicate a specific comment or text group is selected, the color turns to green. Once the comment is set or fixed, the color returns to yellow or to the user selected color.

To delete comments by character, press the **Backspace** key.

To delete all comments and arrow marks, press the **Clear** key twice immediately after entering the comment mode.

To exit the Comment/Library Comment function, press the next function you wish to do.

To move by words or by text group, press the **Tab** key.

Adding Comments to an Image

Comment Retention

Comments from the B-Mode images are retained and carried over when switching to multi-image format or duplex mode.

The position of the comments is adjusted so that it is at the same relative position with respect to the display window in the new format as it was in the single image format.

NOTE: *Comments may not be retained when the image is switched to M-Mode image format depending on the preset.*

Arrow Pointers

Arrow pointers can be used by activating the **F2 (Arrow)** key on the keyboard. When the pointer comes up, it is a GREEN color, indicating it is active and can be moved.

- Move the pointer using the **Trackball** to any place on the screen. The pointer head direction can be controlled by movement of the **Trackball** or **Arrow Rotate** control.
- To readjust the length and thickness of the pointer, use the **Arrow Resize** rotary control. The default for the pointer size can be preset.
- Press **Set** to fix the place of the pointer and direction of the pointer head. The GREEN color turns to YELLOW (or the default color if changed).
- To delete the arrow marks, press the **Clear** key right after pressing the **F2 (Arrow)** key.

NOTE: *This action clears the arrows only, not the comments on the screen. To erase all comments as well as arrows, press the Clear key right after the Comment key.*

NOTE: *To save the arrow angle setting until next changed, select the "Keep arrow angles" preset at Utiltiy -> Annotation -> Comments.*

Text Overlays

There are 2 layers of the text in comments, which can be selected by toggling the **F8 (Text1/Text2)** key on the keyboard. Text1 is the default choice.

By using this function, users can perform a HIDE TEXT/SHOW TEXT, allowing the users to save or printing an image without clearing the typed text.

You can specify to display text 1, text 2, or both. This allows you to have some comments that do not change during the exam while allowing you to change the other comment. Toggle the **F8** key to cycle through the three Text 1/Text 2 states:

1. Text 1 Only -- Only Text 1 displays.
2. Text 2 Only -- Only Text 2 displays. .
3. Text 1 and Text 2 -- Both displayed; only Text 2 comments editable. Only Text 2 comments erased by Clear key. Word Delete only deletes Text 2 comment. Both Text 1 and Text 2 comments erased with new patient, new exam, or probe change.

To preset this, go to **Utility --> Comment --> Comment** and select either **Text 1 and Both** or **Text 1 and Text 2 and Both**.

You select to be able to adjust the Focal Zone Number and Position by adding it to the Annotation Touch Panel in place of the Undo rotary key. To do so, go to **Utility --> Comment --> Comment** and add a checkmark next to **Replace Undo Rotary with B Focus Rotary**.

Annotating an image using the library

To reduce the amount of time spent annotating an image, store frequently-used comments in the Comment Library. As many as 6 libraries are available per study. One of the selected libraries is designated as the default and its entries shall be displayed on the Touch Panel when the comment mode is activated for that study.

Press **Comment** and move the comment cursor location using the **Trackball**.

Select the desired comment from the Touch Panel.

Each Touch Panel key can also be configured to hold a small list of up to 3 comments. The first word in the list is displayed on the Touch Panel and the others can be accessed by toggling the key or pop-up menu. To show the presence of a small list stored under a particular key, a small indicator (>) is placed on the key.

To program your system with specific comments, See 'Comments Libraries Presets' on *page 16-53 for more information*.

Annotating an image with typed words

- Press **Comment** and type the comments where the cursor is currently located (the display's home position) and use the **Trackball** to further place the comment cursor in the desired location.
- Press **Enter** to move to the next line.

NOTE: *Comments wrap to the next line when they are within one character of the right margin if Word Wrapping is selected in the Text Boundary preset. See 'Comments Libraries/Comments Preset Menu' on page 16-56 for more information.*

The word wrap starts one line below the start of that comment.

Comments appear on all prints, photos, and VCR recordings.

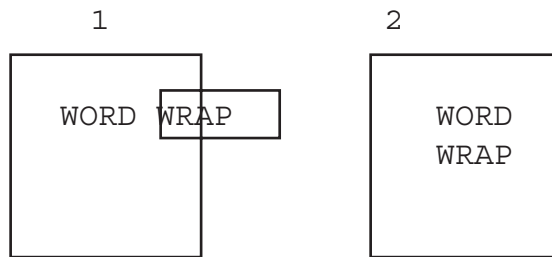


Figure 6-4. Next Line Word Wrap

1. Before

2. After

If the cursor appears at the right edge of the lowest line, or a word cannot be completed in the lower right corner, word wrap cannot be executed.

NOTE: *The same word wrap principles apply for library scripts as typed comments.*

Moving Texts

You have the ability to move comments already on the screen and place them in different locations.

- Place the cursor on the desired text or text group and press **Set**.
- The selected text color turns to green.
- Use the **Trackball** to move the selected text and press **Set**.

Annotating an image with typed words (continued)

Editing while annotating

Backspace over any error(s) made. Blank spaces take the place of the letter(s) that were there. Continue typing the comment after backspacing over all incorrect letters.

To delete previous character(s):

- Press **Backspace** as many times as necessary to make the deletion.
- Once all texts within the selected text group are deleted, then the cursor will find another text group to delete to the upper left direction.
- If there is no more text to delete, the cursor will be located at home position.

To move through the text a word at a time:

- Press **Tab** to move to the right by text group (Preset Keyboard Tab = Word)

NOTE: Press **Shift + Tab** to move to the left.

To activate the last text group typed or selected from the Library:

- Press **F9 (Grab Word)** key. The selected comment will be highlighted.
- To increase/decrease the area of the highlighted selection use HIGHLIGHT rotary.

NOTE: To select all text groups, Press **Shift + F9 (Grab Word)** key.

To cancel the last action:

- Press **Undo** key.

Body Patterns

An additional way to annotate the image display is with body patterns. Body patterns are a simple graphic of a portion of the anatomy that is frequently scanned. The body pattern and probe marker can serve as a reference for a patient and probe positioning when images are archived or scanned.

To activate body patterns, press the **Body Pattern/Ellipse** control. A maximum of 6 body pattern packages are displayed on the Touch Panel according to exam category and preset.

The body pattern packages may be customized to accommodate user preference. Up to 30 individual body patterns in the packages can be changed. See 'Body Pattern Libraries/Applications Preset Menu' on *page 16-65 for more information*.

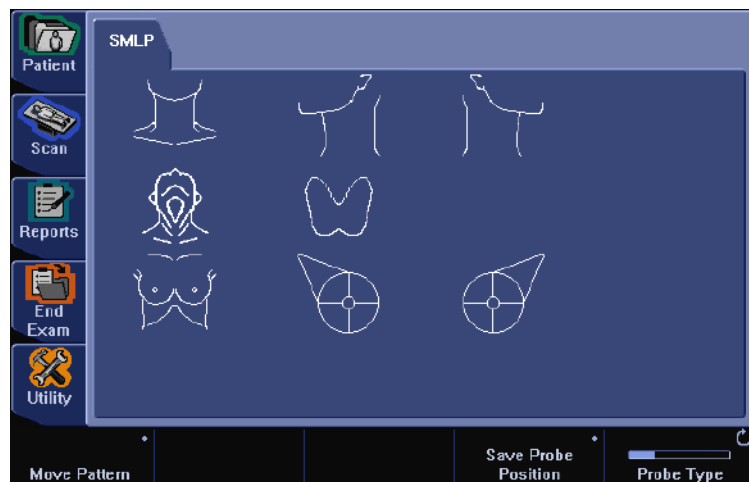


Figure 6-5. Touch Panel displays of Body Pattern

Body Patterns (continued)

Select the desired body pattern on the Touch Panel. The selected body pattern is displayed on the monitor.

Press the **Move Pattern** control on the Touch Panel to reposition the body pattern with the **Trackball** and **Set** controls.

A probe mark is associated with the body patterns and illustrates the probe position on the body pattern. This marker can be placed with the **Trackball** and rotated with the **Ellipse** control.

The probe mark type is selectable by rotating the **Probe Type** control on the Touch Panel. There are different choices available with one being a blank selection.

To select the active side in dual B-Mode, use the **Active Side** rotary control at the bottom of the Touch Panel.

Press the **Body Pattern/Ellipse** control to erase the body pattern. The pattern is erased and the system exits the body pattern mode.

Press **Set** on the keyboard or **Scan** on the Touch Panel to exit without erasing the body pattern.

Select the **Save Probe Position control** to save the probe mark position and angle for each body pattern.

1. Display the body pattern.
2. Move and rotate the probe mark as appropriate.
3. Select **Save Probe Position**.

You can use the **Zoom** control to select the body pattern. If you want to assign the select function to the Zoom control,

Body Patterns (continued)

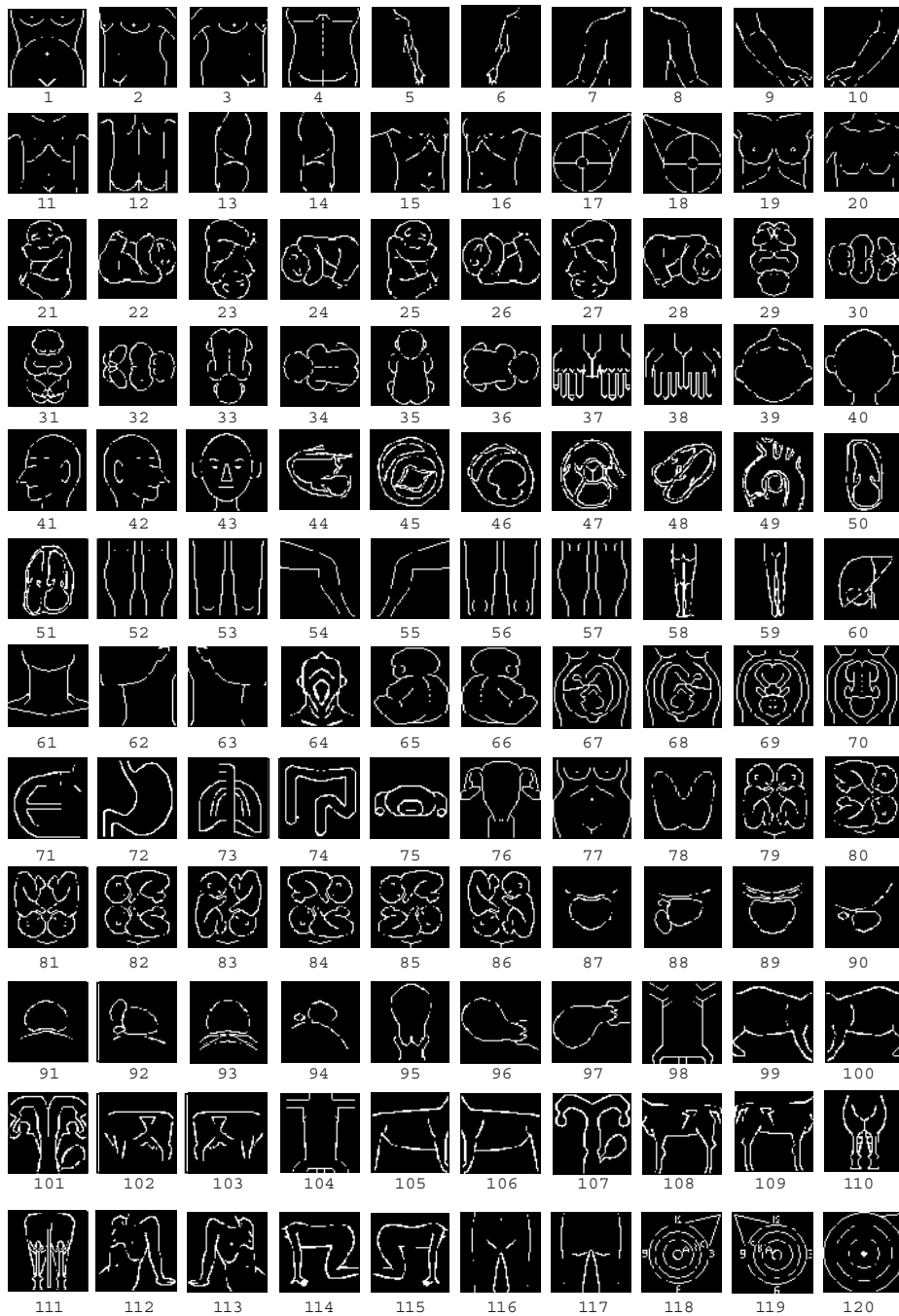


Figure 6-6. Body Patterns Available

Body Patterns (continued)

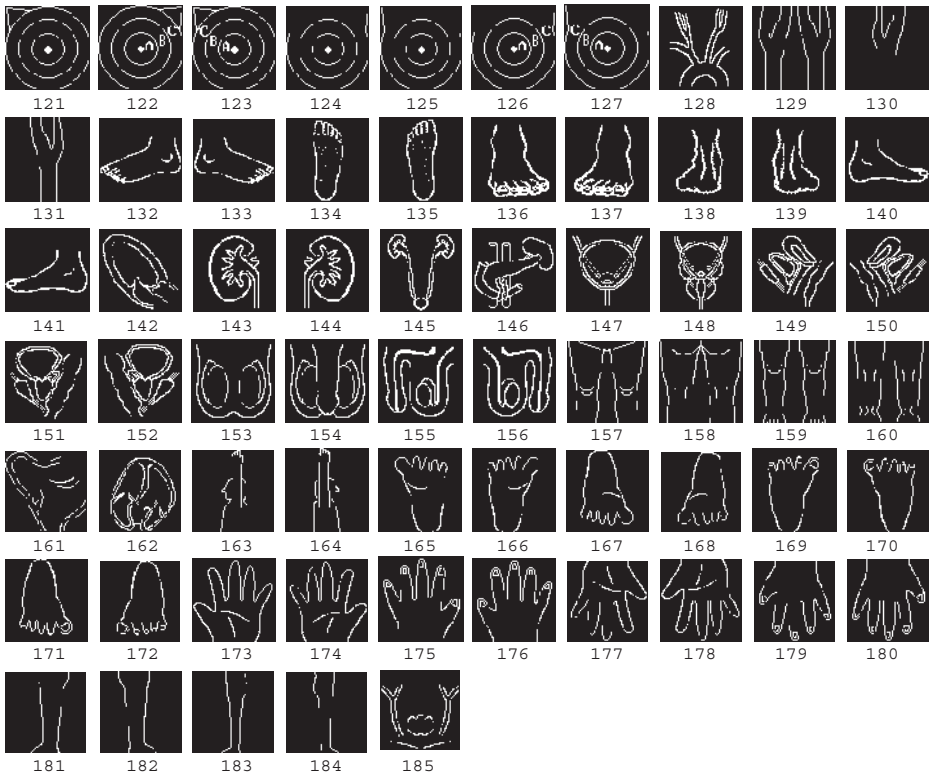


Figure 6-7. Body Patterns Available (continued)

Body Patterns (continued)

1. abdom1	48. heart5	95. uterus1	142. heart9
2. abdom2	49. heart6	96. uterus2	143. organ6
3. abdom3	50. heart7	97. uterus3	144. organ7
4. abdom4	51. heart8	98. vet-cat1	145. organ8
5. arm1	52. leg1	99. vet-cat2	146. organ9
6. arm2	53. leg2	100. vet-cat3	147. uro9-female
7. arm3	54. leg3	101. vet-cow1	148. uro9-male
8. arm4	55. leg4	102. vet-cow2	149. uro10-female-Lt
9. arm5	56. leg5	103. vet-cow3	150. uro10-female-Rt
10. arm6	57. leg6	104. vet-dog1	151. foot4-Lt
11. body1	58. leg7	105. vet-dog2	152. uro10-male-Lt
12. body2	59. leg8	106. vet-dog3	153. uro10-male-Rt
13. body3	60. liver	107. vet-horse1	154. uro11
14. body4	61. neck1	108. vet-horse2	155. uro12
15. body5	62. neck2	109. vet-horse3	156. uro13-Lt
16. body6	63. neck3	110. vet-horse4	157. uro13-Rt
17. breast1	64. neck4	111. vet-horse5	158. leg 9
18. breast2	65. ob1	112. body7-Lt	159. leg 10
19. breast3	66. ob2	113. body7-Rt	160. leg 11
20. chest	67. ob3	114. body8-Lt	161. leg 12
21. fetus1	68. ob4	115. body8-Rt	162. heart 10
22. fetus2	69. ob5	116. body9	163. body11-Lt
23. fetus3	70. ob6	117. body10	164. body11-Rt
24. fetus4	71. organ1	118. breast4-Lt	165. foot6-Lt
25. fetus5	72. organ2	119. breast4-Rt	166. foot6-Rt
26. fetus6	73. organ3	120. breast5-Lt	167. foot7-Lt
27. fetus7	74. organ4	121. breast5-Rt	168. foot7-Rt
28. fetus8	75. organ5	122. breast6-Lt	169. foot8-Lt
29. fetus9	76. pelvis1	123. breast6-Rt	170. foot8-Rt
30. fetus10	77. pelvis2	124. breast7-Lt	171. foot9-Lt
31. fetus11	78. thyroid	125. breast7-Rt	172. foot9-Rt
32. fetus12	79. twin1	126. breast8-Lt	173. hand3-Lt
33. fetus13	80. twin2	127. breast8-Rt	174. hand3-Rt
34. fetus14	81. twin3	128. carotid1	175. hand4-Lt
35. fetus15	82. twin4	129. carotid2	176. hand4-Rt
36. fetus16	83. twin5	130. carotid2-Lt	177. hand5-Lt
37. hand1	84. twin6	131. carotid2-Rt	178. hand5-Rt
38. hand2	85. twin7	132. foot1-Lt	179. hand6-Lt
39. head1	86. twin8	133. foot1-Rt	180. hand6-Rt
40. head2	87. uro1	134. foot2-Lt	181. legs13-a-Lt
41. head3	88. uro2	135. foot2-Rt	182. legs13-a-Rt
42. head4	89. uro3	136. foot3-Lt	183. legs13-Lt
43. head5	90. uro4	137. foot3-Rt	184. legs13-Rt
44. heart1	91. uro5	138. foot4-Lt	185. neck5
45. heart2	92. uro6	139. foot4-Rt	
46. heart3	93. uro7	140. foot5-Lt	
47. heart4	94. uro8	141. foot5-Rt	

Using the Fast Key

Overview

A keyboard Fast Key is now available. F5 activates the Fast Key setup. Press F5, select a key to assign the Fast Key to (a-z, 0-9), press the desired key sequence to be recorded, press F5 to end the Fast Key recording. To execute the Fast Key, press F6, then press the key you programmed (a-z, 0-9).

NOTE: *Ensure that you have a patient selected prior to running the Fast Key operation.*

VoiceScan

Introduction

VoiceScan lets you to speak Ultrasound commands into a headset microphone rather than push the controls directly on the Ultrasound system. This allows you to scan more comfortably in exams where the scanned anatomy requires you to be farther away from the Ultrasound system console.

A wireless microphone and a wired microphone are available for VoiceScan.

The VoiceScan section is divided into two sections: Setting up VoiceScan and Using VoiceScan.



WARNING

DO NOT use the wireless transmitter in the operating room. There is a possibility of harming other medical equipment.

Setting up VoiceScan

The following procedures can be found in this section:

- Wireless Microphone Setup Instructions
- Non-Wireless Microphone Setup Instructions
- VoiceScan Training

Wireless Microphone Setup Instructions

Wireless Microphone Kit

The Wireless Microphone Kit consists of the following:

- Battery Charging Kit, which includes:
 - Battery Recharging Unit with 2 AC adapters (120V and 240V)
 - Rechargeable Battery Packs (2)
 - AC Power Adapter, 120V and 240V (ensure that you use the correct adapter)
 - Universal Adapter Plugs
 - Instructions. **Please Read These Instructions.**
- Wireless Bodypack Transmitter (WBT):
 - Wireless Bodypack Transmitter (referred to as WBT or transmitter)
- Transmitter Microphone Kit, which includes:
 - Headset
 - Collar Clip
 - Extra Windscreen (for mouthpiece)
 - Instructions



Figure 6-8. WBT, Headset, and Antenna

Charging the Batteries/WBT

You need to charge the batteries overnight before using them. To charge the rechargeable batteries,

NOTE: *Only use the L2015 Sennheiser Accu-Charger to charge the BA2015 Accupack batteries.*

1. Remove the battery from its packaging. Remove shrinkwrap from the batteries, if necessary.
2. Connect the Battery Recharging Unit to the correct AC adapter, depending on your location.
3. Plug the AC adapter into an outlet; use adapting plugs if needed.

**WARNING**

Damage to the battery charger can occur if the wrong AC adapter is used.

4. Insert the BA2015 Accupack (or WBT). Check for the correct polarity. If you force the battery and polarity is backwards, you will damage the terminals. Line up the battery pack tab with the charger tab. Press into place.



Figure 6-9. Charging Accupacks

Charging the Batteries/WBT (continued)

5. Charge the BA2015 Accupack overnight.
6. LEDs are red while the batteries are charging. LEDs are green when ready for use.
7. Refer to the instructions included with the charger for additional information.

NOTE: *Recharged batteries operate for approximately eight (8) hours. Keep one battery pack charging at all times. You can leave the pack in the charger at all times. It takes about 2-1/2 hours to charge.*



ONLY charge the accupacks that came with the Sennheiser Accu-Charger L2015 system. DO NOT attempt to charge Alkaline or other types of batteries that are not designed for recharging. Inserting any other batteries may damage the device, destroy the batteries, and cause injury to the user.



ONLY use the AC Adapter that came with the Sennheiser Accu-Charger L2015 system. Using other adapters may cause damage to the charger.

International users, attach the universal adapter to the Battery Recharging Unit.

Preparing the Transmitter

Refer to the illustration below to become familiar with the WBT:

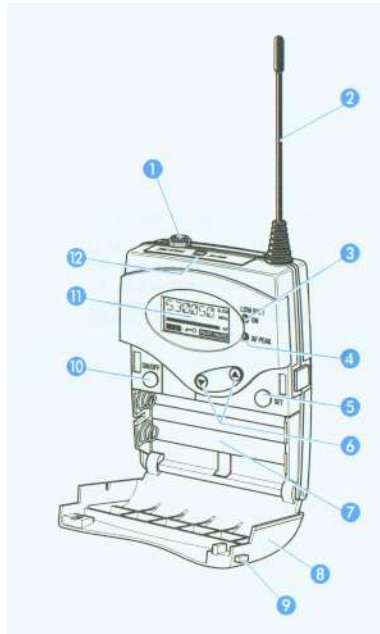


Figure 6-10. Wireless Bodypack Transmitter (WBT)

1. Microphone input jack socket
2. Antenna
3. Red LED for operation and battery status indications (On or blinking for Low Battery)
4. Yellow LED for audio (AF) peak (audio input level excessively high)
5. SET button
6. Rocker button (DOWN/UP)
7. Battery compartment
8. Battery compartment cover
9. Unlocking button
10. ON/OFF button. Press ON/OFF to activate the WBT; press and hold down ON/OFF to deactivate the WBT.
11. LC Display
12. MUTE switch

Preparing the Transmitter (continued)

To attach the antenna and microphone,

1. Attach the antenna by screwing it into the WBT antenna socket (item 2 in previous figure).
2. Attach microphone by plugging it into the WBT socket. If the plug on the microphone is equipped with a locking ring, screw down the locking ring onto the threads on the socket. Adjust the sensitivity of the microphone. See Setting the Sensitivity.

The standard display includes the following:

- Frequency
- ViewMeter (tracks your voice on the bar)
- Accupack/Battery capacity (3 segments=100%, 2 segments=70%, 1 segment=30%, battery icon flashing=low battery)
- Key (locked)
- PILOT
- MUTE

Installing the Microphone's Wind Screen

You need to ensure that the windscreen is placed correctly on the microphone. Follow these instructions when attaching the wind screen to the microphone.

1. The microphone consists of 1) microphone wire, 2) microphone taper, and 3) microphone element. The windshield must fully cover the microphone taper to be correctly installed.



Figure 6-11. Microphone without Windscreen

2. Place the windscreen onto the microphone by slightly twirling the foam in your fingers while threading it onto the microphone element. Once the element has entered the foam, you must push further to fit the plastic cap that is inside the foam onto the element. This takes a little force and rocking.

NOTE: *If the microphone looks like the example below, the windscreen is not yet fully installed. You can see that at (1) below, a portion of the microphone taper is still visible on the (2) windscreen.*

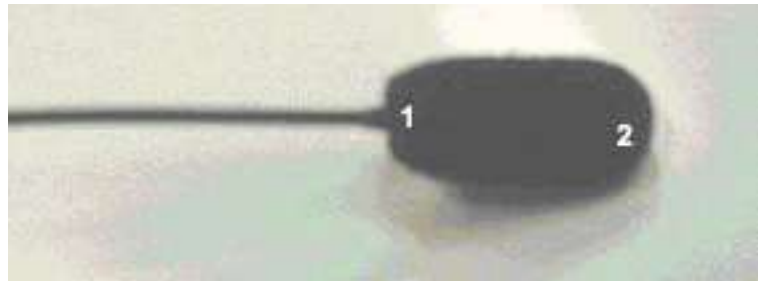


Figure 6-12. Not Fully Installed Windscreen Example

Installing the Microphone' Wind Screen (continued)

3. Push the cap inside the foam all the way as in the example below, to avoid having the windscreen fall off the microphone.

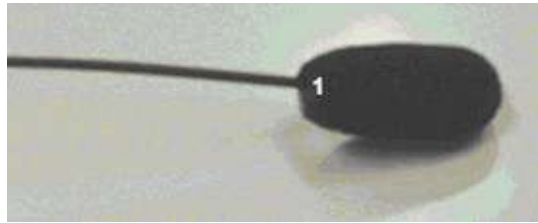


Figure 6-13. Correctly installed Windscreen

Cleaning the Microphone

To clean the microphone, use a damp cloth to wipe down the ear area. DO NOT use anything too wet. You do not want to get moisture in-between the stiff wire and the clear, rubber tubing in the ear area.



DO NOT dip the microphone into any liquid.
DO NOT get any moisture on the microphone element.

Cleaning the Wind Screen

To clean the wind screen,

1. Remove the wind screen from the microphone.
2. Gently wash it with soap and water.
3. Gently rinse and gently squeeze the wind screen between your fingers. Let air dry overnight.

**Installing the
Batteries into the
Wireless
Transmitter**

To install the recharged Sennheiser BA2015 Accupack (or two 1.5V AA batteries) into the WBT,

1. Open the cover.
2. Insert the Accupack (or two AA batteries). Observe the correct polarity when inserting the Accupack/batteries, as shown in the figure below.
3. Close the battery compartment. The battery compartment cover locks into place with an audible click.



Figure 6-14. Inserting BA2015 Accupack into the WBT

Using the Wireless Transmitter

Setting up the WBT,

1. Press the two unlocking buttons to open the battery compartment cover.
2. Turn on the WBT by pressing ON/OFF. The red LED lights up and displays the current frequency or channel, audio signal, battery status, and MUTE/PILOT indicator.

NOTE: *If this LED is flashing red, the Accupack/batteries are low.*

3. Press SET to get into the operating menu. Use SET to access menus and store menu selections.
4. Press the Rocker button to move through menu selections and to adjust the settings. Press ON/OFF to cancel or select EXIT to return to the standard display.
 - **BANK.** Switches between channel banks. There are eight banks + "U", for user bank. Select "U". This allows you to select between eight different frequencies.
 - **LOCK.** DO NOT use this selection.
 - **PILOT.** Leave OFF at all times.
 - **RESET.** DO NOT use this selection.
 - **NAME.** Use the default setting.
 - **DISPLY.** Specifies Display mode (Frequency or Channel).
 - **SENSIT.** Sensitivity in dB. This should be set to -10dB.
 - **TUNE.** Sets the Transmission Frequency for the channel bank ("U" user bank)
 - **CHAN.** Switches between channels.
 - **EXIT.** Returns to standard display.

**Using the Wireless Transmitter
(continued)****Setting the Sensitivity**

1. Press SET. Press the Rocker button until you see SENSIT. Press SET. The current sensitivity value appears.
2. Press the Rocker button until -10dB appears.
3. Press SET. STORED appears.
4. Toggle to EXIT, then press SET to return to the standard display (or press On/Off).

Specifying the Display Mode

1. Press SET. Press the Rocker button until you see DISPLY. Press SET.
2. Press the Rocker button until FREQ appears.
3. Press SET. STORED appears.
4. Toggle to EXIT, then press SET to return to the standard display (or press On/Off).

Tuning the WBT Frequency (One LOGIQ 9 System)

Each LOGIQ 9 system is set to a unique frequency by your institution's Field Engineer. The Field Engineer will have set the frequency you should use on the Utility --> System --> VoiceScan page. This is the value that should be tuned on the WBT. The transmitter's frequency **MUST** match the frequency on the LOGIQ 9 system that you are using.

1. The current frequency appears on the display.
2. Press SET. Press the Rocker button until you see TUNE. Press SET.
3. Press the Rocker button to select the correct frequency (the frequency that matches the frequency on the LOGIQ 9 VoiceScan utility page. The frequencies are tunable in 25kHz steps.
4. Press SET. STORED appears.
5. Toggle to EXIT, then press SET to return to the standard display (or press On/Off).

Using the Wireless Transmitter (continued)

Tuning the WBT Frequency (More than one LOGIQ 9 System)

You can use the same WBT for multiple LOGIQ 9 systems; however, you CANNOT use the same transmission frequency for multiple LOGIQ 9 systems. Each LOGIQ 9 requires a unique frequency. You can tune up to eight (8) frequencies on the WBT. Just toggle to the frequency matching the LOGIQ 9 system you are working on.

- 1. Check the frequency for the LOGIQ 9 system you will be working on via the Utility --> System --> VoiceScan page.
- 2. Press SET. Press the Rocker button until you see CHAN. Press SET. Press the Rocker button to select a channel. Press SET. STORED appears.
- 3. Press the Rocker button until you see TUNE. Press SET.
- 4. Press the Rocker button to select the correct frequency (the frequency that matches the frequency on the LOGIQ 9 VoiceScan utility page. The frequencies are tunable in 25kHz steps.
- 5. Press SET. STORED appears.
- 6. Toggle to EXIT, then press SET to return to the standard display (or press On/Off).



DO NOT set more than one WBT to the same frequency. If you do, neither WBT will work.

Table 6-1: WBT Tracker

LOGIQ 9 Channel								
Unit	U01	U02	U03	U04	U05	U06	U07	U08
Frequency								

Muting the WBT

- Click the MUTE switch to mute the WBT's audio signal. MUTE appears on the display. Remember, muting the transmitter DOES NOT switch off the transmitter.

Using the Wireless Transmitter (continued)

Signal and Battery Status Indicator

The yellow LED on the top of the WBT lights up if the audio signal at the AF input is excessively high (AF peak).

NOTE: *If the yellow light is always on, then lower the sensitivity. Redo the audio set (but you do not need to redo the training).*

The red LED provides information on the battery status.

- If the LED is red, the transmitter is switched on and the battery capacity is sufficient.
- If the LED is flashing red, the battery is too weak. You should replace the battery IMMEDIATELY. With a weak battery, voice recognition accuracy diminishes significantly. Battery capacity is shown on the display: 3 segments=100%, 2 segments=70%, 1 segment=30%, battery icon flashing=low battery.

NOTE: *Please refer to the Installing the Batteries into the Wireless Transmitter instructions above.*

Deactivating the WBT

Turn off the transmitter when it is not in use.

1. To turn off the WBT, open the battery compartment cover, press and hold down the ON/OFF button until OFF appears on the display.
2. After OFF appears on the display, you can then release the ON/OFF button. The red LED turns off.

Attaching the WBT to Clothing

Attach the WBT to your clothing with the belt clip on the back side of the WBT.

Positioning the Microphone

Effective use of VoiceScan is highly dependent upon correctly positioning the microphone, specifically,

1. Use the transmitter clip to place the transmitter on your belt or clothing.
2. Place the microphone on your ear and position the microphone so that it is one inch away from the corner of your mouth. You may need to bend the microphone cable. Refer to the microphone instructions.
3. Attach the cord clip to the microphone cord and attach the clip to your clothing to avoid changing the microphone position during scanning.

Wireless Microphone Setup Instructions (Japan Only)

Wireless Microphone Kit

The Wireless Microphone Kit consists of the following:

- Battery Kit, which includes:
 - Battery Recharging Unit (with North American AC adapter)
 - Rechargeable Batteries (3)
 - Universal Adapter
 - Adapter Plugs
 - Instructions. **Please Read These Instructions.**
- Wireless Bodypack Transmitter (WBT) Kit, which includes:
 - Wireless Bodypack Transmitter (referred to as WBT or transmitter)
 - Headset
 - Antenna
 - Collar Clip
 - Extra Windscreen (for mouthpiece)
 - Instructions



Figure 6-15. WBT, Headset, and Antenna

Charging the Batteries

You need to charge the batteries overnight before using them. To charge the rechargeable batteries,

1. Remove the battery from its packaging. Remove shrinkwrap from the batteries, if necessary.
2. Connect the Battery Recharging Unit to the correct AC adapter, depending on your location.
3. Plug the AC adapter into an outlet; use adapting plugs if needed.

**WARNING**

Damage to the battery charger can occur if the wrong AC adapter is used.

4. Insert the 9 Volt Battery. Check for the correct polarity. If you force the battery and polarity is backwards, you will damage the terminals.



Figure 6-16. Charging the Batteries (US Plug)

Charging the Batteries/WBT (continued)

5. Charge the battery overnight.
6. LEDs are red while the batteries are charging. LEDs are green when ready for use.
7. Refer to the instructions included with the charger for additional information.

NOTE: *Recharged batteries operate for approximately four (4) hours. Therefore, you should recharge two (2) batteries at a time.*



ONLY charge the batteries that came with the 3 in 1 Fast Smart Charger II system (or equivalent Ni-CD / Ni-MH type batteries). DO NOT attempt to charge Alkaline or other types of batteries that are not designed for recharging. Inserting any other batteries may damage the device, destroy the batteries, and cause injury to the user.



ONLY use the AC Adapter that came with the 3 in 1 Fast Smart Charger II system. Using other adapters may cause damage to the charger.

**Charging the
Batteries/WBT
(continued)**

International users, attach the universal adapter to the Battery Recharging Unit.



Figure 6-17. Charging the Batteries (Universal AC Adapter, 100-240Vac)

Preparing the Transmitter

Refer to the illustration below to become familiar with the WBT:

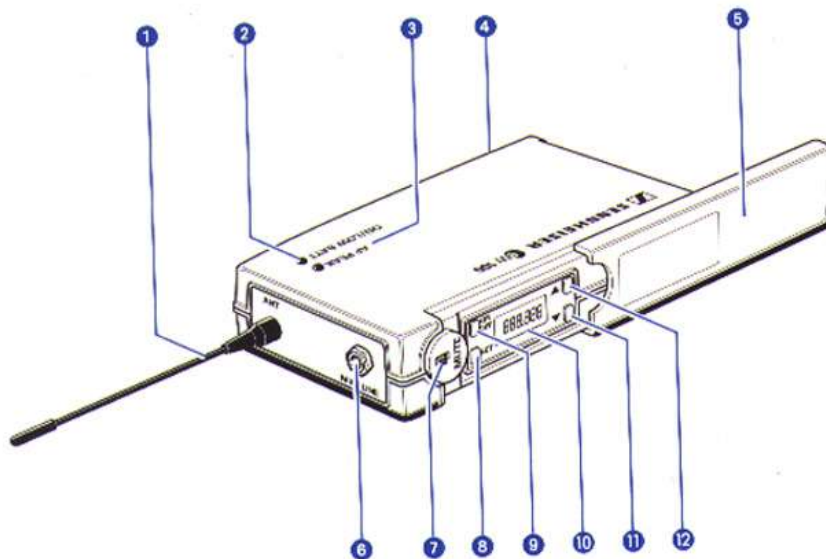


Figure 6-18. Wireless Bodypack Transmitter (WBT)

1. Antenna
2. Red LED for operation and battery status indications (On or Low Battery [LOW BAT])
3. Yellow LED for AF peak
4. Cover plate for battery compartment
5. Cover plate for display and operating controls
6. AF input (MIC/LINE, 3.5mm jack socket) -- microphone jack
7. MUTE switch
8. SET button
9. ON/OFF button
10. Display
11. V button (DOWN)
12. ^ BUTTON (up)

To attach the antenna and microphone,

1. Attach the antenna by screwing it into the WBT antenna socket (#1).
2. Attach microphone by plugging it into the WBT socket. If the plug on the microphone is equipped with a locking ring, screw down the locking ring (#6) onto the threads on the socket.

Installing the Microphone's Wind Screen

You need to ensure that the windscreen is placed correctly on the microphone. Follow these instructions when attaching the wind screen to the microphone.

1. The microphone consists of 1) microphone wire, 2) microphone taper, and 3) microphone element. The windshield must fully cover the microphone taper to be correctly installed.



Figure 6-19. Microphone without Windscreen

2. Place the windscreen onto the microphone by slightly twirling the form in your fingers while threading it onto the microphone element. Once the element has entered the foam, you must push further to fit the plastic cap that is inside the foam onto the element. This takes a little force and rocking.

NOTE: *If the microphone looks like the example below, the windscreen is not yet fully installed. You can see that at (1) below, a portion of the microphone taper is still visible on the (2) windscreen.*

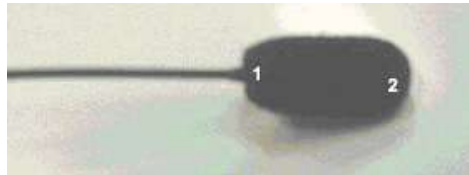


Figure 6-20. Not Fully Installed Windscreen Example

3. Push the cap inside the foam all the way as in the example below, to avoid having the windscreen fall off the microphone.

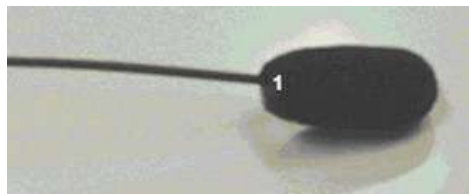


Figure 6-21. Correctly installed Windscreen

Cleaning the Microphone

To clean the microphone, use a damp cloth to wipe down the ear area. DO NOT use anything too wet. You do not want to get moisture in-between the stiff wire and the clear, rubber tubing in the ear area.



CAUTION

DO NOT dip the microphone into any liquid.

DO NOT get any moisture on the microphone element.

Cleaning the Wind Screen

To clean the wind screen,

1. Remove the wind screen from the microphone.
2. Gently wash it with soap and water.
3. Gently rinse and gently squeeze the wind screen between your fingers. Let air dry overnight.

**Installing the
Batteries into the
Wireless
Transmitter**

To install the recharged battery into the WBT,

1. Slide the cover of the battery compartment in the direction of the embossed arrow until it clicks audibly.
2. Open the cover.
3. Insert the 9V battery. Observe the correct polarity when inserting the battery.
4. Close the battery compartment.

NOTE: *To remove a battery, push the small red lever in the battery compartment towards the bottom side of the transmitter.*



Figure 6-22. Inserting Battery into the WBT

Using the Wireless Transmitter

Setting up the WBT,

1. Slide the plastic cover on the WBT to the right.
2. Turn on the WBT by pressing the red ON/OFF button. The LED lights up and displays the current frequency or channel.
3. Toggle through the SET button to understand the different settings:
 - **SEnSit.** Sensitivity in dB. This should be set to -10dB.
 - **tunE.** Transmission Frequency
 - **LOC.** Lock mode
 - **DISPL.** Specifies Display mode (Frequency or Channel).

SET toggles the menu and verifies your menu selection.



Figure 6-23. Sliding the WBT's Cover to Tune WBT

Using the Wireless Transmitter (continued)**Setting the Sensitivity.**

1. Press SET until you see SenSit. The current sensitivity value appears.
2. Press the up or down arrow buttons until -10dB appears.
3. Press SET.

Specifying the Display Mode.

1. Press SET until you see DISPL.
2. Press the up or down arrow buttons until FrEqu appears.
3. Press SET.

Tuning the WBT Frequency (One LOGIQ 9 System).

Each LOGIQ 9 system is set to a unique frequency by your institution's Field Engineer. The Field Engineer will have set the frequency you should use on the Utility --> System --> VoiceScan page. This is the value that should be tuned on the WBT. The transmitter's frequency **MUST** match the frequency on the LOGIQ 9 system that you are using.

1. The current frequency appears on the display.
2. Press the up or down arrow buttons to select the correct frequency (the frequency that matches the frequency on the LOGIQ 9 VoiceScan utility page. The frequencies are tunable in 25kHz steps. The new frequency flashes. Press SET.
3. Press SET to return to the top level menu.

Using the Wireless Transmitter (continued)

Tuning the WBT Frequency (More than one LOGIQ 9 System).

You can use the same WBT for multiple LOGIQ 9 systems; however, you CANNOT use the same transmission frequency for multiple LOGIQ 9 systems. You can tune up to four (4) frequencies on the WBT. Just toggle to the frequency matching the LOGIQ 9 system you are working on.

1. Check the frequency for the LOGIQ 9 system you will be working on via the Utility --> System --> VoiceScan page.
2. Press the up or down arrow buttons to select the correct frequency on the WBT. The new frequency flashes. Press SET.
3. Press SET to return to the top level menu.



DO NOT set more than one WBT to the same frequency. If you do, neither WBT will work.

Locking your Settings (unless you share the same WBT for multiple LOGIQ 9 systems).

1. Press SET until you see LOC.
2. Press the up arrow button, YES appears. (To unlock, press the down arrow button, NO appears.)
3. Press SET.

Muting the WBT.

- Press the MUTE switch to noiselessly mute the WBT's audio signal. Remember, muting the transmitter DOES NOT switch off the transmitter.

Using the Wireless Transmitter (continued)**Signal and Battery Status Indicator**

The yellow LED on the top of the WBT lights up if the audio signal at the AF input is excessively high (AF peak).

The red LED provides information on the battery status.

- If the LED is lit up, the transmitter is switched on and the battery capacity is sufficient.
- If the LED is flashing, the battery is too weak. You should replace the battery IMMEDIATELY. With a weak battery, voice recognition accuracy diminishes significantly.

NOTE: *Please refer to the Installing the Batteries into the Wireless Transmitter instructions above.*

Deactivating the WBT.

Turn off the transmitter when it is not in use.

1. To turn off the WBT, press the ON/OFF button until OFF appears on the display.
2. After OFF appears on the display, you can then release the ON/OFF button. The red LED turns off.

Positioning the Microphone

Effective use of VoiceScan is highly dependent upon correctly positioning the microphone, specifically,

1. Use the transmitter clip to place the transmitter on your belt or clothing. Place the microphone on your ear and position the microphone so that it is one inch away from the corner of your mouth
2. Attach the cord clip to the microphone cord and attach the clip to your clothing to avoid changing the microphone position during scanning.

Voice Training (and Wireless Microphone and Audio Setup)

You need to train VoiceScan to respond more efficiently to your voice commands. This training takes about 15 minutes to complete.

NOTE: *If your system has not yet been set up for VoiceScan, the microphone and audio setup also takes place during the VoiceScan Training session. These instructions assume that you are completing the microphone and audio setup as well as performing the VoiceScan training.*

1. Select Utility-->System-->VoiceScan-->Change User. Press the pull-down arrow and select your User ID. Press OK.

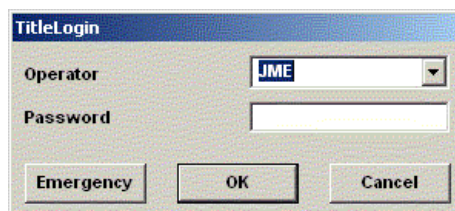


Figure 6-24. Change User

NOTE: *Your User Name only appears on the User Name list if you have first been set up as a user by your LOGIQ 9 administrator via Utility --> Admin --> Users.*

The following message appears: "A voice profile doesn't exist for this user. The audio setup will be performed on the default untrained user." Press OK.

Voice Training (and Wireless Microphone and Audio Setup) (continued)

2. Select Utility-->System-->VoiceScan-->Voice Training --> Train. The ViaVoice User Wizard appears.

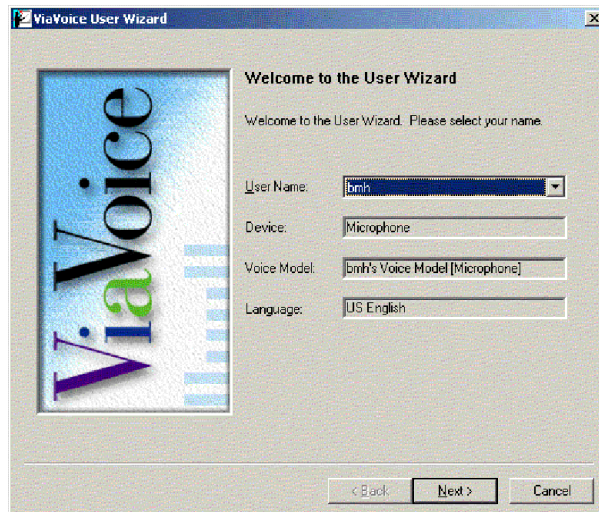


Figure 6-25. Welcome to the User Wizard VoiceScan Training

3. Verify that your User Name appears. If not, press the down arrow and select your User Name. Press *Next*.

NOTE:

If the VoiceScan microphone and audio setup has already been completed, please go to Step 13 below.

Voice Training (and Wireless Microphone and Audio Setup) (continued)

4. Setting Up Your Microphone. Press *Next*.

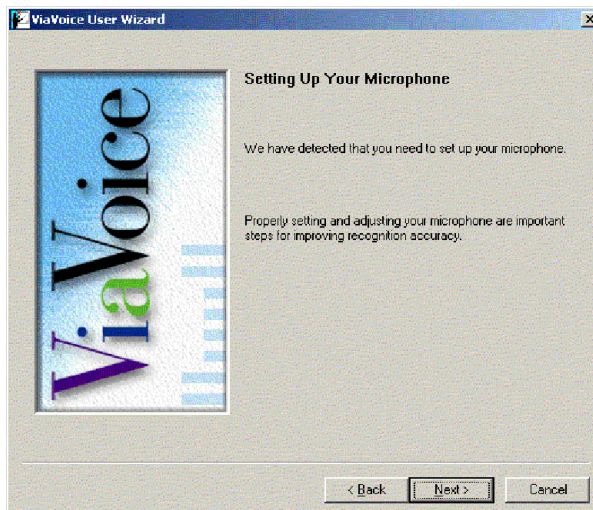


Figure 6-26. Setting Up Your Microphone

5. Setting Up Your Audio Components. Select *Set up the microphone* and *Other microphone*. Press *Next*.

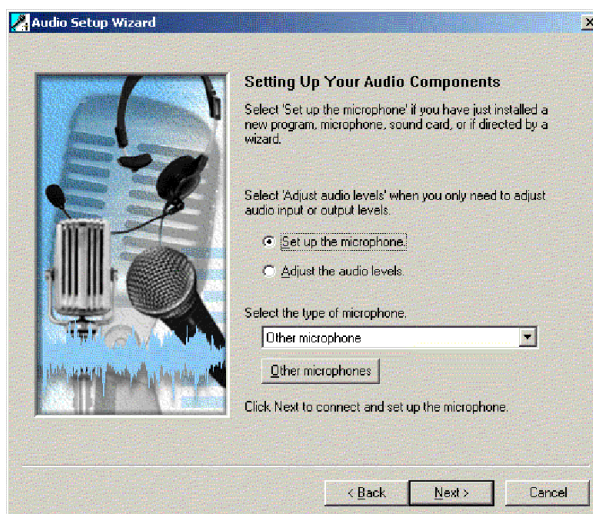


Figure 6-27. Setting up Your Audio Components

Voice Training (and Wireless Microphone and Audio Setup) (continued)

6. Selecting a Sound Card. **Input:** For a wired headset, select *Plantronics Headset*; for a wireless headset, select *AK5370*. **Output:** For systems without an internal MO drive, select *SoundMAX Digital Audio*; for systems with an internal MO drive, select *SB Live! Wave Device*, *SB Live! Audio [D880]*. Press *Next*.

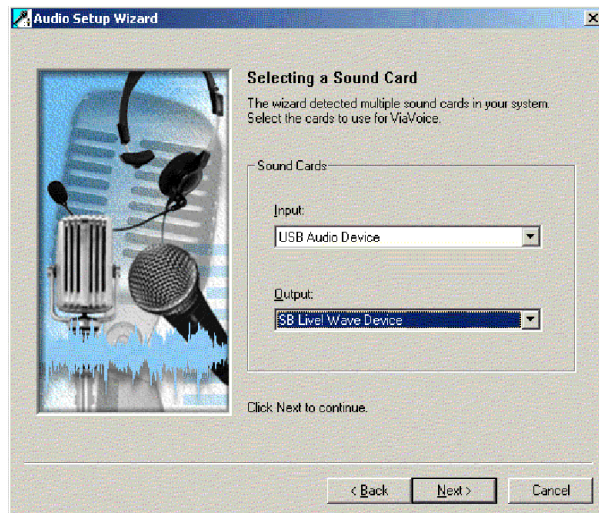


Figure 6-28. Selecting a Sound Card

7. Gathering Information. Select *USB connector*. Press *Next*.

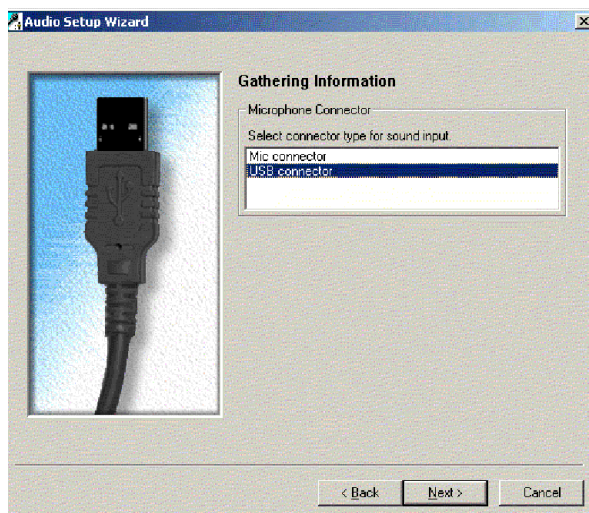


Figure 6-29. Gathering Information, Microphone Connector

Voice Training (and Wireless Microphone and Audio Setup) (continued)

8. Gathering Information. Select *Computer Speaker*. Press *Next*.

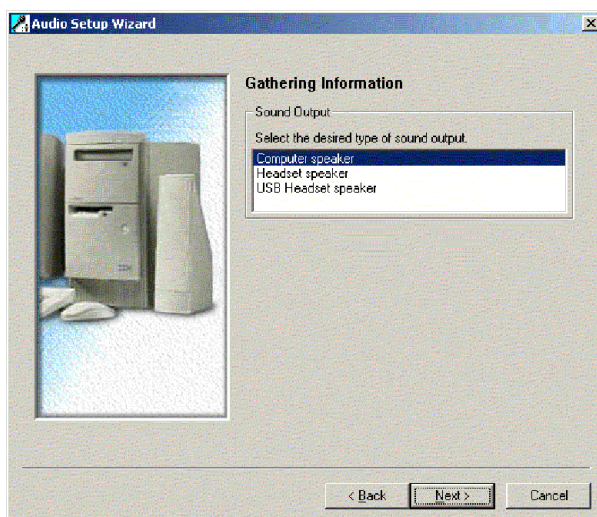


Figure 6-30. Gathering Information, Sound Output

9. Connecting a USB Microphone. Press *Next*.

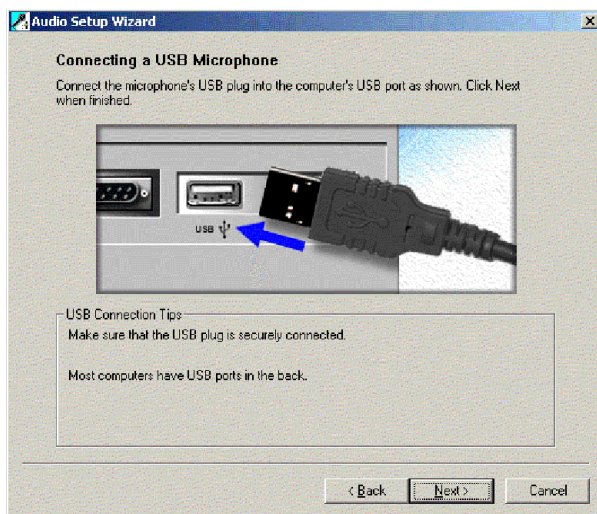


Figure 6-31. Connecting a USB Microphone

Voice Training (and Wireless Microphone and Audio Setup) (continued)

10. Testing the Audio Playback. Press *Test*. If you are able to hear sound from the speakers, press *Next*.

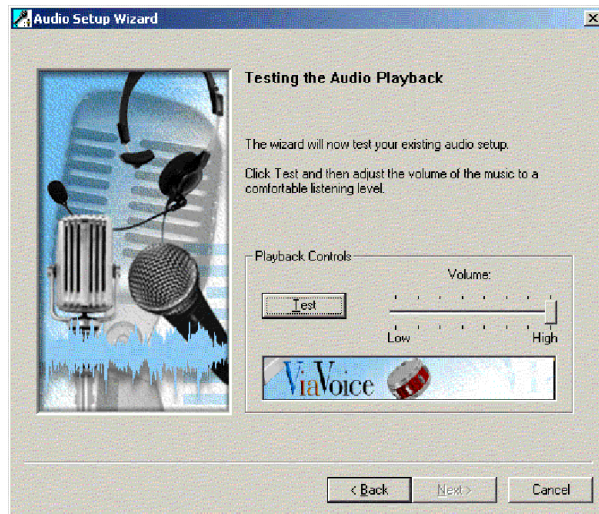


Figure 6-32. Testing the Audio Playback

NOTE:

If you are NOT able to hear sound from the speakers,

- Make sure "SB Live! Wave Device" is selected for the Output.
- Check the PC speaker connections.

Voice Training (and Wireless Microphone and Audio Setup) (continued)

11. Microphone Connection Complete. Press *Next*.

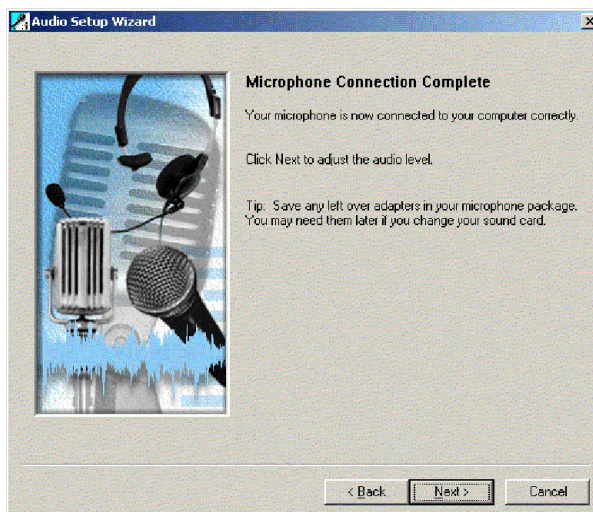


Figure 6-33. Microphone Connection Complete

Voice Training (and Wireless Microphone and Audio Setup) (continued)

12. Adjusting the Microphone Level. Press *Start*, then read the paragraph until you hear a tone. Follow instructions provided on the screen. After successfully completing this screen (audio quality Very Good, or better), press *Next*.

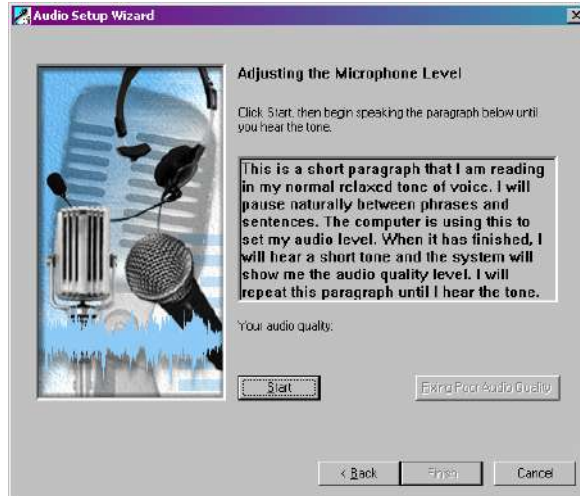


Figure 6-34. Adjusting the Microphone Level

Voice Training (and Wireless Microphone and Audio Setup) (continued)

13. Creating a Personal Voice Model. You are ready to start training when you reach the *Creating a Personal Voice Model* page. Press *Start*. Follow the instructions in the Wizard to create your voice model. You will be asked to read a short story to increase the recognition accuracy for your voice model. Each story takes a different amount of time to complete. When you click on a story, it tells you the estimated time to complete. During training, words become grayed out as the system processes them. If the system does not understand a word, it turns the word blue and you hear a buzz. Repeat the sentence, beginning with the blue word. *Processing...* appears after you have completed reading a section. Press next when the system indicates *Processing finished*. click *Next*.

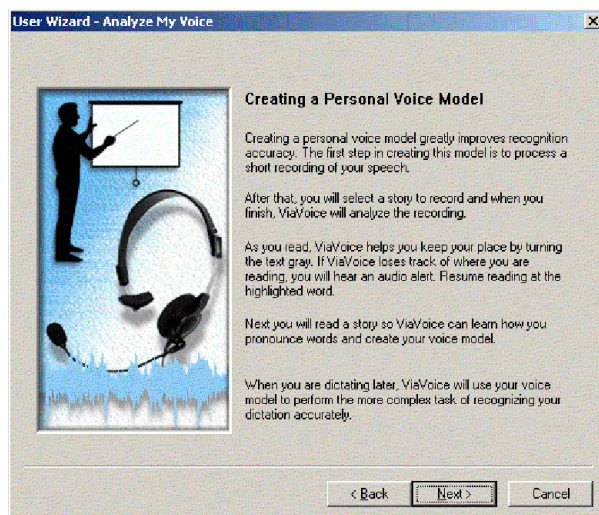


Figure 6-35. Creating a Personal Voice Model

Voice Training (and Wireless Microphone and Audio Setup) (continued)

14. Analyzing My Voice. Select *Record my voice (recommended)*. Press *Next*.

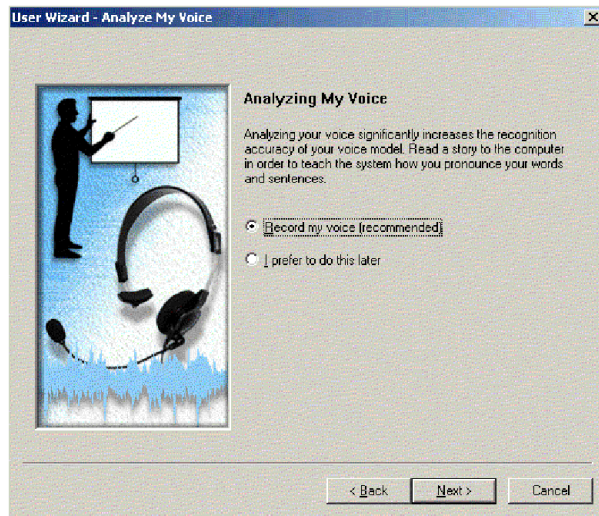


Figure 6-36. Analyzing My Voice

15. Choosing a Story. Select a story, then press *Next*.

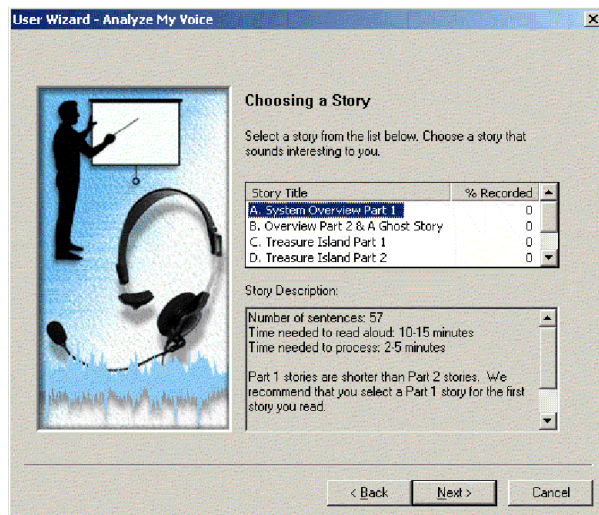


Figure 6-37. Choosing a Story

Voice Training (and Wireless Microphone and Audio Setup) (continued)

16. The story you selected. When you're ready to read the story you selected, press *Start*. As you go along, you can gauge your progress via the *% (percent) recorded* indicator. To pause, press *Click to Pause*. To resume, press *Click to Resume*. If you hear a tone, restart the sentence, starting with the blue, underlined word.

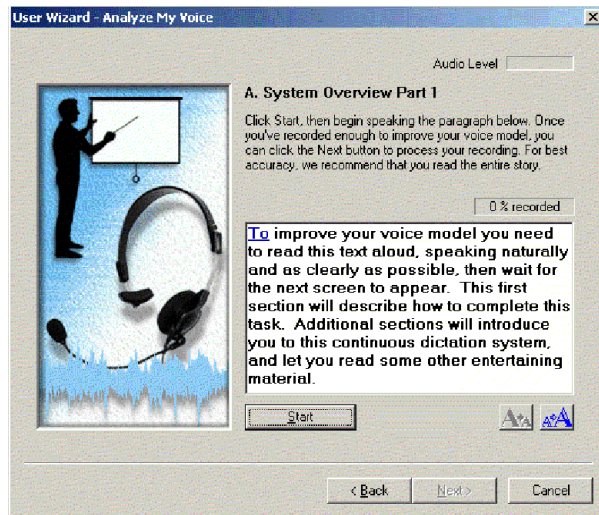


Figure 6-38. The Story You Selected

Voice Training (and Wireless Microphone and Audio Setup) (continued)

At the end of your recitation, the *Story finished* pop-up message appears. Press *Analyze* and wait. The system processes your reading selection. Press *Finish* when the processing has been completed.

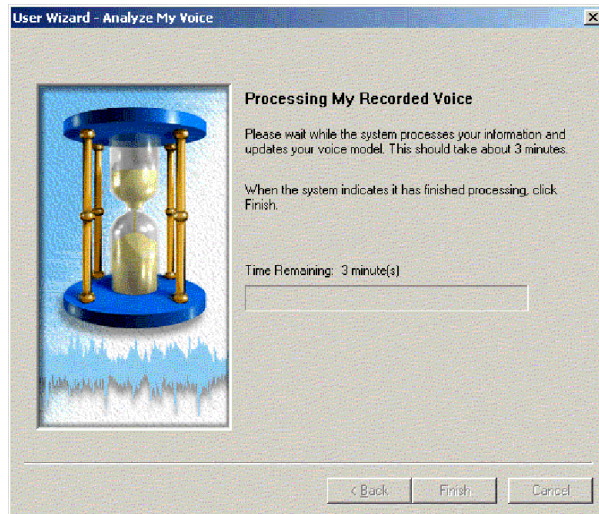


Figure 6-39. Processing My Recorded Voice

NOTE: *If you would like to copy a user's settings to another LOGIQ 9 system, you can use the Backup and Restore to do so. Go to the Utility --> System --> Backup/Restore page to backup Voice Training (Current User Only). Then use Detailed Restore User Defined to restore the Voice Scan Training Profiles to another LOGIQ 9 system.*

NOTE: *All users can be backed up by selecting "User defined configuration."*

Non-Wireless Microphone Setup Instructions

New LOGIQ 9 Systems

Plug in the Microphone to the USB port on the front of the LOGIQ 9 system.

Upgraded LOGIQ 9 Systems

Plug in the Microphone to the USB port located at the back of the LOGIQ 9 system.



Trip Hazard. Be careful when using the wired microphone to ensure that you do not trip over the wire.

Using VoiceScan

To perform an exam using VoiceScan, simply speak the appropriate command verbiage into the headset. There are some rules for the verbiage that you use when voicing a command. Press VoiceScan Help on Utility-->System-->VoiceScan to review the verbiage to use.

Activating/Deactivating VoiceScan

1. Select VoiceScan by pressing VS on the monitor display. The LOGIQ 9 now 'listens' for your voice commands and displays what it 'hears' on the monitor. The system states, "VoiceScan is listening" and "VoiceScan is activated" appears in the message line. The VS indicator has a green lines that move up and down as you speak.



Confirm that the correct command has been received (the message line displays the voice command received from the speaker).

You can verify that the LOGIQ 9 VoiceScan is listening by observing the green status lines on the monitor's VS indicator.

- a. To pause VoiceScan, say "LOGIQ pause." The system sounds a bell, the VS indicator blinks, and "VoiceScan is paused" appears in the message line while VoiceScan is paused .
 - b. To unpause VoiceScan, say "LOGIQ resume." The system sounds a bell, responds, "VoiceScan is listening," and "VoiceScan is activated" appears in the message line.
2. To deactivate VoiceScan, press VS or say "LOGIQ quit". The system states, "Good-bye" and "VoiceScan is deactivated" appears in the message line.

Available VoiceScan Controls

All of the available VoiceScan commands are available by pressing Utility--> System--> VoiceScan--> Command List.

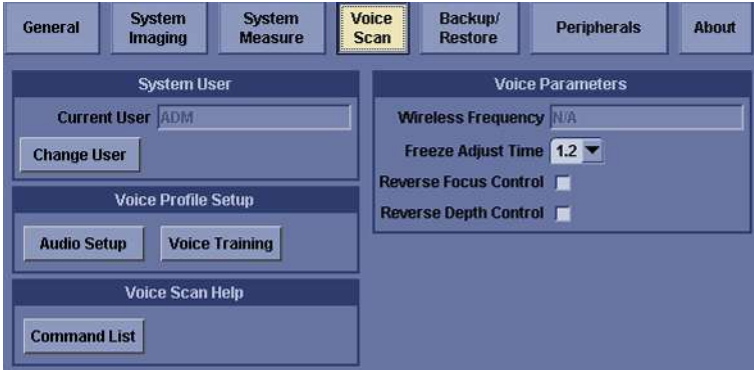


Figure 6-40. VoiceScan Utility Page

VoiceScan Help appears. Navigate to the topic for more information. (Only available in English.)

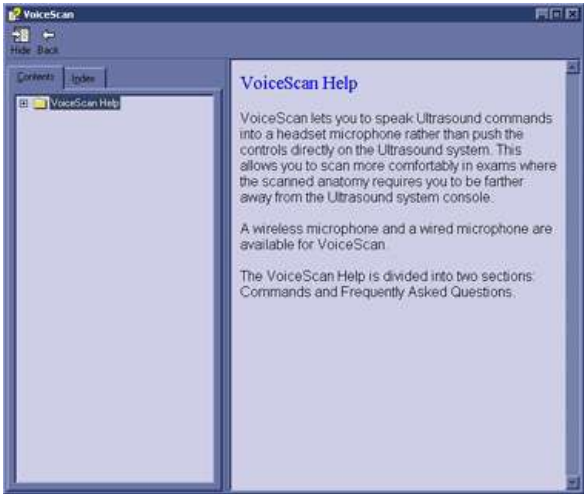


Figure 6-41. VoiceScan Controls Help

Touch Panel Controls

You can adjust Touch Panel controls via VoiceScan. The Touch Panel is divided into a grid, as shown below:



Figure 6-42. Touch Panel VoiceScan Grid

For example, to adjust Rejection, say Touch G2 up; to activate CE, say Touch G4; or to go to the next page of the Touch Panel, say F4. Gray Maps is not yet available via VoiceScan.

Assigning a Calc to a Vessel (Carotid Application Only)

To assign a calculation to a vessel, with Auto Calcs on, after you have completed the measurement, specify the Location and Vessel. For example, say "Vessel Left Prox CCA" or "Vessel Right ICA" or "Vessel Mid Right None".

Graphic
Representations of
Rules

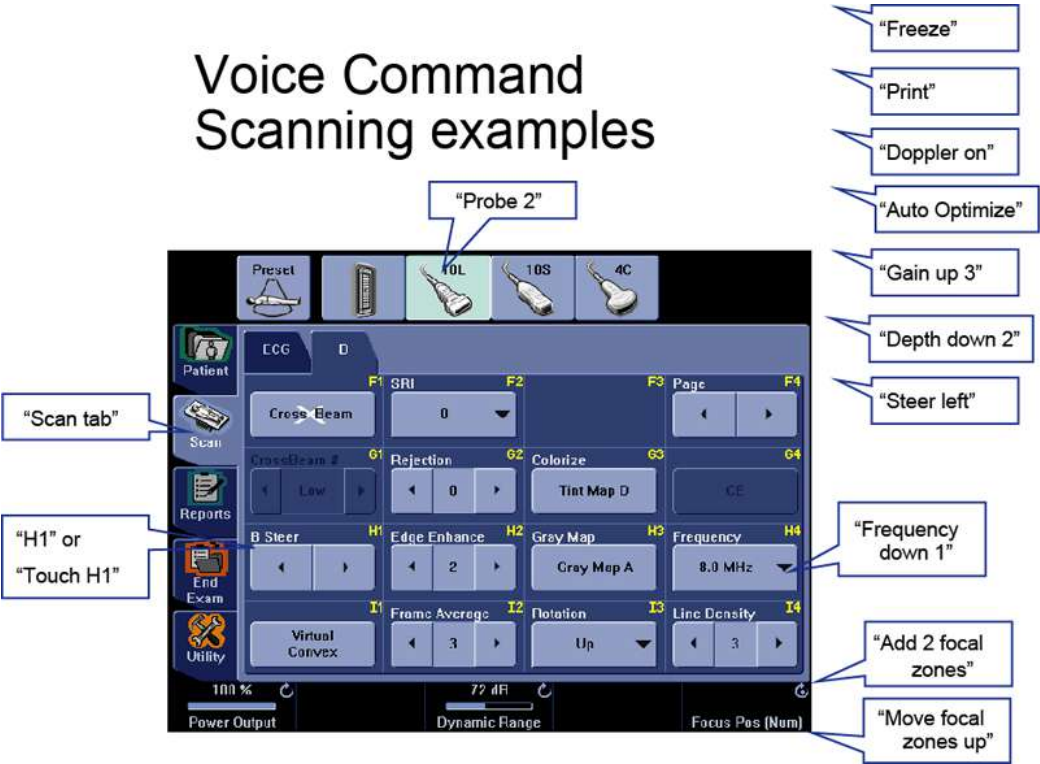


Figure 6-43. Example of VoiceScan Rules

Graphic
Representations of
Rules (continued)

Voice Command
Comment examples

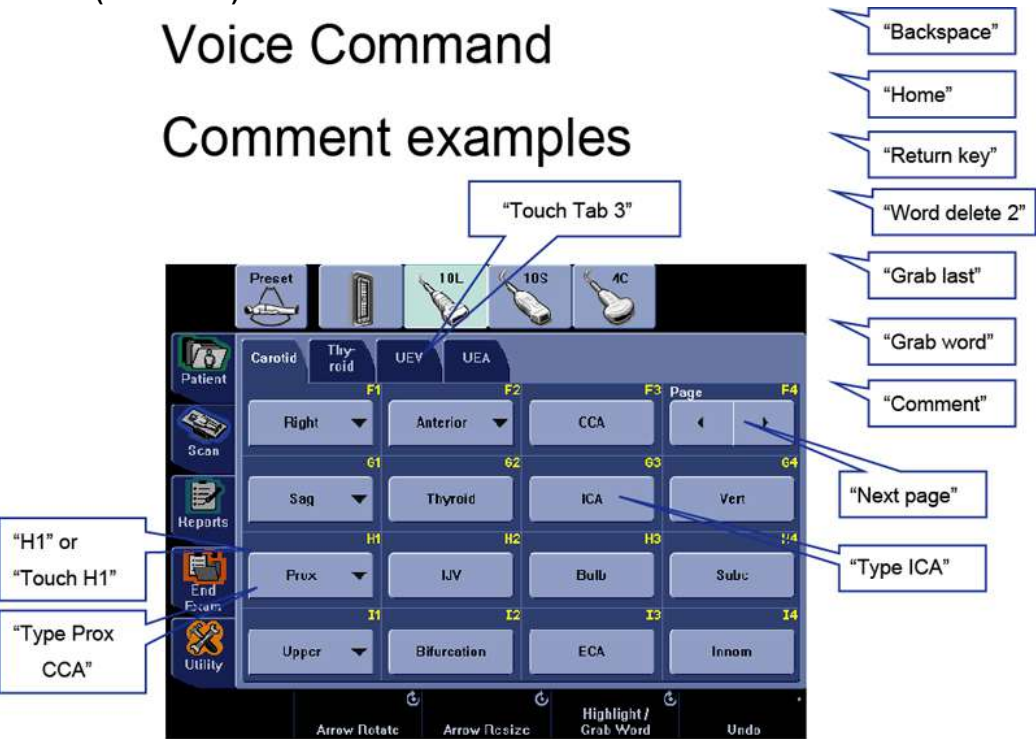


Figure 6-44. Example of VoiceScan Rules

Graphic
Representations of
Rules (continued)

Voice Command AutoCalcs examples

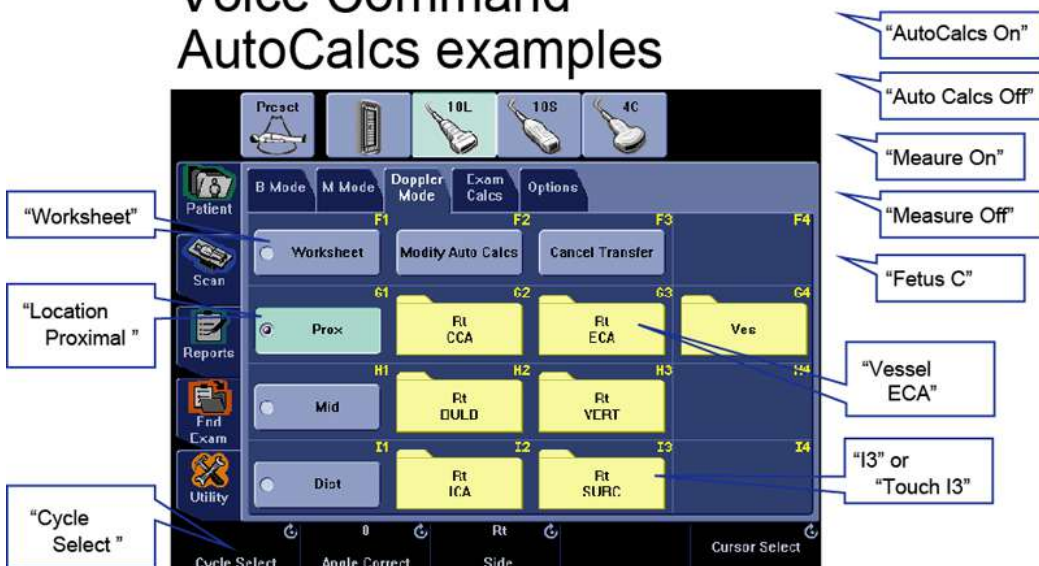


Figure 6-45. Example of VoiceScan Rules



HINTS

- Log in as the correct user so that the proper training file can be used.
- The microphone has a mute button
- The Touch Panel grid appears when you activate VoiceScan.
- The system automatically CINEs back upon Freeze.
- You don't have to wait for the previous command to complete before saying the next command.
- Keep the Doppler volume at its lowest acceptable level.
- Don't speak too loudly.
- The microphone position is very important (off the corner of your mouth, not directly in front of your mouth).
- If accuracy seems diminished, check the batteries. If ok, repeat the audio setup. Generally, you will not have to repeat the training.
- If the yellow light is always on, adjust the sensitivity.

Electronic Documentation

Documentation Distribution

- Release Notes (supplied on paper)
- Quick Guide or User Guide (supplied on paper)
- AIUM Acoustic Output Booklet
- AIUM Acoustic Output Booklet (USA Only)
- Online Help (on the Ultrasound Scanner via F1)
- Electronic media. You can view user documentation on a PC or on the Ultrasound Scanner via the Customer Documentation media, which includes:
 - Basic User Manual
 - Advanced Reference Manual
 - Quick Guide
 - User Guide
 - Quick Card(s)
 - Release Notes and Workarounds
 - Basic Service Manual

NOTE: *All user documentation is provided in multiple languages if translations were available at the time of media publication.*

Using Online Help Via F1

Online Help is available via the F1 key. After pressing F1, Help appears. The Help screen is divided into three sections: navigational tools on the top, left portion of the screen (Hide, Back, Forward), help book navigational tools on the left portion of the screen (Contents, Index, Search, Favorites), and the content portion on the right side of the screen where help topics are displayed.

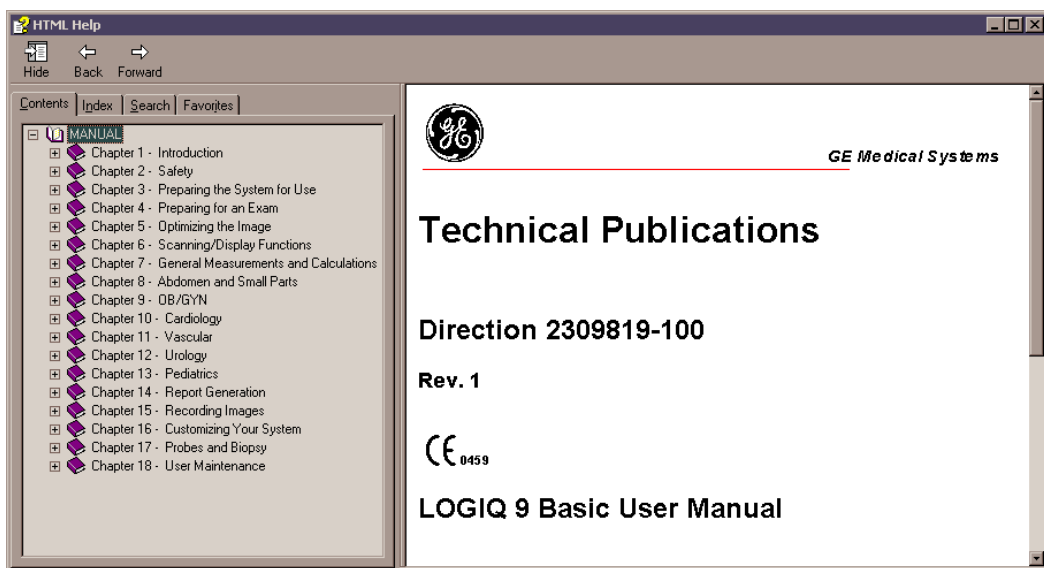


Figure 6-46. Opening Help Screen

Navigating through the Help Book

Online Help is organized like a manual, with individual chapters, sections, and pages. Click on the plus (+) sign next to MANUAL to open up the book. Click on the plus sign next to the chapter you want to view to open up that chapter. Click on the plus sign next to the chapter you want to view to open up that section. Open up the page to view that page's information.

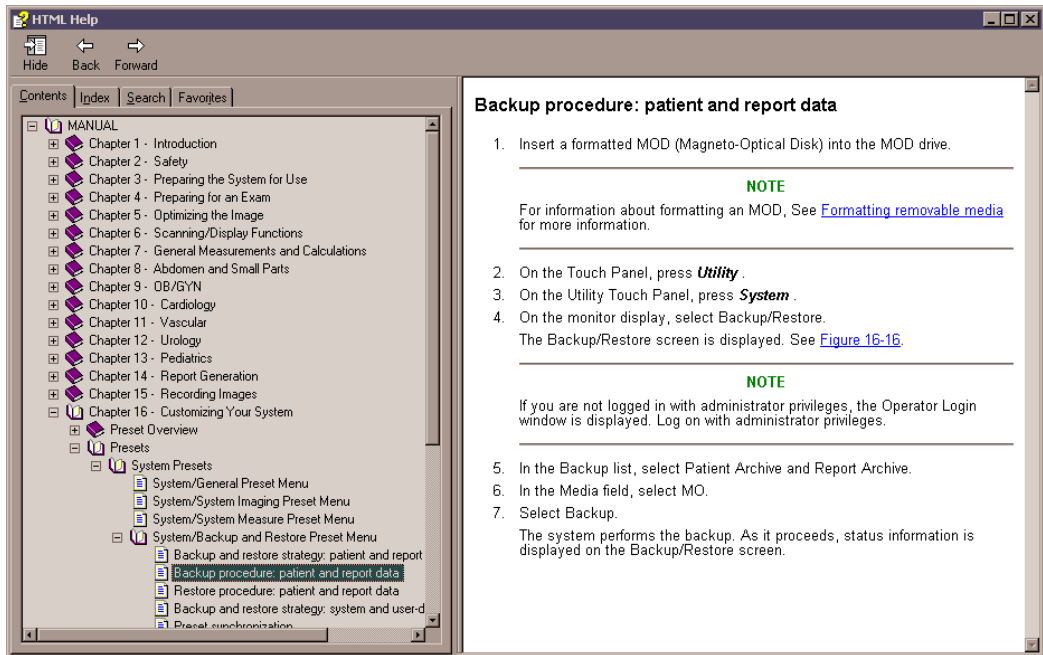


Figure 6-47. Sample Help Topic

The blue, underlined text links you to related topics. Click on the link to move to the new topic.

Links

After you click on a blue, underlined portion of text, the screen updates with this link's content. To go back to the previous screen, press Back. To return to the link, press Forward.

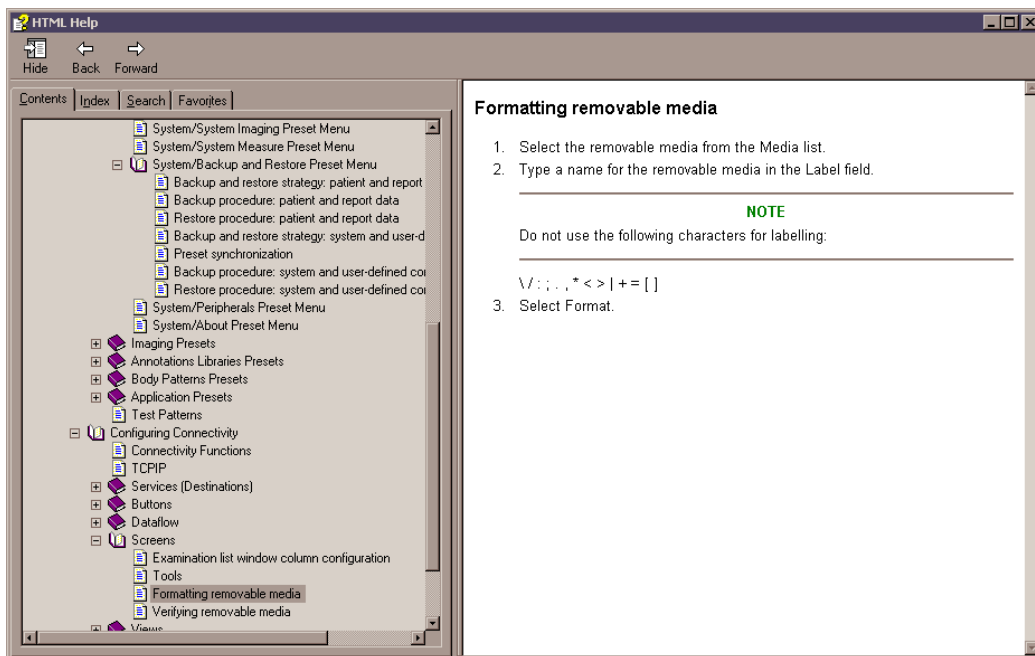


Figure 6-48. Topic Link

Searching for a Topic

To search for a specific topic, click on the Search tab in the left portion of the screen. Type in the topic name in the *Type in the keyword to find:* field. Topics with the word or phrase you typed appear in the *Select Topic to display:* area. Either double click on the topic you want to view or highlight the topic and press the Display button to view this topics.

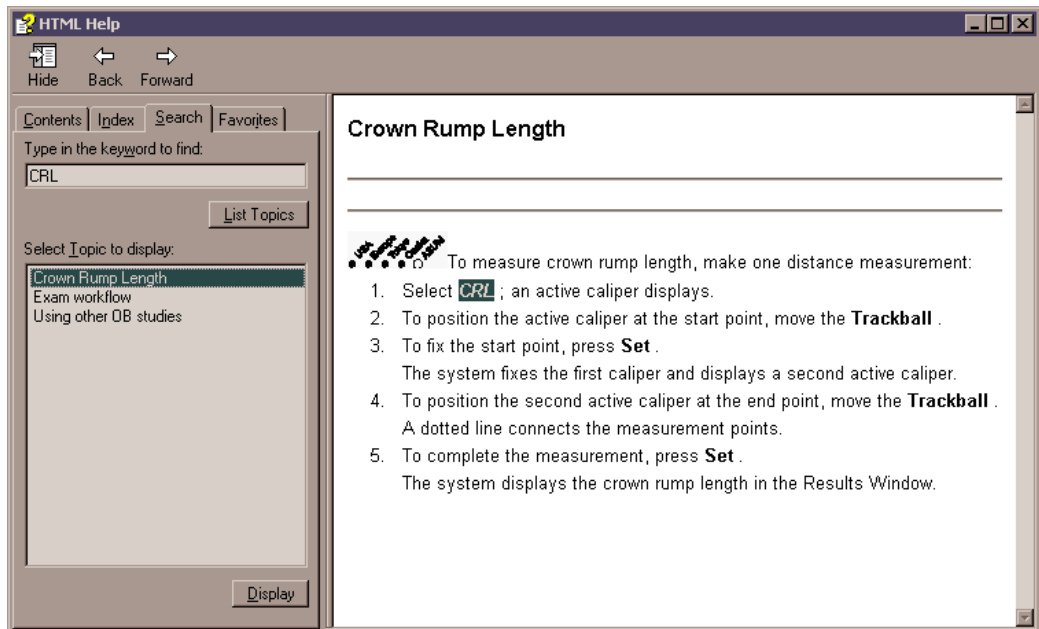


Figure 6-49. Search Results

Saving Favorite Topics

You may find that there are topics you need to refer to often. In this case, it's a good idea to save these topics as Favorites. To save a topic as a favorite, press the Favorites tab, highlight the topic in the Topics window, and press the Add button. You can now view this topic quickly by going to the Favorites help tab.

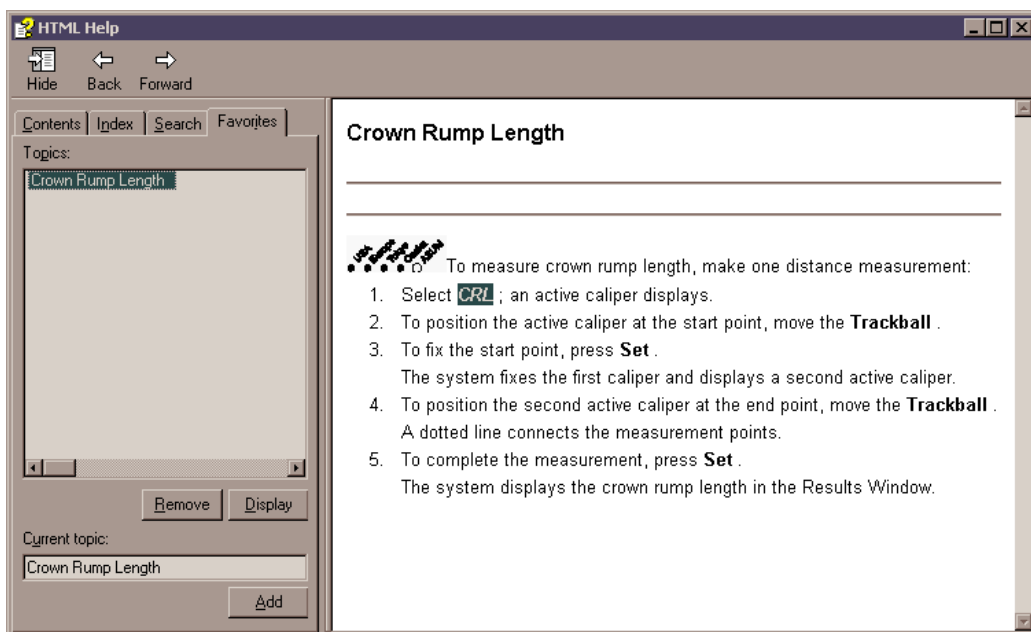


Figure 6-50. Adding Favorites

Using the Index

Or, you can look for topics by using the Index. Press the Index tab, then use the scroll bar to look up a topic.

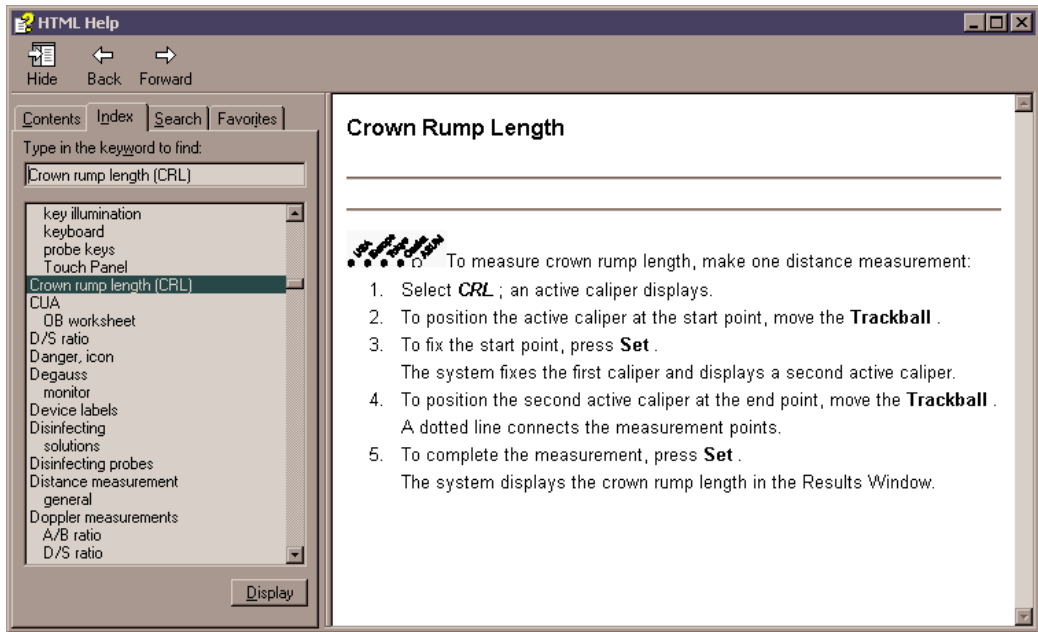


Figure 6-51. Index

Other Help Features

To hide the left side of the screen, press the Hide icon at the upper, left-hand portion of the screen. To view the left side of the screen again, press the Show icon at the upper, left-hand portion of the screen.

To size the Help window, position and hold down the cursor at the corner of the screen while moving the Trackball.

To move the Help window to the Touch Panel display, position and hold down the cursor at the very top of the Help window while moving the Trackball to the Touch Panel display.

Exiting Online Help

To exit Online Help, press the 'X' in the upper, right-hand corner of the Online Help window.

Electronic media

Accessing Documentation Via a PC

To view user documentation on a PC,

1. Insert the media into the media drive.
2. Open the media drive on your desktop.
3. Double click on the 'gedocumentation.html' document.
4. Select the item you want to view (click on the blue, underlined link in the File Name column).

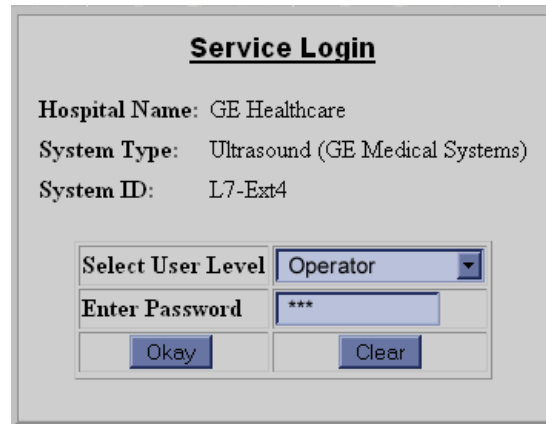
To close the window, click on the 'X' in the upper, right-hand corner of the browser window.

NOTE: *If your PC does not have Adobe Reader, a free download is available on the Adobe website at <http://www.Adobe.com>.*

Accessing Documentation on the Ultrasound Scanner Via the media

To access documentation via the media,

1. Select Utility. Select Service. Wait until the logon screen appears.
2. Logon as 'Operator' next to Select User Level. Enter the following password: 'uls'. Press Okay.



The image shows a 'Service Login' dialog box. At the top, the title 'Service Login' is underlined. Below the title, there are three lines of text: 'Hospital Name: GE Healthcare', 'System Type: Ultrasound (GE Medical Systems)', and 'System ID: L7-Ext4'. Below this text is a table-like structure with two columns. The first column contains the labels 'Select User Level' and 'Enter Password'. The second column contains a dropdown menu with 'Operator' selected, a text box with '***', and two buttons labeled 'Okay' and 'Clear'.

<u>Service Login</u>	
Hospital Name: GE Healthcare	
System Type: Ultrasound (GE Medical Systems)	
System ID: L7-Ext4	
Select User Level	Operator
Enter Password	***
Okay	Clear

Figure 6-52. Operator Service Browser Login

Accessing Documentation on the Ultrasound Scanner Via the media (continued)

3. Press Utilities.

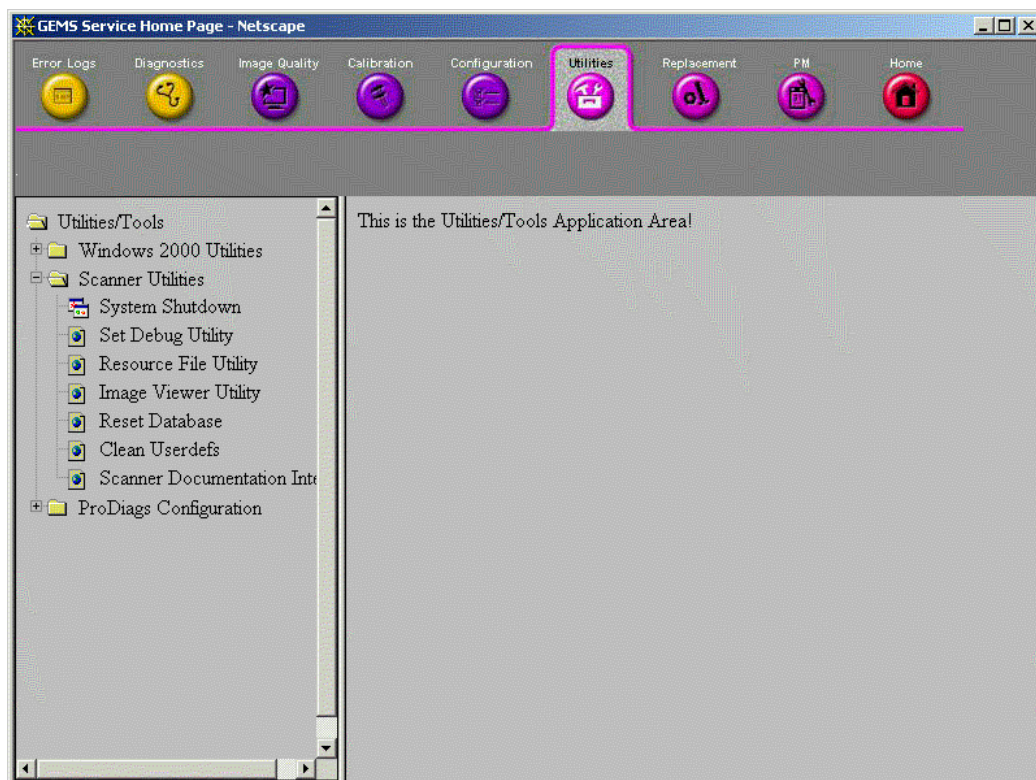


Figure 6-53. Service Browser Home Page

4. Insert the media.

Accessing Documentation on the Ultrasound Scanner Via the media (continued)

5. Select Scanner Utilities.

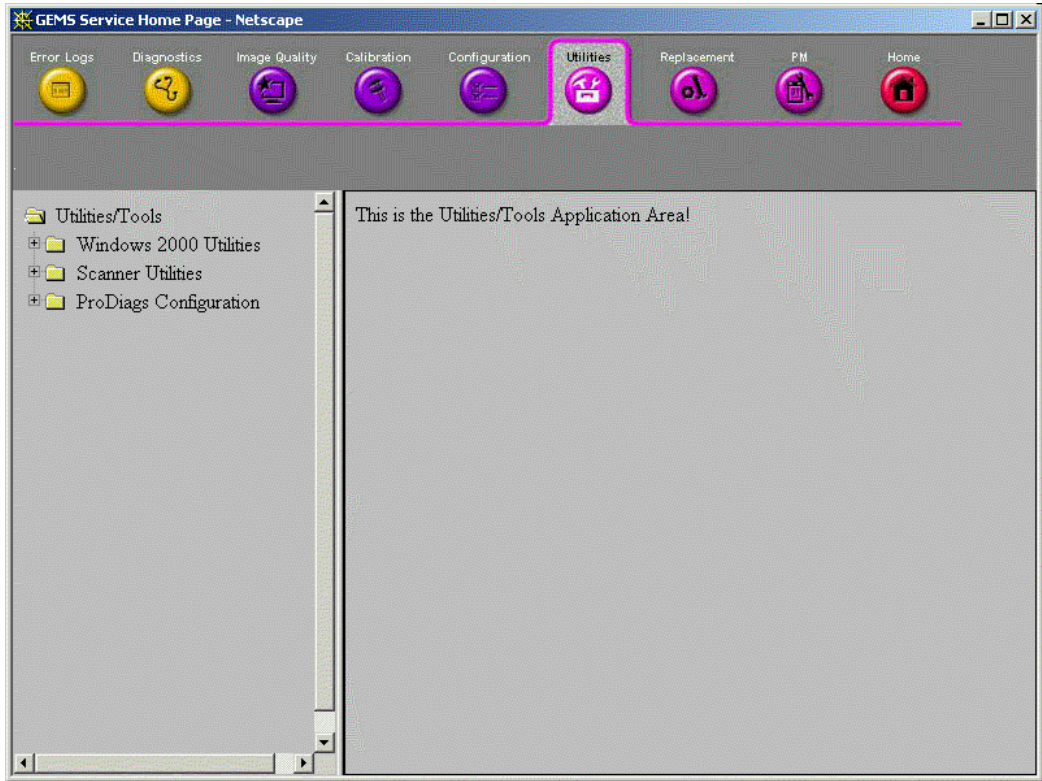


Figure 6-54. Scanner Utility Page

6. Select Scanner Documentation Interface.

Accessing Documentation on the Ultrasound Scanner Via the media
(continued)



Figure 6-55. Document Selection Page

- 7. Scroll to find the document and double click on it to open it.

Accessing Documentation on the Ultrasound Scanner Via the media (continued)

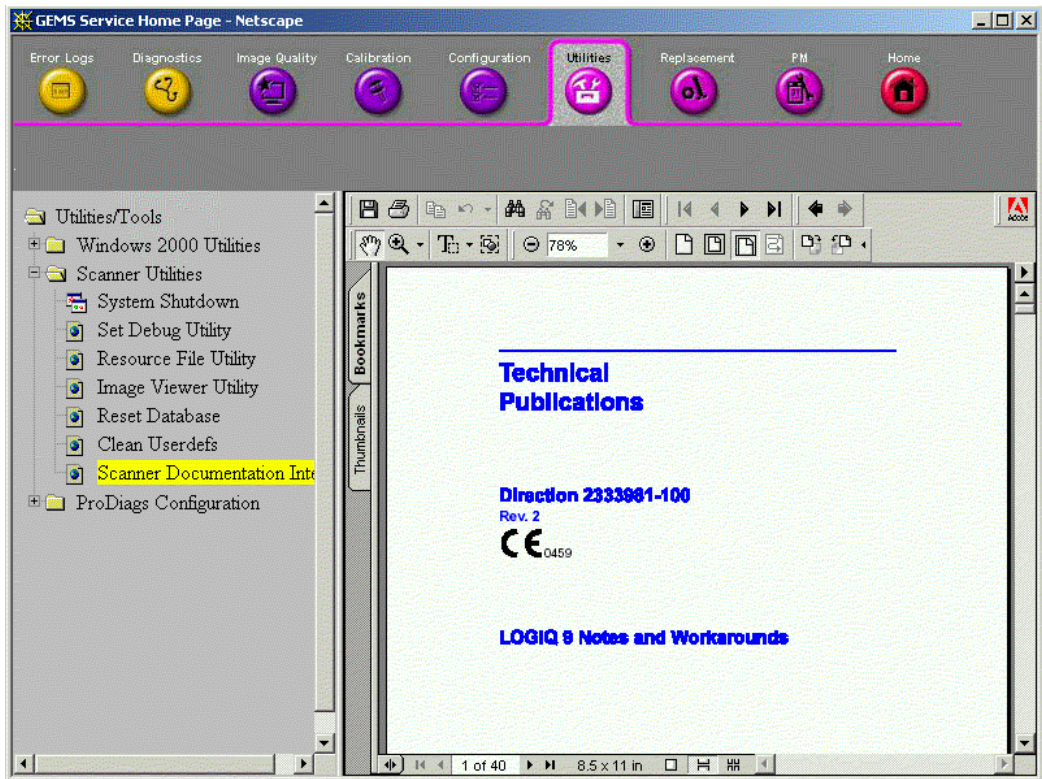


Figure 6-56. Selected User Document

NOTE: You can search through a document, use hyperlinks in the Table of Contents and Index to locate topics, and navigate via bookmarks.

NOTE: In addition to viewing documentation on the Ultrasound system, the Documentation media can be read on any PC.

To exit, press the 'X' in the upper, right-hand corner of the documentation window.

Chapter 7

General Measurements and Calculations

Describes how to perform general measurements and calculations.

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by system accuracy, but also by the use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

Overview

This section provides information about taking measurements and describes calculations available in each mode. It includes the following topics:

- Exam workflow
- Location of measurement controls
- Description of calipers
- List of generic measurements
- General information about taking measurements
- Mode Measurements: Step-by-step instructions for taking specific measurements, organized by mode
- Basic steps to view and edit worksheets

Exam workflow

For each patient, the system organizes information by exam category, study, and measurement. The definitions of these terms are as follows:

- **Exam Category** – categories include the following:
 - Abdomen
 - Obstetrics
 - Gynecology
 - Cardiology
 - Vascular
 - Urology
 - Small Parts
 - Pediatrics
- **Study/Preset** – after you choose an exam category, the system allows you to select a study. For example, when you choose the Obstetrics exam category, you can choose one of the following studies:
 - Generic
 - OB-1
 - OB-2/3
 - OB-General
 - Fetal Heart
 - OB/GYN Vessel
- **Measurement** – the measurements and calculations needed to analyze an item of anatomy. For example, a femur length is a measurement. A measurement can include several pieces of measurement data. For example, to calculate the area of a gestational sac, you need to measure width, length, and depth.

Exam workflow (continued)

The following shows a workflow example for an obstetrics exam.

NEW PATIENT				
(Exam Category)				
Abdomen	OBSTETRICS	Gynecology	Cardiology	Pediatrics
(Study)				
Generic	OB-1	OB-2/3	OB-General	Fetal Heart
(Measurement)				
GS	CRL	BPD	FL	

For details on how to start a new patient, See 'Beginning a New Patient' on page 4-3 for more information.

Location of Measurement Controls

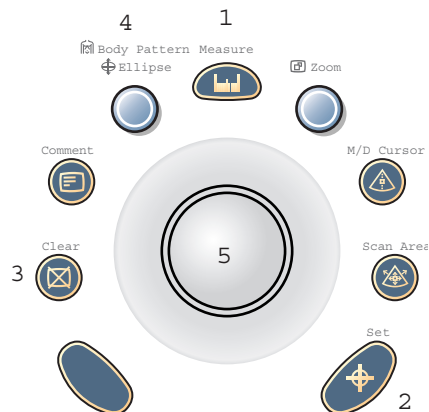


Figure 7-1. Locating Measurement Controls

1. **Measure.** Activates a measurement caliper and the associated calculation package.
2. **Set.** Fixes the caliper for measurements and completes the measurement sequence.
3. **Clear.** During a measurement sequence, erases the measuring caliper and measurement data from the display. When not performing a measurement sequence, clears all calipers and measurements from the display.
4. **Ellipse.** After the first caliper for a distance measurement has been set and the second caliper positioned, **Ellipse** activates the area/ellipse measurement function. During the ellipse adjustment, use the **Trackball** to increase or decrease the size of the ellipse. Select **Cursor Select** to adjust the measurement calipers.
5. **Trackball.** Moves the measurement calipers, selects the measurement on the Summary Window.

Trackball also selects items on the Touch Panel.

Description of calipers

While you are making a measurement, the measurement caliper is either active (open plus sign) or fixed (closed plus sign). An active caliper is green and a fixed caliper is yellow.

The system allows you to identify measurements by number or by unique symbol. If you choose Number as the Cursor Type, after you complete a measurement, it is assigned a number. If you choose Symbol as the Cursor Type, after you complete a measurement, the caliper symbol changes to one of the nine shown below. The symbols are used in sequence as listed. The first symbol is used for the first measurement, the second symbol for the second measurement, and so on. The numbers or symbols also identify measurements in the Results Window.



Figure 7-2. Fixed Caliper Symbols

For information about how to choose Cursor Type of Number or Symbol, See 'System/System Measure Preset Menu' on *page 16-18 for more information.*

Measurement line display

While you are making a measurement, the system displays a dotted line to show the measurement. After you press **Set** to complete the measurement, the dotted line remains on the display if the Show Caliper Line in the Options tab is selected. If Show Caliper Line is not selected, the system erases the dotted line and only the measurement calipers with a number or symbol are displayed. The Show Caliper Line on the Options tab takes precedence over the Cursor Line Display preset, found on the System -> System Measure screen.

List of general measurements

The following table shows the types of general measurements available when you press **Measure** but do not choose a specific calculation. The type of measurement depends on the current scan mode and the number of times **Measure** is pressed.

Table 7-1: General Measurements by Mode

If you press Measure (while frozen)	MODE		
	B and CF	Doppler	M
Once	- Distance - Circumference and area (ellipse)	Velocity	Tissue depth
Twice	Circumference and area (trace, spline trace or open trace)	TAMAX and TAMEAN	Time interval
Three times	- Circumference and area (trace, spline trace or open trace) - Echo level	Two velocities, time and acceleration	Time and velocity
Four times	- Circumference and area (trace, spline trace or open trace) - Echo level	Time	

NOTE: You can preset the sequence of B and CF area measurements can be preset in the Utility -> Measure -> Advanced screen.

NOTE: The type of measurement repeats as you press **Measure**. For example, in M-Mode, when you press **Measure** the fourth time, the Tissue depth measurement is available, and so on.

General Instructions

You can take measurements in all modes and image formats, including real-time, frozen, CINE, or VCR playback. After you select an exam category, the available calculations are displayed on the Touch Panel.

Measurement and calculation results

As you take measurements, each measurement is given a sequential number on the display and in the Results Window. The system can display nine measurements on the screen at one time.

While you are taking a measurement, the value in the Results Window updates until you complete the measurement.

Once the Results Window has nine measurements, if you make any further measurements, the system erases the first measurement and adds the new measurement ("first in, first out").

Measurement graphics are kept while in cine scroll. The measurement graphic is redisplayed on the frame where it is taken, if preset on the Advanced M&A page.

Selecting a calculation

When you take measurements, you can select the calculation before you take the measurement or after you take it. For example, in Obstetrics, if you select the calculation before you take the measurement, the estimated fetal age is displayed as you take the measurement. If you select the calculation after you take the measurement, the estimated fetal age is displayed after you complete the measurement.

NOTE: *After you take a measurement, if you select a calculation and the measurement is not applicable for the calculation, then the system assumes you want to start the calculation. The system then uses the calculation for the next measurement.*

Selecting a calculation (continued)

If there is a measurement listed in the Results Window that has not been assigned a calculation, to assign the measurement:

1. Press **Measure**.
2. To select the measurement in the Results Window, move the **Trackball** to the measurement.
The measurement is highlighted.
3. Press **Set**.
The system displays a list of applicable calculations. For example, if it is a distance measurement, the list includes all distance calculations for the current study.
4. To select an item in the list, move the **Trackball** to highlight the item and press **Set**.
The system assigns the calculation to the measurement.

Selecting a measurement in a different application

While scanning a patient, you may find that you want to measure an item that is not in the current application. To select a calculation from a different application:

1. Press **Measure**.
2. On the Touch Panel, select **Exam Calcs**.
3. On the Touch Panel, select the **Exam Category**.
The system lists the exam categories.
4. Select the exam category that has the calculation you want to make.
The system displays the Touch Panel for the selected exam category.
5. Select the study and the desired measurement.
6. After you complete the measurement, to return to the original application, repeat steps 1–4.

NOTE: *This measurement **DOES NOT** appear on the original application worksheet.*

Specifying measurement and display options

The Options tab allows you to specify the following measurement and display options:

- Whether to display the Results Window and Summary Window
- The cursor size
- The number of heart rate cycles to use to calculate heart rate
- Where to position the Results Window on the monitor display
- Choose whether the Doppler waveform is automatically or manually traced.
- Choose to display the caliper line

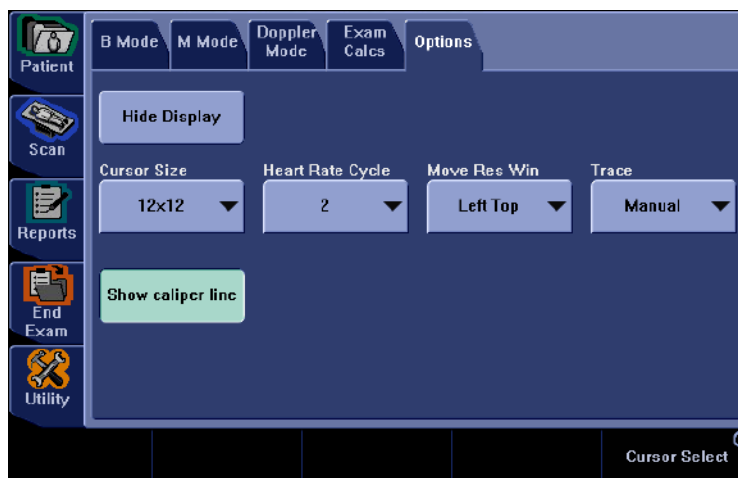


Figure 7-3. Measurement and Display Options

Specifying measurement and display options (continued)

Displaying the Results and Summary Windows

Typically, the Results Window and Summary Window are shown on the monitor display. If desired, you can remove them from the display. To remove them, select the **Options** tab, and then select **Hide Display**. The activation indicator on the Touch Panel key is lit. To re-display them, select **Hide Display** again.

NOTE: Hide Display removes both the Results Window and Summary Window from the monitor display.

NOTE: Hide Display removes both the Results Window and Summary Window from the monitor display.

Selecting cursor size

To select the cursor size, select the **Options** tab, and then select **Cursor Size**. The system displays the following choices:

- 12 x 12
- 9 x 9

Select the size you want.

Specifying heart rate cycles

When you measure heart rate, the system assumes a specific number of cycles are measured. You can change the number of cycles used in the calculation. To specify the number of heart rate cycles, select the **Options** tab, then select **Heart Rate Cycle**. The system displays a list of choices from 1 – 10. Select the number you want.

After you choose a number, the next time you measure heart rate, the system assumes you are measuring the specified number of cycles. That number is used when calculating the heart rate.

After changing patients, the system defaults to the number specified in the Heart Rate Cycle parameter. See 'M&A Advanced Preset' on page 7-52 for more information.

Specifying measurement and display options (continued)

Moving the Results Window

You may want to change where the Results Window is positioned on the monitor display. To move the Results Window, select the **Options** tab, then select **Move Res Win**. The system displays the following choices:

- Left Top
- Right Top
- Right Bottom
- Left Bottom
- Extreme Right Top
- Extreme Right

Select the position you want.

Trace Auto/Manual

To select Auto Doppler Trace or Manual Doppler Trace, select the **Options** tab, then select **Trace**. Toggle between Auto and Manual Doppler Trace by pressing the Touch Panel control.

- Select AUTO – the system traces the Doppler waveform from the begin time to the end time.
- Select MANUAL – trace the waveform manually.

Show Caliper Line

While you are making a measurement, the system displays a dotted line to show the measurement. After you press **Set** to complete the measurement, the dotted line remains on the display if the Show Caliper Line in the Options tab is selected. If Show Caliper Line is not selected, the system erases the dotted line and only the measurement calipers with a number or symbol are displayed. The Show Caliper Line on the Options tab takes precedence over the Cursor Line Display preset, found on the System -> System Measure screen.

Specifying measurement and display options (continued)

Post-assignment for Side/Location

1. Move the cursor to the result box (with the green frame) and press **Set**. The post-assignment menu appears.
2. Select Side and Location as needed.

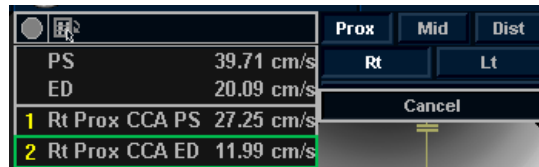


Figure 7-4. Post-assignment menu

NOTE: This feature is only available in manual calcs.

General Instructions (continued)

The following are actions you can take while performing measurements.



HINTS

- Before making measurements, to stop the acquisition of image data, press **Freeze**.
- For measurements such as distance, to make fine adjustments before completing the measurements, press **Measure** to toggle between active calipers.
- Before completing the measurement sequence, to erase the active measuring caliper and the current data measured, press **Clear**.
- After the sequence is complete, to erase all data that has been measured to this point, but not data entered on worksheet pages, press **Clear**.
- When there are several measurements on the display, to rotate through and activate previously fixed calipers, select **Cursor Select**. After a cursor is activated, you can change the measurement.

NOTE: If you want to change a trace measurement, you must erase it and trace again.

- To repeat any measurement, select that measurement again from the Touch Panel.

Calculation formulas are available in the *Advanced Reference Manual*.

Erasing measurements

The following actions erase measurements from the system's memory:



HINTS

- If you adjust the Trackball, unfreeze the image, or press Clear, the system erases all completed measurements and calculations on the display. Measurements and calculations, however, remain on the worksheets.
- If you select New Patient, the system erases all measurements and calculations on the display and clears the worksheets.
- If you make a new measurement that exceeds the maximum number of allowable measurements, the system erases the first (oldest) measurement and adds the new measurement.
- If the second caliper is active, to erase the second caliper and activate the first caliper, press Clear.

Measurement and Calculation Setup

Measurements and studies are organized for typical work flows. If you want, you can change this set up. You can specify which studies are in each exam category, and which measurements and calculations are in each study. You can change the measurements that are available on the Touch Panel. The LOGIQ 9 allows you to quickly and easily set up your system so that you can work most efficiently.

This section describes how to:

- Change a study to include different measurements
- Add a new study or measurement
- Remove a study from an exam category
- Change measurement parameters
- Create a measurement formula to correctly handle unit conversions
- Edit user-defined calculations
- Define application-specific measurement parameters
- Specify the default manual calc measurements for a selected study or folder

Starting Study and Measurement SetUp

You can make changes to studies and measurements in the Measurement & Analysis screen. To open the screen:

1. On the Touch Panel, select **Utility**.
2. On the Touch Panel, select **Measure**.
3. On the monitor display, select the **M & A** tab.

The system displays the Measurement & Analysis screen on the monitor display.

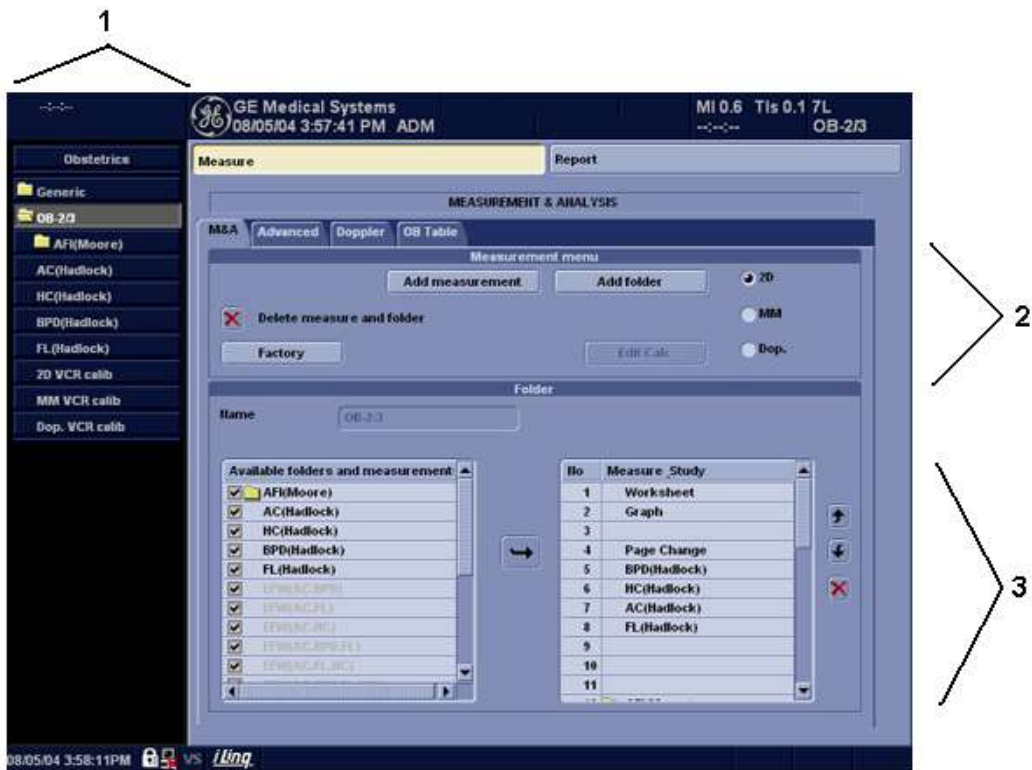


Figure 7-5. Measurement & Analysis screen

1. **Selection menu:** select exam category, study, or measurement.
2. **Measurement menu:** add and delete studies (folders) and measurements; select mode.
3. **Folder or measurement:** define studies and measurements. This section changes between Folder and Measurement, depending on what you select in the Selection menu.

NOTE: In the Measurement menu, the navigational tabs across the top may differ from system to system but the functionality is the same.

Selecting an exam category

When you open the Measurement & Analysis screen, it displays the exam category that was last used on the system. To select the exam category you want to work with:

1. Move the **Trackball** to highlight the exam category at the top of the Selection menu.
2. Press **Set**.
The system displays a list of exam categories.
3. Move the **Trackball** to highlight the exam category you want.
4. Press **Set**.

The Selection menu lists studies and measurements for the selected exam category.



Figure 7-6. Select Exam Category

Selecting the measurement mode

In the Measurement menu section of the Measurement & Analysis screen, select one of the following:

- 2D (B-Mode)
- MM (M-Mode)
- Dop (Doppler Mode)
- Plot (Plot Mode—The measurement on the plot graph of the TIC/QAnalysis)

The Selection menu lists studies and measurements for the selected mode.

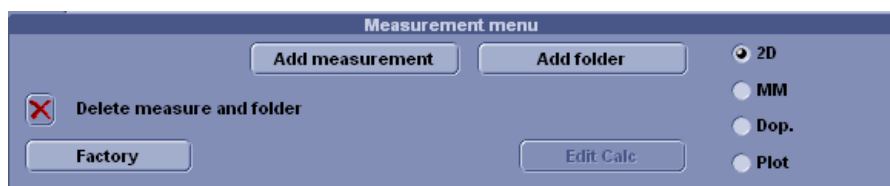


Figure 7-7. Select Mode

Selecting a study or measurement

To work with a folder or measurement, you must first select it in the Selection menu. The Selection menu lists the studies and measurements for an exam category. The studies and measurements are organized in a hierarchy, in the same order that they are organized on the Touch Panel. The following example shows the highest level of the Obstetrics exam category, with the OB studies listed.



Figure 7-8. Selection Menu: Exam Studies

Selecting a study or measurement (continued)

After you select a study, the Selection menu shows all folders and measurements in the study. The Folder section of the Measurement & Analysis screen changes, and lists the measurements. The Selection menu shows all measurements for the OB-1 exam category.

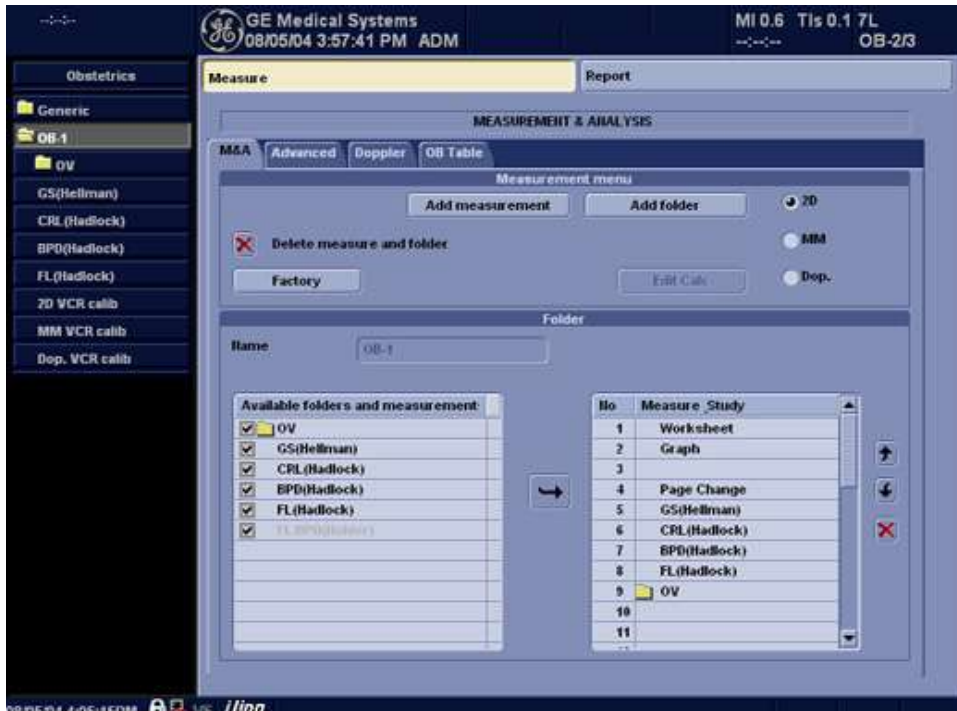


Figure 7-9. Selection Menu: OB-1 Exam Category

Selecting a study or measurement (continued)

The following example shows the Selection menu after the BPD measurement is selected. The Measurement section is now displayed, with information about the BPD measurement.



Figure 7-10. Selection Menu: BPD Measurement

Selecting a study or measurement (continued)

To select a folder or measurement:

1. Move the **Trackball** to the Selection menu and highlight the folder or measurement.
2. Press **Set**.
 - If you selected a folder, the system displays the folder in the Folder section of the Measurement & Analysis screen.
 - If you selected a measurement, the system displays the measurement in the Measurement section of the Measurement & Analysis screen.

NOTE: Items must be selected in the Available folders and measurements list to be in the Selection menu. To move or change an item that is in the Available folders and measurements list but not in the Selection menu, move the **Trackball** to the check box for the item, and press **Set**. The item is now listed in the Selection menu.



Figure 7-11. Available folders and measurements check boxes

Using folders

When you select a folder in the Selection menu, the system displays all folders and measurements that are in the folder. A folder can indicate a study, or can indicate a measurement group that contains related measurements. For example, a calculation such as OB Amniotic Fluid Index (AFI) requires four measurements, one of each quadrant. The AFI folder contains four measurements.

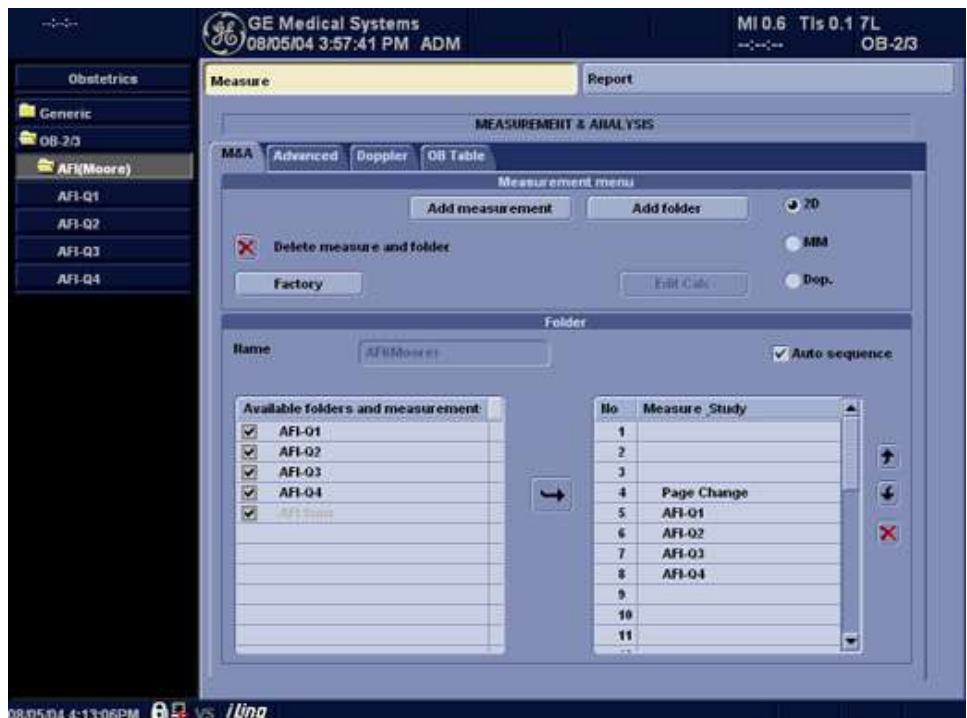


Figure 7-12. AFI Folder

Specifying Which Measurements Go in a Study or Folder

The Folder section of the Measurement & Analysis screen has two lists of folders and measurements. This is where you specify which items go in a study or folder.

- **Available folders and measurements.** The left list contains all possible folders and measurements for the selected study or folder.
- **Measure & Study.** The right list has all folders and measurements currently selected for the study or folder. These are selected from the Available folders and measurements list. These are the folders and measurements you see when you are scanning and choose an exam category. This also defines where the folder or measurement is located on the Touch Panel, based on the number in this list.

Specifying Which Measurements Go in a Study or Folder (continued)

To add an item to the Measure & Study list:

- 1. In the Measure & Study list, move the **Trackball** to highlight which folder you want to put the item in, and press **Set**.



Figure 7-13. Measure & Study list: Selecting item and position

- 2. Move the **Trackball** to highlight an item in the Available folders and measurements list, and press **Set**.

Specifying Which Measurements Go in a Study or Folder (continued)

3. Select the arrow between the lists. The item is copied to the Measure & Study list.



Figure 7-14. Measure & Study list: New item added

The selected item is now displayed in the Touch Panel and the Summary Window.

Specifying Which Measurements Go in a Study or Folder (continued)



Figure 7-15. Touch Panel for measurement added

NOTE: *If an item is already in the Measure & Study list, the system does not allow you to add it again. To move an item within the Measure & Study list, See 'Moving items in the Touch Panel' on page 7-30 for more information.*

Specifying Which Measurements Go in a Study or Folder (continued)

Touch Panel positions

Each Touch Panel has 16 positions, four across and four down. The items in the first row across are numbered 1–4, in the second row 5–8, and so on. This figure shows the OB/GYN Vessel Touch Panel and a portion of the Measurement & Analysis screen for the OB/GYN Vessel. The numbers are shown in the Measure & Study list in the Measurement & Analysis screen. For example, the second row in the Touch Panel shows Prox, Aorta, Umbilical, and Placenta. In the Measurement & Analysis screen, the Measure & Study list shows these as number 5–8.



Figure 7-16. Measure & Study display with Touch Panel positions

NOTE: Positions 1 through 4 are system programmed and cannot be changed.

Position 4 is blank when there is only one Touch Panel page. When a second page is available, "Page" is the fourth position which allows you to advance to the next/previous page.

Moving items in the Touch Panel

To move items that are in the Touch Panel, you move them in the Measure & Study list.

1. Move the **Trackball** to highlight an item in the Measure & Study list.
2. Select the up or down arrow.

The item is displayed at the selected position in the Touch Panel.

NOTE: *Some items cannot be moved. For example, Worksheet Display and Page Change are in the same place on each Touch Panel.*

Removing items from the Touch Panel

To remove items from the Touch Panel, you remove them from the Measure & Study list.

1. Move the **Trackball** to highlight the item in the Measure & Study list and press **Set**.
2. Select the X to the right of the list.

The system removes the item from the Measure & Study list and from the Touch Panel. The item is still listed in the Available folders and measurements list.



Figure 7-17. Measurement & Analysis: Remove item from Touch Panel

Removing items from the Touch Panel (continued)

To remove an item from the Selection menu:

1. Move the **Trackball** to the item in the Available folders and measurements list.
2. To clear the check box for the item, move the **Trackball** to the check box and press **Set**.

The system removes the item from the Selection menu, the Measure & Study list, and from the Touch Panel. It is also not listed in the Summary Window.

Setting up an automatic measurement flow

In some cases, related measurements are put in a measurement folder. This allows you to logically organize measurements. It also allows you to specify that the system automatically start each measurement in a folder, one after the other. This is the automatic sequence feature. To use this feature:

1. In the Selection menu, select the folder that contains the measurements you want.
The system displays the folder and lists the measurements.
2. In the Folder section of the Measurement & Analysis screen, select Auto sequence. For OB/GYN Vessel measurements, Umbilical Doppler measurements are put in the Umbilical measurement folder.



Figure 7-18. Measurement & Analysis screen: Auto sequence

1. Selection Menu

2. Auto Sequence

Changing Measurements

You can make changes to some of the measurements. For example, Head Circumference can be measured with an ellipse, a trace, or two distances. You can specify which measurement type you want the system to use as the default. You specify the measurement type by selecting the tool to use to make the measurement.

- To change the tool used to make a measurement:

In the Measurement section of the Measurement & Analysis screen, select the desired tool from the Tool list. Select the arrow to display the drop-down list.

NOTE: *If the Tool field is gray, it cannot be changed.*

After you choose the tool, this is what the system expects when you scan and choose this measurement.

NOTE: *The diagram to the right of the Tool list shows the measurement type. In the following example, ellipse is selected and the diagram shows an ellipse.*

Changing Measurements (continued)

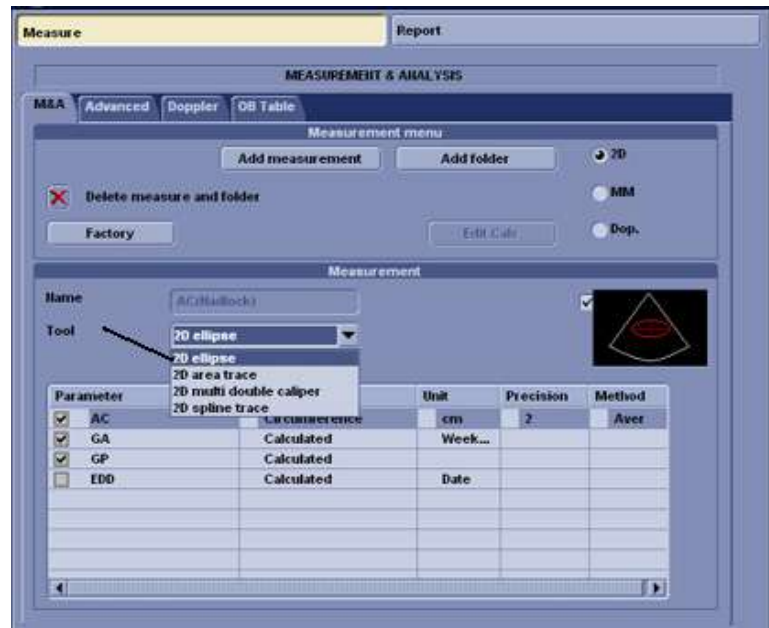


Figure 7-19. Measurement & Analysis screen: Change measurements

Adding Folders and Measurements

Adding a folder

When you add a folder, it can be a study, or a measurement folder that includes related measurements.

1. In the Selection menu, select the study or folder where you want to add the folder.
2. In the Measurement menu section, select Add folder.
 - If you select Blank, the system adds a folder with a name such as USERDEFS1. It is listed in the Selection menu. See Figure 7-21.
 - If you want to use an existing folder, select Insert, and then select a folder from the list. The list includes all folders defined for the current exam category and selected mode. You cannot edit this folder.

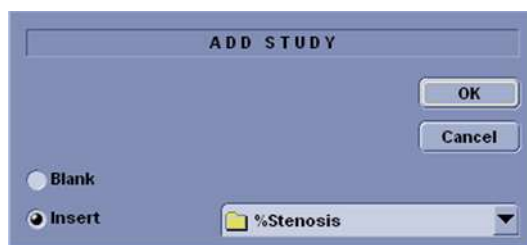


Figure 7-20. Add folder Window

3. Select the user-defined folder in the Selection menu.
The system displays the new folder in the Folder section of the Measurement & Analysis screen.
4. To name the folder, move the **Trackball** to highlight the Name field, press **Set** twice, and type the name.
5. To add measurements to the folder, See 'Adding a user-defined measurement' on *page 7-38 for more information*.

Adding a folder (continued)

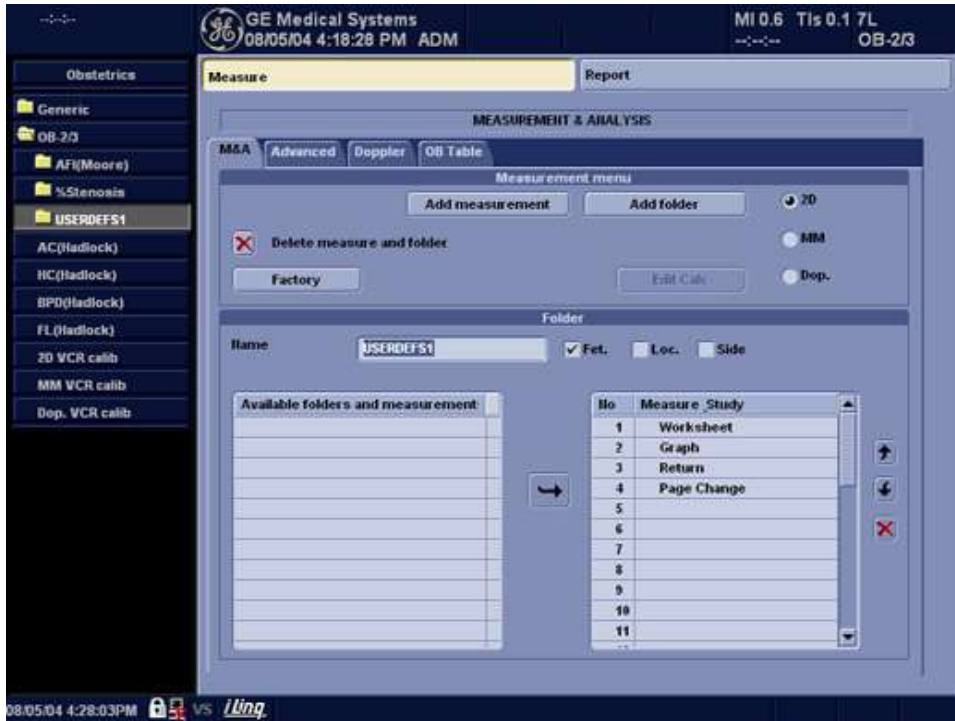


Figure 7-21. Measurement & Analysis: Add folder

Adding a user-defined measurement

You can create a user-defined measurement in a system-defined folder or in a folder you created.

1. In the Selection menu, select the study or folder where you want to add the measurement.
2. In the Measurement menu section, select Add measurement.

The system displays the Add Measurement window.

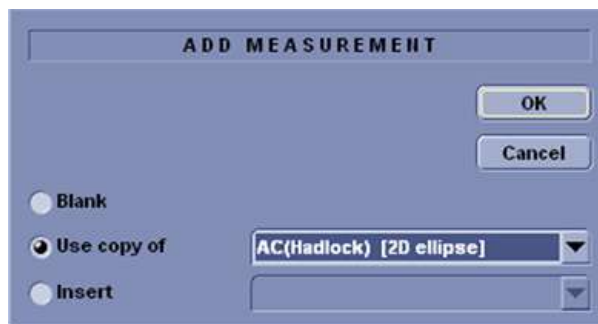


Figure 7-22. Add Measurement window

3. Do one of the following:
 - If you want to create this measurement from a copy of an existing measurement, select Use copy of, and then select a measurement from the list. The list includes all measurements defined for the current exam category and selected mode.

NOTE:

This only applies to OB and Cardiac.

- If you want to use an existing formula, select Insert, and then select a measurement from the list. The list includes all measurements defined for the current exam category and selected mode. You cannot edit this formula.
- If you want to create a blank new measurement, select Blank.

Adding a user-defined measurement (continued)

4. Select OK.
 - If you created a blank measurement, the system adds a measurement with a name such as USERDEFM3.
 - If you created a measurement from a copy of an existing measurement, the system lists the measurement and its parameters in the Measurement section.
5. When you create a new measurement, the measurement name is automatically highlighted. Type a name for the new measurement. You can change the name of a measurement you created from a copy.



Figure 7-23. Measurement & Analysis: Add measurement

Adding a user-defined measurement (continued)

Defining measurement parameters	After you add a measurement, you can add parameters. You may also want to change parameters if you copied an existing measurement. See 'Changing or adding measurement parameters' on <i>page 7-40 for more information</i> .
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Changing or adding measurement parameters

	You can make changes to measurement parameters and you can add measurement parameters.
Changing measurement parameters	To change a measurement parameter: 1. In the Selection menu, select the measurement. 2. To change the name of the Parameter, move the Trackball to the parameter name and press Set twice. Type a name for the parameter. For a description of other measurement changes, See 'Changing Measurements' on <i>page 7-34 for more information</i>.

Adding measurement parameters

To add a measurement parameter:

1. In the Selection menu, select the measurement.
2. To change the tool used to make a measurement:

In the Measurement section of the Measurement & Analysis screen, select the desired tool from the Tool list. Select the arrow to display the drop-down list.

NOTE: *If the Tool field is gray, it cannot be changed.*

3. If necessary, check Fetus (OB only), Location (Loc), or Side:
 - Fetus: If this is an OB measurement, check this box. (Default ON).
 - Location: If this measurement includes a Prox, Mid, or Dist location, check this box.
 - Side: If this measurement includes a Left or Right side, check this box.
4. In the Measurement section, move the **Trackball** to an empty line at the bottom of the Parameter list. Press **Set**. The system adds a parameter with a name of (Name).

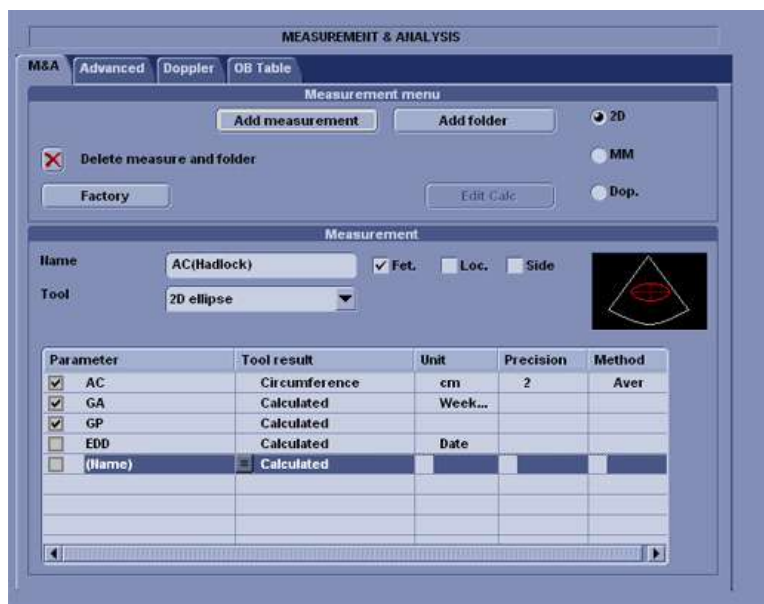


Figure 7-24. Adding a Parameter

Adding measurement parameters (continued)

5. To change the name of the Parameter, move the Trackball to the (Name) and press **Set** twice. Type a name for the parameter.
6. Move the **Trackball** to the Tool result field, and double click the **Set** key.

The Edit Formula window is displayed.

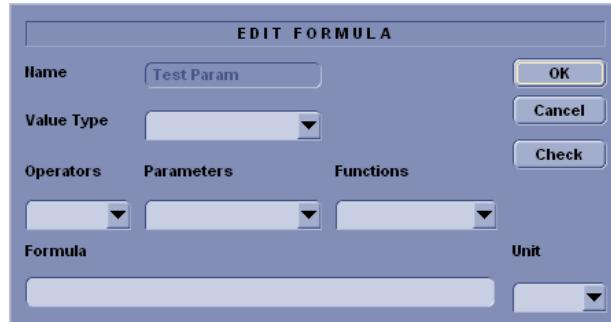


Figure 7-25. Edit Formula

7. To create a formula:
 - a. In the Value Type field, select a value.
 - b. Do one of the following:
 - Type a formula in the Formula field.
 - Select formula components from the Operators, Parameters, and Functions drop-down lists. When you select a component, the system displays it in the Formula field.
8. To test the formula, select Check.

If there are no problems, the system displays "Syntax OK!".

If there are any problems with the formula, the system displays an Error message in place of the Formula field label.
9. When the formula is correct, select OK to save it.

The Edit Formula window closes. The formula is displayed in the Tool result field.

Formula Unit Conversion

When you create a formula, the system changes the calculation result into an output unit as defined in the following table.

Table 7-2: Formula Unit Conversion

Unit		Conversion (coefficient value)
Time		
	s	x1
	ms	x1,000
	min	x0.0167
	h	x0.00027778
Ratio		
	%	x100
Frequency		
	bpm or BPM	x1.0
Angle		
	rad	x1.0
	deg	x57.2958
	grad	x63.6620
Distance		
	cm	x100
	m	x1
	dm	x10
	mm	x1,000
	inch	x39.37
	feet	x39.37
	pixels	x1
Velocity		
	m/s	x1
	dm/s	x10
	cm/s	x100
	mm/s	x1,000
	inch/s	x39.37
Acceleration		

Table 7-2: Formula Unit Conversion

Unit		Conversion (coefficient value)
	m/s ²	x1
	dm/s ²	x10
	cm/s ²	x100
	mm/s ²	x1,000
	inch/s ²	x39.37
Area		
	m ² or m ^{^2}	x1
	dm ²	x100
	cm ² or cm ^{^2}	x10,000
	mm ² or mm ^{^2}	x1,000,000
	inch ²	x1550
Volume		
	m ³	x1
	dm ³	x1,000
	cm ³	x1,000,000
	l	x1,000
	dl	x10,000
	cl	x100,000
	ml	x1,000,000
	gallon	x264,178
	quart	x1056.71
Volume Flow		
	ml/min	0.5 to 1.0 using TAMAX

Table 7-2: Formula Unit Conversion

Unit		Conversion (coefficient value)
	m ³ /s	x1
	dm ³ /s	x1,000
	cm ³ /s	x1,000,000
	l/s	x1,000
	dl/s	x10,000
	cl/s	x100,000
	ml/s	x1,000,000
	m ³ /min	x60
	dm ³ /min	x60,000
	cm ³ /min	x60,000,000
	l/min or L/min	x60,000
	dl/min	x600,000
	cl/min	6,000,000
	ml/min	x60,000,000
	ml/m ²	x1,000,000
Pressure		
	mmHg	x1
	Pa	x133.322
	kPa	x0.133322
	bar	x0.00133322
Pressure/Time		
	mmHg/s	x1
Mass		
	kg	x1
	g	x1,000
	ounce	x35.273962
	pound	x2.2046226
Others		
	mmHG	x100

**Formula Unit
Conversion
(continued)**

For example, when a Volume formula is created:

$$\text{Vol [ml or cm}^3\text{]} = 0.523598 \times \{D1\} \times \{D2\} \times \{D3\}$$

(D1, D2, and D3 indicate a measurement result.)

In this case, the measurement (D1, D2, and D3) is a distance measurement, so the measured data is a meter [m] unit according to the above table.

To change into a milliliter, the system multiplies each measurement value by 100. As a result, it multiplies a formula by 1,000,000.

The standard unit of volume is a cube meter, so the system multiplies the result by 1,000,000.

The system multiplies the calculation result by the coefficient and converts it. To get a correct result, when you define the formula, you must convert the coefficient itself, such as the coefficient of 10^6 .

For example, if you want to define the following formula:

$$\text{efg[g]} = 10^{(1.5662 - 0.0108 \times \{P1\} + 0.0468 \times \{P2\} + 0.171 \times \{D1\} + 0.00034 \times \{P1\} \times \{P1\} - 0.003685 \times \{P2\} \times \{D1\})}$$

D1[cm]: Distance

P1[cm]: Perimeter

P2[cm]: Perimeter

**Formula Unit
Conversion
(continued)**

The system defines the standard value of each measurement as a meter [m]. If the unit of each measurement value of this formula is defined as centimeter [cm], you must define the formula as follows:

$$\text{efw[g]} = 10^{(1.5662 - 0.0108 * \{P1\} * 100 + 0.0468 * \{P2\} * 100 + 0.171 * \{D1\} * 100 + 0.00034 * \{P1\} * \{P1\} * 100 * 100 - 0.003685 * \{P2\} * \{D1\} * 100 * 100)}$$

(This converts each measurement value to a centimeter [cm], since the system standard unit is a meter [m].)

The output unit of this formula is a gram. Since the standard unit of the system is defined as a kilogram [kg], the system multiplies the output by 1,000.

Because the output of this formula is defined as a gram, it is necessary to define the formula as follows.

$$\text{efw[g]} = 10^{(1.5662 - 0.0108 * \{P1\} * 100 + 0.0468 * \{P2\} * 100 + 0.171 * \{D1\} * 100 + 0.00034 * \{P1\} * \{P1\} * 100 * 100 - 0.003685 * \{P2\} * \{D1\} * 100 * 100) / 1,000}$$

As shown, you can obtain an exact calculation result.

Editing Calculations

To modify user-defined calculations:

1. Select Add Measurement from the Measurement menu. The system displays the Add Measurement window.
2. Select Blank and OK.
3. Type the appropriate name and select "Calculation" from the Tool pull-down menu.

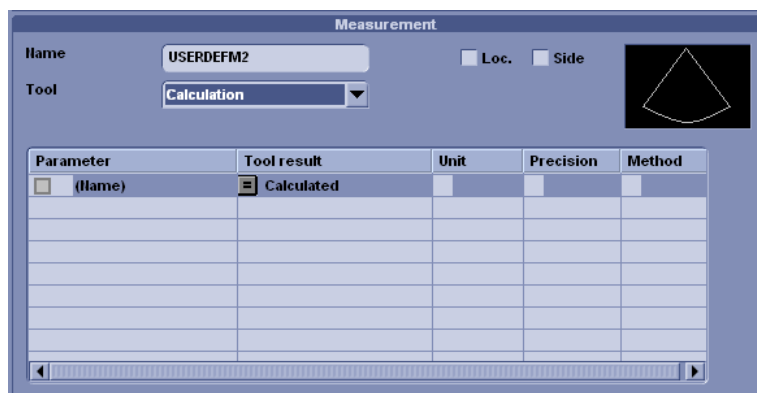


Figure 7-26. Measurement window

4. Type the parameter name.
5. Double click on the = Calculated symbol under Tool Result. The Edit Formula window displays.
6. Select OK.

Editing Calculations (continued)

7. In the Measurement menu section, select *Edit Calc.*

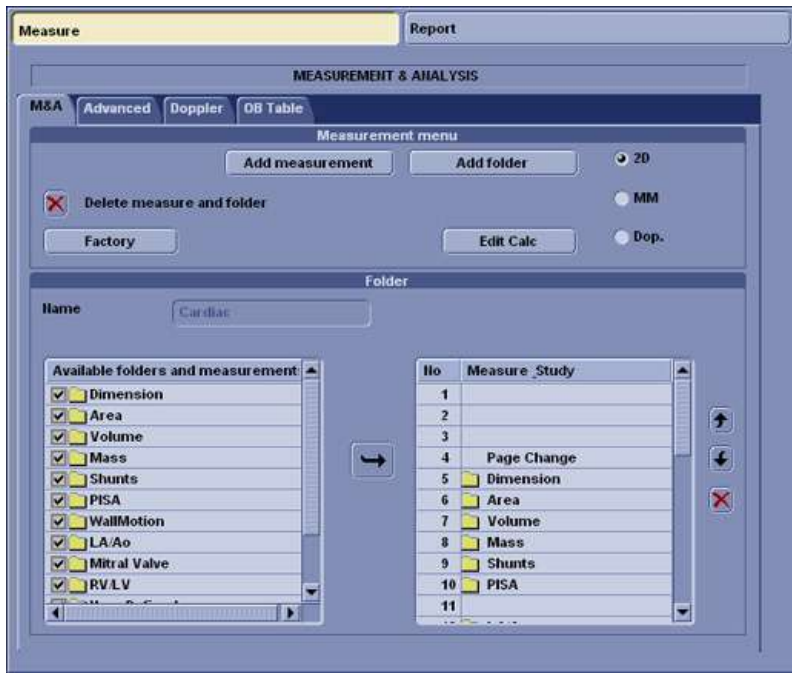


Figure 7-27. Edit Calc

The Modify User CALC window displays.

Editing Calculations (continued)

- 8. In the User Defined list, select the calculation that you want modified, then select OK.

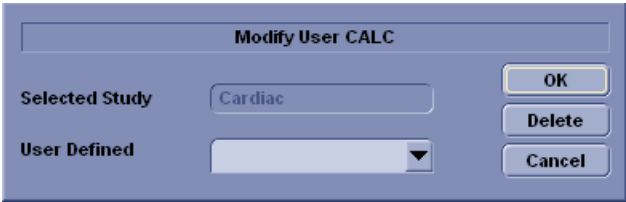


Figure 7-28. Modify User CALC window

- 9. The Measure tab for user-defined calculations displays. Double click on the equals sign symbol under Tool Result for the desired parameter.

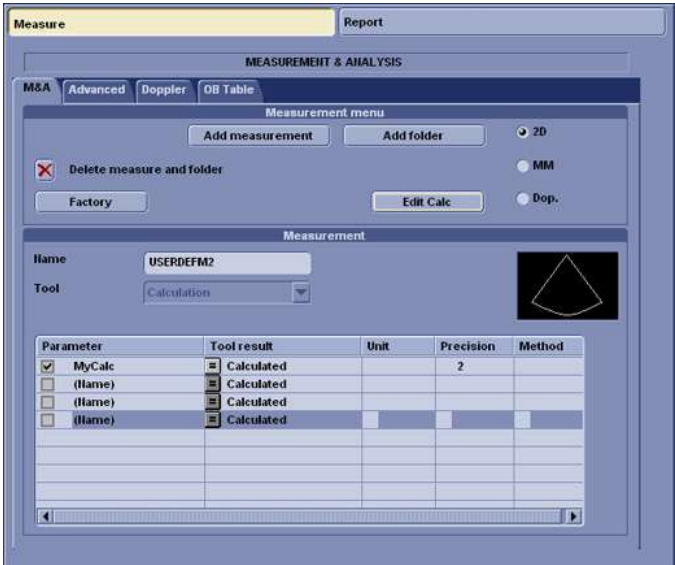


Figure 7-29. Measurement & Analysis M&A tab

- 10. Edit the formula as needed and select OK.

Deleting a Folder or Measurement

NOTE: *You can only delete user-defined folders or measurements. You cannot delete default system folders or measurements.*

1. Select the folder or measurement in the Selection menu.
2. In the Measurement menu section, select the X next to Delete measure and study.

M&A Advanced Preset

The system allows you to specify application-specific values for certain parameters. You specify the parameter values on the Advanced tab of the Measurement & Analysis screen.

- 1. On the Touch Panel, select **Utility**.
- 2. On the Touch Panel, select **Measure**.
- 3. On the monitor display, select the Advanced tab.

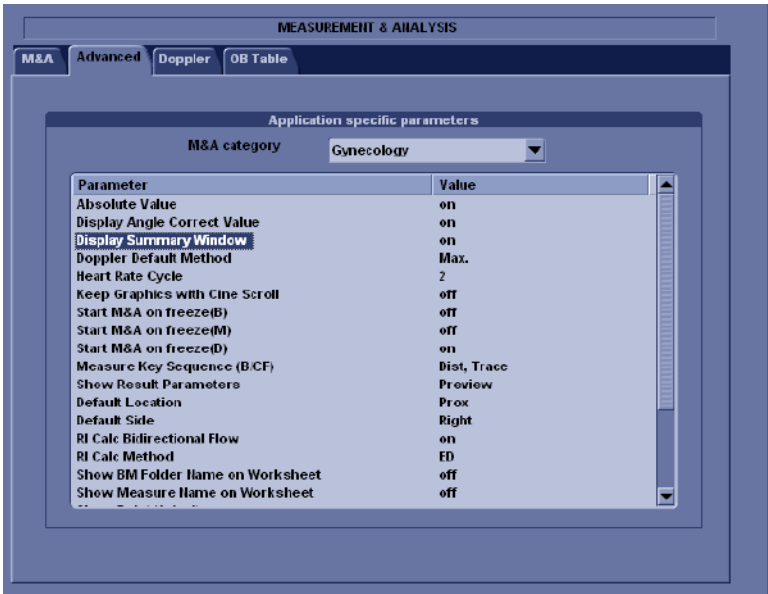


Figure 7-30. M&A Advanced Preset Menu

M&A Category: Display and select current exam category.

Parameter: Lists application specific parameters.

Value: Select the value for a parameter.

- 4. To select an exam category, select it from the M&A category list.

The Parameters list displays parameters for the selected category.

M&A Advanced Preset (continued)

5. To select a value for a parameter, select it from the Value list.

NOTE: *The parameters that appear are category dependent.*

Table 7-3: M&A Advanced

Preset Parameter	Description
Absolute Value	Displays the absolute value of the Doppler Velocity measurement (On or Off)
Display Angle Correct Value	On or Off
Display Summary Window	On or Off
Doppler Default Method	Avg, Max, Min or Last
Heart Rate Cycle	1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 <i>NOTE: For Cardiac, you can select only "1".</i>
Keep Graphics with Cine Scroll	If you select "On", the measurement graphics remain while in CINE scroll. The measurement graphic redisplay on the frame where the measurement was taken in B-Mode.
Start M&A on a Freeze (B)	Off: Select measurement manually on Freeze On: Measurement menu appears automatically on Freeze. Caliper: Measurement menu and caliper appear automatically on Freeze.
Start M&A on a Freeze (M)	
Start M&A on a Freeze (D)	
Measure Key Sequence (B/CF)	2 Sequences: Dist, Trace; Dist, Spline 3 Sequences: Dist, Trace, Spline; Dist, Spline, Trace; Dist, Spline, Intensity; Dist, Trace, Intensity; Dist, Trace, Open Trace; Dist, Spline, Open Trace 4 Sequences: Dist, Trace, Spline, Intensity; Dist, Spline, Trace, Intensity; Dist, Spline, Trace, Open Trace; Dist, Trace, Open Trace, Spline
Show Result Parameters	Preview or After Set cursor where: Preview: Displays while taking the measurement. After Set Cursor: Displays after completing the measurement.
Default Location	Off, Prox, Mid or Dist
Default Side	Left, Right or Off
RI Calc Bidirectional Flow	On or Off
RI Calc Method	MD or ED
Show BM Folder Name on Worksheet	On or Off
Show Measure Name on Worksheet	On or Off
Show Point Velocity	On or Off

Table 7-3: M&A Advanced (Continued)

Preset Parameter	Description
Show Tissue Depth	On or Off
Keep Result Window	On or Off
Trace	Auto or manual
Vol Flow Method	TAMEAN or TAMAX
Vol Flow Compensation with TAMAX	If you select TAMAX as the volume flow method, then you MUST specify the coefficient to use. Select from 0.5 to 1.0.
Worksheet Default Display	Mode/Expand (Abdominal, Small Parts, Obstetrics, Gynecology, Urology and Pediatrics) or Worksheet Summary (Vascular)
Doppler AutoCalc Velocity Unit	Velocity, Hz, Both or Auto
WMS Freeze Loop at ES	On or Off
WMS Segment Model	16 segments or 18 segments
WMS Initial Scoring	Undefined or Normal
WMS Scoring Legend	ASE, European or Asian
Hip Orientation	Cranial-left or Caudal-left

Manual Calcs Presets

The system allows you to preset the parameters for manual calculations. You specify the parameter values on the Doppler tab of the Measurement & Analysis screen.

1. On the Touch Panel, select **Utility**.
2. On the Touch Panel, select **Measure**.
3. On the monitor display, select the Doppler tab.

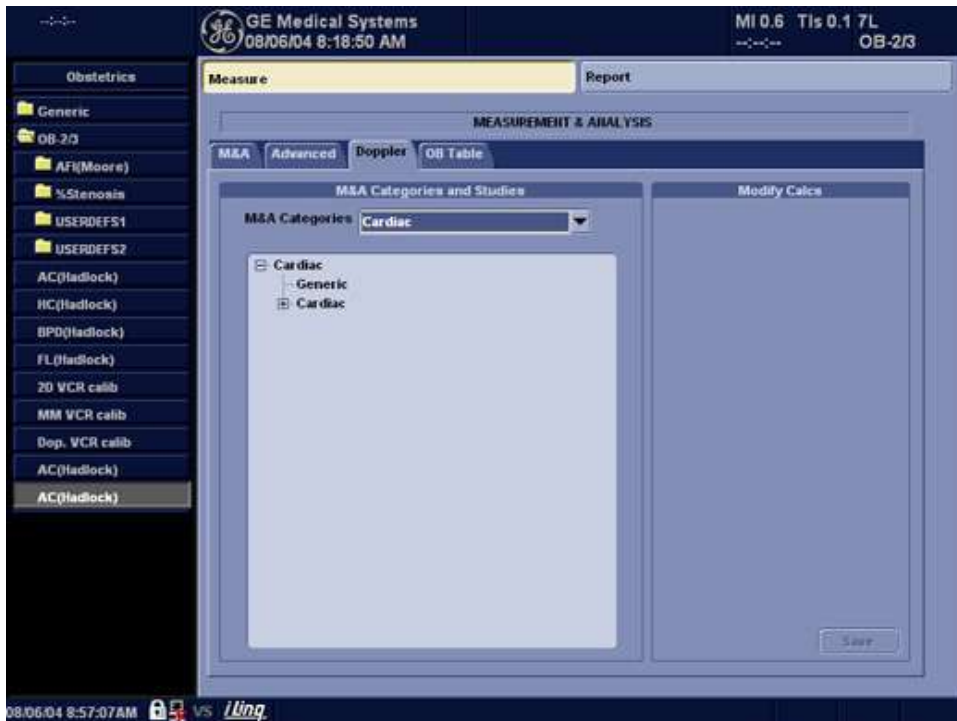


Figure 7-31. M&A Doppler Preset Menu

M&A Category: Display and select current exam category.

Manual Calcs Presets (continued)

4. To select an exam category, select it from the M&A category list.
The system displays a hierarchical view of the exam category and the studies and folders in the category.
5. In the hierarchical view, select a study or folder.
6. In the Modify Calcs column, select the measurements that you want the system to show for manual calcs for the selected study or folder.
7. To save the changes, select the Save button.

Application Measurement Preset

The Application Measurement presets allow different calculation packages to be available under different application presets.

The presets allow you to configure the Measurement Categories and Measurement Exam Calcs. These presets are found on the Utility -> Application -> Measurements screen.



Figure 7-32. Application Measurement

Mode Measurements

B-Mode Measurements

Two basic measurements can be made in B-Mode.

- Distance
- Circumference and Area
 - Ellipse Method
 - Trace Method
 - Spline Method

*NOTE: The following instructions assume that you first scan the patient and then press **Freeze**.*

Distance measurement

To make a distance measurement:

1. Press **Measure** once; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points, if preset accordingly.
5. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window.

The following hints can help you to perform distance measurements:



HINTS

- **Before** you complete a measurement:
 - To toggle between active calipers, press **Measure**.
 - To erase the second caliper and the current data measured and start the measurement again, press **Clear** once.
- **After** you complete the measurement:
 - To rotate through and activate previously fixed calipers, adjust **Cursor Select**.
 - To erase all data that has been measured to this point, but not data entered onto worksheets, press **Clear**.

Circumference and area (ellipse) measurement

You can use an ellipse to measure circumference and area. To measure with an ellipse:

1. Press **Measure** once; an active caliper displays.
2. To position the active caliper, move the **Trackball**.
3. To fix the start point, press **Set**. The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper, move the **Trackball**.
5. Adjust the **Ellipse** control; an ellipse with an initial circle shape displays.
6. To position the ellipse and to size the measured axes (move the calipers), move the **Trackball**.
7. To increase the size, adjust the **Ellipse** control in a clockwise direction. To decrease the size, adjust the **Ellipse** control in a counterclockwise direction.
8. To toggle between active calipers, press **Measure**.
9. To complete the measurement, press **Set**. The system displays the circumference and area in the Results Window.



HINTS

Before you complete the ellipse measurement:

- To erase the ellipse and the current data measured, press **Clear** once. The original caliper is displayed to restart the measurement.
- To exit the measurement function without completing the measurement, press **Clear** a second time.

Circumference and area (trace) measurement

- Trace** To trace the circumference of a portion of the anatomy and calculate its area:
1. Press **Measure** until a trace caliper displays.
 2. To position the trace caliper at the start point, move the **Trackball**.
 3. To fix the trace start point, press **Set**. The trace caliper changes to an active caliper.
 4. To trace the measurement area, move the **Trackball** around the anatomy. A dotted line shows the traced area.
 5. To complete the measurement, press **Set**. The system displays the circumference and the area in the Results Window.

- Open Trace** To trace the circumference of a portion of the anatomy and calculate its length:

- NOTE:** For open trace, Area (A) is not displayed in the result window.
1. Press **Measure** until a trace caliper displays.
 2. To position the trace caliper at the start point, move the **Trackball**.
 3. To fix the trace start point, press **Set**. The trace caliper changes to an active caliper.
 4. To trace the measurement area, move the **Trackball** around the anatomy. A dotted line shows the traced area.
 5. To complete the measurement, press **Set**. The system displays the circumference and the length in the Results Window.



HINTS

Before you complete the trace measurement:

- To erase the line (bit by bit) back from its current point, move the **Trackball** or adjust the **Ellipse** control counterclockwise.
- To erase the dotted line but not the trace caliper, press **Clear** once.
- To clear the trace caliper and the current data measured, press **Clear** twice.

Circumference and area (spline trace) measurement

To trace the circumference of a portion of the anatomy and calculate its area:

1. Press **Measure** until a trace caliper displays.
2. To position the first caliper at the start point, move the **Trackball**.
3. To fix the trace start point, press **Set**. The first caliper turns yellow. The second caliper appears at the same position as the first caliper and is green.

NOTE: *When pressing the **Clear** key once, the second caliper disappears and the first caliper is activated.*

*If **Clear** is pressed again, the first caliper disappears and the Spline trace is cancelled.*

4. To position the second caliper, move the **Trackball** and press **Set**. The third caliper appears at the same position.

NOTE: *The **Clear** key functionality is the same as noted in the previous step.*

The spline trace requires at least three points to draw the trace. Continue setting the points of the trace until the desired points are set.

5. Press **Set** again after the last caliper is fixed to finalize the spline trace. All points are removed from the line and the spline trace turns yellow.

Press **Set** twice to finish this measurement.

If **Clear** is pressed twice when more than 3 points exist on the trace, all points are removed and the first caliper again displays.

NOTE: *Spline trace is not available through the factory default. The system defaults to trace. To enable spline trace, modify the Measure Key Sequence preset found in Utility -> Measure -> Advanced preset menu.*

Circumference and area (spline trace) measurement (continued)

Edit the spline trace

1. Select **Cursor Select**. The spline trace changes to green and all points appear on the trace as yellow.

A pick-caliper appears on the center of the image and the message "Edit spline trace" displays at the bottom of the screen.

NOTE: *The pick-caliper is used to select and move the trace points.*

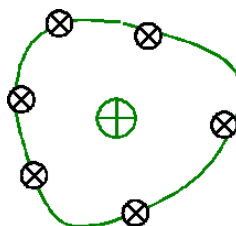


Figure 7-33. Edit spline trace

Select **Cursor Select** again. The trace is deactivated (changes to yellow) and all points, including the pick-caliper, are removed.

If the previous/next fixed caliper exists on the image, it is activated.

NOTE: *Pressing **Clear** at this time removes all points and the trace graphic.*

2. Move the pick-caliper to the desired point and press **Set**. The point is activated and turns green.
3. Move the point to the desired position and press **Set**. The point is fixed and turns yellow. The pick-caliper appears on the center of the image.

NOTE: *The spline trace is updated at run time.*

NOTE: *To remove a point, press **Clear** while moving the point. The trace turns green and the remaining points continue to be shown as yellow. If there are less than three points, the spline trace is removed.*

4. Press **Set** again. All points are removed from the trace and the trace is shown as yellow.

Doppler Mode Measurements

Four basic measurements can be made in Doppler Mode.

- Velocity
- TAMAX and TAMEAN (Manual or Auto Trace)
- Two Velocities with the Time Interval and Acceleration between them
- Time Interval
- Volume Flow

NOTE: *The following instructions assume that you do the following:*

1. In the B-Mode part of the display, scan the anatomy you want to measure.
2. Go to the Doppler Mode part of the display.
3. Press **Freeze**.

Velocity

To measure velocity:

1. Press **Measure**; an active caliper with a vertical dotted line displays.
2. To position the caliper at the desired measurement point, move the **Trackball**.
3. To complete the measurement, press **Set**. The system displays the velocity measurement in the Results Window.

TAMAX and TAMEAN

Manual Trace The value measured depends upon the Vol Flow Method preset. The two selections available are: Peak (TAMAX) and Mean (TAMEAN).

To do a manual trace of TAMAX or TAMEAN:

1. Press **Measure** twice; a trace caliper displays.
2. To position the caliper at the trace start point, move the **Trackball**.
3. To fix the start point, press **Set**.
4. To trace the maximum values of the desired portion of the spectrum, move the **Trackball**.

*NOTE: To edit the trace line, move the **Trackball**.*

5. To complete the measurement, press **Set**. The system displays the measurement values in the Results Window.

Auto Trace The value measured depends upon the Vol Flow Method preset. The selection available is the Peak (TAMAX).

To auto trace TAMAX:

1. Press **Measure** twice; an active caliper with a vertical dotted line displays.
2. To position the caliper at the trace start point in the Doppler spectrum, move the **Trackball**.
3. To fix the start point, press **Set**.
4. To position the vertical caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**. The system automatically fixes both calipers and traces the maximum value between the two points. The system displays this value in the Results Window.

NOTE: When you set the Auto Trace for Both (above and below), the system picks up the maximum power of the signal, NOT the maximum velocity. If the maximum velocity is not the maximum power, the system may not trace accurately. If you want to use maximum velocity, select either Above or Below.

- Edit Trace** Auto Trace can be edited after taking an Auto Trace measurement.
1. After taking an Auto Trace measurement, select the measurement result on the result window. The Edit Trace (Edit Peak or Edit Mean) menu window appears.
- NOTE: If the system cannot take the trace data correctly from the image, Edit Trace does not work.*
2. Select Edit Trace. The first caliper (manual trace caliper) appears on the center of the image. Use the **Trackball** to move the caliper on the trace line to the start point.
- NOTE: To cancel Edit Trace at this time, press **Clear**, **Scan**, or **Freeze**.*
3. Press **Set** to fix the first caliper. The second caliper appears. Edit the trace manually using the second caliper. The Ellipse control is used to edit the trace.
- NOTE: When pressing the **Clear** key once at this time, the second caliper disappears and the first caliper appears in the center of the image.*
- NOTE: If you press **Scan** or **Freeze** at this time, the caliper is automatically fixed and the result window updates.*
4. Press **Set** to fix the second caliper. The trace and the result window update. The trace data (TAMAX and TAMEAN) are updated, though the other points (e.g. PS, ED) are not updated by trace. The points can be edited with **Cursor Select**.
- NOTE: While in Edit Trace, Cursor Select is disabled.*
5. Repeat Edit Trace as needed.

Doppler Auto Calc Average Cycle

When using Auto Calc, a selection is available to average a number of cycles automatically. There is also a preset selection in the Utility Imaging PW page for this feature. When using average cycle:

- Selected cardiac cycle lines display on the image. Point calipers are not displayed.
- When changing the number of cycles from 1 to >1, all the data is reacquired from the image, recalculated and updated.
- When multiple cycles are selected in AutoCalc, the average values calculate and display automatically.
- When selecting Peak Value (PV), average cycle is not available.

NOTE: You cannot edit the lines while in Average Cycle. Cursor Select is not available at that time.

NOTE: Average Cycle data is acquired from the display image area only, for both live and frozen. The average cycle data fails if the setting for the number of cycles is larger than the number of image cycles.

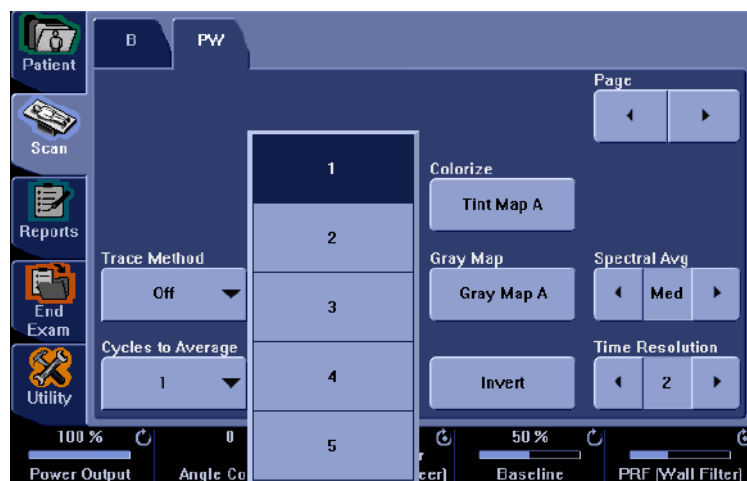


Figure 7-34. PW Touch Panel (Page 2)

Velocity, Time Interval and Acceleration

To measure two velocity values, the time interval (ms), and acceleration (m/s^2):

1. Press **Measure** three times; an active caliper with vertical and horizontal dotted lines displays.
2. To position the caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**. The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**. The system displays the two peak end point velocities, the time interval, and the acceleration in the Results Window.

Time interval

To measure a horizontal time interval:

1. Press **Measure** four times; an active caliper with a vertical dotted line displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**. The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**. The system displays the time interval between the two calipers in the Results Window.

Volume Flow - Manual Calc

You perform a manual Volume Flow measurement using the TAMAX plus a Volume Flow coefficient compensation.

1. To perform the Volume Flow measurement using the TAMAX plus a Volume Flow coefficient compensation, in Utility-->Measure-->Advanced, select the following:
 - Trace = Manual
 - Vol Flow Method = TAMAX [you MUST also select a Volume Flow coefficient for use with TAMAX.]
 - Vol Flow Compensation with TAMAX = [select value from 0.5 to 1.0]
2. Set Auto Calcs to Off via Doppler Mode-->Modify Auto Calcs-->Off.
3. Select a folder in Doppler Mode-->select a calculation folder-->select Show All.
4. Select Volume Flow. You'll notice that TAMAX is automatically selected.

NOTE: *Ensure that you have placed the caliper in the spectral window when selecting the Volume Flow measurement.*

5. Trace the TAMAX. The system prompts you to "Mark the first point on the spectral doppler." Press Set.
6. The system prompts you to "Trace the velocity spectrum boundary." Press Set.

NOTE: *You can back up while tracing the TAMAX by using the Trackball.*

7. Trace the vessel diameter. The system prompts you to "Mark first point of vessel diameter for volume flow calculation." Press Set.
8. The system prompts you to "Mark last point of vessel diameter for volume flow calculation." Press Set.
9. The Volume Flow is calculated in ml/min.

Volume Flow - Auto Calc

You can perform an automatic Volume Flow measurement using TAMEAN or using the TAMAX and a Volume Flow coefficient.

1. To perform the Volume Flow measurement using the TAMEAN, in Utility-->Measure-->Advanced, select the following:

- Trace = Auto
- Vol Flow Method = TAMEAN

OR, to perform the Volume Flow measurement using the TAMAX plus a Volume Flow coefficient compensation, select the following:

- Trace = Auto
- Vol Flow Method = TAMAX [if you use TAMAX, you MUST also select a Volume Flow coefficient for use with TAMAX.]
- Vol Flow Compensation with TAMAX = [select value from 0.5 to 1.0]

2. Set Auto Calcs to Live via Doppler Mode-->Modify Auto Calcs-->Live.
3. Perform the scan.
4. Select Volume Flow via Doppler Mode-->Modify Auto Calcs-->VOLUME FLOW. The system prompts you through the measurement.
5. Take vessel diameter for volume flow calculation. Set the first cursor.
6. Mark last point of vessel diameter for volume flow calculation. Press Set.
7. The calculation automatically completes the Volume Flow measurements as ml/min.

NOTE: *If you change the TAMAX coefficient, the Volume Flow is automatically adjusted when in Auto Calcs (but not in Manual Calcs).*

M-Mode Measurements

Basic measurements that can be taken in the M-Mode portion of the display are:

- Tissue Depth (Distance)
- Time Interval
- Time Interval and Velocity

NOTE: *The following instructions assume that you do the following:*

1. In the B-Mode part of the display, scan the anatomy you want to measure.
2. Go to the M-Mode part of the display.
3. Press **Freeze**.

Tissue depth

Tissue depth measurement in M-Mode functions the same as distance measurement in B-Mode. It measures the vertical distance between calipers.

1. Press Measure once; an active caliper with a vertical and horizontal dotted line displays.
2. To position the active caliper at the most anterior point you want to measure, move the Trackball.
3. To fix the start point, press Set.
The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper at the most posterior point you want to measure, move the Trackball.
5. To complete the measurement, press **Set**.

The system displays the vertical distance between the two points in the Results Window.

Time interval

To measure a horizontal time interval and velocity:

1. Press **Measure** twice; an active caliper with a vertical dotted line displays.
2. To position the caliper at the start point, move the **Trackball**.
3. To fix the first caliper, press **Set**. The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**. The system displays the time interval between the two calipers in the Results Window.

Time interval and Velocity

To measure time and velocity between two points:

1. Press **Measure** three times; an active caliper with a vertical and horizontal dotted line displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays time(s) and slope between the two points in the Results Window.

Viewing and Editing Worksheets

As you complete measurements, the system puts measurement data in the appropriate worksheets.

To view a worksheet

To view a worksheet, select **Worksheet** on the Touch Panel.

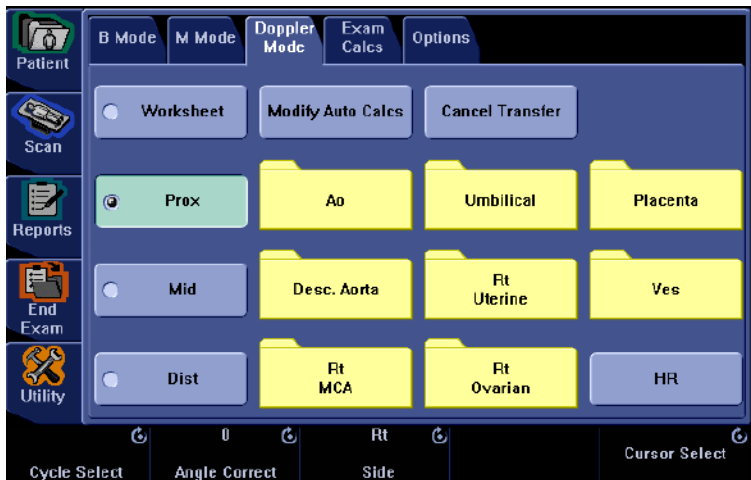


Figure 7-35. Calculation Touch Panel

OR

Select **Worksheet** on the measurement summary window.

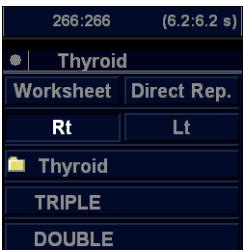


Figure 7-36. Worksheet/Direct Report

To view a worksheet (continued)

The system displays the worksheet for the current study.

GE Medical Systems		patient.			
11/04/04 1:33:34 PM ADM		20041101, GA(LMP)=39w3d		--:--:--	
Origin LMP	LMP 02/02/2004	BBT		GA 39w3d	EDD(LMP) 11/08/2004
Fetus A/1		CUA	18w1d+/- 1w0d		EDD(CUA) 04/06/2005
FetusPos		PLAC		Ref.Physician	Page 1/1

B Mode Measurements							
BPD(Hadlock)	☒	3.44 cm	5.36	2.11	2.83	Avg.	16w4d 15w3d-17w6d
HC(Hadlock)	☒	11.49 cm	6.79	16.18		Avg.	15w4d 14w3d-16w6d
OFD(HC)		4.99 cm	2.84	7.15		Avg.	
AC(Hadlock)	☒	13.28 cm	18.28	8.27		Avg.	18w5d 16w5d-20w6d
FL(Hadlock)	☒	4.03 cm	4.95	3.11		Avg.	23w0d 21w2d-24w6d

2D Calculations			
EFW(AC,BPD,FL,HC)	330.29g +/-49.54g	(12oz)	
EFW(Hadlock)-GP	<3%		
CI(Hadlock)	-> 68.85 (70.00-86.00)	FL/AC(Hadlock)	30.37 (-)
FL/BPD(Hohler)	117.33 (-)	FL/HC(Hadlock)	-> 35.10 (15.84-18.04)
HC/AC(Campbell)	-> 0.87 (1.08-1.27)		

Figure 7-37. OB B-Mode Worksheet

To return to scanning, do one of the following:

- Select **Worksheet**.
- Press **Esc**.
- Select the **Exit** button.

To view a worksheet (continued)

To view a different worksheet, select the worksheet key for the desired worksheet.

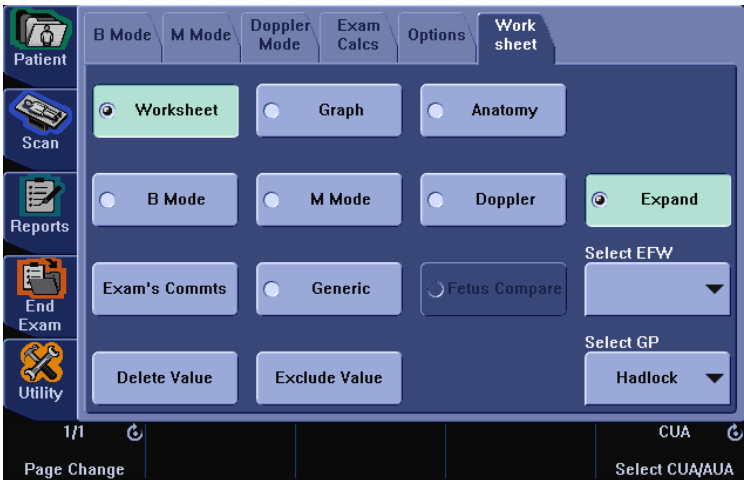


Figure 7-38. Worksheet Display Touch Panel

To view worksheet data for a particular mode, select the key for that mode. To view a worksheet with data for more than one mode, select **Expand**. When Expand is selected, it defaults to view all measurements, noted by mode, on the worksheet.

If a worksheet has more data on a second page, to view the next page, adjust the **Page Change** control.

To edit a worksheet

To change data on a worksheet:

1. To position the cursor at the field you want to change, move the **Trackball**. The field is highlighted.
2. Press **Set**.
3. Type the new data in the field. The new data is displayed in blue to indicate that it was manually entered.

To delete or exclude data on a worksheet:

1. To position the cursor at the field you want to delete or exclude, move the **Trackball**. The field is highlighted.
2. Do one of the following:
 - To delete the field, select **Delete Value**.
 - To exclude the field, select **Exclude Value**.
The data in the field is not visible and is not included in worksheet calculations.
 - To include a value that you previously excluded, select **Exclude Value**.

To type a comment on a worksheet:

1. Select **Examiner's Comments**. The Examiner's Comments window opens.
2. Type comments about the exam.
3. To close the Examiner's Comments window, select **Examiner's Comments**.

To edit a worksheet (continued)

To turn the volume measurement value off:

- Select the method type **Off**. The value field becomes blank.

Parameter	Value	m1	m2	m3	m4	m5	m6	Method
B Mode Measurements								
Ut-L	5.24 cm	5.24						Avg.
Ut-H	5.12 cm	5.12						Avg.
Ut-W	- cm	5.55						Off
Ut Vol	- ml	78.03						

Figure 7-39. Volume Parameter Off



Some fields on the worksheet are view only, and others you can change or select. To easily see which fields you can change or select, move the **Trackball**. As the cursor moves over a field that you can change or select, the field is highlighted.

Delete All Worksheet Values

You can delete all worksheet values on a worksheet.

1. When the Worksheet is displayed on the monitor, press the **Clear** key; the following warning message appears:

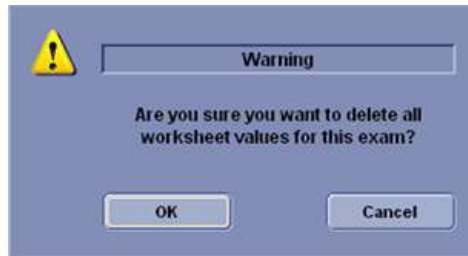


Figure 7-40. Delete All Warning Message

2. Select **OK** to delete all.
Select **Cancel** to cancel the deletion.

Generic Measurements

Overview

Each exam category has a Generic study. The Generic studies provide you quick access to measurements such as volume, angle, A/B ratio, and % stenosis. The particular measurements available in each Generic study vary, depending on the exam category and the mode. This section describes generic measurements, organized by mode.

To access Generic studies:

1. On the Control Panel, press **Measure**.
2. On the Touch Panel, select **Exam Calcs**.
3. On the Touch Panel, select the **Generic** folder.

Calculation formulas are available in the *Advanced Reference Manual*.

Assign a name to the generic measurement

NOTE: Available for any linear and circumference measurement.

1. Move the cursor over the measurement result window and press **Set**.
2. Select **Users...** from the menu. The dialogue window displays.

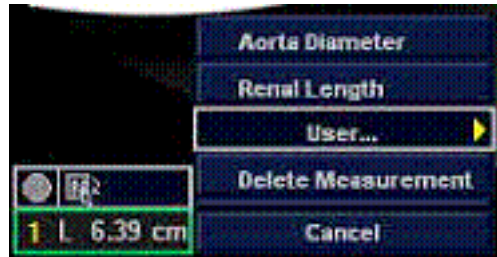


Figure 7-41. Users...

3. Enter the appropriate name and select OK.

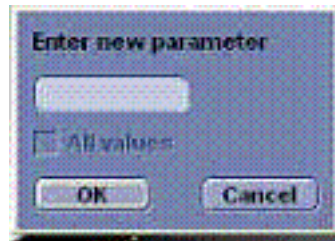


Figure 7-42. Enter new parameter

Copy, move and paste measurement tools

You can copy, move and paste the measurement graphic.

NOTE: *This function is supported with trace, area trace, spline trace, volume trace, ellipse, 3-point ellipse, circle and intensity. The Double tools, like double ellipse or double area trace are not supported.*

Copy and Paste

1. Measure the trace.
2. Press the **left Set** key to display the green arrow cursor on the screen. Move the cursor to the + mark of the measurement graphic. The selected graphic color changes from yellow to green.
3. Press **Set**. The pop-up menu displays. Select **Copy**.



Figure 7-43. Copy and Move Menu

4. Press **Set** on the outer side of the measurement graphic. The pop-up menu displays.
5. Select **Paste**. The copied graphic displays on top of the original graphic in green. Move it to the desired position using the **Trackball** and press **Set** to fix the location.

NOTE: *If the copied graphic is bigger than the pasted area, "Paste" fails and "The copied graphic cannot be pasted to this area" message displays on the status bar.*

Copy and Move

1. Measure the trace.
2. Press the **left Set** key to display the green arrow cursor on the screen. Move the cursor to the + mark of the measurement graphic. The selected graphic color changes from yellow to green.
3. Press **Set**. The popup menu displays. Select **Copy&Move**.
4. The copied graphic displays on top of the original graphic in green. Move it to the desired position using the **Trackball** and press **Set** to fix the location.

Move

1. Measure the trace.
2. Press the **left Set** key to display the green arrow cursor on the screen. Move the cursor to the + mark of the measurement graphic. The selected graphic color changes from yellow to green.
3. Press **Set**. The pop-up menu displays. Select **Move**.
4. Move the selected graphic to the desired position using the **Trackball** and press **Set** to fix the location.

B-Mode Measurements

In B-Mode, the Generic study includes the following measurements:

- % Stenosis
- Volume
- Angle
- A/B Ratio



Figure 7-44. B-Mode Generic Study

NOTE: The following instructions assume that you first scan the patient and then press **Freeze**.

% Stenosis

You can calculate % Stenosis by diameter or by area, depending on the mode.

NOTE: *The LOGIQ 9 automatically activates the % Stenosis with the default selection. If another method is preferred, select it from the Touch Panel.*

Diameter

NOTE: *When you use diameter to calculate the %stenosis, always take the measurement from a cross-sectional view of the vessel.*

To calculate percent stenosis by diameter:

1. From the Generic Touch Panel, select % **Stenosis**.
2. Select %**sten(Diam)**.
The system displays an active caliper.
3. Make a distance measurement of the inner area of the blood vessel.
The system displays an active caliper for the second distance measurement.
4. Make a distance measurement of the outer area of the blood vessel.

The system displays each distance measurement and the % Stenosis in the Results Window.

For details on how to make a distance measurement, See 'Distance measurement' on page 7-58 for more information..

NOTE: *For the diameter calculation, do NOT take a distance measurement from a longitudinal view. This may lead to an inaccurate assessment of % stenosis.*

% Stenosis (continued)

- Area** To calculate percent stenosis by area:
1. From the Generic Touch Panel, select % **Stenosis**.
 2. Select %**sten(Area)**.
The system displays a trace caliper.
 3. Make a trace measurement of the inner area of the blood vessel.

NOTE: To erase an open trace, move the **Trackball**.

4. Press **Set**.
The system displays a second trace caliper.
5. Make a trace measurement of the outer area of the blood vessel.

The system displays the two area measurements and percent stenosis in the Results Window.

See 'Circumference and area (trace) measurement' on page 7-60 for more information.

- Ellipse + Area** To calculate percent stenosis by ellipse and area:

1. From the Generic Touch Panel, select % **Sten[E+A]** folder.
2. Ellipse is selected by default.
The system displays a trace caliper.

NOTE: You can select the trace at this time.

3. Make an ellipse measurement of the inner area of the blood vessel.
4. Press **Set**.
The system displays a trace caliper.
5. Make a trace measurement of the outer area of the blood vessel.

The system displays the two area measurements and percent stenosis in the Results Window.

NOTE: % Stenosis (E+A) is not available through the factory default. To enable %Stenosis (E+A), add "%Steno(E+A)" to the Measure & Study list on the Utility -> Measure -> M&A screen.

Volume

The volume calculation can be made from any of the following measurements:

- One distance
- Two distances
- Three distances
- One ellipse
- One distance and one ellipse

For details on how to make a distance measurement, See 'Distance measurement' on page 7-58 for more information.

For details on how to make an ellipse measurement, See 'Circumference and area (ellipse) measurement' on page 7-59 for more information..

NOTE: *IMPORTANT!! If you want to make a volume calculation using one or two distances, you must select **Volume** BEFORE you make the measurements.*

NOTE: *If you select Fix Caliper by Print Key on the Utility --> System --> System Measure, the print key does not function like the Set key, but instead ends the measurement sequence and initiates the volume calculation based on the number of measurements taken so far.*

To make a volume calculation using one or two distances:

1. Select **Volume**.
2. Make one or two distance measurements.
3. Select **Volume**.

The system displays the distances and the volume in the Results Window.

NOTE: *Use the **Clear** key to erase the green caliper.*

To make a volume calculation using three distances:

1. Make three distance measurements.

NOTE: *Three distances can be done in the dual format mode (side by side images). One measurement is usually made in the sagittal plane and two measurements in the axial plane. To use the dual format mode, press the **L** or **R** key on front panel.*

2. Select **Volume**.

The system displays the distances and the volume in the Results Window.

Volume (continued)

To make a volume calculation using one ellipse:

1. Make one ellipse measurement.
2. Select **Volume**.

The system displays the ellipse measurement and the volume in the Results Window.

To make a volume calculation using one ellipse and one distance:

1. Make one distance measurement and one ellipse measurement.
2. Select **Volume**.

The system displays the distance and ellipse measurement and the volume in the Results Window.



HINTS

- Volumes are most accurate when measurements are taken in the sagittal and axial scan planes.
- To display sagittal and axial plane images simultaneously, use the side-by-side dual format option.

Calculation formulas are available in the *Advanced Reference Manual*.

Volume (continued)

Table 7-4: Volume Calculations

Calc Name	Input Measurements
Volume (spherical)	One distance
Volume (prolate spheroidal)	Two distances, $d1 > d2$
Volume (spheroidal)	Three distances
Volume (prolate spheroidal)	One ellipse: (d1 major axis, d2 minor axis)
Volume (spheroidal)	One distance d1, and one ellipse (d2 major axis, d3 minor axis)

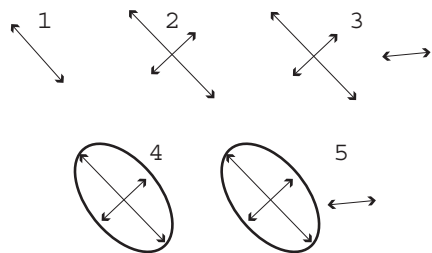


Figure 7-45. Volume Calculation Examples

1. One distance
2. Two distances
3. Three distances
4. One ellipse
5. One distance and one ellipse

Volume (continued)

Post-assignment for General Volume

You can input a unique name for the general volume measurement. You can group the general volume measurements for each application.

1. Complete the volume measurement.
2. Move the caliper to the measurement result box (with green frame) and select **Set**.
3. The volume name menu appears. Select Name Volume.

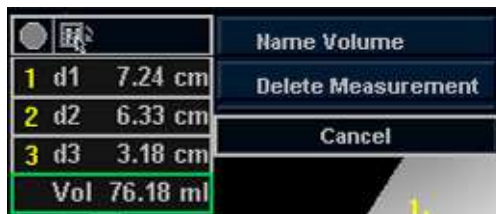


Figure 7-46. Volume Name menu

4. The dialog box displays. Enter a new name or choose the existing name.

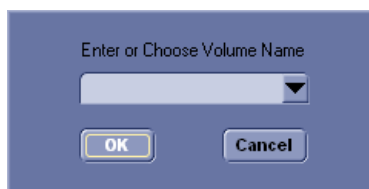


Figure 7-47. Volume Name Dialog box

NOTE: *The factory default volume name cannot be changed (for example, Renal Volume).*

Angle

This function measures the angle between two intersecting planes.

1. From the Generic Touch Panel, select **Angle**.
The system displays an active caliper.
2. To position the caliper, move the **Trackball**.
3. To fix the position of the first caliper, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at the apex of the angle, move the **Trackball**.
5. To fix the position of the second caliper, press **Set**.
The system displays a third active caliper.
6. To position the third caliper, move the **Trackball**.
7. To complete the angle measurement, press **Set**.
The system displays the angle in the Results Window.

NOTE: *To rotate through and activate previously fixed calipers, adjust the **Cursor Select** control.*

A/B Ratio

In B-Mode, you can calculate A/B ratio by diameter or by area.

NOTE: *The LOGIQ 9 automatically activates the A/B Ratio with the default selection. If another method is preferred, select it from the Touch Panel.*

Diameter To calculate A/B ratio by diameter:

1. From the Generic Touch Panel, select **A/B Ratio**.
2. Select **ratio(Diam)**.
The system displays an active caliper.
3. Make a distance measurement of the first diameter.
The system displays an active caliper for the second distance measurement.
4. Make a distance measurement of the second diameter.
The system displays each distance measurement and the A/B ratio in the Results Window.

NOTE: *The first distance is the A diameter. The second distance is the B diameter.*

For details on how to make a distance measurement, See 'Distance measurement' on page 7-58 for more information.

Area To calculate A/B ratio by area:

1. From the Generic Touch Panel, select **A/B Ratio**.
 2. Select **ratio(Area)**.
The system displays a trace caliper.
 3. Make a trace measurement of the A area.
- NOTE:** *To erase an open trace, move the **Trackball**.*
- The system displays a second trace caliper.
4. Make a trace measurement of the B area.

The system displays the two area measurements and the A/B ratio in the Results Window.

For details on how to make a trace measurement, See 'Circumference and area (trace) measurement' on page 7-60 for more information.

M-Mode Measurements

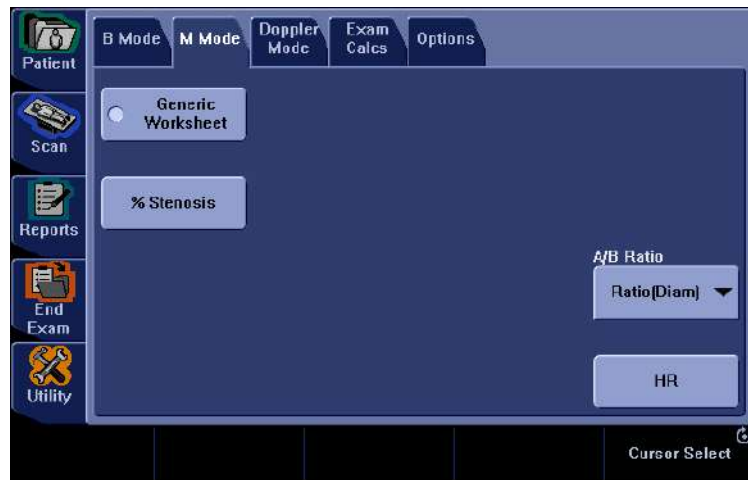


Figure 7-48. M-Mode Generic Study

% Stenosis

See '% Stenosis' on *page 7-83* for more information.

A/B Ratio

In M-Mode you can measure A/B ratio by diameter, time, or velocity.

NOTE: *The LOGIQ 9 automatically activates the A/B Ratio with the default selection. If another method is preferred, select it from the Touch Panel.*

Diameter See 'Diameter' on page 7-90 for more information.

Time To calculate A/B ratio by time:

1. Select **A/B**.
2. Select **ratio(Time)**.
The system displays an active caliper.
3. To position the caliper at the A point, move the **Trackball**.
4. To fix the measure point, press **Set**.
The system displays a second active caliper.
5. To position the second caliper at the B point, move the **Trackball**.
6. To complete the measurement, press **Set**.

The system displays the two time measurements and A/B ratio in the Results Window.

Velocity To calculate AB ratio by velocity:

1. Select **A/B**.
2. Select **ratio(Velocity)**.
The system displays an active caliper with vertical and horizontal dotted lines.
3. To position the caliper at the A velocity, move the **Trackball**.
4. To fix the measure point, press **Set**.
The system displays a second active caliper.
5. To position the second caliper at the B velocity, move the **Trackball**.
6. To complete the measurement, press **Set**.

The system displays the two velocity measurements and the A/B ratio in the Results Window.

Heart Rate

To calculate the heart rate from M-Mode:

1. Obtain an image and press **Measure**. Select **HR**.
The system displays an active caliper.
2. To position the caliper at a recognizable point in the first cycle, move the **Trackball**.
3. To fix the first caliper, press **Set**.
The system displays a second active caliper.
4. To position the caliper at the identical point in the next cycle (depending on preset), you need to move the **Trackball**.

NOTE: *In the message bar at the bottom of the display, the system indicates the number of cycles you should measure.*

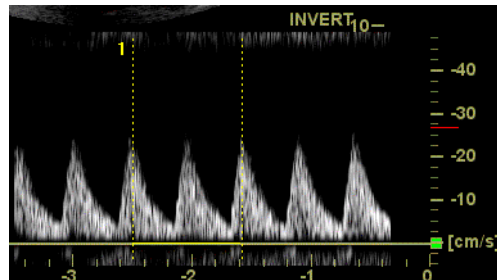


Figure 7-49. Two Heart Beat Reference (example in Doppler mode)

5. To complete the measurement and transfer the calculation to the worksheet, press **Set**.

NOTE: *For information about how to specify the number of heart beats that the system will use, See 'Specifying measurement and display options' on page 7-10 for more information.*

Doppler Mode Measurements

In Doppler Mode, the Generic study includes the following measurements:

- PI (Pulsatility Index)
- RI (Resistive Index)
- PS/ED Ratio or ED/PS Ratio
- A/B Ratio
- Max PG (Pressure Gradient)
- Mean PG (Pressure Gradient)
- SV (Stroke Volume)
- HR (Heart Rate)

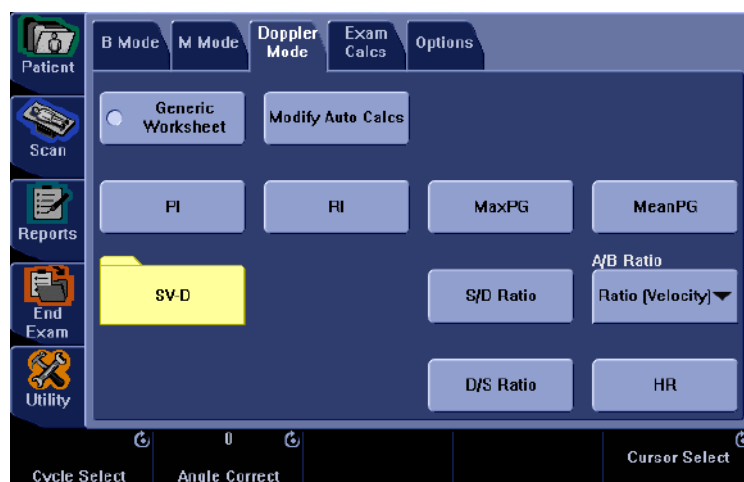


Figure 7-50. Doppler Mode Generic Study

NOTE: The following instructions assume that you do the following:

1. In the B-Mode part of the display, scan the anatomy you want to measure.
2. Go to the Doppler Mode part of the display.
3. Press **Freeze**.

Pulsatility Index (PI)

For auto trace:

1. Select **PI**.
The system displays a caliper and a vertical dotted line.
2. Position the caliper at the beginning of the waveform.
3. To fix the start point, press **Set**.
The system displays a second active caliper.
4. Position the caliper at the end of the waveform.
5. To complete the measurement, press **Set**.
The system displays peak systole, minimum diastole, end diastole, TAMAX, and PI in the Results Window.

For manual trace:

1. Select **PI**.
The system displays a trace caliper and a vertical dotted line.
2. Position the caliper at the beginning of the waveform.
3. To fix the start point, press **Set**.
The system displays a second active caliper.
4. Manually trace the entire waveform.
5. To complete the measurement, press **Set**.
The system displays peak systole, minimum diastole, end diastole, TAMAX, and PI in the Results Window.

Resistive Index (RI)

1. From the Doppler Generic Touch Panel, select **RI**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the peak systolic velocity, move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at the end diastolic velocity, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays PS, ED, and RI in the Results Window.

PS/ED or ED/PS Ratio

To calculate the Peak Systole/End Diastole ratio or End Diastole/Peak Systole ratio:

1. Select **PS/ED** or **ED/PS**.

The system displays an active caliper with vertical and horizontal dotted lines.

2. To position the caliper at peak systole (PS) or end diastole (ED), move the **Trackball**.

3. To fix the measure point, press **Set**.

The system displays a second active caliper.

4. To position the second caliper at end diastole (ED) or peak systole (PS), move the **Trackball**.

5. To complete the measurement, press **Set**.

The system displays the peak systole, end diastole, and PS/ED or ED/PS ratio in the Results Window.

A/B Ratio

In Doppler Mode you can measure A/B ratio by velocity, time, or acceleration.

NOTE: *The LOGIQ 9 automatically activates the A/B Ratio with the default selection. If another method is preferred, select it from the Touch Panel.*

Velocity See 'Velocity' on page 7-92 for more information.

Time See 'Time' on page 7-92 for more information.

Acceleration To measure A/B ratio by acceleration:

1. Select **A/B**.
2. Select **ratio(Acc)**.
The system displays an active caliper.
3. Make a distance measurement of the A acceleration point.
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window and displays an active caliper for the second distance measurement.
4. To make a distance measurement of the B acceleration point, repeat steps a–d.
The system displays the two acceleration measurements and the A/B ratio in the Results Window.

Max Pressure Gradient (PG)

To measure Max PG:

1. From the Doppler Generic Touch Panel, select **Max PG**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the first pressure gradient point, move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at second pressure gradient point, move the **Trackball**.
5. To complete the measurement, press **Set**.

The system displays the two $V_{\max}$ measurements and the Max PG in the Results Window.

Set the first caliper on $V_{\max}$ and the second on V_d .

Automatic Calculation of Max PG

When you select Max PG, if you previously measured and calculated velocity in Doppler Mode, the system automatically calculates Max PG from the $V_{\max}$ and V_d measurements taken for velocity.

Mean Pressure Gradient (PG)

To measure Mean PG (Mean Pressure Gradient):

1. From the Doppler Generic Touch Panel, select **Max PG**.
The system displays a trace caliper.
2. To position the caliper at the beginning of the waveform ($V_{\max}$), move the **Trackball**.
3. To fix the start point, press **Set**.
The system displays a second active caliper.
4. To trace the waveform from $V_{\max}$ to V_d move the **Trackball**.
5. To complete the measurement, press **Set**.

The system displays TAMAX and Mean PG in the Results Window.

If Trace Auto is selected, the waveform is automatically traced after the second caliper is fixed.

If Trace Auto is not selected, manually trace the desired portion of the waveform.

Automatic Calculation of Mean PG

If velocity was previously measured and calculated in Doppler Mode, Mean PG is calculated from the velocity value when Mean PG is first selected in the Touch Panel.

The waveform is automatically traced. Velocity and Mean PG are also displayed on the screen.

If Trace Auto is off, the Auto Trace function is off and traces can be performed manually.

Heart Rate

To measure heart rate, See 'Heart Rate' on *page 7-93 for more information.* or select any of the following measurements.

SV Diam (Stroke Volume Diameter)

To measure SV Diam:

1. From the Doppler Generic Touch Panel, select the SV folder.
2. Select SV Diam.

SV TAMEAN (Stroke Volume TAMEAN)

To measure SV TAMEAN:

1. From the Doppler Generic Touch Panel, select the SV folder.
2. Select SV TAMEAN.

CO HR (Cardiac Output Heart Rate)

To measure CO HR:

1. From the Doppler Generic Touch Panel, select the SV folder.
2. Select CO HR.

Auto vs. Manual Calculations

The same calculations can be performed using either manual or auto calcs.

Manual Calcs

To perform manual calcs:

1. To turn Auto Calcs off and perform manual measurements, choose **Auto Calcs -> OFF** on the PW tab of the Touch Panel.
2. After obtaining a waveform, press **Measure**. Choose the appropriate vessel folder or calculation. The system walks you through the measurement.

NOTE: To program which calculations are done under manual calcs when using measurement folders for measuring specific vessels, press the Utility key. Select Measure -> Doppler and program your manual calcs (Auto Calcs OFF). Each vessel must be programmed individually and saved after each change.

Auto Calcs

To perform auto calcs:

1. Ensure that the auto calcs function is on by choosing **Auto Calcs -> Frozen** or **Live** on the PW tab of the Touch Panel.
2. After obtaining a waveform, press **Measure**. Choose the appropriate vessel folder, side and location. The measurements that are pre-programmed are performed automatically and entered in the worksheet.

To modify auto calcs:

1. Select **Modify Auto Calcs** on the Touch Panel.
2. Choose the measurements to be performed with this preset.
3. To save these measurements:
 - If this is a temporary change, press **Return**.
 - If this is a permanent change, select **Save as default**.

The measurements are saved and can be performed with the auto calcs function.

Edit Auto Calcs

Auto Calcs can be edited after taking an Auto Trace measurement.

1. After taking an Auto Calc with a trace, select the measurement result on the result window. The Edit Trace menu window appears.

NOTE: If the system cannot take the trace data correctly from the image, Edit Trace does not work.

2. Select Edit Trace. The first caliper (manual trace caliper) appears on the center of the image. Use the **Trackball** to move the caliper on the trace line to the start point.

*NOTE: To cancel Edit Trace at this time, press **Clear**, **Scan**, or **Freeze**.*

3. Press **Set** to fix the first caliper. The second caliper appears. Edit the trace manually using the second caliper. The Ellipse control is used to edit the trace.

*NOTE: When pressing the **Clear** key once at this time, the second caliper disappears and the first caliper appears in the center of the image.*

*NOTE: If you press **Scan** or **Freeze** at this time, the caliper is automatically fixed and the result window updates.*

4. Press **Set** to fix the second caliper. The trace and the result window are updated. The data is retaken from the trace and updated.

NOTE: While in Edit Trace, Cursor Select is disabled.

The trace data (TAMAX and TAMEAN) is updated, but the other selections (e.g. PS, ED) are not updated by trace. The points can be edited using **Cursor Select** if needed.

5. Repeat Edit Trace as needed.

Helpful hints



HINTS

The following hints can help when making a measurement

- Prior to making measurements, use the Cine function, if necessary, to display the best image.
- As you take measurements, each measurement is given a sequential number on the display and in the Results Window. Nine measurements can be displayed in the Results Window at one time.
- Once the Results Window has nine measurements, if you make any further measurements, the system erases the top (first) measurement and adds the new measurement last ("first in, first out").
- While you are taking a measurement, the value in the Results Window updates until you complete the measurement.

Chapter 8

Abdomen and Small Parts

Describes how to perform Abdomen and Small Parts measurements and calculations.

Abdomen/Small Parts Exam

Preparation

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by the system accuracy, but also by use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

Calculation formulas are available in the *Advanced Reference Manual*.

General Guidelines

New Patient information must be entered before beginning an exam. See 'Beginning a New Patient' on *page 4-3 for more information*.

Any measurement can be repeated by selecting that measurement again from the Touch Panel.

Abdomen

Introduction

Abdominal measurements offer a few different types of measurement studies:

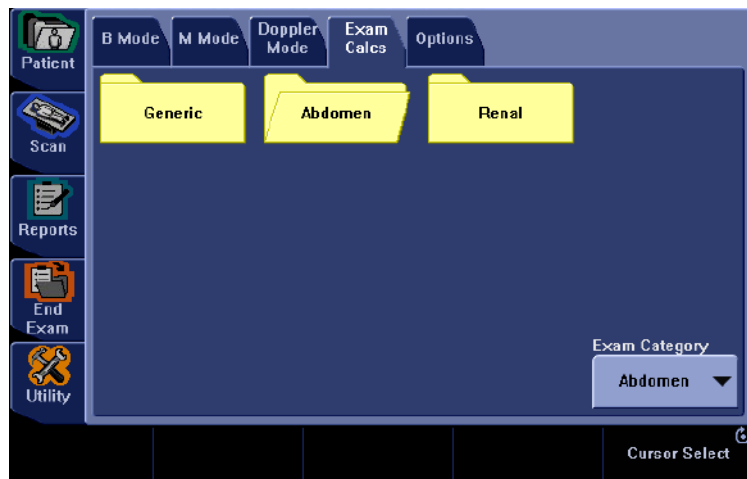


Figure 8-1. Abdomen Exam Category Touch Panel

- Generic—Common to all applications. See ‘Generic Measurements’ on *page 7-78 for more information.*
 - Abdomen
 - Renal
1. Press **Measure**.
 2. Select the Touch Panel key.
 3. Press **Exam Calcs**.
The Abdomen exam category allows you to choose from the displayed studies.
 4. To choose another study, select that desired study.

B-Mode Measurements

The following measurements are located in the Exam Calcs for the Abdomen exam category. Some measurements are only available in certain Exam Calcs. Those specific measurements (splenic length, splenic width, splenic height, Aorta Diameter, and Renal Length) are listed on the following pages.

In B-Mode, the Generic Exam Calcs for Abdomen includes the following measurements:

- % Stenosis
- Volume
- Angle
- Renal Volume
- A/B Ratio

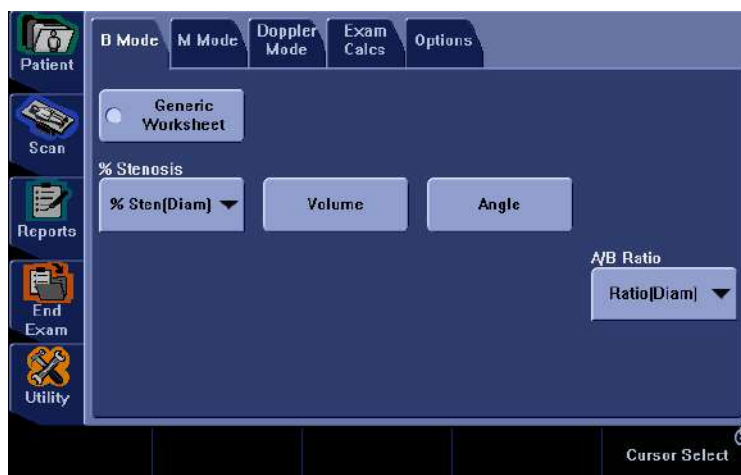


Figure 8-2. Generic Calculations B-Mode Touch Panel

See 'B-Mode Measurements' on *page 7-82* for more information.

Aorta Diameter

This measurement is found in the Abdomen Exam Calcs. To measure aorta diameter, make one distance measurement:

1. Select **Aorta Diameter**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the aorta diameter in the Results Window.

Renal Length

This measurement is found in the Renal Exam Calcs. To measure renal length, make one distance measurement:

1. Select the appropriate orientation (side): Right or Left.
2. Select **Renal Length**; an active caliper displays.
3. To position the active caliper at the start point, move the **Trackball**.
4. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
5. To position the second active caliper at the end point, move the **Trackball**.

A dotted line connects the measurement points, if preset accordingly.

6. To complete the measurement, press **Set**.

The system displays the renal length in the Results Window.

M-Mode Measurements

In M-Mode, all Exam Calcs include the following measurements:

- % Stenosis
- A/B Ratio
- Heart Rate

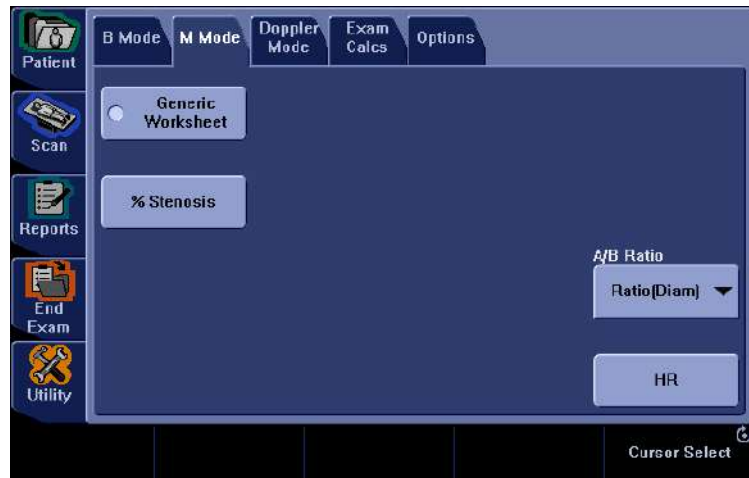


Figure 8-3. Abdominal Calculations M-Mode Touch Panel

See 'M-Mode Measurements' on page 7-91 for more information.

Doppler Mode Measurements

Generic Exam Calcs

In Doppler Mode, the Generic Exam Calcs includes the following measurements:

- PI (Pulsatility Index)
- RI (Resistive Index)
- S/D Ratio
- D/S Ratio
- A/B Ratio
- Max PG (Pressure Gradient)
- Mean PG (Pressure Gradient)
- SV (Stroke Volume)
- Heart Rate

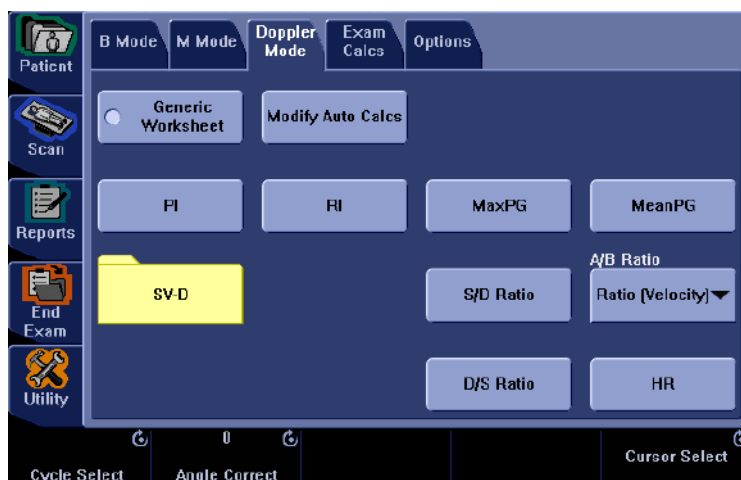


Figure 8-4. Doppler Mode Touch Panel: Abdominal Calculations

See 'Doppler Mode Measurements' on *page 7-94 for more information.*

Abdomen and Renal Exam Calcs

Your system is set up to show the measurements that you usually make for each region of interest. To make a measurement that is not shown for the selected region:

1. Select the folder for the region you want to measure.

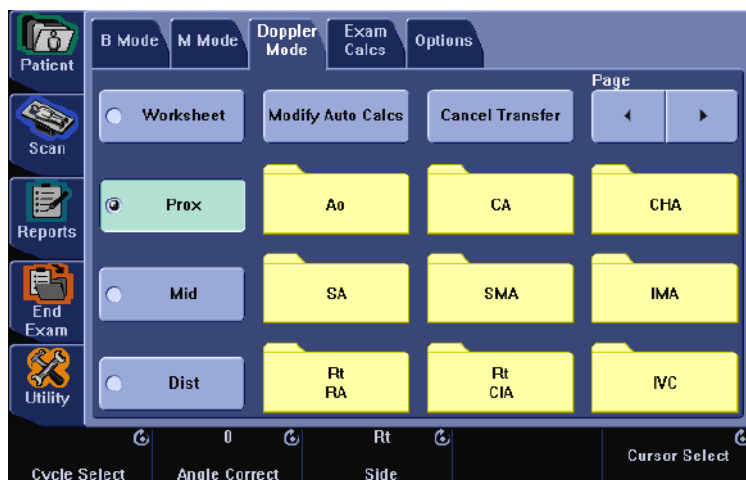


Figure 8-5. Abdomen Exam Calcs Folder (Doppler Mode) in Abdomen exam category

Abdomen and Renal Exam Calcs (continued)

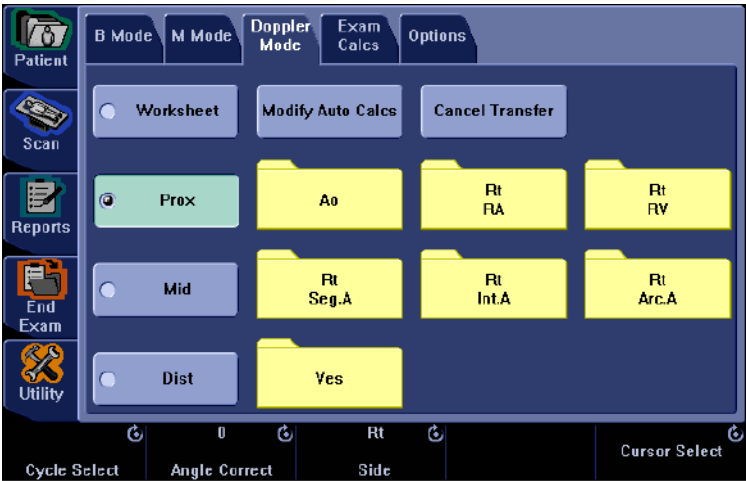


Figure 8-6. Renal Exam Calcs (Doppler Mode)

2. Press Show All.

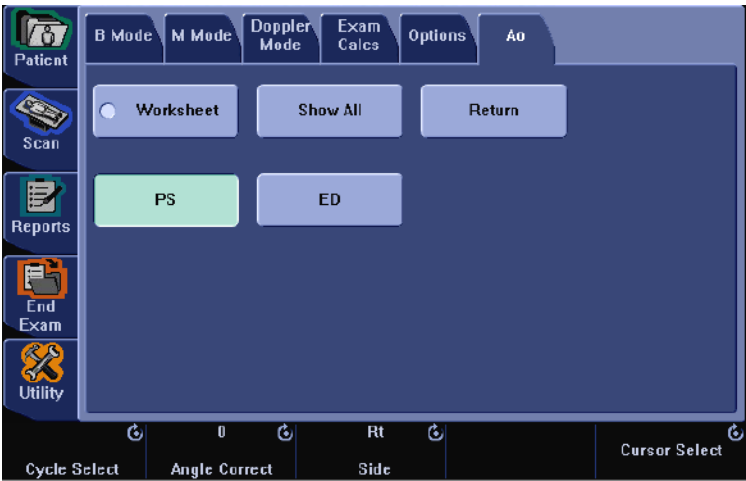


Figure 8-7. Aorta folder

Abdomen and Renal Exam Calcs (continued)

The system displays all possible measurements.

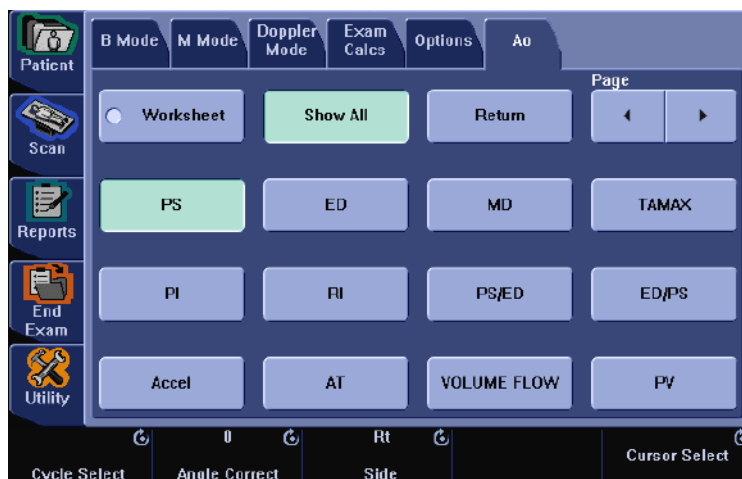


Figure 8-8. Show All measurements

3. Select the desired measurement.

NOTE: *The following instructions assume that you first scan the patient and then press **Freeze**.*

Abdomen and Renal Exam Calcs (continued)

Acceleration

1. Select **Accel**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at peak systole, move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at end diastole, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, acceleration time, and acceleration in the Results Window.

Acceleration Time (AT)

1. Select **AT**.
The system displays an active caliper and a vertical dotted line.
2. To position the caliper at the start point, move the **Trackball**.
3. To fix the first caliper, press **Set**.
The system displays a second active caliper.
4. To position the caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the acceleration time in the Results Window.

Heart Rate (HR)

See 'Heart Rate' on *page 7-93 for more information*.

Abdomen and Renal Exam Calcs (continued)

Peak Systole (PS), End Diastole (ED), or Minimum Diastole (MD)

To calculate the peak systole, end diastole, or minimum diastole:

1. Select **PS**, **ED**, or **MD**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the measurement point, move the **Trackball**.
3. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, or minimum diastole in the Results Window.

Pulsatility Index (PI)

See 'Pulsatility Index (PI)' on *page 7-95 for more information*.

PS/ED or ED/PS Ratio

To calculate the Peak Systole/End Diastole ratio or End Diastole/Peak Systole ratio:

1. Select **PS/ED** or **ED/PS**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at peak systole (PS) or end diastole (ED), move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at end diastole (ED) or peak systole (PS), move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, and PS/ED or ED/PS ratio in the Results Window.

Resistive Index (RI)

See 'Resistive Index (RI)' on *page 7-96 for more information*.

TAMAX

See 'TAMAX and TAMEAN' on *page 7-64 for more information*.

Small Parts

B-Mode Measurements

The Small Parts exam category includes the following two folders:

- Generic, measurements listed on the following page
- Small Parts, which includes the thyroid measurements described in this section

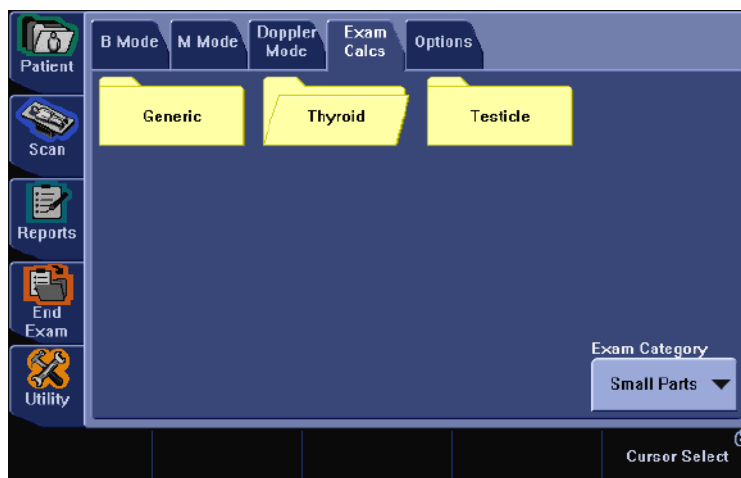


Figure 8-9. Small Parts Exam Category Touch Panel

Generic Measurements

In B-Mode, the Generic Exam Calcs for Small Parts includes the following measurements:

- % Stenosis
- Volume
- Angle
- A/B Ratio



Figure 8-10. Generic Calculations B-Mode Touch Panel

See 'B-Mode Measurements' on *page 7-82* for more information.

Thyroid length, width, and height

Each of these is a standard distance measurement. Length and height are typically measured in the sagittal plane. Width is measured in the transverse/axial plane.

To measure thyroid length, width, or height:

1. On the **Exam Calcs**, select **Small Parts**.
2. Select **Thyroid**.
3. Select the appropriate orientation (side): Right or Left.
4. Select **Thyroid L**, **Thyroid W**, or **Thyroid H**.

An active caliper displays.

5. Perform a standard distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper, if preset accordingly.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window.
6. To make the second and third distance measurement, repeat steps 2–3.

After you complete the third distance measurement, the system displays the thyroid volume in the Results Window.

M-Mode Measurements

In M-Mode, all Exam Calcs include the following measurements:

- % Stenosis
- A/B Ratio
- Heart Rate

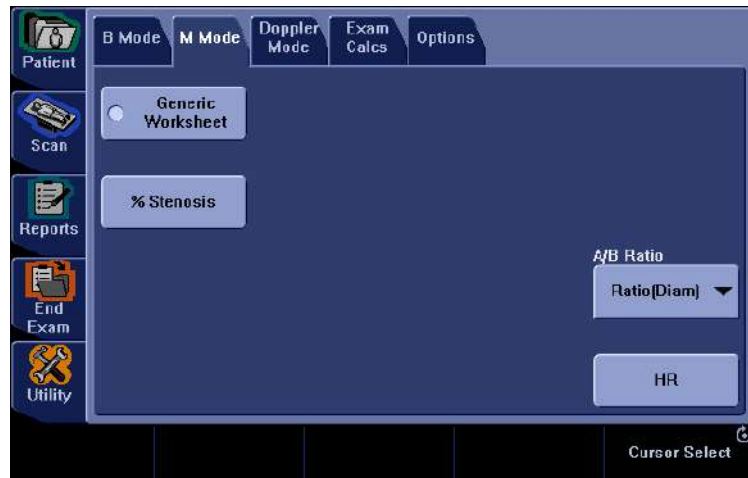


Figure 8-11. Abdominal Calculations M-Mode Touch Panel

See 'M-Mode Measurements' on page 7-91 for more information.

Doppler Mode Measurements

In Doppler Mode, the Exam Calcs include the following measurements:

- PI (Pulsatility Index)
- RI (Resistive Index)
- S/D Ratio
- D/S Ratio
- A/B Ratio
- Max PG (Pressure Gradient)
- Mean PG (Pressure Gradient)
- Heart Rate

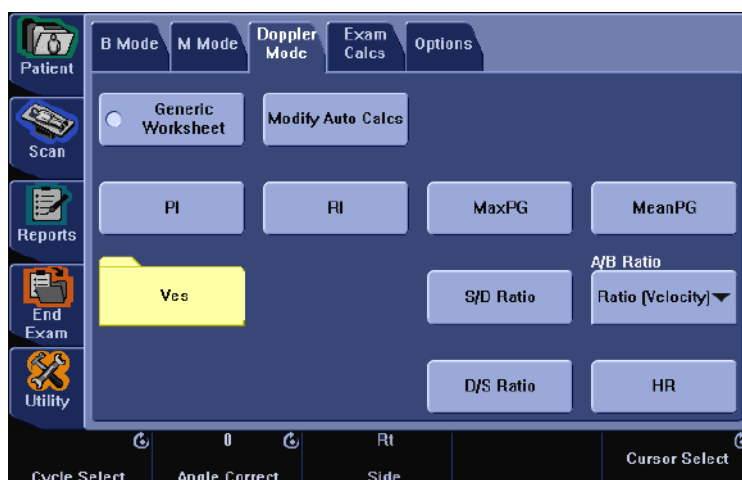


Figure 8-12. Doppler Mode Touch Panel: Abdominal Calculations

See 'Doppler Mode Measurements' on page 7-94 for more information.

Calculation formulas are available in the *Advanced Reference Manual*.

Chapter 9

OB/GYN

Describes how to perform obstetric and gynecology measurements and calculations, and how to use OB graphs and worksheets.

Exam Preparation

Prior to an ultrasound examination, the patient should be informed of the clinical indication, specific benefits, potential risks, and alternatives, if any. In addition, if the patient requests information about the exposure time and intensity, it should be provided. Patient access to educational materials regarding ultrasound is strongly encouraged to supplement the information communicated directly to the patient. Furthermore, these examinations should be conducted in a manner and take place in a setting which assures patient dignity and privacy.

- Prior material knowledge and approval of the presence of nonessential personnel with the number of such personnel kept to a minimum.
- An intent to share with the parents per the physician's judgement, either during the examination or shortly thereafter, the information derived.
- An offer of choice about viewing the fetus.
- An offer of choice about learning the sex of the fetus, if such information becomes available.

Ultrasound examinations performed solely to satisfy the family's desire to know the fetal sex, to view the fetus, or to obtain a picture of the fetus should be discouraged.

Acoustic Output Considerations

General warning

The LOGIQ 9 system is a multi-use device which is capable of exceeding FDA Pre-enactment acoustic output (spatial peak-temporal average) intensity limits for fetal applications.



It is prudent to conduct an exam with the minimum amount and duration of acoustic output necessary to optimize the image's diagnostic value.

Concerns surrounding fetal exposure

Always be aware of the acoustic output level by observing the Acoustic Output Display. In addition, become thoroughly familiar with the Acoustic Output Display and equipment controls affecting output.

Training

It is recommended that all users receive proper training in fetal Doppler applications before performing them in a clinical setting. Please contact a local sales representative for training assistance.

To Start an Obstetrics Exam

NOTE: *Calculation formulas are listed in the Advanced Reference Manual.*

To begin an Obstetrics exam, you enter patient data or, if the patient data from a previous exam is saved in the system, find the patient information.

1. On the control panel, press **Patient**.
The Patient Data Entry screen is displayed.
2. On the Patient Data Entry screen, select New Patient.
3. To choose an Obstetrics exam, move the **Trackball** to highlight Obstetrics, then press **Set**.
The obstetric fields are listed in the Exam Information section of the Patient Data Entry screen.
4. Do one of the following:
 - If the patient data is already stored in the system, search for the data. Use the search fields in the bottom section of the Patient Data Entry screen. For information about how to search for patient data, See 'Changing Patient Information or an Exam' on page 4-27 for more information.

When the correct patient data is listed in the search list, move the **Trackball** to highlight the patient name and press **Set**. The system displays the patient data.

NOTE: *To change patient data, use the **Trackball** to move the cursor to the field and press **Set**. Press **Backspace** to delete the data, and then type the correct data.*

- If the patient data is not stored in the system, enter the data. To enter data in a field, move the **Trackball** to highlight the field and then press **Set**. Use the **Tab** key to move between fields. Obstetric patient fields are listed in Table 9-1.

NOTE: *For information about entering general patient data such as Patient ID and name, See 'Beginning a New Patient' on page 4-3 for more information.*

To Start an Obstetrics Exam (continued)

Table 9-1: Obstetric fields

Field	Description
LMP	Last Menstrual Period; enter the date that the patient started her last menstrual period, in mm/dd/yyyy format. You must enter 4 digits for the year. When you type the month and day, the system fills in the /.
BBT	Basal Body Temperature.
EDD by LMP	Estimated Delivery Date by LMP; the system fills in the date after you enter the LMP.
GA by LMP	Gestational Age by LMP; the system fills in the age after you enter the LMP.
Gravida	Number of pregnancies.
Para	Number of births.
AB	Number of abortions.
Ectopic	Number of ectopic pregnancies.
Fetus #	Number of fetuses; default is 1. Can be 1-4.
Accession #	Exam number used with hospital information system (DICOM). This is a tracking number from the worklist.
Exam Description	Describe the type of exam.
Perf Physician	The physician who performs the exam. Choose from the list or type the name.
Ref. Physician	The physician who requested the exam. Choose from the list or type the name.
Operator	The person (not a physician) who performs the scan. Choose from the list.

NOTE: To fill in the following information, move the **Trackball** to highlight the **Detail** button and press **Set**.

Table 9-2: Obstetric fields: Detail

Field	Description
Indications	Why the patient needs the ultrasound exam.
Comments	Comments about the exam.

To Start an Obstetrics Exam (continued)

After you complete the patient information, you can begin the scan.

1. To change from the Patient Data Entry screen to the Scan screen, do one of the following:
 - On the keyboard, press **Esc**.
 - On the Touch Panel, select **Scan**.
 - On the Control Panel, select **Patient** or **Freeze**.
 - On the Control Panel, press the **B-Mode** key.

The system displays the Scan screen.

2. To choose the appropriate probe, select the probe icon on the Touch Panel.
3. On the Control Panel, press **Measure**.

The default Obstetrics study is displayed on the Touch Panel.

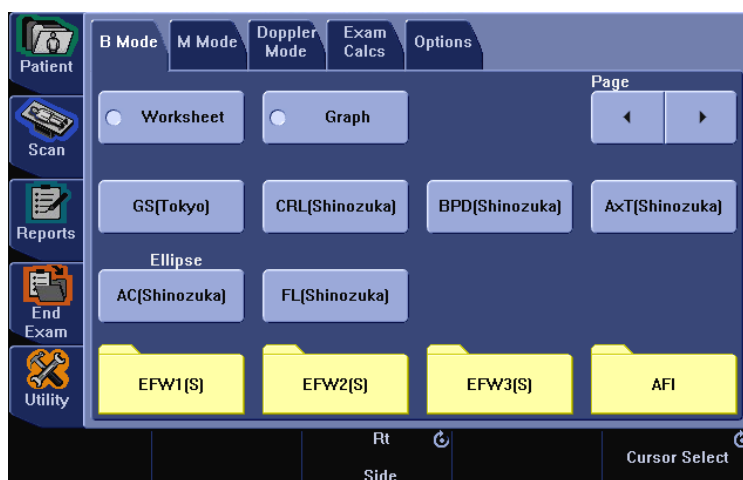


Figure 9-1. OB-General Study

To choose a study

1. To change the study on the exam category, select **Preset**.
The Obstetrics exam category allows you to choose from the following studies:
 - Generic
 - OB-1
 - OB-2/3
 - OB-General
 - Fetal Heart
 - OB/GYN Vessel
2. To select a study, select the appropriate study on the Touch Panel.

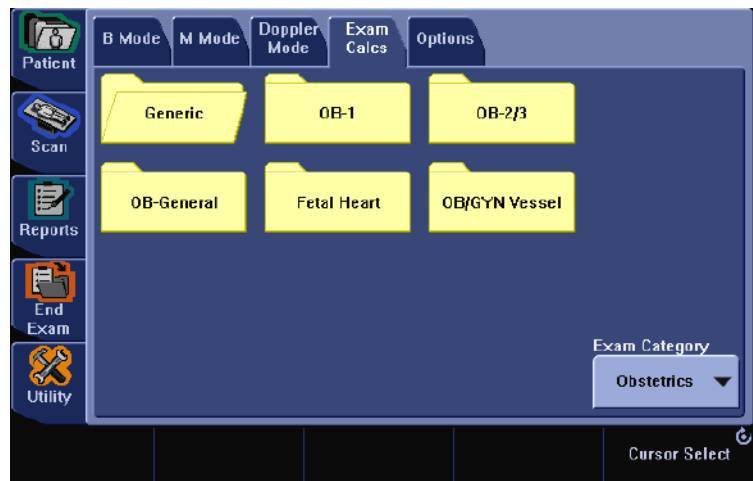


Figure 9-2. OB Study Touch Panel

NOTE: *The folders you see on the Touch Panel may be different if your system has been customized.*

OB Measurements and Calculations

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by the system accuracy, but also by use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

Introduction (continued)

When you make measurements, you can select the calculation before you make the measurement or after you make it. If you select the calculation before you make the measurement, the Results Window shows the estimated fetal age as you make the measurement. If you select the calculation after you make the measurement, the estimated fetal age is displayed after you complete the measurement. The measurements steps in this section tell you to select the calculation before you make the measurement.

The following pages describe how to make OB measurements and calculations. The measurements are organized by mode, and within mode are listed in alphabetical order.

Out of Range – If the system indicates that a measurement is out of range (OOR), it means one of the following:

- The measurement is out of the normal range based on the gestational age that is calculated from the LMP. The system determines OOR from the ultrasound age compared to the gestational age. The gestational age is calculated from the last menstrual period or the estimated delivery date.
- The measurement is outside of the range for the data used in the calculation. That means that the measurement is either less than or more than the range of measurements used to determine fetal age based on the measurement.

NOTE: *Calculation formulas are listed in the Advanced Reference Manual.*

B-Mode Measurements

This section describes all B-Mode measurements that you typically find in OB studies. Additional OB measurements follow the typical ones.

Abdominal Circumference (AC)

To calculate abdominal circumference, you make an ellipse, trace, spline trace or two distance measurements.

- Ellipse**
1. Select **AC**; an active caliper displays.
 2. If necessary, select **AC** until *Ellipse* is displayed.
 3. To position the active caliper, move the **Trackball**.
 4. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 5. To position the second caliper, move the **Trackball**.
 6. Adjust the **Ellipse** control; an ellipse with an initial circle shape displays.
 - To position the ellipse and to size the measured axes (move the calipers), move the **Trackball**.
 - To increase the size, adjust the **Ellipse** control in a clockwise direction.
 - To decrease the size, adjust the **Ellipse** control in a counterclockwise direction.
 - To toggle between active calipers, press **Measure**.
 7. To complete the measurement, press **Set**.
The system displays the circumference in the Results Window.



HINTS

Before you complete the ellipse measurement:

- To erase the ellipse and the current data measured, press **Clear** once. The original caliper is displayed to restart the measurement.
- To exit the measurement function without completing the measurement, press **Clear** a second time.

Abdominal Circumference (AC) (continued)

- Trace**
1. Select **AC**; an active caliper displays.
 2. If necessary, select **AC** until *Trace* is displayed.
 3. To position the trace caliper, move the **Trackball**.
 4. To fix the trace start point, press **Set**.
The trace caliper changes to an active caliper.
 5. To trace the measurement area, move the **Trackball** around the anatomy.
 6. To complete the measurement, press **Set**.
The system displays the circumference in the Results Window.



HINTS

Before you complete the ellipse measurement:

- To erase the line (bit by bit) back from its current point, move the **Trackball** or adjust the **Ellipse** control counterclockwise.
- To erase the dotted line but not the trace caliper, press **Clear** once.
- To clear the trace caliper and the current data measured, press **Clear** twice.

Abdominal Circumference (AC) (continued)

- Spline Trace**
1. Select **AC**; an active caliper displays.
 2. If necessary, select *Spline*.
 3. To position the trace caliper, move the **Trackball**.
 4. To fix the trace start point, press **Set**. The second caliper appears at the same position.
 5. To position the second caliper, move the **Trackball** and press **Set**. The third caliper appears at the same position.
 6. To position the third caliper, move the **Trackball** and press **Set**.

NOTE: *The spline trace requires at least 3 points to draw a trace. Continue setting the points of the trace until the desired points are set (by moving the Trackball to the appropriate position and pressing Set).*

7. To finalize the spline trace, press **Set** twice at the last caliper position.

The system displays the circumference in the Results Window.

Abdominal Circumference (AC) (continued)**Two distances**

1. Select **AC**; an active caliper displays.
2. If necessary, select **AC** until *Caliper* is displayed.
3. To make the first distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points, if preset accordingly.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window. After the first measurement, the system displays an active caliper.
4. To make the second distance measurement, repeat steps a–d above.
The system displays the abdominal circumference in the Results Window.

**HINTS**

- Before you complete a measurement (press **Set**):
 - To toggle between active calipers, press **Measure**.
 - To erase the second caliper and the current data measured and start the measurement again, press **Clear** once.
- To rotate through and activate fixed calipers from other measurements, adjust **Cursor Select**.
- After you complete the measurement, to erase all data that has been measured to this point, but not data entered onto worksheets, press **Clear**.

Biparietal Diameter



To measure biparietal diameter, make one distance measurement:

1. Select **BPD**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the biparietal diameter in the Results Window.

Crown Rump Length



To measure crown rump length, make one distance measurement:

1. Select **CRL**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the crown rump length in the Results Window.

Femur Length



To measure femur length, make one distance measurement:

1. Select **FL**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the femur length in the Results Window.

Gestational Sac



To calculate the gestational sac, you make three distance measurements in two scan planes. To display two scan planes, press the **L** or **R** key. Get an image in each scan plane and press **Freeze**.

1. Select **GS**; an active caliper displays.
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.

The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.

A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.

The system displays the distance value in the Results Window and displays an active caliper.
2. To make the second and third distance measurement, repeat steps a–d.

After you complete the third distance measurement, the system displays the gestational sac measurement in the Results Window.

Head Circumference (HC)



To calculate head circumference, you make an ellipse, trace, spline trace or two distance measurements.

Ellipse

1. Select **HC**; an active caliper displays.
2. If necessary, select **HC** until *Ellipse* is displayed.
3. To position the active caliper, move the **Trackball**.
4. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
5. To position the second caliper, move the **Trackball**.
6. Adjust the **Ellipse** control; an ellipse with an initial circle shape displays.
 - To position the ellipse and to size the measured axes (move the calipers), move the **Trackball**.
 - To increase the size, adjust the **Ellipse** control in a clockwise direction.
 - To decrease the size, adjust the **Ellipse** control in a counterclockwise direction.
 - To toggle between active calipers, press **Measure**.
7. To complete the measurement, press **Set**.
The system displays the circumference in the Results Window.



HINTS

Before you complete the ellipse measurement:

- To erase the ellipse and the current data measured, press **Clear** once. The original caliper is displayed to restart the measurement.
- To exit the measurement function without completing the measurement, press **Clear** a second time.

Head Circumference (HC) (continued)

- Trace**
1. Select **HC**; an active caliper displays.
 2. If necessary, select **HC** until *Trace* is displayed.
 3. To position the trace caliper, move the **Trackball**.
 4. To fix the trace start point, press **Set**.
The trace caliper changes to an active caliper.
 5. To trace the measurement area, move the **Trackball** around the anatomy.
 6. To complete the measurement, press **Set**.
The system displays the circumference in the Results Window.

**HINTS**

Before you complete the ellipse measurement:

- To erase the line (bit by bit) back from its current point, move the **Trackball** or adjust the **Ellipse** control counterclockwise.
- To erase the dotted line but not the trace caliper, press **Clear** once.
- To clear the trace caliper and the current data measured, press **Clear** twice.

Head Circumference (HC) (continued)

- Spline Trace**
1. Select **HC**; an active caliper displays.
 2. If necessary, select *Spline*.
 3. To position the trace caliper, move the **Trackball**.
 4. To fix the trace start point, press **Set**. The second caliper appears at the same position.
 5. To position the second caliper, move the **Trackball** and press **Set**. The third caliper appears at the same position.
 6. To position the third caliper, move the **Trackball** and press **Set**.

NOTE: *The spline trace requires at least 3 points to draw a trace. Continue setting the points of the trace until the desired points are set (by moving the Trackball to the appropriate position and pressing Set).*

7. To finalize the spline trace, press **Set** twice at the last caliper position.

The system displays the circumference in the Results Window.

Head Circumference (HC) (continued)**Two distances**

1. Select **HC**; an active caliper displays.
2. If necessary, select **HC** until *Caliper* is displayed.
3. To make the first distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.

The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.

A dotted line connects the measurement points, if preset accordingly.
 - d. To complete the measurement, press **Set**.

The system displays the distance value in the Results Window. After the first measurement, the system displays an active caliper.
4. To make the second distance measurement, repeat steps a–d above.

The system displays the abdominal circumference in the Results Window.

**HINTS**

- Before you complete a measurement (press **Set**):
 - To toggle between active calipers, press **Measure**.
 - To erase the second caliper and the current data measured and start the measurement again, press **Clear** once.
- To rotate through and activate fixed calipers from other measurements, adjust **Cursor Select**.
- After you complete the measurement, to erase all data that has been measured to this point, but not data entered onto worksheets, press **Clear**.

Amniotic Fluid Index (AFI)



To calculate the amniotic fluid index, you make measurements of the four quadrants of the uterine cavity. The system adds these four measurements together to calculate the Amniotic Fluid Index.

NOTE: *The four quadrants can be measured with distance (caliper) or circumference (circle) measurements. Press the appropriate AFI quadrant Touch Panel key to toggle between caliper and circle.*

1. Select **AFI**.

The first distance measurement, AFI-Q1, is already selected.

2. Make a standard distance measurement for the first quadrant:

a. To position the active caliper at the start point, move the **Trackball**.

b. To fix the start point, press **Set**.

The system fixes the first caliper and displays a second active caliper.

c. To position the second active caliper at the end point, move the **Trackball**.

A dotted line connects the measurement points.

d. To complete the measurement, press **Set**.

The system displays the distance value in the Results Window.

3. When the measurement of the first quadrant is completed, unfreeze and move to the second quadrant.

4. After you obtain the image, press **Freeze** and then **Measure**.

The system prompts you to continue with the AFI measurements. Make sure that the next quadrant has been selected.

Amniotic Fluid Index (AFI) (continued)

5. Perform a standard distance measurement for the second, third, and fourth quadrants (see step 2).

When all four quadrants have been measured, the system calculates the AFI total and displays it in the Results Window.

**HINTS**

- If you unfreeze the image after doing an AFI measurement, the system does not delete the previous measurements. Unfreeze and change scan planes as necessary.
- To specify that an unassigned distance measurement be used for an AFI measurement:
 - Select **AFI**.
 - Press **Measure**.
 - Move the **Trackball** to highlight the unassigned distance measurement in the Results Window.
 - Select the AFI measurement on the Touch Panel.
- If the fluid in a pocket is zero, set the second caliper on top of the first one to give it a zero value.
- You can measure an AFI quadrant that is zero (0) by pressing **Set** twice.

A/B Ratio

In B-Mode you can calculate A/B ratio by diameter or by area.
See 'A/B Ratio' on *page 7-90 for more information.*

Angle

See 'Angle' on *page 7-89 for more information.*

Antero-Postero Trunk Diameter & Transverse Trunk Diameter (APTD-TTD)



Make two distance measurements, one of the antero-postero trunk diameter and one of the transverse trunk diameter.

1. Select **APTD_TTD**; an active caliper displays.
2. Make a distance measurement of the antero-postero trunk diameter:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window and displays an active caliper.
3. To make a distance measurement of the transverse trunk diameter, repeat steps a–d above.

The system displays the antero-postero trunk diameter and the transverse trunk diameter in the Results Window.

Antero-Postero Trunk Diameter by Transverse Trunk Diameter (AxT)



Make two distance measurements, one of the antero-postero trunk diameter and one of the transverse trunk diameter.

1. Select **AxT**; an active caliper displays.
2. Make a distance measurement of the antero-postero trunk diameter:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window and displays an active caliper.
3. To make a distance measurement of the transverse trunk diameter, repeat steps a–d above.

The system displays the antero-postero trunk diameter, the transverse trunk diameter, and AxT in the Results Window.

Cardio-Thoracic Area Ratio (CTAR)



To calculate cardio-thoracic area ratio, you make two ellipse measurements.

1. Select **CTAR**; an active caliper displays.
2. Make an ellipse measurement of the cardiac area:
 - a. To position the active caliper, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second caliper, move the **Trackball**.
 - d. Adjust the **Ellipse** control; an ellipse with an initial circle shape displays.
 - To position the ellipse and to size the measured axes (move the calipers), move the **Trackball**.
 - To increase the size, adjust the **Ellipse** control in a clockwise direction.
 - To decrease the size, adjust the **Ellipse** control in a counterclockwise direction.
 - To toggle between active calipers, press **Measure**.
 - e. To complete the ellipse measurement, press **Set**.
The system displays the cardiac area measurement in the Results Window
3. To make an ellipse measurement of the thoracic area, repeat steps a–e.

The system displays the cardio-thoracic area ratio in the Results Window.

Estimated Fetal Weight (EFW)



To measure estimated fetal weight, you make several OB measurements. These measurements can vary, based on how your system is set up. Measurements can include biparietal diameter, fetal trunk area, femur length, antero-postero trunk diameter and transverse trunk diameter, abdominal circumference, head circumference and spinal length.

1. Select **EFW**.

The system displays the required measurements.

2. Make each measurement.

The system displays each measurement and the estimated fetal weight in the Results Window.

NOTE: *For a description of any of the required measurements, refer to that measurement.*

Fetal Trunk Area (FTA)



To measure fetal trunk area, you make an ellipse, trace, spline trace or two distance measurements.

Ellipse

1. Select **FTA**; an active caliper displays.
2. To position the active caliper, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second caliper, move the **Trackball**.
5. Adjust the **Ellipse** control; an ellipse with an initial circle shape displays.
 - To position the ellipse and to size the measured axes (move the calipers), move the **Trackball**.
 - To increase the size, adjust the **Ellipse** control in a clockwise direction.
 - To decrease the size, adjust the **Ellipse** control in a counterclockwise direction.
 - To toggle between active calipers, press **Measure**.
6. To complete the measurement, press **Set**.
The system displays the measurement in the Results Window.



HINTS

Before you complete the ellipse measurement:

- To erase the ellipse and the current data measured, press **Clear** once. The original caliper is displayed to restart the measurement.
- To exit the measurement function without completing the measurement, press **Clear** a second time.

Fetal Trunk Area (FTA) (continued)

- Trace**
1. Select **FTA**; a trace caliper displays.
 2. To position the trace caliper at the start point, move the **Trackball**.
 3. To fix the trace start point, press **Set**.
The trace caliper changes to an active caliper.
 4. To trace the measurement area, move the **Trackball** around the anatomy.
A dotted line shows the traced area.
 5. To complete the measurement, press **Set**.
The system displays the measurement in the Results Window.

**HINTS**

Before you complete the trace measurement:

- To erase the line (bit by bit) back from its current point, move the **Trackball** or adjust the **Ellipse** control counterclockwise.
- To erase the dotted line but not the trace caliper, press **Clear** once.
- To clear the trace caliper and the current data measured, press **Clear** twice.

Fetal Trunk Area (FTA) (continued)

- Spline Trace**
1. Select **FTA**; an active caliper displays.
 2. If necessary, select *Spline*.
 3. To position the trace caliper, move the **Trackball**.
 4. To fix the trace start point, press **Set**. The second caliper appears at the same position.
 5. To position the second caliper, move the **Trackball** and press **Set**. The third caliper appears at the same position.
 6. To position the third caliper, move the **Trackball** and press **Set**.

NOTE: *The spline trace requires at least 3 points to draw a trace. Continue setting the points of the trace until the desired points are set (by moving the Trackball to the appropriate position and pressing Set).*

7. To finalize the spline trace, press **Set** twice at the last caliper position.

The system displays the circumference in the Results Window.

Fetal Trunk Area (FTA) (continued)**Two distances**

1. Select **FTA**; an active caliper displays.
2. Make the first distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points, if preset accordingly.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window and displays an active caliper.
3. To make the second distance measurement, repeat steps a–d above.
The system displays the measurement in the Results Window.

**HINTS**

- Before you complete a measurement:
 - To toggle between active calipers, press **Measure**.
 - To erase the second caliper and the current data measured and start the measurement again, press **Clear** once.
- To rotate through and activate previously fixed calipers, adjust **Cursor Select**.
- After you complete the measurement, to erase all data that has been measured to this point, but not data entered onto worksheets, press **Clear**.

Foot Length



To measure foot length, make one distance measurement:

1. Select **Ft**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the foot length in the Results Window.

Humerus Length



To measure humerus length, make one distance measurement:

1. Select **HL**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the humerus length in the Results Window.

Nuchal Translucency (NT)

To measure nuchal translucency, make one distance measurement:

1. Select **NT**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the nuchal translucency in the Results Window.

NOTE: *Nuchal Translucency is not available through the factory default. To enable Nuchal Translucency, add NT to the measurement folder in Utility -> Measure -> M&A -> Add measurement (Insert).*

Occipitofrontal Diameter



To measure occipitofrontal diameter, make one distance measurement:

1. Select **OFD**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the occipitofrontal diameter in the Results Window.

% Stenosis

In B-Mode, you can calculate % Stenosis by diameter or by area. See '% Stenosis' on *page 7-83 for more information*.

Spinal Length (SL)



To measure spinal length, make one distance measurement:

1. Select **SL**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the spinal length in the Results Window.

Transverse Abdominal Diameter



To measure transverse abdominal diameter, make one distance measurement:

1. Select **TAD**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the transverse abdominal diameter in the Results Window.

Transverse Cerebellar Diameter



To measure transverse cerebellar diameter, make one distance measurement:

1. Select **TCD**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the transverse cerebellar diameter in the Results Window.

Thorax Transverse Diameter



To measure thorax transverse diameter, make one distance measurement:

1. Select **ThD**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the thorax transverse diameter in the Results Window.

Tibia Length



To measure tibia length, make one distance measurement:

1. Select **Tibia**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the tibia length in the Results Window.

Ulna Length



To measure ulna length, make one distance measurement:

1. Select **Ulna**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the ulna length in the Results Window.

Volume

See 'Volume' on page 7-85 for more information.

M-Mode Measurements

In M-Mode you can measure % stenosis, A/B ratio, and heart rate.

% Stenosis

In M-Mode, you measure % Stenosis by diameter. See '% Stenosis' on *page 7-83 for more information*.

A/B Ratio

In M-Mode you can measure A/B ratio by diameter, time, or velocity. See 'A/B Ratio' on *page 7-92 for more information*.

Heart Rate

See 'Heart Rate' on *page 7-93 for more information*.

Doppler Mode Measurements

You can use Doppler mode to study fetal blood flow in the heart, umbilical cord, placenta, and middle cerebral arteries. OB/GYN Doppler mode also allows you to study uterine and ovarian blood flow.

The OB/GYN vessel study includes the following vessels:

- Aorta (Ao)
- Desc. Aorta
- Middle Cerebral Artery (MCA) (right and left)
- Ovarian (right and left)
- Placenta
- Umbilical
- Uterine (right and left)

For each of these studies, you can make any of the following measurements:

- Peak Systole (PS)
- End Diastole (ED)
- Minimum Diastole (MD)
- Heart Rate
- TAMAX
- Pulsatility Index (PI)
- Resistive Index (RI)
- PS/ED Ratio
- ED/PS Ratio
- Acceleration
- AT
- TAMEAN
- Volume Flow
- PV

To select OB/GYN vessel measurements

OB/GYN Vessel measurements use the auto sequence feature. With this feature, when you select a folder for the vessel you want to measure, the system automatically starts the first measurement. It then continues with each of the other measurements in that study.

1. Select the folder for the vessel you want to measure.
The system shows all the measurements for that vessel. The caliper for the first measurement is automatically displayed.
2. Make the measurement.
After you complete each measurement, the system starts the next measurement. After the last measurement is complete, the system returns to the OB/GYN Vessel Touch Panel.

Your system is set up to show the measurements that you usually make for each vessel. To make a measurement that is not shown for the selected vessel:

1. Select the folder for the vessel you want to measure.
2. Select Show All.
The system display all possible vessel measurements.
3. Select the desired measurement.

The following pages describe the steps to make each specific measurement in the OB/GYN Vessel study.

Acceleration

1. Select **Accel**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the onset of systole, move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at early systolic peak, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the acceleration time and acceleration in the Results Window.

Acceleration Time (AT)

1. Select **AT**.
The system displays an active caliper and a vertical dotted line.
2. To position the caliper at the start point, move the **Trackball**.
3. To fix the first caliper, press **Set**.
The system displays a second active caliper.
4. To position the caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the acceleration time in the Results Window.

Heart Rate

This measurement is available in the Fetal Heart study and the OB/GYN Vessel study. See 'Heart Rate' on *page 7-93 for more information*.

Peak Systole (PS), End Diastole (ED), or Minimum Diastole (MD)

To calculate the peak systole, end diastole, or minimum diastole:

1. Select **PS**, **ED**, or **MD**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the measurement point, move the **Trackball**.
3. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, or minimum diastole in the Results Window.

PS/ED or ED/PS Ratio

To calculate the Peak Systole/End Diastole ratio or End Diastole/Peak Systole ratio:

1. Select **PS/ED** or **ED/PS**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at peak systole (PS) or end diastole (ED), move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at end diastole (ED) or peak systole (PS), move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, and PS/ED or ED/PS ratio in the Results Window.

Pulsatility Index (PI)

See 'Pulsatility Index (PI)' on *page 7-95 for more information.*

Resistive Index (RI)

See 'Resistive Index (RI)' on *page 7-96 for more information.*

TAMAX

See 'TAMAX and TAMEAN' on *page 7-64 for more information.*

OB Worksheet

The OB Worksheet lists patient information, and all measurement and calculation data.

To view the OB Worksheet:

1. Press **Measure**.
2. Select **Worksheet Display**.

GE Medical Systems patient, 11/04/04 1:33:34 PM ADM 20041101, GA(LMP)=39w3d --:--:--

Origin LMP LMP 02/02/2004 BBT GA 39w3d EDD(LMP) 11.08/2004

Fetus A/1 CUA 18w1d+/- 1w0d EDD(CUA) 04.06/2005

FetusPos PLAC Ref.Physician Page 1/1

B Mode Measurements

BPD(Hadlock)	☒	3.44 cm	5.36	2.11	2.83	Avg.	16w4d	15w3d-17w6d
HC(Hadlock)	☒	11.49 cm	6.79	16.18		Avg.	15w4d	14w3d-16w6d
OFD(HC)		4.99 cm	2.84	7.15		Avg.		
AC(Hadlock)	☒	13.28 cm	10.28	0.27		Avg.	18w5d	16w5d-20w6d
FL(Hadlock)	☒	4.03 cm	4.95	3.11		Avg.	23w0d	21w2d-24w6d

2D Calculations

EPW(AC,BPD,FL,HC)	330.29g +/- 49.54g	{ 12oz }	
EPW(Hadlock)-GP	<3%		
CI(Hadlock)	> 68.85 (70.00-86.00)	FL/AC(Hadlock)	30.37 (-)
FL/BPD(Hohler)	117.33 (-)	FL/HC(Hadlock)	> 35.10 (15.84-18.04)
HC/AC(Campbell)	> 0.87 (1.08-1.27)		

Meas.Transf

Figure 9-3. OB Worksheet

The OB Worksheet has three sections of information:

1. Patient data
2. Measurement information
3. Calculation information

Patient data

The Patient data section, at the top of the worksheet, lists information from the Patient Data Entry screen.

You can select the following fields:

- FetusNo – if this is a multi-gestational patient, you can select the fetus in this field. You can also adjust the **Fetus** selection to change the fetus.
- CUA/AUA – select the ultrasound age calculation method
 - Composite Ultrasound Age (CUA) – regression calculation
 - Average Ultrasound Age (AUA) – an arithmetic average

You can select the method in this field, or adjust the **Select CUA/AUA** control.

You can enter information in the following fields:

- FetusPos – type information about the fetus position.
- PLAC – type information about the placenta.

Measurement information

This section lists the results of all measurements.

- CUA or AUA – If this field is checked, the system uses the measurement to calculate the ultrasound age.
- Value – The measured value. If more than one measurement was made for an item, the system uses the specified method (average, maximum, minimum, or last) to determine this value.
- m1–m3 – Up to three measurement values for each item. If you make more than three measurements, the worksheet uses the last three.
- Method – When there is more than one measurement for an item, this specifies the method used to calculate the measurement value listed in the Value column. Choices are average, maximum, minimum, or last. To change the method:
 - a. Move the **Trackball** to the Method field.
 - b. Press **Set**.
 - c. Move the **Trackball** to select from the list.
 - d. Press **Set**.
- AGE – The fetal age for this measurement.
- Range – The typical range of fetal age for this measurement.

Calculation information

This section of the worksheet provides calculation choices and lists calculation results.

- EFW – lists the parameters used to calculate EFW. This is followed by the calculation result.

To change which parameters are used:

- a. Select this field or press **Select EFW**.
- b. Select the desired parameters.

- EFW GP – lists the source used to calculate EFW-GP (growth percentile). This is followed by the growth percentile.

To change the source:

- a. Select this field or press **Select GP**.
- b. Select the desired source.

The remaining calculation information shows ratios for several measurements, and the Cephalic Index (CI).

The worksheet shows if any of the ratios are out of range (OOR). Out of range indicates one of the following:

- The measurement is out of the normal range based on the gestational age that is calculated from the LMP. The system determines OOR from the ultrasound age compared to the gestational age. The gestational age is calculated from the last menstrual period or the estimated delivery date.
- The measurement is outside of the range for the data used in the calculation. That means that the measurement is either less than or more than the range of measurements used to determine fetal age based on the measurement.

For more information about how to use the worksheet, See 'Viewing and Editing Worksheets' on *page 7-72 for more information*.

OB/GYN

Anatomical Survey

Overview

The Anatomical Survey page provides a checklist that indicates which anatomy was imaged and its appearance.

[illegible]

Figure 9-4. Anatomical Survey

Editing

- To activate the Anatomical Survey, select **Anatomy** on the OB Worksheet Touch Panel.

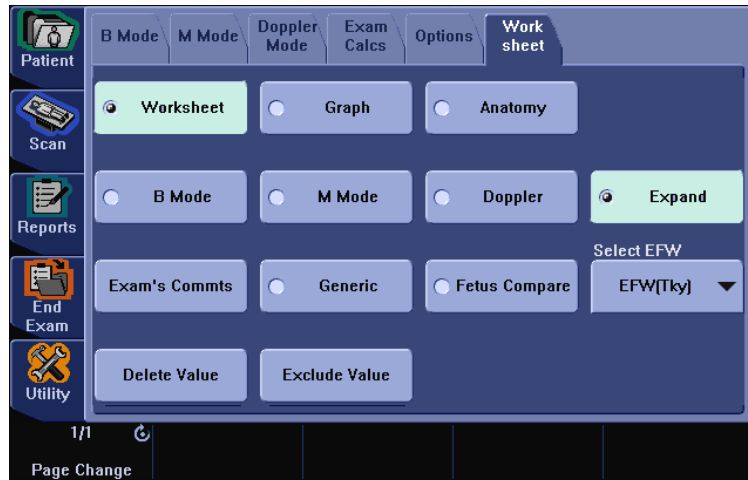


Figure 9-5. OB Worksheet Touch Panel

- Fill the required field.

Table 9-3: Anatomical Survey

Field	Description
Fetus Pos	Indicate the fetal position within the uterus.
PLAC	Identify the location of the placenta.
ANATOMY	Enter the following information for each part of the anatomy imaged: 1. Imaged? : Check the box that you did image this part of the anatomy. 2. Appearance : If you checked the Imaged? box, indicate whether the appearance was normal or abnormal.
	You can add the required anatomy up to 9 additional anatomy. Move the Trackball to the blank field. Enter the anatomy name.
BIOPHYSICAL	The score is _ of 10 possible total points, depending upon the number of parameters entered. Enter the following information to assess the fetus's biophysical well-being.
Movement	Type 0, 1 or 2
Tone	Type 0, 1 or 2
Breathing	Type 0, 1 or 2

Table 9-3: Anatomical Survey (Continued)

Field	Description
Fluid	Type 0, 1 or 2
Reactive NST (Reactive non-stress test)	Type 0, 1 or 2
COMMENTS	Free text

Select Exit to return to the Scan screen.

Select Worksheet to return to the Worksheet or Graph to enter the OB Graph screen on the Touch Panel.



Figure 9-6. Anatomical Survey Touch Panel

NOTE: The patient specific contents input on the Anatomical Survey page are returned to the factory default settings after starting a new patient.

OB Graphs

Overview

OB Graphs allow you to assess fetal growth compared to a normal growth curve. When a patient has completed two or more ultrasound exams, you can also use the graphs to look at fetal trending. For multi-gestational patients you can plot all fetuses and compare the growth on the graphs.

The LOGIQ 9 provides the following two basic types of graphs:

- **Fetal Growth Curve graphs** – show one measurement per graph. These graphs show the normal growth curve, positive and negative standard deviations or applicable percentiles, and ultrasound age of the fetus using the current measurement. For multi-gestational pregnancies, you can view all fetuses. If previous exam data is available, the graph can show fetal trending.
- **Fetal Growth Bar graph** – shows the ultrasound age and the gestational age based on patient data. Plots all measurements on one graph.

To View OB Graphs

To view OB graphs:

1. Press **Measure**.
2. Select **Graph**.

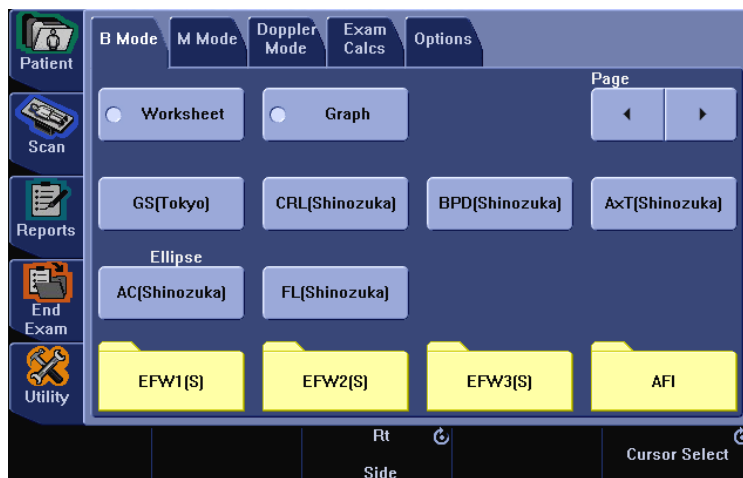


Figure 9-7. B-Mode OB Measurement Touch Panel

To View OB Graphs (continued)

After you select **Graph Display**, the system displays the OB Graph keys on the Touch Panel.

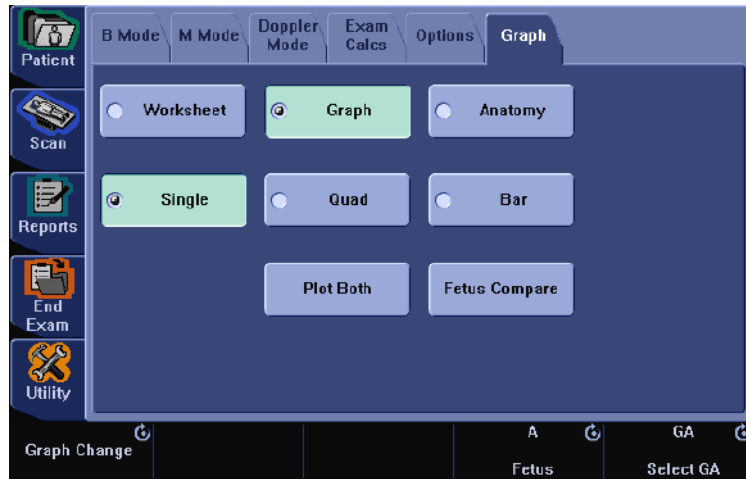


Figure 9-8. OB Graph keys on Touch Panel

To View OB Graphs (continued)

The monitor display shows the Fetal Growth Curve Graph.

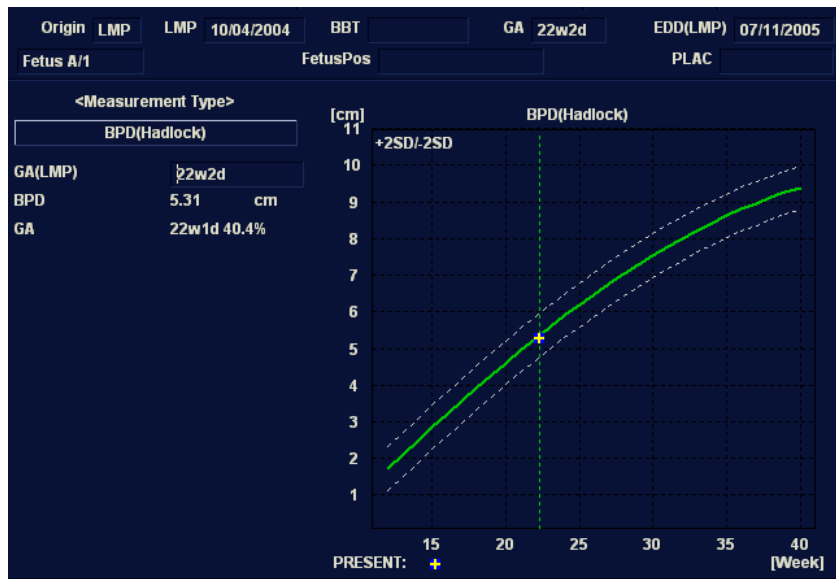


Figure 9-9. Fetal Growth Curve Graph

Fetal Growth Curve Graph

The horizontal axis shows the fetal age in weeks. The system determines this age from the data on the Patient Data Entry screen. The vertical axis shows one of the following:

- For measurements, mm or cm
- For ratios, percent
- For fetal weight, grams

The Fetal Growth Curve Graph shows the following information for the selected measurement:

- The normal growth curve
- The standard deviations or relevant percentiles
- The gestational age of the fetus, using patient data (vertical dotted line)
- Using the current ultrasound measurement data, where the fetus is on the growth curve

The legend at the bottom of the graph shows the symbols and colors that indicate data for fetal trending (Past and Present) and multiple gestation (Fetus).

To select the measurement

To select which measurement you want to display on the Fetal Growth Curve Graph, do one of the following:

- To select a specific measurement:
 - a. On the graph display, move the **Trackball** to the Measurement Type field and press **Set**.
The system displays a list of measurements.
 - b. Move the **Trackball** to select the desired measurement and press **Set**.

The system displays the Fetal Growth Curve Graph for the selected measurement.
- To scroll through all Fetal Growth Curve Graphs, adjust the **Graph Change** control.

- To select the age to use

To plot the fetus age, the system allows you to use the gestational age (GA) from the LMP, or to use the composite ultrasound age (CUA). To select, adjust the **Select GA** control. The information in the left column changes between CUA and GA(EDD), and the data may change.
- To view a single or four graphs

You can view either a single Fetal Growth Curve Graph or you can view four graphs at the same time. To select each view, press **Single** or **Quad**.

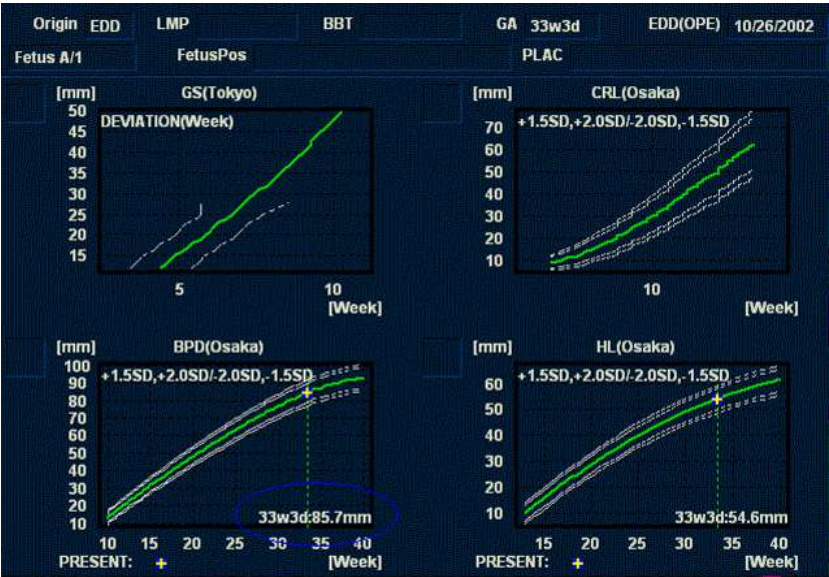


Figure 9-10. Fetal Growth Curve Graph: Quad View

The measurement values are displayed at the bottom of the graph.

**To change
measurements in
quad view**

When you view four graphs simultaneously, you can select which four you want to see. To change each graph in quad view:

1. On the graph display, use the **Trackball** to move the cursor to the small box that is upper left of each graph, then press **Set**.

The system displays a list of measurements.

2. Move the **Trackball** to select the desired measurement and press **Set**.

The system displays the Fetal Growth Curve Graph for the selected measurement.

To scroll through all Fetal Growth Curve Graphs, adjust the **Graph Change** control.

The order of a quad graph view can be saved by selecting **Save**.

Fetal Trending

When you have ultrasound data for more than one exam for a patient, you can use the data to look at fetal trending on the Fetal Growth Curve Graphs.

1. Select **Graph Display** and select the desired Fetal Growth Curve Graph.
2. Select **Plot Both**.

The system automatically finds the data from previous ultrasound exams, and displays it on the graph with the present data.

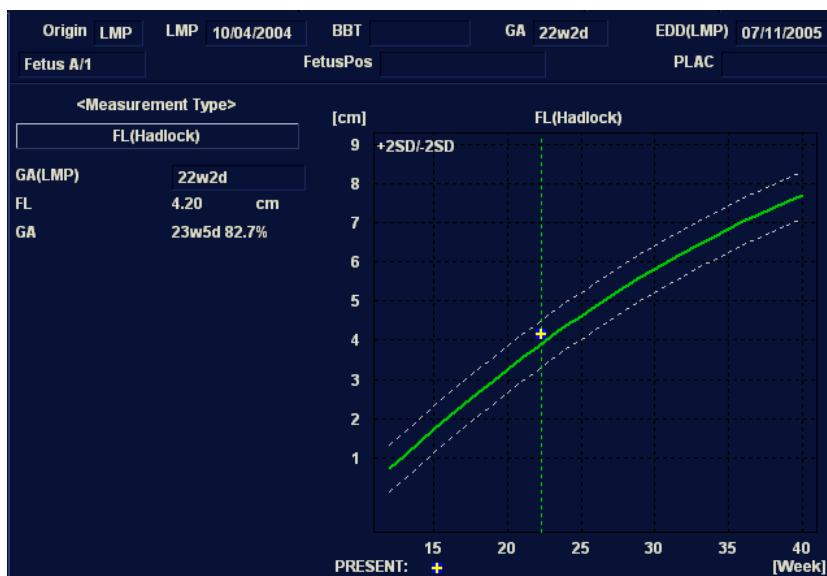


Figure 9-11. Fetal Trending on Fetal Growth Curve Graph: FL

The legend at the bottom of the graph shows the symbols and colors that indicate Past and Present data.

Fetal Trending (continued)

To manually enter past exam data

If you have data from a previous ultrasound exam that you want to use for fetal trending, but it is not in the system, you can manually enter the data.

1. After you have registered the patient for this exam, on the Patient Data Entry screen, in the Exam Information (Obstetrics) section, select Past Exam.
The system displays the Input Past Exam screen. See Figure 9-12.
2. Enter the data from previous exams.
3. To enter data on page 2, select Next.
4. After you enter the previous exam data, select Exit to Save.

The system saves the previous exam data. When you view the Fetal Growth Curve Graphs, select **Plot Both** to view fetal trending. The system automatically uses the data you entered.

Fetus A

Prev Next

Page 1 of 2

Input Past Exam

Exam Date (mm/dd/yyyy)	EFW g Hadlock	BPD cm Hadlock	HC cm Hadlock	AC cm Hadlock	FL cm Hadlock

Past Exam Data is used for plotting on Fetal Trend Graph

Cancel Exit to Save

Figure 9-12. Input Past Exam screen, page 1

Figure 9-13. Input Past Exam screen, page 2

To edit patient data

When you are working with graphs, you can change or enter the following patient data.

- GA(LMP) – this field is computed using the LMP date on the Patient Data Entry screen. To change this field:
- NOTE:** *You can only change this field on the Fetal Growth Curve Graph in single view.*

- a. Move the **Trackball** to the field, which is left of the graph. To select the field, press **Set**.
The system displays a window with the GA weeks and days.
- b. To select each field, move the **Trackball** to the field and press **Set**.
- c. Type the correct weeks or days.
- d. Select OK.

The system makes the following changes:

- GA (LMP) is now GA (GA) and shows the age you entered.
- In the Patient Data section, the GA changes.
- In the Patient Data section, The EDD (LMP) changes to EDD(GA) and shows an updated date, using the GA you entered.

The LMP is erased.

- FetusPos – type information about the fetus position.
- PLAC – type information about the placenta.

To return from a graph to the scan display

After viewing graphs, to return to the scan display, do one of the following:

- On the graph display, select Exit.
- On the Touch Panel, select **Graph Display**.

Fetal Growth Bar Graph

The fetal growth bar graph shows current exam measurements and the normal growth range based on the gestational age. It shows all measurements on one graph.

To view the Fetal Growth Bar Graph:

- 1. Press **Measure**.
- 2. Select **Graph**.
- 3. Select **Bar**.

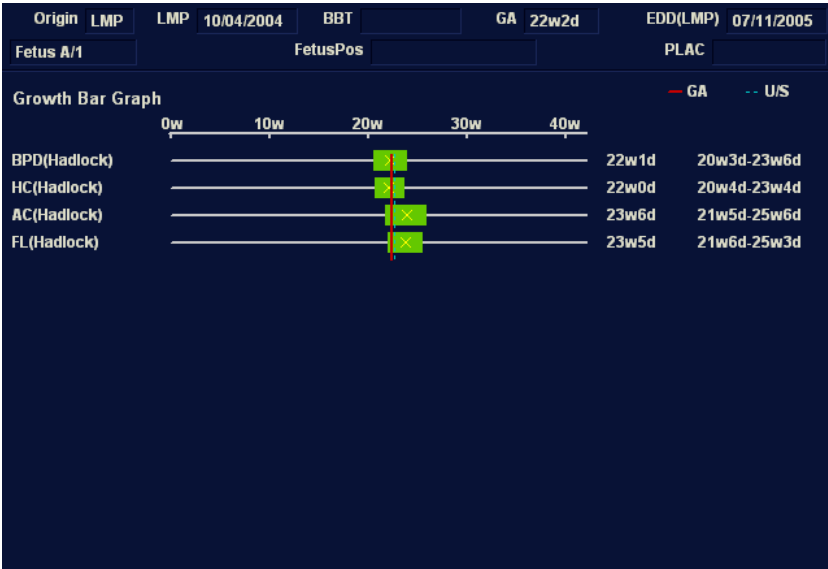


Figure 9-14. Fetal Growth Bar Graph

Fetal Growth Bar Graph (continued)

- The horizontal axis shows the gestational weeks.
- The red vertical line shows the gestational age using the patient data.
- The blue dotted vertical line shows the ultrasound age using the current measurements.
- The yellow x shows the ultrasound age for each measurement.
- The green rectangle shows the normal age range for the measurement.

You cannot do fetal trending or view multiple gestation data on the bar graph.

OB-Multigestational

Using other OB studies

The LOGIQ 9 system includes measurements for the following studies: USA, Europe, Tokyo, Osaka, and ASUM. The following figures show the OB Touch Panel displays for the Tokyo and Osaka studies and for the ASUM studies. For information about how to change studies, refer to “Measurement and Analysis (M&A) System Presets” on page 16-10.

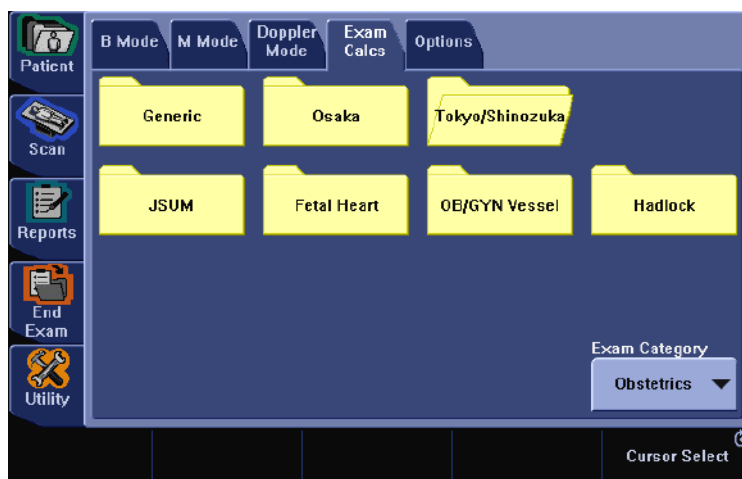


Figure 9-15. OB Touch Panel: Tokyo and Osaka studies

NOTE: Tokyo and Osaka studies have a folder for JSUM measurements. JSUM measurements include AC, BPD, CRL, and FL.

Using other OB studies (continued)

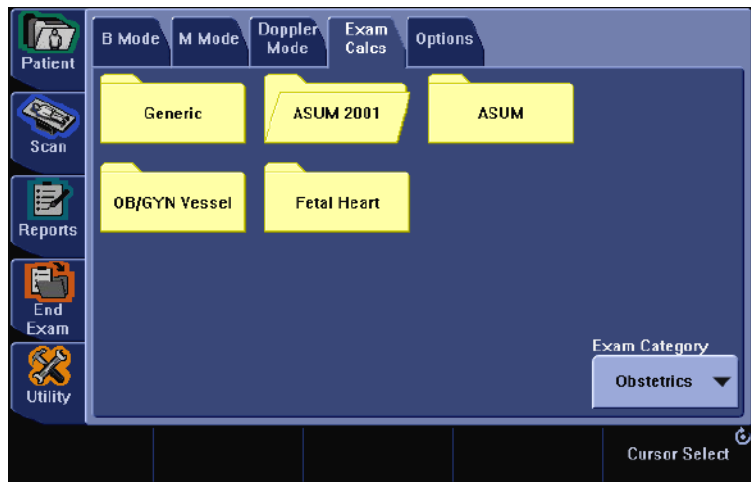


Figure 9-16. OB Touch Panel: ASUM studies

NOTE: ASUM studies include the following measurements:

- ASUM: AC, BPD, and CRL
- ASUM 2001: AC, BPD, CRL, FL, HC, HL, and OFD

Multiple Fetus

LOGIQ 9 allows you to measure and report multiple fetus development. The system can report a maximum of four fetuses.

To enter the number of fetuses

If more than one fetus is imaged during the exam, enter the number of fetuses in the Patient Data Entry Menu.

The screenshot shows the 'OB' tab selected in the top navigation bar. The 'Fetus #' field is a dropdown menu currently set to '1', which is circled in red. Other fields include LMP, BBT, EDD by LMP, GA by LMP, Gravida, Para, AB, Ectopic, Accession #, Exam Description, Perf. Physician, Ref. Physician, and Operator (set to ADM). Buttons for 'Past Exam', 'Images', and 'Clear' are at the bottom.

Figure 9-17. Fetus Number

When you start an OB exam, the system automatically fills in the Fetus # field with 1. To change the number:

1. Move the cursor to the fetus number and press **Set** twice.
The number is highlighted.
2. Type the correct number and press **Set**.
The system displays a message to confirm that you want to change the fetus number.
3. Select Yes.

To identify each fetus

For measurements, calculations, and worksheet displays, the system labels each fetus A, B, C, or D. Each fetus is identified by a letter and the total number of fetuses. For example, fetus A/3 is fetus A from a total of 3.

When scanning, you can enter information about the fetus position and placenta location. You can enter the information in the Patient Data section of the worksheets and the graphs. You can type up to 23 characters in the FetusPos and PLAC fields.

Origin	LMP	LMP	10/04/2004	BBT		GA	22w3d	EDD(LMP)	07/11/2005
Fetus B/3				FetusPos				PLAC	

Figure 9-18. Patient Data section of the OB Worksheet

To select a fetus

During measurements and calculations, to change between fetuses, do one of the following:

- Adjust the **Fetus** selection.
- Move the **Trackball** to the Summary Window and select the fetus.



Figure 9-19. Summary Window: Multiple fetus

You can change between fetuses at any time during the exam.

NOTE: *After you change to the next fetus, any measurements you make are recorded and reported to that fetus. If you have any active measurement or calculation that is not completed when you change the fetus, the system cancels the measurement or calculation.*

To view multiple fetuses data on graphs

You can view multiple gestation data on fetal growth curve graphs. After you have made measurements for each fetus, select **Graph Display**.

- To view the graph for each fetus, do one of the following:
 - Adjust the **Fetus** selection.
 - In the Patient Data section, move the **Trackball** to highlight the FetusNo field. In the list of fetuses, move the **Trackball** to select the fetus you want, and press **Set**.
- To display data for multiple fetuses on the same graph, select **Fetus Compare**.

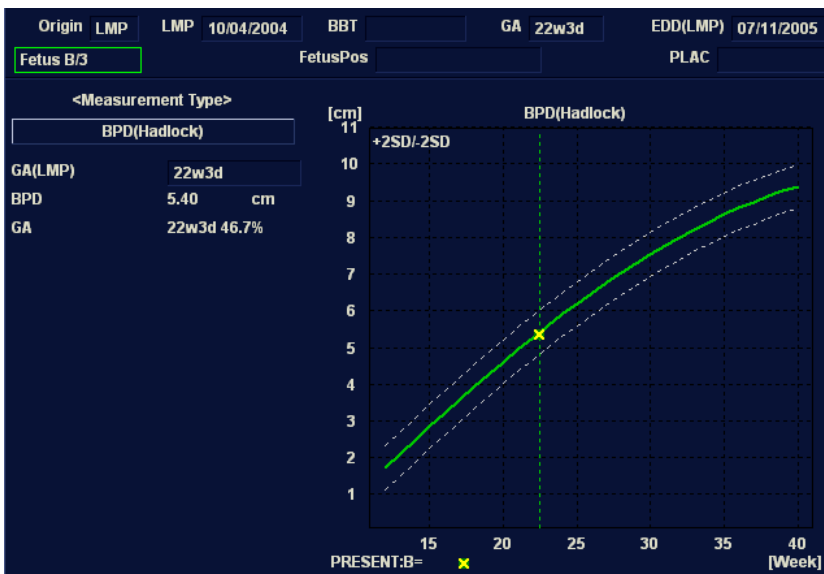


Figure 9-20. Fetal Growth Curve Graph: Fetus Compare

The legend at the bottom of the graph shows the symbols and colors that represent each fetus.

To compare multiple fetus data on a worksheet

With multiple fetuses, you can list and compare measurements of the fetuses on the worksheet.

Select **Worksheet Display**, then select **Fetus Compare**.

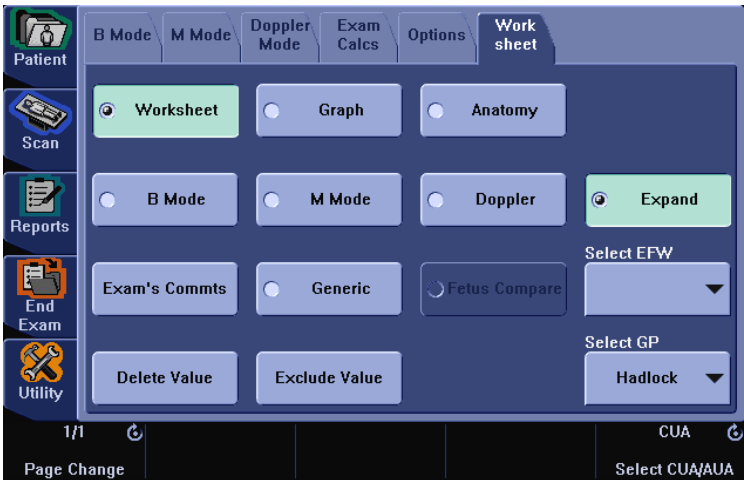


Figure 9-21. OB Worksheet Display Touch Panel

To compare multiple fetus data on a worksheet (continued)

When you select **Fetus Compare**, the system lists the measurement results for each fetus on the Worksheet.

Origin	LMP	LMP	07/01/2002	BBT		GA	21w4d	EDD(GA)	04/07/2003
Fetus Compare		A		B		C			
CUA									
EDD(CUA)									
EFW									
AFI(Moore)			11.24			15.82			
BPD(Hadlock)			2.29 cm			3.75 cm			
FL(Hadlock)			3.32 cm			4.94 cm			

Figure 9-22. Worksheet Display with Fetus Compare

To Show Fetal Trending for Multiple Fetuses

When you have data for more than one exam, you can show fetal trending and compare fetuses on one graph.

To view fetal trending for multiple fetuses:

1. Select **Graph Display**.
2. Select **Fetus Compare**.
3. Select **Plot Both**.

NOTE: *You can only view fetal trending for multiple fetuses in single graph display.*



Figure 9-23. Multigestation Fetal Growth Curve Graph

The symbol key for fetal trending and multiple fetuses is shown at the bottom of the graph.

OB Table Editor

You can add user programmable OB tables to the system.

OB Table Settings Menu

You add OB Tables in the Measurement & Analysis menu. To open the menu:

1. On the Touch Panel, select **Utility**, then select **M&A**.
2. Check the Exam Category on the far left of the monitor screen. Make sure that Obstetrics is selected.

If it is not selected, select Obstetrics and continue selecting the folders until the appropriate area is selected as to where this new OB Table will be entered. For example, select Obstetrics, then select OB-2/3. If there are further folders within OB-2/3, select that appropriate folder.

OB Table Settings Menu (continued)

- 3. On the monitor display, select the OB Table tab.
The system displays the OB Table settings menu.



Figure 9-24. OB Table settings

OB Table Settings Menu (continued)

4. The OB Table settings menu lists OB Table parameters. Specify the following parameter values as appropriate:
 - **Study:** Shows the study to which this measurement table belongs.
 - **New/Edit:** To create a new OB table, select New Table. To edit an existing user-programmable OB table, select Edit Table.

NOTE: *You cannot edit the system's OB Tables.*

- **OB Table Template:** To create a new OB table, select the Template (1 - 7) which you want to use as the basis of the user programmable OB Table. See 'OB Table Templates' on *page 9-76 for more information*.
To edit an existing user OB table, select the desired OB table to edit.
- **Tool Type:** Select the type of measurement: Distance or Circumference.
- **Measure Name:** Type the name of measurement that will display on the Touch Panel.
- **Author Name:** Type the author's name.
- **Table Type:** If necessary, select the Table Type: Fetal Age or Fetal Growth.
- **Measure Type:** Select a measurement type that can be used to calculate EFW, for example BPD.

NOTE: *Measure Type is used only when calculating EFW.*

NOTE: *The following items are display only: Table Format, Table Unit, SD/GP Range, and Graph Range. The system determines these values automatically, based on the type of OB table you are creating.*

5. After specifying all parameter values, move the **Trackball** to **Edit Table** and press **Set**.

The system displays the Edit Menu.

NOTE: *If any of the OB table parameters are not correct, the Edit Menu is not displayed.*

OB Table Templates

Tool Type:

- Distance: 2D Caliper
- Circumference: 2D Ellipse, 2D Trace, 2D Caliper

Template 1

Table 9-4: Template 1 (based on Hadlock)

Template 1: SD Range Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	Week	Week
	Table Range	1SD		
	Graph Range	1SD		
Measurement Result	Value [cm]			
	GA [#w#d]			
	Min [#w#d]			
	Max [#w#d]			
Fetal Growth Table	Table Format	AGE	MEAN	SD
	Table Unit	Week	mm	Week
	Others are same as above			

Template 2

Table 9-5: Template 2 (based on Tokyo)

Template 2: SD Range Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	Day	Day
	Table Range	1SD		
	Graph Range	1SD		
Measurement Result	Value [cm]			
	GA [#w#d]			
	SD: day(+/-)			
Fetal Growth Table	Table Format	AGE	MEAN	SD
	Table Unit	Day	mm	Day
	Others are same as above			

Template 3

Table 9-6: Template 3 (based on Osaka)

Template 3: SD Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	Day	mm
	Table Range	1SD		
	Graph Range	1SD		
Measurement Result	Value [cm]			
	GA [#w#d]			
	SD: sd=(mv-pv)/sd			
Fetal Growth Table	Table Format	AGE	MEAN	SD
	Table Unit	Day	mm	Day
	Others are same as above			

Template 4

Table 9-7: Template 4 (based on several European tables)

Template 4: 5%-95% Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	WeekDay	mm
	Table Range	5%:95%		
	Graph Range	5%:95%		
Measurement Result	Value [cm]			
	GA [#w#d]			
	GP [%]			
	GP is calculated by Fetal Growth Table. If you did not edit Growth Table, GP is not calculated by the system,			
Fetal Growth Table	Table Format	AGE	MEAN	SD
	Table Unit	WeekDay	mm	Day
	Table Range	5%:95%		
	Graph Range	5%:95%		

Template 5

Table 9-8: Template 5 (based on several European tables)

Template 5: 5% - 95% Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	WeekDay	mm
	Table Range	1SD		
	Graph Range	5%:95%		
Measurement Result	Value [cm]			
	GA [#w#d]			
	GP [%] GP is calculated by Fetal Growth Table. If you did not edit Growth Table, GP is not calculated by the system,			
Fetal Growth Table	Table Format	AGE	MEAN	MAX
	Table Unit	WeekDay	mm	mm
	Table Range	1SD		
	Graph Range	5%:95%		

Template 6

Table 9-9: Template 6 (based on several European tables)

Template 6: 10% - 90% Table					
Fetal Age Table	Table Format	MEAS	MIN	MEAN	SD
	Table Unit	mm	WeekDay	WeekDay	WeekDay
	Table Range	10%-90%			
	Graph Range	10%-90%			
Measurement Result	Value [cm]				
	GA [#w#d]				
	GP [%] GP is calculated by Fetal Growth Table. If you did not edit Growth Table, GP is not calculated by the system,				
Fetal Growth Table	Table Format	AGE	MIN	MEAN	SD
	Table Unit	WeekDay	mm	mm	mm
	Table Range	10%-90%			
	Graph Range	10%-90%			

Template 7

Table 9-10: Template 7 (Based on several european tables)

Template 7: 10% - 90% Table				
Fetal Age Table	Table Format	MEAS	MEAN	SD
	Table Unit	mm	WeekDay	mm
	Table Range	1SD		
	Graph Range	10%:90%		
Measurement Result	Value [cm]			
	GA [#w#d]			
	GP [%] GP is calculated by Fetal Growth Table. If you did not edit Growth Table, GP is not calculated by the system,			
Fetal Growth Table	Table Format	AGE	MEAN	MAX
	Table Unit	WeekDay	mm	mm
	Table Range	1SD		
	Graph Range	10%:90%		

OB Table Edit Menu

The data you enter in the OB Table Edit Menu depends on whether the table type is Fetal Age or Fetal Growth.

Fetal Age Table

If you are creating or editing a Fetal Age table, the OB Table Edit Menu is as follows:

	MEAS mm	MEAN Day	SD Day
1	10.00		
2	15.00		
3	20.00		
4	25.00		
5	30.00		
6	35.00		
7	40.00		
8	45.00		
9	50.00		
10	55.00		
11	60.00		
12	65.00		

Figure 9-25. OB Table Edit Menu: Fetal Age Table

- Complete the field**
1. Input value to Min, Max and Interval of the Parameters field.
The system automatically fills in the MEAS column.
Input value to the columns of MEAN and SD.

NOTE: To move between the fields in the table, use the up, down, left, and right arrow keys.

NOTE: You must enter a minimum of two rows of data. Any lines with a blank cell are not saved.

To save the Table Data, move the **Trackball** to Exit to Save and press **Set**. If you want cancel this table, move the **Trackball** to Cancel and press **Set**.

Fetal Growth Table

If you are creating or editing a Fetal Growth table, the OB Table Edit Menu is as follows:

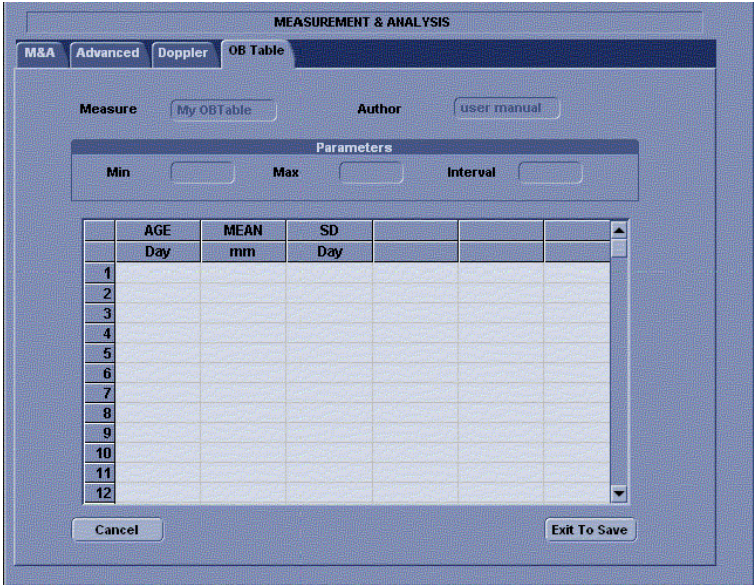


Figure 9-26. OB Table Edit Menu: Fetal Growth Table

Complete the field

- NOTE: To move between the fields in the table, use the up, down, left, and right arrow keys.
- NOTE: You must enter a minimum of two rows of data. Any lines with a blank cell are not saved.

- Input value to the required columns.
- To save the Table Data, move the **Trackball** to Exit to Save and press **Set**. If you want cancel this table, move the **Trackball** to Cancel and press **Set**.

After you complete the OB table, it is now available for the selected study. To use the measurement, you must assign it to a Touch Panel. See 'Measurement and Calculation Setup' on page 7-16 for more information.

EFW for OB User Table/Formula Editor

EFW Table Editor

You can edit an EFW Formula at the OB Table Editor.

1. Select Utility -> Measure -> OB Table.
2. Select the appropriate parameters and press **Edit Table**.
 - New/Edit: Select “New Table”
 - OB Table Template: Select appropriate template.
 - Tool Type: Select “Weight”
 - Measure Name: Enter measurement name.
 - Author Name: Enter author’s name.
 - Table Type: Select “Fetal Age”



Figure 9-27. OB Table Tab Screen

EFW Table Editor (continued)

- 3. Edit the table data and press **Exit To Save**.

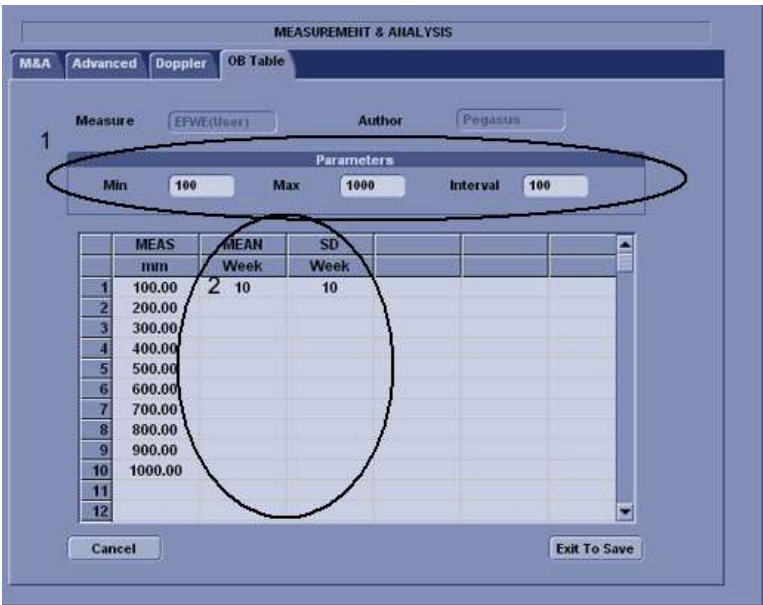


Figure 9-28. OB Table Editor Screen

EFW Formula Editor

1. Select the M&A tab and select **Edit Calc**. The Modify User CALC window displays.

Select the added table [e.g. EFW (User)(Pegasus)] from the User Defined pull-down menu and press **OK**.

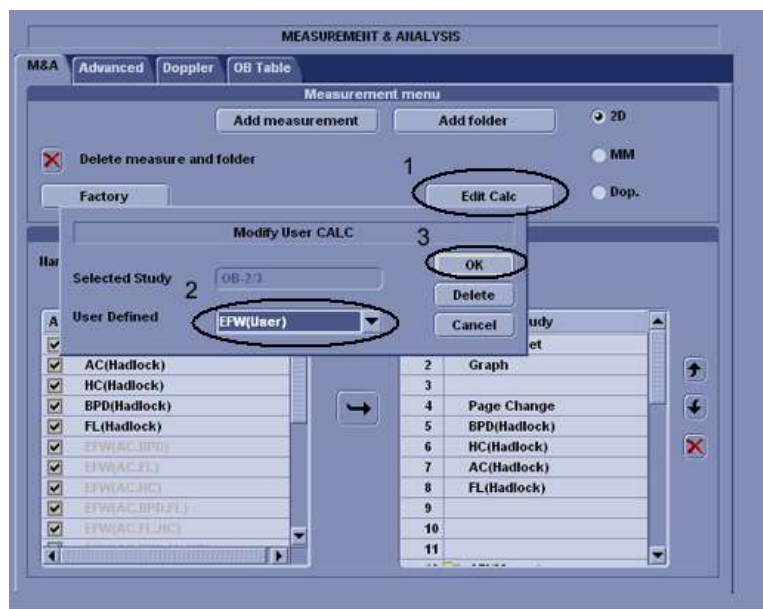


Figure 9-29. Modify User CALC window

EFW Formula Editor (continued)

- 2. Select the “Calculated” button for the EFW parameter.

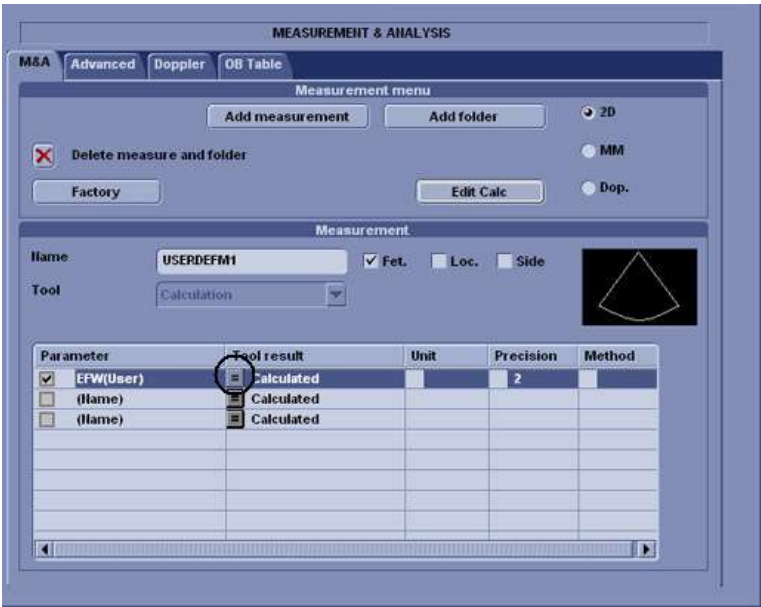


Figure 9-30. M&A Tab Screen

EFW Formula Editor (continued)

- The EDIT FORMULA window displays. Edit the formula and select **OK**.

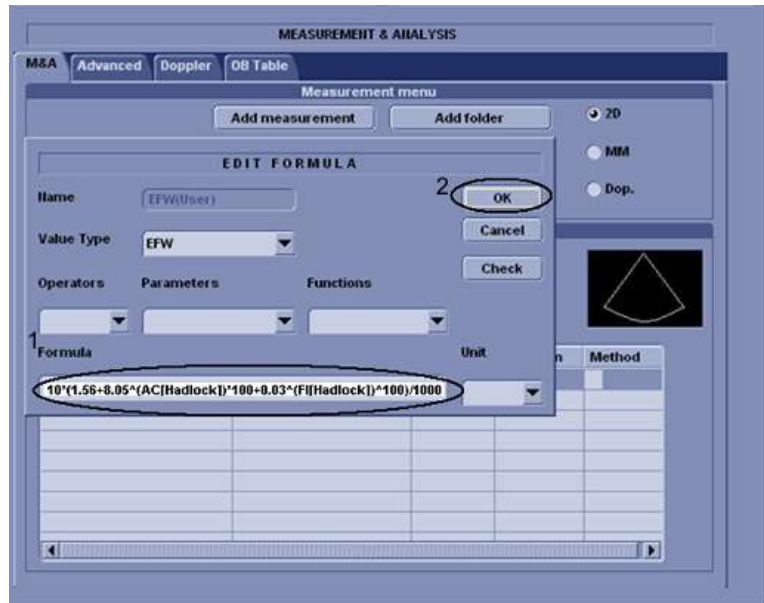


Figure 9-31. EDIT FORMULA window

NOTE: When you edit a formula, be careful of the following points.

- If you want to calculate EFW by centimeter, add “ *100 ” to the {parameter}.
 - If EFW is calculated in grams, add “ /1000 ” to the formula.
- or example,

$$10^{(1.56\{AC[Hadlock]\} * 100 + 0.08\{FL[Hadlock]\} * 100) / 1000}$$

GYN Measurements

Introduction

The Gynecology exam category includes the following three studies:

- Generic. This study is common to all exam categories. See 'Generic Measurements' on *page 7-78 for more information*.
- General Gynecology. This study includes uterine, ovarian, ovarian follicle, and endometrium measurements.
- OB/GYN Vessel. This study includes the following vessels: uterine, ovarian, umbilical, middle cerebral artery, aorta, placenta, and descending aorta.

NOTE: *The calculation formulas are listed in the Advanced Reference Manual.*

To Start a Gynecology Exam

To begin a gynecology exam, you enter patient data or, if the patient data from a previous exam is saved in the system, find the patient information.

For details about how to start an exam, See 'To Start an Obstetrics Exam' on *page 9-4 for more information*.

After you complete the patient information, you can begin the scan.

1. To change from the Patient Data Entry screen to the Scan screen, do one of the following:
 - On the keyboard, press **Esc**.
 - On the Touch Panel, select **Scan**.
 - On the Control Panel, select **Patient** or **Freeze**.
 - On the Control Panel, press the B-Mode key.

The system displays the Scan screen.

2. To choose the appropriate probe, select the probe icon on the Touch Panel.
3. On the Control Panel, press **Measure**.

The default Gynecology study is displayed on the Touch Panel.

B-Mode Measurements

In B-Mode, you make the measurements in the General Gynecology study. These measurements include:

- Uterine length, width, and height
- Ovarian length, width, and height
- Ovarian follicle
- Endometrium thickness

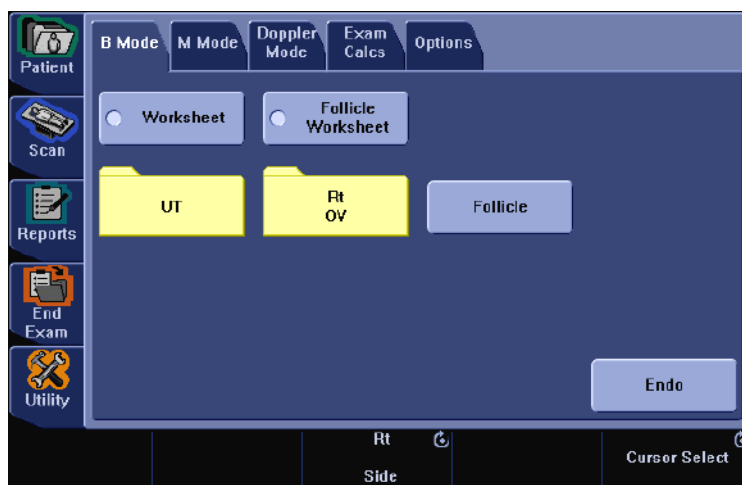


Figure 9-32. General Gynecology study

Follicle measurements

You can make left and right ovary follicle measurements from one, two, or three distances.

One distance

1. To select the left or right, adjust the **Side** selection.
2. Select **Follicle**; an active caliper displays.
3. To position the active caliper at the start point, move the **Trackball**.
4. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
5. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
6. To complete the measurement, press **Set**.
7. Press **Clear**.

The system displays the ovary follicle measurement in the Results Window.

Follicle measurements (continued)

- Two distances**
1. To select the left or right, adjust the **Side** selection.
 2. Select **Follicle**; an active caliper displays.
 3. Make the first distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window and displays an active caliper.
 4. To make the second distance measurement, repeat steps a–d above.
 5. Press **Clear**.
The system displays the ovary follicle measurement in the Results Window.

Follicle measurements (continued)**Three distances**

1. To select the left or right, adjust the **Side** selection.
2. Select **Follicle**; an active caliper displays.
3. Make the first distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window. After the first and second distance measurement, the system displays an active caliper.
4. To make the second and third distance measurement, repeat steps a–d above.

After the third measurement, the system displays the ovary follicle measurement in the Results Window.

Endometrium thickness (Endo)

To measure the endometrium thickness, make one distance measurement.

1. Select **Endo**; an active caliper displays.
2. To position the active caliper at the start point, move the **Trackball**.
3. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
4. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
5. To complete the measurement, press **Set**.
The system displays the endometrium thickness in the Results Window.

Ovary length, width, and height

You can measure the length, width, and height of the left and right ovaries. Each measurement is a typical distance measurement made in the appropriate scan plane.

Typically, length and height are measured on the sagittal plane while the width is measured on the axial/transverse plane.

To measure ovarian length, width, or height:

1. Scan the patient's right or left ovary in the appropriate plane.
2. To select left or right, adjust the **Side** selection.
3. Select the **OV** folder, then select **OVL**, **OVW**, or **OVH**.
4. Perform a standard distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.

The system fixes the first caliper and displays a second active caliper, if preset accordingly.
 - c. To position the second active caliper at the end point, move the **Trackball**.

A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.

The system displays the distance value in the Results Window. After the first and second measurement, the system displays an active caliper for the next measurement.
5. To make the second and third distance measurements, repeat steps 3–4.

After you complete the length, width, and height measurements, the system displays the ovarian volume in the Results Window.

Uterus length, width, and height

Each of these is a standard distance measurement. Typically, length and height are measured on the sagittal plane while the width is measured on the axial/transverse plane.

To measure uterus length, width, or height:

1. Scan the patient in the appropriate scan plane.
2. Select the **UT** folder, then select **UT L**, **UT W**, or **UT H**.
An active caliper displays.
3. Perform a standard distance measurement:
 - a. To position the active caliper at the start point, move the **Trackball**.
 - b. To fix the start point, press **Set**.
The system fixes the first caliper and displays a second active caliper.
 - c. To position the second active caliper at the end point, move the **Trackball**.
A dotted line connects the measurement points.
 - d. To complete the measurement, press **Set**.
The system displays the distance value in the Results Window. After the first and second measurement, the system displays an active caliper for the next measurement.
4. To make the second and third distance measurement, repeat steps 2–3.

After you complete the third distance measurement, the system displays the uterine volume in the Results Window.

M-Mode Measurements

M-Mode measurements for the Gynecology exam are identical to M-Mode measurements for the Obstetrics exam. These measurements include % stenosis, A/B ratio, and heart rate.

For details regarding these measurements, See 'M-Mode Measurements' on *page 9-38 for more information.*

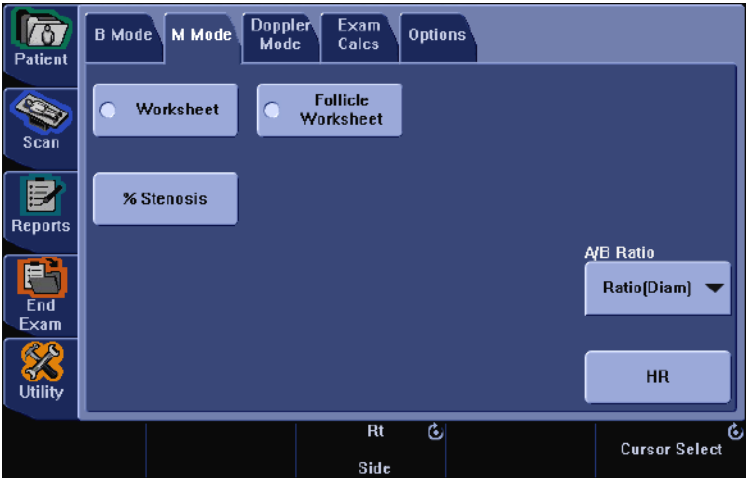


Figure 9-33. M-Mode Generic study

Doppler Mode Measurements

Doppler measurements for the Gynecology exam are identical to Doppler measurements for the Obstetrics exam. These measurements include the following vessels: uterine, ovarian, umbilical, middle cerebral artery, aorta, placenta, and descending aorta. For each vessel, you can make any of the following measurements: peak systole, minimum diastole, end diastole, heart rate, TAMAX, pulsatility index, resistive index, acceleration, PS/ED, ED/PS, and acceleration time.

For details regarding these measurements, See ‘Doppler Mode Measurements’ on *page 9-39 for more information.*

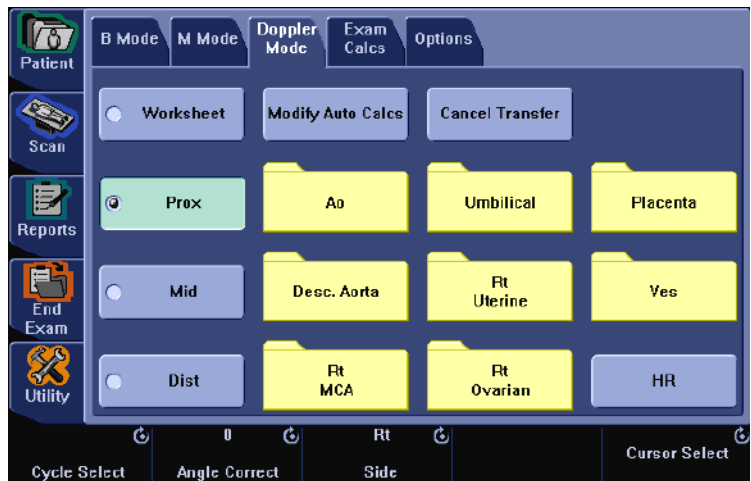


Figure 9-34. Doppler Mode General Gynecology study

Chapter 10

Cardiology

Describes how to perform cardiac measurements and calculations.

This feature is not available at this time.

Chapter 11

Vascular

Describes how to perform Vascular measurements and calculations.

Vascular Exam Preparation

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by the system accuracy, but also by use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

General Guidelines

New Patient information must be entered before beginning an exam. See 'Beginning a New Patient' on *page 4-3 for more information*.

Any measurement can be repeated by selecting that measurement again from the Touch Panel Menu.

Vascular Measurements

Introduction

Vascular measurements offer several different types of measurement studies:

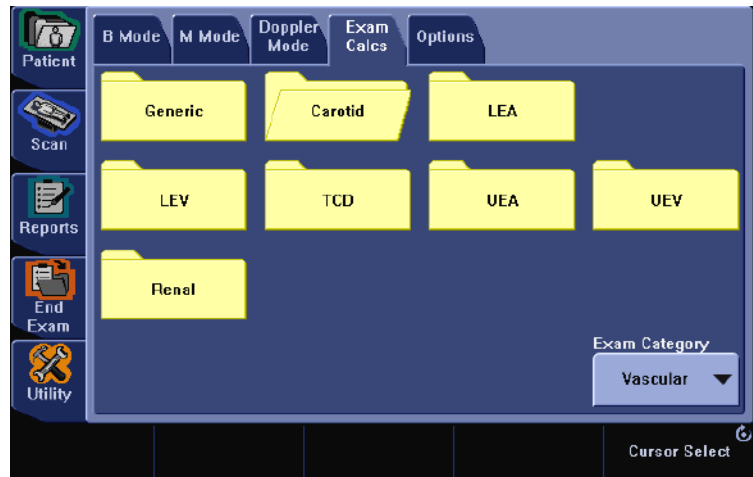


Figure 11-1. Vascular Exam Category Touch Panel

- Generic – Common to all applications. See 'Generic Measurements' on *page 7-78 for more information.*
- Carotid
- LEA (Lower Extremity Artery)
- LEV (Lower Extremity Vein)
- TCD (Trans Cranial Doppler)
- UEA (Upper Extremity Artery)
- UEV (Upper Extremity Vein)
- Renal

Introduction (continued)

To change an exam calc:

1. Select **Exam Calcs**.

The Vascular exam category allows you to choose from the displayed exam calcs.

2. To select another exam calcs, press the desired exam calcs folder.

A vascular study is a group of particular vessels. You can customize the vessel exam calcs in the configuration menu. See 'Measurement and Calculation Setup' on *page 7-16 for more information*.

When you use Auto Vascular calculation, you use the vessel keys on the Touch Panel to post-assign vascular calculations. When you are not using Auto Vascular calculation, the vessel key is used for manual measurement.

B-Mode Measurements



Figure 11-2. B-Mode Touch Panel

NOTE: The following instructions assume that you first scan the patient and then press **Freeze**.

% Stenosis

See '% Stenosis' on page 7-83 for more information.

Volume

See 'Volume' on page 7-85 for more information.

A/B Ratio

See 'A/B Ratio' on page 7-90 for more information.

IMT Measurement

You can measure the average of the intima media thickness for use as the index of arterial sclerosis.

NOTE: *The IMT measurement is available with linear probes only.*

NOTE: *Due to the physical properties of ultrasound imaging, the posterior IMT measurement is generally more accurate than the anterior IMT measurement.*

1. Before you measure the IMT, add the IMT measurement to the Carotid folder via the Measurement & Analysis screen. There are three types of IMT measurements:
 - **IMT:** Three vertical lines are parallel. Place the start point on the line and place the end point anywhere.
 - **IMT2:** Each vertical line can be rotated with the **Ellipse** control. You must place the start and end points on the line.
 - **5mm Scale:** The horizontal line can be rotated with the **Ellipse** control. A maximum of 20 distance values which produce one average value can be taken.

IMT and IMT2 have three kinds of measurements:

- **IMT --+/IMT2 --+:** Measure from right to left.
 - **IMT -+/-IMT2 -+/-:** First measure at the center, then right and left are last.
 - **IMT+-/IMT2 --+:** Measure from left to right.
2. Scan the carotid artery, then press **Freeze**.

IMT Measurement (continued)

3. Press **Measure**, then select **IMT**. An active caliper displays.

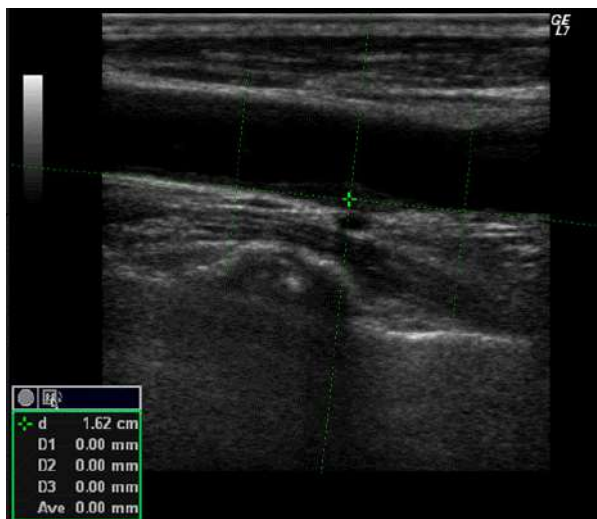


Figure 11-3. IMT caliper (Example)

4. Use the **Trackball** to move the caliper and the **Ellipse** control to adjust the angle. Press **Set** to fix the caliper.

NOTE: *The interval of the vertical line is 1cm.*

5. Measure the thickness of three points.

NOTE: *The caliper moves to the next point automatically.*

IMT Measurement (continued)

- 6. After you complete the measurement, the system calculates the average automatically.

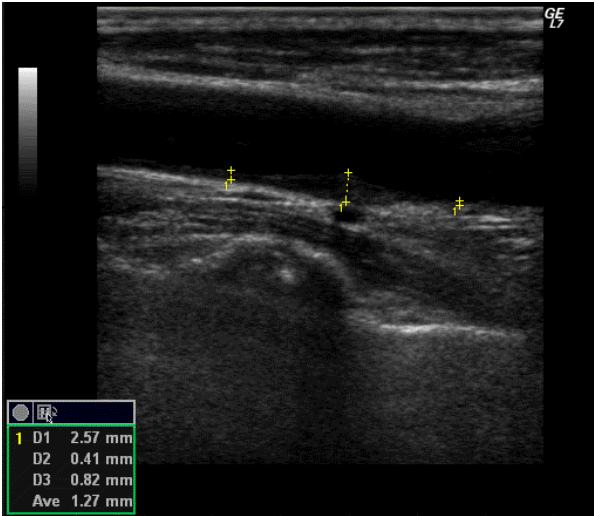


Figure 11-4. IMT Measurement

Plaque Score Tool

1. Before you measure, add Plaque Score to the Carotid folder at the Utility -> Measure -> Measurement & Analysis screen.
2. Scan the carotid artery and press **Freeze**. Display dual images to measure across split screen images.
3. Press **Measure** and select **Plaque Score Tool**. An active caliper, one horizontal line and four vertical lines display. The interval of the vertical lines are 1.5 cm.

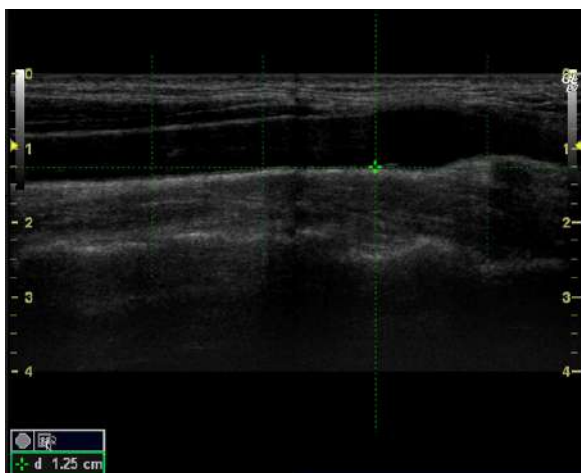


Figure 11-5. Plaque score example

4. Use the **Trackball** to move the caliper and the **Ellipse** control to adjust the angle. Press **Set** to fix the caliper.

Plaque Score Tool
(continued)

- 5. Measure the thickness (up to twenty times) on any place as necessary.

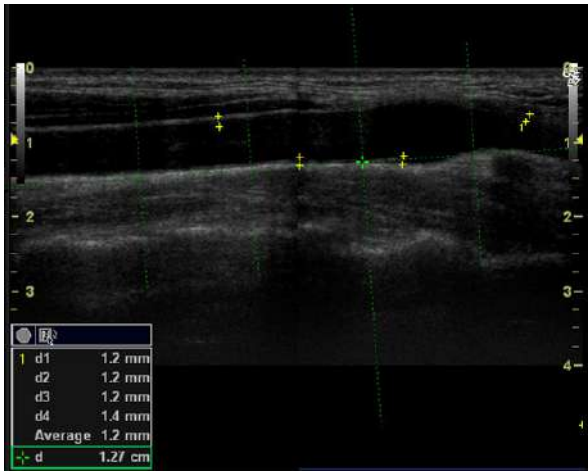


Figure 11-6. Plaque Score Example 2

- 6. The system displays each distance value, all addition values and each addition value (larger than 1.1 mm values in each area).

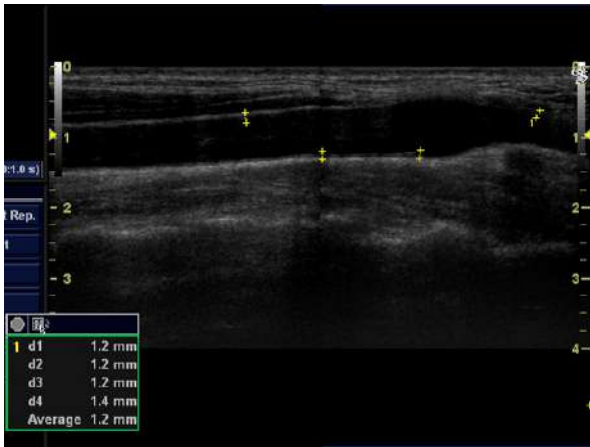


Figure 11-7. Plaque Score Example 3

NOTE: Only calculates the value larger than 1.1mm.

- 7. Double click the **Set** key to finish the measurement.

M-Mode Measurements

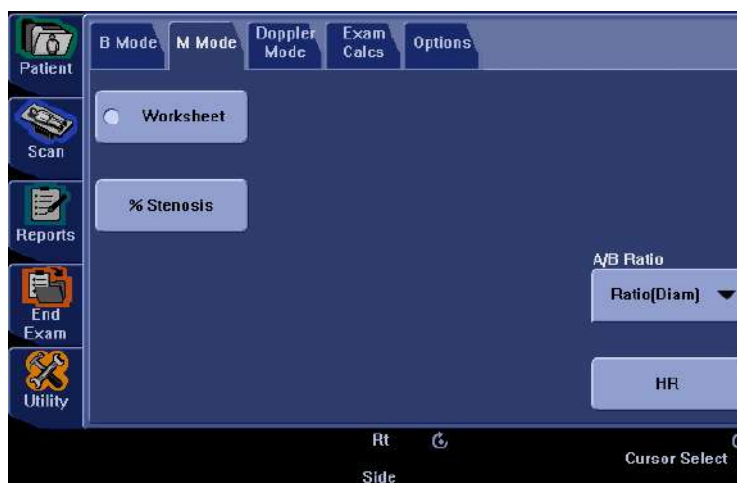


Figure 11-8. M-Mode Touch Panel

NOTE: The following instructions assume that you first scan the patient and then press **Freeze**.

% Stenosis

See '% Stenosis' on page 7-83 for more information.

A/B Ratio

See 'A/B Ratio' on page 7-92 for more information.

Heart Rate

See 'Heart Rate' on page 7-93 for more information.

Doppler Mode Measurements

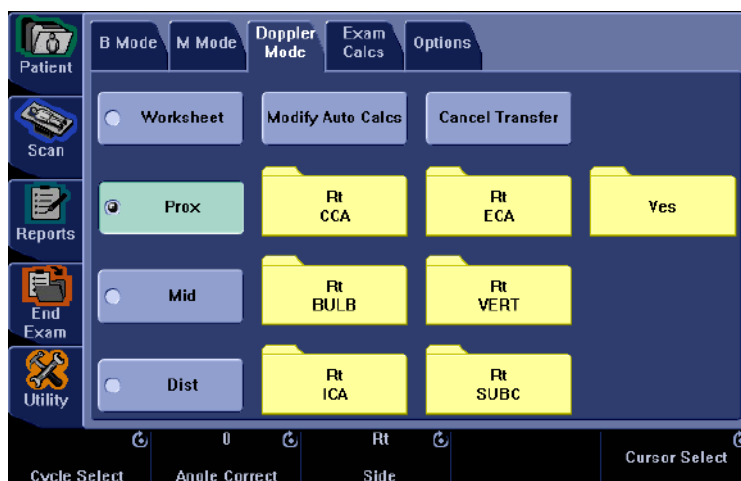


Figure 11-9. D-Mode Touch Panel Example

Control Assignment

Cancel Transfer After the Auto Vascular calculation results are assigned to a particular vessel, the user can cancel the assignment and assigned parameters are removed from Worksheet and Report page.

When Cancel Transfer occurs, a message appears on the screen to indicate the value was erased from Worksheet and Report page.

Vessel location If the vessel has a location, you can select one of the following:

- Proximal (**Prox**)
- Middle (**Mid**)
- Distal (**Dist**)

NOTE: *If you do not wish to assign a vessel location, press the lit location, then no location is assigned. Choose the folder you want the value assigned to.*

Side Rt/Lt The system has measurements for the patient's right and left side. To select right or left, adjust the Touch Panel knob.

Control Assignment (continued)

Modify Auto Calcs

When you select this key, the Modify Calculation menu is displayed as below. In this menu, you select parameters to display in the Auto Vascular Calculation window. Only parameters that can be used by the calculation are displayed.

Select **Save as Default** to save the selected parameters as the default calculations for this application.

Select **Return** to return to the previous Touch Panel screen.

If you select **PV**, all selected parameters are turned off. When you deselect **PV**, the system returns to the previously selected calculation.

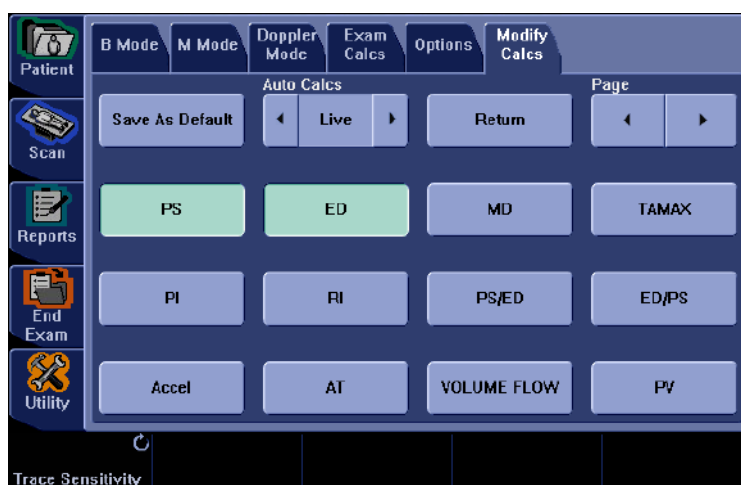


Figure 11-10. Modify Auto Calculation Menu (Page1)

Naming format for vessels

When you want to measure a vessel, on the Touch Panel you select the folder for the vessel. Many vessel folders are labeled with an abbreviation. The following table lists abbreviations used for naming vascular vessels.

Table 11-1: Vascular Vessel Abbreviations

Acronym	Name
ACA	Anterior Cerebral Artery
ACoMA	Anterior Communicating Artery
ATA	Anterior Tibial Artery
ATV	Anterior Tibial Vein
Axill	Axillary Artery
Axill V	Axillary Vein
BA	Basilar Artery or Brachial Artery
Ba V	Basilic Vein
Br V	Brachial Vein
CCA	Common Carotid Artery
Ceph V	Cephalic Vein
CFV	Common Femoral Vein
CHA	Common Hepatic Artery
Com Femoral	Common Femoral Artery
CIA	Common Iliac Artery
CIV	Common Iliac Vein
Com Iliac A	Common Iliac Artery
DFA	Deep Femoral Artery
DFV	Deep Femoral Vein
Dors Pedis	Dorsalis Pedis
DPA	Dorsalis Pedis Artery
ECA	Exterior Carotid Artery
EIA	External Iliac Artery
EIV	External Iliac Vein
FV	Femoral Vein

Table 11-1: Vascular Vessel Abbreviations

Acronym	Name
GSV	Greater Saphenous Vein
ICA	Internal Carotid Artery (Transcranial Doppler)
ICA	Interior Carotid Artery (Carotid Artery)
IJV	Internal Jugular Vein
IMA	Inferior Mesenteric Artery
Inn	Innominate
IVC	Inferior Vena Cava
LSV	Lesser Saphenous Vein
MCA	Middle Cerebral Artery
McuB V	Median Cubital Vein
Mid Hep V	Middle Hepatic Vein
MRA	Main Renal Artery
PCA	Posterior Cerebral Artery
PComA	Posterior Communicating Artery
Peron	Peroneal
POP	Popliteal
PTA	Posterior Tibial Artery
PTV	Posterior Tibial Vein
RA	Radial Artery
SMA	Superior Mesenteric Artery
SMV	Superior Mesenteric Vein
SUBC	Subclavian Artery
SUBC V	Subclavian Vein
SFA	Superficial Femoral Artery
TCD	Transcranial Doppler
TIPS	Transjugular Intrahepatic Portosystemic Shunt
UA	Ulnar Artery
VERT	Vertebral Artery

Auto Vascular Calculation Overview

Auto Vascular Calculation enables the LOGIQ 9 to detect and identify a cardiac cycle. It allows you to assign measurements and calculations during live timeline imaging, while the image is frozen, or in CINE. Peak values are detected for venous flow.

During cardiac cycle detection, the system identifies the cardiac cycle using calipers, vertical bars, and/or highlighting of timeline data. Use of identifiers is based on measurements and calculations selected by an operator for the current application. The system may place calipers at early systolic peak, peak systole, minimum diastole and end diastole. Vertical bars may also be placed to indicate the beginning and end of the cardiac cycle. The peak and/or mean trace may be highlighted. You can edit the cardiac cycle identified by the system or select a different cardiac cycle.

You can select the calculations to be displayed in the M&A Result window during live scanning or on a frozen image. These calculations are displayed at the top of M&A Result Window located adjacent to the image. These calculations are presettable by application which means you can set up the default calculations to be displayed for each application.

Auto Vascular Calculation

Activating Auto Vascular Calculation

To activate Auto Vascular Calculation, select the **Auto Calc** Touch Panel key to select Live (calculations displayed on the real-time image), or Freeze (calculations displayed on the frozen image).

To deactivate Auto Vascular calculation, select Off.

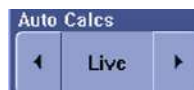


Figure 11-11. Auto Calculation Touch Panel key

Setting up Auto Vascular Calculation Parameters

- **Selecting Auto Trace**
You can select to have a continuous auto trace of the max or mean velocities.
 - Select Max or Mean using the **Trace Method** Touch Panel pull-down menu.
- **Selecting Trace Detection**
Trace Detection lets you use peak timeline data above, below, or composite (above and below) the baseline.
 - Select Positive, Negative, or Both to set the peak timeline data.
- **Modify Calculation**
 - a. Select the **Modify Calcs** Touch Panel key.
The Modify Calculation menu is displayed.
 - b. Select which measurements and calculations are to be displayed in the Auto Vascular calculation window.

You can select the following parameter: PS, ED, MD, HR, TAMAX, PI, RI, Accel, PS/ED, ED/PS, AT, Volume Flow, PV.

Auto Vascular Calculation (continued)

- Auto Vascular Calculation Exam
1. Preset the system.

2. Perform the scan and press **Freeze**.

3. Activate Auto Vascular Calculation.

The system performs a calculation automatically.



Figure 11-12. Auto Vascular Calculation

Auto Vascular Calculation Exam (continued)

The Auto Vascular calculation is assigned to particular vessel measurements.

1. Press **Measure** to display the Measurement menu.
2. Select the location of the vessel (Prox, Mid, or Dist) and Side (Right or Left).
3. Select the desired vessel name from the Touch Panel.

Selected vessel measurements are automatically assigned with the Auto Vascular calculation. The results are then displayed in the Results Window.



Figure 11-13. Assigned Vessel

NOTE: When you want to cancel the assignment, you can use the **Cancel Transfer** Touch Panel key. See 'Cancel Transfer' on page 11-12 for more information.

Auto Vascular Calculation (continued)

During the course of an exam, the cardiac cycle may be indicated between two yellow bars; the peak trace and the mean trace may appear in green; calculation indicators appear on the spectral trace as a caliper identifier (these vary, depending on the selected calculation in the Results Window).

The right-most, most complete cycle is typically chosen to be the selected cardiac cycle. You can select a different cardiac cycle.

To select a different cardiac cycle:

- Move through CINE memory with the Trackball until the desired cardiac cycle is selected by the system.
- Use the **Cycle Select** control to cycle to a different cardiac cycle.

NOTE:

You need several good cycles in front of the new cardiac cycle for this to be successful. Oftentimes, this is problematic near a freeze bar.

To move the systole or diastole position:

- Use the **Cursor Select** control to move the start systole position or the end diastole position..

Manual Vascular Calculation

You can perform the following calculations manually when Auto Doppler Calculation is not activated.

1. Press **Measure**.

If necessary, you can select another Exam Calc and then select parameters from Modify Calculation.

2. Select the location of the vessel (Prox, Mid, or Dist) and Side (Right or Left).
3. Select the desired vessel folder.

The Measurement menu is displayed.

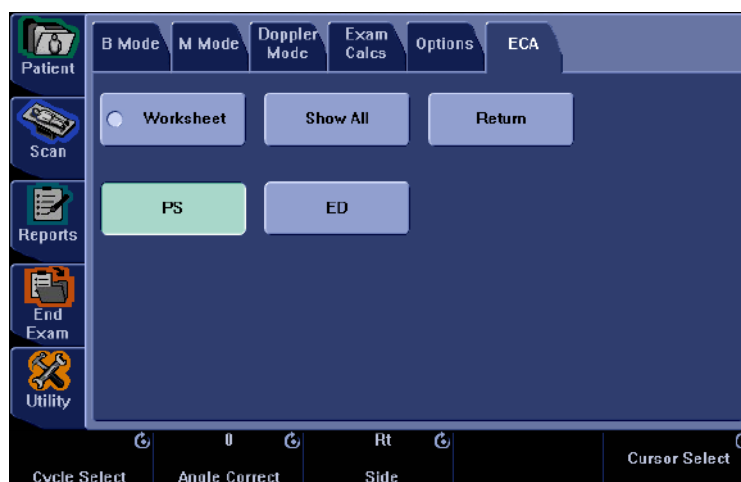


Figure 11-14. Measurement Menu Example

4. Make the required measurements according to the system, or select your preferred measurements.

Manual Vascular Calculation (continued)

For each vessel in Doppler mode, you can make any of the following measurements:

- Peak Systole (PS)
- End Diastole (ED)
- Minimum Diastole (MD)
- Heart Rate
- TAMAX
- TAMEAN
- Pulsatility Index (PI)
- Resistive Index (RI)
- S/D Ratio
- D/S Ratio
- Acceleration (Accel)
- Acceleration Time (AT)
- Volume Flow
- Peak Value (PV)

To select vascular measurements

Your system is set up to show the measurements that you usually make for each vessel. To make a measurement that is not shown for the selected vessel:

1. Select the folder for the vessel you want to measure.

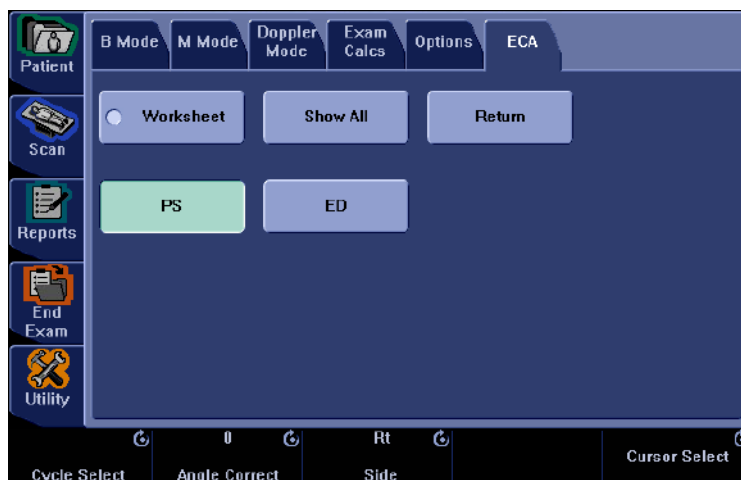


Figure 11-15. ICA folder

2. Select **Show All**.

The system displays all possible vessel measurements.

To select vascular measurements (continued)

3. Select the desired measurement.

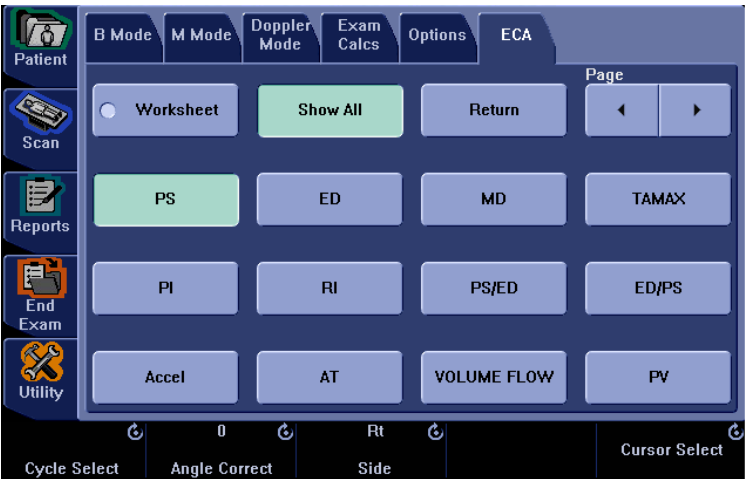


Figure 11-16. Show All measurements

NOTE: The following instructions assume that you first scan the patient and then press **Freeze**.

Acceleration

1. Select **Accel**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at peak systole, move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at end diastole, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, acceleration time, and acceleration in the Results Window.

Acceleration Time (AT)

1. Select **AT**.
The system displays an active caliper and a vertical dotted line.
2. To position the caliper at the start point, move the **Trackball**.
3. To fix the first caliper, press **Set**.
The system displays a second active caliper.
4. To position the caliper at the end point, move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the acceleration time in the Results Window.

Heart Rate

See 'Heart Rate' on page 7-93 for more information.

Peak Systole (PS), End Diastole (ED), or Minimum Diastole (MD)

To calculate the peak systole, end diastole, or minimum diastole:

1. Select **PS**, **ED**, or **MD**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at the measurement point, move the **Trackball**.
3. To complete the measurement, press **Set**.
The system displays the peak systole, end diastole, or minimum diastole in the Results Window.

ED/PS or PS/ED Ratio

To calculate the End Diastole/Peak Systole ratio or Peak Systole/End Diastole ratio:

1. Select **ED/PS** or **PS/ED**.
The system displays an active caliper with vertical and horizontal dotted lines.
2. To position the caliper at end diastole (ED) or peak systole (PS), move the **Trackball**.
3. To fix the measure point, press **Set**.
The system displays a second active caliper.
4. To position the second caliper at peak systole (PS) or end diastole (ED), move the **Trackball**.
5. To complete the measurement, press **Set**.
The system displays the end diastole, peak systole, and ED/PS or PS/ED ratio in the Results Window.

Pulsatility Index (PI)

See 'Pulsatility Index (PI)' on *page 7-95 for more information.*

S/D or D/S Ratio

See 'PS/ED or ED/PS Ratio' on *page 7-97 for more information.*

Resistive Index (RI)

See 'Resistive Index (RI)' on *page 7-96 for more information.*

TAMAX and TAMEAN

See 'TAMAX and TAMEAN' on *page 7-64 for more information.*

Volume Flow

See Chapter 7 for more information.

Vascular Worksheet

The vascular worksheet is structured to automatically display vascular measurements made at specific anatomical sites. The worksheet can also display an average, last, maximum, or minimum value of the latest three measurements. Calculated ratios are automatically summarized and displayed.

To view the Vascular Worksheet

- 1. Press **Measure**.
- 2. Select **Worksheet Display**.

The system displays the worksheet.

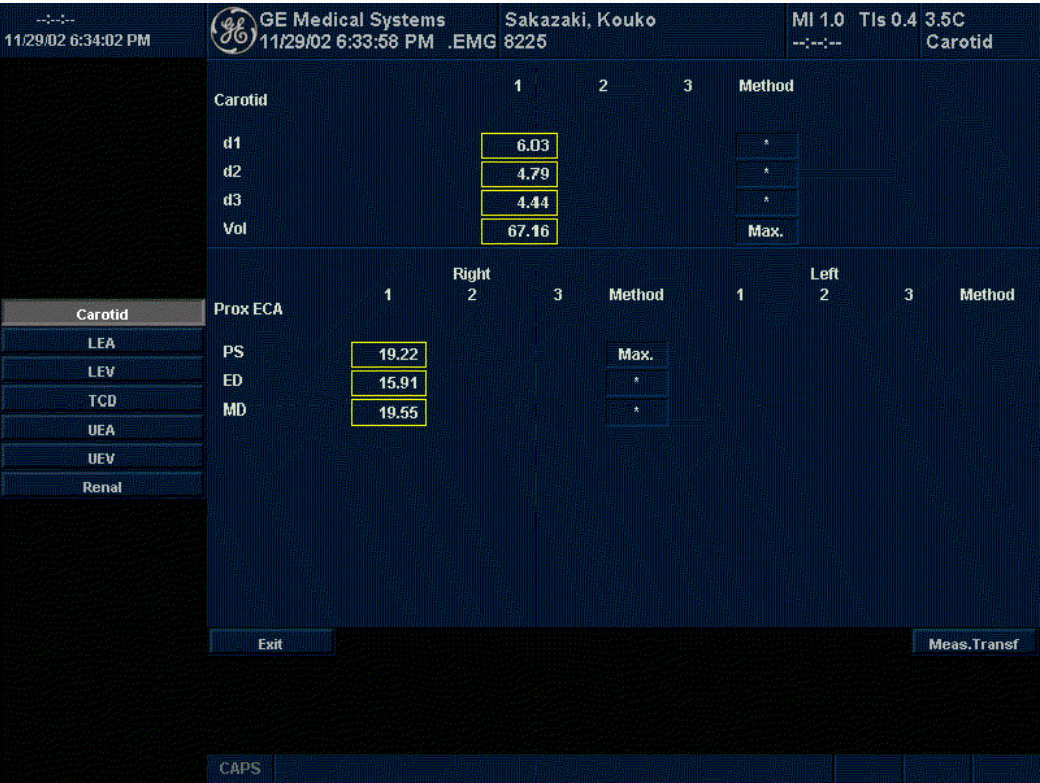


Figure 11-17. Vascular Worksheet Example

To view the Vascular Worksheet (continued)

Only measured parameters are displayed. Location information is labeled with vessel name first. Measured parameters of the vessel are grouped under the vessel label.

Selected value by method is highlighted, however, when the average method is selected, the highlighted cursor is removed.

When an entire vessel measurement does not have sides (left or right), the side label is not displayed in that vessel study worksheet.



HINTS

Some fields on the worksheet are view only, and others you can change or select. To easily see which fields you can change or select, move the **Trackball**. As the cursor moves over a field that you can change or select, the field is highlighted.

Worksheet Display Touch Panel

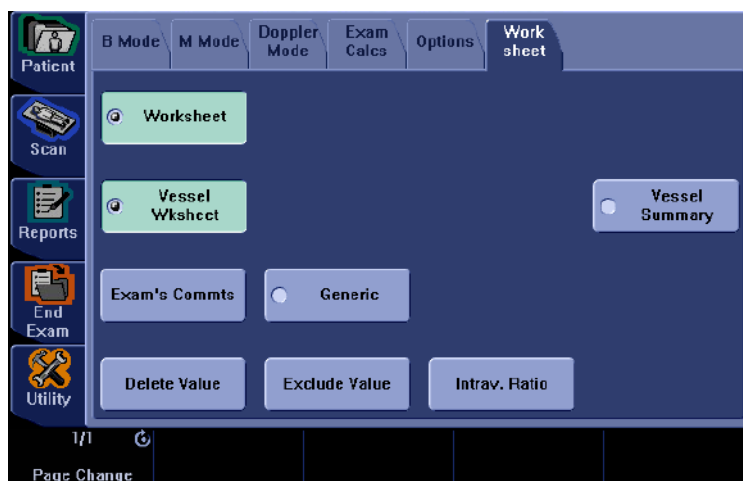


Figure 11-18. Worksheet Display Touch Panel

1. **Worksheet Display:** To exit the worksheet display and return to scanning.
2. **Vessel Worksheet:** Select this key to display the Vessel Worksheet when the Vessel Summary is displayed.
3. **Vessel Summary:** Select this key to display the Vessel Summary when the Vessel Worksheet is displayed.
4. **Examiner's Comment:** Select this key to display the Examiner's comment window. See 'Examiner's Comments' on page 11-35 for more information.
5. **Generic Worksheet:** Select this key to display the Generic Worksheet. Generic study measurements/calculations, such as volume and velocity, are displayed on this worksheet.
6. **Delete Value:** Use to delete a value (each measurement value). See 'To edit a worksheet' on page 11-31 for more information.
7. **Exclude Value:** Use to exclude a value from the result line. See 'To edit a worksheet' on page 11-31 for more information.
8. **Intravessel Ratio:** Select this key to display the Intravessel Ratio Calculation window. See 'Intravessel ratio' on page 11-36 for more information.
9. **Page Change (knob):** If a worksheet has more data, to view the next page, adjust the Page Change knob.

To edit a worksheet

To change data on a worksheet:

1. Select **Worksheet Display** from any page of the Vascular Calculation Touch Panel.
2. To position the cursor at the field you want to change, move the **Trackball**.
The field is highlighted.
3. Type the new data in the field and move the cursor to another place, then the new data is displayed in blue and asterisk are appended to value and resultant value to indicate that it was manual entered.

The average measurements, calculations and ratios are automatically updated to reflect the edited values.

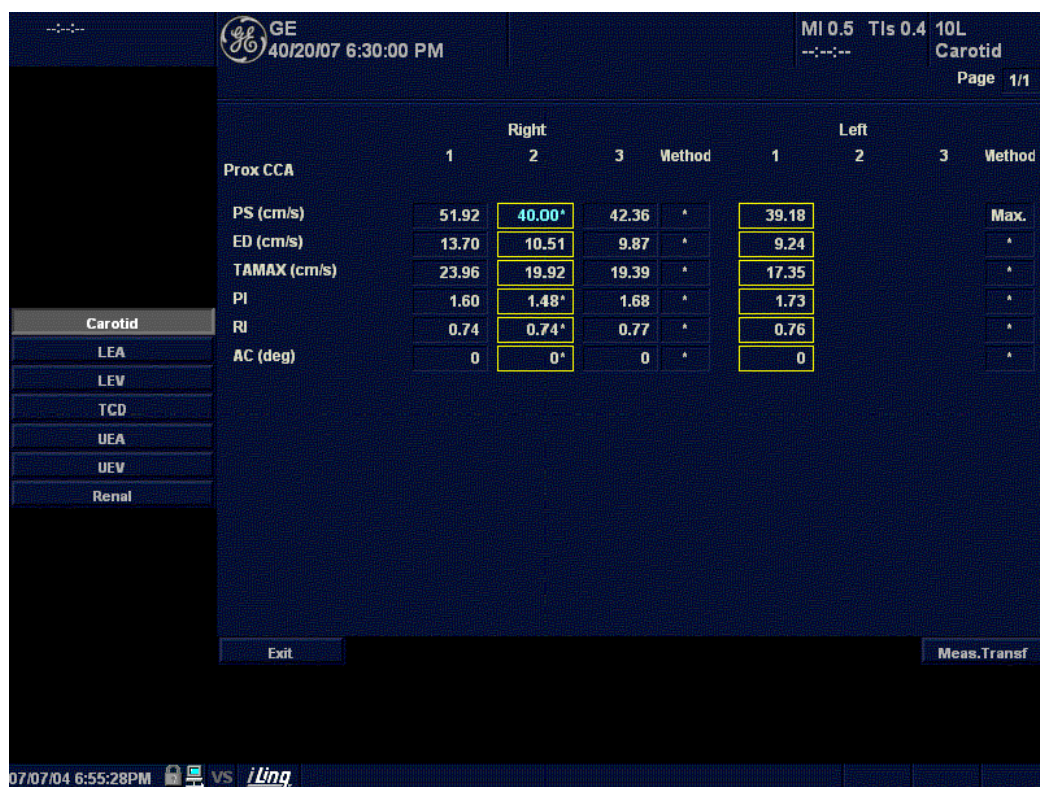


Figure 11-19. Display of the edited value

NOTE: If the user moves the cursor to the edited value and presses the **Set** key once, the value returns to the original value before the edit.

To edit a worksheet (continued)

To delete data:

Delete Value key is used to delete value (each measured value). Whenever particular value is deleted, the group (set) of the measurement/study value is also deleted from the worksheet as a group (set). When PS value is deleted, ED and RI is also deleted if this value was measured as the set of RI measurement.

1. Select **Worksheet** from any page of the Vascular Calculation Touch Panel Menu.
2. To position the cursor at the field you want to delete or exclude, move the **Trackball**.
The field is highlighted.
3. Select **Delete Value**.

For Example:

1. If the user measured RI 4 times, however, latest 3 sets of measurements were displayed in the worksheet.

Result Number	#2	#3	#4
PS	0.500	0.600	0.700
ED	0.100	0.200	0.300
RI	0.800	0.667	0.571

1. Then, the user deleted PS value of #3 from the worksheet.
2. The whole set of #3 measurements is deleted from the worksheet and #1 set of measurements is shifted and displayed as below.

Result Number	#1	#2	#4
PS	0.400	0.500	0.700
ED	0.000	0.100	0.300
RI	1.000	0.800	0.571

To edit a worksheet (continued)

To exclude data:

When the user selects a particular value on the Worksheet and selects **Exclude Value**, this value is excluded from result line and resultant value is re-calculated without this value and also calculation values using this value is 'blank'.

1. Select **Worksheet Display** from any page of the Vascular Calculation Touch Panel Menu.
2. To position the cursor at the field you want to delete or exclude, move the **Trackball**.
The field is highlighted.
3. Select **Exclude Value**.
4. The data in the field is not visible and is not included in worksheet calculations as below.
5. To include a value that you previously excluded, select **Exclude Value**.

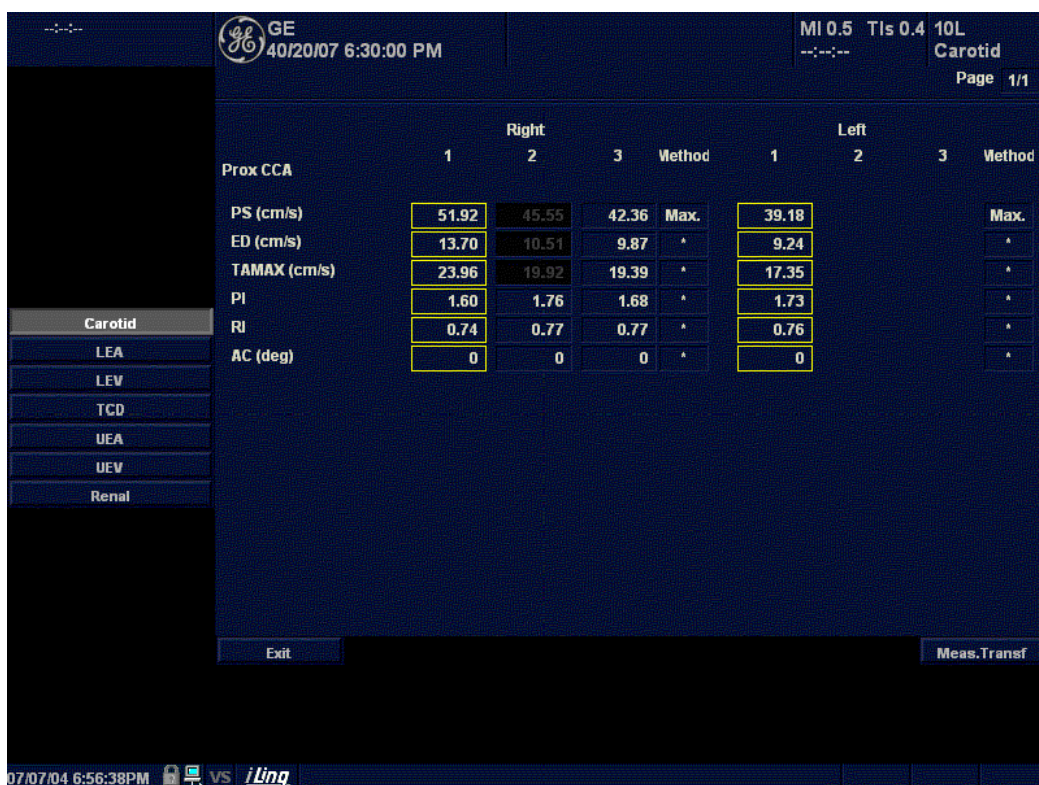


Figure 11-20. Display of the excluded value

To edit a worksheet (continued)

To select the method:

The user can select the method for calculating the cumulative value.

This value is only calculated by using displaying values. If the user takes parameters more than 3 times, latest 3 values are used for this calculation.

- 1. Move cursor to method column and press **Set**.
- 2. The pull-down menu is displayed. Move to cursor any one of methods and press **Set**. The selected method is displayed in the column.

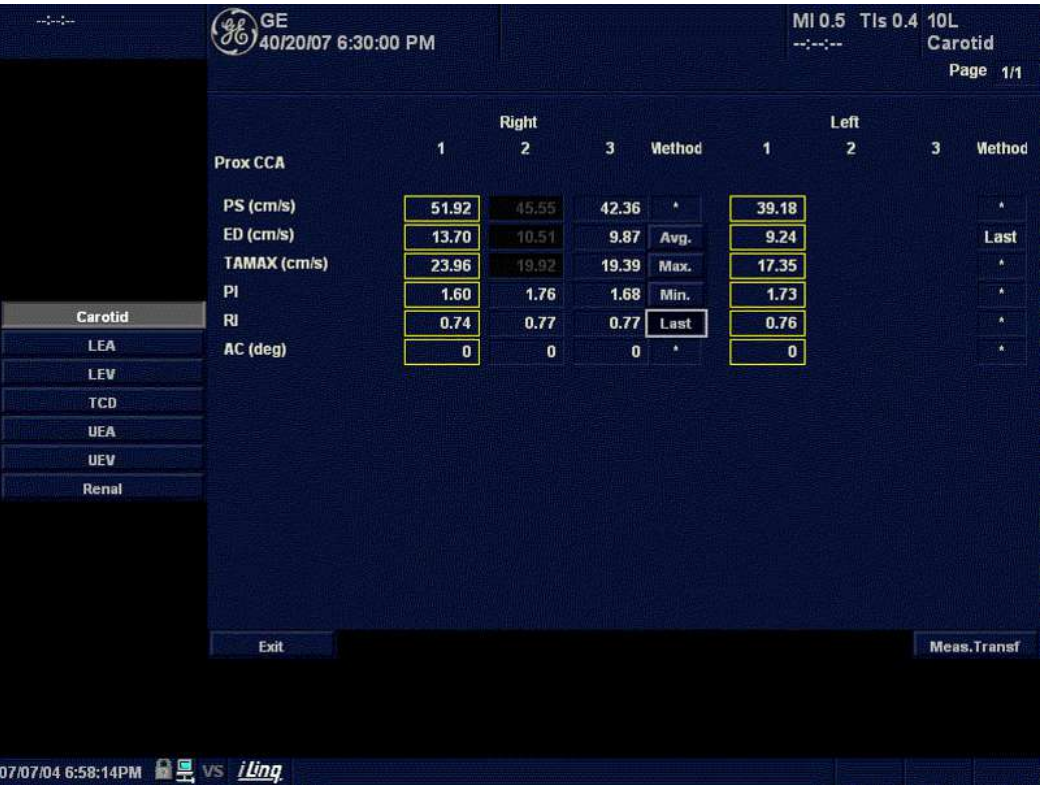


Figure 11-21. Pop-up menu of methods

Examiner's Comments

To type a comment on a worksheet:

1. Select **Examiner's Comments**.
The Examiner's Comments window opens.
2. Type comments about the exam.
3. To close the Examiner's Comments window, select **Examiner's Comments**.



Figure 11-22. Examiner's comment field

Intravessel ratio

To calculate Intravessel ratio, you need a measurement of accessing pressure and stenotic velocities.

- 1. Select **Intravessel** to display the pop-up window in the header section of the worksheet.

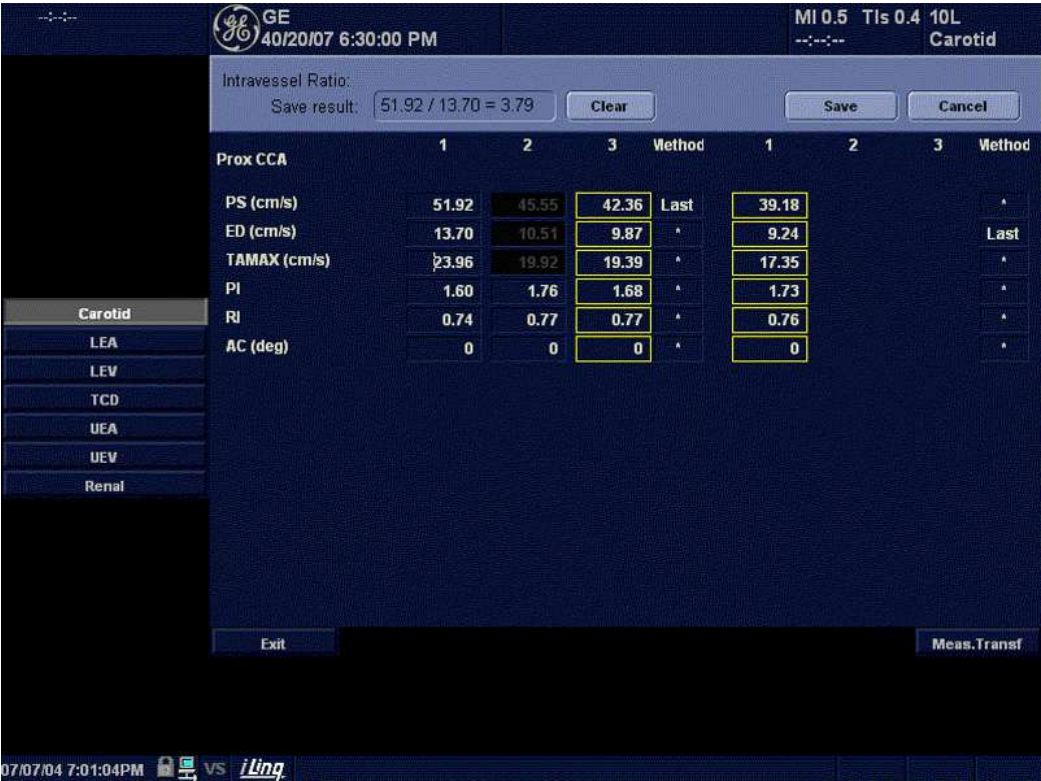


Figure 11-23. Intravessel Pop-up Window

- 2. Select the first velocity.
The value is displayed in the window.



Figure 11-24. Intravessel ratio one

Intravessel ratio (continued)

3. Select the second velocity.

The second value and Result value are displayed in the window.



Figure 11-25. Intravessel ratio two

- To save the Intravessel ratio to the Vessel Summary, move the cursor to **Save** and press **Set**.
- To clear values, move the cursor to **Clear** and press **Set**.
- To cancel and exit Intravessel ratio, move the cursor to **Cancel** and press **Set**.

NOTE: *Intravessel Ratio is only displayed and saved in the Vessel Summary.*

Vessel Summary

The Vessel Summary is designed to automatically display measurements made at specific anatomical sites. Calculated ratios are automatically summarized and displayed.

The Vessel Summary can be displayed at any time during the exam by selecting **Vessel Summary** from the Vascular Worksheet Touch Panel.

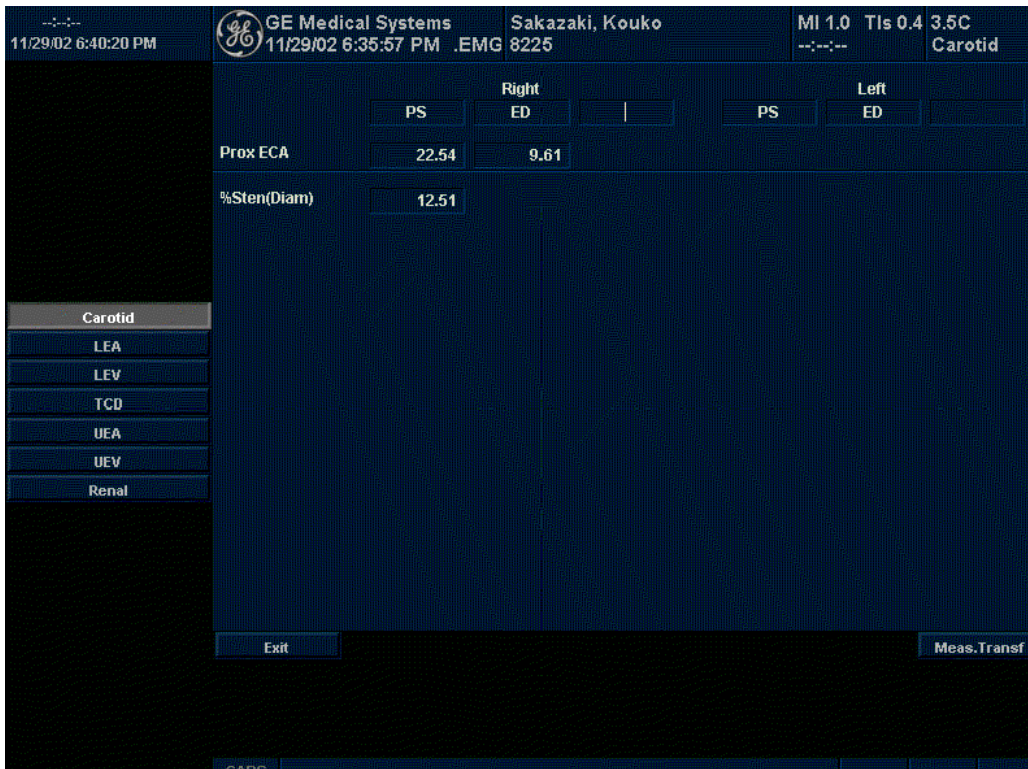


Figure 11-26. Vessel Summary Example

Vessel Summary (continued)

1. The first row, indicating Right or Left, is not displayed when the side is not defined in the vessel. In the third column on the second line, you select the calculations. Move the cursor to the third column, and the pop-up menu is displayed as in Figure 11-27. The selected parameter is displayed in every third column.

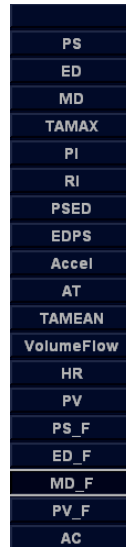


Figure 11-27. Pop-up menu

2. Vessel Name with location information.
3. Check Box. Use to select the vessel velocity for calculating the vessel ratio (ex. ICA/CCA). You can only select one location (position) in a vessel.
4. Result value column. This value cannot be changed or excluded from this page.
5. Calculation name and result. ICA/CCA: The ICA/CCA ratio selects the highest systolic ICA and CCA velocities when calculating this ratio, and displays the velocities.

Carotid Study

The vertebral vessel also has systole and diastole selections. In the summary page, there is a box to select flow reversal for vertebral flows. The choices are Ante (Antegrade), Retr (Retrograde), and Abs (Absent).

To select the method:

Move cursor to the box and press **Set**. After the pop-up menu (Blank, Ante, Retr, Abs) is displayed, select from a menu of choices. The selected choice is displayed in the column.

The box is independent of Left and Right.

Renal Artery Study

For renal arteries, you can calculate RENAL/AORTIC ratio (RAR) based on peak systolic velocities.

You can combine the two renal summary pages, and have a heading to separate the different measurements (main renal, intra renal). You can scroll between the content. The most commonly used, the main renal artery, is the default.

Lower Extremity Artery Study

For the lower extremity artery, you need an intra vessel ratio (assessing pre vs. stenotic velocities). You can specify which (ratio is stenotic/pre).

The intra-vessel ratio needs to be available for all vascular measurements. This appears on the worksheet only if used.

Recording Worksheet

The worksheet can be saved as you would any ultrasound image. Once it is displayed on the screen, it can be recorded on the VCR, printed on the B/W or color page printer, stored on media with the Image Archive option, or placed on regular paper with a line printer.

Chapter 12

Urology

Describes how to perform Urology measurements and calculations.

Urology Exam Preparation

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by the system accuracy, but also by use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

General Guidelines

New Patient information must be entered before beginning an exam. See 'Beginning an Exam' on *page 4-2 for more information*.

Any measurement can be repeated by selecting that measurement again from the Touch Panel Menu.

The system retains as many as eight measurements, but the worksheet retains only the last six measurements of each type.

The three report page measurements can be averaged and the average used in other calculations.

Urology Calculations

Introduction

Urology measurements offer two different types of measurement studies:

- Generic—Common to all applications. See ‘Generic Measurements’ on *page 7-78 for more information.*
- Urology
 - This chapter describes Urology B-Mode measurements.
 - The Urology M-Mode measurements are common to other applications. See ‘M-Mode Measurements’ on *page 7-91 for more information.*
 - The Urology Doppler measurements are common to other applications. See ‘Doppler Mode Measurements’ on *page 7-94 for more information.*

To change a study:

1. Press **Exam Calcs**.
The Urology exam category allows you to choose from the displayed studies.
2. To select another study, select the desired study folder.

Urology B-Mode Measurements

In B-Mode, the Generic Exam Calcs for Urology includes the following measurements:

- % Stenosis
- Volume
- Angle
- A/B Ratio

See 'B-Mode Measurements' on *page 7-82 for more information.*

The following measurements are located specifically in the Urology Exam Calcs. Those specific measurements (Bladder Volume, Prostate Volume and Renal Volume) are listed on the following pages.

Select the Urology Exam Calcs. The following Touch Panel is displayed.

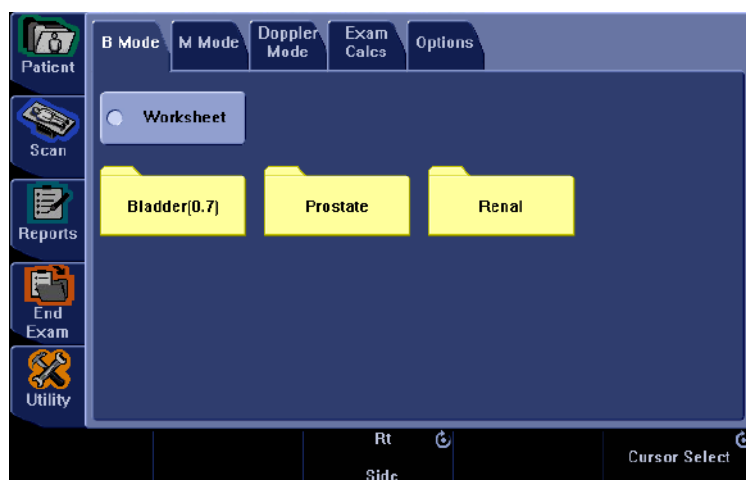


Figure 12-1. Urology Exam Calcs B-Mode Touch Panel

NOTE: *Bladder, Prostate Vol, Renal Vol and Bladder Vol can be displayed on the Touch Panel if preset at the Utility -> Measure screen.*

Bladder Volume

This calculation uses a standard distance measurement. Length is typically measured in the sagittal plane. Width and height are measured in the axial plane.



Figure 12-2. Bladder Volume Touch Panel

To measure Bladder Volume:

1. Scan the patient in the appropriate scan plane.
2. Select the **Bladder** folder, an active caliper displays.
3. Perform a standard distance measurement.
The system displays the distance value in the Results Window.
4. To make the second and third distance measurement, repeat steps 2–3.

After you complete the third distance measurement, the system displays the bladder volume in the Results Window.

Prostate Volume

This calculation uses a standard distance measurement. Length is typically measured in the sagittal plane. Width and height are measured in the axial plane.

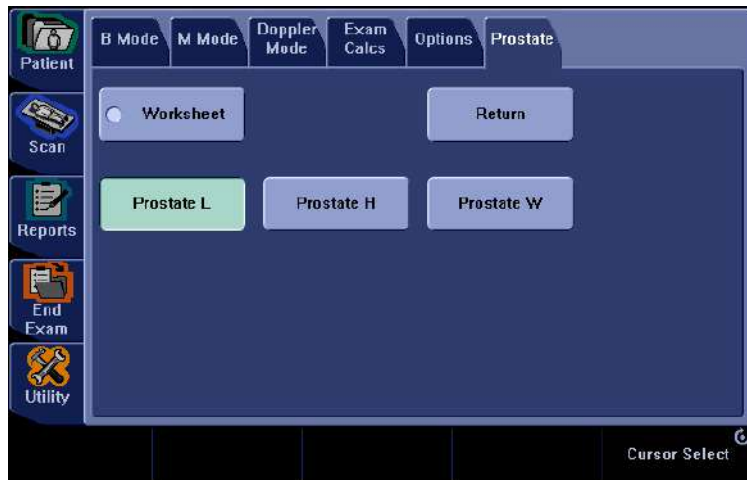


Figure 12-3. Prostate Volume Touch Panel

To measure Prostate Volume:

1. Scan the patient in the appropriate scan plane.
2. Select the **Prostate** folder, an active caliper displays.
3. Perform a standard distance measurement.
The system displays the distance value in the Results Window.
4. To make the second and third distance measurement, repeat steps 2–3.

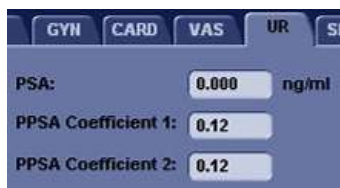
After you complete the third distance measurement, the system displays the prostate volume in the Results Window.

Prostate Volume (continued)

PSA Measurement

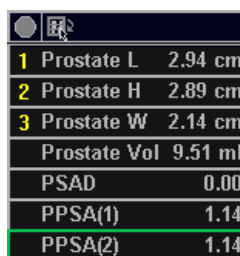
If you enter the value of PSA (Prostatic Specific Antigen) and PPSA Coefficient at the Urology Patient screen, PSAD and PPSA are automatically calculated.

The values are displayed on the Worksheet and Report (if set appropriately on the Report Designer page).



The screenshot shows a software interface with tabs labeled GYN, CARD, VAS, UR, and SM. Below the tabs, there are three input fields: 'PSA:' with a value of 0.000 and unit 'ng/ml', 'PPSA Coefficient 1:' with a value of 0.12, and 'PPSA Coefficient 2:' with a value of 0.12.

Figure 12-4. Urology Patient Screen



1	Prostate L	2.94 cm
2	Prostate H	2.89 cm
3	Prostate W	2.14 cm
	Prostate Vol	9.51 ml
	PSAD	0.00
	PPSA(1)	1.14
	PPSA(2)	1.14

Figure 12-5. Measurement result window

PSAD: Prostatic Specific Antigen (PSA) Density – defined as:
 $PSAD = PSA/Volume$

PPSA: Predicted Prostate Specific Antigen – defined as: $PPSA = Volume \times PPSA \text{ Coefficient}$

Prostate Volume (continued)

- Worksheet
- For the prostate volume calculation, you can select the method “m1”, “m2” or “m3” in addition to Avg., Max., Min. and Last.
 - The value of PSA and PPSA are displayed.

PSA 0.000ng/ml		PPSA Coef(1) 0.12		PPSA Coef(2) 0.12		Page 1/1		
Parameter	Value	m1	m2	m3	m4	m5	m6	Method
B Mode Measurements								
Prostate L	0.40 cm	0.40						Avg.
Prostate H	0.29 cm	0.29						Avg.
Prostate W	0.28 cm	0.28						Avg.
Prostate Vol	0.02 ml	0.02						
PSAD	0.00	0.00						
PPSA(1)	0.00	0.00						
PPSA(2)	0.00	0.00						

Figure 12-6. Urology Worksheet

Renal Volume

This calculation uses a standard distance measurement. Length is typically measured in the sagittal plane. Width and height are measured in the axial plane.

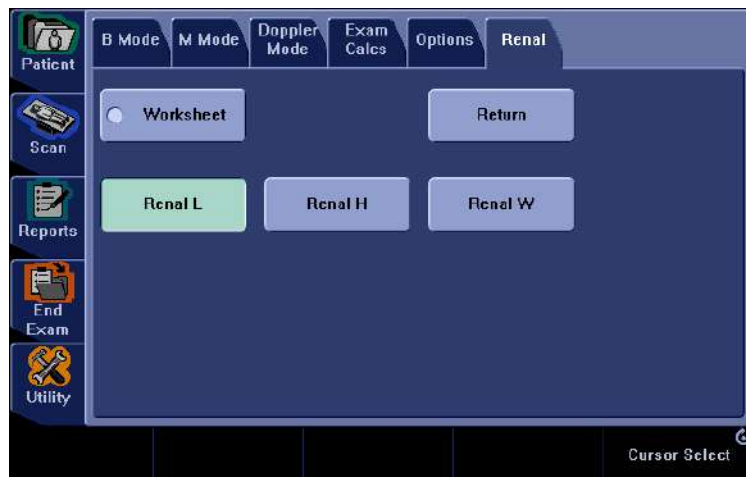


Figure 12-7. Renal Volume Touch Panel

To measure Renal Volume:

1. Scan the patient in the appropriate scan plane.
2. Select the **Renal** folder, an active caliper displays.
3. Perform a standard distance measurement:
The system displays the distance value in the Results Window.
4. To make the second and third distance measurement, repeat steps 2–3.

After you complete the third distance measurement, the system displays the renal volume in the Results Window.

Chapter 13

Pediatrics

Describes how to perform Pediatrics measurements and calculations.

Pediatrics Exam Preparation

Introduction

Measurements and calculations derived from ultrasound images are intended to supplement other clinical procedures available to the attending physician. The accuracy of measurements is not only determined by the system accuracy, but also by use of proper medical protocols by the user. When appropriate, be sure to note any protocols associated with a particular measurement or calculation. Formulas and databases used within the system software that are associated with specific investigators are so noted. Be sure to refer to the original article describing the investigator's recommended clinical procedures.

General Guidelines

New Patient information must be entered before beginning an exam. See 'Beginning an Exam' on *page 4-2 for more information*.

Any measurement can be repeated by selecting that measurement again from the Touch Panel.

The system retains as many as eight measurements, but the worksheet retains only the last six measurements of each type.

The six worksheet measurements can be averaged and the average used in other calculations.

Pediatrics Calculations

Overview

Pediatrics measurements offer two different types of measurement studies:

- Generic. The Generic Calculations study is common to all applications. See 'Generic Measurements' on *page 7-78 for more information*.
- Pediatric Hip (PedHip).
 - This chapter describes Pediatrics B-Mode measurements.
 - The Pediatrics M-Mode measurements are common to other applications. See 'M-Mode Measurements' on *page 7-91 for more information*.
 - The Pediatrics Doppler measurements are common to other applications. See 'Doppler Mode Measurements' on *page 7-94 for more information*.

Pediatrics

B-Mode Measurements

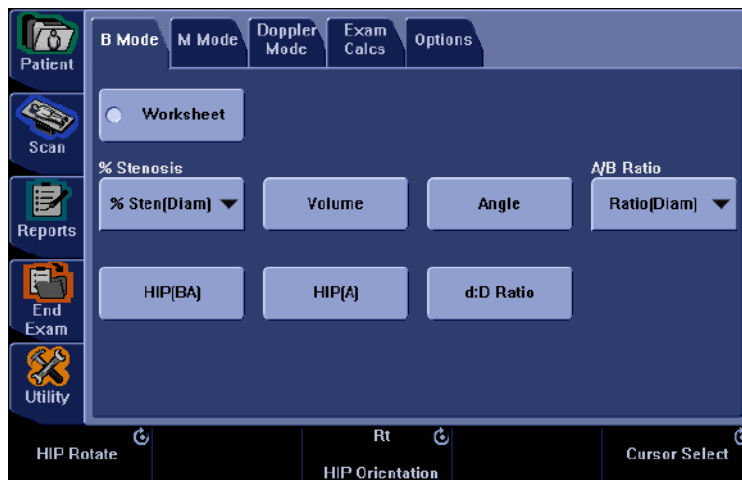


Figure 13-1. Pediatrics B-Mode Measurement Touch Panel

The following generic measurements are common to other exam applications:

- %Stenosis
- Volume
- Angle
- A/B Ratio

See 'B-Mode Measurements' on page 7-82 for more information.

B-Mode Measurements (continued)

Hip Dysplasia Measurement

The HIP calculation assists in assessing the development of the infant hip. In this calculation, three straight lines are superimposed on the image and aligned with the anatomical features. The two angles are computed, displayed, and can be used by the physician in making a diagnosis.

The three lines are:¹

1. The baseline connects the osseous acetabulum convexity to the point where the joint capsule and the perichondrium unite with the iliac bone.
2. The inclination line connects the osseous convexity to labrum acetabulare.
3. The Acetabulum roof line connects the lower edge of the osilium to the osseous convexity.

The α (Alpha) angle is the supplement of the angle between 1 and 3. It characterizes the osseous convexity. The β (Beta) angle is the angle between lines 1 and 2. It characterizes the bone supplementing additional roofing by the cartilaginous convexity.

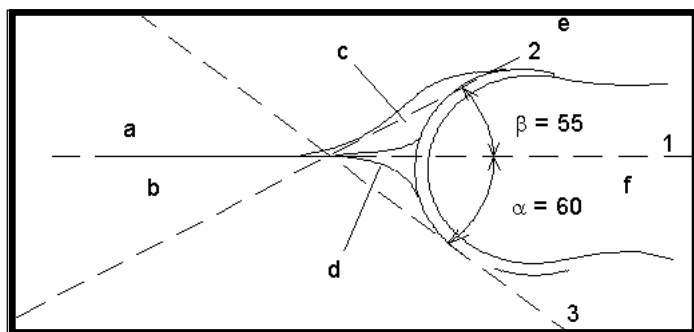


Figure 13-2. Hip Dysplasia

Anatomical Landmarks	
a. Ilium	d. Bony Roof
b. Iliac Bone	e. Cartilaginous acetabular roof
c. Labrum	f. Femoral Head

¹Source: R GRAF, journal of Pediatric Orthopedics, 4: 735-740(1984)

B-Mode Measurements (continued)

Hip Dysplasia Measurement (continued)

To make a Hip Dysplasia measurement:

1. From the Touch Panel, select either the **right** or **left side** (orientation) and then select **HIP**.

A horizontal dotted line displays.

2. To place the baseline, move the **Trackball**. Position the crosshairs edge at the osseous convexity of the ilium.
3. To rotate or change inclination, adjust the **Ellipse** control or **Hip Rotate**.
4. To fix the baseline, press **Set**.
The system displays a second dotted line at an angle.
5. To place the line along the inclination line of the osseous convexity to labrum acetabulare, move the **Trackball**.
6. To rotate or change inclination, adjust the **Ellipse** control or **Hip Rotate**.
7. To fix the second measurement line, press **Set**.
The system displays a third dotted line at an angle.
8. To place the caliper along the acetabular roof line, move the **Trackball**.
9. To rotate or change inclination, adjust the **Ellipse** control or **Hip Rotate**.
10. To fix the third measurement line and complete measurement, press **Set**.

The system displays the hip measurements (α and β) in the Results Window.

B-Mode Measurements (continued)

- Alpha HIP** The Alpha HIP measurement measures the angle between the iliac baseline and the bony roof line. To make an Alpha HIP measurement:
1. From the Touch Panel, select either the **right** or **left side** (orientation) and then select **Alpha HIP**.
A horizontal dotted line displays.
 2. To place the baseline, move the **Trackball**. Position the crosshairs edge at the osseous convexity of the ilium.
 3. To rotate or change inclination, adjust the **Ellipse** control or **Hip Rotate**.
 4. To fix the baseline, press **Set**.
The system displays a second dotted line at an angle.
 5. To place the caliper along the acetabular roof line, move the **Trackball**.
 6. To rotate or change inclination, adjust the **Ellipse** control or **Hip Rotate**.
 7. To fix the second measurement line, press **Set**.
The system displays the alpha hip measurement (α) in the Results Window.

B-Mode Measurements (continued)

d:D Ratio Measurement

The d:D Ratio measurement measures the percentage of the femoral head coverage under the bony roof. To make this measurement:

1. From the Touch Panel, select either the **right** or **left side** (orientation) and then select **d:D Ratio**.
A horizontal dotted line displays.
2. Use the **Trackball** to place the baseline along the ilium. Position the crosshairs edge at the osseous convexity of the ilium.
3. Use the **Ellipse** control to adjust or change inclination or **Hip Rotate**.
4. Press **Set** to fix the baseline.
5. The system displays a circle representing the femoral head. Use the **Trackball** to position the circle.
6. Use the **Ellipse** control to size the femoral head circumference.
7. Press **Set** to fix the femoral head circumference.

The system displays the d:D ratio for the femoral head in the Results Window.

Chapter 14

ReportWriter

Describes how to generate reports.

Standard Report Pages

Introduction

The LOGIQ 9 enables the generation of patient reports based on the examination performed and the analyses that were made during the exam. The reports are generated using the data stored in the system with pre-selected templates.

You may edit a report while performing the exam; customize, delete, or add measurements; and save changes until you use the Store command. Once Stored, the reports are read-only.

It is recommended that the data be saved often, and then carefully reviewed before the report is Stored. Use the worksheet to facilitate the review and adjust data before storing a report. The final report can be printed on a standard printer.

Creating a report

Reports summarize the data obtained in the examination. They can contain data, images, and cine loops.

Once generated, the report can be viewed, images can be added, and the patient's personal data can be modified. The examination data itself CANNOT be changed.

The screenshot displays a GE Medical Systems report interface. At the top left is the GE logo and the text "GE Medical Systems". Below this is a patient information section with a table containing the following data:

Name : GTest	Patient ID : 123-4567	Date
--------------	-----------------------	------

Below the patient information section are six ultrasound images arranged in a 3x2 grid. Each image shows a different view of a scan, with various parameters and settings visible on the right side of each image. At the bottom of the report is a "Comments" section with a text area containing the text "Test exam".

Figure 14-1. Report Example

Activating the Report

1. Select **Reports** on the Touch Panel.
2. The system displays the default report for the current application on the monitor.

The information entered during the examination is automatically filled in the appropriate fields (e.g. demographic, diagnosis, comments).

The preview image appears when the cursor is over the clipboard image.

NOTE: *The template is the skeleton of your report. It is composed of different objects that can be customized by the user.*

3. Adjust the **Page Change** control to move down the page.

GE Healthcare Hospital
05/07/22 02:03:18 PM ADM

MI 0.4 TIs 0.0 3.5C
OB-2/3

GE Medical Systems

Hospital name
Address
City, State Zip code
Phone

Patient Information

Name:
Birthdate:
Fetus #:
Ref. Physician:

Patient Id:
LMP:
BBT:
Perf. Physician:

Date: 2005/07/19
GA:
EDD:
Operator:

Indications: Comments:

Measurements

PRINT STORE RETRIEVE TEMPLATE INSERT TEXT More

05/07/22 02:04:51 PM

Figure 14-2. Report Page Example

Activating the Report (continued)

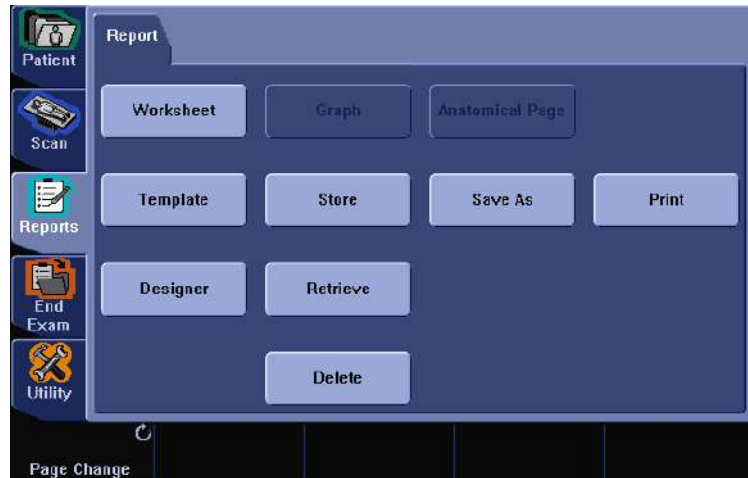


Figure 14-3. Report Touch Panel

Table 14-1: Report Touch Panel Controls

Button	Description
Print	Prints out the report to the default printer.
Store	Stores the report page into Archive as CHM* file.
Retrieve	Retrieves the report page from Archive. Stored Date/Time is appended to the name of stored report.
Template	Selects template from the list of selected applications.
Designer	Accesses template editor screen.
Save As	Exports the report page to storage media as CHM or PDF format.
Delete	Deletes the report page from Archive.
Worksheet	Accesses Worksheet Page.
Graph	Accesses OB Graph page (applies only to OB).
Anatomical Page	Accesses Anatomical Survey page (applies only to OB).
More	Toggles between other pages.
*CHM is a compressed HTML help file.	

Selecting another template

You can select another template for the current patient:

1. Select **TEMPLATE** at the bottom of the monitor display or the Touch Panel.
2. A list of available templates and exam categories displays.



Figure 14-4. Application List Example

Selecting another template (continued)

3. Select the desired template using the **Trackball** and press **Set**.

The selected template displays on the monitor.

NOTE: *If you select another exam category, the template list of the selected category displays. Select the desired template.*

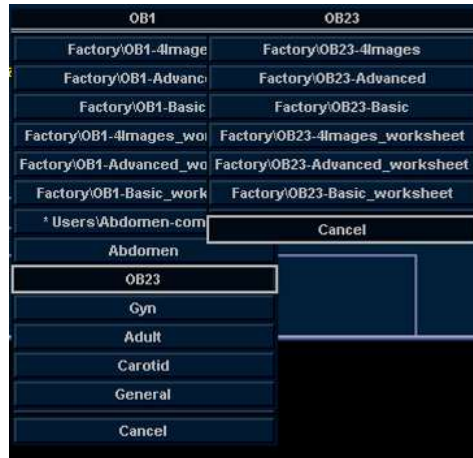


Figure 14-5. Available Template list

4. Select the desired template name and press **Set**.
5. The report changes to the selected template.

Factory Templates

The system has factory templates for each application. You can modify these templates or create user-defined templates. You need to save revised/new templates with unique names.

A template may include one or more of the following:

- Measurements
- Worksheet or Vessel Summary Images
- Anatomical Surveys or Biophysical Profiles
- Anatomical Graphics
- Graphs
- Images areas
- Score Boxes

NOTE: *Additional factory templates can be added from the Utility-->Report menu (OB for multiple gestation, Renal, etc.).*

Editing a Report

Entering the hospital address

When using a factory template, the area for the hospital information is usually placed in the upper portion of the report.

To make a new area, See 'Fixed Text' on *page 14-50 for more information*.

To modify the factory template:

1. Select **Reports**.
2. Select **Designer**.
3. Double-click on the area for the hospital information in the template. The Fixed Text dialogue box displays.



Figure 14-6. Fixed Text Dialogue Box

4. Make changes as necessary.
 - a. Enter the hospital address in the text area.
 - b. Modify Box Properties (box width, box border line width, text align, box height, box left margin, and font).
5. Select OK.

Entering the hospital address (continued)

6. Save the template.

To keep the same template name:

- Select Save from the File menu, and press Set. The Save Template dialog box opens.
- Select Yes. The template retains the same name and appends “[user]”. For example, OB23-Basic[user].

To save the template with a new name:

- Select Save As from the File menu, and press Set. The Save Template As dialog box opens.
- Enter the name of the new template, and press Set. The template receives the new name and appends “[user]”. For example, NewReport[user].

7. Exit the Report Designer. The report with the hospital address displays.

Inserting the hospital logo

When using a factory template, the area for the logo is usually placed in the upper left portion of the report.

To make a new area, See 'Fixed Text' on *page 14-50 for more information*.

To modify the factory template:

1. Save the preferred hospital logo in either a jpeg or bmp format on a CD.
2. Insert the CD into the CD/DVD drive.
3. Select **Reports**.
4. Select **Designer**.
5. Double-click the GE Logo so that the frame is highlighted. The logo box displays.



Figure 14-7. Logo Box

Inserting the hospital logo (continued)

6. Select Import Logo (1). Select the CD drive first and then the hospital logo.
7. Select OK. The hospital logo displays in the logo list (2). Click the logo to select.

NOTE: *Scroll the logo list using the left/right arrow key (3).*

8. Modify Appearance (4).
9. Select OK.
10. Save the template.

To keep the same template name:

- Select Save from the File menu, and press Set. The Save Template dialog box opens.
- Select Yes. The template retains the same name and adds “[user]”. For example, OB23-Basic[user].

To save the template with a new name:

- Select Save As from the File menu, and press Set. The Save Template As dialog box opens.
- Enter the name of the new template, and press Set. The template receives the new name and adds “[user]”. For example, NewReport[user].

11. Exit the Report Designer. The template with the hospital logo displays.

Changing the Archive Information

When using a factory template, the Archive Information is usually placed below the hospital name and logo.

The contents of the Archive Information is inserted through the related page automatically. If you want to change the description, such as Information or Comments that was entered in the patient menu:

1. Double-click the yellow text to be changed, e.g. Information or Comments.

The area where the description was entered (e.g. Patient menu) displays.

2. Change the existing data as necessary.
3. Select **Report** to return to the report.

Patient Information

Name: Yokogawa, Hana	Patient Id: 20040318	Date: 03/18/2004
Birthdate: 12/13/1987	Sex: Female	Accession #:
Ref.Physician:	Perf.Physician:	Operator: ADM
Indications:		
Comments:		

Figure 14-8. Patient Information Area (Example)

Modifying the displayed objects of Archive Information

1. Select **Designer**.
2. Double click on the area for the Archive Information in the template. The Archive Information Box displays.

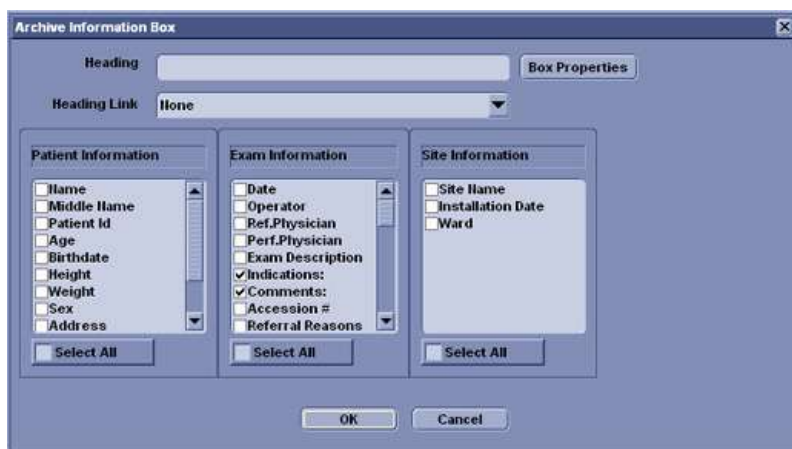


Figure 14-9. Archive Information Box

3. Click the checkboxes to select and deselect the objects. Objects with checkmarks will appear in the report template.
4. Select Box Properties to change the font, font size, font color, or box size, and select OK.
5. Select OK to return to the Report Designer.
6. Save the template.

To keep the same template name:

- Select Save from the File menu, and press Set. The Save Template dialog box opens.
- Select Yes. The template retains the same name and adds "[user]". For example, OB23-Basic[user].

To save the template with a new name:

- Select Save As from the File menu, and press Set. The Save Template As dialog box opens.
- Enter the name of the new template, and press Set. The template receives the new name and adds "[user]". For example, NewReport[user].

7. Select File -> Exit to leave the Report Designer.

Entering free text

You can enter free text to the report using the alphanumeric keyboard.

The factory template terms their text area as “Summary or Comments”.

1. Move the cursor to the text field and press **Set**.

NOTE: *You can enter the text only to the field set as free text in the Report Designer.*

2. Type the text.

Inserting Text

1. Select **Designer**.
2. Move the cursor where the text is to be inserted and press **Set**.
3. Select the Text Field from the Insert menu. The Text Field dialog box displays.

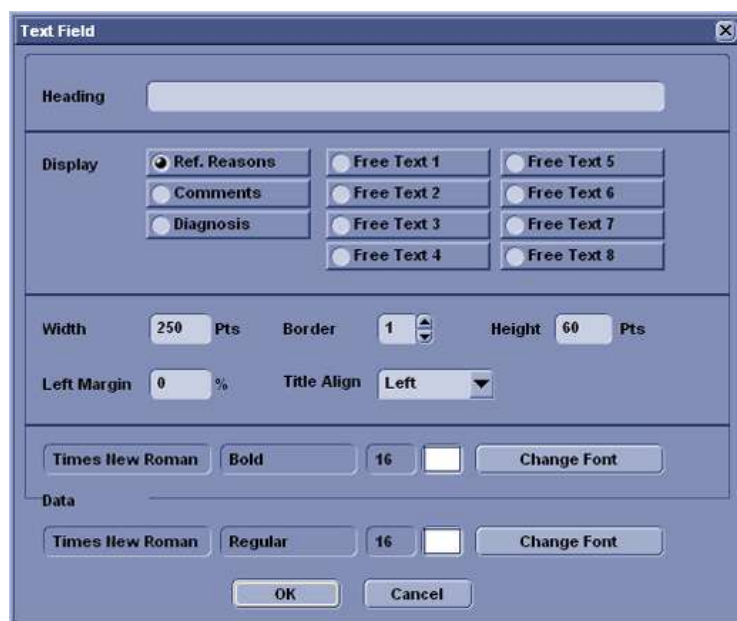


Figure 14-10. Text Field Dialogue Box

4. Select the appropriate display items:
 - Ref. Reasons: Retrieves this information from the Direct Report
 - Comments: Retrieves this information from the Comment field of the patient screen and the Exam Comment field of the worksheet.
 - Diagnosis: Retrieves this information from the Direct Report
 - Free Text 1 - 8

Inserting Text (continued)

5. Type the heading Text.
6. Modify box properties, the heading text and font, and data.
7. Select OK or Cancel.
8. Save the template.

To keep the same template name:

- Select Save from the File menu, and press Set. The Save Template dialog box opens.
- Select Yes. The template retains the same name and adds “[user]”. For example, OB23-Basic[user].

To save the template with a new name:

- Select Save As from the File menu, and press Set. The Save Template As dialog box opens.
- Enter the name of the new template, and press Set. The template receives the new name and adds “[user]”. For example, NewReport[user].

Inserting an image to the report

Some factory templates include an image area. If you want to insert or modify the image area, See 'Image Display Fields' on page 14-41 for more information.

To insert images from clipboard into the image field of the report:

- NOTE:** *The preview image appears when the cursor is over a clipboard image.*
1. Move the cursor to the desired image on the clipboard.
 2. Press and hold down the **Set** key and drag the selected image to the report by using the **Trackball** or double click the **Set** key on the desired image.
 3. To move images between image areas, press and hold down the **Set** key and using the **Trackball**, drag the selected image to the new location.

To remove an image from the report, press and hold down the **Set** key and using the **Trackball**, drag the select image back to the clipboard.

Measurement result section

Measurement results for the current patient display automatically if you have the measurement section in the report template.

The factory template has an appropriate measurement result area. If you want to insert or modify the measurement area, See 'Measurements' on page 14-47 for more information..

GE Medical Systems		Yokogawa, Hana	MI 0.8	TIs 0.3	3.5C
03/18/04 2:29:31 PM ADM		20040318, LMP=02/10/04	--:--:--		Gyn
Measurements					
2D					
Ut-L	7.97	cm			
Ut-H	4.56	cm			
Ut-W	4.50	cm			
Ut Vol	85.71	ml			
Rt Ov-L	5.88	cm			
Rt Ov-H	4.82	cm			
Rt Ov-W	3.10	cm			
Rt Ov Vol	46.07	ml			
Lt Ov-L	4.95	cm			
Lt Ov-H	4.22	cm			
Lt Ov-W	2.98	cm			
Lt Ov Vol	32.65	ml			
Rt Fo-D1 01	6.00	cm			
Rt Fo-D2 01	5.15	cm			
Rt Fo-D3 01	2.99	cm			
Rt Fo-Mean 01	4.72	cm			
Rt Fo Vol 01	48.44	ml			
Lt Fo-D1 01	1.56	cm			
Lt Fo-D2 01	1.12	cm			
Lt Fo-D3 01	0.80	cm			

Figure 14-11. Measurement section

Inserting the worksheet

You can insert the worksheet (like you can insert an image) to the image display field. To set a image display field in the report template, See 'Image Display Fields' on *page 14-41 for more information*.

1. Display the worksheet on the monitor display.
2. Save the worksheet using the **Print** key.
3. Press **Report**.
4. Drag the worksheet into the report.
 - a. Move the cursor to the desired worksheet on the clipboard.
 - b. Press and hold down the **Set** key. Use the **Trackball** to drag the selected worksheet into the Image Display Field.
 - c. Release **Set**.

NOTE: You can also move the cursor to the desired worksheet on the clipboard, double-click the worksheet, move the cursor to the Image Display Field, and select Set.

5. The worksheet displays on the report.

NOTE: You can double-click the worksheet in the report to change the background color to white to save ink during printing. Double-click the worksheet again to return the worksheet to the original color.

Placing objects side-by-side

If you want to place images, the image and comment, anatomical graphic and comment, etc. side-by-side, you must first place a table, which has two (or more) columns, into the report template.

1. Select **Report**.
2. Select **Designer** to display the Report Designer.
3. Place the cursor where you want to insert the object.
4. Select Table from the Insert menu. Insert Table box displays.

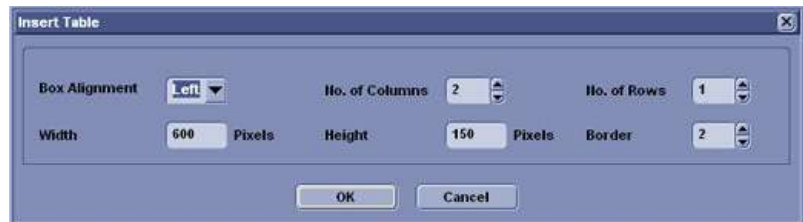


Figure 14-12. Insert Table Box

5. Set the number of columns to 2 (or more, as required) and change the table parameters, if needed. Select OK.

NOTE: *If you do not need a table border, set the Border to 0. Add additional rows if required.*

6. Place the cursor in the column and select the desired items from the Insert menu (e.g. logo, image, free text). Specify those items.
7. Repeat step 6 for each column as required.

Placing objects side-by-side (continued)

8. Save the template.

To keep the same template name:

- Select Save from the File menu, and press Set. The Save Template dialog box opens.
- Select Yes. The template retains the same name and adds “[user]”. For example, OB23-Basic[user].

To save the template with a new name:

- Select Save As from the File menu, and press Set. The Save Template As dialog box opens.
- Enter the name of the new template, and press Set. The template receives the new name and adds “[user]”. For example, NewReport[user].

You can insert the images in the order preferred, by row or column, on the factory templates. See ‘Inserting the Table’ on *page 14-34 for more information.*

Retrieving an Archived Report

1. Select **Retrieve**. The Retrieve menu displays.



Figure 14-13. Retrieve Menu (Prefix “User1\” may not appear)

2. Select the desired report and press **Set**.

NOTE: *The retrieved report cannot be edited.*

Deleting a Report from Archive

1. Select **Delete**. The Retrieve menu appears on the screen.

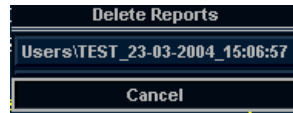


Figure 14-14. Delete Reports Menu (Prefix “User1\” may not appear)

2. Select the report to delete and press **Set**.

Storing the Report

1. Select **Store**.

The Report is saved as a CHM file to Archive.

NOTE: *The archived report cannot be edited; therefore, it is recommended that the data is carefully reviewed before the report is saved.*

Printing the Report

To preview the Print Layout before printing, See ‘Preview the Print Layout’ on *page 14-31 for more information.*

NOTE: *Double-click the worksheet and/or image in the report to change the background color to white. A white background will save ink during printing. Double-click the worksheet or image again to return to the original color.*

1. Select **Print** to print out the report.

The Report is printed on the default printer.

Exporting the Report to Media

1. Select **More**.
2. Select **Save As**.
The Save As dialog box appears on the screen.
3. Enter the Report title and select the file format.
4. Select the media to export the Report. The system supplies a name (numeric DICOM UID, unique identifier).



Figure 14-15. Save as Dialog Box

5. Select **Save**.

Exiting the report

1. Select **Store** to save the report.

NOTE: *If the user is working on a report and leaves the report screen for any reason, all information added to the report is automatically saved without loss of data.*

2. Select another key to close the report page.

Accessing Worksheet, OB Graph and Anatomical Survey Pages

If the Worksheet, OB Graph, and/or Anatomical Survey pages have been saved for the current patient, you can access these pages from the report page.

NOTE: *OB Graph and Anatomical Survey pages apply only to OB.*

1. Select either **Worksheet**, **Graph** or **Anatomical Page** on the Touch Panel.

NOTE: *There is also Fixed Text set up as hyperlinks for these pages. Cursor to the fixed text and press **Set**.*

2. The system displays the appropriate page (Worksheet, OB Graph or Anatomical Survey) with the corresponding Touch Panel.
3. Select **Report** to return to the Report page.

Designing Your Own Template

Template Designer

You can design and create your own customized template from a blank template page, or you can use an existing template (factory or user-defined) and save the changes.

Display the desired template and select **Designer** to open the Template Designer page.

The screenshot shows a window titled "Report Designer" with a close button (X) in the top-left corner. The form is divided into several sections:

- Hospital Information:** A box containing labels for "Hospital name", "Address", "City, State Zip code", and "Phone".
- Patient Information:** A section with a title bar. It contains a grid of labels for data entry:
 - Row 1: Name, Patient Id, Date
 - Row 2: Birthdate, LMP, GA
 - Row 3: Fetus #, Perf.Physician, EDD
 - Row 4: Ref.Physician, (empty), Operator
- Indications:** A label followed by a text input field.
- Comments:** A label followed by a text input field.
- Measurements:** A section with a title bar, followed by a table with 4 columns and 3 rows of input fields.

Figure 14-16. Report Designer

File Menu

Table 14-2: File Menu

	Description
New	Creates a new template. A blank template appears.
Save	Overwrites the existing template.
Save As	Saves as a new name.
Page Setup	Enters Print Layout screen.
Print Preview	Executes print preview.
Exit	Exits Report Designer page.

Create a new template

To design a new template without using a pre-existing factory template:

1. Select **Designer** to open the Report Designer.
2. Select New from the File menu, and press **Set**.
The blank template displays.
3. Create the report template as needed.
4. Select Save from the File menu, and press **Set**.
The Save Template As dialogue box displays.
5. Enter a template name and click OK.
6. To exit Report Designer, select **Exit** from the File menu, and press **Set**.
 - Yes: Saves changes and exits Report Designer.
 - No: Does not save changes and exits Report Designer.
 - Cancel: Returns to Report Designer.

Create a new template and save as a factory template name

To design a new template by modifying an existing factory template and keeping the same name of the factory template:

1. Select and display the existing factory template.
2. Select **Designer** to open the Report Designer.
3. Modify the report template as needed.
4. To save changes, select **Save** from the File menu, and press **Set**.

The Save Template dialog box displays.

- Yes: Saves changes.
- No: Does not save changes.
- Cancel: Returns to Report Designer.

NOTE:

The template name displays in the template list, retains the same name, and adds "[user]". For example, "OB23-Basic[user]". You do not lose the original factory template.

5. To exit Report Designer, select **Exit** from the File menu, and press **Set**.
 - Yes: Saves changes and exits Report Designer.
 - No: Does not save changes and exits Report Designer.
 - Cancel: Returns to Report Designer.



HINTS

Save changes frequently as you modify your template. Saving often reduces the risk of losing all your changes.

Create a new template and save with a new name

To design a new template by modifying or copying an existing factory template and saving it with a new name:

1. Select and display the existing factory template.
2. Select **Designer** to open the Report Designer.
3. Modify the report template as needed.
4. Select **Save as** from File menu and press **Set**.
The Save Template As dialog box displays.
5. Type the new template name and click OK.
6. Select Exit from the File menu and **Set**.
7. The Report Designer closes and returns to the Report Page.

NOTE: *The template receives the new name and adds “[user]”. For example, NewReport[user].*

Page Setup

1. Modify the factory template as necessary in **Designer**.
2. Select **Page Setup** from File menu and press **Set**.
3. Change the paper size or orientation to fit the print layout, as necessary.

To define the header and footer for the printed report, type text and enter the required variables listed in the table below. Select “Different for first page” and enter a specific header/footer for that page.

Table 14-3: Variable and Definition

Variable	Definition	Variable	Definition
{pid}	Patient ID	{prt}	Current time (printing time)
{pnm}	Patient name	{cp}	Current page
{pbd}	Patient date of birth	{tp}	Page count
{exd}	Examination date	{c}	Subsequent text is centered
{prd}	Current date (printing date)	{r}	Subsequent text is right aligned.
{inm}	Institution name		

NOTE: *Default is left aligned. Report will appear as black ink on white background.*

- Page Setup
(continued)
4. Select OK or Cancel.



Figure 14-17. Page Setup with Header Example

File Menu (continued)

Preview the Print Layout

1. Select **Template** to choose the Report Template.
2. The Print Preview screen displays.
If changes need to be made, select **Close** to exit the Preview page. Modify the template or return to the Report and modify the contents.
3. Select **Print Preview** from File menu, and press **Set**.
The Print Preview screen displays.

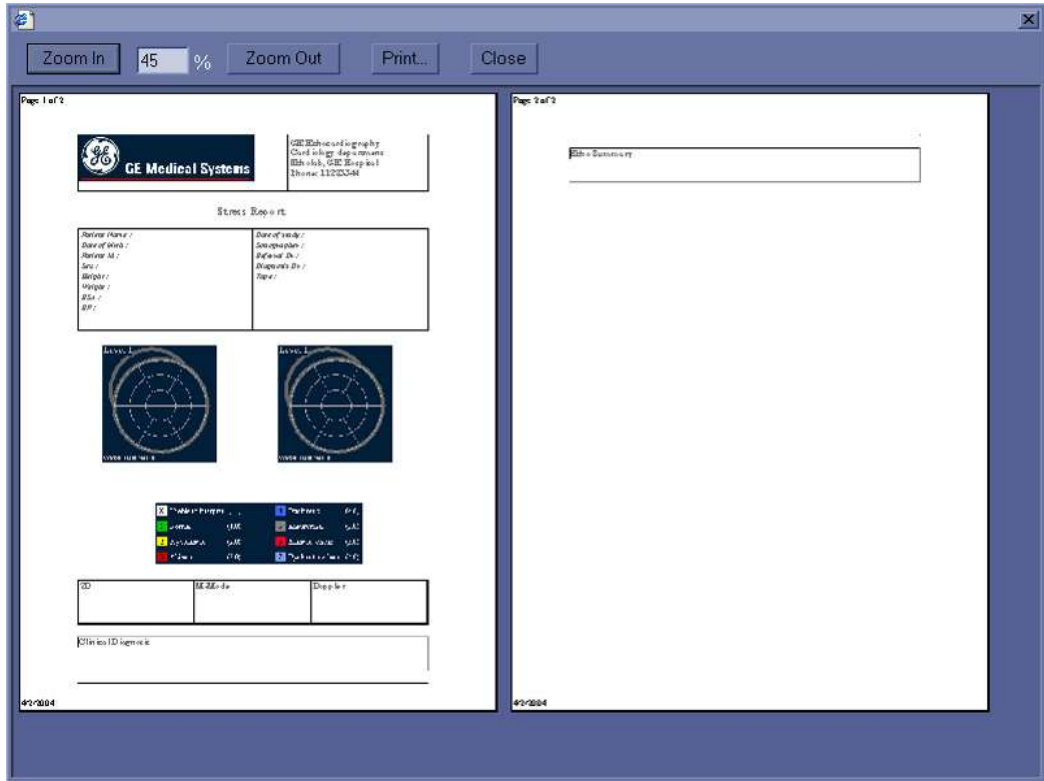


Figure 14-18. Print Preview

4. Select **Print**. The report is printed on the default printer.
If changes need to be made, select **Close** to exit the Preview page. Modify the template or return to the Report and modify the contents.

Edit Menu

Table 14-4: Edit Menu

	Description
Delete	Deletes the selected object from the report template.
Undo	Restores the previous state(s) of the report template.

Deleting a template object

1. Select the object to be deleted.
2. Select Delete from the Edit menu, and press Set. The object is deleted from the template.

Undoing the operation

1. Select Undo from the Edit menu, and press **Set**.
2. Repeat as required.

Insert Menu

Table 14-5: Insert Menu

	Description
Page Break	Inserts a Page Break.
Table	Inserts a Table.
Logo	Inserts a Logo Bitmap File.
Archive Info	Inserts Archive Information.
Anatomical Graphics	Selects anatomical graphics by category to be inserted into a field.
Image	Inserts the image display field to the template.
Wall Motion Analysis	Selects Cut Planes, Bull's Eye, or Score Table Box.
OB Graph	Selects Graph or Bar.
Measurements	Inserts the measurement display field in the template.
Text Field	Edits text field.
Fixed Text	Enters any comments as Fixed Text.

Inserting the Page Break

1. Place the cursor where the Page Break is to be inserted and press **Set**.
2. Select Page Break from the Insert menu and press **Set**. The page break line displays on the template.

NOTE: *To edit the page break line, select the line and double click the **Set** key.*

Inserting the Table

1. Place the cursor where the table is to be inserted and press **Set**.
2. Select Table from the Insert menu and press **Set**. The Insert Table dialog box displays.



Figure 14-19. Insert Table Dialog

3. Specify each parameter as required.

NOTE: *To set the table border as not visible, set "Border" parameter to 0 (zero)*

4. Select OK to insert the table or Cancel.

NOTE: *To insert/delete a row/column from the table or access table properties, double click the **Set** key in any empty area inside the table. A table menu appears with those options.*

Inserting Images in a Table

You can choose the order in which images are inserted into tables: by row (default) or by column.

Image Order by Row

The system default inserts images in the cells of the first row, then to the next row.



Figure 14-20. Image Order—Row Preference
(System Default)

1. Follow the instruction for inserting a table. When specifying parameters, specify:
No. of Columns=2; No. of Rows=2
2. After inserting the table, insert an image box in each cell of the table.
 - a. Move the cursor to the first cell and select Insert -> Image.
 - b. Repeat this step for each cell in the table.

After the template is saved and you are working in the ReportWriter, when you select images to be inserted in the table, they are placed in the default order.

Inserting Images in a Table (continued)

Image Order by Column

If you prefer to have the image placement by column, images are inserted in each cell of the first column, then the next column.

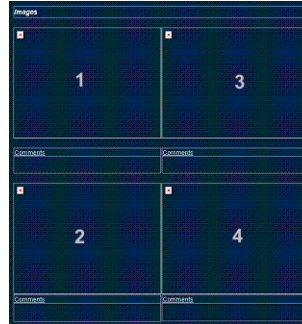


Figure 14-21. Image Order—Column preference

In order to achieve the column preference, you need to create a table with 2 columns and 1 row. In each cell of this table, you need to insert another table.

1. Follow the instructions for inserting a table. When specifying parameters, specify:
No. of Columns=2; No. of Rows=1
2. After inserting the table, create a table inside each of the existing table's cells.
 - a. Move the cursor to the left column's cell and press **Set**.
 - b. Select Table from the Insert menu and press **Set**.
 - c. When specifying parameters, specify:
No. of Columns=2; No. of Rows=1; Width=290 pixels.
Select **OK**.
 - d. Repeat steps a-c for the right column.
3. Insert an image box to each table cell.
 - a. Move the cursor to the first cell and select Insert -> Image.
 - b. Repeat this step for each cell in the 2 tables.

After the template is saved and you are working in the ReportWriter, when you select images to be inserted in the table, they are placed with your column preference.

Inserting Logos

1. Place the cursor where you want to insert the logo and press **Set**.
2. Select Logo from the Insert menu and press **Set**. The Logo Box displays.



Figure 14-22. Logo Box

3. Select a logo that you want to insert (1) or import a bmp or jpg file from the removable media (2). Scroll the images using the arrow key (3). Specify the appearance (4).
4. Select OK to insert the logo or Cancel.

Changing a logo:

1. Place the cursor on the logo to be changed and press **Set** twice. The Logo Box displays.
2. Select a different logo. If the desired logo is not shown, select Import Logo to import a different logo.
3. Specify the appearance.
4. Select OK or Cancel.

Inserting Archive Information

Archive information contains all the objects from the different information menus (Patient, Exam, and Site Information). This box accumulates different information menu selections that can be grouped together and displayed in one table.

1. Place the cursor where you want to enter the archive information and press **Set**.

If you use a factory template, double click on the current archive information area to display the Archive Information Box.

2. Select Archive Info from the Insert menu and press **Set**. The Archive Info Box displays.



Figure 14-23. Archive Information Box

3. Type the Heading, select a heading link from the pull-down menu, and select the parameters you want to display in the report.

Inserting Archive Information (continued)

NOTE:

4. Select Box Properties to change the Font, Alignment, Appearance, etc.

To set the same font to all fields, select *Set All fields*.



Figure 14-24. Table Properties

5. Select OK or Cancel. The contents of the Archive Information is inserted to the related page automatically.

Patient Information		
Name: Yokogawa, Hana	Patient Id: 20040318	Date: 03/18/2004
Birthdate: 12/13/1967	Sex: Female	Accession #:
Ref.Physician:	Perf.Physician:	Operator: ADM
Indications:		Comments:

Figure 14-25. Patient (Archive) Information Example

Editing displayed Archive Information:

1. Select **Designer**.
2. Move the cursor to Archive Information field to be edited.
3. Press **Set** twice. The Archive Information Box displays.
4. Edit the heading, the Heading Link and Information parameters, as necessary.
5. Select OK to save or Cancel.

Anatomical graphics

1. Place the cursor where you want to insert the Anatomical Graphics and press **Set**.
2. Select Anatomical Graphics from the Customize menu.

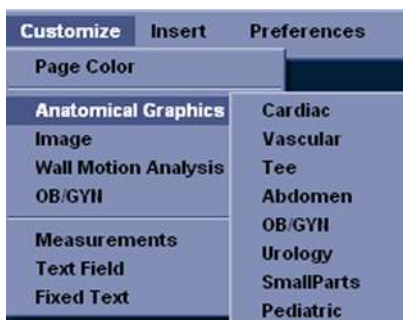


Figure 14-26. Anatomical Graphics Menu Example

3. Select the desired category and press **Set**. The graphic box displays.

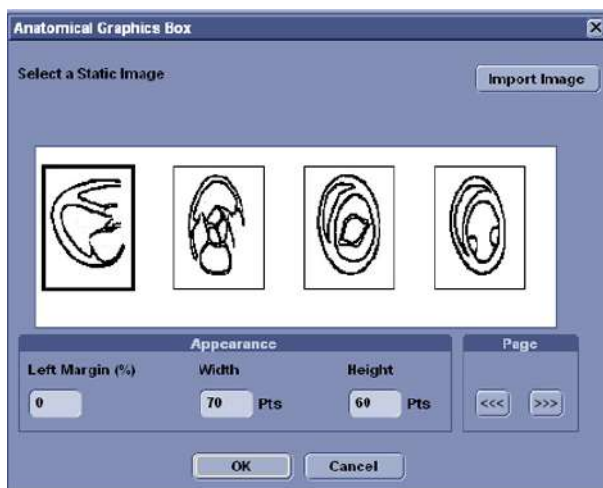


Figure 14-27. Anatomical Graphics Box Example

4. Select the graphic to be inserted to the template or import a bmp or jpg file from the removable media. Scroll the images using the arrow key.
5. Select Appearance.
6. Select OK or Cancel.

**Image Display
Fields**

1. Place the cursor where you want to insert the image.
2. Select image from the Insert menu and press **Set**. The Ultrasound Image Box displays.



Figure 14-28. Ultrasound Image Box

3. Type the Heading text, modify the box properties, and change the heading text font, as necessary.
To keep the monitor image appearance, the ratio of width to height (W:H) should be 4:3. So, basically 640:480 for large images and 300:225 for two side-by-side images.
4. Select OK or Cancel.

**Cardiac Studies
Wall Motion
Analysis**

1. Place the cursor where you want to insert the wall motion analysis and press **Set**.
2. Select Wall Motion Analysis from the Customize menu.

Cardiac Studies
Wall Motion
Analysis
(continued)

3. Select and set up the desired parameter.
- Bull's Eye



Figure 14-29. Bull's Eye Dialog Box

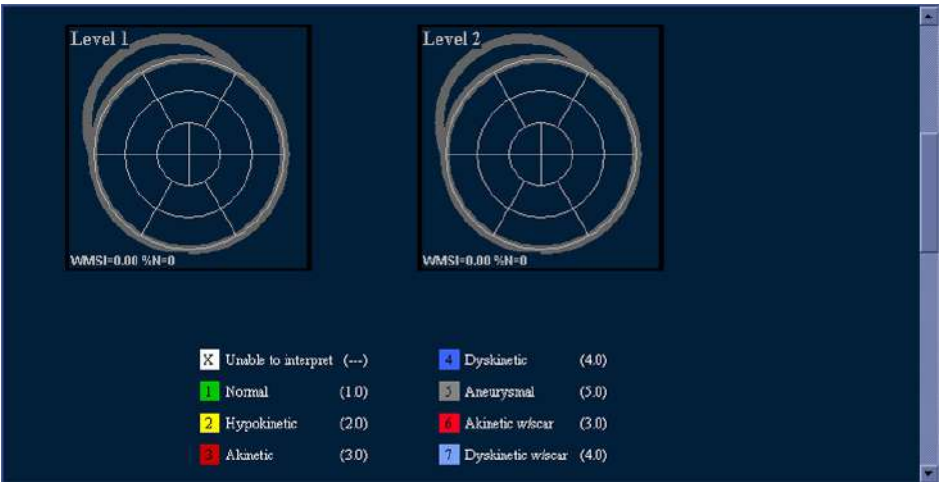


Figure 14-30. Bull's Eye Report Example

Cardiac Studies
Wall Motion
Analysis
(continued)

- Cut Planes

NOTE:

The Cut Planes dialog box parameters are similar to the Bull's Eye Dialog Box shown previously.

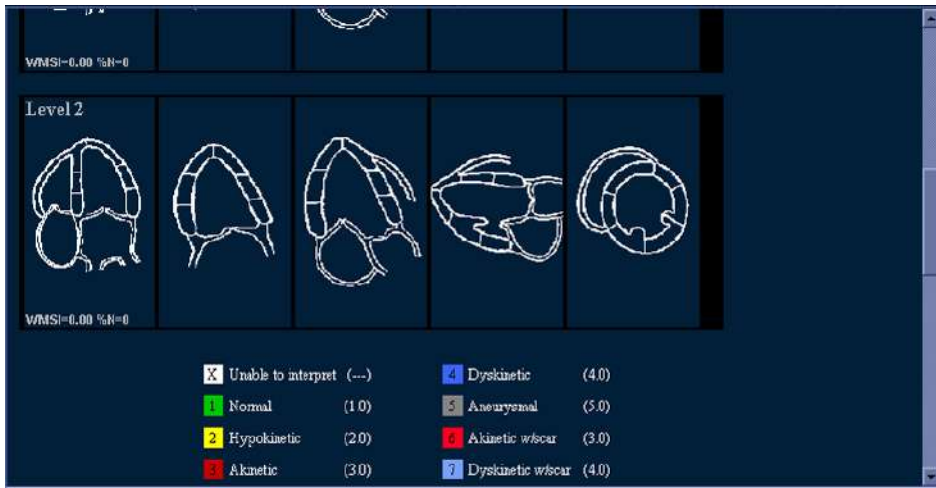


Figure 14-31. Cut Planes Report Example

- Score Table Box



Figure 14-32. Score Table Box Dialog Box

4. After you finish the setup, select OK or Cancel.

OB/GYN (OB and GYN Only)

The OB Graph can be entered into the Report.

1. Place the cursor where you want to insert the graph or anatomy and press **Set**.
2. From the Customize menu, select OB/GYN and then OB Graph. The Graph/Bar selection menu displays.
3. Select the appropriate item as necessary. A dialog box displays.



Figure 14-33. Graph/Bar Selection Menu

- OB Graph

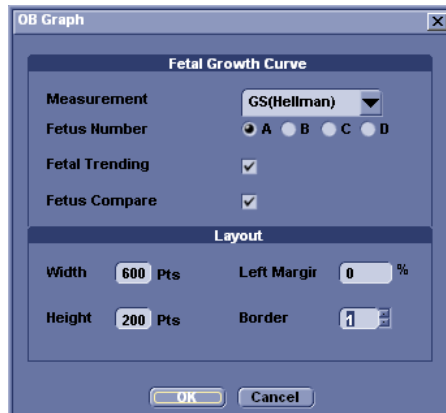


Figure 14-34. OB Graph Dialog Box

- a. Select the Measurement and Fetus Number.
- b. Check Fetus Trending and Fetus Compare, if appropriate.
- c. Modify the Layout, if necessary.
- d. Select OK.

OB/GYN (OB and
GYN Only)
(continued)

- Bar Graph

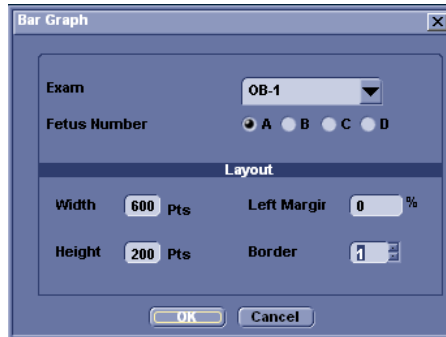


Figure 14-35. Bar Graph Dialog Box

- a. Select the exam and fetus number.
- b. Modify the Layout, if necessary.
- c. Select OK.

NOTE: *The Bar Graph already contains default application measurements.*

OB/GYN (OB and
GYN Only)
(continued)

- Anatomy

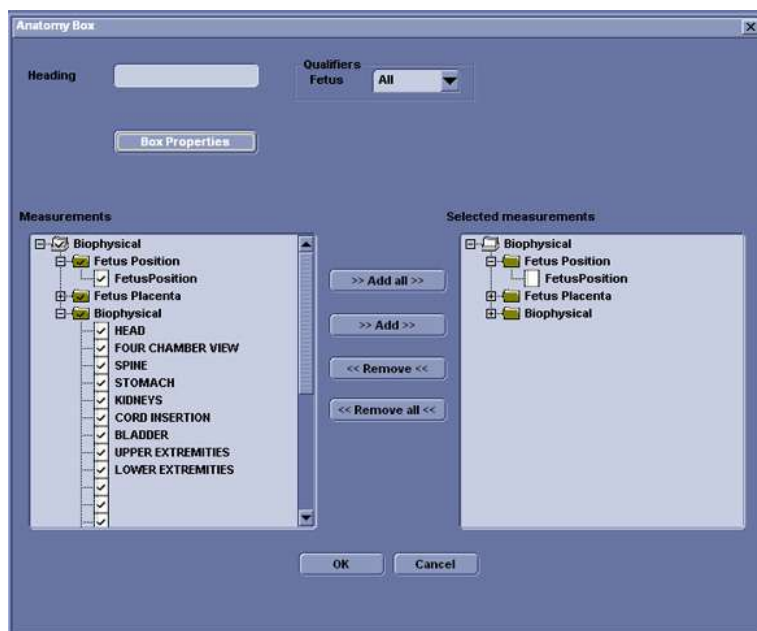


Figure 14-36. Anatomy Dialog Box

- Type the Heading.
- Select qualifiers from the pull-down menu.
- Select "Add all" to copy all measurements to the right column
- Check the box in front of the measurement you need in the left column and select "Add". The select measurements copy to the right column.
- To remove measurements you do not need, check the boxes in front of those measurements in the right column, and select "Remove" or "Remove all".
- If you want to modify the properties, select Box Properties and set required parameters.

Measurements

Insert a field to display the measurements. The measured parameters displayed in the measurement display field are configured.

1. Place the cursor where you want to insert the measurement and press **Set**.
2. Select Measurements from the Insert menu and press **Set**. The Measurements Box displays.

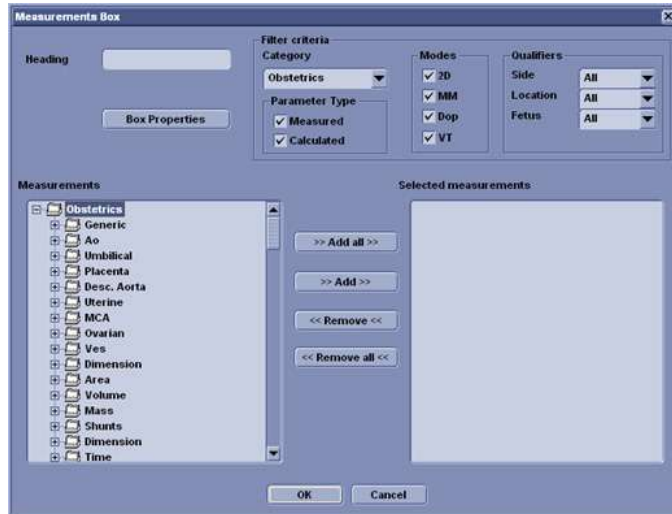


Figure 14-37. Measurement Box

3. Type the Heading text, select the Filter Criteria and measurements from the tree, as necessary.
4. Select OK or Cancel.

- Text Fields**
1. Place the cursor where you want to insert the text and press **Set**.
 2. Select Text Field from the Insert menu and press **Set**. The Text Field dialog box displays.



Figure 14-38. Text Field Dialog Box

3. Type the Heading Text, if desired.
4. Select Display item.
 - Ref.Reason: Gets information from Direct Report.
 - Comments: Gets information from the Comment field of the Patient screen and the Exam Comment field of the Worksheet.
 - Diagnosis: Gets information from Direct Report.
 - Free Text: 1 - 8
5. Specify the border of the Text Field and Font as necessary.
6. Select OK or Cancel.

The text is saved automatically into the corresponding area selected on this dialog box.

**Text Fields
(continued)**

Editing an existing text field:

1. Move the cursor to the Text Field to be edited.
2. Press **Set** twice. The Text Field dialog box displays.
3. Edit the heading, the settings, or font, as necessary.
4. Select OK or Cancel.

- Fixed Text**
1. Place the cursor where you want to insert the fixed text and press **Set**.
 2. Select Text Field from the Customize menu and press **Set**. The Fixed Text dialog box displays.



Figure 14-39. Fixed Text Dialog Box

3. Type the text (e.g. hospital information, report title, or table title) and specify the border and font.
4. Select OK or Cancel.

Editing existing Fixed Text:

1. Move the cursor to the Fixed Text to be edited.
2. Press **Set** twice. The Fixed Text dialog box displays.
3. Edit the text, the border or font, as necessary.
4. Select OK or Cancel.

Customize Menu

Table 14-6: Customize Menu

	Description
Page Color	Changes the template color.
Preference	The Preference menu for Archive Information field displays.

- Page Color
1.

To change the page color, select Page Color from the Customize Menu and press **Set**. The Color dialog box displays.

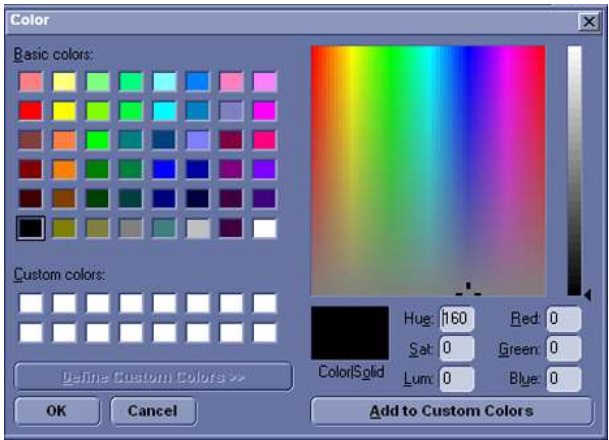


Figure 14-40. Color Dialog

2.

Choose the desired color or create a new color.
3.

Select OK or Cancel.
4.

Save the template.

Setting Preferences

To set preferences for the Archive Information:

1. Select Preferences from the Preferences menu and press **Set**. The Preference Box displays.

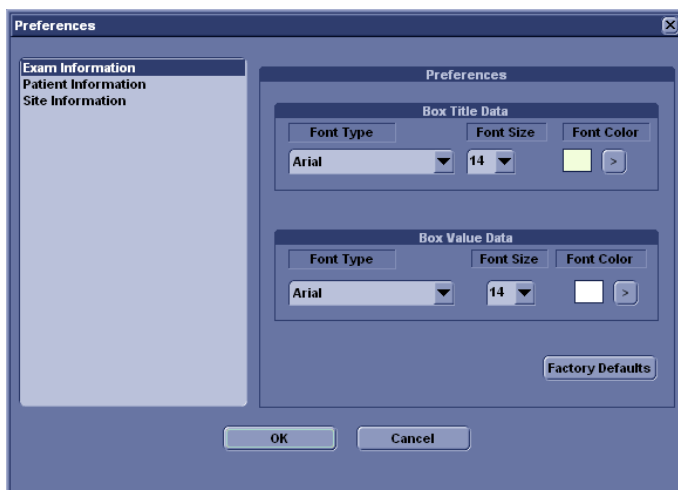


Figure 14-41. Preferences Box



HINTS

Changes are not globally applied; changes are applied AFTER you've changed the preference settings. So be sure that you update preferences BEFORE making other changes to the template.

2. Select the information to be modified and set the desired preferences.
3. Select OK or Cancel.
4. Save the template.

Direct Report

Direct Report

You can use Direct Report to enter Comments, Diagnosis, and Referral Reasons at any time during the examination that will be part of the final report. The comments are reflected on the Report if the Report is configured for those parameters.

1. Select **Direct Rep.** on the measurement Summary window. The Direct Report displays on the left side of the monitor display.

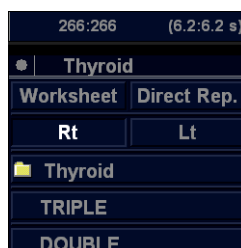


Figure 14-42. Measurement Summary Window

OR

Place the cursor on Menu at the bottom, right-hand corner of the screen. Press **Set**. The System Menu displays.

2. Select Direct Report and press **Set**.

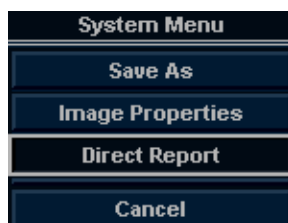


Figure 14-43. System Menu

Direct Report (continued)

3. The Direct Report displays on the left side of the screen.



Figure 14-44. Direct Report

NOTE:

- a. Select the type of information
*Comments entered under Diagnosis appear in the Clinical Diagnosis section of the final report.
 Comments entered under Referral Reasons appear in the Referral Diagnosis section of the final report.*
- b. Create/insert pre-defined text
- c. Text field
- d. List of measurements completed
 The measurement results display on the Measurement Overview field.
- e. Exits the Direct report

Direct Report (continued)

4. Select the appropriate parameter and type the free text with the alphanumeric keyboard or use Insert Text.

NOTE: You can configure the pre-defined text at the Utility Report screen.

- a. Select **Insert Text** to display the Insert Text Window.
- b. Use the **Trackball** to select the text to be inserted.
- c. Press **Set**. The selected text displays on the Direct Report.



Figure 14-45. Full Insert Text Window

- New: Enters the new text
 - Edit: Edits the existing text
 - Delete: Deletes the existing text
 - More: Displays the Full Insert Text Window
 - Close: Closes the insert text window
 - <<Less: Minimizes the insert text window
 - Move up/Move down: Moves the text up or down
5. Move the cursor over the measurement result displayed in the overview window, and double click the **Set** key.

Direct Report (continued)

6. Select **Done** at the bottom of Direct Report to exit.

If you configure the field of comment, diagnosis, referral reasons or Measurement on the Report, the text and/or measurement results entered in the Direct Report are automatically displayed on the Report.



Figure 14-46. Direct Report and Report (Example)

Report Presets

Utility Report Page

You can edit the report template, diagnosis code, and text on the Utility Report page.

Templates

Left Column: The list of all templates (Factory Default, User defined, etc)

Right Column: The list of templates displayed on the template list.

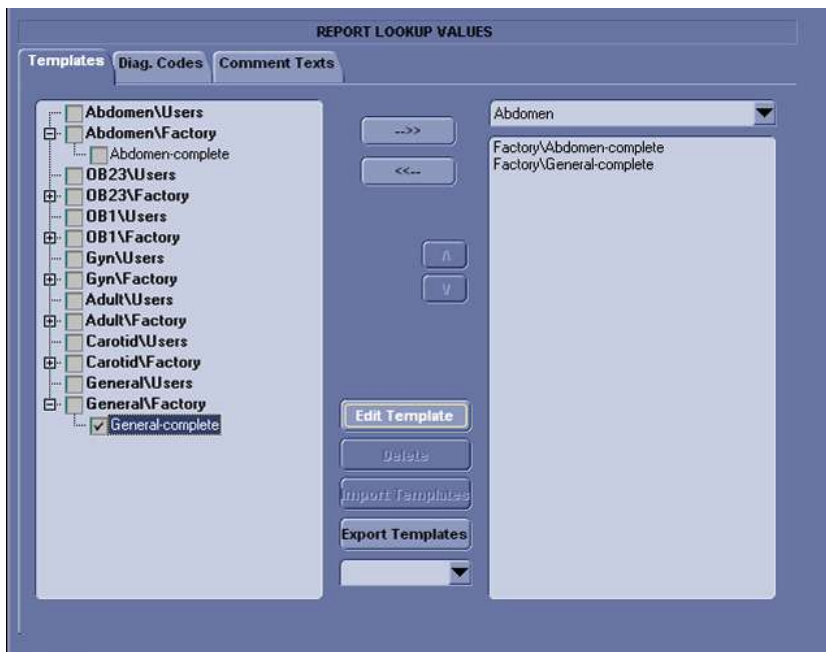


Figure 14-47. Report Template Tab

Templates (continued)

- To insert the template on the template list:
 - a. Select the application which you want to insert into the template from the pull-down menu above the right column.
 - b. Select the category (categories) and/or the template(s) in the left column by the check the box.
 - c. Select the right arrow to copy the template to the right column.
- To remove the template from the template list but not from the system):
 - a. Select the template in the right column.
 - b. Use the left arrow to remove the template from the right column.
- To edit the template or to make a new template:
 - a. Enter Utility -> Report -> Template tab.
 - b. Select the appropriate template in the left column.
 - c. Select **Edit Template**. The Template Designer page displays.
 - d. Edit the template and save or save as with a new name.

If you use Save As with a new name, the new template is added to the left column. See 'Designing Your Own Template' on *page 14-26 for more information*.
- To delete the template:
 - a. Select the template to be deleted.
 - b. Press **Delete**.

Templates (continued)

- To export the template:



HINTS

Export templates to removable media (CDs, DVDs) so, at a later time, you can import those templates to a different system or a system with a different software version. Export only works on templates, not data.

- Insert the removable media in the drive.
- Move the cursor to “Export Templates” and press **Set**. The available user-defined templates display in the Export Templates window.

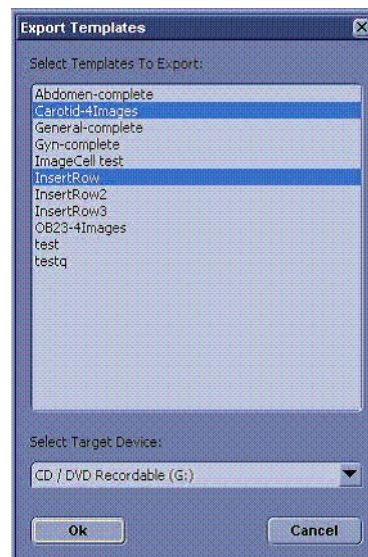


Figure 14-48. Export Templates

- NOTE:**
- Select the template(s) to be exported.
*To select multiple templates, use the **Ctrl** or **Shift** keys.*
 - Select the desired removable media under the Select Target Device field.
 - Select **OK**.
 - Press **F3** to eject the media.

Templates (continued)



HINTS

- To import the template:

Import templates from a different system or a system with a different software version. Import only works on templates, not data.

- a. Insert the removable media with the report template(s) to be imported.
- b. Select **Import Template**. The Import Template window displays.

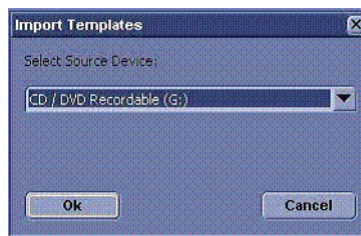


Figure 14-49. Import Templates

- c. Select the Source Device from the pull-down menu. Select **OK**.
- d. Press **F3** to eject the media.

NOTE:

Imported templates are stored in the User defined templates\General directory.

- To move the template from the left column to the right, or from the right to the left:
 - a. Select the template to be moved.
 - b. Select the Right Arrow or Left Arrow button.
- To move the template up or down in the right column:
 - a. Select the template to be moved.
 - b. Press the Up Arrow or Down Arrow button.

Diagnosis Code

You can edit the diagnosis code on the Diag. Codes Tab.

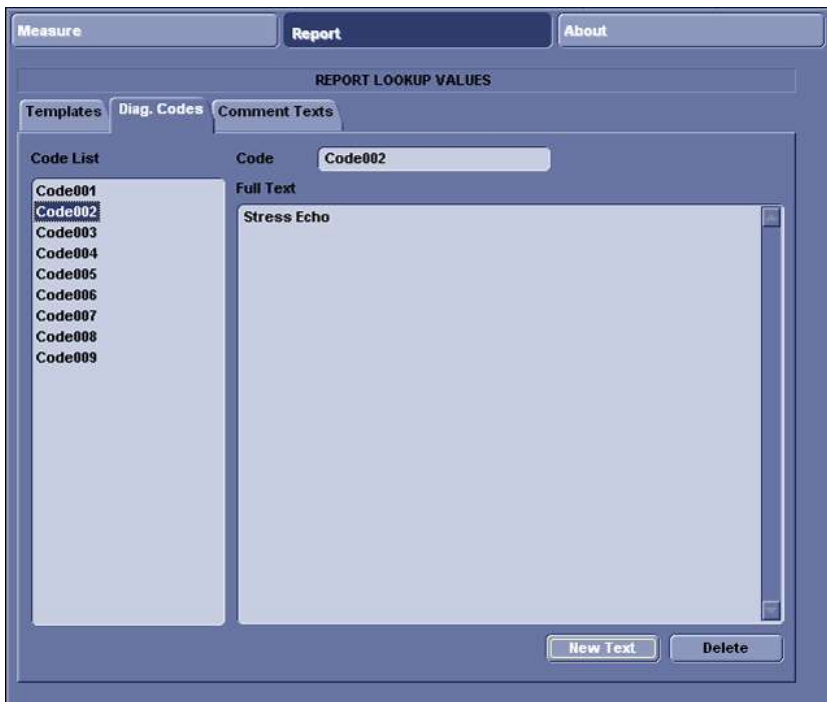


Figure 14-50. Diag. Codes Tab

- To insert the new code:
 - a. Select **New Text**. The code number is added to the Code List and a “?” is displayed in the Full Text field.
 - b. Move the cursor over the “?” and then double-click Set key.
 - c. With the “?” highlighted, enter the text manually.
 - d. Repeat steps a.-c. to create another Diagnostic Code.
- To delete the existing code:
 - a. Select the existing code that you want to delete.
 - b. Select **Delete**.

Comment Texts

You can edit the comment text on the Comment Texts tab.

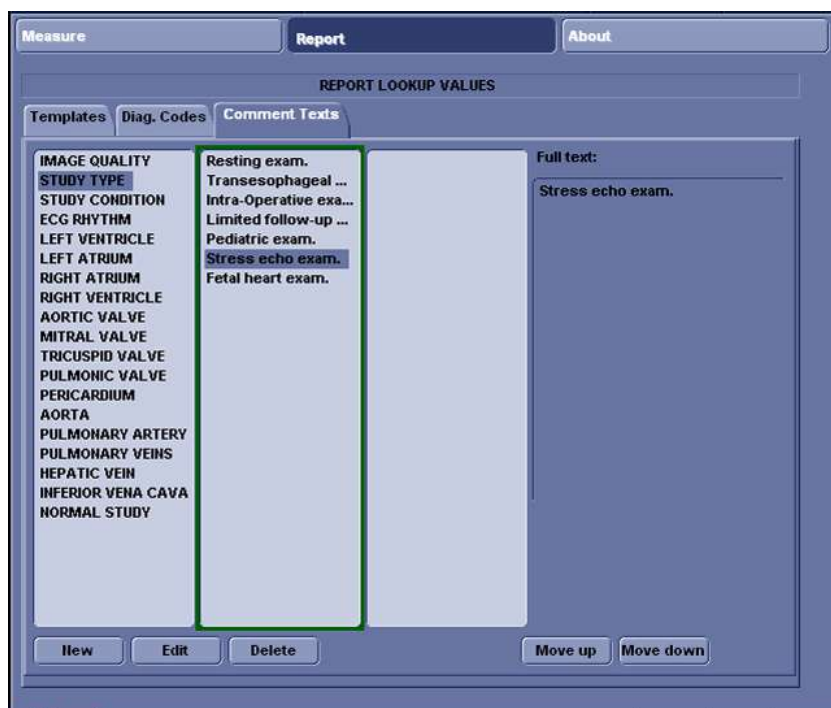


Figure 14-51. Comment Texts Tab

- New: Enters the new comment.
- Edit: Edits the existing comment.
- Delete: Deletes the existing comment.
- Move up/Move down: Moves the comment up or down.

Backup/Restore Report Templates

Backup moves templates to removable media (CDs, DVDs).

Restore moves templates, that were backed up onto media, to a similar system or a system with the same software version. For example, it allows templates to move from one LOGIQ 9 to another LOGIQ 9, or from one system with R6.x.x software to another R6.x.x.

NOTE: *To move reports with patient data, use Backup/Restore Report Archive and Patient Archive.*

To backup the report template:

1. Select **Utility**.
2. Select **System** and select the Backup/Restore tab.
3. Select the media.
4. Check User Defined configuration box of the Backup field.
5. Select **Backup**.
6. Select **Save** and eject the media.

To restore the report template:

1. Insert the media.
2. Select **Utility**.
3. Select **System** and select Backup/Restore tab.
4. Check Report Template box of the Detailed Restore of User Defined.
5. Select **Restore**.
6. After the system reboot, select **Utility** and **Report**.
7. Select the Template tab.
8. Select the appropriate template (See 'Templates' on page 14-57 for more information.)

For preset specifics, see Chapter 16.

Chapter 15

Recording Images

Describes how to record images.

Getting Set Up to Record Images

Overview

A typical workflow for connectivity might be as follows (this setup varies by each user setup):

- Select the dataflow, worklist for example.
- Start a new exam. Select the patient.
- Perform the patient scan.
- Store images as multi-frame CINE Loops and Raw DICOM data via the P1 button.
- Store secondary capture for DICOM print via P2.
- Store images to the color printer via P3.
- Store images to the black and white printer via P4.
- Check the DICOM Job Spooler via F4 to verify delivery.
- End the exam.
- Permanently store images via the Patient menu.



WARNING

Imaging functions may be lost without warning. Develop emergency procedures to prepare for such an occurrence.

Overview (continued)

Dataflow combines archive, data, DICOM, and onboard records into one coherent workflow. Destination devices are configured and assigned to the print buttons. You select the appropriate dataflow (Portable, etc.) according to your requirements. You manage the patient database (local, shared, or via a worklist broker).

- DO NOT use the internal hard drive for long-term image storage. Daily backup is recommended. External storage media is recommended for image archive.
- NOTE: DICOM images are stored to external media storage devices separately from patient data, which also needs to be backed up to a dedicated database-formatted external storage media.*
- You need to set up a protocol for locating images stored to external storage media for easy recall
 - GE Medical Systems IS NOT responsible for lost data if you do not follow suggested back-up procedures. GE Medical systems WILL NOT aid in the recovery of lost data.

Refer to the Customizing Your System chapter for instructions on setting up your system's connectivity.

Adding Devices

To add a destination device (printer, worklist server, etc.) to this system, see Chapter 16.

To verify a DICOM device, see Chapter 16.

Adding a Dataflow

To add a new dataflow to this system, see Chapter 16.

Adding Devices to a Print Button

To add devices/dataflows to a print button, see Chapter 16.

Formatting Removable Media

To format removable media, see Chapter 16.

Using the DICOM Spooler

To monitor/control DICOM jobs, press F4. You can view, resend, redirect, and delete images from the DICOM spooler by selecting a job, then specifying the action to be performed on this job.

Table 15-1: Spooler status description

Status	Description
Hold	Waiting for user activity. Select Resend or Send-To to complete the job.
Pending	Waiting for the previous job(s) to finish (a previous job may be Active or Pending). No user interaction required.
Append	Not completed. Example 1: Direct Storing job. Waiting for more images or the end of the exam (by selecting New Patient or End Current Patient). Example 2: Print job with 3x3 images has only 8 images. Waiting for one more image or the end of the exam (by selecting New Patient or End Current Patient).
Active	Signifies network activity (or connection attempt).
Success	Sent successfully.
Failed	Unsuccessful job attempt. Job stays in spooler. Select Retry or Delete to complete the job.
Done	Finished successfully.

Troubleshooting

To troubleshoot connectivity on this system, see Chapter 16.

To verify that a service is connected to the network, see Chapter 16.

Image/Data Management

Reviewing Patient Images

Please refer to Chapter 4 for more information on the following:

- Retrieving and editing archived information
 - Searching for a patient
 - Reviewing a patient exam
 - Reviewing an image
 - Deleting a patient, exam, or image

Clipboard

The clipboard displays thumbnail images of the acquired data for the current exam. Images from other exams are not displayed on the current patient's clipboard.

The active image is stored and placed on the clipboard when you press the print key (this assumes that you have already set up a print key to do this). The clipboard contains preview images with enough resolution to clearly indicate the contents of the image. If an image has a triangle indicator on the lower, left-hand corner, then something has been done to the image (printed, sent to DICOM, etc.). Single images are indicated by a '1'; CINE Loops have no special identifier.

The clipboard fills from left to right, starting in the left-hand corner. Once the top row is full, the second row starts to fill. Once both rows are full, the next image stored starts to fill a 'third' row (the first row disappears from the clipboard display, with the second row now becoming the first row, and the third row now becoming the second row).

Clipboard (continued)

All of the images can be viewed in the Active Images screen or in the Analyze screen, available from the display or from the New Patient menu.



Figure 15-1. Clipboard

Previewing Clipboard Images

1. Select the unmarked key (opposite the **Set** key) to obtain a cursor arrow.
2. Move the **Trackball** to position the pointer over the clipboard image you want to recall.
3. An enlarged preview of the image is displayed on the left-hand side of the monitor.

Recalling Images from the Clipboard

To recall images from the clipboard,

1. Select the unmarked key (opposite the **Set** key) to obtain a cursor arrow.
2. Move the **Trackball** to position the pointer over the clipboard image you want to recall.
3. Press **Set** to recall the image.

Storing an Image

To store an image,

1. While scanning, press Freeze.
2. Scroll through the CINE Loop and select the desired image.
3. Press the appropriate Print key.

The selected image is stored (per your preset instructions) and a thumbnail is displayed on the clipboard.

Storing a CINE Loop

A CINE Loop is a sequence of images recorded over a certain timeframe. The stored CINE Loops are displayed chronologically on the clipboard.

CINE Loops can be stored at any time during scanning. You can choose to preview the CINE Loop before storage and save the CINE Loop directly, as described below.

Previewing and Storing a CINE Loop

1. While scanning, press Freeze.
2. Determine the best CINE Loop to store.
3. Play the CINE Loop to review it.
4. Press the appropriate Print key.

Depending on whether the system has been configured to enable or disable “Preview Loop before store” (see Chapter 16), the following procedures enable the CINE Loop to be stored directly.

Storing a CINE Loop Without Preview

If “Preview Loop before store” is disabled,

1. While scanning, press the appropriate Print key.
2. The last valid CINE Loop is stored in the archive and a thumbnail is displayed on the clipboard.

Storing a CINE Loop With Preview

If “Preview Loop before store” is enabled,

1. While scanning, press the appropriate Print key.
2. The last valid CINE Loop is previewed.
3. Adjust the CINE Loop, as necessary.
4. Press the appropriate Print key.

The thumbnail is displayed on the clipboard.

Using the Monitor Display Controls to Manage Images

You can also manage images from the display. In the lower, right-hand corner of the monitor display, there are four menu icons: Active Images (1), Delete (2), Previous/Next Clipboard Image (3), and Menu (4).



Figure 15-2. Menu Icons

Active Images Page

Press Active Images Page to go to the Patient Active Images page.

Analyzing Images

To review archived images, select the image, then select Review. The archived images is displayed with the date and time of archival.

To compare the analyzed image to a live image, press L / R. Now both the archived and live images appear on the monitor display. Unfreeze the live image area.

Viewing Two Different Studies from the Same Patient

To view images side-by-side from two different studies on the same patient,

1. Select the patient.
2. Go to the Image History page.
3. Select the first image.
4. Select the next image from the other exam.
5. Press **Review**

Using the Monitor Display Controls to Manage Images (continued)

Delete (Trash Can)

You can use this to delete an image from the clipboard.

1. Place the cursor on the clipboard image you want to delete, then press **Set** to select the image.
2. Place the cursor on the Delete icon and press **Set**.

Previous/Next Clipboard Image

Press the left arrow to move to the previous image; press the right arrow to move to the next image.

Menu

Place the cursor on Menu and press Set. Select from the following:

- Save As
- Image Properties
- Direct Report
- Cancel

Save As: Save the image as a RawDicom (.dcm), Dicom (.dcm), Avi (.avi), Jpeg (.jpg), or SR format. To save an image as a specific format,

1. Select the media.
2. Select the save as format (.dcm, .avi, .jpg, or SR).
3. Name the image.

NOTE: *Image only saves the image screen; secondary capture saves the entire screen.*

Image Management Guide

Save As to View on any PC

Use this to save images in a computer-friendly format so you can view it on any PC.

EZBackup/Move Images to Archive

Use this to take images off your Ultrasound system onto removable media for long-term archive. This is the way to free up hard disk space, rather than deleting images.

Export/Import Data/Images Between Systems

Use this to copy both patient data and images for specified patient(s) from one system to another.

Media Requirements

Only use CD-R CD-ROMs and DVD-R DVDs, DO NOT use CD-RW CD-ROMs or DVD-RAM or DVD+R or mini discs. Also, the CD-R CD-ROM MUST support 16x writing speed or greater.

Media Handling Tips

To eject the media, always press **F3**. **DO NOT** press the eject button on the drive.

Always finalize the media. This allows you to read them on any other PC.

DO NOT finalize the media if you want to add more images.

Save As (Saving Images to the media to View on a Windows PC)

To save images to the media:

NOTE: *ALWAYS format the media PRIOR to saving data/reports.*

1. Format the media: Insert the media. Go to Utility --> Connectivity --> Removable Media. Select the Media. Label the media. Press Format to format the media.
2. On the scan screen, press the left Set key. The arrow cursor displays.
3. Place the cursor on the image or CINE Loop in the clipboard to be saved and press Set. The image displays on the screen.
4. Select Menu in the lower, right-hand corner of the screen. The system menu displays.

NOTE: *If you save the image as an .avi file, run the CINE Loop before you select Menu.*

5. Select Save As. The SAVE AS menu appears.

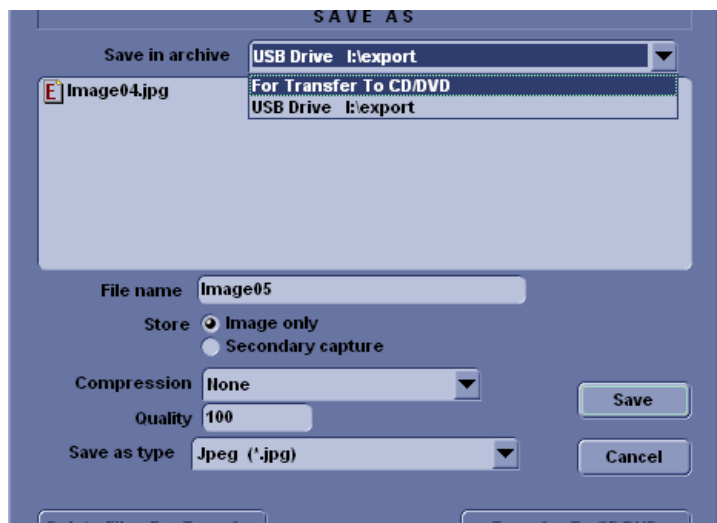


Figure 15-3. Save As Menu



HINTS

DO NOT use the following special characters when saving images: slash (/), dash (-), or asterisk (*).

Save As (Saving Images to the media to View on a Windows PC) (continued)

6. Save in archive: Select the media.

7. File name: Type the file name.

NOTE: *If you select to save as DICOM or RawDICOM, the file name is automatically generated to follow the DICOM standard.*

8. Store: Select Image only or Secondary capture.

- Image only: Saves only the ultrasound image area
- Secondary capture: Saves the ultrasound image area, title bar, and scan information area. Not available for DICOM or RawDICOM images.

9. Compression: Specify Compression.

10. Quality: Specify image quality (between 10-100). A high quality setting gives a lower compression.

11. Save as type: Select one of the following.

- RawDICOM: saves the still image or CINE Loop in both GE raw format and DICOM format.
- DICOM: saves the still image or CINE Loop in pure DICOM format.
- AVI: Saves the CINE Loop in avi format.
- JPEG: Saves the still image in jpeg format.

12. Press Save.

13. When you have put all the images you want on the media, press **F3** to eject the media.

Save As (Saving Images to the media to View on a Windows PC) (continued)

14. Specify whether you want to finalize the media. Finalize the media by selecting Yes. If you want to add more images to the media, select No. The media is then ejected from the system.

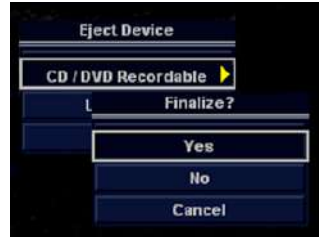


Figure 15-4. Finalize CD-ROM Menu

NOTE: **Save As: To CD/DVD** - When saving an image(s) from the clipboard to CD/DVD, select 'For Transfer to CD/DVD'. When you select 'Save,' the image(s) are moved to a temporary directory. You can then select to either clear the temporary directory ('Delete Files for Transfer') or to transfer the contents of this temporary directory to CD/DVD ('Transfer to CD/DVD').

Storing Images with More Resolution

To store images with more resolution than is available with the JPEG selection, select Save As and select AVI as the Save As Type. You can save single images as .avi files.

Table 15-2: Store Options

Image Type	Store as Image Only	Store as Secondary Capture
CINE Loop	Gives you a loop of just the image (no title bar and scan information).	Gives you a single image of the video area. DO NOT DO THIS BECAUSE YOU DO NOT KNOW WHICH IMAGE FROM THE LOOP THAT YOU ARE GETTING.
Still Image	Gives you a single image (no title bar and scan information).	Gives you a single image of the video area.

USB Drives



WARNING

DO NOT use the USB Hard Disk Drive for patient storage. The USB HDD is not considered a permanent storage device.

If you connect the USB HDD to a virus-infected computer, the USB HDD may also be infected with a computer virus.



CAUTION

Before removing the USB drive from the USB port, press Eject (F3) and select USB Drive from the pull-down menu.

Disconnect the USB drive after the success dialogue is displayed. If the unsuccessful dialogue is displayed, retry after a while.

Connect only one USB drive (USB HDD/USB flash drive) to the system in total. DO NOT connect two or more devices at the same time.



CAUTION

When exporting a large quantity of patient data to the USB-HDD, the system may crash. If this occurs, the export is not completed properly. First, delete the data remaining in the USB-HDD, and then try again with a smaller number of patients.



CAUTION

DO NOT use "Select All" when you export the patient data to the USB-HDD.

NOTE: *Do not create multiple partitions. Use USB Drive with single partition.*

NOTE: *Connect the USB drive to the front USB port located near the CD/DVD drive. This drive supports USB2.0. The USB port on the rear panel supports USB1.1 and the transfer rate is much slower than USB2.0.*

NOTE: *USB1.1 port is not adequate to transfer large volume of data.*

USB Flash Drive

The USB Flash Drive is an ESD-sensitive device. GE only supports the NDL 256MByte 066E0690 USB Flash Drive.



Before removing the USB Flash Drive from the USB port, press Eject (F3) and select USB Drive from the pull-down menu to disconnect the USB Flash Drive.

NOTE: *Before using the USB flash drive, format it in Utility -> Connectivity -> Removable media.*

Formatting the USB Flash Drive

To format the USB Flash Drive

1. Insert the USB Flash Drive into the front USB port located near the CD/DVD drive.
2. Select Utility--> Connectivity--> Removable Media.
3. Select USB Drive from the Removable Media pull-down menu.
4. Type the USB Flash Drive label.
5. Press Format.

USB Flash Drive (continued)

MPEGvue (Data Transfer)

To transfer a patient/exam to the USB Flash Drive,

1. Insert the USB Flash Drive into the front USB port located near the CD/DVD drive.
2. On the Patient menu, select Data Transfer, then MPEGvue. Specify USB Drive in the transfer To: pull-down menu. Select the patient/exam you want to transfer. Press Transfer.
3. When the transfer has been completed, press **Eject (F3)**. Select USB Drive from the pull-down menu to disconnect the USB Flash Drive.
4. Remove the USB Flash Drive from the USB port.



CAUTION

DO NOT use MPEGvue for diagnostic purposes. MPEGvue only transfers exams with images; exams without images are not transferred.

**Save As (Saving
Images to the USB
Flash Drive)**

To save images to the USB Flash Drive,

1. Insert the USB Flash Drive into the front USB port located near the CD/DVD drive.
2. Select the image(s) to be saved.
3. Press Menu in the lower, right-hand corner of the screen. Select Save As. Select the USB Drive as the archive media.
4. Specify: Image only or Secondary Capture, type of compression, quality, and image save format (Raw Dicom, Dicom, Avi, Jpeg, or SR).
5. Press Save. When the images have been saved, press **Eject (F3)**. Select USB Drive from the pull-down menu to disconnect the USB Flash Drive.
6. Remove the USB Flash Drive from the USB port.

Backup/Restore

To Backup/Restore to/from a USB Flash Drive,

1. Insert the USB Flash Drive into the front USB port located near the CD/DVD drive.
2. Press Utility--> Backup/Restore. Select the USB Drive as the media.
3. Follow instructions for Backup/Restore in Chapter 16.
4. When the backup/restore have been completed, press **Eject (F3)**. Select USB Drive from the pull-down menu to disconnect the USB Flash Drive.
5. Remove the USB Flash Drive from the USB port.

USB Hard Disk Drive

MPEGvue (Data Transfer)

To transfer a patient/exam to the USB Hard Disk,

1. Insert the USB cable into the front USB port located near the CD/DVD drive.
2. On the Patient menu, select Data Transfer, then MPEGvue. Specify USB Drive in the transfer To: pull-down menu. Select the patient/exam you want to transfer. Select Transfer.
3. When MPEGvue has completed, press **F3**. Select USB Drive from the pull-down menu.
4. Remove the USB HDD from the USB port after the success dialogue displays.

Save As (Saving Images to the USB Hard Disk)

To save images to the USB Hard Disk,

1. Insert the USB cable into the front USB port located near the CD/DVD drive.
2. Select the image(s) to be saved.
3. Press Menu in the lower, right-hand corner of the screen. Select Save As. Select the USB Drive as the archive media.
4. Specify: Image only or Secondary Capture, type of compression, quality, and image save format (Raw Dicom, Dicom, Avi, Jpeg, or SR). Press Save.
5. When SaveAs has completed, press F3. Select USB Drive from the pull-down.
6. Remove the USB HDD from the USB port after the success dialogue displays.

- Backup/Restore** To Backup/Restore to/from a USB Hard Disk,
1. Insert the USB cable into the front USB port located near the CD/DVD drive.
 2. Press Utility--> Backup/Restore. Select the USB Drive as the media.
 3. Follow instructions for Backup/Restore in Chapter 16.
 4. When Backup/Restore has completed, press **F3**. Select USB Drive from the pull-down menu.
 5. Remove the USB HDD from the USB port after the success dialogue displays.
- Export/Import** To export/import exams using the USB Hard Disk,
1. Insert the USB cable into the front USB port located near the CD/DVD drive.
 2. On the Patient menu, select Data Transfer, then Export/Import. Specify USB Drive in the transfer To: pull-down menu. Select the patient/exam you want to transfer. Press Transfer.
 3. When Export/Import has completed, press **F3**. Select USB Drive from the pull-down menu.
 4. Remove the USB HDD from the USB port after the success dialogue displays.

EZBackup/EZMove

Please refer to Chapter 16 for instructions on how to configure and use EZBackup/EZMove.

Data Transfer

The user can select and access the 5 Exam Transfer services from the Exam Data Transfer screen.

- Import
- Export
- Worklist
- Q/R (Query/Retrieve)
- MPEGvue

GE Medical Systems

Patient ID: Last Name: First Name:

Search key: Exam Description string: Clear

Patient ID	Last Name	First Name	Last Exam	Copi...	Status Msg	Img. size

Task: ☒ Import ☐ Export ☐ Worklist ☐ Q/R ☐ MPEGvue

From: No Archive

To: Local Archive - Int ID

Exit

Search key: Patient ID string: Clear

Patient ID	Last Name	First Name	Birthdate	Sex	Last Exam	Img. size
061803-75148-PM				N	06/18/2003	45.7 MB
q				N	09/26/2003	None
111	1	1		N	07/09/2003	1.84 MB
230802-1303-00			01/25/1965	F	03/26/2004	None
1	DOG	1		N	11/25/2002	785 MB
2	DOG	1 SONOVUE		N	11/21/2002	645 MB
3	DOG	2 HCC		N	11/21/2002	838 MB
4	DOG	MAY		N	11/21/2002	593 MB

Show: ☒ All ☐ 19 / 19 Hide

04/15/04 5:07:51PM Please select 'From' Source.

Figure 15-5. Exam Data Transfer Screen

Data Transfer (continued)

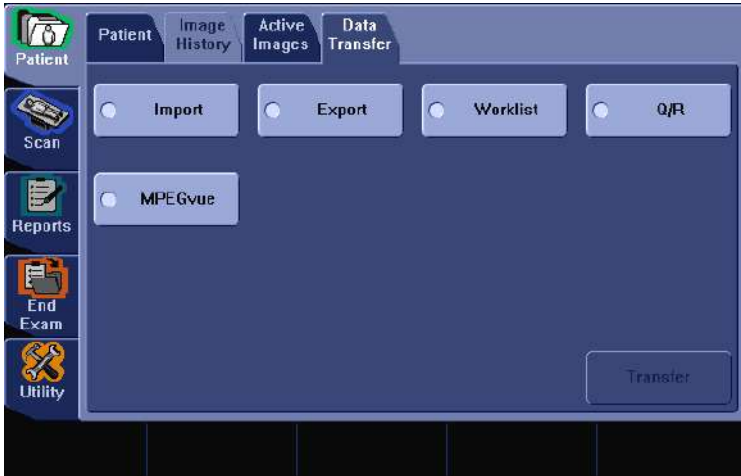


Figure 15-6. Exam Data Transfer Touch Panel

Export/Import (Moving Data Between Systems or Back Up/Retrieve data)

To move exams from one Ultrasound system to another system or to back up/retrieve exam information, you need to export/import exam information.

NOTE: *Both database information and images are exported. No data is deleted from the local archive when exporting data.*

NOTE: *Export/Import patient records may take more than ten (10) minutes. Please allow sufficient time to export/import patients.*

NOTE: *If you use a DVD for Export/Import, select Removable CD Archive.*

NOTE: *You MUST verify the media you use BEFORE performing Export/Import. You must do this once each session. If you encounter problems, eject the media and then re-insert the media; then try the Export/Import again.*

NOTE: *You cannot export locked images. If an image that you want to export appears locked, reselect the patient, then export the exam OR select "Unlock All" at Utility -> Admin -> Logon.*

Exporting Data

To export an exam(s) to a compatible Ultrasound system:

1. Format and label the removable media. Answer Yes/OK to the messages.
2. Press **Patient** and select **Data Transfer**.
3. The Data Transfer screen is displayed. Select Export.
4. "Local Archive-Int.HD" displays on the Transfer From pull-down menu and the patient list included in the Local Archive displays.

Exporting Data (continued)

5. Select the patient(s) to export by using the Transfer From search field (the upper field).

You can use Windows commands to select more than one patient.

To select a consecutive list of patients, click the cursor on the first name, move the cursor to the last name, then press and hold down the Shift+right Set key to select all the names.

To select a non-consecutive list of patients, click the cursor at the first name, move the cursor to the next name, then press and hold down the Ctrl+right Set key, move the cursor to the next name, then press and hold down the Ctrl+right Set key again, etc.

You can also search for patients via the Search key and string.

Or, use Select All to select all patient.

NOTE: *You need to use your best judgment when moving patients' images. If there are lots of images or loops, then only move a few patients at a time.*

6. Select the destination at the Transfer To pull-down menu.
7. Press **Transfer**. The progress bar displays during the transfer.
8. Press **F3** to eject the media. Specify that you want to finalize the media.

NOTE: *To display exported DICOM or Raw DICOM images on a PC, you need the dedicated viewer.*

Importing Data

To import an exam(s) to another Ultrasound system:

1. At the other Ultrasound system, insert the media.
2. Press **Patient** and select **Data Transfer**.
3. The Data Transfer screen displays. Press Import.
4. Select the media from the Transfer From pull-down menu.
5. The Transfer From search field shows the patients available for import from the removable media you just loaded onto the system.
6. Select the patient(s) or the exam(s) from the list to be imported.
7. Press Transfer. The progress bar displays during the transfer.
8. Please wait for the patient information to be copied to this Ultrasound system. Informational messages appear while the import is taking place.
9. Press **F3** to eject the media.

NOTE: Use Import to restore EZBacked up and/or EZ Moved images.

NOTE: You can retrieve from the media to the Local HDD, playback, or process exam information on the system as Raw Data.

Query/Retrieve (Search and retrieve the data from DICOM device)

NOTE: *For Query/Retrieve to find a patient, the patient **MUST** have a Patient ID.*

- Query**
1. Press **Patient** and select **Data Transfer**. The Data Transfer screen displays.
 2. Select Q/R. The patient/exam list in the Local Archive displays in the Transfer To section.

NOTE: *Only "Local Archive - Int.HD" is enabled for Transfer To.*

3. Select the Query/Retrieve server from the Transfer From pull-down menu.

NOTE: *The server is configured in the Utility screen. Multiple servers are able to be configured.*

4. Press Query in the Transfer From section. The Query is performed.
5. The server's patient list displays.

NOTE: *Press Query again to refresh the list.*

- Retrieve**
1. Select the patient(s) or the exam(s) to be retrieved from the patient list.
 2. Press Transfer. Retrieve the data from the Query/Retrieve Server. The progress bar displays during the transfer.

Worklist (Search and retrieve the Patient/Exam information)

1. Press **Patient** and select **Data Transfer**. The Data Transfer screen displays.
2. Select Worklist. The patient/exam list in the Local Archive displays in the Transfer To section.

NOTE: Only "Local Archive - Int.HD" is enabled for Transfer To.

3. The Worklist used last time is displayed on the monitor display. Press Refresh to refresh the list or select another Worklist server from the Transfer From pull-down menu.

NOTE: The worklist server is configured in the Utility screen. Multiple servers are able to be configured.

NOTE: You can configure whether the auto-refresh worklist has been enabled/disabled in the Utility screen. The system automatically refreshes the list when the exam data transfer accesses the Worklist server or changes the Worklist server.

4. Select the patient(s) or the exam(s) from the list.
5. Press Transfer. The progress bar displays during the transfer.

MPEGvue (Save the data to CD-R to see on the PC)

You can see the exam data on the PC by using MPEGvue.



DO NOT transfer more than 50 patients at one time.

1. Format the removable media. Label the removable media. Answer Yes/OK to the messages.
2. Press **Patient** and select **Data Transfer**. The Data Transfer screen displays.
3. Select MPEGvue. The Patient list which has image in the Local Archive is displayed in the Transfer From section.

NOTE: *Only "Local Archive - Int.HD" is enabled for the Transfer From.*

4. Select CD-R from the Transfer To pull-down menu.
5. Select the patient(s) or the exam(s) from the list.

NOTE: *If you press Clear in the Transfer From and Transfer To section, all the search criteria clears and the list is refreshed accordingly.*

6. Press Transfer. The progress bar displays during the transfer.



DO NOT use MPEGvue for diagnostic purposes. MPEGvue only transfers exams with images; exams without images are not transferred.

MPEGvue (Save the data to CD-R to see on the PC) (continued)

On your PC An MPEG exam can be read from any computer with Windows 98/2000/XP, provided that DirectX 8.1 or later and Windows Media Player 7.1 or later are installed.

If a warning message displays, press **OK** to continue.

NOTE: *Before you read the media on the PC, finalize the media on the LOGIQ 9.*

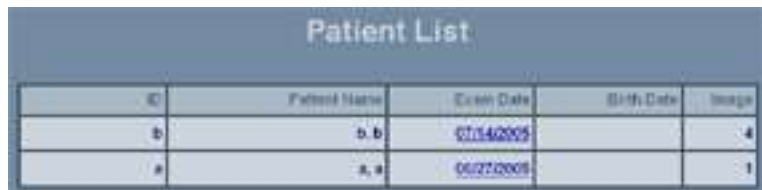
NOTE: *Measurement graphics from the exam performed on the system are maintained with the MPEG exam.*

1. To read MPEG exams stored on a CD:

- Insert the MPEGVue CD in the computer CD drive. The MPEGVue Patient list is displayed.

To read MPEGVue exams stored on other media:

- Insert the media containing the MPEG exams and double-click on the file Start_MPEGvue.bat. The MPEGVue Patient list is displayed.



ID	Patient Name	Exam Date	Birth Date	Image
b	b, b	07/14/2005		4
a	a, a	06/27/2005		1

Figure 15-7. Patient list

2. Select the desired examination date to display the images. The MPEGVue screen is displayed.

On your PC
(continued)

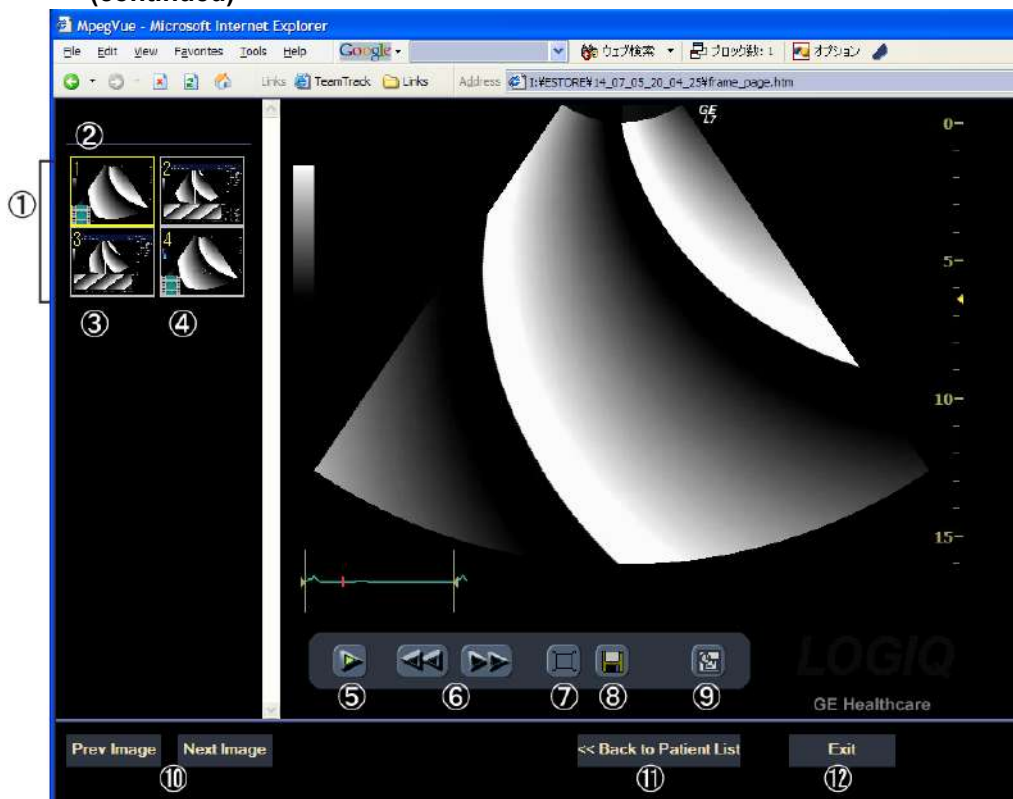


Figure 15-8. MPEGvue screen

- | | |
|---|---|
| 1. Clipboard: select the image to display | 7. Display full screen |
| 2. Selected image | 8. Save image as .wmv, .bmp, jpeg or .avi |
| 3. Single frame image | 9. E-mail support |
| 4. Cineloop | 10. Display previous/next image |
| 5. Freeze/run cineloop | 11. Display the MPEGVue patient list |
| 6. Scrolling tool when in Freeze | 12. Exit |

MPEGvue (Save the data to CD-R to see on the PC) (continued)

E-mail support Sending images or examinations by e-mail.

NOTE: E-mail support and a desktop icon cannot be used on a non-English PC.

The selected image or the entire examination can be sent by e-mail as an attachment, providing the computer has a mail client application (e.g. MS Outlook, MS Outlook Express).

To send an image

1. Select the image to send on the clipboard and select the e-mail button. The e-mail dialogue window displays.
2. Check Send current image. The e-mail address window displays.
3. Select an existing address or enter a new address. Up to 10 addresses can be stored.
4. Select Send.
5. The e-mail with the image is sent and a Confirmation window displays. Select OK.

The person receiving the e-mail can open the image in Windows Media Player.

To send an examination

Examinations sent by e-mail as attachments are zip-compressed, encrypted and password protected. If the size of the exam is too large to be sent as a single attachment, it is divided into several zip file attachments and sent in several e-mails.

1. Open the examination to send in MPEG viewer and select the e-mail button. The e-mail dialogue window displays.
2. Check Send current exam. The e-mail address window displays.
3. Select an existing address or enter a new address.
4. Select Send. The Password window displays.
5. Enter a password and select OK. The exam is sent and a confirmation window displays. Select OK.

E-mail support (continued)

To open an MPEGvue exam from an e-mail

1. In the MPEG viewer press the E-mail button. The E-mail dialogue window is displayed.
2. Check Receive exam. The Password window is displayed.
3. Enter the password and select OK. The exam is uncompressed and opened in the viewer and the E-mail(s) containing the MPEG examination is deleted.

Send To (Send the image to the DICOM Device)

1. Search and select the patient and press **Review**. The Patient Exam screen displays.

GE Medical Systems

Patient ID: 0001 Last Name: patient name DOB: First Name: Age: Middle Name: Sex: ☐ female ☐ male

ABD OB GYN CARD VAS UR SM P PED

Accession #: Exam Description: Ref.Physician: Operator: ADM

Images Clear

Patient View Exam View

Date	Category	Exam Description	Img. size	Disk	MA	DiagCode
09/01/2004	Small Parts		16.8 MB	Local HD		

New Exam Delete Exam Send To

09/02/04 8:35:11AM VS Ling

Figure 15-9. Patient Exam Screen

2. Select the exam which has the images and press **Send to**.

NOTE:

You can only select "Local Archive - Int.HD" for Workflow.
The Send To Dialogue box displays.

Send To (Send the image to the DICOM Device) (continued)

3. Select the destination device and press **OK**.

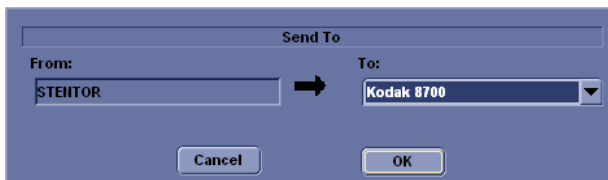


Figure 15-10. Send To Dialogue Box

NOTE: *The destination device is configured in the Utility screen. Multiple devices are able to be configured.*

The successful/unsuccessful message is displayed at the bottom of the screen.

Daily Maintenance

Patient, Report, and User-Defined Backup Protocols

Back up patient data AFTER you've archived (via EZBackup/Move) images so that the pointers to the patient's images reflect that the images have been moved to removable media and are no longer on the hard drive.

1. Insert the backup media. Format the backup media. Select **Utility--> Connectivity--> Removable Media**. Label the media.
2. The Ultrasound system displays a pop-up menu when the formatting has been completed. Press Ok to continue. Verify that the format was successful.



Figure 15-11. Format Successful Pop-up Menu

3. Select **Utility**. Select System, then Backup/Restore.

Patient, Report, and User-Defined Backup Protocols (continued)

4. Select the media. Select everything under Backup by placing a check mark in front of Patient Archive, Report Archive, and User Defined Configuration. Then press Backup.

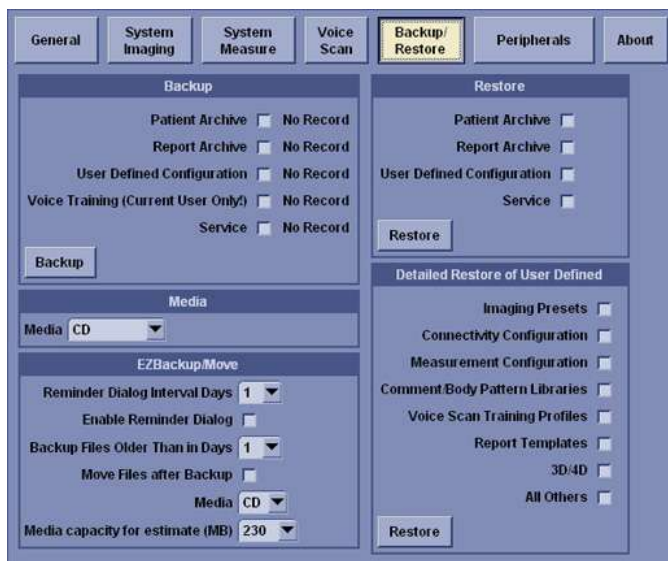


Figure 15-12. Backup/Restore Menu

NOTE: *The detailed section of this menu decouples the user defined configuration above. This allows you to selectively restore what you want to restore across multiple machines. Refer to Chapter 16 for more information.*

5. Answer 'Ok' to the following pop-up message as many times as the number of items you are backing up.

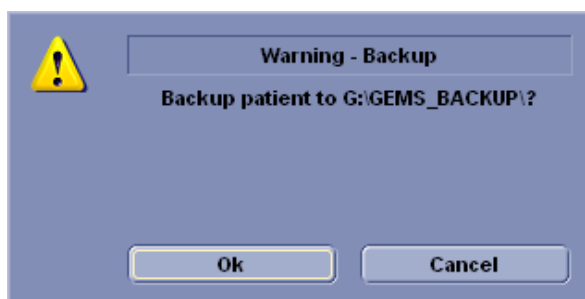


Figure 15-13. Back-up Confirmation Message

Other Printing Options

Connecting to a Standard Computer Printer

We currently support the following:

- HP970 CXI (obsolete)
- HP990 CXI (going obsolete)
- HP 6122
- HP6540
- HP5650 (HP970 and HP990 replacement)
- Xerox 8200DP
- Xerox 8400

To connect to a standard computer printer via the USB connection (Service).

1. Connect the printer to an outer USB port at the back of the system (in the Service port).
2. Install the driver, if necessary. Please refer to the Basic Service Manual for more information.
3. Set the printer as the default printer via Utility --> System --> Peripherals page. Next to Default Printer, select the printer from the drop-down list.
4. Assign one of the print keys to the printer via Utility --> Connectivity --> Buttons. Select the print button, then select the printer from the Destinations drop-down list. Select Add.
5. Bring up the worksheet you want to print on the monitor. Any page displayed on the monitor can be printed.
6. Press the appropriate print key that you assigned above.

NOTE: *You can also set a preset to override the automatic area print and force the entire screen to print by pressing Utility -> System --> Peripherals and setting Print Full Screen.*

NOTE: *Worksheets and Graphs are printed in reverse video for easier readability. You can also set a preset to override this presentation by pressing Utility --> System --> Peripherals and setting the Standard Printer to Enable Video Invert.*

NOTE: *The Field Service Engineer can connect a different printer if you have the driver for it on a CD-ROM (*.inf file).*

External Printers

You can also connect an external printer through the Printer 1/
Printer 2 connections on the back of the system.

To use a remote printer, assign a print key via Utility -->
Connectivity --> Buttons and use the Video Captuer device
service.

Setting up the Off-Line Paper Printer

You can connect an off-line paper printer via the USB connection.

We recommend the following printers.

- HP DeskJet 5650/6540/6122/990



DO NOT place an off-line paper printer inside the patient environment.

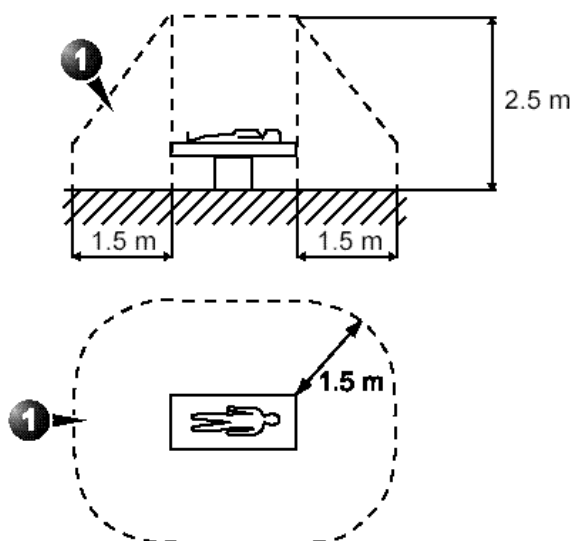


Figure 15-14. Patient Environment

Setting up the off-line paper printer

NOTE: The printer driver is customized for the LOGIQ 9 at the factory; you do not need to change the settings.

1. Connect the printer to the USB port.
2. Select **Utility--> Connectivity--> Service**. Add the **Standard Print** service.

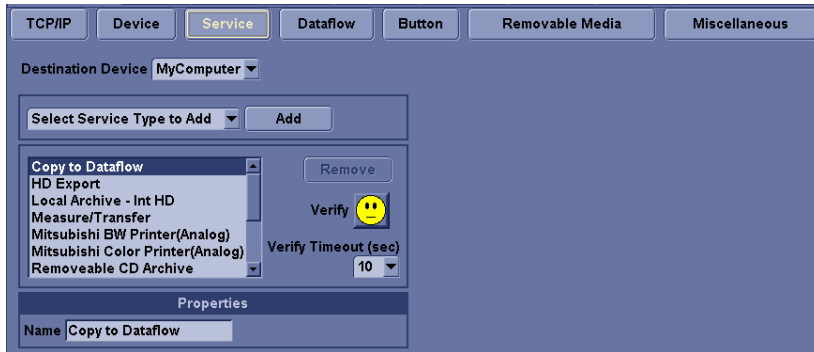


Figure 15-15. Connectivity -> Service Screen

3. Select the printer from the Printer pull-down Properties menu.

NOTE: After selecting the printer, the field turns white.

4. Set the following parameters in Properties:
 - Rows=3
 - Columns=2
 - Orientation=Portrait
 - Right Margin (mm)=10

5. Type the printer name in the Name field.

NOTE: This name is used on the Button screen.

Setting up the off-line paper printer (continued)

6. Press Save, then select the Button tab.

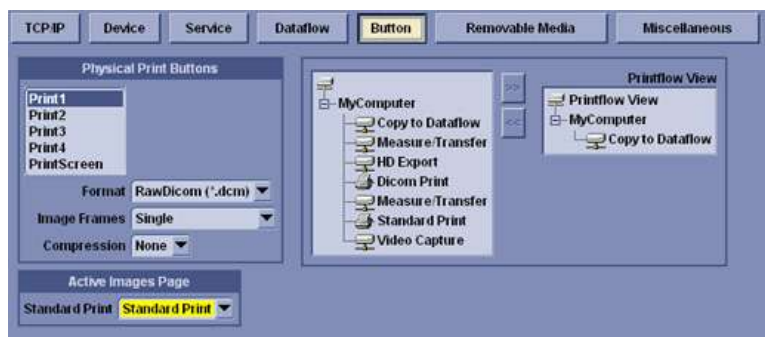


Figure 15-16. Connectivity -> Button Screen

7. Select the appropriate print key (Print1, Print2...) from the Physical Print Buttons section.
8. Select the printer from the MyComputer column and press ">>" to move it to the Printflow View column.
9. If you want to assign this printer to the Standard Print Button on the Active Image Screen, select this printer at the Active Image Printer section.
10. Press Save.

NOTE: *If you want settings other than 1 image per sheet, or 2x3 sheet, or to improve the image quality, refer to the manual that came with the printer.*

Setting up the Printer to Print Reports

To set up the Off-Line Printer to print reports,

1. Press **Utility**--> **System**--> **Peripherals** and select **Printers** under Setup.
2. Select the printer in the Name column.
3. Press the **Left Set** key. Select "Set as Default Printer" and press Set.

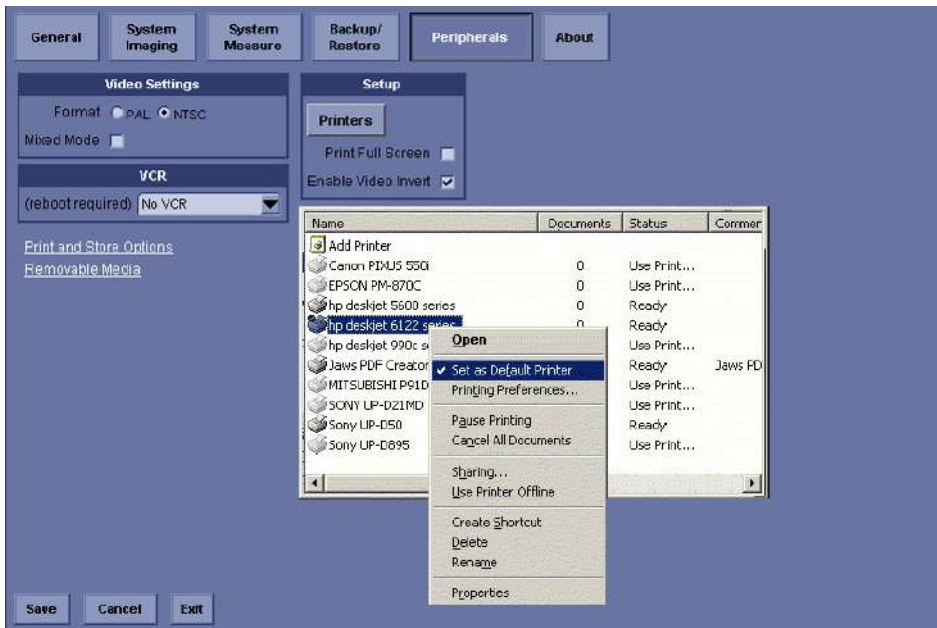


Figure 15-17. Report Printer Setup

4. Press Save.
5. Press Print on the Report screen to print the report.

Setting Up a Network Printer

The LOGIQ 9 supports the Tally, Xerox 8200/8400 network printers.

Setting up the Network Printer

To set up the network printer on the LOGIQ 9 (ensure that Rev 5 or greater Base System Software Load Image is loaded),

1. Press and hold down the Test button on the back of the printer until the data light blinks. Wait for the pages to print out.
2. Select **Utility**. Select **System**. Select Peripherals. Select Printers.
3. Double click Add Printer. Press Next.
4. Deselect *Automatically detect and install my Plug and Play printer*. Select Next.
5. Select Create a New Port. Select Standard TCP/IP Port. Select Next twice.
6. Type in the printer's IP address in the Printer Name or IP Address Box (IP Address=192.28.124.201 and Subnet Mask=255.255.252.0). If desired, you can request an IP Address and Subnet Mask from the local network administrator. Select Next. Select Finish.
7. Make sure the correct printer is highlighted. Select Next.
8. Select *Keep existing Driver (recommended)*. Select Next. Select Yes. Select Next three times. Select Finish.
9. Verify that the test page printed. Select OK.
10. Assign the network printer to be the default printer (Utility-->System-->Peripherals).

NOTE: *The network printer can ONLY be installed in a non-medically used room. If the network printer is used in a patient environment, AC power to the printer needs to be supplied through an isolation transformer.*

Setting up Digital Peripherals

You set up digital peripherals from the Utility --> System --> Peripherals menu. The following digital printers are supported: Sony UP-D21, UP-D23, UP-D50 (only in the USA), UP-D55, UP-D895, and Mitsubishi P91D.

NOTE: *Printing using a standard printing service overrides the orientation and N-up feature of the printer preferences. Printer preferences are set up in the printer folder (via Utility-->System-->Peripherals).*

Digital Printer Setup (LOGIQ 9)

There are two steps to do when setting up a digital printer:
1) follow the procedure below for each printer, then 2) set up specific properties for each printer (specific instructions are provided for each printer following this section).



- Before powering on the LOGIQ 9, connect the printer via the USB cable and turn the printer power on.
- DO NOT remove the cable while the LOGIQ 9 is powered on.

Follow this procedure for each printer:

1. Select Utility--> Connectivity--> Service. Add the Standard Print service. Select the printer from the Printer pull-down Properties menu. For the UP-897 printer, select "Portrait" as orientation.
2. Type the printer name in the Name field. This name is used on the Button screen. After you select the printer from the Printer pull-down Properties menu again, it turns white. Press Save.
3. Select Button. Select the appropriate print key (Print1, Print2...) from the Physical Print Buttons section. Select the printer from the MyComputer column and press >> to move it to the Printflow View column. Press Save.

**Sony UP-D21
Instructions**

Follow these steps to set up the Sony UP-D21 printer.

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D21 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.
2. Select the Layout Tab and select Landscape.
3. Select the Paper Tab and select UPC-21L as the Paper Size.
4. Select the Gray Balance Tab. Set both the Gray Balance A and B to 5.
5. Select the Graphics Tab. Select Gamma Select from the Color Adjust pull-down menu. Set Gamma 3. Select Lightness from the Color Adjust pull-down menu. Set Light to 3, Dark to 0, Sharpness to 7, and Gamma to 6. Press Apply. Press OK.
6. Press Apply. Press OK.
7. Press Save, then Exit.

NOTE: *Landscape values set up in Utility--> Connectivity--> Standard Print take precedence over this setting.*

Sony UP-D23 Instructions

Follow these steps to set up the Sony UP-D23 printer.

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D23 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.
2. Select Landscape.
3. Select the Paper Tab and select UPC-21L as the Paper Size.
4. Select the Gray Balance Tab. Set both the Gray Balance A and B to 5.
5. Select the Graphics Tab. Select Gamma Select from the Color Adjust pull-down menu. Set Gamma 3. Select Lightness from the Color Adjust pull-down menu. Set Gamma to 4. Press Apply. Press OK.
6. Press Apply. Press OK.
7. Press Save, then Exit

NOTE: *Landscape values set up in Utility-->Connectivity-->Standard Print take precedence over this setting.*

**Sony UP-D50
Instructions**

Follow these steps to set up the Sony UP-D50 printer.

NOTE: *Set up the UP-D50 printer as 1-up. If you set it up as 2-up or 4-up and there are pages pending for the printer, you may get an error message and the printer will not print until the error is cleared out.*

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D50 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.
2. On the Layout Tab, select Landscape, then select 1 Page(s) Per Sheet. Press Advanced at the bottom of the Layout page.
3. Select Print Speed and select High Speed from the pull-down menu. Select Color Adjust. Select Properties. Select Soft from the Gamma pull-down menu. Do not adjust any color settings (keep everything set at zero). Set Dark to -5, set Light to -5, and set Sharpness to 6.

NOTE: *High Speed takes approximately 35 seconds with minimal print quality degradation; Normal Speed takes approximately 55 seconds.*

4. Press OK twice. Press Apply. Press OK.
5. On the Properties page, select the Sharing Tab. Ensure that Do Not Share This Printer is selected.
6. Select the Ports Tab. Select UPD50 as the Port.
7. Select the Advanced Tab. Select Always Available, Priority 1, Driver for Sony UP-D50, Spool print documents so program finishes printing faster, Start printing immediately, Print spooled documents first, and Enable advanced printing features.
8. Press Apply. Press OK.
9. Press Save, then Exit.

NOTE: *To set up the UP-D50 printer to print 4 images on one page via the Active Images page, select Rows=2, Columns=2 on the Service Connectivity page; select UP-D50 as the Active Image Printer on the Button Connectivity page.*

Sony UP-D55 Instructions

Follow these steps to set up the Sony UP-D55 printer.

NOTE: *Set up the UP-D55 printer as 1-up. If you set it up as 2-up or 4-up and there are pages pending for the printer, you may get an error message and the printer will not print until the error is cleared out.*

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D55 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.
2. On the Layout Tab, select No Division from the Split-Screen pull-down menu.
3. On the Paper Tab, select High Speed.

NOTE: *High Speed takes approximately 30 seconds with minimal print quality degradation; Normal Speed takes approximately 35 seconds.*

4. On the Graphics Tab, select Lightness from the Color Adjust pull-down menu. Set Sharpness to 7 and Gamma to 3 (keep all other settings at 0). Select Gamma Select from the Color Adjust pull-down menu and select Gamma 2.
5. Press Apply. Press OK.
6. On the Properties page, select the Sharing Tab. Ensure that Do not share this printer is selected.
7. Select the Ports Tab. Select UPD55 as the Port.
8. Select the Advanced Tab. Select Always Available, Priority 1, Driver for Sony UP-D55, Spool print documents so program finishes printing faster, Start printing immediately, Print spooled documents first, and Enable advanced printing features.
9. Press Apply. Press OK.
10. Press Save, then Exit

NOTE: *To set up the UP-D55 printer to print 4 images on one page via the Active Images page, select Rows=2, Columns=2 on the Service Connectivity page; select UP-D55 as the Active Image Printer on the Button Connectivity page.*

**Sony UP-D895
Instructions**

Follow these steps to set up the Sony UP-D895 printer.

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D895 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.

2. Select the Layout Tab and select Portrait.

NOTE: *Change Landscape value to Portrait in Utility-->Connectivity--> Standard Print, which takes precedence.*

3. Press the Advanced button at the bottom of the Layout page. Select 1980x1280 from the Paper Size pull-down menu. Select Density Adjust. Select Properties. Set Gamma to SOFT (TONE3). Set Dark to 0, set Light to 0, and set Sharpness to 7. Press Save. Save the file to the recommended location. Press Save. Press OK twice. Press Apply. Press OK.

4. On the Properties page, select the Sharing Tab. Ensure that 'Do not share this printer' is selected.

5. Select the Ports Tab. Select the USB Port for Sony UP-D895. Select Enable bidirectional support.

6. Select the Advanced Tab. Select Always Available, Priority 1, Driver for Sony UP-D895, Spool print documents so program finishes printing faster, Start printing immediately, Print spooled documents first, and Enable advanced printing features. Press OK.

7. Select the Security Tab. Specify permissions for each user group.

8. Press OK.

9. Press Save and Exit.

Sony UP-D897 Instructions

Follow these steps to set up the Sony UP-D897 printer.

1. Press Utility-->System-->Peripherals. Select Printers. Select the UP-D897 printer, then right click (usually with the left Set key) and select Properties. Select Printing Preferences at the bottom of the page.

2. Select the Layout Tab and select Portrait.

NOTE: *Change Landscape value to Portrait in Utility-->Connectivity--> Standard Print, which takes precedence.*

3. Press the Advanced button at the bottom of the Layout page. Select 1980x1280 from the Paper Size pull-down menu. Select Density Adjust. Select Properties. Set Gamma to TONE2. Set Dark to -12, set Light to -26, and set Sharpness to 7. Press Save. Save the file to the recommended location. Press Save. Press OK twice. Press Apply. Press OK.
4. On the Properties page, select the Sharing Tab. Ensure that 'Do not share this printer' is selected.
5. Select the Ports Tab. Select the USB Port for Sony UP-D897. Select Enable bidirectional support.
6. Select the Advanced Tab. Select Always Available, Priority 1, Driver for Sony UP-D897, Spool print documents so program finishes printing faster, Start printing immediately, Print spooled documents first, and Enable advanced printing features. Press OK.
7. Press OK.
8. Press Save and Exit.

Standard Print

A Standard Print button has been added to the Active Images screen on the Patient menu. You need to configure the printer to the Standard Print button via **Utility --> Connectivity --> Button**.



Figure 15-18. Active Images Screen

Standard Print (continued)

To print an image,

1. Select the image you want to print from the Active Images screen. You can print one (1) image per sheet or 2x3 images per sheet.
2. Press Standard Print.

NOTE: *If the printer is not assigned to the button, you will get a message telling you to Check Printer Button Configuration.*

NOTE: *There is no warning to let you know that the printer is not functioning. Check the printer.*

To assign a printer to the Active Images screen,

- Configure the printer to the Standard Print Button on the Utility--> Connectivity--> Button screen.
- Select the printer at Standard Print under the Active Images Page section.

Transferring Patient Data to a PC

Transferring OB/GYN Patient Data to a PC

The LOGIQ 9 now supports the transfer of OB, GYN and Vascular patient data to a PC.

NOTE: *The LOGIQ 9 currently supports Viewpoint, Sonultra, and CustomLab patient data communications software for the PC.*

Connecting the LOGIQ 9 to the PC

You need to connect an RS232C Converter and RS232C Cross Cable (cross-over 9-pin serial cable, not provided) to the USB port to connect a serial peripheral device to the system.

NOTE: *To send the data, the LOGIQ 9 must be connected to the PC using a serial interface.*

Transferring Patient Data

After you perform a scan, you can send the patient and measurement data to a PC that accepts patient data.

1. Ensure that the peripheral device is on.
2. For the current patient, on the Worksheet screen, select Meas. Transf.

Origin	LMP	LMP	BBT	GA	EDD(LMP)
Fetus A/I		CUA	13w6d+/- 1w0d		EDD(CUA) 06/30/2005
FetusPos		PLAC		Ref.Physician	Page 1/1

B Mode Measurements							
BPD(Hadlock)	☒	2.05 cm	2.05		Avg.	13w2d	12w1d-14w3d
HC(Hadlock)	☒	7.47 cm	8.80	6.13	Avg.	13w1d	12w0d-14w2d
OFD(HC)		2.80 cm	3.46	2.14	Avg.		
AC(Hadlock)	☒	6.69 cm	6.69		Avg.	13w2d	11w5d-15w0d
FL(Hadlock)	☒	1.20 cm	1.20		Avg.	13w4d	12w1d-14w6d

2D Calculations							
EFW(AC,BPD,FL,HC)		75.28g +/- 11.29g		(3oz)			
AFI(Moore)		7.01 cm	1.77	1.46	2.07	1.71	
CI(Hadlock)		73.33 (70.00-86.00)		FL/AC(Hadlock)		17.91 (-)	
FL/BPD(Hohler)		58.35 (-)		FL/HC(Hadlock)		16.04 (-)	
HC/AC(Campbell)		1.12 (1.10-1.34)					

Figure 15-19. Transferring OB Worksheet Data to a PC

Transferring Patient Data (continued)

3. The system transfers the data. A beep on the peripheral device indicates a successful transfer. If there is a problem with the connection or transfer, the system displays an error message.

NOTE: *If you currently do a heart rate measurement without specifying a vessel, as in the case of a fetal heart, the measurement IS NOT automatically transferred to the PC. You need to enter this measurement manually on the PC.*

NOTE: *Deleted measurements from the worksheet are still sent to the off-line reporting PC.*

VCR / DVR

Setting up the VCR or DVD Recorder

To set up the VCR or DVD Recorder,

1. Select the **Utility --> System --> Peripherals** on the Touch Panel.
2. Select the VCR or DVD Recorder from the pull-down menu.
3. Select **Save**, then select **Exit**.
4. Turn off the LOGIQ 9, then turn it back on.



Be sure to connect the converter and cable to VCR1 before booting up the system. If this is not done, the LOGIQ 9 does not recognize the peripheral device.



DO NOT set up the VCR if there is not a VCR connected.

NOTE: *Record does not work on a finalized DVD-R. On a finalized DVD-R on the LOGIQ 9 with the LQ-MD800, Fast-forward/Fast-rewind functions operate differently. Increasing the search speed or using the Fast-forward/Fast-rewind functions do not continue across Titles. The speed returns to the regular play speed.*

NOTE: *You cannot play the media which was recorded by another DVD Recorder on the LQ-MD800.*

NOTE: *The media recorded by the LQ-MD800 cannot be played by the DVD-Drive on the LOGIQ 9. This is applicable for both finalized and non-finalized DVD-R/DVD-RAM.*

NOTE: *Make sure to connect the cable to the VCR1 connection on the LOGIQ 9 rear panel.*

Using the VCR/DVD Recorder

To use the VCR/DVD Recorder, press VCR on the Control Panel. The following Touch Panel appears.



Figure 15-20. VCR/DVD Touch Panel Page 1

Table 15-3: Touch Panel Keys

Touch Panel Key	Explanation
Video Source	Use to change Scan/Video.
[Microphone]	Microphone On/Off.
Reverse Skip	Title-by-Title Reverse Skip at REC/ STOP. Useful to skip by title.
Forward Skip	Title-by-Title Reverse Skip at REC/ STOP. Useful to skip by title.
[Red Circle - Record Icon]	Press to start recording.
[Two Lines - Pause Icon]	Press to pause playback.
[Eject Icon]	Press to Eject media.
[<< - Fast Reverse Icon]	Press to Fast Reverse media.
[> - Play Icon]	Press to begin playback.
[Box - Stop Icon]	Press to stop.
[>> - Fast Forward Icon]	Press to Fast Forward media.

Using the VCR/DVD Recorder (continued)



Figure 15-21. VCR/DVD Touch Panel Page 2

Table 15-4: Touch Panel Keys

Touch Panel Key	Explanation
Edit Title	Use to change the name of the recorded title.
Format	Use to format media -- or to erase data from the DVD-RAM.
Finalize	Use to finalize the DVD-R.

Recorder Touch Panel Menu

When you use Record/Stop or Record / New Patient, the DVD recorder creates divisions called "Title." These divisions of titles manage the Search, Reverse Skip, and Forward Skip Touch Panel keys.

As you use Record/Stop or Record / New Patient, the system shows the message, "Now writing....." During this time, you cannot scan or perform any system operation. Please wait until this message disappears to continue. This message does not record on the media.

Using the VCR/DVD Recorder (continued)

Table 15-5: VCR Functions and Appropriate Steps

VCR Function	How to Do
To record an image	1. Insert a tape/DVD. The Search dialog displays. 2. Press Video Source. Press VCR. Press Record. 3. Press MIC On/OFF to activate and control the microphone. DVD NOTE: A record start (REC) to a record stop (STOP) is treated as one (1) title. You can record up to 99 titles per DVD. Maximum recording time is two (2) hours per DVD.
To pause recording	Press Pause.
To begin recording again	Press Record again.
To stop recording	Press Stop/Eject.
To rewind	Press Rewind.
To playback	Press Play.
To adjust playback speed	Move the Trackball to the left/right to decrease/increase playback speed.
To pause playback	Press Pause.
To begin playback again	Press Play.
To stop playback	Press Stop.
To turn the microphone on/off	Press MIC ON/OFF.
To adjust playback audio volume	Adjust Volume left/right to lower/increase audio playback volume.
To fast forward	Press Fast Forward.
To fast rewind	Press Fast Rewind.
To skip the title (DVD)	When you playback the title, • Press Forward Skip to move to the next title. • Press Reverse Skip to move to the previous title.
To search for a recorded sequence	For VCR: • You can search VCR tapes exam by Counter value. Press Search Forward/Reverse to move forward through the tape. For DVD Recorder: • Press Search. The title list displays. Select the desired title.
To adjust search speed	Adjust Search Speed left/right to lower/increase search speed.
To reset the counter (VCR)	Press Counter Reset. ONLY available for VCR.
To eject the VCR tape/DVD	Press Eject. • Use the Eject control on the LOGIQ 9 Touch Panel when ejecting the media from the DVD-Recorder. • When inserting the media into the DVD-Recorder, use the Open/Close button on the DVD-Recorder.

Table 15-5: VCR Functions and Appropriate Steps (Continued)

VCR Function	How to Do
To format the DVD-RAM	1. Press Format (Page 2 on Video Touch Panel). The dialog box appears. 2. Press OK. The initialization is performed. 3. The termination dialog box appears. Press OK to complete. NOTE: All the existing images are deleted from the DVD-RAM. NOTE: This can take up to 70 minutes. NOTE: The DVD-RAM is rewritable.
To finalize DVD-R	1. Press Finalize (Page 2 on Video Touch Panel). The dialog box appears. 2. Press OK. Finalize the DVD-R. 3. The termination dialog box appears. Press OK to complete. You can play the finalized DVD-R through the DVD Video player. The list of recorded images is created. NOTE: You cannot add data to the finalized DVD. NOTE: You cannot perform any other system function while finalizing a DVD-R, which usually takes 4-5 minutes (unless you finalize manually from the operation panel of the DVR - see Finalize -- Creating the DVD-Video (DVD-R) below).
To edit the title (DVD Recorder)	1. Press Edit Title (on Page 2) 2. Select the title from the list. 3. Type the new title in the Edit Box. 4. Press Change to save. NOTE: Auto Title: The current patient ID is automatically recorded as a title at the start of the recording. You can also edit this title.
To change the source of the monitor image	Press Video Source. • VCR: Selects the VCR. • Printer: Selects the onboard analog color video printer (UP-2900MD or UP-51MD). This option is used for on-screen setup and configuration of the color printer. • Off: Returns to ultrasound display.
To set up a print key to Record/Pause	1. Go to Utility-->Connectivity-->Service. 2. Destination Device = My Computer. 3. Add Service Device = Video Capture. 4. In right Properties box, select VCR Record Pause. 5. In lower Properties box, rename Video Capture to Panasonic LQ MD800. 6. Go to Utility-->Connectivity-->Button 7. Added VCR record/pause to print key (this can be added to any print key).

NOTE: *The VCR counter is displayed in two places: in the video area and outside the video area. The playback counter is in the video area and the current counter is outside the video area.*

Available media and maximum recording time

Available Media	• DVD-RAM Video 120 min (with or without cartridge) • DVD-RAM 4.2GB (with or without cartridge) • DVD-R Video 120min • DVD-R Data 4.2GB
Maximum Recording Time	2 hours, if you use the media listed above.

DVD Recorder Settings

Clock Setup Set the clock manually via the LQ-MD800 control panel:

1. Press **VCR** to enter the video input mode on the LOGIQ 9.
2. Press **MENU** (SHIFT + MENU SEARCH) on the LQ-MD800 front panel to display the menu screen on the LOGIQ 9 monitor display.

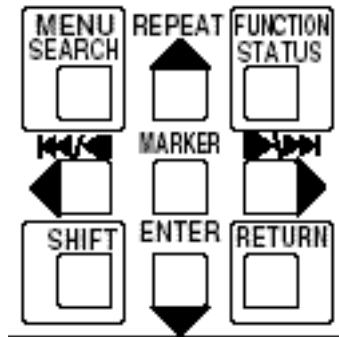


Figure 15-22. LQ-MD800 Operation Panel

3. Use the arrow keys to select **SETUP**, then press **ENTER**.
4. Use the up or down arrow key to select **Settings**, then press the left or right arrow key.
5. Use the up or down arrow key to select **Clock**, then press **ENTER**.
6. Use the left or right arrow key to select the item you want to change.
7. Use the left or right arrow key to change the setting, then press **ENTER**.

Record Mode Setup

Set the record mode manually via the LQ-MD800 control panel:

1. After performing steps 1-3 of the Clock Setup, do the following steps.
2. Use the up or down arrow key to select **Picture**, then press the left or right arrow key.
3. Use the up or down arrow key to select **Rec. Mode**, then press **ENTER**.
4. Use the up or down arrow key to select the recording mode, then press **ENTER**.

Table 15-6: Recording modes and approximate recording times in hours

Mode	Recording time (Single-sided)	Quality
XP	1	Above DV25
SP	2	S-VHS
LP	4	VHS Normal Mode

Setting up the System Settings

Under the System tab:

1. Change On-Screen Status mode to Off. The Off setting disables the Status Window message, and prevents a popup from covering the right upper corner of the monitor while scanning.
2. Set the finalizing speed to 4x.
3. Press the Enter button to save the selection.

Setting up the DVR Connection Setting

Specify NTSC or PAL (Europe).

1. Under the Connection setting, confirm the TV system is set to NTSC (or PAL for Europe).
2. When all settings are correct, press the Return button on the DVR to close the Setup screen.

Finalize -- Creating the DVD-Video (DVD-R)

If you choose to finalize the DVD-R manually using the operation panel on the DVR, you can perform normal system operation while finalization completes. Finalize the DVD-R manually via the LQ-MD800 control panel in order to be able to view it on a standard PC or compatible DVD player:

NOTE: *You cannot operate the LOGIQ 9 while finalizing the DVD-R. It takes 4 to 5 minutes to finalize a DVD-R.*

1. After performing steps 1-2 of the Clock Setup, do the following steps.
2. Use the arrow keys to select **DISK INFORMATION**, then press **ENTER**.
3. Use the up or down arrow key to select **Finalize**, then press the left or right arrow key.
4. Use the right arrow key to select **Yes**, then press **ENTER**.
5. Use the right arrow key to select **Start**, then press **ENTER**.

NOTE: *You can finalize a DVD-R from the Touch Panel of the LOGIQ 9. See "To finalize DVD-R" on Table 15-5 for more information.*

Cleaning the VCR/DVR

To clean the VCR/DVR:

1. Turn off the power. If possible, disconnect the power cord.
2. Wipe the external surfaces of the unit with a soft, clean, dry cloth.

NOTE: *Do not use a wet cloth or any cleaning fluid because it may enter and damage the unit.*

Returning to Scanning

To return to scanning, select **Video Source** on the Touch Panel or press **B-Mode**.

Portable Exam

To perform a portable exam (using the worklist),

1. Go to the worklist and get the patient(s) you will need for this portable exam(s).
2. Ensure that images will be saved to the local hard drive.
3. Press the Power On/Off switch and select Standby.
4. Proceed to perform the exam(s). Press Power On/Off. Select the patient, do the exam. Store images. Images are held in the spooler.
5. If there are additional patients to scan, repeat steps 4 and 5. After you have completed all portable exams, press Power On/Off and select Standby.
6. When you have returned from performing the portable exam(s), reconnect to the network. Press Power On/Off.
7. Press F4. Resend all of the images in the spooler to the printer/storage device.
8. Perform an EZBackup/Move to remove the images from the hard drive.

Chapter 16

Customizing Your System

Describes how to create system, user, and exam presets.

Overview

Preset Menus provides the following functionality:

- **System presets.** View and update general system configuration settings, measurement and analysis settings, and video settings; backup and restore data and configuration files.
- **Imaging presets.** View and update exam and imaging parameters.
- **Comment library presets.** Set up comment libraries by application.
- **Body Pattern library presets.** Set up body pattern libraries by application.
- **Application and User Defined presets.** Configure application- and user-specific settings.
- **Test patterns.** Helps configure system settings.
- **Connectivity Setup.** Define connection and communication setup, including exam dataflow information.
- **Measurement and Analysis presets.** Customize exam studies, create measurements, set up manual sequencing, and create OB Tables.
- **Reports Presets.** Allows you to edit the report template, diagnosis codes, and report comments. Please refer to Chapter 14 for more information.
- **System Administration presets.** Perform system administrator activities such as setting up user IDs and logon formats.
- **3D/4D.** Real-time 4D and Static 3D scanning.
- **Service.** Activates the Service Browser.
- **Search.** You can search for a parameter on the Utility pages (Measure, Reports, and Service pages cannot be searched.)

To access these functions, select the **Utility** tab on the Touch Panel, then select the appropriate Touch Panel key.

Overview (continued)

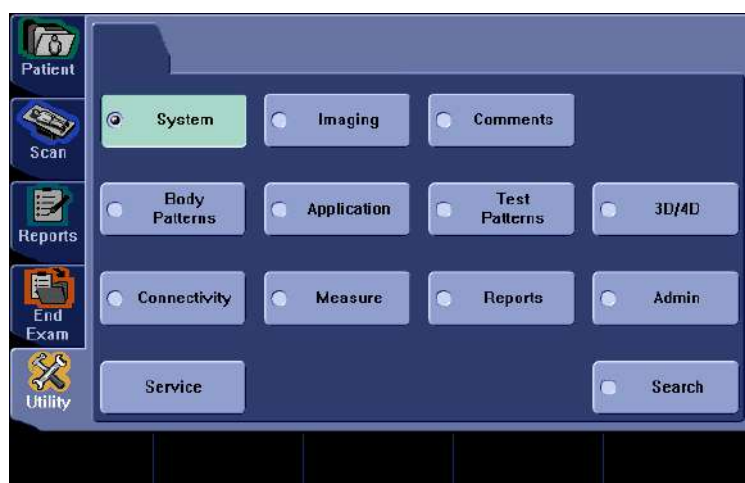


Figure 16-1. Utility Touch Panel

System Presets

Overview

System presets allows you to view or change the following parameters

- **General** – Location, Date/Time, Patient Info, Key Usage, and Utility configuration
- **System Imaging** – CINE Loop Store, Cardiac, Biopsy Guides, and Image Control and Display configuration
- **System Measure** – Measurement, Cursor, and Results Window configuration
- **VoiceScan** – System User, Voice Profile Setup, and VoiceScan Parameters
- **Backup/Restore** – Backup, Media, EZBackup/EZMove, Detailed Restore of User Defined.
- **Peripherals** – Video Settings, VCR, Print and Store Options, and Setup configuration
- **About** – System software, patent, and image information
- **Service** – Use to view the documentation CD-ROM or to start the Service platform

Changing system parameters

To change system parameters:

1. On the Touch Panel, select **Utility**.
2. On the Touch Panel, select **System**.
The System screen is displayed.
3. On the monitor display, move the **Trackball** to select the tab that has the information you want to change.
4. Select values for the parameters you want to change.
5. To save the changes, select the **Save** button. Select **Exit** to return to scanning. In some cases, you may need to reboot the system for the change to take effect.

NOTE: *In some cases, you may need to reboot the system for the change to take effect.*

System/General Preset Menu

The System/General screen allows you to specify hospital name and system date and time.

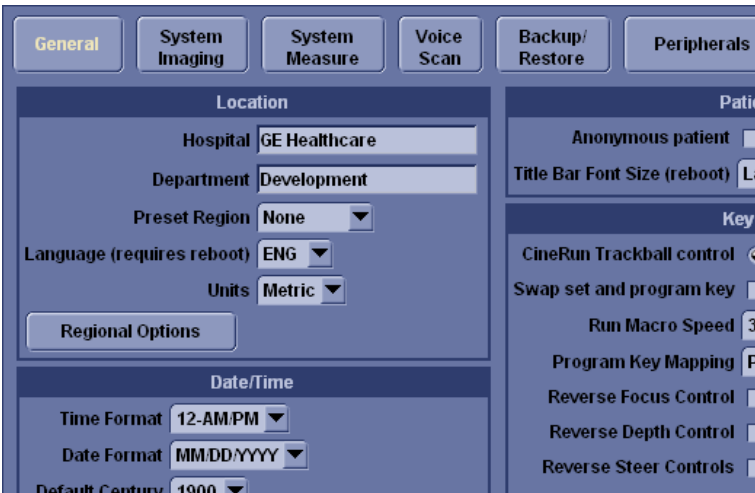


Figure 16-2. System/General Preset Menu

Table 16-1: Location

Preset Parameter	Description
Hospital	Type the <i>institution's</i> name.
Department	Type the institution's department name.
Preset Region	Unique imaging presets are available for each region (Americas, Asia, Europe, or None). A message appears, "This will result in loss of Imaging Presets. Do you want to continue? Select OK, or Cancel." Probe, Application, and 3D/4D imaging presets will be overwritten if you select OK. User-named presets, however, are not overwritten if you select OK.
Language	Select the appropriate language from the drop-down list.
Units	Select metric or US units of measurement.
Regional Options	Select to set up Russian/Greek keyboard.

System/General Preset Menu (continued)

Table 16-2: Date and Time

Preset Parameter	Description
Time Format	Select the time format: 12 Hr. AM/PM or 24 Hr.
Date Format	Select the date format: dd/mm/yyyy, mm/dd/yyyy, or yyyy/mm/dd.
Default Century	Select the default century for the system to use.
Date/Time	Select to display the Date/Time Properties window, to specify the system date, time, time zone, and to auto adjust for daylight savings time.

Table 16-3: Patient Information

Preset Parameter	Description
Anonymous patient	When selected, no patient information is displayed on the scanning screen Title bar.
Title Bar Font Size (reboot)	Select to display patient information in the title bar using a small, medium, or large font size. You need to reboot the system for this change to take effect.
Title Bar Font Size - High Resolution (reboot)	For the LCD monitor. Select to display patient information in the title bar using a small, medium, or large font size. You need to reboot the system for this change to take effect.

System/General Preset Menu (continued)

Table 16-4: Key Usage

Preset Parameter	Description
CineRun Trackball control	Select Frame by Frame or Loop Speed. Frame by Frame manually steps through CINE playback by using the Trackball. Loop Speed controls the speed of the CINE Loop playback. Roll the Trackball to the left to slow down playback; roll the Trackball right to speed up playback. The Loop Speed displays in the lower, right-hand corner of the screen.
Swap set and program key	Swap the function of the Programmable Set key and the Set key.
Program Key Mapping	Allows you to program the left set key's functionality (Pointer, Set, or Freeze). You need to reboot the system after reprogramming the Set and Select keys.
Reverse Focus Control	Select to reverse Focus key operation.
Reverse Depth Control	Select to reverse Depth key operation.
Reverse Steer Control	Select to reverse Steer key operation.
Reverse Baseline Rotaries	Select to reverse Baseline key operation.
Trackball	Field Engineer use only. Select to display the Mouse Properties window (faster, slower, snap-to default).

Table 16-5: Screen Resolution

Preset Parameter	Description
Screen Resolution (requires reboot)	Select default for CRT monitor. Select High for the LCD monitor.

Table 16-6: General User Interface

Preset Parameter	Description
Color Level (requires reboot)	Select System Color according to the condition of the room.

Table 16-7: Utility

Preset Parameter	Description
Prompt for Save on Cancel or Exit	If selected, the system prompts you to save data when you select cancel or exit without saving.
Utility Font Size	Select the font size you want to use to view the Utility menus: Small, Medium, or Large.

Foreign Language Keyboard Setup

Keyboard Setup for non-Russian/ Greek Languages

To set up the keyboard for non-Russian/Greek languages:

NOTE: *You must apply the changes on each setup page before moving to the next page.*

1. In Utility--> System--> General, set the Language as desired. Save this setting, but do not reboot the system yet.
2. Press Regional Options, select the Language tab, press Details, under Installed Services press Add to set the Keyboard layout/IME to United States-International, press OK, set the Default input language to English (United States) - United States International, press Apply, Press OK.

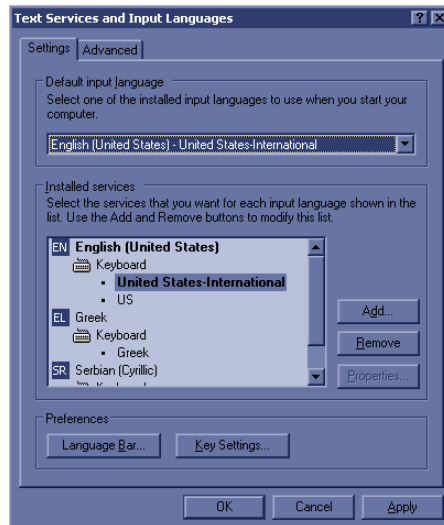


Figure 16-3. Selecting the International Keyboard

Keyboard Setup for non-Russian/ Greek Languages (continued)

3. Select the Advanced tab, then select the language in the Language for non-Unicode programs pull-down menu. Press Apply. Answer Yes to use files already loaded on the hard disk, then answer No to not reboot the system yet, press OK. Press Save and Exit the Utility screen.

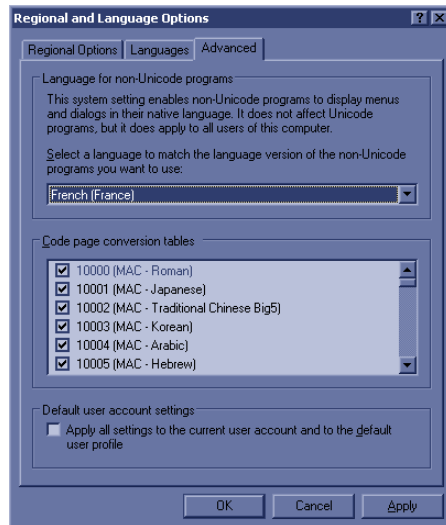


Figure 16-4. Set Language

4. Reboot the system. When your system restarts, the system appears in the selected language.
5. To type foreign characters, press Ctrl+Shift to change the keyboard to the international keyboard, then press the Alt GR+appropriate keyboard key.

**Keyboard Setup
Procedure for
Russian and Greek**

1. In Utility--> System--> General, set the Language as Russian or Greek. Save this setting, but do not reboot the system yet.



Figure 16-5. Changing the System Language to Russian/
Greek

Keyboard Setup Procedure for Russian and Greek (continued)

2. Press Regional Options, under Standards and Formats select Russian or Greek, under Location select Russia or Greece. Press Apply.

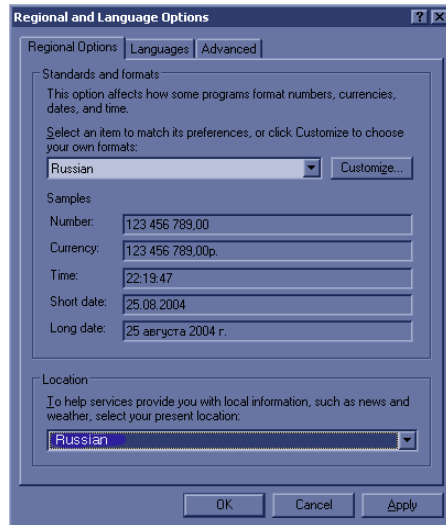


Figure 16-6. Regional Options

3. Select the Language tab, press Details, under Installed Services select the Russian or Greek keyboard, under Default input language select Russian - Russian or Greek - Greek, press Apply, Press OK.

Keyboard Setup Procedure for Russian and Greek (continued)

4. Select the Advanced tab, then select Russian or Greek in the Language for non-Unicode programs pull-down menu. Press Apply. Answer Yes to use files already loaded on the hard disk, then answer No to not reboot the system yet, press OK. Press Save and Exit the Utility screen.

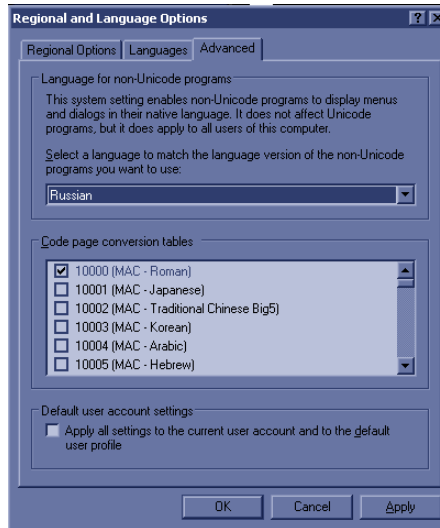


Figure 16-7. Set Language

5. Reboot the system. When your system restarts, the system appears in the selected language.
6. To switch between the English and Russian (or Greek) keyboard, press Alt+Shift to change the keyboard to the Russian or Greek keyboard.
7. Apply the changes by pressing Apply. Press OK TWICE.

NOTE: *To have the settings take effect, you **MUST** turn off the system and turn it back on.*

Keyboard Setup Procedure for Russian and Greek (continued)

Here is an example of the Russian keyboard:

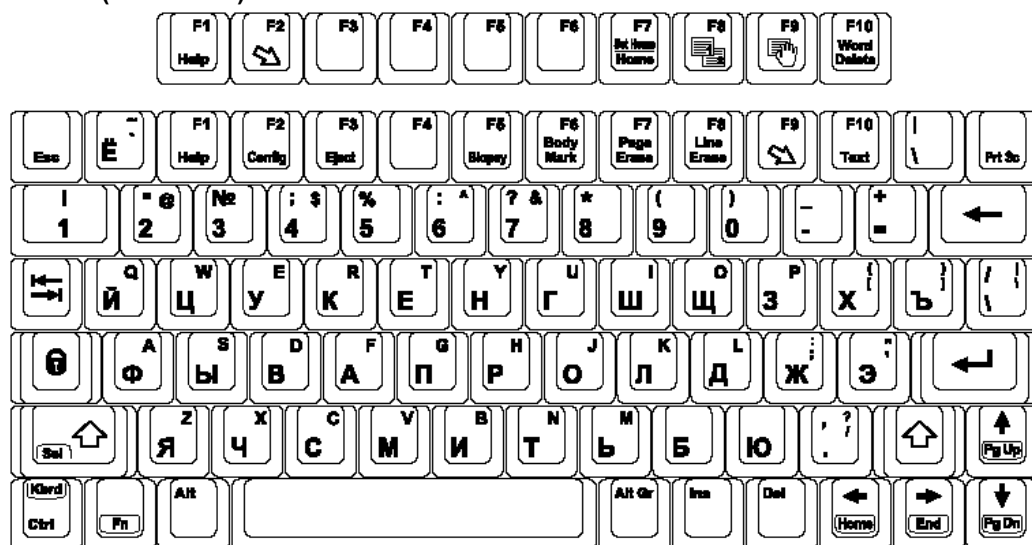


Figure 16-8. Russian Keyboard

Here is an example of the Greek keyboard:

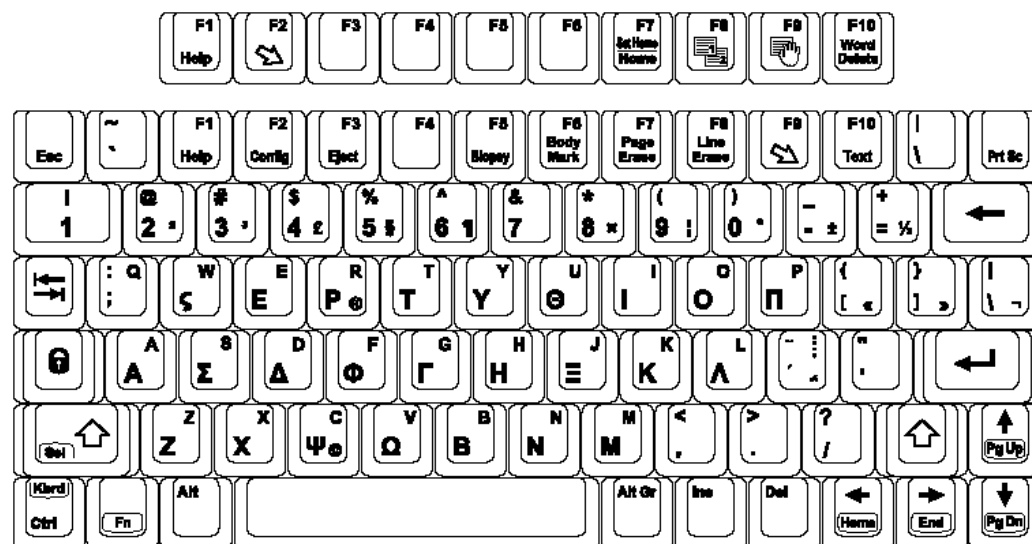


Figure 16-9. Greek Keyboard

System/System Imaging Preset Menu (LOGIQ 9)

The System --> System Imaging screen allows you to specify parameters for cine loop storage, biopsy guide, and image control and display presets.



Figure 16-10. System/System Imaging Preset Menu

Table 16-8: Cine Loop Store

Preset Parameter	Description
Time span (no ECG) [s]	Select the number of seconds of CINE Loop storage (no ECG). The default is 3 seconds (approximately 77 B/W frames, 97 Color Frames).
3D/4D Timespan (no ECG) [s]	Select the number of seconds of CINE Loop storage (no ECG) during 3D/4D.
Preview loop before store	When selected, allows you to review cine loops before storage.

System/System Imaging Preset Menu (LOGIQ 9) (continued)

Table 16-9: Cardiac

Preset Parameter	Description
Time before heart cycle [ms]	Sets the total storage time span of the cine loop in ECG mode.
Time after heart cycle [ms]	Sets the total storage time span of the cine loop in ECG mode.
Time span of systole [ms]	The default systolic time to enable synchronization of cine loops.

Table 16-10: Biopsy Guides

Preset Parameter	Description
Show Center Line	Displays center biopsy guideline.
Show Outer Lines	Displays outer biopsy guidelines.
Enable 0.5cm markers	Activates biopsy depth markers every 0.5cm.
Show Biopsy Mark on CFM Simultaneous Mode	Displays the Biopsy Guideline on the Color Flow Mode image while in Simultaneous Mode.

System/System Imaging Preset Menu (LOGIQ 9) (continued)

Table 16-11: Image Control and Display

Preset Parameter	Description
Horizontal Scale	Select to display width markers.
TGC Display	Select to display TGC curve.
PW Velocity Units in cm/s	Select to change scale on timeline from centimeters per second to meters per second.
Split screen full height display of linear images	When selected, while in split screen and using a linear probe, the system clips the sides of images so that the image can fill the available display.
Use full height in Zoom on linear images	Select to use full height zoom while scanning on a linear image.
Display image size	Specify display size, Small, Medium, or Large. After you save this feature, go back to scanning, press B-Mode or reboot the system. The image size changes.
Auto Invert on Linear Steer	When selected, for auto calcs, automatically inverts the timeline if needed when using ASO.
Auto Invert on ASO	Automatically inverts the spectrum with ASO.
Audio Volume	Adjusts the Doppler audio volume via a drop-down menu (for example, 0=softer; 31=louder). Tied to Doppler Volume. Cannot adjust separately.
Auto Freeze (15 minutes)	Automatically freezes the system after 15 minutes of inactivity.
Image Parameter Size - Default Resolution	Choose Small, Medium, or Large for the CRT monitor. Must reboot the system.
Image Parameter Size - High Resolution	Choose Small, Medium, or Large for the LCD monitor. Must reboot the system.
Countdown Time For Contrast (Sec)	Specify time for the Contrast Clock to countdown during a contrast study, 0, 3, or 5 seconds.

System/System Measure Preset Menu

The System/System Measure screen allows you to specify measurement parameters such as the type of default OB measurements and calculations. You can also define cursor and Results Window default functionality.

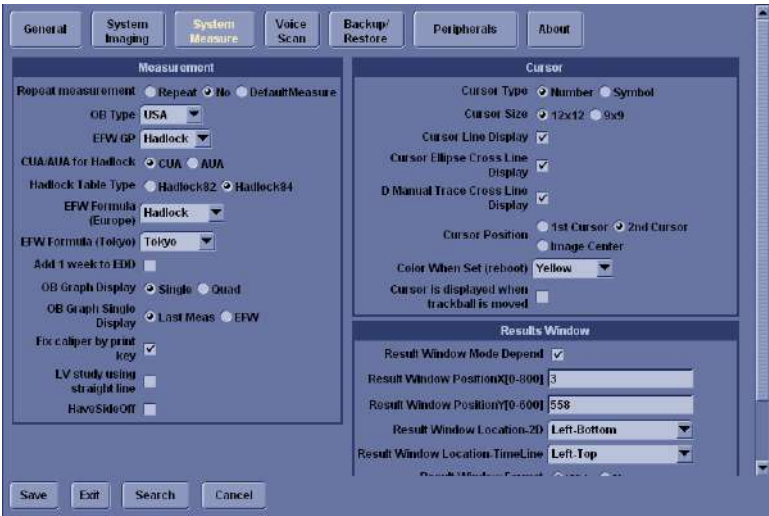


Figure 16-11. System/System Measure Preset Menu

System/System Measure Preset Menu (continued)

Table 16-12: Measurement

Preset Parameter	Description
Repeat Measurement	Select Off, Repeat, DefaultMeas
OB Type	Select which OB measurements and calculations studies to use: USA, Europe, Tokyo, Osaka, or ASUM.
EFW GP	Select the source used to calculate EFW-GP Estimated Fetal Weight-Growth Percentile)
CUA/AUA for Hadlock	Select to use CUA (Composite Ultrasound Age) or AUA (Average Ultrasound Age) as the default
Hadlock Table Type	Select Hadlock 82 or Hadlock 84 tables
EFW Formula (Europe)	Select the source used to calculate EFW (Europe) (Estimated Fetal Weight), Shepard, Merz, Hadlock, German, Rich
EFW Formula (Tokyo)	Select the source used to calculate EFW (Tokyo) (Estimated Fetal Weight)
Add 1 week to EDD	Select to add additional week to estimated date of delivery
OB Graph Display	Select Single or Quad for displaying OB Graphs.
OB Graph Single Display	Select Last Meas or EFW Single OB Graph displayed by default.
Fix Caliper by Print key	Select to use the Print key like the Set key. Note that if you select this, during a generic volume measurement, the print key does not function like the Set key, but instead ends the measurement sequence and initiates the volume calculation based on the number of measurements taken so far.
LV Study using straight line	Sets straight line as the default for 2D LV studies.
Have Side Off	Select to deactivate right/left side.

Table 16-13: Cursor

Preset Parameter	Description
Cursor Type	Select whether to mark measurements with numbers or symbols.
Cursor Size	Specify 12x12 or 9x9.
Cursor Line Display	If selected, after you press Set to complete a measurement, the cursor line is displayed. If not selected, after you press Set to complete a measurement, only the cursor number or symbol is displayed.
Cursor Ellipse Cross Line Display	Check box to display the cross line in Ellipse.
D Manual Trace Cross Line Display	Check box to display the cross line with the caliper.
Cursor Position	Select 1st Cursor, 2nd Cursor, or Image Center.
Color When Set (reboot)	Select white, yellow, bright red, or orange.

Table 16-13: Cursor (Continued)

Preset Parameter	Description
Cursor is Displayed when Mouse is Moved	The active cursor does not display until you move the mouse. This assumes the following presets are set: Repeat Measurement, Repeat, Default Measurement, and Cursor.

Table 16-14: Results Window

Preset Parameter	Description
Result Window Mode Depend	Select this if you want the measurement result window to be repositioned, depending on the mode.
Result Window Position X[0-800]	You can set the coordinates for the measurement result window when you do not have the result window set to be mode dependent. This is the X coordinate (left/right)
Result Window Position Y[0-600]	You can set the coordinates for the measurement result window when you do not have the result window set to be mode dependent. This is the Y coordinate (up/down)
Result Window Location-2D	Select the Result Window location on the Monitor Display: Left-Bottom, Left-Top, Right-Bottom, or Right-Top.
Result Window Location-Timeline	Select the Result Window location: Left-Bottom, Left-Top, Right-Bottom, or Right-Top.
Result Window Format	Select Wide or Narrow.
Font Color (reboot)	Select White, Off White, Yellow, Bright Red or Orange (reboots the system)
Font Size - Default Resolution (For CRT)	Select small, medium, or large (reboots the system)
Font Size - High Resolution (For LCD)	Select small, medium, or large (reboots the system)

VoiceScan Preset Menu

Please refer to the VoiceScan section in Chapter 6 for more information.

System/Backup and Restore Preset Menu

The backup and restore procedures described in this section are divided into two parts. The first part describes procedures to backup and restore patient data. The second part describes procedures to backup and restore system and user-defined configurations.



GE Medical Systems is not responsible for lost data if the suggested backup procedures are not followed and will not aid in the recovery of lost data.

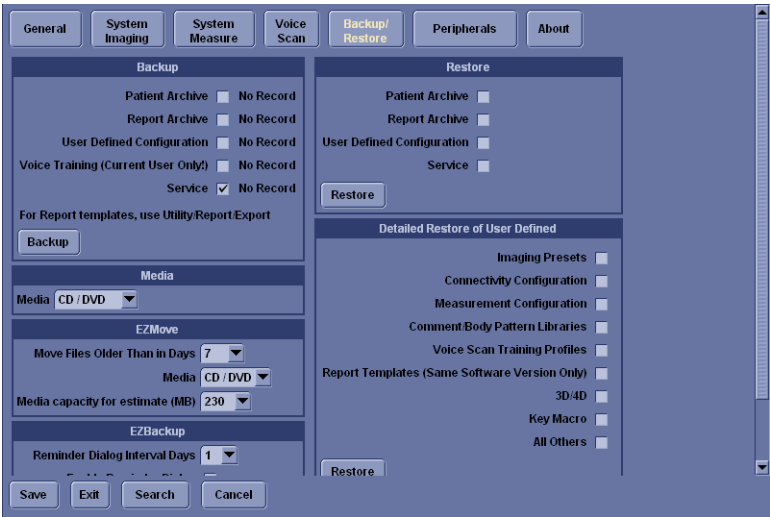


Figure 16-12. System/Backup/Restore Preset Menu

System/Backup and Restore Preset Menu (continued)

Table 16-15: Backup

Preset Parameter	Description
Patient Archive	Select to back up patient data.
Report Archive	Select to back up report data.
User Defined Configuration	Select to back up the user-defined configuration settings.
Voice Training (Current User Only)	Select to back up the current user's voice training.
Service	Select to back up Service (iLinq and Network) settings.
Backup	Select to begin the backup.
For Report Templates, Use Utility/Report/Export	Go to Utility-->Reports-->Export Templates to export report templates.

Table 16-16: EZBackup/EZMove

Preset Parameter	Description
EZMove	
Move Files Older Than in Days	The system will move (and delete) images older than the number of days specified here. If you enter a zero (0), then all of the images from today on will be moved.
Media	Select media type.
Media capacity for estimate (MB)	Specify the capacity of the backup media.
EZBackup	
Reminder Dialog Interval days (EZBackup only)	Specify the number of days after the last backup that you want the system to prompt you to perform an EZBackup/EZMove procedure (only for moving images).
Enable Reminder Dialog (EZBackup only)	Select to activate the EZBackup/EZMove reminder pop-up dialog.
Media	Select media type.
Media capacity for estimate (MB)	Specify the capacity of the backup media.

Table 16-17: Media

Preset Parameter	Description
Media	Select media type to use for backup and restore.

Table 16-18: Restore

Preset Parameter	Description
Patient Archive	Select to restore patient data.
Report Archive	Select to restore report data.
User Defined Configuration	Select to restore the user-defined configuration settings.
Service	Select to restore service iLinq and Network settings. CAUTION: DO NOT restore service presets on to a different LOGIQ 9 system. Only restore service presets to the same system.
Restore	Select to begin the restore process for the selected configuration files.

System/Backup and Restore Preset Menu (continued)

The detailed section of this menu allows you to restore one area at a time from the user defined configuration. This allows you to selectively restore what you want to restore across multiple machines. Check the box(es) you want to restore, insert the CD-ROM/MOD, and press Restore.

Table 16-19: Detailed Restore

Preset Parameter	Description
Imaging Presets	Select to restore imaging presets.
Connectivity Configuration	Select to restore connectivity configurations.
Measurement Configuration	Select to restore measurement configurations.
Comment/Body Pattern Libraries	Select to restore comment and body pattern configurations.
VoiceScan Training Profiles	Select to restore VoiceScan training (all users).
Report Templates (Same Software Version Only)	Select to restore Report templates.
3D/4D	Select to restore 3D/4D settings.
Key Macro	Select to restore Key Macro.
All Others	Select to restore all other configurations not listed in the Detailed Restore section. This includes parameters defined on the System preset menus.
Restore	Select to begin the restore process for the selected configuration files.

Backup and restore strategy: patient data

To minimize accidental loss of data, perform backup of the patient archives stored on the local hard drive **DAILY** as described in this section. Use a formatted Backup/Restore disk to back up patient archives from the hard drive, using the backup procedure described in this section. Data from the Backup/Restore disk may be restored to the local hard drive using the restore procedure.



The restore procedure overwrites the existing database on the local hard drive.

NOTE: *To perform backup and restore procedures, you must login with administrator privileges.*

Backup procedure: patient data

1. Insert media into the media drive.

NOTE: For information about formatting media, see “Formatting removable media”.

2. On the Touch Panel, press **Utility**.
3. On the Utility Touch Panel, press **System**.
4. On the monitor display, select Backup/Restore.
The Backup/Restore screen is displayed.

NOTE: If you are not logged in with administrator privileges, the Operator Login window is displayed. Log on with administrator privileges.

5. In the Backup list, select Patient Archive and Report Archive.
6. In the Media field, select CD.
7. Select Backup.

The system performs the backup. As it proceeds, status information is displayed on the Backup/Restore screen.

EZBackup/EZMove

EZBackup/EZMove allows you to manage hard disk space (move images off the hard drive) while maintaining the patient database on the scanner, as well as to back up the patient database and images.



HINTS

PLEASE READ THIS

Ensure that you have established a data management protocol for your office/institution. You **MUST** manage the backup media by keeping a log and by creating a media filing system.

For example, if you need to back up 500 MB/day, or 2.5 GB/week, then you need to back up 5 CDs/week, or ~250CDs/year.

Generally speaking, you should back up the system when you have 10 GB of images to back up.

You should assign the person who is in charge of performing the backups. Backups will vary by the volume of your work. You need to track how long it takes your office/institution to get to 10 GB, and set the back-up parameters accordingly.

Your office/institution needs to determine your backup strategy, for instance, backup weekly and move monthly. It should be an easy strategy to perform and to remember. And follow this same strategy/schedule consistently.

It's also useful to keep your more recent information on the hard drive since it's easier to recall that way.



CAUTION

DO NOT use DVD-RAM for EZBackup/EZMove.

Use a CD or DVD-R for EZBackup/EZMove.

EZBackup/EZMove (continued)



If you use EZBackup/EZMove as a “true” patient archive, you must maintain a separate backup of the patient database. If for any reason the Local Archive - Int HD gets corrupted or the base system software has to be reloaded, then the patient archive is the **ONLY** way to rebuild the EZBackup/EZMove patient archive.



After copying the image file to the media, EZMove deletes the image file from the Local HD.

EZBackup does not back up the exams which were previously backed up once by EZBackup or Export. If you want to back up those exams by EZBackup again, select “Full Backup Option” on the EZBackup wizard. Even when “Full backup” is selected, the system only backs up the exams within the range specified in the Utility -> System -> Backup/Restore -> Backup Files Older Than in Days preset.



“Archived” information is saved to each exam. When you perform EZBackup or Export, the system backs up the exams except for the archived exam.



When EZBackup requires more than one disk (CD-R or DVD-R) for backup, a message appears when the first disk is full. If you select “Cancel” to stop the backup procedure and later try EZBackup again, all the data may not be backed up.

Select “Full Backup” on the first EZBackup wizard screen if the last time you were performing EZBackup you selected “Cancel”.

EZBackup/EZMove (continued)



DO NOT turn off the power while EZBackup is running. The data may be lost. It may take several hours for EZBackup to finish, depending on the amount of data being backed up.

The following may give the impression of a lockup, but EZBackup is continuing in the background.

- The progress bar does not move.
- The screen may become white.
- The hourglass icon keep turning.

Basically, when you perform the EZBackup/EZMove procedure, you insert the media (or connect USB HDD if applicable), the system backs up/moves the images, and creates a reference between the patient database and the media's volume.

1. Prepare unformatted media or the USB HDD before starting EZBackup/EZMove.

NOTE: *BEFORE starting the EZBackup, select "Unlock All" in Utility --> Admin --> Logon.*

2. Specify the EZBackup/EZMove setup on the Utility --> System --> Backup/Restore page.

EZBackup/EZMove (continued)

- To start the EZBackup/EZMove procedure, go to the Patient menu and select the EZBackup/EZMove button at the bottom of the Patient list. The EZBackup/EZMove Wizard starts.

The screenshot shows the GE Healthcare software interface. On the left is a sidebar with icons for 'Patient' and 'Data Transfer'. Below these are buttons for 'New Patient', 'Register', and 'Details'. At the bottom of the sidebar is an 'EZBackup' button. The main window has a 'Patient ID' field and a 'Last Name' field. Below these are fields for 'First Name', 'Middle Name', 'DOB', 'Age', and 'Sex' (radio buttons for 'female' and 'male'). There are tabs for 'ABD', 'OB', 'GYN', 'CARD', 'VAS', 'UR', 'SM P', and 'PED'. The 'ABD' tab is selected. Below the tabs are fields for 'Height' (cm), 'Weight' (kg), 'BSA' (m^2), 'Accession #', 'Exam Description', 'Perf. Physician', 'Ref. Physician', and 'Operator' (set to 'ADM'). There are 'Images' and 'Clear' buttons. Below this is a 'Patient View' section with a search key dropdown set to 'Patient ID' and a 'string' field. Below the search is a table of patient records.

Patient ID	Last Name	First Name	Birthdate	Sex	Last Exam	Img. size
062706-023032-1				N	06/27/2006	2.08 MB
062706-035413-1				N	06/27/2006	12.5 MB
062706-091421-1				N	06/27/2006	14.9 MB
062706-092629-1				N	06/27/2006	3.44 MB
062906-024352-1				N	06/29/2006	13.4 MB
062906-022657-1				N	06/29/2006	None
063006-073023-1				N	06/30/2006	24.3 MB
063006-065434-1				N	06/30/2006	143 MB
063006-103129-1				N	06/30/2006	21.7 MB
070406-062252-1				N	07/04/2006	171 MB
070406-095630-1				N	07/04/2006	27.2 MB
070606-065454-1				N	07/06/2006	3.64 MB

At the bottom of the table are buttons for 'Review', 'New Exam', 'Delete', and 'Lock'. There is also a 'Show:' dropdown set to 'All' and a page indicator '26 / 26'.

Figure 16-13. New Patient EZBackup/EZMove Example

NOTE: *EZBackup/EZMove can take up to 20 minutes (or longer, depending on the size of the backup). Make sure to schedule this at the same time daily, when no patients are scheduled.*

EZBackup/EZMove (continued)

4. Verify the information on the first page of the EZBackup/ EZMove Wizard, then press *Next*.

Full backup options display on the first page of the EZBackup wizard. If you want to backup all of the exams in the range (even if the exam was previously backed up, check this option).

NOTE: *You can set the range in Utility --> System --> Backup/ Restore --> Backup files older than in days.*

If you uncheck this option, the system only backs up exams which have not yet been backed up.

NOTE: *If you update an exam which is already backed up, the exam is also backed up.*



Figure 16-14. EZBackup/EZMove Wizard, Page 1

NOTE: *You do not need to format the media prior to performing an EZBackup/EZMove. You cannot use formatted media for EZBackup/EZMove.*

EZBackup/EZMove (continued)

NOTE: *EZBackup/EZMove cannot span a single image across two (2) or more media. Therefore, if EZBackup/EZMove encounters an image that is greater than the capacity of the media, it skips the oversized image.*

NOTE: *EZBackup/EZMove does not store images to media in sequential order. Instead it maximizes the most amount of images per media.*

NOTE: *The calculation for the number of backup CDs is only an estimate. Allow for one additional CD when performing an EZBackup/EZMove.*

NOTE: *If you use the USB HDD, some wizards and the pop-up messages DO NOT appear.*

5. Verify the information on the EZBackup/EZMove Wizard, Page 2. The backup may span multiple media. This page tells you how many media you need to do this backup. After you have gathered the media (allow for one extra media, just in case), you are ready to begin the backup. Press Next.

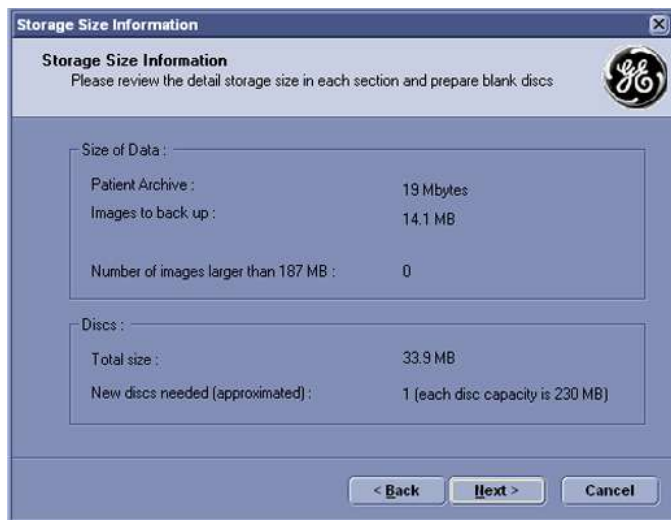


Figure 16-15. EZBackup/EZMove Wizard, Page 2

EZBackup/EZMove (continued)

- 6. A pop-up message appears that provides you with the media label. Label the media, then insert the media. Press OK.

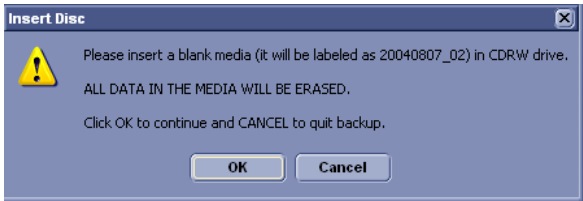


Figure 16-16. Insert Media Message

- a. Ensure that you label the media with not only the volume name indicated on the Insert Media Message, but with the name of the LOGIQ 9 system where this backup/move procedure was done.
- b. Update the EZBackup/EZMove log with this information the volume information and the location of the media.
- c. After the backup/move has been completed, file the media.

Table 16-20: Typical EZBackup/EZMove Log

Date	Scanner ID Name	Backup Images Y/N	Older than __ Days	Move Images Y/N	Media Label (and Scanner ID)

NOTE: You can cancel the backup/move at any time. The system completes backing up the current media and then cancels the operation.



CAUTION Only cancel the backup/move in case of an emergency.

EZBackup/EZMove (continued)

NOTE: *This message appears if you press Next without inserting the backup media: "Please insert a blank media...". Insert the media and continue.*

7. The status menu appears. When the backup/move has been completed, press *Next*.

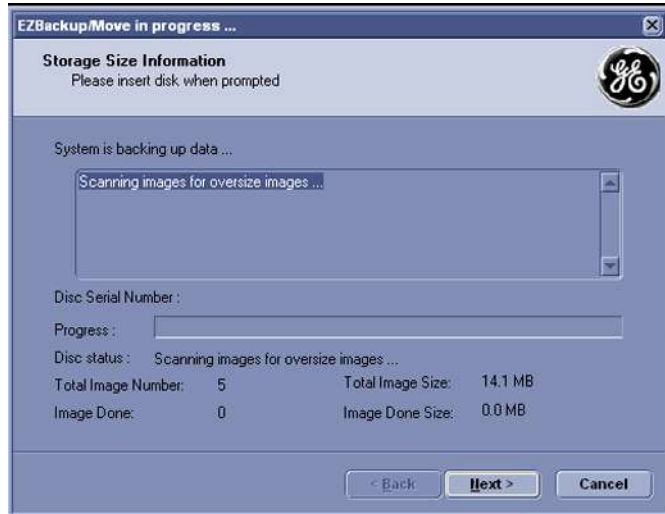


Figure 16-17. EZBackup Wizard Page 3

NOTE: *When/if you need to insert the next media, a message appears providing you with the media label. Label the media, then insert the next media and press OK.*

EZBackup/EZMove (continued)

- When the backup is complete, the completed wizard page appears. Press *Finish*.

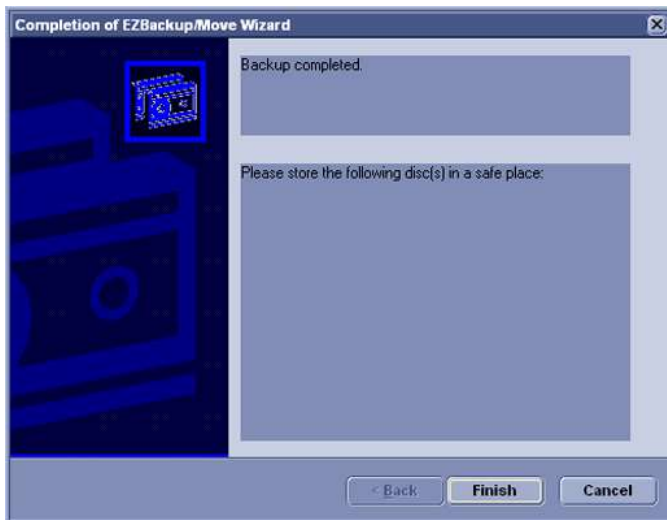


Figure 16-18. EZBackup/EZMove Wizard, Page 4

- Do a patient archive after each EZBackup/EZMove (move).

NOTE: Use *Import* to restore EZBackup images.



CAUTION NEVER restore the patient archive from media made previous to the last move. You can still do a backup/move daily; but ALWAYS do a patient archive backup after each move.

To View Backed Up/Moved Images

You can view backed up media via the Patient Menu, Import, and the DICOM CD View dataflow.

1. Select the patient on the Patient Menu (on the same system where the backup/move was performed).
2. Insert the media volume indicated on the Patient Menu.
3. View the exam from the media.

NOTE: You may need to insert a media volume prior to or after the recommended media.

NOTE: If the patient is split over multiple media, images on the previous or next media are displayed as triangles.

NOTE: To view the whole patient on the system, use Import, from as many media as you have for that patient. However, take care not to import studies over existing studies; duplicate or missing images may result. Delete the existing exam first.

Restore procedure: patient data



The restore procedure overwrites the existing database on the local hard drive. Make sure to insert the correct CD.

1. Insert the Backup/Restore CD into the CD-RW drive.
2. On the Touch Panel, press **Utility**.
3. On the Utility Touch Panel, press **System**.
4. On the monitor display, select Backup/Restore.

The Backup/Restore screen is displayed.

NOTE: *If you are not logged in with administrator privileges, the Operator Login window is displayed. Log on with administrator privileges.*

5. In the Restore list, select Patient Archive.
6. In the Media field, select the Backup/Restore CD.
7. Select Restore.

The system performs the restore. As it proceeds, status information is displayed on the Backup/Restore screen.

Backup and restore strategy: user-defined configurations

In addition to generating a safety copy, the backup/restore function of the user-defined configuration (presets) can be used to configure several LOGIQ 9 systems with identical presets (preset synchronization).

Preset synchronization

The procedure for preset synchronization of several scanners is as follow:

1. Make a backup of the user-defined configurations on a removable media from a fully configured LOGIQ 9 system.
2. Restore user-defined configurations from the removable media to another LOGIQ 9 system (you can restore all the user-defined presets or select specific presets to restore via Detailed Restore).

Backup procedure: user-defined configurations



To avoid the risk of overwriting the local patient and report archives on the system to update, DO NOT check Patient Archive when making a backup of configuration settings.

1. Insert media into the media drive.
2. On the Touch Panel, press **Utility**.
3. On the Utility Touch Panel, press **System**.
4. On the monitor display, select Backup/Restore.
The Backup/Restore screen is displayed.

NOTE: *If you are not logged in with administrator privileges, the Operator Login window is displayed. Log on with administrator privileges.*

5. In the Backup list, select User Defined Configuration.
6. In the Media field, select CD.
7. Select Backup.
The system performs the backup. As it proceeds, status information is displayed on the Backup/Restore screen.
8. After the backup is complete, press **F3** to eject the CD.

Restore procedure: user-defined configurations



The restore procedure overwrites the existing database on the local hard drive. Make sure to insert the correct CD-ROM/MOD.



To avoid the risk of overwriting the local patient and report archives, DO NOT check Patient Archive when restoring user-defined configurations.

1. Insert the CD with the user-defined configurations into the CD-RW drive.
2. On the Touch Panel, press **Utility**.
3. On the Utility Touch Panel, press **System**.
4. On the monitor display, select Backup/Restore.
The Backup/Restore screen is displayed.

NOTE: *If you are not logged in with administrator privileges, the Operator Login window is displayed. Log on with administrator privileges.*

5. In the Restore list, select from the following, depending on which configurations you want to restore:
 - User Defined Configuration,
OR, in the Detailed Restore section select one of the following:
 - Imaging Presets
 - Connectivity Configuration
 - Measurement Configuration
 - Comment/Body Pattern Configuration
 - All Others
6. In the Media field, select the system and user-defined configurations CD.
7. Select Restore.

The system performs the restore. As it proceeds, status information is displayed on the System/Backup/Restore screen.

System/Peripherals Preset Menu

The System/Peripherals screen allows you to specify video and system setup parameters.



Figure 16-19. System/Peripherals Preset Menu

Table 16-21: Video Settings

Preset Parameter	Description
Format	Select the video format, PAL or NTSC.
Mixed Mode	Allows you to select the PAL format, but still operate the color printer in NTSC mode. This setting has no effect on NTSC mode.

Table 16-22: VCR

Preset Parameter	Description
VCR / DVR	Select the type of VCR / DVR, or no VCR / DVR.

System/Peripherals Preset Menu (continued)

Print and Store Options. Press Print and Store Options to go to the Utility --> Connectivity --> Miscellaneous setup page.

Removable Media. Press Removable Media to go to the Utility --> Connectivity --> Removable Media page.

Table 16-23: Setup

Preset Parameter	Description
Printers	Press this key to add an additional standard printer via the USB serial port and to configure digital printers. This activates the Windows Add Printer wizard. NOTE: Most printer drivers are available via Windows; however, newer printers may require you to load the manufacturer-supplied print driver (must be on CD-ROM). Refer to the Basic Service Manual for more information.
Print Full Screen	Select this if you want the standard printer to print the full screen.
Enable Video Invert	Select this is you want the standard printer to print black on white rather than white on black.

System/About Preset Menu

The System/About screen lists information about the system software.

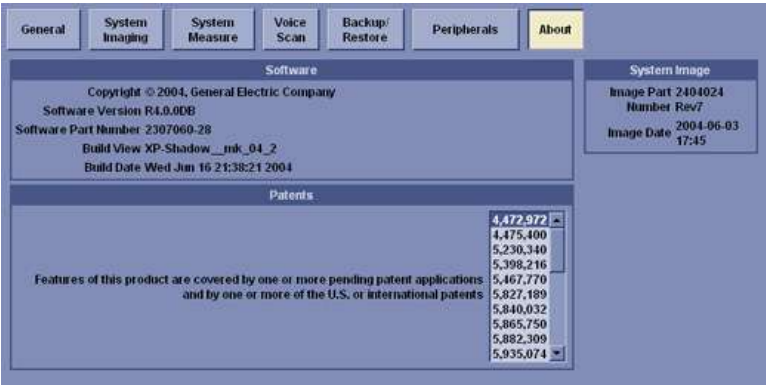


Figure 16-20. System About Preset Menu

Table 16-24: Software

Preset Parameter	Description
Software Version	The current software version on this system.
Software Part Number	The software part number.
Build View	The software build view.
Build Date	The software build date.

Table 16-25: System Image

Preset Parameter	Description
Image Part Number	The image part number (ghost part number).
Image Date	The image date (ghost date).

Table 16-26: Patents

Preset Parameter	Description
Patents	Lists system patents.

Imaging Presets

Overview

Imaging screens allow you to specify parameters for the following:

- B-Mode (B)
- Color Flow Mode (CF)
- Power Doppler Imaging (PDI)
- M-Mode (M)
- Pulse Wave Mode (PW)
- Continuous Wave Mode (CW)
- Harmonics (HAR)
- B Flow (BF)
- B Contrast
- TruAgent Detection
- General

Changing imaging presets

To change imaging presets:

1. On the Touch Panel, select **Utility**.
2. On the Touch Panel, select **Imaging**.
The system displays the Imaging screens. See the Imaging screens on the following pages.
3. In the Preset list, select the exam.
4. In the Probe list, select the probe.
5. In the row across the top of the screen, select the mode.
The system displays two sets of parameters and settings. The left column lists all settings for the exam (for example, Abdomen). The right column lists settings that apply only to the exam and probe combination.
6. To change a parameter, do one of the following:
 - Select the value from a list
 - Select one value from a choice of two or more buttons
 - Select or clear a check box
7. After changing the parameters, to save the changes, select the Save button.

NOTE: *When you Save changes to imaging parameters, the system saves changes to all modes, not just the mode currently displayed.*

NOTE: *If you have problems with imaging, you can return parameters back to the original settings. Select the exam, probe, and mode, and then select Reload Factory Defaults. The system returns the selected parameters to the original settings.*

For information about the specific parameters, refer to Chapter 5 Optimizing the Image.

LOGIQ 9 Imaging Presets

B-Mode

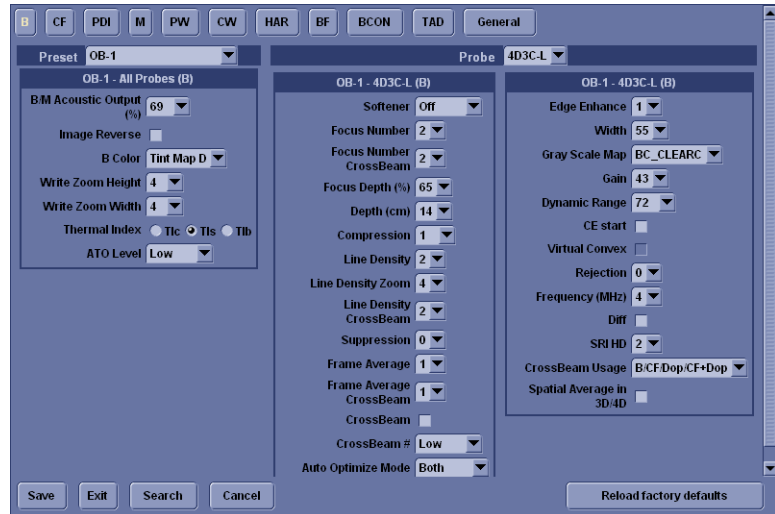


Figure 16-21. B-Mode parameters

Color Flow Mode

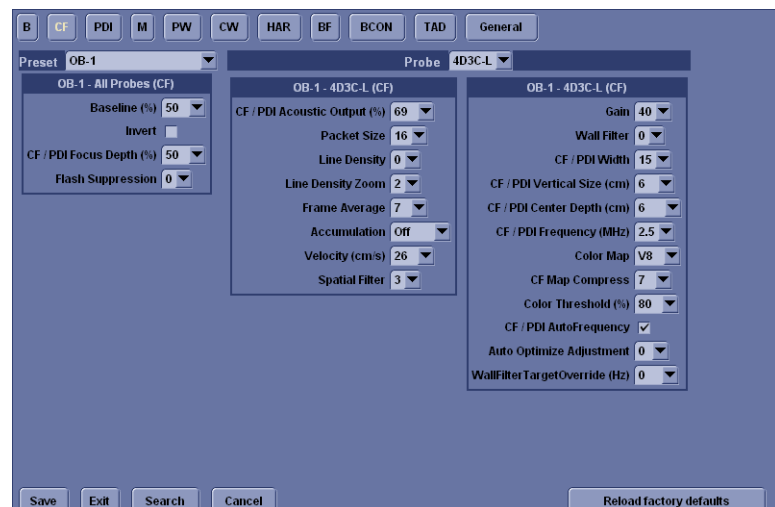


Figure 16-22. Color Flow parameters

Power Doppler Imaging (PDI)

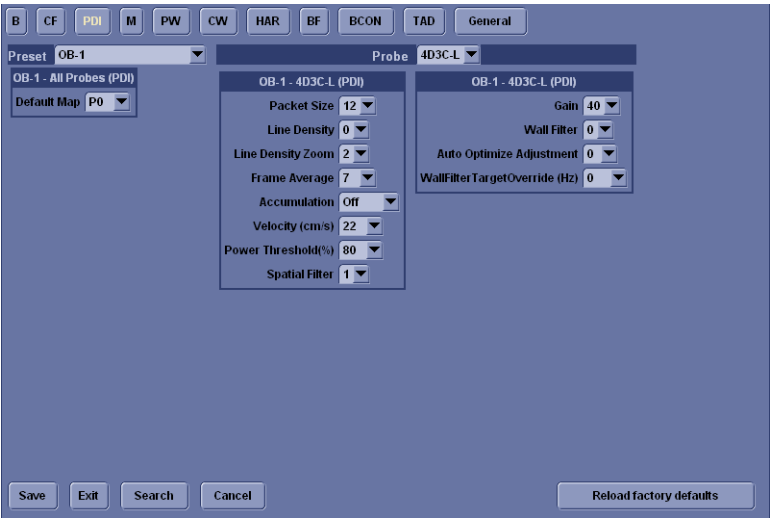


Figure 16-23. PDI parameters

M-Mode

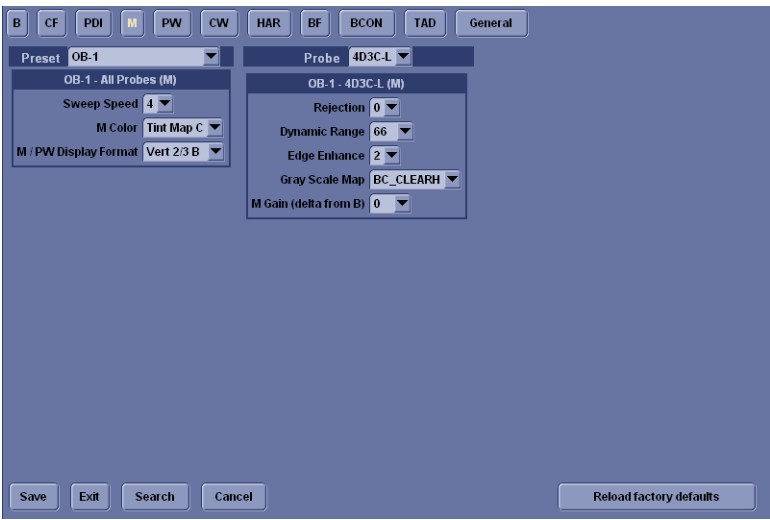


Figure 16-24. M-Mode parameters

Pulse Wave (PW)

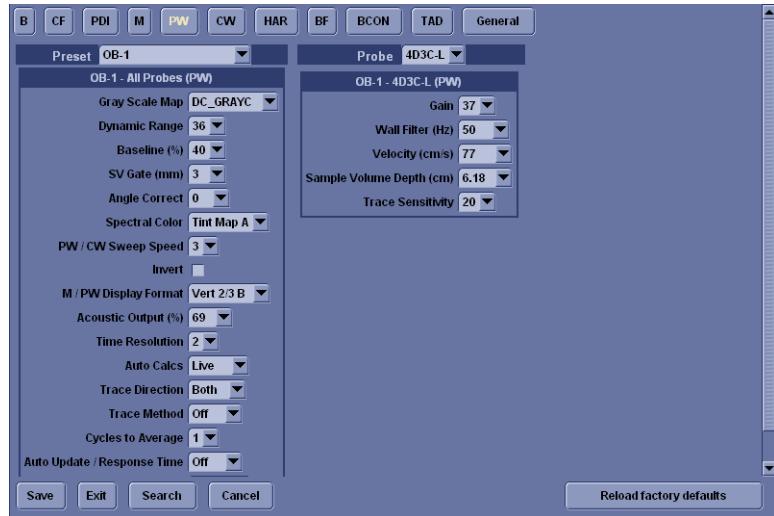


Figure 16-25. PW parameters

Continuous Wave (CW)

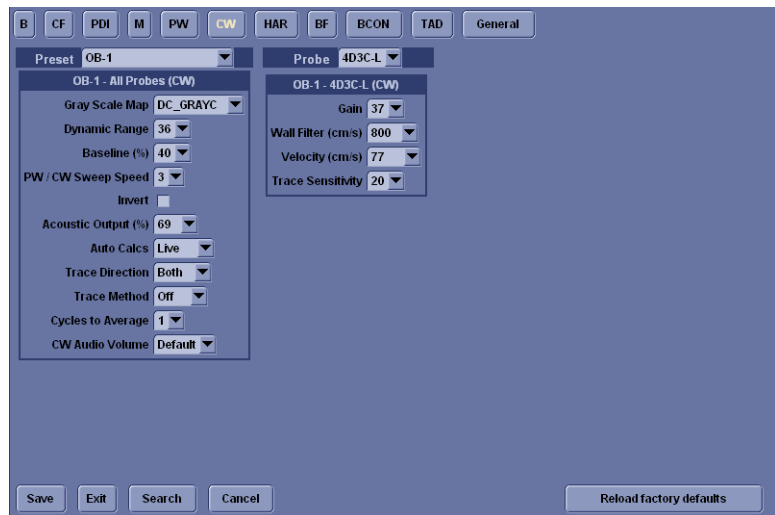


Figure 16-26. CW parameters

Harmonics

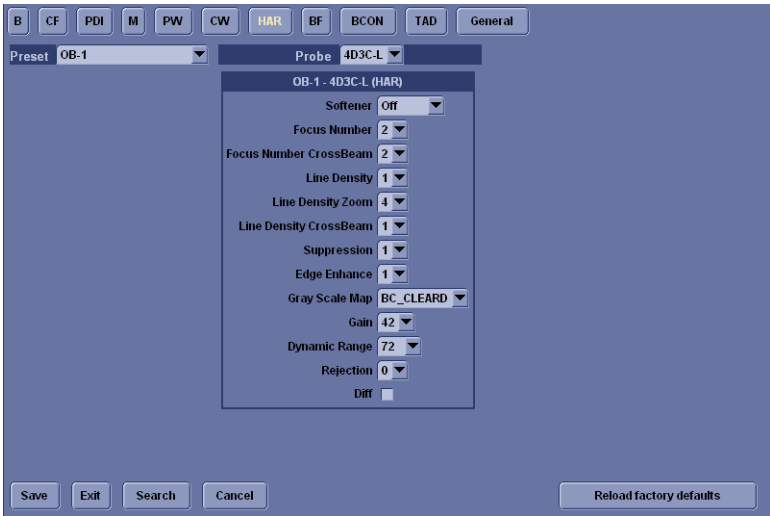


Figure 16-27. Harmonics parameters

B Flow

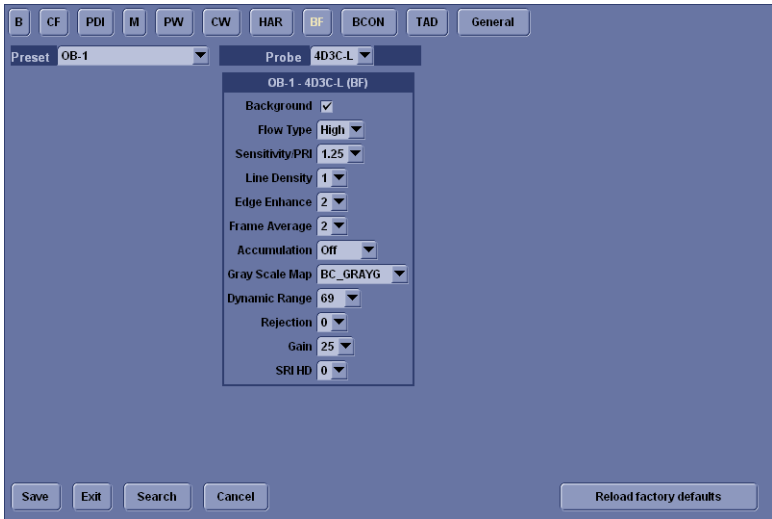


Figure 16-28. B Flow parameters

B Contrast

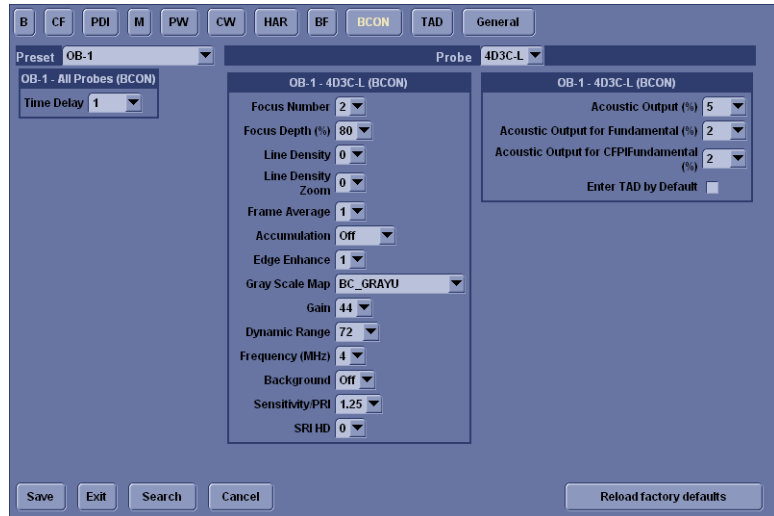


Figure 16-29. B Contrast parameters

TruAgent Detection

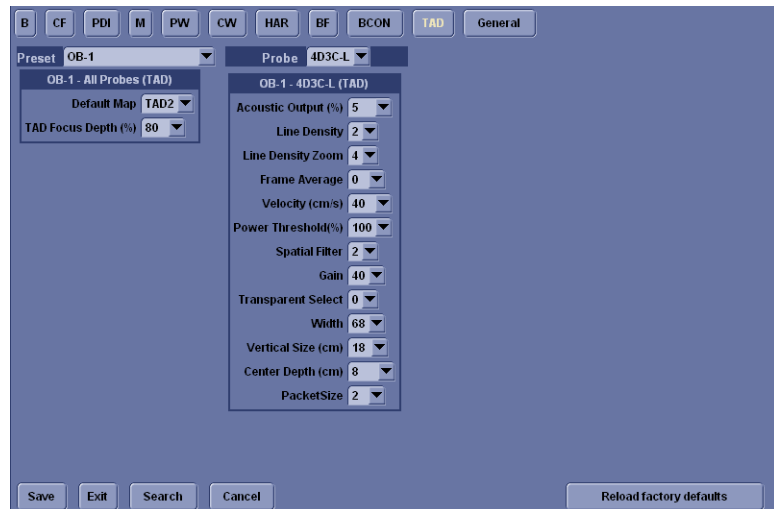


Figure 16-30. TruAgent Detection Imaging Presets

General

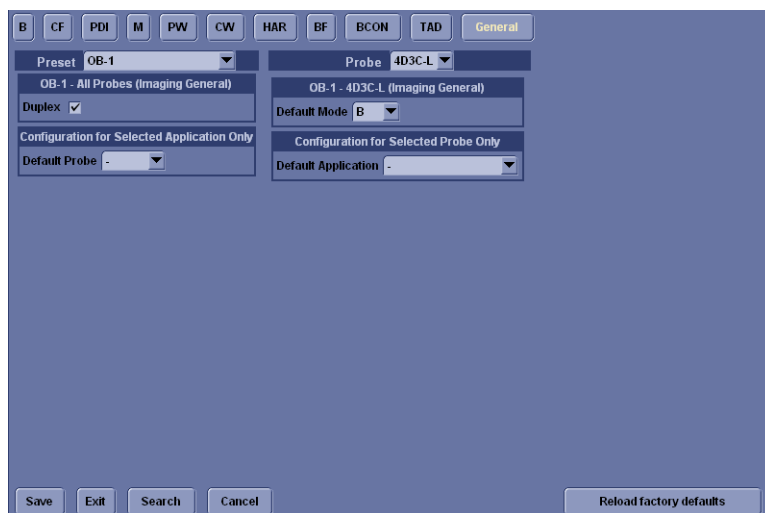


Figure 16-31. General parameters

You can specify a default probe per application and a default application per probe.

Default probe per application

1. To specify a default probe per application, select Utility --> Imaging --> General.
2. Under Preset, specify the desired probe from the pull-down menu.

Default probe per application

1. To specify a default application per probe, select Utility --> Imaging --> General.
2. Under Probe, specify the desired application from the pull-down menu.

Comments Libraries Presets

Overview

Comment screens allow you to specify comment text and pointer options, to define comment libraries, and assign comment libraries to applications.

Comments Libraries/Libraries Preset Menu

On the comments *Libraries* tab, you can change and create comment libraries. A comment library is a list of comments that are associated with a specific application. The comments are listed in the library in the order in which they display on the Touch Panel. For each library, you can define two Touch Panel displays of comments (Page1 and Page2), with 15 comments on each Touch Panel.

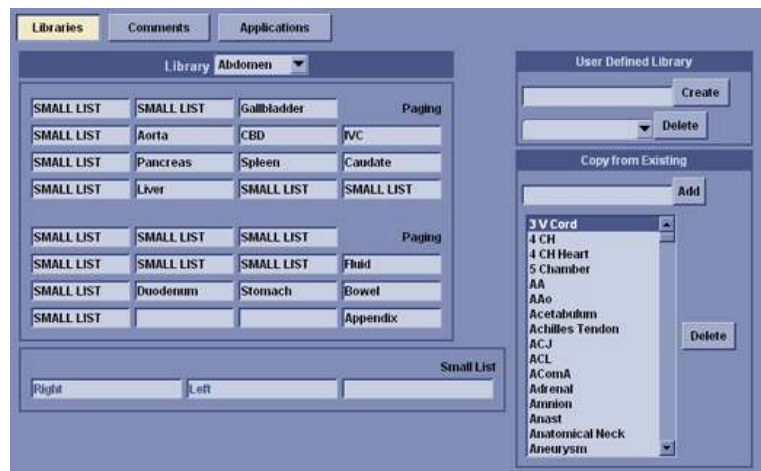


Figure 16-32. Comment Libraries Preset Menu

Comments Libraries/Libraries Preset Menu (continued)

Table 16-27: Libraries

Preset Parameter	Description
Library	The name of the comment library.
Paging	The first (or second) Touch Panel page of comments for the selected library.
Small List	Fields where you define a small list.
User Defined Library	The name of a new comment library that you want to create/delete.
Copy from Existing	You can add to or delete from the selection of comments.

Defining Comments

1. In the *Library* field, select the library you want.
The system displays all comments for the library. You can have two Touch Panel displays of comments for each library. The comments are listed in the order that they are shown on the Touch Panel when you use comments.
2. To change or add an comment, select the comment or blank location and press **Set**, then do one of the following:
 - Type the comment.
 - Select the comment in the *Copy from Existing* list, and press **Set**.
3. To save the changes, select the Save button.

Creating a new comments library

1. In the *User Defined Library* field, type a name for the library, then select Create.
The system creates a new library.
2. Enter comments as described in step 2 above.
3. To save the changes, select the Save button.

Deleting a user defined library

1. Select the library name which you want to delete from the pull-down menu.
2. Press **Delete**.
3. Press **Save** to save the changes.

Creating a small list

A small list is a list of up to three comments attached to one comment location on the Touch Panel. You can use a small list to group similar comments, such as those indicating a probe location. For example, you can specify that a small list include the following comments: Long, Transverse, and Coronal. To make comments easier to use, you can define the small list in the same location in each comment library.

To define a small list:

1. Move the **Trackball** to the comment field on Page1 or Page2 where you want to create a small list, and press **Set**.
2. Move the **Trackball** to the first field in the *Small List* section, and press **Set**.
3. To enter comments in the fields in the Small List section, select the field and press **Set**, then do one of the following:
 - Type the comment
 - Select the comment in the Copy from Existing list, and press **Set** twice.

You can enter up to three comments. When you enter an comment in the first field of the Small List section, the selected comment field on Page1 or Page2 changes to SMALL LIST.

4. To save the changes, select the Save button.

NOTE: *The small list can be displayed as a pop-up window or as a toggle field. The Small List Operation field on the General tab allows you to specify how it is displayed.*

Comments Libraries/Comments Preset Menu

On the Comments tab, you specify text and pointer options.



Figure 16-33. Comment/Comments Preset Menu

Comments Libraries/Comments Preset Menu (continued)

Table 16-28: Text

Preset Parameter	Description
Text Font Size	Specify the font size. The font size increases as the number increases.
Text color	Select the color for comment text.
Text Boundary	Select Group Move or Word Wrapping.
Small List Operation	Select whether you want small list options to display in a pop-up window or by a toggle function.
Enable Type Over Mode	Select to type over existing comments. Position the cursor over the text to be changed, then start typing.
Reset Small List	Select to indicate that small lists should not be reset to the first item.

Table 16-29: Arrow

Preset Parameter	Description
Arrow Length	Select the default pointer length.
Arrow Size	Select the default pointer size.
Keep Arrow Angle	Keep the angle of arrow pointer head until next change.

Table 16-30: General

Preset Parameter	Description
Retain while entering or leaving timeline mode	If selected, the system keeps the comment(s) on the monitor display when you enter or leave timeline mode.
TextOverlay in Multiple Image	When selected, and you select the F8 key to hide or show comments, if you are in multiple image, the system hides the text in both images. When cleared, the system only hides the text for the active image.
Erase When the image is unfrozen	Deletes comments when you unfreeze the image.
Erase When the probe or application is changed	Deletes annotations when you change the application or probe.

After you change comment options, select **Save** to save the changes.

Comments Libraries/Applications Preset Menu

The Comments Libraries/Applications tab is a link to the Applications preset menu. The Applications preset screen allows you to specify which libraries belong to an application. You also specify which is the default library that displays when you use comments.



Figure 16-34. Applications/Comments Link

The Applications/Comments screen can be accessed through either the Comments Libraries or Applications Touch Panel keys.



Figure 16-35. Applications/Comments Preset Menu

Table 16-31: Applications

Preset Parameter	Description
Preset	The name of the application preset.
Tabs	A list of libraries for the application. You can select up to six libraries.
Default Tab	The default library that the system displays when you use comments.

Specifying which libraries belong to an application

1. On the Applications tab, in the Application field, select the application.
2. In the Library Group Tabs fields, select the libraries for this application. You can select up to six libraries.
3. In the Default Library Group field, select the default library you want the system to display when you use comments.

NOTE: *When you use comments, the default library is displayed. To use other libraries for the application, press the tab for the library.*

4. To save the changes, select the Save button.

Using comments from a library

To use comments, press the **Comment** key on the Control Panel. Comments are then displayed on the Touch Panel.

To select a comment library, press the appropriate tab (e.g. the tabs are OB23 and OB23_1).

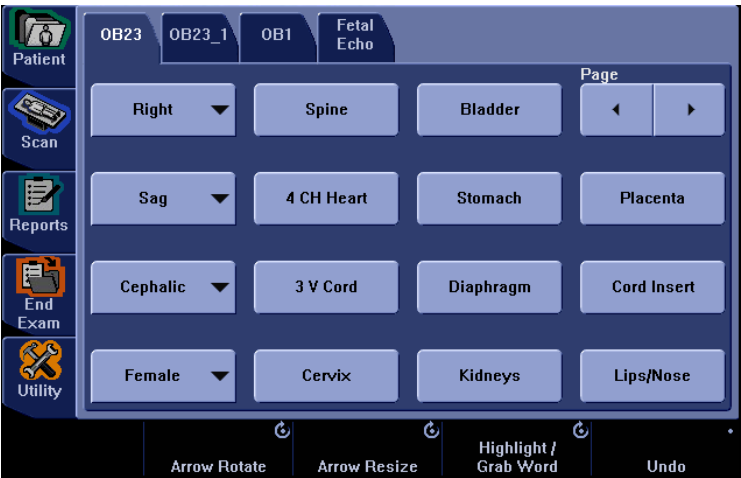


Figure 16-36. OB 2/3 Comments Touch Panel

Body Patterns Presets

Overview

Body patterns screens allow you to specify body pattern options, to define body pattern libraries, and assign body pattern libraries.

Body Pattern Libraries/Libraries Preset Menu

On the Body Patterns Libraries tab, you can change and create body pattern libraries. A body pattern library is a list of body patterns that are associated with a specific application. The body patterns are listed in the library in the order in which they display on the Touch Panel. For each library, you can define two Touch Panel displays of body patterns (Page1 and Page2), with 15 body patterns on each Touch Panel.

Body Pattern Libraries/Libraries Preset Menu (continued)

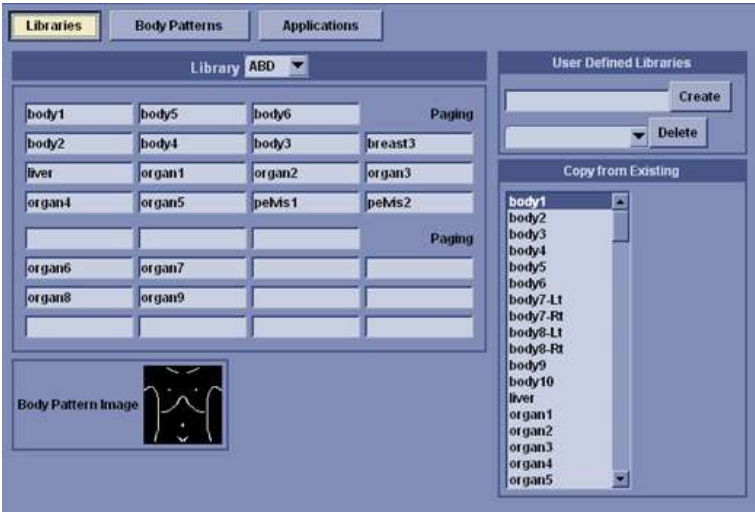


Figure 16-37. Body Patterns Libraries Preset Menu

Table 16-32: Body Patterns Libraries

Preset Parameter	Description
Library	The name of the body pattern application library.
Paging	The first (or second) Touch Panel page of body patterns for the selected library.
Body Pattern Image	Displays the image of the currently selected body pattern.
User Defined Libraries-Create	The name of a new body pattern application library that you want to create.
User Defined Libraries-Delete	Allows the selection of the user defined library to be deleted.
Copy from Existing	A list of body patterns you can use to create an application library.

Defining body patterns

1. In the *Library* field, select the application library you want.
The system displays all body patterns for the library. You can have two Touch Panel displays of body patterns for each library. The body patterns are listed in the order that they are shown on the Touch Panel.
2. To change or add a body pattern, select the body pattern or blank location and press **Set**, then do one of the following:
 - Type the body pattern name.
 - Select the body pattern in the *Copy from Existing* list, and press **Set**.

NOTE: *When you select a body pattern name in a Touch Panel location or in the Copy from Existing list, the system displays the pattern in the lower left corner of the screen.*

3. To save the changes, select the Save button.

Creating a new body pattern library

1. In the New Library field, type a name for the library, then select Create.
The system creates a new library.
2. Enter body patterns as described in step 2 above.
3. To save the changes, select the Save button.

Body Pattern Libraries/Body Patterns Preset Menu

On the Body Patterns tab, you specify body pattern options.

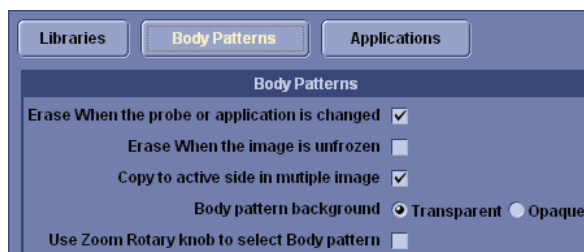


Figure 16-38. Body Patterns General Preset Menu

Table 16-33: Body Patterns

Preset Parameter	Description
Erase When the probe or application is changed	If checked, when you change probes or applications, the system erases the body pattern.
Erase When the image is unfrozen	If checked, when you unfreeze the image, the system erases the body pattern.
Copy to active side in multiple image	If checked, when you use dual B-Mode, the system copies the body pattern to the active side of the dual image.
Body pattern background	Select whether you want the body pattern background to be Transparent or Opaque.
Use Zoom Rotary knob to select Body pattern	If selected, you can scroll through the body patterns with the Zoom control.

After you change body pattern options, select Save to save the changes.

Body Pattern Libraries/Applications Preset Menu

The Body Patterns Library/Applications tab is a link to the Applications preset menu. The Body Patterns Applications tab allows you to select body pattern application libraries. You also specify which is the default library that displays when you use body patterns.



Figure 16-39. Applications/Body Patterns Link

The Applications/Body Patterns screen can be accessed through either the Body Pattern Libraries or Applications Touch Panel keys.



Figure 16-40. Body Patterns Applications Preset Menu

Table 16-34: Applications

Preset Parameter	Description
Preset	Defines the Body Pattern option.
Tabs	A list of body pattern applications.
Default Tab	The default library that the system displays when you use body patterns.

Selecting body pattern application libraries

1. On the Applications tab, in the Application field, select the body pattern.
2. In the Library Group Tabs fields, select the application libraries for Body Patterns. You can select up to six libraries.
3. In the Default Library Group field, select the default application library you want the system to display when you use body patterns.

NOTE: *When you use body patterns, the default library is displayed. To use other application libraries, press the tab for the library.*

4. To save the changes, select the Save button.

Using body pattern application libraries

See the following Body Patterns Small Parts Touch Panel.

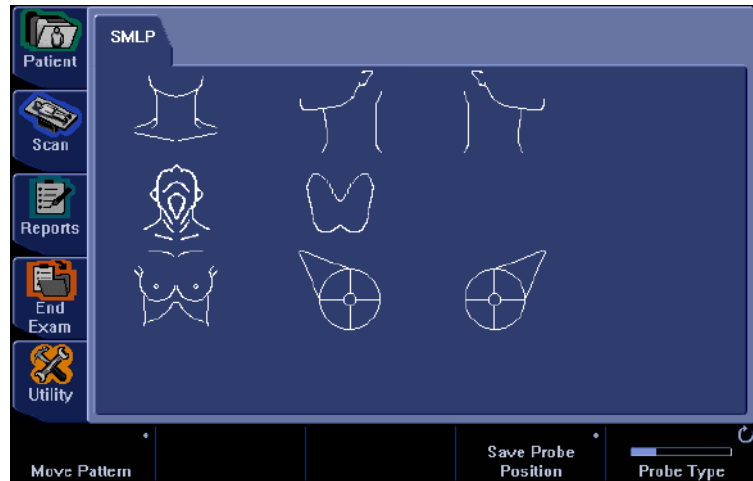


Figure 16-41. Body Patterns Small Parts Touch Panel

To select a body pattern library, select the tabs (for example, ABD or OB).

To select body patterns, use the **Ellipse/Body Pattern** control on the Control Panel.

Application Presets

Overview

Application Settings presets allow you to configure the application-specific settings (presets).

The other two tabs, Comments and Body Patterns were described earlier in this chapter.

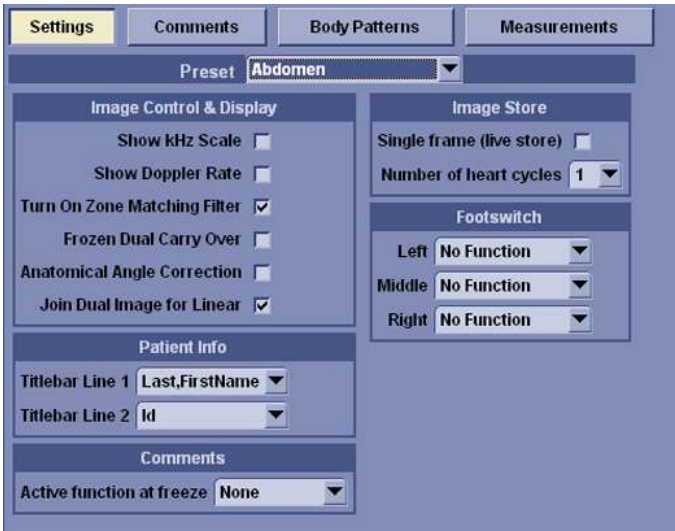


Figure 16-42. Application Settings Preset Menu

Table 16-35: Preset

Preset Parameter	Description
Preset	Select the application that you want to specify the presets. Along with the various applications available on the system, there are four user-defined application presets that can be set.

Overview (continued)

Table 16-36: Image Control & Display

Preset Parameter	Description
Show kHz scale	When selected, displays the kHz scale on the left side of the Doppler spectrum.
Show Doppler Rate	When selected, displays the Doppler rate (mm/s) below the Doppler spectrum.
Turn on Zone Matching Filter	Select to make the image more uniform throughout. Exit the application and re-select the application to activate the zone matching filter. Not recommended for use in applications containing near field hypo-echoic vessels, as in carotid studies.
Frozen Dual Carry Over	Select to carry over the live image's imaging parameters to the other image after pressing Freeze.
Anatomical Angle Correction	Select to keep the angle constant with regard to the anatomy.
Join Dual Image for Linear	Select to place linear probe dual images directly next to each other.

Table 16-37: Patient Info

Preset Parameter	Description
Titlebar Line 1	Select the patient information to display on the scanning screen Title bar.
Titlebar Line 2	Select the patient information to display on the scanning screen Title bar.

Table 16-38: Comments

Preset Parameter	Description
Active function at Freeze	Select None, Body Pattern, or Comments. If Body Pattern or Comment is selected, the Body Pattern or Comment is activated automatically when freezing the system.

Table 16-39: Image Store

Preset Parameter	Description
Single frame (live store)	If selected, stores single frame images only. If not selected, stores cine loop.
Number of heart cycles	Select the number of heart cycles to store. (Must be de-selected for single frame.)

Table 16-40: Footswitch

Preset Parameter	Description
Left	Configure the left footswitch pedal for the selected application. Select the functionality from the list.

Table 16-40: Footswitch (Continued)

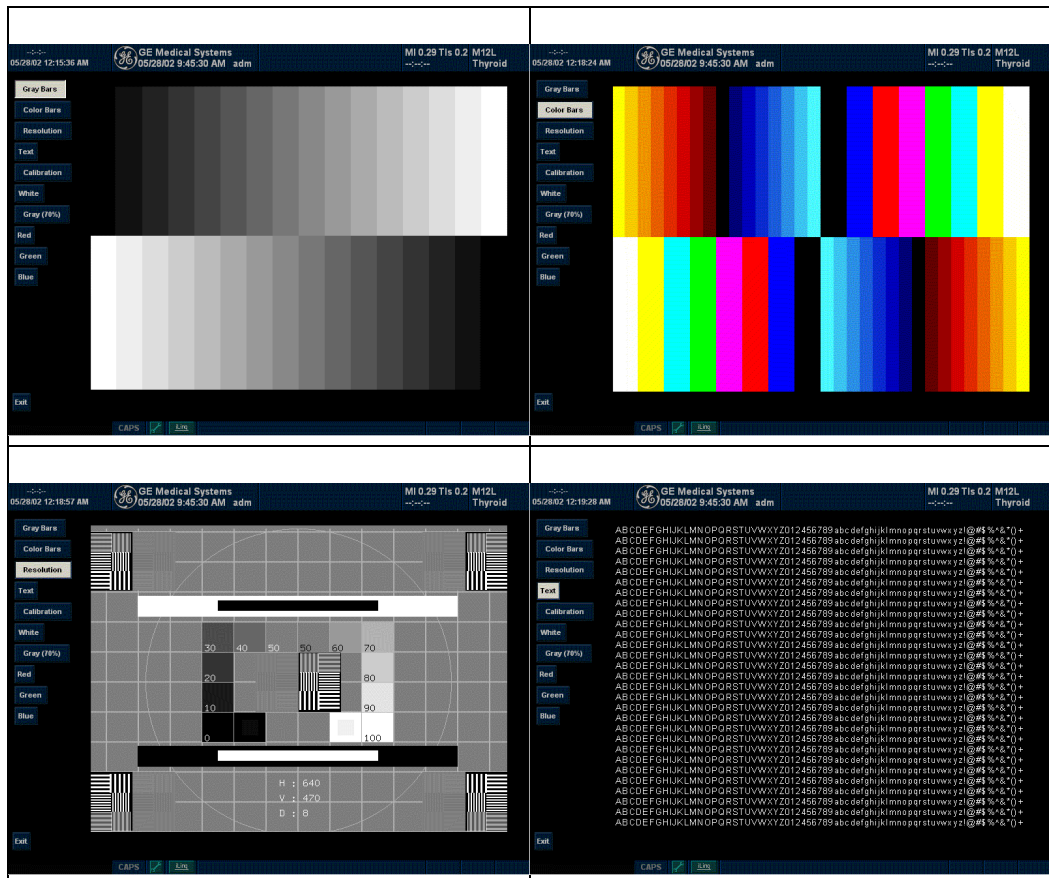
Preset Parameter	Description
Middle	Configure the middle footswitch pedal for the selected application. Select the functionality from the list.
Right	Configure the right footswitch pedal for the selected application. Select the functionality from the list.

Test Patterns

LOGIQ 9 Overview

There are different test patterns available: Gray Bars, Color Bars, Resolution, Text, Brightness Calibration, White, Gray, Red, Green and Blue.

Table 16-41: Available Test Patterns



Brightness Calibration

To calibrate the monitor,

1. Select **Test Patterns** from the Utility Touch Panel.; then select Brightness Calibration. The test pattern consists of a small box inside a larger box.

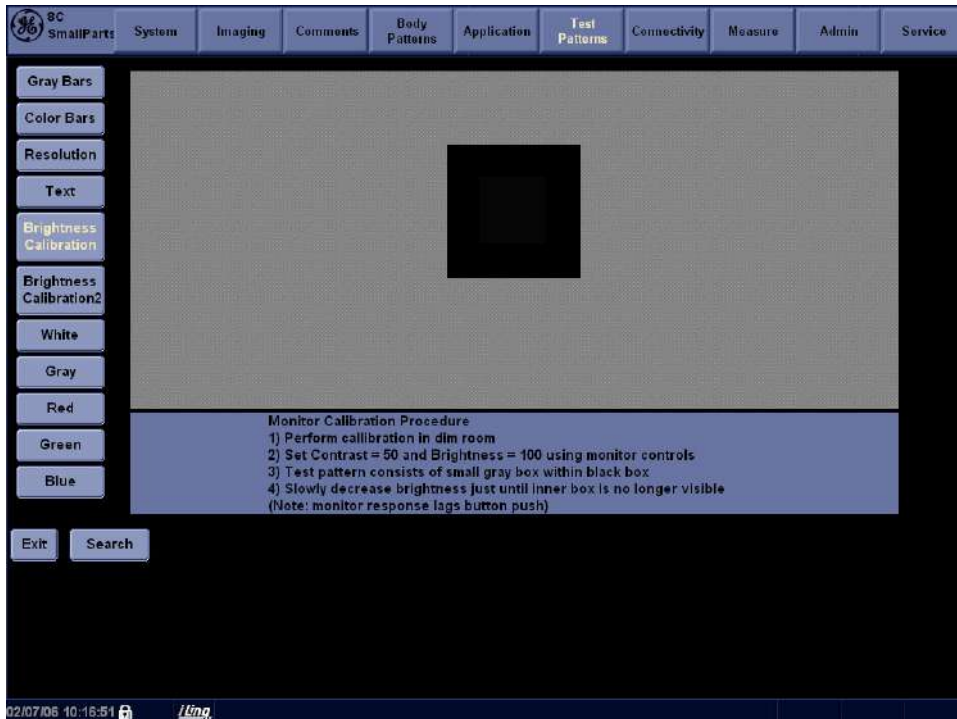


Figure 16-43. Brightness Calibration Test Pattern

2. Press the Brightness/Contrast control on the front of the monitor.
3. Set the Brightness at 100. Then slowly decrease the Brightness until the inner box is no longer visible. Please note that the monitor response to this adjustment lags behind the button push.

NOTE: *After you calibrate the monitor you may need to adjust peripheral settings.*

Overview

4D presets allow you to set up application-specific settings (presets) for each 4D image acquisition type. You can define different application-specific settings for each probe. Refer to Chapter 5 for more information.

4D Presets

To set up 4D presets:

1. On the Touch Panel, select **Utility**.
2. On the Touch Panel, select **3D/4D**.
The system displays the 4D Presets screen.
3. To select a probe, click on the plus sign (+) that appears next to the desired probe.
4. To select the application, click on the plus sign (+) that appears next to the desired application.
5. To select the acquisition type, click on the plus sign (+) that appears next to the desired application.
6. Double-click the desired application under the acquisition type. The Display tab is selected.

4D Presets (continued)

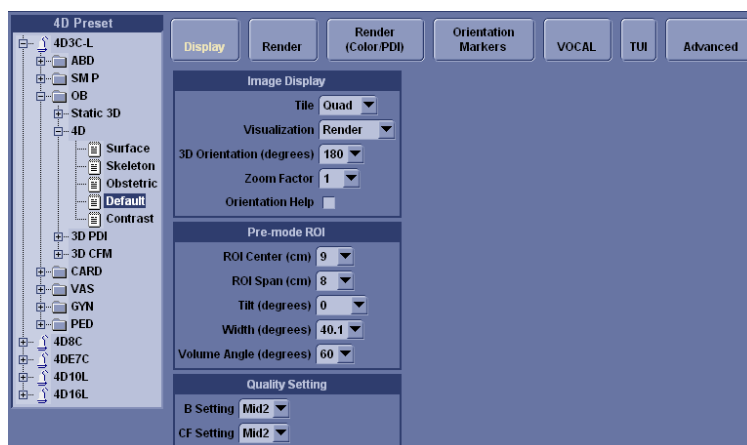


Figure 16-44. 4D Preset Parameters

Table 16-42: Display

Preset Parameter	Description
Tile	Determines the number of display windows. Values include: 1 (Single), 2 (Dual), and 4 (Quad).
Visualization	Determines the method of display for working with images. Values include: 0 (Sectional) and 1 (Render)
3D Orientation (degrees)	Determines the orientation of the ROI on the monitor display. Values include: 0, 90, 180, 270.
Zoom Factor	Determines the magnification factor of the zoom. Values include: 0.3 through 4.0, in .01 increments.
Orientation Help	Activate Orientation Help.
ROI Center (cm)	Determines the vertical center of the region of interest, 1-25cm.
ROI Span (cm)	Determines the height of the region of interest, 1-30cm.
Tilt (degrees)	Determines the degree of tilt from the vertical center location of the ROI, -28.7 to 28.7 degrees.
Width (degrees)	Determines the width of the ROI, 5.7 to 63 degrees.
Volume Angle (degrees)	Set the range of the volume sweep, 18-75 degrees.
B Setting	Set quality setting -- balances speed with line density. Selections are Low, Mid1, Mid2, Hi1, Hi2. High combines the highest density with the slowest speed. Low combines the lowest density with the highest speed.
CF Setting	Set quality setting -- balances speed with line density. Selections are Low, Mid1, Mid2, Hi1, Hi2. High combines the highest density with the slowest speed. Low combines the lowest density with the highest speed.

Render Tab



Figure 16-45. Render tab

Table 16-43: Render

Preset Parameter	Description
Render Mode 1:Mode2	Set render mode values. Surface Smooth, Surface Texture, Transp Max, Transp X-Ray, or TransMin (Render 1). Surface Smooth, Light, Gradient Light, Transp Max, Transp X-Ray, or Transp Min (Render 2).
Mix (% Render Mode 2) --	Set mix of Render 1 / Render 2 Mode, 0-100.
Lower Threshold	Set a lower threshold below which weaker echoes are removed, 0-255.
Transparency	Set the transparency of the image, 10/20-250. The higher the number, the more transparent the gray scale information.
Render View Direction	Set the direction in which the ROI is viewed.
VCI Static Slice Thickness	Set slice thickness, 2-10.

Color / PDI Render tab

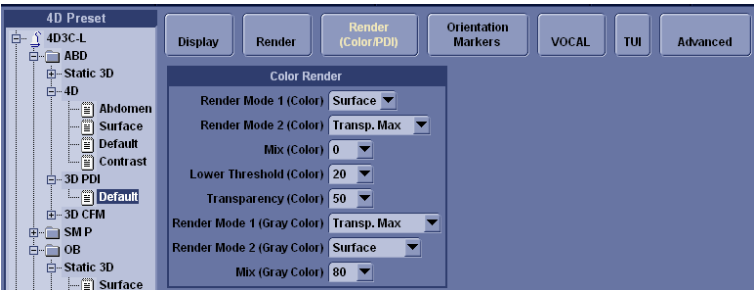


Figure 16-46. Color / PDI Render tab

Table 16-44: Color / PDI Render

Preset Parameter	Description
Render Mode 1 (Color) / Render Mode 2 (Color)	Determine the render mode, selected from Render Mode 1 and Render Mode 2.
Mix (Color)	Set the percentage of Render Mode 1 to be mixed with Render Mode 2.
Lower Threshold (Color)	Set the lower threshold below which weaker echos are removed.
Transparency (Color)	Determine the transparency of the image. The higher the number, the more transparent the gray scale information. Values: 0 to 255.
Render Mode 1 (Gray Color) / Render Mode 2 (Gray Color)	Determine the render mode, selected from Render Mode 1 and Render Mode 2.
Mix (Gray Color)	Set the percentage of Render Mode 1 to be mixed with Render Mode 2.

Orientation Markers tab

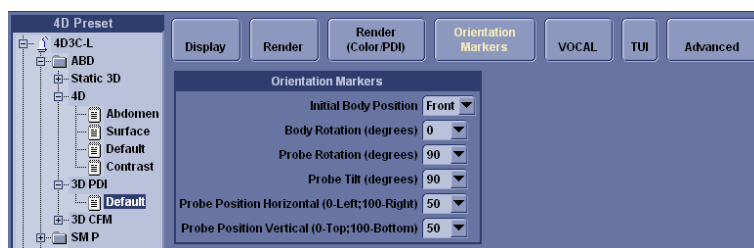


Figure 16-47. Orientation Markers tab

Table 16-45: Orientation Markers

Preset Parameter	Description
Initial Body Position	Set Head, Feet, Front, or Back
Body Rotation (degrees)	Set 0, 45, 90, 135, 180, 225, 270, or 315.
Probe Rotation (degrees)	Set 0, 45, 90, 135, or 180.
Probe Tilt (degrees)	Set 0-100.
Probe Position Horizontal (0-Left;100-Right)	Set 0-100.
Probe Position Vertical (0-Top;100-Bottom)	Set 0-100.

VOCAL tab



Figure 16-48. VOCAL tab

Table 16-46: VOCAL

Preset Parameter	Description
Vocal Method	Set Sphere, Manual, Contour Detect, or Semi-Auto Contoour Detect.
Vocal Semi-Auto-Detect type	Set Hypo, Cystic, or Hyper/Iso.
Vocal Rotation step	Set 6, 9, 15, or 30.

TUI tab



Figure 16-49. TUI Tab

Table 16-47: TUI

Preset Parameter	Description
Display Format	Set 1x1, 1x2, 2x2, or 3x3.
Total Slices	Set 3, 5, 7, 9, 11, 13, 15, 17, or 19.
Slice Distance (mm)	Set 0.5-40 (0.1 step increments).

Advanced tab

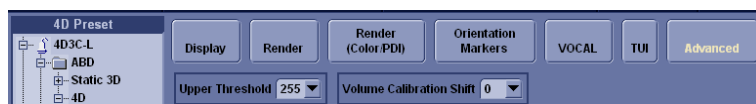


Figure 16-50. Advanced tab

Table 16-48: Advanced

Preset Parameter	Description
Upper Threshold	Sets the higher threshold above which weaker echos are removed.
Volume Calibration Shift	See a Field Service Engineer for information on this parameter.

Configuring Connectivity

Overview

You use Connectivity functionality to set up the connection and communication protocols for the ultrasound system. This page gives an overview of each of the Connectivity functions. Each function is described in detail in the following pages.

Structured Reporting

DICOM Structured Reporting provides the results of a procedure as structured data elements (well-defined fields) as opposed to unstructured data (large amounts of text undifferentiated by individual fields). This greatly improves query capability. DICOM Structured Reporting creates coded clinical data that can be used for clinical research, outcomes analysis, and disease management.

NOTE: *Structured Reporting is ONLY available with the LOGIQworks workstation.*

Supported parameters

range of lower measurement uncertainty	Est RVSP	OB/ASUM2000/BPD	Origin of vessel
% Area Reduction	estimated	OB/ASUM2000/BPD-GA	other
% Diameter Reduction	ESV(bp el)	OB/ASUM2000/BPD-Max	Ovarian Artery
range of upper measurement uncertainty	ESV(bullet)	OB/ASUM2000/BPD-Min	Ovarian vein
range of measurement uncertainty	ESV(mod sim)	OB/ASUM2000/CRL	P_Vein A
2 sigma deviation	expiration	OB/ASUM2000/CRL-GA	P_Vein A Dur
2D	External Carotid Artery	OB/ASUM2000/FL	P_Vein D
2D/%FS	External Iliac Artery	OB/ASUM2000/HC	P_Vein D VTl
2D/%IVS Thck	External Iliac Vein	OB/ASUM2000/HC-GA	P_Vein S
2D/%LVPW Thck	fetal biometry	OB/ASUM2000/HL	P_Vein S VTl
2D/Ao Arch Diam	fetal biometry ratios	OB/ASUM2000/HL-GA	P_Vein S/D Ratio
2D/Ao Asc Diam	fetal cranium	OB/ASUM2000/OFD	PA
2D/Ao Desc Diam	fetal long bones	OB/ASUM2000/OFD-GA	pat characteristics
2D/Ao Isthmus	fetus summary	OB/Biophysical/breathing	Peak Systolic Velocity
2D/Ao Root Diam	Fifth Lumbar Artery	OB/Biophysical/fluid	pelvis and uterus
2D/Ao/LA	finding site	OB/Biophysical/movement	percentile ranking
2D/ASD Diam	findings	OB/Biophysical/reactiveNst	Perforating Artery of Kidney
2D/AV Area	First Lumbar Artery	OB/Biophysical/score	Peroneal Artery
2D/AV Cusp	flow direction	OB/Biophysical/tone	Peroneal Vein
2D/AV Diam	Fourth Lumbar Artery	OB/Campbell/Ratio HC/AC	phase
2D/AVA Planimetry	Full Bernoulli	OB/Campbell/Ratio HC/AC-Max	PHT
2D/AVA/AV Diam	Gastric Artery	OB/Campbell/Ratio HC/AC-Min	PISA
2D/CI(Cube)	Gastric vein	OB/Eriksen/TAD	PISA/AR/ERO

2D/CI(Teich)	Gastrocnemius vein	OB/Eriksen/TAD-GA	PISA/AR/Flow
2D/CO(Cube)	Gastroduodenal Artery	OB/Eriksen/TAD-Max	PISA/AR/RF
2D/CO(Teich)	Giacomini vein	OB/Eriksen/TAD-Min	PISA/AR/RV
2D/EDV(A-L)	Great Saphenous Vein	OB/Goldstein/TCD	PISA/AR/Vmax
2D/EDV(Cube)	growth percentile rank	OB/Goldstein/TCD-GA	PISA/AR/VTI
2D/EDV(MOD)	growth Z-score	OB/Hadlock/AC	PISA/MR/ERO
2D/EDV(Teich)	Gyn/2D/Endo	OB/Hadlock/AC-GA	PISA/MR/Flow
2D/EF(A-L)	Gyn/2D/OvFoVolume/ Distance1	OB/Hadlock/AC-GP	PISA/MR/RF
2D/EF(Cube)	Gyn/2D/OvFoVolume/ Distance2	OB/Hadlock/AC-Max	PISA/MR/RV
2D/EF(MOD)	Gyn/2D/OvFoVolume/ Distance3	OB/Hadlock/AC-Min	PISA/MR/Vmax
2D/EF(Teich)	Gyn/2D/OvFoVolume/ MeanDistance	OB/Hadlock/BPD	PISA/MR/VTI
2D/ESV(A-L)	Gyn/2D/OvFoVolume/ Volume	OB/Hadlock/BPD-GA	PISA/PR/ERO
2D/ESV(Cube)	Gyn/2D/UtH	OB/Hadlock/BPD-GP	PISA/PR/Flow
2D/ESV(MOD)	Gyn/2D/UtL	OB/Hadlock/BPD-Max	PISA/PR/RV
2D/ESV(Teich)	Gyn/2D/UtVolume	OB/Hadlock/BPD-Min	PISA/PR/Vmax
2D/IVC	Gyn/2D/UtW	OB/Hadlock/CI	PISA/PR/VTI
2D/IVC Diam Exp	Heart Rate	OB/Hadlock/CRL	PISA/TR/ERO
2D/IVC Diam Ins	Hepatic Vein	OB/Hadlock/CRL-GA	PISA/TR/Flow
2D/IVSd	Highly significant	OB/Hadlock/CRL-Max	PISA/TR/RV
2D/IVSs	Hilar Artery	OB/Hadlock/CRL-Min	PISA/TR/Vmax
2D/LA	ICA/CCA velocity ratio	OB/Hadlock/ EFW(AC,BPD,FL)	PISA/TR/VTI
2D/LA/Ao	identifier	OB/Hadlock/ EFW(AC,BPD,FL)- EQUATION	Planimetry
2D/LAA systole	Ileal vein	OB/Hadlock/ EFW(AC,BPD,FL,HC)	Plantar Arterial Arch
2D/LAX/Trans AoD diastole	image library	OB/Hadlock/ EFW(AC,BPD,FL,HC)- EQUATION	Plax
2D/LAX/Trans AoD systole	Inferior	OB/Hadlock/ EFW(AC,FL)	Popliteal Artery

2D/LAX/Trans AVA diastole	Inferior Mesenteric Artery	OB/Hadlock/EFW(AC,FL)-EQUATION	Popliteal Vein
2D/LAX/Trans AVA systole	Inferior Mesenteric Vein	OB/Hadlock/EFW(AC,FL,HC)	population description
2D/LPA	Inferior Right Hepatic Vein	OB/Hadlock/EFW(AC,FL,HC)-EQUATION	Portal Vein
2D/LVA diastole	Inferior Vena Cava	OB/Hadlock/EFW(AC,HC)	Posterior arch vein
2D/LVA systole	Infra-renal Aorta	OB/Hadlock/EFW(AC,HC)-EQUATION	Posterior Cerebral Artery
2D/LVd Mass	Innominate Artery	OB/Hadlock/FL	Posterior Communicating Artery
2D/LVd Mass/ASE	Innominate vein	OB/Hadlock/FL-GA	Posterior Tibial Artery
2D/LVIDd	inspiration	OB/Hadlock/FL-GP	Posterior Tibial Vein
2D/LVIDs	Interlobar Artery of Kidney	OB/Hadlock/FL-Max	pound
2D/LVOT Area	Internal Carotid Artery	OB/Hadlock/FL-Min	PR Dec Slope
2D/LVOT Diam	Internal Carotid Artery	OB/Hadlock/HC	PR Dec Time
2D/LVPWd	Internal Iliac Artery	OB/Hadlock/HC-GA	PR maxPG
2D/LVPWs	Internal Jugular vein	OB/Hadlock/HC-GP	PR meanPG
2D/LVs Mass	IVCT	OB/Hadlock/HC-Max	PR PHT
2D/LVs Mass/ASE	IVRT	OB/Hadlock/HC-Min	PR Vmax
2D/MPA	LA	OB/Hadlock/OFD(HC)	PR Vmean
2D/MV Annulus Diam	LAAs(A2C)	OB/Hadlock/Ratio FL/AC	PR VTl
2D/MV Area	LAAs(A4C)	OB/Hadlock/Ratio FL/HC	PRend maxPG
2D/MVA Planimetry	LAESV(A-L A2C)	OB/Hansmann/AC	PRend Vmax
2D/PV Annulus Diam	LAESV(A-L A4C)	OB/Hansmann/AC-GA	presence undetermined
2D/PV Area	LAESV(MOD A2C)	OB/Hansmann/AC-Max	present
2D/RPA	LAESV(MOD A4C)	OB/Hansmann/AC-Min	Procedural Scope of Findings
2D/RVAWd	LALd(A4C)	OB/Hansmann/BPD	procedure reported
2D/RVAWs	LALs(A4C)	OB/Hansmann/BPD-GA	Profunda Femoris Artery

2D/RVIDd	Lateral	OB/Hansmann/BPD-Max	Profunda Femoris Vein
2D/RVIDs	Lateral calf perforator	OB/Hansmann/BPD-Min	Proper Hepatic Artery
2D/RVOT Area	Lateral mitral annulus	OB/Hansmann/CRL	Proximal
2D/RVOT Diam	Laterality	OB/Hansmann/CRL-GA	Psax
2D/SAX/LVA diastole	Left	OB/Hansmann/CRL-Max	Psaxao
2D/SAX/LVA systole	Left Hepatic Vein	OB/Hansmann/CRL-Min	Psaxchord
2D/SAX/MVA	left lower	OB/Hansmann/FL	Psaxmv
2D/SAX/Trans AoD diastole	Left Main Branch of Portal Vein	OB/Hansmann/FL-GA	Psaxpap
2D/SAX/Trans AoD systole	left upper	OB/Hansmann/FL-Max	Pulsatility Index
2D/SAX/Trans AVA diastole	Lesser Saphenous Vein	OB/Hansmann/FL-Min	PV
2D/SAX/Trans AVA systole	LIMP	OB/Hansmann/GS	PV Acc Time
2D/SI(A-L)	Lobar Artery	OB/Hansmann/GS-GA	PV Acc Time/ET Ratio
2D/SI(Cube)	Lumbar Artery	OB/Hansmann/HC	PV HR
2D/SI(MOD)	LV	OB/Hansmann/HC-GA	PV maxPG
2D/SI(Teich)	LV Mass-by-M-mode	OB/Hansmann/HC-Max	PV meanPG
2D/SV(A-L)	LV Mass-by-TE	OB/Hansmann/HC-Min	PV Vmax
2D/SV(Cube)	LVAd(A2C)	OB/Hansmann/OFD	PV Vmax P
2D/SV(MOD)	LVAd(A4C)	OB/Hansmann/OFD-GA	PV Vmean
2D/SV(Teich)	LVAd(LAX)	OB/Hansmann/OFD-Max	PV VTI
2D/SVC Diam Exp	LVAd(sax MV)	OB/Hansmann/OFD-Min	PVA (Vmax)
2D/SVC Diam Ins	LVAd(sax PM)	OB/Hansmann/TAD	PVA (Vmax)P
2D/TV Annulus Diam	LVAd(sax)	OB/Hansmann/TAD-GA	PVA (VTI)
2D/TV Area	LVAs(A2C)	OB/Hansmann/TAD-Max	PVein
2D/VSD Diam	LVAs(A4C)	OB/Hansmann/TAD-Min	PVET
5-point	LVAs(LAX)	OB/Hansmann/ThD	PW
A2C	LVAs(sax MV)	OB/Hansmann/ThD-GA	Qp/Qs
A4C	LVAs(sax PM)	OB/Hansmann/ThD-Max	quart

AA	LVA(sax)	OB/Hansmann/ThD-Min	RA
Abnormal	LVd Mass(A-L)	OB/Hansmann/TTD	RAAs(A4C)
Abnormally High	LVEDV(A-L A2C)	OB/Hansmann/TTD-GA	rad
Abnormally Low	LVEDV(A-L A4C)	OB/Hellman/GS	Radial artery
absent	LVEDV(A-L LAX)	OB/Hellman/GS-GA	Radial vein
Acceleration Index	LVEDV(MOD A2C)	OB/Hellman/GS-Max	RALd(A4C)
Acceleration Time	LVEDV(MOD A4C)	OB/Hellman/GS-Min	RALs(A4C)
Accessory Renal Artery	LVEDV(MOD BP)	OB/Hill/TCD	RAP
Accessory Renal Artery	LVEDV(MOD BP)_03	OB/Hill/TCD-GA	recent
Adult Echocardiography Procedure Report	LVEDV(MOD LAX)	OB/Hill/TCD-Max	reference authority
age	LVESV(A-L A2C)	OB/Hill/TCD-Min	regurgitant
AI	LVESV(A-L A4C)	OB/Hohler/FL	Renal Artery
AL	LVESV(A-L LAX)	OB/Hohler/FL-GA	Renal Artery/Aorta velocity ratio
anatomical site modifier	LVESV(MOD A2C)	OB/Hohler/Ratio FL/BPD	Renal Vein
antegrade	LVESV(MOD A4C)	OB/Jeanty/AC	Resistivity Index
Anterior Cerebral Artery	LVESV(MOD BP)	OB/Jeanty/AC-GA	respiration state
Anterior Communicating Artery	LVESV(MOD BP)_03	OB/Jeanty/AC-Max	Right
Anterior Tibial Artery	LVESV(MOD LAX)	OB/Jeanty/AC-Min	Right Hepatic Vein
Anterior Tibial Vein	LVET	OB/Jeanty/BPD	right lower
Anterior-Middle Cerebral Artery Bifurcation	LVLd(A2C)	OB/Jeanty/BPD-GA	Right Main Branch of Portal Vein
Anterior-Posterior Cerebral Artery Bifurcation	LVLd(A4C)	OB/Jeanty/BPD-GP	right upper
Aorta	LVLd(apical)	OB/Jeanty/BPD-Max	RIMP
Aorta	LVLs(A2C)	OB/Jeanty/BPD-Min	RV
Aortic Valve Ring	LVLs(A4C)	OB/Jeanty/CRL	RVET
AP/LVOT Diam	LVLs(apical)	OB/Jeanty/CRL-GA	RVIT

AP/LVOT VTI	LVOT	OB/Jeanty/CRL-Max	RVLd(A4C)
Apex	LVOT CI	OB/Jeanty/CRL-Min	RVLs(A4C)
Aplax	LVOT CO	OB/Jeanty/FIB	RVOT
AR Dec Slope	LVOT HR	OB/Jeanty/FIB-GA	RVOT
AR Dec Time	LVOT maxPG	OB/Jeanty/FIB-GP	RVOT CI
AR maxPG	LVOT meanPG	OB/Jeanty/FL	RVOT CO
AR meanPG	LVOT SI	OB/Jeanty/FL-GA	RVOT HR
AR PHT	LVOT SV	OB/Jeanty/FL-GP	RVOT maxPG
AR Vmax	LVOT Vmax	OB/Jeanty/FL-Max	RVOT meanPG
AR Vmean	LVOT Vmax P	OB/Jeanty/FL-Min	RVOT SI
AR VTI	LVOT Vmean	OB/Jeanty/HC	RVOT SV
Arcuate Artery of the Kidney	LVOT VTI	OB/Jeanty/HC-GA	RVOT Vmax
Area-Length Biplane	LVPEP	OB/Jeanty/HC-GP	RVOT Vmax P
Area-length single plane	LVs Mass(A-L)	OB/Jeanty/HC-Max	RVOT Vmean
ARend maxPG	MA	OB/Jeanty/HC-Min	RVOT VTI
ARend Vmax	Main	OB/Jeanty/HL	RVPEP
Artery of Abdomen	manual	OB/Jeanty/HL-GA	s (Scarred Myocardium)
Artery of Abdomen-Unilateral	MAS	OB/Jeanty/HL-Max	Saphenofemoral Junction
Artery of Lower Extremity	MCO	OB/Jeanty/HL-Min	scope of findings
Artery of neck	mean	OB/Jeanty/Radius	scoring scheme
Artery of Upper Extremity	measurement group	OB/Jeanty/Radius-GA	SD/Q-to-PV close
AS	Medial	OB/Jeanty/Radius-GP	SD/Q-to-TV open
ASD	Medial mitral annulus	OB/Jeanty/TIB	Second Lumbar Artery
ASD maxPG	Median Cubital vein	OB/Jeanty/TIB-GA	Segmental Artery
ASD Vmax	method	OB/Jeanty/TIB-Max	selection
atm	method citation	OB/Jeanty/TIB-Min	sex
AV	MI	OB/Jeanty/ULNA	SI(A-L A2C)
AV Acc Slope	Middle Cerebral Artery	OB/Jeanty/ULNA-GA	SI(A-L A4C)
AV Acc Time	Middle Hepatic Vein	OB/Jeanty/ULNA-Max	SI(A-L LAX)

AV Acc Time/ET Ratio	Mid-longitudinal	OB/Jeanty/ULNA-Min	SI(Biplane)
AV CI	min	OB/JSUM/AC	SI(Biplane)_03
AV CO	Minimum Diastolic Velocity	OB/JSUM/BPD	SI(bp el)
AV Dec Slope	Mitral Annulus	OB/JSUM/CRL	SI(bullet)
AV Dec Time	ML	OB/JSUM/FL	SI(MOD A2C)
AV HR	MM	OB/Kurtz/BPD	SI(MOD A4C)
AV maxPG	MM/%FS	OB/Kurtz/BPD-GA	SI(MOD LAX)
AV meanPG	MM/%IVS Thck	OB/Kurtz/BPD-Max	SI(mod sim)
AV SI	MM/%LVPW Thck	OB/Kurtz/BPD-Min	significance
AV SV	MM/Ao Root Diam	OB/Mayden/IOD	Significance Undetermined
AV Vmax	MM/Ao/LA	OB/Mayden/IOD-GA	Significant
AV Vmax P	MM/AV Cusp	OB/Mayden/OOD	Simplified Bernoulli
AV Vmean	MM/AV Diam	OB/Mayden/OOD-GA	Single Plane Ellipse
AV VTI	MM/CI(Cube)	OB/Mercer/Ft	Sixth Lumbar Artery
AVA (Vmax)	MM/CI(Teich)	OB/Mercer/Ft-GA	Soleal vein
AVA (Vmax)2	MM/CO(Cube)	OB/Mercer/Ft-Max	Splenic artery
AVA (Vmax)P	MM/CO(Teich)	OB/Mercer/Ft-Min	Splenic Vein
AVA (Vmax)P2	MM/EDV(Cube)	OB/Merz/AC	standard deviation
AVA (VTI)	MM/EDV(Teich)	OB/Merz/AC-GA	Subclavian Artery
aver	MM/EF(Cube)	OB/Merz/AC-GP	Subclavian artery
AVET	MM/EF(Teich)	OB/Merz/AC-Max	Subclavian vein
Axillary artery	MM/EPSS	OB/Merz/AC-Min	Subxlax
Axillary vein	MM/ESV(Cube)	OB/Merz/BPD	Subxsax
BA	MM/ESV(Teich)	OB/Merz/BPD-GA	summary
bar	MM/IVSd	OB/Merz/BPD-GP	Superficial Femoral Artery
BAS	MM/IVSd/LVPWd	OB/Merz/BPD-Max	Superficial Femoral Vein
Basilar Artery	MM/IVSs	OB/Merz/BPD-Min	Superficial Palmar Arch
Basilic vein	MM/LA	OB/Merz/FL	Superficial Palmar Venous Arch
BI	MM/LA/Ao	OB/Merz/FL-GA	Superior

biometry group	MM/LAAo/Ao Root Diam	OB/Merz/FL-GP	Superior Mesenteric Artery
biophysical profile	MM/LAAo/Ao/LA	OB/Merz/FL-Max	Superior Mesenteric Vein
Biplane Ellipse	MM/LAAo/LA/Ao	OB/Merz/FL-Min	Superior Vena Cava
BL	MM/LVd Mass	OB/Merz/HC	Supra-renal Aorta
Blood Vessel of Head	MM/LVd Mass/ASE	OB/Merz/HC-GA	SuprastLaA
Blood Vessel of Head-Unilateral	MM/LVET	OB/Merz/HC-GP	SuprastSA
BP	MM/LVIDd	OB/Merz/HC-Max	SV(A-L A2C)
Brachial artery	MM/LVIDs	OB/Merz/HC-Min	SV(A-L A4C)
Brachial vein	MM/LVPEP	OB/Moore/AFI Sum	SV(A-L LAX)
BS	MM/LVPWd	OB/Moore/AFI1	SV(Biplane)
BSA	MM/LVPWs	OB/Moore/AFI2	SV(Biplane)_03
BSA formula	MM/LVs Mass	OB/Moore/AFI3	SV(bp el)
calculated	MM/LVs Mass/ASE	OB/Moore/AFI4	SV(bullet)
Carotid Bifurcation	MM/MV D-E Excursion	OB/Nelson/CRL	SV(MOD A2C)
Carotid Bulb	MM/MV E/A Ratio	OB/Nelson/CRL-GA	SV(MOD A4C)
Carotid Siphon	MM/MV E-F Slope	OB/NT	SV(MOD LAX)
CE	MM/Q-to-PV close	OB/Osaka/BPD	SV(mod sim)
CE-by-MV	MM/Q-to-TV open	OB/Osaka/BPD-GA	systole
CE-by-PV	MM/RVAWd	OB/Osaka/BPD-Max	systolic BP
CE-by-VTI	MM/RVAWs	OB/Osaka/BPD-Min	Systolic to Diastolic Velocity Ratio
Celiac Axis	MM/RVET	OB/Osaka/CRL	table of values
Central Retinal Artery	MM/RVIDs	OB/Osaka/CRL-GA	table of values citation
Central Retinal Vein	MM/RVOT	OB/Osaka/CRL-Max	TCO
Cephalic vein	MM/RVPEP	OB/Osaka/CRL-Min	Teichholz
CF	MM/SI(Cube)	OB/Osaka/EFW	Testicular Artery
CFM/AR Signal Area	MM/SI(Teich)	OB/Osaka/EFW-EQUATION	Testicular Vein
CFM/MR Signal Area	MM/SV(Cube)	OB/Osaka/FL	Thigh perforator
CFM/PR Signal Area	MM/SV(Teich)	OB/Osaka/FL-GA	Third Lumbar Artery
CFM/TR Signal Area	MOD biplane	OB/Osaka/FL-Max	Time averaged mean velocity

CI(A-L A2C)	MOD single plane	OB/Osaka/FL-Min	Time averaged peak velocity
CI(A-L A2C)/AutoHR	mode	OB/Osaka/FTA	TR maxPG
CI(A-L A4C)	Most significant	OB/Osaka/HL	TR meanPG
CI(A-L A4C)/AutoHR	MP	OB/Osaka/HL-GA	TR Vmax
CI(A-L LAX)	MP/LVOT Diam	OB/Osaka/HL-Max	TR Vmean
CI(A-L LAX)/AutoHR	MP/LVOT VTI	OB/Osaka/HL-Min	TR VTI
CI(Biplane)	MPA Vmax	OB/Paris/BPD	Transjugular Intrahepatic Portosystemic Shunt
CI(Biplane)_03	MR Acc Slope	OB/Paris/CRL	Tricuspid Annulus
CI(bp el)	MR dp/dt	OB/Paris/FL	TV
CI(bullet)	MR maxPG	OB/Paris/Ft	TV A Velocity
CI(MOD A2C)	MR meanPG	OB/Paris/TAD	TV Acc Slope
CI(MOD A2C)/AutoHR	MR Vmax	OB/Rempen/BPD	TV Acc Time
CI(MOD A4C)	MR Vmean	OB/Rempen/BPD-GA	TV Dec Slope
CI(MOD A4C)/AutoHR	MR VTI	OB/Rempen/BPD-GP	TV Dec Time
CI(MOD LAX)	MS	OB/Rempen/CRL	TV E Velocity
CI(MOD LAX)/AutoHR	MV	OB/Rempen/CRL-GA	TV E/A Ratio
CI(mod sim)	MV A Dur	OB/Rempen/CRL-GP	TV HR
CO(A-L A2C)	MV A Velocity	OB/Rempen/CRL-Max	TV maxPG
CO(A-L A2C)/AutoHR	MV Acc Slope	OB/Rempen/CRL-Min	TV meanPG
CO(A-L A4C)	MV Acc Time	OB/Rempen/GS	TV PHT
CO(A-L A4C)/AutoHR	MV Acc Time/MV Dec Time	OB/Rempen/GS-GA	TV Vmax
CO(A-L LAX)	MV CI	OB/Rempen/GS-GP	TV Vmax P
CO(A-L LAX)/AutoHR	MV CO	OB/Rempen/GS-Max	TV Vmean
CO(A-L)	MV Dec Slope	OB/Rempen/GS-Min	TV VTI
CO(Biplane)	MV Dec Time	OB/Robinson/CRL	TVA
CO(Biplane)_03	MV E Velocity	OB/Robinson/CRL-GA	TVA (Vmax)
CO(bp el)	MV E/A Ratio	OB/Robinson/CRL-Max	TVA (Vmax)P
CO(bullet)	MV Eann Velocity	OB/Robinson/CRL-Min	TVA (VTI)
CO(MOD A2C)	MV HR	OB/Shepard/ EFW(AC,BPD)	Ulnar artery

CO(MOD A2C)/ AutoHR	MV maxPG	OB/Shepard/ EFW(AC,BPD)- EQUATION	Ulnar vein
CO(MOD A4C)	MV meanPG	OB/Shinozuka/AC	Umbilical Artery
CO(MOD A4C)/ AutoHR	MV PHT	OB/Shinozuka/AC-GA	Umbilical Vein
CO(MOD LAX)	MV SI	OB/Shinozuka/AC-Max	undetermined
CO(MOD LAX)/ AutoHR	MV SV	OB/Shinozuka/AC-Min	Unilateral
CO(mod sim)	MV Vmax	OB/Shinozuka/APTD	Uterine Artery
Collateral branch of vessel	MV Vmean	OB/Shinozuka/BPD	Vascular Structure of Kidney
comment	MV VTI	OB/Shinozuka/BPD-GA	Vascular Ultrasound Procedure Report
Common Carotid Artery	MVA (PHT)	OB/Shinozuka/BPD- Max	VC
Common Femoral Artery	MVA (VTI)	OB/Shinozuka/BPD-Min	Vein of Abdomen
Common Femoral Vein	myocardial wall	OB/Shinozuka/CRL	Vein of Abdomen- Unilateral
Common Hepatic Artery	Normal	OB/Shinozuka/CRL-GA	Vein of Lower Extremity
Common Hepatic artery	normal range authority	OB/Shinozuka/CRL-Max	Vein of Upper Extremity
Common Iliac Artery	normal range description	OB/Shinozuka/CRL-Min	Velocity ratio
Common Iliac Artery	Normal Range Lower Limit	OB/Shinozuka/EFW1	Vertebral Artery
Common Iliac Vein	Normal Range Upper Limit	OB/Shinozuka/EFW1- EQUATION	Vessel lumen cross- sectional area
Common iliac vein	normality	OB/Shinozuka/EFW2	Vessel lumen diameter
Congenital	Normality Undetermined	OB/Shinozuka/EFW2- EQUATION	Vessel outside diameter
CS	Not significant	OB/Shinozuka/EFW3	view
Cube	Ob/2D/OvH	OB/Shinozuka/EFW3- EQUATION	Volume flow
CW	Ob/2D/OvH-left	OB/Shinozuka/FL	VSD
Deceleration Time	Ob/2D/OvH-right	OB/Shinozuka/FL-GA	VSD maxPG

Deep Palmar Arch of Radial Artery	Ob/2D/OvL	OB/Shinozuka/FL-Max	VSD Vmax
Deep Palmar Venous Arch	Ob/2D/OvL-left	OB/Shinozuka/FL-Min	Wall Motion Score -1.0
derivation	Ob/2D/OvL-right	OB/Shinozuka/TTD	Wall Motion Score 0.0
diastole	Ob/2D/OvVolume	OB/Tokyo/APTD	Wall Motion Score 1.0
diastolic BP	Ob/2D/OvVolume-left	OB/Tokyo/BPD	Wall Motion Score 1.5
Distal	Ob/2D/OvVolume-right	OB/Tokyo/BPD-GA	Wall Motion Score 2.0
Doppler Volume Flow	Ob/2D/OvW	OB/Tokyo/BPD-Max	Wall Motion Score 2.5
Dorsalis Pedis Artery	Ob/2D/OvW-left	OB/Tokyo/BPD-Min	Wall Motion Score 3.0
early gestation	Ob/2D/OvW-right	OB/Tokyo/CRL	Wall Motion Score 4.0
EDV(bp el)	OB/AFI Sum	OB/Tokyo/CRL-GA	Wall Motion Score 5.0
EDV(bullet)	OB/AFI1	OB/Tokyo/CRL-Max	Wall Motion Score 6.0
EDV(mod sim)	OB/AFI2	OB/Tokyo/CRL-Min	Wall Motion Score 7.0
EF(A-L A2C)	OB/AFI3	OB/Tokyo/EFW	wall segment
EF(A-L A4C)	OB/AFI4	OB/Tokyo/EFW-EQUATION	wms
EF(A-L LAX)	OB/ASUM/AC	OB/Tokyo/FL	wms finding
EF(Biplane)	OB/ASUM/AC-GA	OB/Tokyo/FL-GA	wms morphology
EF(Biplane)_03	OB/ASUM/AC-Max	OB/Tokyo/FL-Max	wmsi
EF(bp el)	OB/ASUM/AC-Min	OB/Tokyo/FL-Min	X_OB/Hansmann/OFD-GP
EF(bullet)	OB/ASUM/BPD	OB/Tokyo/GS	X_OB/Jeanty/AC-GP
EF(MOD A2C)	OB/ASUM/BPD-GA	OB/Tokyo/GS-GA	X_OB/Jeanty/CRL-GP
EF(MOD A4C)	OB/ASUM/BPD-Max	OB/Tokyo/GS-Max	Z-score
EF(MOD LAX)	OB/ASUM/BPD-Min	OB/Tokyo/GS-Min	
EF(mod sim)	OB/ASUM/CRL	OB/Tokyo/LV	
end diastole	OB/ASUM/CRL-GA	OB/Tokyo/TTD	
End Diastolic Velocity	OB/ASUM2000/AC	OB/Yarkoni/CLA	
end systole	OB/ASUM2000/AC-GA	OB/Yarkoni/CLA-GA	
equation	OB/ASUM2000/AC-Max	OB-GYN Ultrasound Procedure Report	
equation citation	OB/ASUM2000/AC-Min	Ophthalmic Artery	

Connectivity Functions

To set up your institution's connectivity, you must login with administrator privileges.

1. **TCPIP:** allows you to configure the Internet Protocol.
2. **Device:** allows you to set up devices.
3. **Service:** allows you to configure a service (for example, DICOM services such as printers, worklist, and other services such as video print and standard print) from the list of supported services. This means that the user can configure a device with the DICOM service(s) that particular device supports.
4. **Dataflow:** allows you to adjust the settings of the selected dataflow and associated services. Selecting a dataflow customizes the ultrasound system to work according to the services associated with the selected dataflow.
5. **Button:** allows you to assign a pre-configured output service (or a set of output services) to the Print keys on the control panel.
6. **Removable Media:** enables formatting (DICOM, database, or blank formatting) and DICOM verification of removable media.
7. **Miscellaneous:** allows you to set up the patient exam menu options, print and store options, and the order of the columns in the examination list on the Patient menu.

Configure these screens from left to right, starting with the Tcpiip tab first.

NOTE: *The ultrasound system is pre-configured for many services, with default settings selected. You can change these services and settings as needed.*



You must restart the LOGIQ 9 (shutdown) after making any changes to connectivity settings in the Utility menus. This includes any changes on the TCPIP or dataflow setup screens.

TCPIP

This configuration category enables users with administrative rights to set the TCPIP for the system and connected remote archive.

1. Type the name of the Ultrasound system in the Computer Name field.
2. In the IP settings section, identify the ultrasound system to the rest of the network by one of the following:
 - DO NOT enable DHCP.
 - Type the IP-Address (acquire unique static IP address from hospital network administrator), Subnet Mask, and Default Gateway (if applicable).

NOTE: *Do not set up the system with DHCP. The IP address MUST BE static for the diagnostic and DICOM to function correctly.*

3. Select Save settings.
4. Re-boot the ultrasound system.

NOTE: *TCPIP settings do not get restored when restoring backups. This is per system design. The LOGIQ 9 IP address MUST BE unique.*

TCPIP (continued)



Figure 16-51. Connectivity TCPIP Preset Menu

Table 16-49: Computer Name

Preset Parameter	Description
Computer Name	Type the unique name for the Ultrasound system (no spaces in name).

Table 16-50: IP settings

Preset Parameter	Description
Enable DHCP	DO NOT select this box to enable dynamic IP Address selection.
IP-Address	Type the IP Address of the Ultrasound system. NOTE: IP stands for Internet Protocol. Every device on the network has a unique IP address.
Subnet Mask	Type the subnet mask address. NOTE: The Subnet Mask is an IP address filter that eliminates communication/messages from network devices of no interest to your system.
Default Gateway	Type the default gateway address.
Network Speed	Select the network speed (Auto Detect, 10Mbps/Half/Full Duplex, or 100 Mbps/Half/Full Duplex

NOTE: *Reboot the system to activate any changes saved from this page.*

Device

To add a new device,

- 1. Press Add.
- 2. Type the device name in the Name field.
- 3. Type the device's IP address in the IP Address field.



Figure 16-52. Connectivity Device Preset Menu

Table 16-51: Device

Preset Parameter	Description
Add/Remove	Press Add to add a new device; press Remove to delete a device.
Ping	Press Ping to confirm that a device is connected.
Properties: Name	Type the name of the device.
Properties: IP Address	Type the device's IP address.

To ping a device,

- 1. Select the device.
- 2. Press Ping. If the smiley face smiles, then the connection has been confirmed. If the smiley face frowns, then the connection has not been made. Check the device name and IP address.

Service

For each Device that you added to the system, you need to set up the service(s) that device supports (you must be an administrator to update these screens).

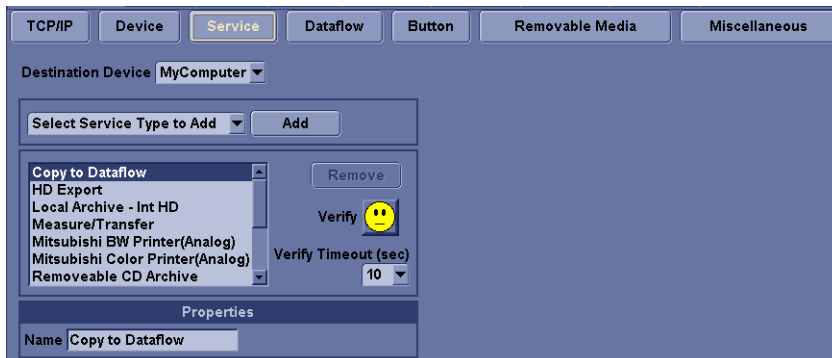


Figure 16-53. Connectivity Services Preset Menu

The Services screen has the following sections of information:

1. **Destination Device** - lists information about destination devices. You can select from a list of currently existing devices.
2. **Service Type to Add** - lists information about services for the destination device. You can add services, select from a list of currently existing services, and remove services.
3. **Service Parameters** - lists parameters for the service currently selected in the Services section. The name and parameters in this section change, depending on what service is currently selected. In the above figure, this section shows DICOM Print parameters.

Adding a service to a destination device

1. Select the service from the pull-down menu. Press Add.
2. Specify the properties for this service. Press Save.
3. Verify the service.

Removing a service

1. Select the service. Press Remove.
2. Press Save.

Changing parameters for a service

There are certain parameters that may need to be set up for each service:

Table 16-52: Service Parameters: Common Service Parameters

Preset Parameter	Description
Name	Free text: give a descriptive name to the device.
AE Title	The Application Entity Title for the service.
Port Number	The port number of the service.
Maximum Retries	Max # – the maximum number of times to try establishing a connection to the service..
Retry Interval (sec)	Specify how often (in seconds) the system should try to establish a connection to the service.
Timeout	The amount of time after which the system will stop trying to establish a connection to the service.

Changing parameters for a service (continued)

Many service parameters are specific to each type of service. The parameters are described on the following pages:

- DICOM Image Storage
- DICOM Performed Procedure
- DICOM Print
- DICOM Query/Retrieve
- DICOM Storage Commitment
- DICOM Worklist
- Echo Database V3.0
- Measure/Transfer
- Standard Print
- Video Capture

DICOM Image Storage

DICOM Image Storage allows the system to send or receive ultrasound images in a format that can be interpreted by PACS.

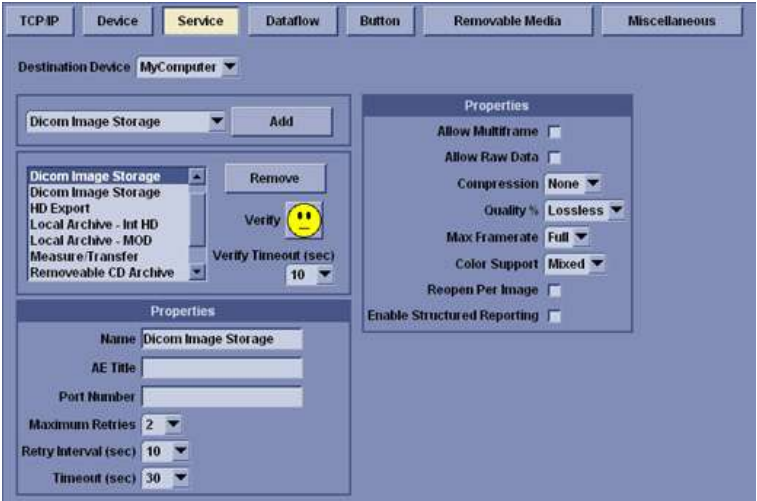


Figure 16-54. DICOM Image Storage Service

Table 16-53: DICOM Image Storage

Preset Parameter	Description
Allow Multiframe	Select to allow cine loop storage.
Allow raw data	Select to save data in both TruAccess (raw data) and DICOM format.Clear to save in DICOM format only.
Compression	Select the compression type: None, Rle, or Jpeg.
Quality %	Set picture quality from 1 to 100% and Lossless. A low picture quality level allows high data compression, while a high picture quality restrains the compression.
Max Framerate	Select the maximum frame rate: Full, 25, or 30.
Color Support	Select: Mixed, Gray or Color
Reopen per image	Reopen per image
Enable Structured Reporting	Select for Structured Reporting.

DICOM Performed Procedure

DICOM Performed Procedure provides an acknowledgement that a study has been performed.

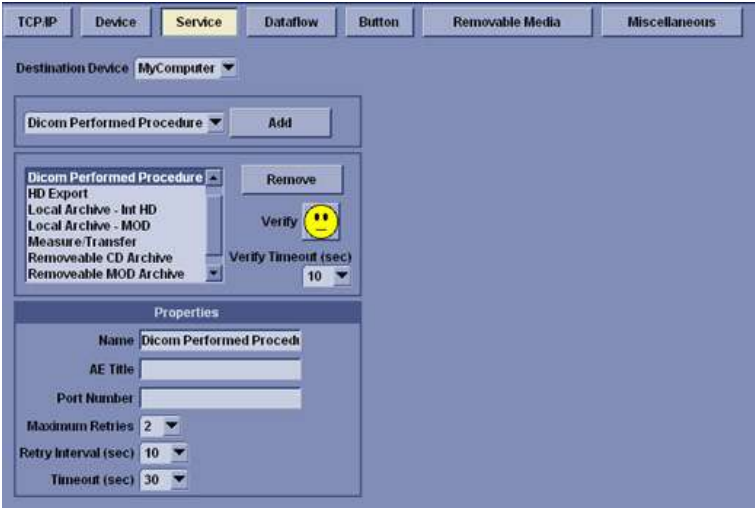


Figure 16-55. DICOM Performed Procedure Service

DICOM Print

DICOM Print provides the ability to send or receive ultrasound image data to DICOM printers.

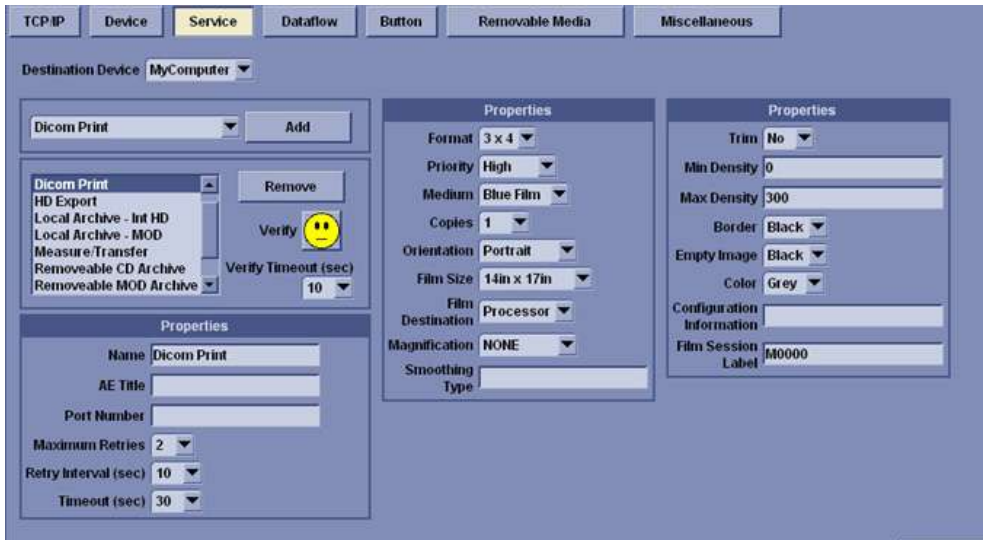


Figure 16-56. DICOM Print Service

Table 16-54: Properties

Preset Parameter	Description
Format	Indicates how many prints to print per page, for example, 1,1, 1,2, 1,3, up to 7,5. Partial prints are displayed as one print job.
Priority	Specify the print job priority: High, Medium, or Low.
Medium	Select the print medium: Clear Film, Paper, or Blue Film.
Copies	Enter the number of copies.
Orientation	Specify whether to print the image Portrait (vertically) or Landscape (horizontally).
Film Size	Specify the dimensions of the film size.
Film Destination	Specify the film destination for the exposed filmMagazine – Store in a film magazineProcessor – Develop in a film processor
Magnification	Specify how the printer magnifies the image to fit it onto the film.Replicate – Interpolated pixel are copies of the adjacent pixelsBilinear – Interpolated pixels are created by bilinear interpolations between the adjacent pixelsCubic – Interpolated pixels are created by cubic interpolations between the adjacent pixelsNone – No interpolation
Smoothing Type	Specify the printer's magnification interpolation for the output.
Trim	Specify whether you want a trim box to be printed around each image on the film: Yes or No.

Table 16-54: Properties (Continued)

Preset Parameter	Description
Min Density	Enter a number indicating the minimum density level of the film.
Max Density	Enter a number indicating the maximum density level of the film.
Border	Select to have the border area surrounding and between the images of the film: Black or White.
Empty Image	Select to have a Black or White empty image.
Color	Select whether to have the image Color or Grey.
Configuration Information	Enter vendor-specific image quality settings.
Film Session Label	Type a name for the group of film labels associated with the print job.

DICOM Query/Retrieve

DICOM Query/Retrieve provides a list of patients sorted by query parameters.

NOTE: Some PACS vendors only offer Query/Retrieve as an option. Please confirm that this service is available.

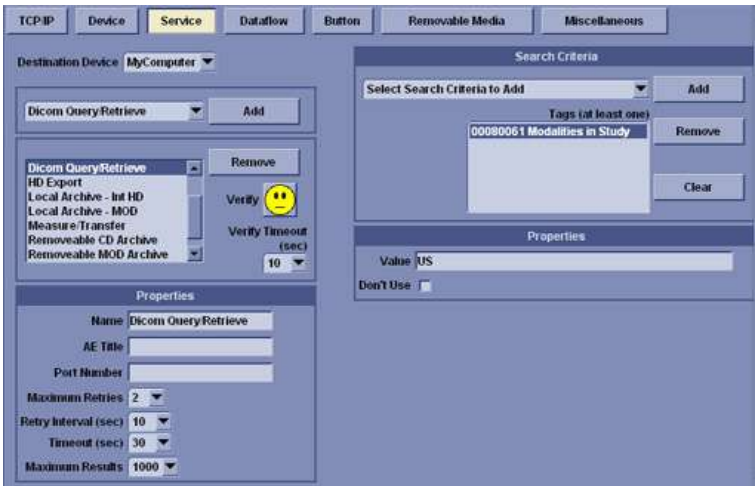


Figure 16-57. DICOM Query/Retrieve Service

Table 16-55: DICOM Query/Retrieve

Preset Parameter	Description
Max. Result	Specify the maximum number of patient records you want the system to retrieve when searching the patient database.
Search Criteria	Displays the Search Criteria window, where you can enter search parameters for the system to use when searching the patient database.

DICOM Query/Retrieve (continued)

Table 16-56: DICOM Query/Retrieve Search Criteria

Preset Parameter	Description
Select Search Criteria to Add	Select the type of information that you want to define for search parameters. The following searches are allowed: Patient Name, Patient ID, Modality, Scheduled Procedure Start Date, Start Time, and End Time.
Tags (at least one)	The name of a tag selected to use for search criteria.
Properties: Value	Type the value of the Selected Tag item. For example, if you select Referring Physician's Name in the Select Tag field, you can enter the name of the physician in the Value field.
Properties: Don't Use	Select to turn off the selected search criteria. To exclude a tag from the worklist query, select Don't Use and then select Add to List.
Add	Select to add the tag and value to the list of search criteria.
Remove	Select to remove the tag and value from the list of search criteria.
Clear	Clears all tags.

DICOM Storage Commitment

DICOM Storage Commitment provides acknowledgement from PACS that the study has been accepted into archive.

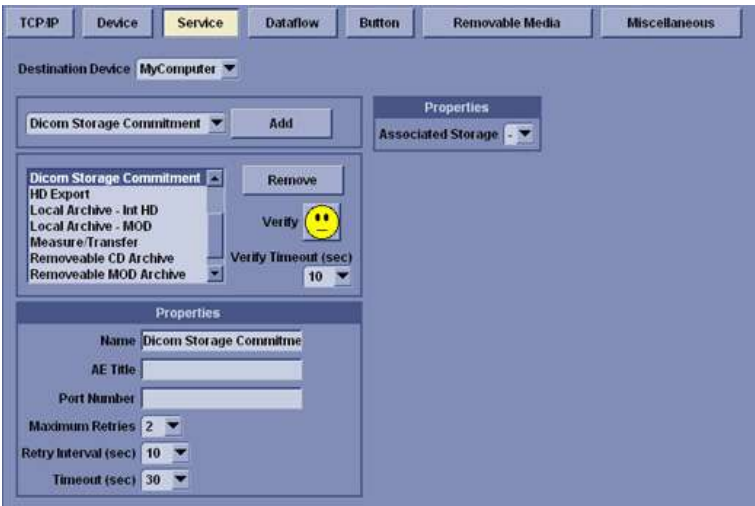


Figure 16-58. DICOM Storage Commitment Service

Table 16-57: DICOM Storage Commitment

Preset Parameter	Description
Associated Storage	This selection is based on the services entered by the user.

DICOM Worklist

DICOM Worklist provides a list of patients sorted by query parameters.

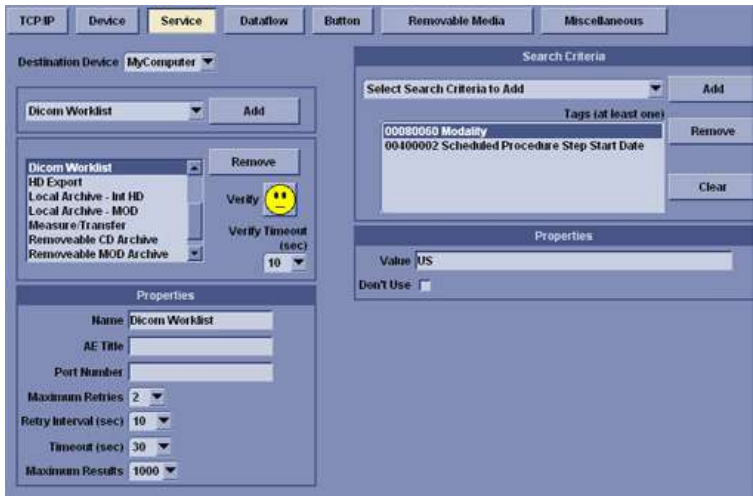


Figure 16-59. DICOM Worklist Service

Table 16-58: DICOM Worklist

Preset Parameter	Description
Max. Result	Specify the maximum number of patient records you want the system to retrieve when searching the patient database.
Search Criteria	Displays the Search Criteria window, where you can enter search parameters for the system to use when searching the patient database.

DICOM Worklist (continued)

Table 16-59: DICOM Worklist Search Criteria

Preset Parameter	Description
Select Search Criteria to Add	Select the type of information that you want to define for search parameters. The following searches are allowed: Patient Name, Patient ID, Modality, Scheduled Procedure Start Date, Start Time, and End Time.
Tags (at least one)	The name of a tag selected to use for search criteria.
Properties: Value	Type the value of the Selected Tag item. For example, if you select Referring Physician's Name in the Select Tag field, you can enter the name of the physician in the Value field.
Properties: Don't Use	Select to turn off the selected search criteria. To exclude a tag from the worklist query, select Don't Use and then select Add to List.
Add	Select to add the tag and value to the list of search criteria.
Remove	Select to remove the tag and value from the list of search criteria.
Clear	Clears all tags.

Echo Database V3.0

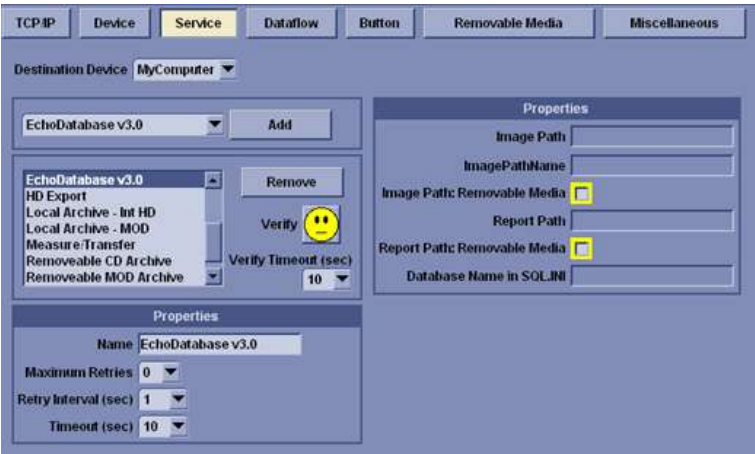


Figure 16-60. Echo Database V3.0 Service

Table 16-60: EchoDatabase ODBC

Preset Parameter	Description
Image Path	Enter the path to the image archive. If the image archive is on a removable disk, select Removable Media.
Image Path Name	Type the path's name.
Report Path	Enter the path to the report archive. If the report archive is on a removable disk, select Removable Media.
Image Path/Report Path: Removable Media	Select media.
Database Name in SQL.INI	The name of the database.

Measure/Transfer

You can use the Measure/Transfer service for sending exam information to another PC where data management information gets downloaded to.

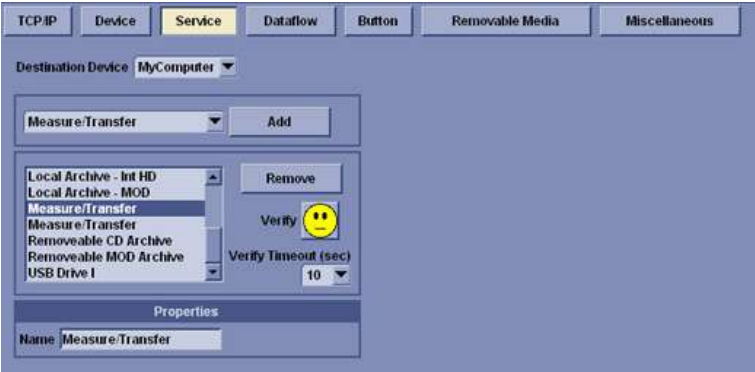


Figure 16-61. Measure/Transfer Service

Standard Print

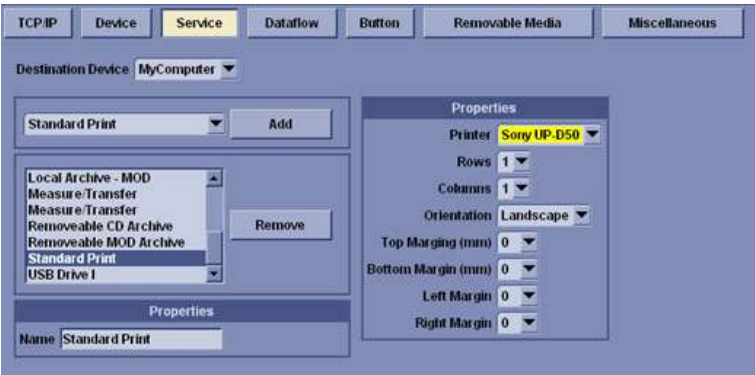


Figure 16-62. Standard Print Service

Table 16-61: Standard Print

Preset Parameter	Description
Printer	Select the printer.
Rows	Specify 1-5.
Columns	Specify 1-5.
Orientation	Specify Landscape/Portrait
Top Margin (mm)	Specify the top margin (0-51mm)
Bottom Margin (mm)	Specify the bottom margin (0-51mm)
Left Margin	Specify the left margin (0-51mm)
Right Margin	Specify the right margin (0-51mm)

Video Capture

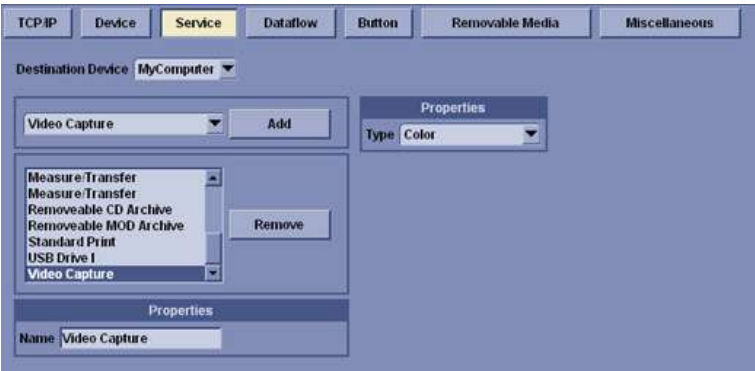


Figure 16-63. Video Capture Service

Table 16-62: Standard Print

Preset Parameter	Description
Color	Specify Color, BW, VCR Record/Pause, Ext1, or Ext2.

Setting up a Printer

There are two service types that pertain to printers: Standard Print and Video Capture Device.

Standard Print. Use Standard Print for digital peripherals. These are printers with either a USB interface or Ethernet interface (Sony UP-D23MD, for example).

Video Capture Device. Use Video Capture Device for devices that are triggered by a contact closure, typically analog devices (color printer, black/white printer, VCR Record/Pause, EXT1 and EXT2, for example).

On the Utility --> Connectivity --> Button page, select the Print key in the upper, left-hand corner of the display. In the middle portion of the page, under Available Input/Outputs, select the printer you want to configure. Next, press the two right arrows (>>) in the upper, right-hand corner of the page to move this printer into the Printflow View.

You can also configure the Standard Print button that appears on the New Patient Active Images screen.

Example: For instance a video capture device, on the Utility --> Connectivity --> Service page, in the Service Type to Add box, and press Add. In the properties box on the upper, right-hand side, select the type of device and in the Properties box in the lower, left-hand side, type in a unique descriptive name for this device.

Dataflow

A dataflow is a set of pre-configured services. When you select a dataflow, the ultrasound system automatically works according to the services associated with the dataflow. The Dataflow tab allows you to select and review information about dataflows. You can also create, change, and remove dataflows.

Set up dataflows for the services. Workflow MUST be PRIMARY in a dataflow. Also, Query/Retrieve MUST be the only service in a dataflow -- also, set it to Hidden so that it cannot be selected from the Patient menu.

NOTE: You must be logged on as Administrator to use the Dataflow tab.

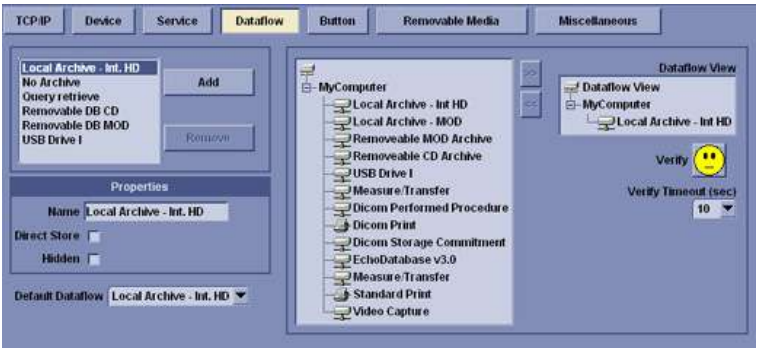


Figure 16-64. Dataflow Preset Menu

Table 16-63: Dataflow

Preset Parameter	Description
Name	Select the dataflow from the list.
Direct Store	Select to store data directly to archive (no buffer storage).
Hidden	Select so that this dataflow does not appear as a Dataflow on the Patient menu.
Default Dataflow	Select to use this dataflow as the default dataflow when you start the system.

Button

You can assign print buttons via the Utility --> Connectivity --> Button page.

Assigning print buttons. First select the print button to configure on the upper, left corner of the page. Then select the device you want to add in the middle part of the page, under Available Input/Outputs. Then click on the right arrow in the top right corner of the page.

- NOTE:

You can configure each print key to multiple output devices/ dataflows.
- NOTE:

Only attach one DICOM service per print key (e.g., PACS and DICOM printer). Multiple DICOM devices should be configured via a dataflow.
- NOTE:

When using a print key to send an image directly to a DICOM device, this causes a single DICOM association per image. Most devices (all known printers) work fine with this. However, some storage devices, such as ALI, Kodak Access, and Cemax, assume that the end of each association is the end of the exam and can result in a new folder for each image. In the Utility menu, select a single association or open PR for the desired DICOM storage device.

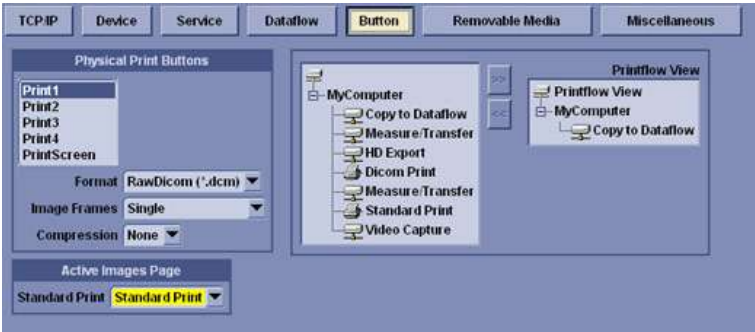


Figure 16-65. Button Preset Menu

Table 16-64: Button

Preset Parameter	Description
Format	RawDICOM, DICOM, or M&A.
Image Frames	Single (captures a DICOM single frame image), Multiple (captures a DICOM multiframe, CINE) -- only select this if the PACS supports multiframe, and Secondary Capture (forces all DICOM images to Secondary Capture).
Compression	Always set to None.
Active Images Page	Lets you send to a Windows-based printer.

Table 16-64: Button (Continued)

Preset Parameter	Description
Single Association	Only visible with DICOM image storage destinations. When selected, the spooler sends multiple images in one store job.

Removable Media

The Removable Media tab allows you to:

- Verify the DICOM directory on removable media.
- Verify the free space of the media.
- Verify that the media is finalized or unfinalized.
- Verify that the media is formatted or unformatted.
- Format removable media (rewritable CD/DVD or USB device).

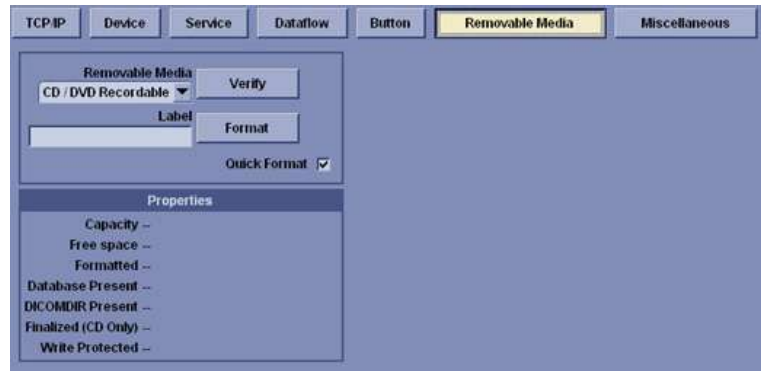


Figure 16-66. Removable Media Preset Menu

Removable Media (continued)

Table 16-65: Tools

Preset Parameter	Description
Verify	• Select to verify DICOM directory on removable DICOM disk. • Verify the free space of the media. • Verify that the media is finalized or unfinalized. • Verify that the media is formatted or unformatted.
Format	Select to format removable media.
Quick Format (MOD)	To format the MOD quickly, check this box. If you uncheck this box, the MOD is formatted with a full format. New MODs should always be formatted with a full format.
Media	Select the removable media to format or verify.
Label	Type a label for a new removable media (free text).

The bottom of the screen lists properties of the selected media.

Formatting removable media

1. Select the removable media from the Media list.
 2. Type a name for the removable media in the Label field.
- NOTE:** *Do not use the following characters for labelling:*
- \ / : ; . , * < > | + = []
3. Select Format. Confirm **OK** or **Cancel**.
 4. An information window confirms when the format has been completed. Select **OK** to exit.

NOTE: *Select Full Format (do not check Quick Format box) when formatting a new MOD for the first time.*

Verifying removable media

1. Select the removable media from the Media list.
2. Select Verify.

Miscellaneous

The Miscellaneous tab allows you to configure tools related to patient management and print and store options. You can specify default system functionality, such as whether patient ID is required when you archive data, or if you want the system to automatically search the archive for a patient when you enter patient data.

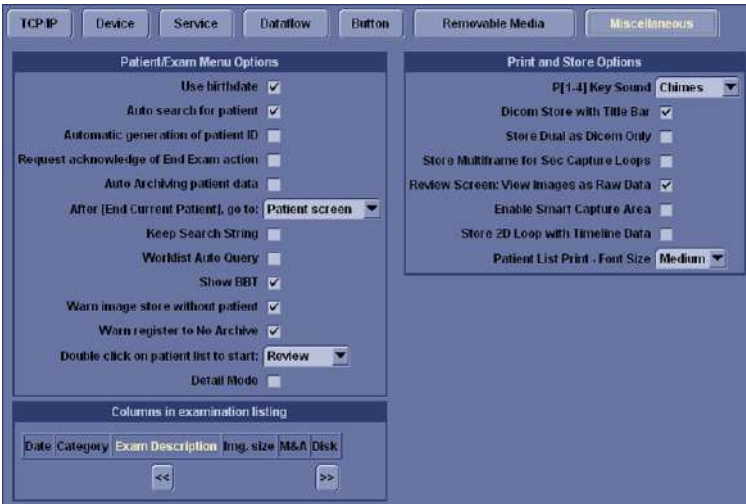


Figure 16-67. Miscellaneous Preset Menu

Table 16-66: Patient/Exam Menu Options

Preset Parameter	Description
Use birthdate	In the Patient information window, enter either the patient age or the birth date:When selected, enter birth date, then the age is calculated.When cleared, enter age (birth date field not available).
Auto search for patient	In the Search/Create Patient window:When selected, the system automatically searches through the selected patient archive, while the user enters patient information.When cleared, the automatic search tool is turned off. If you are trying to keep the past patient data confidential, DO NOT use this feature.
Automatic generation of patient ID	In the Search/Create Patient window:When selected, the Patient ID is not required when entering a new patient in the archive. The system automatically generates an ID number.When cleared, the Patient ID is required when entering a new patient in the archive.
Request acknowledge of End Exam action	When selected, the user is asked to confirm action when ending an examination.
Auto Archiving Patient Data	When selected, the patient data is automatically archived when you select End Exam.

Table 16-66: Patient/Exam Menu Options (Continued)

Preset Parameter	Description
After [End Current Exam], go to:	Select Worklist screen or Patient screen.
Keep Search String	Select so that search tool 'remembers' last search string.
Worklist Auto Query	When selected and you enter Worklist, the system queries the worklist automatically from the server. When deselected and you enter Worklist, the system queries the worklist in the local hard drive.
Show BBT	Select to input basal body temperature on the OB Patient screen.
Warn Image Store without Patient	Select to receive a warning when you press the Print key without an active patient.
Warn Register to No Archive	Select to receive a warning when you register a patient to the "No Archive" data flow. Select a different data flow for permanent storage of patient data.
Double Click on Patient List to Start	Select Review or New Exam to display each time you double click on the patient name in the patient list on the Patient menu.
Detail Mode	Select to display Detail Mode, rather than Exam View, when you select the patient name in the patient list on the Patient menu. You can also type comments while in Detail Mode.

Measure

Please refer to Chapter 7, General Measurements and Calculations for more information on setting up Measurement and Analysis Presets.

Reports

Refer to Chapter 14 for more information.

System Administration

Overview

The Admin screen has the following three sections:

- **System Administration** – lists all the options implemented in the system.
- **Users** – allows you to define user IDs, specify operator's registration, operator's rights, and registration of staff related to an examination (for example, referral doctors and sonographers).
- **Logon** – defines logon procedures.

System Admin

The System Admin screen has information about any options implemented for the system.

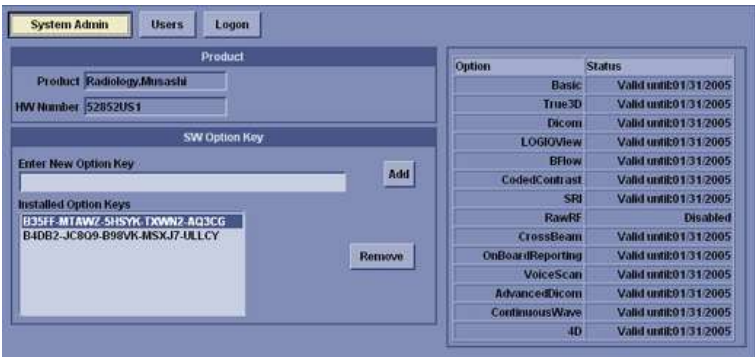


Figure 16-68. Administrative System Admin Preset Menu

Table 16-67: System Administration

Preset Parameter	Description
Product	The name of the product.
HW Number	The hardware number of the product.
SW Option Key	The software option key section.
Enter New Option Key	Type the key for the option you wish to add and press Add.
Installed Option Keys	Lists the key for the installed options.
Remove	To remove a software option key, select the key in the SW Option Key list, and then select Remove.
Options	A list of the option name and status.
Status	Lists each option's effectivity.

Users

The Users screen allows you to define user IDs. It also allows you to specify operators registration, operator’s rights setting, and registration of staff related to an examination (for example, referring and interpreting physicians).



Figure 16-69. Users Preset Menu

Table 16-68: User List

Preset Parameter	Description
User List	Lists the user ID for all system users.
Identity	Type the operator's user ID, Password, Prefix, Last Name, First Name, Middle Name, Suffix, Phone Number.
Group Membership	Select the user's group:Operator (sonographers, doctors, or any person using the ultrasound system); Ref.Phys. (referring physician can be associated to the patient examination in the extended Patient information window(; Perf.Phys. – physician performing the exam can be associated to the patient examination in the extended Patient information window.
Operator Rights	Admin – If selected, the operator has extended rights with access to the administrative setup functionality. The operator can also perform advanced operations

Creating a user

1. Press New.
2. Type the user ID. ENSURE that you DO NOT include the following characters in a user's ID: slash (/), dash (-), asterisk (*), question mark (?), an underscore (_), or blank spaces. Also, DO NOT set up a user with the same initials/signifier.
3. Type the user's information in the Identity section.
4. Select the user's group(s).
5. If the user needs full configuration and advanced operations access, select Admin.
6. Press Save.

NOTE: *DO NOT add users with the same initials/signifier. The system allows you to do this; however, the first user is erased and only the second remains.*

NOTE: *When adding a new user, press Add first. Then edit the ID from the default of "NewUser" and edit the other fields. DO NOT press Add again unless you actually want to create another user. Press Save after adding one or more users. The user listed as NewUser on the list will be updated with the edited ID when you re-enter this screen.*

NOTE: *You need to first set up a user here prior to setting up a user for VoiceScan.*

Changing a user configuration

1. Move the **Trackball** to a user ID in the User List.
2. Make the desired changes.

Deleting a user

1. Move the **Trackball** to a user ID in the User List.
2. Select Remove.

The user is removed from the User List.

Logon

The Logon section defines log on procedures.

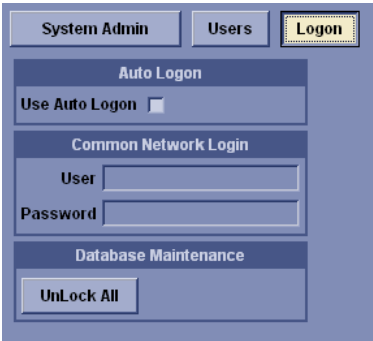


Figure 16-70. Administrative Logon Preset Menu

Table 16-69: Logon

Preset Parameter	Description
Auto Logon	Specifies logon procedures: • When blank, the user must select a user ID and enter a password when logging on. • When selected, the system is started automatically, using the last user logon.
Common Network Login	Specifies the user ID and password used to access the network. User – User ID for network access Password – Password for network access
Database Maintenance	Select to release all patients.

Service

Press Service to activate the Service browser interface.

Search

Opens up a search window to find a parameter on the utility pages.

To search for a utility parameter,

- 1. Press Search.
- 2. Type in the search string. For instance, if you're searching for *Turn on Zone Matching Filter*, you could just type 'zone'.



Figure 16-71. Search Example

- 3. A list of possible matches appears to the right. Select the correct match.

NOTE: You cannot perform a search on the Measure, Reports, or Service utility pages.

Chapter 17

Probes and Biopsy

This chapter consists of the information of each probe and describes some special concerns, biopsy kits and accessories as well as basic procedures for attaching a biopsy guide to the different types of probes.

Probe Overview

Ergonomics

Probes have been ergonomically designed to:

- Handle and manipulate with ease
- Connect to the system with one hand
- Be lightweight and balanced
- Have rounded edges and smooth surfaces.
- Stand up to typical wear by cleaning and disinfectant agents, contact with approved gel, etc.

Cables have been designed to:

- Connect to system with appropriate cable length

Cable handling

Take the following precautions with probe cables:

- Keep free from wheels
- Do not bend the cable acutely
- Avoid crossing cables between probes.

Probe orientation

Each probe is provided with an orientation marking (refer to Figure 17-1). This mark is used to identify the end of the probe corresponding to the side of the image having the orientation mark on the display.



Figure 17-1. Orientation Marking on Probe (Example)

1. Orientation Mark

Labeling

Each probe is labeled with the following information:

- Seller's name and manufacturer
- Operating frequency (not shown on all probes)
- GE part number
- Probe serial number
- Month and year of manufacture
- Probe designation-provided on the probe grip and the top of the connector housing, so it is easily read when mounted on the system and is also automatically displayed on the screen when the probe is selected.

NOTE: *Below is a representative sample of how a probe is labeled; however, not all probes are labeled in this manner.*

Labeling (continued)

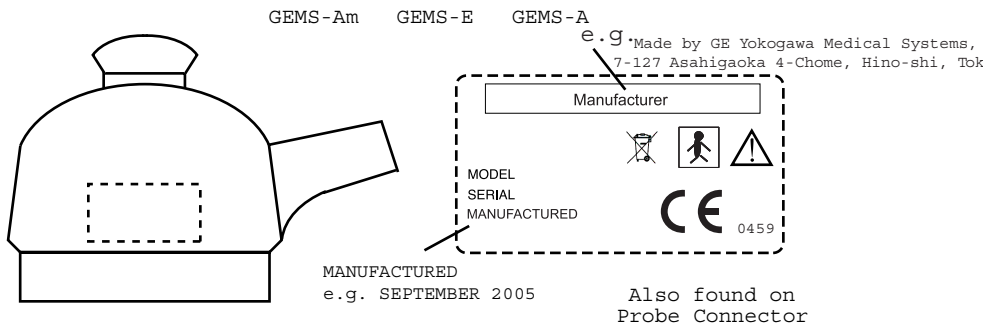


Figure 17-2. Probe Adapter Label

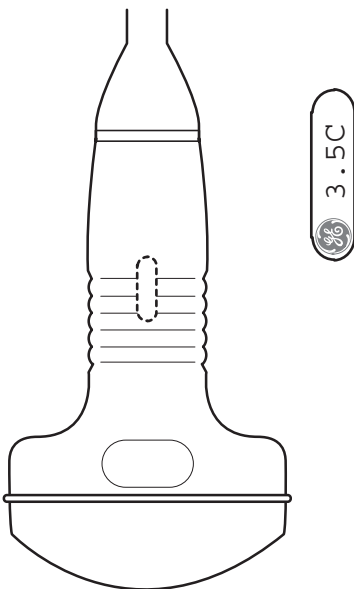


Figure 17-3. Probe Handle Labels

Labeling (continued)

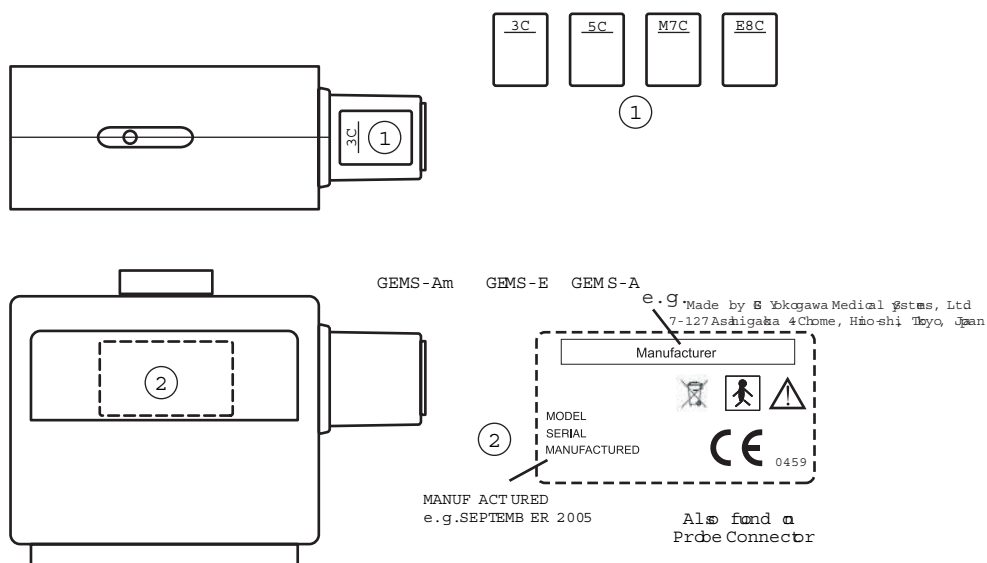


Figure 17-4. Probe Connector Labels

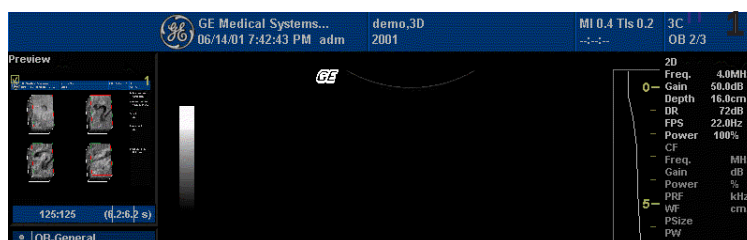


Figure 17-5. Displayed Probe Information

1. Probe Information Location

LOGIQ 9 Applications

Table 17-1: Probe Indications for Use

Applications	3.5C	3.5 CS	4C	M 7C	E 8C	8C	7L	9L	10L	M 12L	i 12L	i 739
Abdomen	X	X	X	X				X				
Small Parts							X	X	X	X	X	X
Periph. Vasc.	X	X	X				X	X	X	X	X	X
OB/GYN	X	X	X	X	X							
Pediatrics				X		X			X	X	X	
Neonatal						X			X	X		
Urology	X	X	X	X								
Surgery										X	X	X
Endocavitary					X							
Transcranial												
Vascular												

Table 17-2: Probe Indications for Use

Applications	t 739	3S	4S	10S	P5D	P8D	4D 3C-L	4D 8C	4D 10L	4D 16L	4D E7C
Abdomen		X	X	X			X				
Small Parts	X			X					X	X	
Periph. Vasc.	X				X	X			X		
OB/GYN		X	X				X				X
Pediatrics				X				X	X		
Neonatal				X				X			
Urology											X
Surgery	X										
Endocavitary											X
Transcranial		X	X								
Vascular					X	X					

LOGIQ 9 Applications (continued)

Table 17-3: Probe Indications for Use with 4D scanning

Applications	4D 3C-L	4D 8C	4D 10L	4D 16L	4D E7C
Abdomen	X				
Small Parts			X	X	
Periph. Vasc.			X	X	
Obstetrics	X				X
Gynecology					X
Pediatrics		X	X	X	
Neonatal		X			
Urology					X
Surgery					
Endocavitary					X
Transcranial					
Musculoskeletal (conventional)			X	X	
Intraoperative			X		

LOGIQ 9 Features

Table 17-4: Probe Features

Probe Application	3.5C	3.5CS	4C	M7C	E8C	8C
Coded Excitation				X	X	X
Coded Harmonics	X	X	X	X	X	X
B-Flow	X	X	X	X	X	X
SRI HD	X	X	X	X	X	X
CrossXBeam	X	X	X	X	X	X
Ultra Coded Harmonics	X (5MHz)		X (5MHz)	X (8MHz)		
LOGIQ View	X	X	X	X	X	X
Virtual Convex						
Easy 3D	X	X	X	X	X	X
Advanced 3D	X	X	X	X	X	X
Tru3D Magnetically Tracked (3D Sensor Based)	X	X	X	X	X	
Anatomical M Mode	X	X	X	X	X	
M Color Flow	X	X	X	X		
TruAccess	X	X	X	X	X	X
Multi-Level Codes	CHI and Contrast Coded PI	CHI and Contrast Coded PI	CHI and Contrast Coded PI	Contrast Coded PI		
Contrast Imaging	X	X	X	X		
Biopsy	X	X	X	X	X	X

LOGIQ 9 Features (continued)

Table 17-5: Probe Features

Probe Application	7L	9L	10L	M12L	i12L	i739	t739	3S
Coded Excitation			Available in Abdominal & OB/ GYN					
Coded Harmonics	X	X	X	X	X	X	X	X
B-Flow	X	X	X	X	X	X	X	X
SRI HD	X	X	X	X	X	X	X	X
CrossXBeam	X	X	X	X	X	X	X	
Ultra Coded Harmonics			X (9.5MHz)	X (13/ 14MHz)				
LOGIQ View	X	X	X	X	X	X	X	X
Virtual Convex	X	X	X	X	X	X	X	X
Easy 3D	X	X	X	X	X	X	X	X
Advanced 3D	X	X	X	X	X	X	X	X
Tru3D Magnetically Tracked (3D Sensor Based)			X	X				
Anatomical M Mode								
M Color Flow								
TruAccess	X	X	X	X	X	X	X	X
Multi-Level Codes	Contrast Coded PI		Fund in Sm Pts and PV	Fund and CHI in Sm Pts, PV, and Ped				
Contrast Imaging	X	X	X	X			X	
Biopsy	X	X	X	X				

LOGIQ 9 Features (continued)

Table 17-6: Probe Features

Probe Application	4S	10S	4D 3C-L	4D 8C	4D 10L	4D 16L	4D E7C	P5D	P8D
Coded Excitation		X							
Coded Harmonics	X	X	X	X	X	X	X		
B-Flow			X	X	X	X	X		
SRI HD	X	X	X	X	X	X	X		
CrossXBeam			X	X	X	X	X		
Ultra Coded Harmonics			X (5 MHz)				X (8 MHz)		
LOGIQ View	X	X	X	X	X	X	X		
Virtual Convex	X	X	X	X	X	X	X		
Easy 3D	X	X							
Advanced 3D	X	X							
Tru3D Magnetically Tracked (3D Sensor Based)	X								
Anatomical M Mode			X				X		
M Color Flow			X						
TruAccess	X	X	X	X	X	X	X		
Multi-Level Codes	Fund and CHI and Contrast Coded PI								
Contrast Imaging	X				X				
Biopsy									
CW								X	X

LOGIQ 9 Features (continued)

Table 17-7: Contrast Technique Availability by Probe (LOGIQ 9)

Contrast Technique	3.5C	3.5 CS	4C	M 7C	7L	9L	10L	M 12L	t 739	4S
B Fundamental	X	X	X	X	X	X	X	X	X	X
Coded PI	X	X	X	X	X	N/A	N/A	N/A	N/A	X
Coded Harmonic Angio	X	X	X	X	X	X	X	X	X	X
TruAgent Detection High MI	X	X	X	N/A	N/A	N/A	N/A	N/A	N/A	X
TruAgent Detection Low MI	X	N/A	X	N/A	X	X	X	N/A	X	N/A

Probe Usage

For details on connecting, activating, deactivating, disconnecting, transporting and storing the probes, See 'Probes' on page 3-47 for more information.

Care and Maintenance

Inspecting probes

**Perform After
Each Use**

Inspect the probe's lens, cable, casing, and connector. Look for any damage that would allow liquid to enter the probe. If any damage is found, do not use the probe until it has been inspected and repaired/replaced by a GE Service Representative.

NOTE: *Keep a log of all probe maintenance, along with a picture of any probe malfunction.*

Environmental Requirements

Probes should be operated, stored, or transported within the parameters outlined below.



Ensure that the probe face temperature does not exceed the normal operation temperature range.

Table 17-8: Probe Environmental Requirements

	Operational	Storage	Transport
Temperature	10° - 40° C 50° - 104° F	-10° - 60° C 14° - 140° F	-40° - 60° C -40° - 140° F
Humidity	30 - 85 % non-condensing	30 - 90% non-condensing	30 - 90% non-condensing
Pressure	700 - 1060hPa	700 - 1060hPa	700 - 1060hPa

Table 17-9: 4D Probe Environmental Requirements

	Operational	Storage	Transport
Temperature	18° to 30° C 50° - 104° F	-10° to 60° C 14° to 140° F	-40° to 60° C -40° to 140° F
Humidity	30 to 75 % non-condensing	30 to 90% non-condensing	10 to 100% non-condensing
Pressure	700 to 1060hPa	700 to 1060hPa	700 to 1060hPa

Probe Safety

Handling precautions



Ultrasound probes are highly sensitive medical instruments that can easily be damaged by improper handling. Use care when handling and protect from damage when not in use. **DO NOT** use a damaged or defective probe. Failure to follow these precautions can result in serious injury and equipment damage.

Electrical shock hazard



The probe is driven with electrical energy that can injure the patient or user if live internal parts are contacted by conductive solution:

- **DO NOT** immerse the probe into any liquid beyond the level indicated by the immersion level diagram. Refer to the immersion illustration in the Probe Cleaning Process section. Never immerse the probe connector or probe adaptors into any liquid.
- **DO NOT** drop the probes or subject them to other types of mechanical shock or impact. Degraded performance or damage such as cracks or chips in the housing may result.
- Prior to each use, visually inspect the probe lens and case area for cracks, cuts, tears, and other signs of physical damage. **DO NOT** use a probe which appears to be damaged until you verify functional and safe performance. You must perform a more thorough inspection, including the cable, strain relief, and connector, each time you clean the probe.
- Before inserting the connector into the probe port, inspect the probe connector pins. If a pin is bent, do not use the probe until it has been inspected and repaired/replaced by a GE Service Representative.
- **DO NOT** kink, tightly coil, or apply excessive force on the probe cable. Insulation failure may result.
- Electrical leakage checks should be performed on a routine basis by GE Service or qualified hospital personnel. Refer to the service manual for leakage check procedures.

Mechanical hazards



A defective probe or excessive force can cause patient injury or probe damage:

- Observe depth markings and do not apply excessive force when inserting or manipulating intercavitary probes.
- Inspect probes for sharp edges or rough surfaces that could injure sensitive tissue.
- DO NOT apply excessive force to the probe connector when inserting into the probe port. The pin of a probe connector may bend.

Special handling instructions

Using protective sheaths



Protective barriers may be required to minimize disease transmission. Probe sheaths are available for use with all clinical situations where infection is a concern. Use of legally marketed, sterile probe sheaths is strongly recommended for intra-cavitary and intra-operative procedures. Use of legally marketed, sterile, pyrogen free probe sheaths is **REQUIRED** for neurological intra-operative procedures.

Instructions. Custom made sheaths are available for each probe. Each probe sheath kit consists of a flexible sheath used to cover the probe and cable and elastic bands used to secure the sheath.

Sterile probe sheaths are supplied as part of biopsy kits for those probes intended for use in biopsy procedures. In addition to the sheath and elastic bands, there are associated accessories for performing a biopsy procedure which are included in the kit. Refer to the biopsy instructions for the specific probes in the Discussion section of this chapter for further information.

Reordering. To reorder sheaths, please contact your local distributor or the appropriate support resource.



Devices containing latex may cause severe allergic reaction in latex sensitive individuals. Refer to FDA's March 29, 1991 Medical Alert on latex products.



Do not use pre-lubricated condoms as a sheath. In some cases, they may damage the probe. Lubricants in these condoms may not be compatible with probe construction.



DO NOT use an expired probe sheath. Before using probe sheaths, verify whether the term of validity has expired.

Endocavitary Probe Handling Precautions

If the sterilization solution comes out of the endocavitary probe, please follow the cautions below.



Sterilant Exposure to Patient (e.g., Cidex)—Contact with a sterilant to the patient's skin or mucous membrane may cause an inflammation. If this happens, refer to the sterilant's instruction manual.

Sterilant Exposure from Probe Handle to Patient (e.g., Cidex)—DO NOT allow the sterilant to contact the patient. Only immerse the probe to its specified level. Ensure that no solution has entered the probe's handle before scanning the patient. If sterilant comes into contact with the patient, refer the the sterilant's instruction manual.

Sterilant Exposure from Probe Connector to Patient (e.g., Cidex)—DO NOT allow the sterilant to contact the patient. Only immerse the probe to its specified level. Ensure that no solution has entered the probe's connector before scanning the patient. If sterilant comes into contact with the patient, refer the the sterilant's instruction manual.

Endocavitary Probe Point of Contact—Refer to the sterilant's instruction manual.

Probe handling and infection control

This information is intended to increase user awareness of the risks of disease transmission associated with using this equipment and provide guidance in making decisions directly affecting the safety of the patient as well as the equipment user.

Diagnostic ultrasound systems utilize ultrasound energy that must be coupled to the patient by direct physical contact. Depending on the type of examination, this contact occurs with a variety of tissues ranging from intact skin in a routine exam to recirculating blood in a surgical procedure. The level of risk of infection varies greatly with the type of contact.

One of the most effective ways to prevent transmission between patients is with single use or disposable devices. However, ultrasound transducers are complex and expensive devices that must be reused between patients. It is very important, therefore, to minimize the risk of disease transmission by using barriers and through proper processing between patients.



Risk of Infection. ALWAYS clean and disinfect the probe between patients to the level appropriate for the type of examination and use FDA-cleared probe sheaths where appropriate. See Chapter 18 for ordering information.



Adequate cleaning and disinfection are necessary to prevent disease transmission. It is the responsibility of the equipment user to verify and maintain the effectiveness of the infection control procedures in use. Always use sterile, legally marketed probe sheaths for intra-cavitary and intra-operative procedures.

For neurological intra-operative procedures, use of a legally marketed, sterile, pyrogen free probe sheath is **REQUIRED**. Probes for neuro surgical use must not be sterilized with liquid chemical sterilants because of the possibility of neuro toxic residues remaining on the probe.

Probe Cleaning Process

Cleaning probes

Perform After Each Use

To clean the probe:

1. Disconnect the probe from the ultrasound console and remove all coupling gel from the probe by wiping with a soft cloth and rinsing with flowing water.
2. Wash the probe with mild soap in lukewarm water. Scrub the probe as needed using a soft sponge, gauze, or cloth to remove all visible residue from the probe surface. Prolonged soaking or scrubbing with a soft bristle brush (such as a toothbrush) may be necessary if material has dried onto the probe surface.



Take extra care when handling the lens face of the Ultrasound transducer. The lens face is especially sensitive and can easily be damaged by rough handling. NEVER use excessive force when cleaning the lens face.

3. Rinse the probe with enough clean potable water to remove all visible soap residue.
4. Air dry or dry with a soft cloth.



To minimize the risk of infection from blood-borne pathogens, you must handle the probe and all disposables which have contacted blood, other potentially infectious materials, mucous membranes, and non-intact skin in accordance with infection control procedures. You must wear protective gloves when handling potentially infectious material. Use a face shield and gown if there is a risk of splashing or splatter.

Cleaning probes (continued)

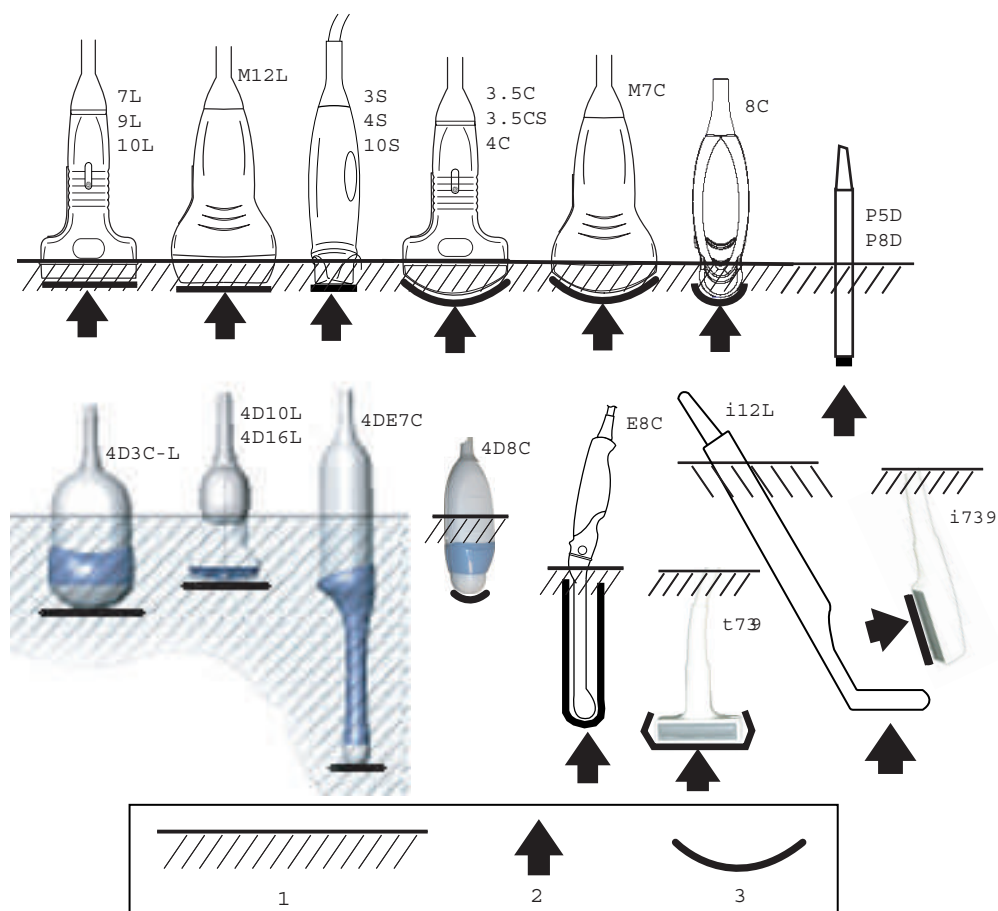


Figure 17-6. Probe Immersion Levels

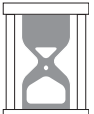
1. Fluid Level
2. Aperture
3. Contact face within Patient Environment

Disinfecting probes

Perform After Each Use

Ultrasound probes can be disinfected using liquid chemical germicides. The level of disinfection is directly related to the duration of contact with the germicide. Increased contact time produces a higher level of disinfection.

Table 17-10: Description of Pictogram on Care card

Pictogram	Description
	“ATTENTION” - Consult accompanying documents” is intended to alert the user to refer to the operator manual or other instructions when complete information cannot be provided on the label.
	“CAUTION” - Dangerous voltage” (the lightning flash with arrowhead) is used to indicate electric shock hazards.
	Biohazard - Patient/user infection due to contaminated equipment. Usage • Cleaning and care instructions • Sheath and glove guidelines
	Ultrasound probes are highly sensitive medical instruments that can easily be damaged by improper handling. Use care when handling and protect from damage when not in use.
	Do not immerse the probe into any liquid beyond the level specified for that probe. Refer to the user manual of the ultrasound system.
	Since there is a possibility of having negative effects on the probe, observe the specified immersing time by the germicide manufacturer strictly. Do not immerse the probe in liquid chemical germicides more than the time prescribed in the care card.

Disinfecting probes (continued)

Review the probe care card that is packed with each probe. The following website contains the most current and up-to-date recommendations:

http://www.gehealthcare.com/us/en/ultrasound/products/probe_care.html

The following high level disinfectant agents have been approved for use with all probes:

- Cidex OPA
- Cidex (except P5D and P8D)

Cidex Plus has been approved for the 8C, 4DE7C, 4D8C, 4D3C-L, and 4D16L probes.

Sporox II high level disinfectant has been approved for the 7L, i/t739, 9L, 10L, 11L, 12L, 4S, 8C, and E8C probes.

Pera Safe high level disinfectant has been approved for the 7L, i/t739, 9L, 10L, 11L, 12L, M7C, M12L, E8C, 4DE7C, 4D8C, 4D3C-L, 4D10L, and 4D16L probes.

The following low level disinfect agents have been approved for use with all probes:

- Ster Bac Blu (except P5D and P8D)
- Sani-Cloth HB (Wipes)

T-Spray and T-Spray II low level disinfectant has been approved for the 7L, i/t739, 9L, 10L, 12L, M7C, M12L, 4S, E8C, P5D, P8D, 4DE7C, 4D8C, 4D3C-L, 4D10L, and 4D16L probes.

Virex II 256 low level disinfectant has been approved for the 7L, i/t739, 9L, 10L, 12L, M7C, M12L, and E8C probes.

Disinfecting probes (continued)



In order for liquid chemical germicides to be effective, all visible residue must be removed during the cleaning process. Thoroughly clean the probe, as described earlier before attempting disinfection.

You **MUST** disconnect the probe from the LOGIQ 9 prior to cleaning/disinfecting the probe. Failure to do so could damage the system.

DO NOT soak probes in liquid chemical germicide for longer than is stated by the germicide instructions for use. Extended soaking may cause probe damage and early failure of the enclosure, resulting in possible electric shock hazard.

1. Prepare the germicide solution according to the manufacturer's instructions. Be sure to follow all precautions for storage, use and disposal.
2. Place the cleaned and dried probe in contact with the germicide for the time specified by the germicide manufacturer. High-level disinfection is recommended for surface probes and is required for endocavitary and intraoperative probes (follow the germicide manufacturer's recommended time).



Probes for neuro surgical intra-operative use must **NOT** be sterilized with liquid chemical sterilants because of the possibility of neuro toxic residues remaining on the probe. Neurological procedures must be done with the use of legally marketed, sterile, pyrogen free probe sheaths.

3. After removing from the germicide, rinse the probe following the germicide manufacturer's rinsing instructions. Flush all visible germicide residue from the probe and allow to air dry.

Disinfecting probes (continued)



WARNING

CREUTZFELD-JAKOB DISEASE

Failure of the probe sheath or direct contact of the probe with dura or any intra-cranial tissue of patients with Creutzfeld-Jakob disease requires that the probe be destroyed. There is no effective means for decontamination of the probe. For more information, see the Center of Disease Control and Prevention <http://www.cdc.gov/ncidod/hip/sterile/cjd.htm>.



WARNING

Ultrasound transducers can easily be damaged by improper handling and by contact with certain chemicals. Failure to follow these precautions can result in serious injury and equipment damage.

- Do not immerse the probe into any liquid beyond the level specified for that probe. Never immerse the transducer connector or probe adapters into any liquid.
- Avoid mechanical shock or impact to the transducer and do not apply excessive bending or pulling force to the cable.
- Transducer damage can result from contact with inappropriate coupling or cleaning agents:
 - Do not soak or saturate transducers with solutions containing alcohol, bleach, ammonium chloride compounds or hydrogen peroxide
 - Avoid contact with solutions or coupling gels containing mineral oil or lanolin
 - Avoid temperatures above 60°C.
- Inspect the probe prior to use for damage or degeneration to the housing, strain relief, lens and seal. Do not use a damaged or defective probe.

Coupling gels



Do not use unrecommended gels (lubricants). They may damage the probe and void the warranty.

Applying

In order to assure optimal transmission of energy between the patient and probe, a conductive gel or couplant must be applied liberally to the patient where scanning will be performed.



Do not apply gel to the eyes. If there is gel contact to the eye, flush eye thoroughly with water.

Precautions

Coupling gels should not contain the following ingredients as they are known to cause probe damage:

- Methanol, ethanol, isopropanol, or any other alcohol-based product
- Mineral oil
- Iodine
- Lotions
- Lanolin
- Aloe Vera
- Olive Oil
- Methyl or Ethyl Parabens (para hydroxybenzoic acid)
- Dimethylsilicone

Planned Maintenance

The following maintenance schedule is suggested for the system and probes to ensure optimum operation and safety..

Table 17-11: Planned Maintenance Program

Do the Following	Daily	After Each Use	As Necessary
Inspect the Probes	X		X
Clean the Probes		X	X
Disinfect Probes		X	X

Returning/Shipping Probes and Repair Parts

US Department of Transportation and GE Medical Systems policy requires that equipment returned for service **MUST** be clean and free of blood and other infectious substances.

When you return a probe or part for service (Field Engineer or customer), you need to clean and disinfect the probe or part prior to packing and shipping the equipment.

Ensure that you follow probe cleaning and disinfection instructions provided in the Basic User Manual.

This ensures that employees in the transportation industry as well as the people who receive the package are protected from any risk.

Probe Discussion

Introduction

The LOGIQ 9 supports the following types of probes:

- **Active Matrix Array.** Active Matrix Array probes are usually designated by the prefix/suffix "M".
- **Curved Array (Convex).** Curved Array (Convex) probes, including 'micro' convex, are usually designated by the prefix/suffix "C"; the endocavitary probe is designated by the prefix/suffix "E".
- **Linear Array.** Linear Array probes are designated by the prefix/suffix "L"; the linear intra-operative probes are designated by the prefix/suffix "I" or "T".
- **Phased Array Sector.** Phased Array Sector probes are designated by the prefix/suffix "S"; the biplane TEE probe is designated by the prefix/suffix "B"; the multi-plane TEE probe is designated by the prefix/suffix "T".
- **Continuous Wave Doppler.** Pencil probes for Continuous Wave Doppler are designated by the prefix/suffix "P".
- **4D.** 4D probes are designated by the prefix "4D".

LOGIQ 9 Convex Probes

Table 17-12: Active Matrix Array (Convex) Probes

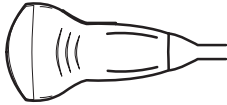
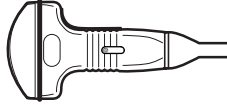
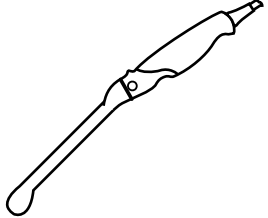
Probe	Intended Uses	Capabilities and Features	Illustration
M7C	• OB/GYN • Abdomen • Fetal Echo	• Wide field of view • Biopsy capability	

Table 17-13: Curved Array (Convex) Probes

Probe	Intended Uses	Capabilities and Features	Illustration
3.5C	• Abdominal • OB/GYN	• Wide field of view • Biopsy capability	
3.5CS	• Abdominal • OB/GYN • Urology	• Wide field of view • Small footprint • Biopsy capability	
4C	• Abdominal • OB/GYN	• Wide field of view • Small footprint • Biopsy capability	
8C	• Neonatal • Pediatrics	• Wide field of view • Small footprint	
E8C	• Transvaginal • Transrectal	• Wide field of view • Small headshell and probe shaft • Biopsy capability	

LOGIQ 9 Linear Probes

Table 17-14: Active Matrix Array (Linear) Probes

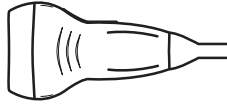
Probe	Intended Uses	Capabilities and Features	Illustration
M12L	• Small Parts • Peripheral Vascular • Pediatrics • Neonatal • Surgery	• Excellent near field resolution • Improved contrast resolution • Improved penetration • Biopsy capability	

Table 17-15: Linear Array Probes

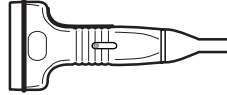
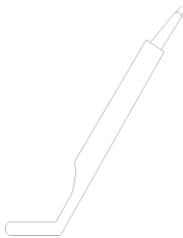
Probe	Intended Uses	Capabilities and Features	Illustration
7L	• Small Parts • Peripheral Vascular	• Wide field of view • Slant scan • Wide field of view for B-Mode resolution & homogeneity • Biopsy capability	
9L	• Abdomen • Small Parts • Vascular • Pediatrics	• Wide field of view • Slant scan • Biopsy capability	
10L	• Small Parts • Peripheral Vascular	• Wide field of view • Slant scan • Wide field of view for B-Mode resolution & homogeneity • Biopsy capability	

Table 17-15: Linear Array Probes

Probe	Intended Uses	Capabilities and Features	Illustration
i12L	• Intraoperative • Small Parts • Vascular • Pediatrics	• Low height I-shaped micro-case • Long, light cable • Wide field of view • High resolution • Compound scan	
i739	• Intraoperative • Small Parts • Vascular • Pediatrics	• Low height I-shaped micro-case • Long, light cable • Slant scan • Wide field of view • High resolution • CFM/Doppler sensitivity	
t739	• Intraoperative • Superficial imaging for organs with space constraints	• Low height T-shaped micro-case • Long and light probe cable • Slant scan • Wide field of view for B-Mode resolution and homogeneity • CFM/Doppler sensitivity • Ergonomics for scanning and cleaning	

LOGIQ 9 Sector Probes

Table 17-16: Sector Probes

Probe	Intended Uses	Capabilities and Features	Illustration
3S	• Cardiology • Transcranial • Abdomen	• Small footprint • Wide field of view for B-Mode resolution and homogeneity • CFM Doppler sensitivity	
4S	• Transcranial • Abdomen	• Small footprint • Dedicated handle design for Transcranial • Wide field of view for B-Mode resolution and homogeneity	
10S	• Transcranial • Abdomen	• Small footprint • Dedicated handle design for Transcranial • Wide field of view for B-Mode resolution and homogeneity	
Some versions of the sector probe have an LED which is not activated with the LOGIQ 9.			

LOGIQ 9 CW Probes

Table 17-17: CW Probes

Probe	Intended Uses	Capabilities and Features	Illustration
P5D	CW Doppler for Peripheral Vascular	• High sensitivity • T-bar support for holding • Ergonomics for scanning and cleaning	
P8D	CW Doppler for Peripheral Vascular	• High sensitivity • Ergonomics for scanning and cleaning	

4D Probes

Table 17-18: 4D Probes

Probe	Intended Uses	Capabilities and Features	Illustration
4D3C-L	3C	• 3D/4D real-time imaging • Wide field of view • CFM, Power, and PW Doppler • Broad bandwidth • Multi-frequency • Harmonics	
4D8C	• Neonatal • Pediatrics	• 3D/4D real-time imaging • Wide field of view • CFM, Power, and PW Doppler • Broad bandwidth • Multi-frequency • Harmonics	
4D10L	• Small parts • Pediatrics • Peripheral vascular • Musculoskeletal	• 3D/4D real-time imaging • Wide field of view • CFM, Power, and PW Doppler • Broad bandwidth • Multi-frequency • Harmonics	
4D16L	• Small Parts • Peripheral vascular • Pediatrics	• 3D/4D real-time imaging • Wide field of view • CFM, Power, and PW Doppler • Broad bandwidth • Multi-frequency • Harmonics	
4DE7C	• Gynecology • Obstetrics • Urology • Neonatal Cephalic	• 3D/4D real-time imaging • Wide field of view • CFM, Power, and PW Doppler • Broad bandwidth • Multi-frequency • Harmonics	

Biopsy Special Concerns

Precautions Concerning the Use of Biopsy Procedures



WARNING

Do not freeze the image during a biopsy procedure. The image must be live to avoid a positioning error.

Biopsy guidezones are intended to assist the user in determining optimal probe placement and approximate the needle path. However, actual needle movement is likely to deviate from the guideline. Always monitor the relative positions of the biopsy needle and the subject mass during the procedure.



CAUTION

The use of biopsy devices and accessories that have not been evaluated for use with this equipment may not be compatible and could result in injury.



CAUTION

The invasive nature of biopsy procedures requires proper preparation and technique to control infection and disease transmission. Equipment must be cleaned as appropriate for the procedure prior to use.

- Follow the probe cleaning and disinfection procedures and precautions to properly prepare the probe.
- Follow the manufacturer's instructions for the cleaning of biopsy devices and accessories.
- Use protective barriers such as gloves and probe sheaths.
- After use, follow proper procedures for decontamination, cleaning, and waste disposal.

Precautions Concerning the Use of Biopsy Procedures (continued)



Improper cleaning methods and the use of certain cleaning and disinfecting agents can cause damage to the plastic components that will degrade imaging performance or increase the risk of electric shock.

See 'Probe Safety' on *page 17-14* for more information.

Preparing for a Biopsy

Displaying the Guidezone

Activate the Biopsy Kit by selecting it from the B-Mode Touch Panel.

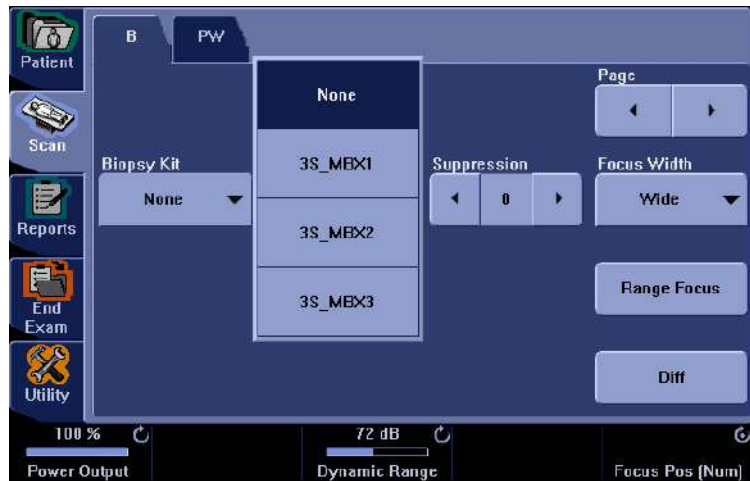


Figure 17-7. B-Mode Touch Panel Menu

The available biopsy options appear when Biopsy Kit is selected. There are fixed and adjustable angle biopsy kits available with the LOGIQ 9 depending on the probe. Select the desired biopsy kit.

Displaying the Guidezone (continued)



Figure 17-8. Biopsy Guidezones for the 10L Probe

1. 1 cm increments
2. 5 cm increments

The biopsy guidezone represents a path of the needle. The dots which make up the guidezones is the depth readout where:

- Yellow represent 1 cm increments.
- Red represents 5 cm increments.

The display should be carefully monitored during a biopsy for any needle deviation from the center line or guidezone.

NOTE: To set up biopsy guidezones, refer to Table 16-10 for more details.

Displaying the Guidezone (continued)

The needle may vary from the center line or guidezone for various reasons:

- Needle barrel to needle clearance or strength.
- Bracket manufacturing tolerance.
- Needle deflection due to tissue resistance.
- Needle size chosen. Thinner needles may deflect more.

Table 17-19: Biopsy Guide Availability

Probe	Fixed Angle	Multi-Angle		
		MBX1	MBX2	MBX3
3.5C	7.0			
5C	7.0	4.0	5.5	7.0
M7C		3.5	5.5	7.5
E8C	5.0			
7L	4.0	4.0	5.5	7.0
10L	2.0	2.0	4.0	6.0
M12L		1.5	2.5	3.5



Failure to match the guidezone displayed to the guide may cause the needle to track a path outside the zone.

It is extremely important that when using the adjustable angle biopsy guides, the angle displayed on the screen matches the angle set on the guide, otherwise the needle will not follow the displayed guidezone which could result in repeated biopsies or patient injury.

NOTE: *Although the multi-angle guides are compatible with the Civco Ultrapro and Ultrapro II, it is recommended the multi-angle guides only be used with the Ultrapro II.*

Preparing the Biopsy Guide Attachment

Convex, Sector and Linear probes have optional biopsy guide attachments for each probe. The guide consists of a non-disposable bracket to attach to the probe, disposable needle clip to attach to the bracket, sheath, gel (sterile gel if necessary) and disposable needle barrels.

The disposable needle barrels are available for a variety of needle sizes.



Please refer to the manufacturer's instructions included in the biopsy kit.

Fixed Needle Biopsy Guide Assembly

1. Identify the appropriate biopsy guide bracket by matching the label on the bracket with the probe to be used.
2. Orient the bracket so that the needle clip attachment will be on the same side as the probe orientation mark (ridge).

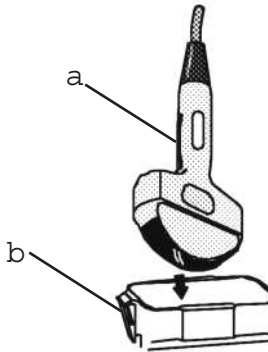


Figure 17-9. Probe/Bracket Alignment

- a. Probe Orientation Mark
 - b. Bracket
3. Attach the biopsy bracket to the probe by sliding the bracket over the end of the probe until it clicks or locks in place.
 4. Place an adequate amount of coupling gel on the face of the probe.
 5. Place the proper sanitary sheath over the probe and biopsy bracket. Use the rubber bands supplied to hold the sheath in place.

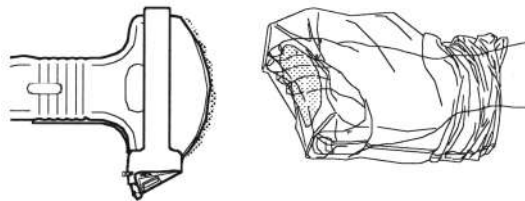


Figure 17-10. Applying Sanitary Sheath

Fixed Needle Biopsy Guide Assembly (continued)

6. Snap the fixed or adjustable needle clip onto the biopsy guide bracket.

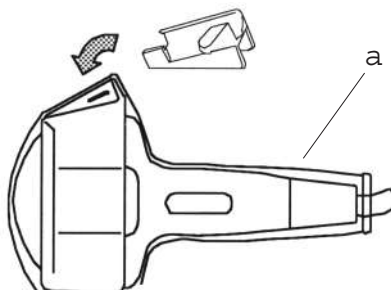


Figure 17-11. Fixed Needle Clip Attachment

a. Sheath

7. Push the locking mechanism towards the bracket to secure the lock. Make sure the needle guide is firmly attached to the bracket.



Figure 17-12. Locking the Needle Clip

8. Choose the desired gauge (size) needle barrel. Twist it back and forth to remove it from the plastic tree.



Figure 17-13. Needle Barrel Selection

Fixed Needle Biopsy Guide Assembly (continued)

9. Place the needle barrel into the needle clip with the desired gauge facing the needle clip and snap into place.

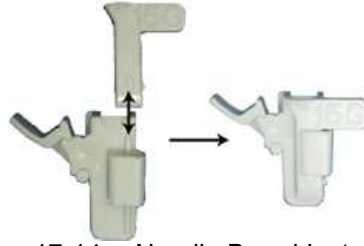


Figure 17-14. Needle Barrel Installation



CAUTION

Ensure that all guide parts are seated properly prior to performing a biopsy.

Multi Angle Biopsy Guide Assembly



WARNING

DO NOT attempt to use the biopsy bracket and needle guide until the manufacturer's instructions, provided with the biopsy bracket and needle guide in the kit, have been read and thoroughly understood.

1. Scan the patient and identify the target for biopsy. Move the probe to locate the target to the center of the image. Enable the system biopsy guidezone and try guidezone angles A1 to A3 to decide the best angle setting for needle path.
2. Identify the appropriate biopsy guide bracket by matching the label on the bracket with the probe to be used.

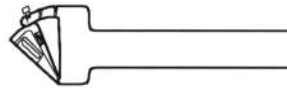


Figure 17-15. Multi-Angle Biopsy Guide Bracket

3. Orient the bracket so that the needle clip attachment will be on the same side as the probe orientation mark (ridge).

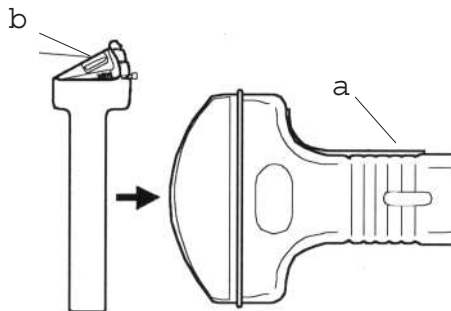


Figure 17-16. Probe/Bracket Alignment

- a. Probe Orientation Mark
 - b. Bracket
4. Attach the biopsy bracket to the probe by sliding the bracket over the end of the probe until it clicks or locks in place.

Multi Angle Biopsy Guide Assembly (continued)

5. Pull up on the knob to freely move the needle guide attachment. Align the knob with the selected position of the needle guide attachment from MBX1, MBX2 and MBX3, to match the guidezone display on the ultrasound system.

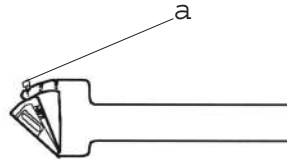


Figure 17-17. Select the angle position

- a. Pull up
6. Push the knob down into the desired slot to secure the angle position of the needle guide attachment.

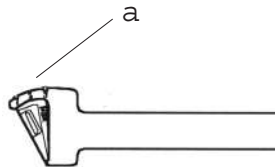


Figure 17-18. Fix the angle position

- a. Push



Hold the bracket in place on the probe when pushing the knob to secure the angle position of the needle guide attachment. Excessive force may cause the bracket to release from the probe.

Multi Angle Biopsy Guide Assembly (continued)

7. Place an adequate amount of coupling gel on the face of the probe.
8. Place the proper sanitary sheath tightly over the probe and biopsy bracket. Use the rubber bands supplied to hold the sheath in place.

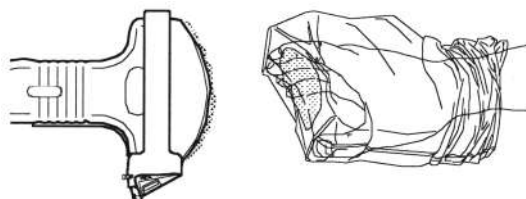


Figure 17-19. Applying Sanitary Sheath

9. Snap the needle clip onto the biopsy guide bracket.

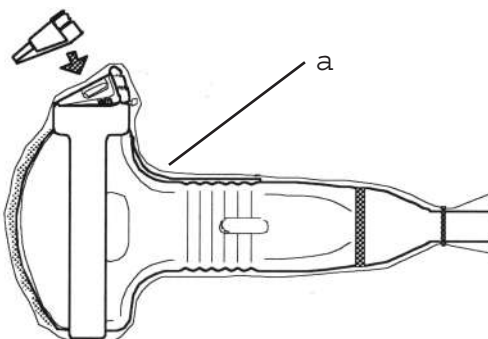


Figure 17-20. Fixing the Needle Clip Attachment

a. Sheath

10. Push the locking mechanism towards the bracket to secure the lock. Make sure the needle guide is firmly attached to the bracket.



Figure 17-21. Locking the Needle Clip

Multi Angle Biopsy Guide Assembly (continued)

11. Choose the desired gauge (size) needle barrel. Twist it back and forth to remove it from the plastic tree.



Figure 17-22. Needle Barrel Selection

12. Place the needle barrel into the needle clip with the desired gauge facing the needle clip and snap into place.

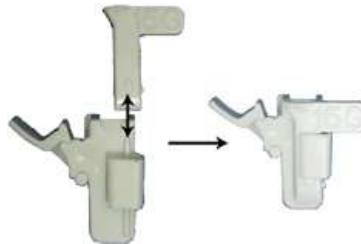


Figure 17-23. Needle Barrel Installation



Ensure that all guide parts are seated properly prior to performing a biopsy.

Releasing the needle

According to the following procedure, you remove the needle from a probe and an assembly without moving the needle.

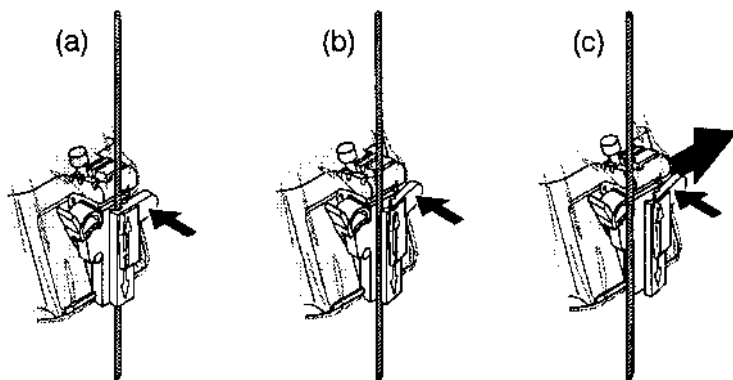


Figure 17-24. Release the needle from assembly

- a. Push the knob portion of a sleeve in the direction of the arrow.
- b. The needle is released from the assembly.
- c. Push the probe and the assembly in the direction of the larger arrow to remove the needle.

Endocavitary Probe Biopsy Guide Assembly

NOTE: The reusable biopsy guide fits with 16G needle or thinner.

E8C Preparation To prepare the E8C for use:

1. Remove the probe from the box and carefully examine it for any damage.
2. If the biopsy guide is to be attached, use the filling removal tool to clean out the attachment area on the probe head.

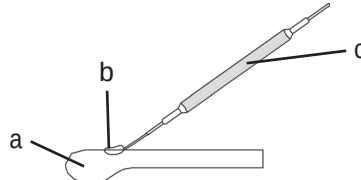


Figure 17-25. Attachment Filling Removal

- a. Probe Head
 - b. Attachment
 - c. Filling Removal Tool
3. Clean, then disinfect the probe.

NOTE: Ensure that protective gloves are worn.

Installing the sheath

To install the sheath:

1. Remove the sheath from its package. Do not unroll the sheath.
2. Place an adequate amount of ultrasound gel inside the sheath tip (the gel is between the sheath inner surface and the probe aperture).

NOTE: *Remember to rinse all sanitary probe sheaths of powder before placing on the probe. Powder can degrade the displayed image.*

NOTE: *Ensure that only acoustic coupling gel is used for this purpose.*

3. Place the sheath tip over the probe aperture and then pull the sheath end toward the probe handle.
4. Inspect the sheath for nicks, cuts or tears.

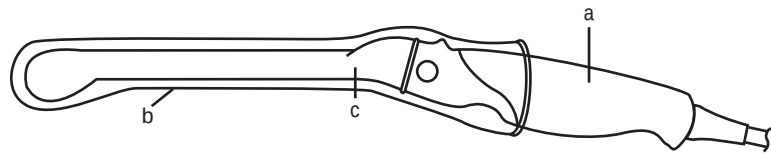


Figure 17-26. E8C Probe with Sheath



Figure 17-27. BE9C Probe with Sheath

- a. Probe Handle
- b. Sanitary Sheath
- c. Probe Body

5. Rub a finger over the tip of the probe to ensure all air bubbles have been removed.

E8C Biopsy Guide Preparation



1. If a biopsy is to be performed, snap the metal or plastic biopsy guide on to the probe over the sheath.

Patient injury or repeated biopsies may result. The needle placement will not be as intended if the needle guide is not properly seated and secure.

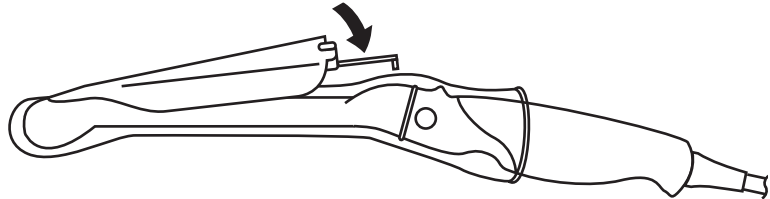


Figure 17-28. Civco Disposable Biopsy Guide 5° Angle

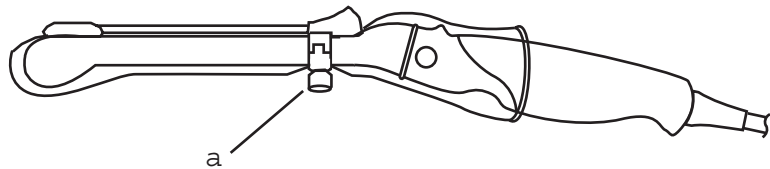


Figure 17-29. Convex Biopsy Guide for E8C (Reusable)

a. Fix with a screw

2. Place an adequate amount of ultrasound gel on the gel-filled sheath tip's outer surface.
3. Ensure the guide is properly seated and secure by pushing forward on the needle insertion end of the guide until the attachment node is firmly in place in it's hole.



You **MUST** select E8C_RU from the Touch Panel when using the reusable E8C biopsy guide.

DO NOT use the needle with the catheter (soft tube). There is a possibility of breaking the catheter in the body.

Biopsy Needle Path Verification

To verify that the path of the needle is accurately indicated within the guidezone on the system monitor, perform the following:

- Properly install the bracket and biopsy guide.
- Scan in a container filled with water (47° C).
- Display the biopsy guidezone on the monitor.
- Ensure that the needle echo falls within the guidezone markers.

The Biopsy Procedure



WARNING

Biopsy procedures must only be performed on live images.

1. Place coupling gel on the scanning surface of the probe/sheath/biopsy guide assembly.
2. Activate the biopsy guidezone on the system through the B-Mode Touch Panel. When using multi-angle guides, ensure that the proper guidezone angle is displayed.

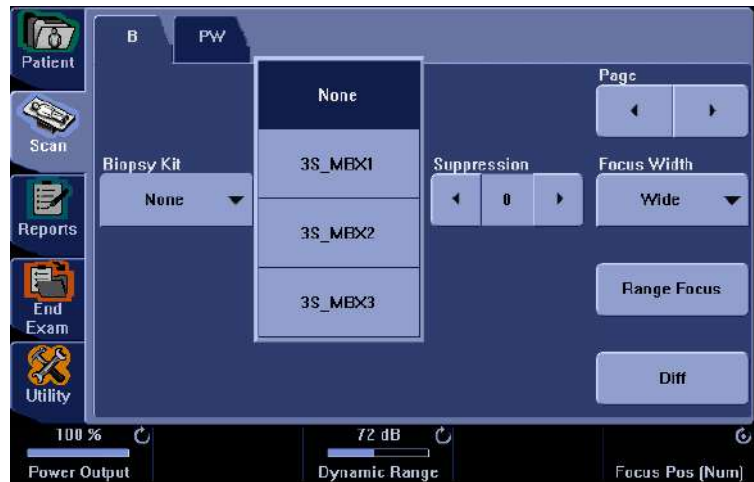


Figure 17-30. B-Mode Touch Panel

3. Scan to locate the target. Center the target in the electronic guidezone path.

NOTE: *Enabling color flow would allow for visualization of the vascular structure around the area to be biopsied.*

4. Place the needle in the guide between the needle barrel and needle clip. Direct it into the area of interest for specimen retrieval.

Post Biopsy

When the biopsy is complete, remove the needle barrel, needle clip and probe sheath. Properly dispose of these items in accordance with current facility guidelines.

Clean and disinfect the probe. See 'Probe Cleaning Process' on *page 17-19 for more information*.

The biopsy bracket can be cleaned and disinfected in a recommended disinfecting agent and reused.



When the biopsy needle guide kit (UP, UP2 or UP2+) is opened, all parts must be discarded after the procedure whether they have been used or not.

Surgery/Intra-operative Use

Preparing for Surgery/Intra-operative Procedures

Preparing the transducer for intra-operative use follows the same sterile procedure as for biopsy use except that no biopsy attachments are used. See 'Preparing the Biopsy Guide Attachment' on *page 17-38 for more information*. Sterile gel is applied to the transducer face and a sterile sheath completely covers the transducer and cable which has first undergone a thorough cleaning and high-level disinfection.

The invasive nature of biopsy procedures requires proper preparation and technique to control infection and disease transmission. Equipment must be cleaned as appropriate for the procedure prior to use.



For surgery/intra-operative procedures, a sterile environment is required. Therefore, both the operator and probe needs to be sterile.

Preparing for Surgery/Intra-operative Procedures (continued)

To ensure a sterile environment during the procedure, it is recommended that this be a two-person job.

1. Perform a high level disinfection of the probe.
2. The scanner (surgeon, sonographer, etc.) should be sterile and gloved.
3. Place an adequate amount of sterile coupling gel on the face of the probe.
4. Place the proper sterile sheath over the probe and cord.

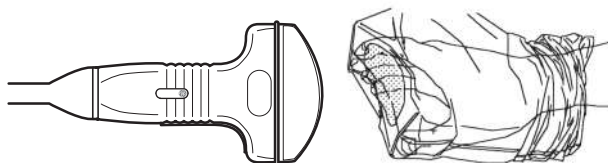


Figure 17-31. Applying Sterile Sheath

5. Depending on the type of procedure, use either sterile water or sterile gel on the sheath cover.

NOTE: *Follow your institutions guidelines on post surgery/intra-operative procedures for probe cleaning and disinfection.*

Chapter 18

User Maintenance

This chapter supplies system data, assistance information, and system care and maintenance instructions.

System Data

Features/Specifications

Table 18-1: Physical Attributes

Physical Attributes	
<u>Dimensions and Weight</u> • Height: Minimum 140.3cm (55.25 in). Minimum LCD: 127cm (50 in). Maximum 160.7cm (63.25 in) • Width: 64 cm (25.2 in) • Depth: 89.9 cm (35.4 in) • Weight with CRT: approx. 185 kg (408 lb.) • Weight with LCD: approx. 171 kg (376 lb.) <u>Keyboard</u> • Floating keyboard adjustable in three dimensions • Adjustable Keyboard Height: 76-96.5cm (30-38in) • Rotation: adjustable +/- 75 degrees from center • Extension: 31.8cm (12.5in) from console • Full alphanumeric keyboard • Ergonomic hard key operations • Interactive Back-Lighting • 10.4" LCD touch panel • Integrated recording keys for remote control • User Define, User Programmable Keys • Integrated gel warmer <u>Electrical Power</u> • Voltage: 100-120 Vac or 220-240 Vac • Frequency: 50/60 Hz • Power: Max. 1.2 kVA with built-in and on-board peripherals	<u>Console Design</u> • 4 Active Probe Ports • Integrated HDD (80 GB) • Integrated DVD-R Drive • Probe connection area designed to allow free movement of legs • On-board storage for peripherals: Black/white printer, color printer, VCR. • Rear Storage Cover for Peripheral Connectors and Cables • Probe Holder, removable for cleaning and washing • Gel Holder, removable for cleaning and washing • Wheels. Wheel diameter: 175 mm. Rotating flexible wheels with protective cover. Locking Mechanism for wheels. <u>Monitor</u> • Single 17 High-Resolution monitor Non Interlace Scan Flat CRT • SVGA Formats (800x600) • Tilt/Rotate Adjustable Monitor: Tilt Angle: Up 10 degrees, Down 10 degrees. Rotate Angle: + 90 degrees. <u>LCD Monitor Option</u> • 17" High-Resolution LCD • Integrated Speakers • Fold-down Lock Mechanism for Transport • Height adjustable +/-15cm (+/-6 in) • Side to side +/- 50cm (+/-19 in)

Features/Specifications (continued)

Table 18-2: System Overview

<u>Applications</u> • Abdominal • Obstetrical • Gynecological • Musculoskeletal • Vascular • Urological • Small parts and Superficial • Breast • Pediatric and Neonatal • Orthopedic • Intraoperative • Transcranial <u>Scanning Methods</u> • Electronic Sector • Electronic Convex • Electronic Linear • Mechanical Volume Sweep <u>Operating Modes</u> • B-Mode • M-Mode • PW Doppler • High PRF Doppler • Color Flow Doppler (CFM) • Power Doppler Imaging (PDI) • Continuous Wave Doppler • B Flow Mode • M Color Flow Mode • Volume Modes (3D/4D) • 3D Static • 4D Real Time <u>Display Modes</u> • Simultaneous Capability [B/D; B/CFM (PDI); B/M; B + CFM/M; Realtime Triplex Mode (B + CFM (PDI)/D); Dual B (B/B); B(b/w)/B + CFM (PDI)] • Selectable alternating Modes [B/M; B/D; B + CFM (PDI)/M; B + CFM (PDI)/D; 3D-Mode (optional); 3D-Color Mode (optional)] • Colorized Image - Colorized B - Colorized M - Colorized D • Time line Display [Independent Dual B/D Display; 2 Display Formats; Top/Bottom or Side/Side selectable Format; Size: 1/2: 1/2; 1/3: 2/3 (full format, switchable after freeze); 2 Time line Methods: Scrolling or Moving Bar] • Virtual Convex <u>Transducer Types</u> • Sector Phased Array • Convex Array • Linear Array • Pencil probe • Active Matrix Array • Volume Probes (4D)	<u>Standard Features</u> • Standard CINE Memory • Realtime Triplex Mode at any depth and any PRF • Automatic Optimization • Flash Suppression • Virtual Convex • Built-in patient archive with images, patient information • Auto Doppler Calculation • Vascular Calcs • OB Calcs • Fetal Trending • Multi gestational Calcs • Gynecological Calcs • Urological Calcs • Renal Calcs • Coded Harmonics • Coded Excitation • SRI HD • Anatomical M-mode • Standby Mode • Fine Angle Steer • iLinq Capability • Integrated CD/DVD • Advanced 3D • Raw Data Analysis • Hip Dysplasia Calcs • On-Board Electronic Documentation <u>Options</u> • LOGIQ View • B-Flow • CrossXBeam • 4D • VOCAL • VCI Static • CW • ReportWriter • VoiceScan • Coded Contrast • Tru3D • DICOM 3.0 connectivity • LCD • TV Probe Holder • Footswitch • Insite capability • UPS (Uninterruptible Power Supply) <u>Peripheral Standard</u> • CD/DVD-RW <u>Peripheral Options</u> • B/W Printer • Color Printer • S-VHS Video Cassette Recorder • MO Drive / HDD Drive
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Features/Specifications (continued)

Table 18-3: System Parameters

<u>Post-Processing with Raw Data</u> • B/M-Mode (Gray Map; Rejection, Colorized B and M; Frame Averaging, Gain, Compression, Anatomical M Mode) • PW-Mode (Gray Map, Colorized D, Gain, Baseline Shift, Sweep Speed, Invert, Compression, Display Format) • Color Flow (Color Map, CFM Display Threshold, Gain, CFM Velocity Tag, Invert) • 4D (Map, Gain, Display) • SRI HD • CrossXBeam • 3D Reconstruction • Anatomical M Mode • Auto Optimize <u>Pre-Processing</u> • B/M-Mode (Gain; TGC; Dynamic Range; Transmission Focus Position; Transmission Focus Number; Transmission Frequency; Edge Enhancement; Line Density; Smoothing; Sweep Speed for M-Mode, # of CrossXBeam Angles, Acoustic Output) • PW-Mode (Gain; Dynamic Range; Transmission Frequency; Velocity Scale/PRF; Wall Filter; Baseline Shift; Sweep Speed, Acoustic Output, Spectral Averaging, SV Gate--length & Depth) • Color Flow (CFM Gain; CFM Velocity Range; Packet Size; Frame Rate Control; CFM Spatial Filter; CFM Frame Averaging; Baseline Shift; CFM Flash Suppression, Acoustic Output, Wall Filter, CFM Line Resolution, Frequency/Velocity Baseline Shift)	<u>Image Processing and Presentation</u> • Image Reverse: Right/ Left • Image Rotation: 4 steps, Rotation: 0, 90, 180 and 270 degrees • Imaging Depth: 0-30cm [Minimum: 0-1 cm (Zoom); Maximum: 0-30 cm] • Transmission Focus [1-8 Focus Points selectable; Focus Position: 8 steps • Receiving Focus [CDA/CDF (Continuous Dynamic Focus/Continuous Dynamic Aperture)] • 256 Shades of Gray <u>Zoom</u> • Smart Zoom (Acoustic Write Zoom up to 4x): • Reference Image for Smart Zoom • Read Zoom (available in B/ M and CFM up to 8x) <u>Image Archive</u> • Archiving Format: DICOM • Possibly exported to other format <u>CINE Memory/Image Memory</u> • Dual Image CINE Display • CINE Gauge and CINE Image number display • CINE Review Loop • Selectable CINE Sequence for CINE Review (by Start Frame and End Frame) • Side Change in dual CINE Mode • Measurements/Calculations & Annotations on CINE & VCR Playback Images
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Features/Specifications (continued)

Table 18-4: Measurements and Calculations

<u>B-Mode</u> • Depth • Distance • Circumference/ Area • Volume (Ellipsoid) • Angle • Ratios • % Stenosis <u>M-Mode</u> • M-Depth • Distance • Time • Slope • Heart Rate <u>Doppler Measurements/Calculations</u> • Velocity • Frequency • Time • Acceleration • Heart Rate • Auto Doppler Trace function with automatic calculations • Time Averaged Max./ Mean Velocity • Ratios • PI (Pulsatility Index) • RI (Resistivity Index) • PS, ED, PS/ED, ED/PS • AT • Volume Flow	<u>Vascular Measurements/Calculations</u> • Summary Worksheet • Summary Report <u>Obstetrics Measurements/Calculations</u> • Gestational Age Calculation • Multi-Gestational Calculation • EFW Calculation • Summary Worksheet • Summary Report • Fetal Trend Graph <u>Gynecology Measurements/Calculations</u> • Summary Worksheet • Summary Report <u>Urology Measurements/Calculation</u> • Summary Worksheet • Summary Report ReportWriter • Customized Templates
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Features/Specifications (continued)

Table 18-5: Probes

• 3.5C Convex Probe (Applications: Abdomen, OB/ GYN, Urology) • 3.5CS Convex Probe (Applications: Abdomen, PV, OB/GYN, Urology) • 4C Convex Probe (Applications: Abdomen, OB/ GYN, Urology, Vascular) • M7C Convex AMA Probe (Applications: Abdomen, OB/GYN, Pediatrics) • 8C Convex Probe (Applications: Pediatrics, Neonatal) • E8C Micro Convex Endocavity -Probe (Applications: Transvaginal, Transrectal, Obstetrics, Gynecology, Urology) • 3S Phased Array Sector Probe (Applications: Abdomen, Transcranial) • 4S Phased Array Sector Probe (Applications: Transcranial) • 10S Phased Array Sector Probe (Applications: Neonatal, Small Parts, Abdomen) • 7L Linear Probe (Applications: Vascular, Pediatrics, Small Parts) • i739 Linear Probe (Applications: Intraoperative, Small Parts, Vascular, Pediatrics)	• t739 Linear Probe (Applications: Intraoperative, Superficial imaging) • 9L Linear Probe (Applications: Abdomen, Small Parts, PV, OB-GYN, Pediatrics, Neonatal) • 10L Linear Probe (Applications: Vascular, Small Parts, Musculoskeletal, Mammasonography) • M12L Linear AMA Probe (Applications: Small Parts, Musculoskeletal, Mammasonography) • i12L Linear Probe (Applications: Surgery, Small Parts, Peripheral Vascular, Pediatrics) • 4D3C Convex Volume Probe (Applications: Abdomen, OB) • 4DE7C Convex Volume Probe (Applications: OB/ GYN, Urology) • 4D8C Convex Volume Probe (Applications: Neonatal, Pediatrics) • 4D10L Linear Volume Probe (Applications: Small Parts, Peripheral Vascular, Pediatrics) • 4D16L Linear Volume Probe (Applications: Small Parts, Peripheral Vascular, Pediatrics) • PCW 5.0 and 8.0 non-imaging Peripheral CW Pencil Probe (Applications: Peripheral Vascular)
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Table 18-6: Inputs and Outputs Signal

• Video In (Composite Color; S-Video) • Video Out (RGB; SVGA; Composite Color; Composite b/w; S-Video) • Audio Stereo Out	• Audio Stereo In • Connectors [Remote for black/white Printer (2); Remote for color printer or camera; external microphone; Footswitch; USB; Ethernet; iLinq-Connection; Power by peripherals (5)]
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Clinical Measurement Accuracy

Basic Measurements

The following information is intended to provide guidance to the user in determining the amount of variation or measurement error that should be considered when performing clinical measurements with this equipment. Error can be contributed by equipment limitations and improper user technique. Be sure to follow all measurement instructions and develop uniform measurement techniques among all users to minimize the potential operator error. Also, in order to detect possible equipment malfunctions that could affect measurement accuracy, a quality assurance (QA) plan should be established for the equipment that includes routine accuracy checks with tissue mimicking phantoms.

Please be advised that all distance and Doppler related measurements through tissue are dependent upon the propagation velocity of sound within the tissue. The propagation velocity usually varies with the type of tissue, but an average velocity for soft tissue is assumed. This equipment is designed for, and the accuracy statements listed on are based on, an assumed average velocity of 1540 m/s. The percent accuracy when stated applies to the measurement obtained (not the full scale range). Where the accuracy is stated as a percent with a fixed value, the expected inaccuracy is the greater of the two.

Basic Measurements (continued)

Table 18-7: System Measurements and Accuracies

Measurement	Units	Useful Range	Accuracy	Limitations or Conditions
Depth	mm	Full Screen	±3% or 1 mm	
Distance:				
Axial	mm	Full Screen	±3% or 1 mm	
Lateral	mm	Full Screen	±5% or 1 mm	Linear Probes
Lateral	mm	Full Screen	±5% or 4 mm	Convex Probes
Lateral	mm	Full Screen	±5% or 1 mm	Sector Probes
LOGIQView	Measurement error is within 5% of the distance you measured for all linear probes.			
Circumference:			±5% or 1 mm	
Trace	mm	Full Screen	±5% or 1 mm	
Ellipse	mm	Full Screen	±5% or 1 mm	
Area:			±6% or 1.5 mm ²	
Trace	mm ²	Full Screen	±6% or 1.5 mm ²	
Ellipse	mm ²	Full Screen	±6% or 1.5 mm ²	
Static 3D/4D Volume Accuracy			±9%	
Time	s	Timeline Display	±5% or 10 ms	M or Doppler Mode
Slope	mm/s	Timeline Display	±5% or 1 mm/s	M-Mode Only
Doppler SV Position	mm	Image depth <3cm Image depth 3-10cm Image depth 10-20cm Image depth 20-30cm Image depth 30-36cm	± 0.3 mm ± 0.7 mm ± 1.5 mm ± 2.2 mm ± 2.5 mm	Any Direction
Velocity	cm/s	Full Ranges	4% Velocity Scale*1.6%	PW Doppler Mode
Doppler Angle Correction	cm/s	From 0-59° From 60-90°	±1% ±2%	For each 1 degree error in cursor alignment, add indicated percent to above Doppler accuracy.

Clinical Calculation Accuracy

Estimate the overall inaccuracy of a combined measurement and calculation by including the stated inaccuracy from the basic measurement accuracy statements.



Diagnostic errors may result from the inappropriate use of clinical calculations. Review the referenced source of the stated formula or method to become familiar with the intended uses and possible limitations of the calculation.

Calculation formulas and databases are provided as a tool to assist the user, but should not be considered an undisputed database, in making a clinical diagnosis. The user is encouraged to research the literature and judge the equipment capabilities on an ongoing basis in order to assess its utility as a clinical tool.

System Care and Maintenance

Overview

Refer to Section 10 of the LOGIQ 9 Service Manual for any additional maintenance guidance.

Contact the local Service Representative for parts or periodic maintenance inspections.

Inspecting the System

Examine the following on a monthly basis:

- Connectors on cables for any mechanical defects.
- Entire length of electrical and power cables for cuts or abrasions.
- Equipment for loose or missing hardware.
- Control panel and keyboard for defects.
- Casters for proper locking operation.



To avoid electrical shock hazard, do not remove panels or covers from console. This servicing must be performed by qualified service personnel. Failure to do so could cause serious injury.



Biological
Hazard

If any defects are observed or malfunctions occur, do not operate the equipment but inform a qualified service person. Contact a Service Representative for information.

Weekly Maintenance

The system requires weekly care and maintenance to function safely and properly. Clean the following:

- Monitor
- Operator control panel
- Footswitch
- Video Cassette Recorder (VCR)
- Video Page Printer

Failure to perform required maintenance may result in unnecessary service calls.

Cleaning the system

Prior to cleaning any part of the system:

1. Turn off the system power. If possible, disconnect the power cord. See 'Power Off' on *page 3-24 for more information.*

System Cabinet

To clean the system cabinet:

1. Moisten a soft, non-abrasive folded cloth with a mild, general purpose, non-abrasive soap and water solution.
2. Moisten a soft, non-abrasive folded cloth.
3. Wipe down the top, front, back, and both sides of the system cabinet.

NOTE: *Do not spray any liquid directly into the unit.*

Monitor

To clean the monitor face:

Use a soft, folded cloth. Gently wipe the monitor face.

Do NOT use a glass cleaner that has a hydrocarbon base (such as Benzene, Methyl Alcohol or Methyl Ethyl Ketone) on monitors with the filter (anti-glare shield). Hard rubbing will also damage the filter.

NOTE: *When cleaning the monitor, make sure not to scratch the monitor.*

LCD Monitor (option)

NOTE: *Never use thinner, benzene, alcohol (ethanol, methanol, or isopropyl alcohol), abrasive cleaners, or other strong solvents, as these may cause damage to the cabinet or LCD panel.*

NOTE: *DO NOT scratch or press on the panel with any sharp objects, such as pencils or pens, as this may result in damage to the panel.*

To clean the cabinet:

- To remove stains, wipe the cabinet with a soft, lightly moistened cloth using a mild detergent. Do not spray wax or cleaner directly into the cabinet.

To clean LCD panel:

- The LCD surface can be cleaned with a soft cloth, such as cotton or lens paper.
- If necessary, stubborn stains can be removed by moistening part of a cloth with water to enhance its cleaning power.

You can use the "Read Right Alcohol - Free Screen Kleen" wipe to clean the LCD display.

Operator Controls

To clean the operator control panel:

1. Moisten a soft, non-abrasive folded cloth with a mild, general purpose, non-abrasive soap and water solution.
2. Wipe down operator control panel.
3. Use a cotton swab to clean around keys or controls. Use a toothpick to remove solids from between keys and controls.

NOTE: *When cleaning the operator control panel, make sure not to spill or spray any liquid on the controls, into the system cabinet, or in the probe connection receptacle.*

NOTE: *In case of SARS, use bleach, alcohol, or Cidex in a normal diluted form for cleaning/disinfecting the operator panel.*

NOTE: *DO NOT use T spray or Sani Wipes on the control panel.*

Footswitch

To clean the footswitch:

1. Moisten a soft, non-abrasive folded cloth with a mild, general purpose, non-abrasive soap and water solution.
2. Wipe the external surfaces of the unit then dry with a soft, clean, cloth.

VCR

To clean the VCR:

1. Turn off the VCR power. If possible, disconnect the power cord.
2. Wipe the external surfaces of the unit with a soft, clean, dry cloth.

NOTE: *Do not use a wet cloth or any cleaning fluid because it may enter and damage the unit.*

1. Clean the record and playback heads with a soft, non-abrasive cleaning system, according to the manufacturer's instructions.

For more information, see the VCR's Operator Manual.

Discs

To clean the discs:

- If the disc is dirty, you can clean it with a commercially available dust removal spray. Be careful not to scratch the disc when wiping it off.

NOTE: *CD type cleaner cannot be used.*



HINTS

Keeping your media disc in an original media case or caddy all the time will prevent it from becoming dirty or damaged.

Touch Panel

For the touch screen, use a soft cloth without any cleaning agents. DO NOT use Windex, Screen-Clean, etc.

Video Page Printer

To clean the external surface of the video page printer:

1. Turn off the power. If possible, disconnect the power cord.
2. Wipe the external surfaces of the unit with a soft, clean, dry cloth.
3. Remove stubborn stains with a cloth lightly dampened with a mild detergent solution.

NOTE: *Never use strong solvents, such as thinner or benzine, or abrasive cleansers because they will damage the cabinet.*

No further maintenance, such as lubrication, is required.

To clean the surface of the print head:

1. Run the cleaning sheet (provided with the printer) through the printer.

For more information, see the Video Page Printer's Operator Manual.

Other Peripheral Maintenance

Refer to the peripheral manuals for more information.

Other Maintenance

Replacing illuminated key caps/lamps

Contact a local Service Representative when a key cap or lamp needs to be replaced.

Uninterruptible Power Supply (UPS) Option

Lifetime The lifetime of a UPS is limited and varies according to circumstances. It lasts for three years under normal conditions; however, it may last one year if the UPS is used over twelve hours per day and if the room temperature is over 30 degrees C.

Possible damage may occur with the HDD, system software and user data by the electrical power outage or by disconnecting the power plug after the battery is dead.

Battery Replacement and Disposition Battery replacement every three years is recommended.
Contact a local Service Representative for the replacement of the battery. Used batteries will be discarded appropriately by GE.

NOTE: Disposing of the battery should meet local law and regulatory requirements.

NOTE: Dispose of the system according to local law and regulatory requirements.

Cleaning the air filter

Clean the system's air filters to ensure that a clogged filter does not cause the system to overheat and reduce system performance and reliability. It is recommended the filters be cleaned quarterly (once every three months).



Be sure to lock the wheels before cleaning the air filters to avoid injury by any unexpected movement of the system.

Locating

The LOGIQ 9 filter is located at the back of the system.

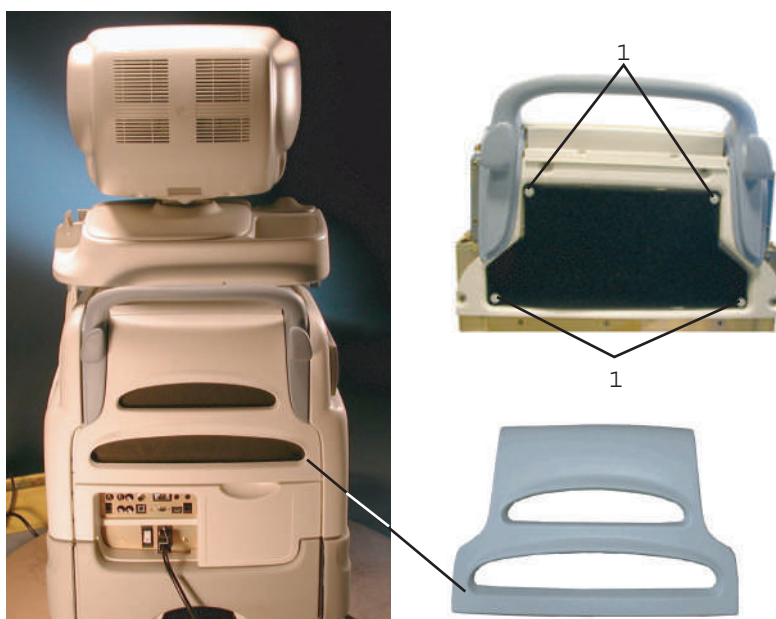


Figure 18-1. Location of Air Filters

1. Plastic retainers

Cleaning

To clean the filter:

1. Be sure to shake the filter in an area away from the system.
2. Wash the filter in a mild soapy solution, rinse and air dry or dry with a cloth.

NOTE: Allow wet filter to dry thoroughly before installing.

Installing

1. Slide the filter back into the system.
2. Secure the filter with the Velcro tab.

Quality Assurance

Introduction

A good Quality Assurance Evaluation program consists of periodic systematic actions that provide the user with adequate confidence that their diagnostic ultrasound system will produce consistently high quality images and quantitative information.

Therefore, it is in the best interests of every ultrasound user to routinely monitor equipment performance.

The frequency of Quality Assurance evaluations should be based on user's specific needs and clinical practice.

Periodic monitoring is essential in order to detect the performance changes that occur through normal aging of system components. Routine equipment evaluations may also reduce the duration of exams, number of repeat exams, and maintenance time required.

For details on system and peripheral routine preventive maintenance instructions, See 'System Care and Maintenance' on page 18-10 for more information.

Typical Tests to Perform

Quality assurance measurements provide results relating to system performance. Typically these are:

- Axial Measurement Accuracy
- Lateral Measurement Accuracy
- Axial and Lateral Resolution
- Penetration
- Functional & Contrast Resolution
- Gray Scale Photography.

With these tests, a performance baseline can be set at installation with the phantom in your department. Future test results can be compared to the baseline in order to maintain a record of system performance trends.

Frequency of tests

Quality assurance tests are used to determine whether a scanner is providing the same level of performance from day to day.

The frequency of testing varies with the amount of system usage and modes to be tested. It is recommended that the user perform quality assurance tests at least every three months or every 400 patient studies. Tests should also be performed when a question about system performance exists.

A mobile system may require more frequent tests.

Image quality should also be tested immediately after the following events:

- Service calls
- System upgrades/modifications
- Dropped probe, power surge, etc.

Phantoms

Quality Assurance Evaluations should be done with phantoms and test objects that are applicable to the parameters being evaluated or to the user's clinical practice.

Typical phantoms are composed of material that acoustically mimic human tissue. Pins, anechoic and echogenic targets are physically positioned to provide information for a variety of tests.

Doppler phantoms are currently expensive and complicated to deal with on the user level. If a problem with any Doppler parameters or measurement is suspected, contact a local service representative for evaluation.

The RMI 403GS phantom is still available. Due to the superior penetration and resolution capabilities of GE ultrasound systems, the RMI 405GSX is recommended. It is the most current one available to our field service personnel and will provide the targets and extended life necessary for consistent system testing.

Phantoms (continued)

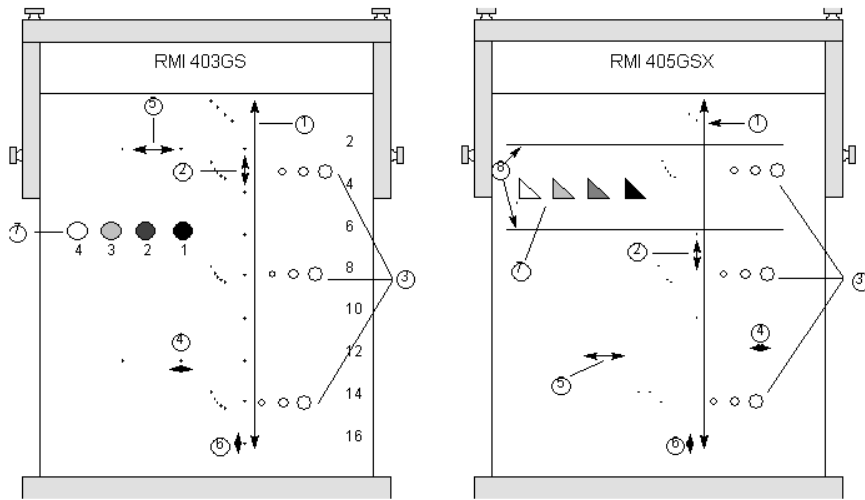


Figure 18-2. Phantoms

1. Penetration
2. Axial Distance Measurement
3. Functional Resolution
4. Lateral Resolution
5. Lateral Distance Measurement
6. Axial Resolution
7. Contrast Resolution and Gray Scale Photography
8. Gray Scale Plane Targets

Baselines

An absolute necessity for a quality assurance program is establishing baselines for each test or check. Baselines are established after the system has been verified to be working properly at installation or after a repair. If a probe or major assembly is replaced, new baselines should be generated.

Baselines can be made by adjusting system parameters to prescribed levels or to the best possible image. The key factor to remember is reproducibility. The same conditions must be reproduced for each periodic check.

All system parameters not displayed on the monitor should be recorded for the permanent record.

Periodic Checks

Periodic checks should be performed in accordance with your facility's quality assurance requirements. For the data to be valid, periodic checks should mimic the baseline setup parameters.

The resulting image, when scanning the phantom exactly as before, should be recorded and compared to the baseline. When a matching image is obtained, it can be assumed that the system performance has not degraded from the baseline.

If a significant difference between the baseline and periodic check is noted, double check the system setup and repeat the test. If the difference between the baseline and periodic check persists, contact a local Service Representative.

Failing to reproduce the control settings as in the baselines will introduce errors in the data and potentially invalidate the results.

Results

Lack of standardization among test instruments, the wide range of acceptance criteria, and incomplete knowledge regarding the significance of certain performance parameters prohibit the establishment of absolute performance criteria for these tests.

Quality Assurance Evaluation results should be compared to previously-recorded results.

Performance trends can then be detected. Unacceptable performance or diminishing trends should be identified for maintenance or repair before a malfunction or inappropriate diagnosis occurs.

The user should determine the best method for recording and archiving the baseline and periodic checks. In most cases the choice is hard copy.

It is important to maintain good consistent records for inspections that may arise, as well as to detect system performance trends.

System Setup

The user should tailor the tests to their particular needs. It is certainly not necessary to make all checks with all probes. A representative example, with the probes used most often by the customer, should be adequate in judging system performance trends.

Use a gray scale phantom as the scan object for the tests. Commercial phantoms are supplied with its own operator manual. Be familiar with proper phantom operating procedures prior to use for quality assurance evaluations.

1. Adjust image monitor. Brightness and contrast should be set to the normal viewing of a good gray scale image.
2. Check all recording devices for proper duplication of image monitor. Ensure that what is seen is what is recorded.
3. Annotate non-displayed image processing controls.
4. Set TGC slide pots to center (detent) position.
5. Place focal zone marker(s) in area of interest for an optimum image.

Test Procedures

The following are recommended Quality Assurance tests. A brief description of the test, the benefit it provides and steps to accomplish the test are supplied.

The importance of recording scan parameters and consistent record keeping cannot be stressed enough. Reproducibility to monitor system trends is the key to quality assurance evaluations.

Using the system's dual image display format is often very convenient and saves recording media.

Axial distance measurements

Description	Axial measurements are the distance measurements obtained along the sound beam. See Figure 18-2 for more information.
Benefit	The accurate measurement of the size, depth and volume of a structure is a critical factor in determining a proper diagnosis. Most imaging systems use depth markers and/or electronic calipers for this purpose.
Method	Axial distance should be measured in the near, mid and far fields as well as in zoom. If necessary, different depths or fields of view can be tested.
Procedure	To measure axial distance: 1. Scan a test phantom with precisely-spaced vertical pin targets. Adjust all scan controls, as necessary, for the best image of the pin targets to typical depths for the probe being used. 2. Press Freeze to stop image acquisition and perform a standard distance measurement between the pins at different points in the image. Record all images for archiving. 3. Scan the vertical pins in zoom or at different depth/scale factors. 4. Press Freeze to stop image acquisition; repeat the distance measurements between pins and record the images for archiving. 5. Document the measurements for reference and future comparison. Contact a Service Engineer if vertical measurements differ by more than 1.50% of the actual distance.

Lateral distance measurements

Description	Lateral measurements are distance measurements obtained perpendicular to the axis of the sound beam. See Figure 18-2 for more information.
Benefit	The purpose is the same as vertical measurements. Precisely-spaced horizontal pin targets are scanned and results compared to the known distance in the phantom.
Method	Lateral distance should be measured in the near, mid and far fields as well as in zoom. If necessary, different depths of fields of view can be tested.
Procedure	To measure lateral distance: 1. Scan a test phantom with precisely-spaced horizontal pin targets. Adjust all scan controls, as necessary, for the best image of the pin targets from side to side. 2. Press Freeze to stop image acquisition and perform a standard distance measurement between the pins at different points in the image. Record all images for archiving. 3. Scan the horizontal pins in zoom or at different depth/scale factors. 4. Press Freeze to stop image acquisition; repeat the distance measurements between pins and record the images for archiving. 5. Document the measurements for reference and future comparison.

Contact a Service Engineer if horizontal measurements differ by more than 3mm or 3% of that depth, whichever is greater.

Axial resolution

Description Axial resolution is the minimum reflector separation between two closely-spaced objects to produce discrete reflections along the axis of the sound beam. It can also be monitored by checking the vertical size of known pin targets. See Figure 18-2 for more information.

Axial resolution is affected by the transmitting section of the system and the probe.

Benefit In clinical imaging, poor axial resolution displays small structures lying close together as a single dot. This may lead to improper interpretation of the ultrasound image.

Procedure To measure Axial resolution:

1. Scan a test phantom with precisely-spaced vertical pin targets.
2. Adjust all scan controls, as necessary, for the best image of the pin targets to typical depths for the probe being used.
3. Press **Freeze** to stop image acquisition.
4. Perform a standard distance measurement of the pin vertical thickness at different points in the image. Record all images for archiving.
5. Scan the vertical pins in zoom or at different depth/scale factors.
6. Press **Freeze** to stop image acquisition; repeat the vertical thickness measurements of the pins and record the images for archiving.
7. Document the measurements for reference and future comparison.

Axial resolution should remain stable over time. Contact a Service Engineer if any changes are observed.

Lateral resolution

Description	Lateral resolution is the minimum reflector separation between two closely spaced objects to produce discrete reflections perpendicular to the axis of the sound beam. It can also be monitored by checking the horizontal size of known pin targets. See Figure 18-2 for more information. Lateral resolution is dependent upon the beam width produced by the probe. The narrower the beam, the better the lateral resolution. The beam width is affected by the frequency, degree of focusing, and distance of the object from the face of the probe.
Benefit	Clinically, poor lateral resolution will display small structures lying close together as a single dot. This may lead to improper interpretation of the ultrasound image.
Procedure	To measure lateral resolution: 1. Scan a test phantom with precisely-spaced horizontal pin targets. 2. Adjust all scan controls, as necessary, for the best image of the pin targets from side to side. 3. Press Freeze to stop image acquisition and perform a standard distance measurement of the horizontal thickness of a pin at different points in the image. Record all images for archiving. 4. Scan the horizontal pins in zoom or at different depth/scale factors. 5. Press Freeze to stop image acquisition; repeat the horizontal thickness measurements of the pins and record the images for archiving. 6. Document the measurements for reference and future comparison. Pin width should remain relatively constant over time ("1mm). Dramatic changes in pin width may indicate beamforming problems. Contact a Service Engineer if beam width changes consistently over 2 to 3 periodic tests.

Penetration

Description	Penetration is the ability of an imaging system to detect and display weak echoes from small objects at large depths. See Figure 18-2 for more information. Penetration can be affected by the system's: • Transmitter/receiver • Degree of probe focusing • Attenuation of the medium • Depth and shape of reflecting object • Electromagnetic interference from local surroundings.
Benefit	Weak reflecting echoes are commonly produced from the internal structure of organs. Definition of this tissue texture is important in the interpretation of the ultrasound findings.
Method	Scan a phantom to see how echoes begin to fade as depth is increased. The maximum depth of penetration is the point at which homogeneous material in the phantom begins to lose brightness.
Procedure	To measure penetration: 1. Set the front panel TGC slide pots to their center (detent) position. 2. Gain and acoustic output can be adjusted, as necessary, since these values are displayed on the monitor. 3. Scan a test phantom along the vertical pin targets to typical depths for the probe being used. 4. Perform a standard distance measurement from the top of the image displayed to the point at which homogeneous material in the phantom begins to lose brightness. 5. Document the depth measurement for reference and future comparison. Contact a Service Engineer if the depth of penetration shifts more than one centimeter (1cm) when using the same probe and same system settings.

Functional resolution

Description	Functional resolution is an imaging system's ability to detect and display the size, shape, and depth of an anechoic structure, as opposed to a pin target. See Figure 18-2 for more information. The very best possible image is somewhat less important than reproducibility and stability over time. Routine tests at the same settings should produce the same results.
Benefit	The data obtained will give a relative indication of the smallest structure the system is capable of resolving at a given depth.
Procedure	To measure functional resolution: 1. Set the front panel TGC slide pots to their center (detent) position. 2. Gain and acoustic output can be adjusted as necessary, since these values are displayed on the monitor. 3. Scan a test phantom with a vertical row of anechoic cyst targets to typical depths for the probe being used. 4. Evaluate the cysts at various depths for a good (round) shape, well-defined borders and no fill in. Remember, TGC slide pots are centered and should remain fixed. This may NOT provide optimal cystic clearing. 5. Document all results for future reference and comparison. Contact a Service Engineer if a greatly distorted image is obtained.

Contrast resolution

Description	Contrast resolution is the ability of an imaging system to detect and display the shape and echogenic characteristics of a structure. See Figure 18-2 for more information. Specific values measured are less important than stability over time. Routine tests at the same settings should produce the same results.
Benefit	A correct diagnosis is dependent upon an imaging system's ability to differentiate between a cystic or solid structure versus echo patterns from normal surrounding tissue.
Method	A phantom with echogenic targets of different sizes and depths should be used.
Procedure	To measure contrast resolution: 1. Set the front panel TGC slide pots to their center (detent) position. Set dynamic range to 54 db. 2. Gain and acoustic output can be adjusted, as necessary, since these values are displayed on the monitor. 3. Scan a test phantom with echogenic targets at the depths available. 4. Evaluate the echogenic targets for contrast between each other and between the surrounding phantom material. Remember, TGC slide pots are centered and should remain fixed. This may NOT provide an optimal scan image. 5. Document all results for future reference and comparison. Contact a Service Engineer if the echogenic characteristics or shapes of the targets appear distorted.

Gray Scale photography

Description	Poor photography will cause loss of low level echoes and the lack of contrast between large amplitude echoes. See Figure 18-2 for more information.
Benefit	When photographic controls and film processors are properly adjusted, weak echoes, as well as strong echoes, are accurately recorded on film.
Procedure	1. Adjust the camera according to the manufacturer's instructions until the hard copy and video display are equal. 2. Scan the phantom and it's echogenic contrast targets. 3. Make a hard copy photograph of the display and compare it to the image on the video monitor for contrast and weak echo display. 4. Document all results for future reference and comparison. Contact a Service Engineer if camera cannot duplicate what is on the image monitor.
NOTE:	<i>Optimization of brightness/contrast controls on the display monitor is imperative in order to make sure that the hardcopy and monitor look alike.</i> The display monitor is adjusted first. The hardcopy camera or printer is adjusted to match the display monitor.

Setting up a Record Keeping System

Preparation

The following is needed:

- Quality Assurance binder.
- Hard copy or electronic file of images.
- Quality Assurance Checklists.
- Display the following information while testing quality assurance:
 - Acoustic Output
 - Gain
 - Depth
 - Probe
 - Dynamic Range
 - Set up new patient to be the name of the test.
- Annotate the following:
 - Any control where its value is **NOT** displayed.
 - Significant phantom information.

Record Keeping

Complete the following:

1. Fill out the Ultrasound Quality Assurance Checklist for each probe, as scheduled.
2. Make a hard copy or archive the image.
3. Compare images to baseline images and acceptable values.
4. Evaluate trends over previous test periods.
5. File hard copy or electronic file of images and checklist in Quality Assurance binder.

Ultrasound Quality Assurance Checklist

Table 18-8: Ultrasound Quality Assurance Checklist (Part 1)

Performed By		Date
System		Serial Number
Probe Type	Probe Model	Serial Number
Phantom Model	Serial Number	Room Temperature
Acoustic Output	Gain	Focal Zone
Gray Map	TGC	Depth
Monitor Setting		
Peripheral Settings		
Other Image Processing Control Settings		

Table 18-9: (Part 2)

Test	Baseline Value Range	Tested Value	Image Hardcopy/ Archived	Acceptable? Yes/No	Service Called (Date)	Date Resolved
Vertical Measurement Accuracy						
Horizontal Measurement Accuracy						
Axial Resolution						
Lateral Resolution						
Penetration						
Functional Resolution						
Contrast Resolution						
Gray Scale Photography						

Assistance

Supplies/Accessories



DO NOT connect any probes or accessories without approval by GE.

Not all features or products described in this document may be available or cleared for sale in all markets.

The following supplies/accessories have been verified to be compatible with the system:

Peripherals

Table 18-10: Peripherals and Accessories

Accessory	Units
3.5" MO Drive	Each
Sony VCR Model SVO-9500MD/9500MDP	Each
Panasonic VCR Model AG-MD835/MD835P/MD835E	Each
Sony B/W Printer Model UP-895MDW	Each
Sony Digital B/W Printer Model UP-D895/897MD	Each
Sony Color Printer Model UP-2950MD/2850P	Each
Sony Color Printer Model UP-D21/23MD	Each
Sony Color Printer Model UP-50/55MD	Each

Console

Table 18-11: Console Accessories

Accessory	Units
Footswitch	Each
TV Probe Holder	Each
UPS (Battery Pack)	Each
4D Probe Holder	Each

Probes

Table 18-12: Probes and Accessories

Accessory	Units
3.5C	Each
3.5CS	Each
4C	Each
M7C	Each
E8C	Each
8C	Each
7L	Each
i739	Each
t739	Each
9L	Each
10L	Each
M12L	Each
3S	Each
4S	Each
10S	Each
i12L	Each
4D3C	Each
4DE7C	Each
4D8C	Each
4D10L	Each
4D16L	Each
P5D	Each
P8D	Each

VoiceScan Accessories

Table 18-13: Accessories

Accessory	Units
Beltpack Transmitter (WBT), H4903WC	Each
Battery Recharger, H4903WD	Each
Wireless Microphone, H4903WE	Each
Wired Microphone, H4903WK	Each
Wireless Microphone Care Kit, 2386142	Each

Gel

Table 18-14: Gel

Accessory	Units
Thermasonic Gel Warmer	Holds three plastic bottles (250ml or 8 oz)
Aquasonic 100 Scan Gel	5 liter jug
	250 ml plastic bottles (12/ case)
Scan Ultrasound Gel	8 oz plastic bottles (12/ case)
	1 gallon plastic jug
	Four 1-gallon plastic jugs

Disinfectant

Table 18-15: Disinfectant

Accessory	Units
Cidex Activated Dialdehyde	16/1 quart bottles
	4/1 gallon bottles
	2/2.5 gallon bottles

Ultrasound Probe and Cord Sheath Sets

Table 18-16: Probe and Cord Sheath Sets

Accessory	Units
Sterile Ultrasound Probe Sheath Set	20 Per Set
Sterile Ultrasound Cord Sheath Set	20 Per Set
Sanitary Rectal/Vaginal Probe Cover	20 Per Set
Sterile Combination Probe and Cord Cover Set	12 Per Set
Sterile Ultrasound Probe Sheath Set for Wide (2.5 and 3.5) Aperture Sector Probes	20 Per Set

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